

Gauge Theory Lecture Notes

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February 5, 2026

Contents

1	Path Integral Quantization for a Single Particle	1
1.1	Lagrangian mechanics	1
1.2	The idea of quantization by path integrals	2
1.3	Path integral for a single free particle	3
1.4	The Hamiltonian for a free particle	6
1.5	The probabilistic approach to quantum mechanics	9
1.6	Heuristics for the Schrödinger equation with a potential	13
1.7	The Wick rotation heuristic for a single particle system	17
1.8	Validity of the Wick rotation heuristic	19
2	Abstracting the Probabilistic Approach	27
2.1	The Markov process heuristic	27
2.2	The simple harmonic oscillator	31
2.3	Heuristic definition of vacuum expectation values	34
2.4	Rigorous definition of vacuum expectation values	36
2.5	Calculating vacuum expectation values	38
2.6	The two-point function for the simple harmonic oscillator	41
2.7	The Osterwalder–Schrader approach	42
3	The Free Quantum Field	47
3.1	Classical Klein–Gordon fields	47
3.2	Heuristic definition of the free quantum field	48
3.3	Wick rotation for the free quantum field	48
3.4	Rigorous construction of the Euclidean free field	50
3.5	Time slices of the Euclidean free field	57
3.6	Markov property of the Euclidean free field	59
3.7	Rigorous construction of the free quantum field	62
3.8	The Feynman propagator	64
3.9	An alternate form of the Feynman propagator	68
3.10	The Feynman propagator in momentum coordinates	71
3.11	Wightman functions of the free quantum field	72

4	From Fields to Particles	75
4.1	Lorentz transforms	75
4.2	Special relativity	76
4.3	The mass shell	79
4.4	Integration on mass shells	80
4.5	Fourier transform of the Euclidean free field	84
4.6	Bosonic Fock space	87
4.7	From free field to Fock space	88
4.8	Time evolution on Fock space	92
5	Perturbative Renormalization	95
5.1	The heuristic algorithm	95
5.2	A toy example	97
5.3	The paradoxical nature of perturbative renormalization	99
5.4	A rigorous criterion	100
5.5	An example and a counterexample	101
6	Heuristics for φ_4^4 Theory	103
6.1	Perturbative expansion of φ_4^4 theory	103
6.2	First order of perturbation theory	105
6.3	Wick rotation	107
6.4	Regularization by hard cutoff	108
6.5	Renormalizing to first order	109
6.6	Renormalized Wightman functions	110
7	Heuristics for φ_3^4 Theory	115
7.1	Perturbative expansion of φ_3^4 theory	115
7.2	First order of perturbation theory	116
7.3	Regularization by hard cutoff	118
7.4	Renormalizing the two-point function to all terms	119
7.5	Renormalization to all orders	122
7.6	Renormalized Wightman functions	123
8	The Yang–Mills Action	127
8.1	Yang–Mills action of a connection	127
8.2	Covariant derivative associated with a connection	128
8.3	The Yang–Mills equation	131
8.4	Pseudo-Riemannian manifolds	134
8.5	Maxwell’s equations from U(1) Yang–Mills theory	135
8.6	Gauge transformations	140

9	The Free Electromagnetic Field	147
9.1	The Lorenz gauge condition	147
9.2	Rigorous construction of the free electromagnetic field	150
9.3	The photon propagator in the Feynman gauge	153
9.4	Expected value of a gauge invariant observable	154
9.5	The photon propagator in momentum coordinates	155
10	Grassmann Fields	157
10.1	Grassmann variables	157
10.2	Differentiating Grassmann functions	158
10.3	Berezin integrals	158
10.4	Exponentials of Grassmann functions	159
10.5	Gaussian integration with Grassmann variables	160
10.6	Grassmann-valued Gaussian random variables	161
10.7	Heuristic description of the Grassmann-valued free field	164
10.8	Fermionic Fock space	165
10.9	Rigorous construction of the Grassmann-valued free field	165
11	Faddeev–Popov Ghosts	167
11.1	The heuristic	167
11.2	A rigorous formulation in finite dimensions	169
11.3	A toy example	176
12	Dimensional Regularization	179
12.1	The main idea	179
12.2	A key integral	180
12.3	Some examples	181
13	Heuristics for Pure Yang–Mills Theories	185
13.1	Gauge fixing and ghost fields	185
13.2	Covariant gauge fixing	189
13.3	The action in momentum coordinates	192
13.4	First order of perturbation theory	195
13.5	Second order of perturbation theory	198
13.6	The Feynman parameter	206
13.7	Dimensional regularization of Yang–Mills theories	208
13.8	Renormalization to first order	209
14	The Wightman Axioms	211
14.1	The Poincaré group	211
14.2	The framework of relativistic quantum mechanics	211
14.3	The Wightman axioms	212
14.4	Verifying the axioms for the free quantum field	214

15 Introduction to Lattice Gauge Theories	215
15.1 Euclidean Yang–Mills theories	215
15.2 Lattice gauge theories	216
15.3 Gauge transformations	221
15.4 Gauge invariant observables	224
15.5 Gauge fixing	227
15.6 Mass gap and quark confinement	229
15.7 Mass gap in 2D lattice gauge theories	230
15.8 Area law in 2D lattice gauge theories	232
16 Lattice Gauge Theories in $d \geq 3$	235
16.1 The strong and weak coupling regimes	235
16.2 Correlation decay under Dobrushin’s condition	235
16.3 Mass gap at strong coupling	239
16.4 Area law at strong coupling	240
16.5 Confinement in 3D lattice gauge theories	241
16.6 The Villain action	251
16.7 Deconfinement in 4D U(1) theory	258
A Measure Theory and Analysis	297
A.1 Basics of measure theory	297
A.2 L^p spaces	322
A.3 Fourier transforms	330
B Probability Theory	343
B.1 Basics of probability theory	343
B.2 Gaussian random variables	350
B.3 Some other distributions	359
B.4 Conditional probability	363
B.5 Markov chains and Markov processes	369
B.6 Brownian motion	377
C Differential Geometry	383
C.1 Review of topics from linear algebra	383
C.2 Review of basic differential geometry	387
C.3 Lie groups and Lie algebras	393
C.4 Differential forms	401
C.5 Principal bundles	409
C.6 Riemannian manifolds	411
C.7 Haar measure on matrix Lie groups	415
C.8 Discrete differential geometry	419

CONTENTS

D Functional Analysis	433
D.1 Some basic results	433
D.2 Self-adjoint operators	438
D.3 Stone's theorem	442
D.4 Semigroups of symmetric operators	454

Chapter 1

Path Integral Quantization for a Single Particle

1.1 Lagrangian mechanics

Suppose that we have a system of N particles, with masses $m_1, \dots, m_N$, interacting in such a way that if their positions are at $x_1, \dots, x_N \in \mathbb{R}^3$, then the potential energy of the system is a smooth function $V(x_1, \dots, x_N)$. Newton's equation of motion tells us that in the absence of outside influences, the trajectory of the system $\gamma(t) = (\gamma_1(t), \dots, \gamma_N(t))$ through time must satisfy the system of equations

$$m_i \ddot{\gamma}_i(t) = -\partial_i V(\gamma(t)), \quad i = 1, \dots, N, \quad (1.1.1)$$

where $\partial_i V$ is shorthand for $\frac{\partial V}{\partial x_i}$ and $\ddot{\gamma}_i$ is shorthand for $\frac{d^2 \gamma_i}{dt^2}$. (Note that $\ddot{\gamma}_i(t)$ and $\partial_i V(\gamma(t))$ are both elements of $\mathbb{R}^3$.) Lagrange's formulation of classical mechanics arrives at this system of equations in the following way. Given an arbitrary smooth trajectory $\gamma = (\gamma(t))_{-\infty < t < \infty}$ of the system (that is, γ is any smooth function from $\mathbb{R}$ into $(\mathbb{R}^3)^N$), the 'action' of the trajectory is defined as the formal integral

$$S(\gamma) := \int_{-\infty}^{\infty} \left(\frac{1}{2} \sum_{i=1}^N m_i \|\dot{\gamma}_i(t)\|^2 - V(\gamma(t)) \right) dt. \quad (1.1.2)$$

This integral is formal because it is typically not convergent for a given γ . However, we can still define the notion of trajectories that are critical points of this action, as follows. Suppose that γ and θ are two smooth trajectories that are equal outside a bounded time interval. Then although $S(\gamma)$ and $S(\theta)$ may not be well-defined, the difference $S(\gamma) - S(\theta)$ is well-defined, because the integrand is smooth and vanishes outside a bounded time interval. Thus, if η is a trajectory that satisfies $\eta(t) = 0$ outside a bounded time interval, then it makes sense to talk about the directional derivative

$$\lim_{\epsilon \rightarrow 0} \frac{S(\gamma + \epsilon \eta) - S(\gamma)}{\epsilon}.$$

In fact, a simple calculation shows that this limit exists and equals

$$\int_{-\infty}^{\infty} \left(\sum_{i=1}^N (m_i \dot{\gamma}_i(t) \cdot \dot{\eta}_i(t) - \eta_i(t) \cdot \partial_i V(\gamma(t))) \right) dt.$$

Applying integration by parts on the first term (together with the fact that η vanishes outside a bounded interval) shows that the above integral is equal to

$$- \int_{-\infty}^{\infty} \left(\sum_{i=1}^N \eta_i(t) \cdot (m_i \ddot{\gamma}_i(t) + \partial_i V(\gamma(t))) \right) dt.$$

A trajectory is a critical point of the action S if these directional derivatives vanish for all η . From the above expression, it is easy to show that a trajectory is a critical point if and only if it satisfies Newton's equations (1.1.1).

Instead of considering all possible paths, one can also consider the set of paths that start from some point $x_0 \in (\mathbb{R}^3)^N$ at time t_0 , or the set of paths that start from x_0 at time t_0 and end at x_1 at time t_1 . The equations of motion will be the same, together with these additional constraints.

For a detailed introduction to the mathematical aspects of Lagrangian mechanics, we refer to the classic text of Arnol'd [2, Part II].

1.2 The idea of quantization by path integrals

While the trajectory of a classical system is deterministic, quantum mechanics says that it is actually random; the randomness 'goes away' in the classical limit, where we look at systems that are so large that quantum effects are negligible. The process of 'converting' a classical system into a quantum system is called quantization. There are several equivalent approaches to quantization. Arguably the most useful approach is quantization by path integrals. The rough idea is as follows.

Instead of the probability density functions that we have in classical probability theory, the randomness in quantum systems is captured by *probability amplitudes*. For example, in the N -particle system described in the previous section, the state of the system at some given point in time would be described by a complex-valued function $\psi \in L^2((\mathbb{R}^3)^N)$.¹ Strictly speaking, the state of the system is not the function ψ itself, but the equivalence class of all functions that are nonzero scalar multiples of ψ . This is true for a general quantum system: The states of the system are *rays* (i.e., one-dimensional subspaces sans the origin) of a Hilbert space $\mathcal{H}$. In practice we do not explicitly make this distinction, and treat states as points in $\mathcal{H}$ instead of rays.

The value $\psi(x)$ is called the probability amplitude at x . For an N -particle system in $\mathbb{R}^3$, the interpretation of the state ψ is that the probability density function of the system is proportional to the function $|\psi(x)|^2$. That is, the probability that the system is in some set

¹See Section A.2.3 in Appendix A for an introduction to L^p spaces.

$A \subseteq (\mathbb{R}^3)^N$ is given by the integral of $|\psi(x)|^2$ over A , divided by its integral over the whole of $(\mathbb{R}^3)^N$. Note that if ψ_1 and ψ_2 are nonzero scalar multiples of each other, then they give rise to the same probability density function. Thus, there is a well-defined probability density function corresponding to any state.

The path integral formulation of quantum mechanics says the following. Suppose that we are given an action S for the trajectories of an N -particle system, such as the one displayed in equation (1.1.2). Suppose that the system is in a state ψ_0 at time 0, as above. Then at any time $t > 0$, the state of the system is the function

$$\psi_t(x) = \frac{1}{Z(t)} \int_{\gamma: \gamma(t)=x} \psi_0(\gamma(t_0)) e^{\frac{i}{\hbar} S(\gamma)} \mathcal{D}\gamma, \quad (1.2.1)$$

where $\int_{\gamma: \gamma(t)=x} \dots \mathcal{D}\gamma$ is some notion of ‘integration over the space of trajectories’ $\gamma = (\gamma(s))_{0 \leq s \leq t}$ that are at x at time t , $\hbar$ is Planck’s constant, $i = \sqrt{-1}$, and $Z(t)$ is a time-varying ‘normalization factor’. The value of $Z(t)$ is not important, because any value will give the same state at time t . Typically, $Z(t)$ is chosen such that the L^2 norm of ψ_t remains the same for all t .

The above definition usually suffers from severe mathematical difficulties, because integration on the space of trajectories is typically hard to define, and even if it can be defined, integrating an oscillatory function like $e^{\frac{i}{\hbar} S(\gamma)}$ on this space is often outside the reach of rigorous mathematics.

The heuristic justification of quantum systems approaching a classical limit is that when we are working in units where Planck’s constant $\hbar$ is very small, rapid oscillations force the integration to concentrate around trajectories that are critical points of the action. Again, this is usually extremely hard to justify rigorously.

For an introduction to path integrals and the way they are used in physics, one can look in the comprehensive treatise by Kleinert [14]. For rigorous mathematical treatments, some basic references are the books by Glimm and Jaffe [10], Simon [28], and Albeverio et al. [1].

1.3 Path integral for a single free particle

As the simplest example, consider a single particle on the real line (instead of $\mathbb{R}^3$), with mass 1, unaffected by any external force. By the formula (1.1.2), the action of a trajectory γ from time 0 to time $t > 0$ is then

$$S(\gamma) = \frac{1}{2} \int_0^t \dot{\gamma}(t)^2 dt.$$

Thus, the heuristic formula (1.2.1) becomes

$$\psi_t(x) = \frac{1}{Z(t)} \int_{\gamma: \gamma(t)=x} \psi_0(\gamma(0)) \exp\left(\frac{i}{2\hbar} \int_0^t \dot{\gamma}(t)^2 dt\right) \mathcal{D}\gamma, \quad (1.3.1)$$

where we want to choose $Z(t)$ such that the L^2 norm of $\psi(t, \cdot)$ is invariant.

We will now make sense of this in a rigorous way. The first step is to discretize time as $0 = t_0 < t_1 < \dots < t_n = t$. A path γ with $\gamma(t) = x$ is then just a sequence of points $x_j = \gamma(t_j)$, $j = 0, \dots, n$, with $x_n = x$. The time derivative $\dot{\gamma}(t_j)$ should be replaced by the discrete time derivative

$$\frac{x_{j+1} - x_j}{t_{j+1} - t_j}.$$

Lastly, the integral $\mathcal{D}\gamma$ should be replaced by integration over $x_0, x_1, \dots, x_{n-1}$, leaving out x_n since it is fixed to be equal to x . With these substitutions, the discretized version of the path integral displayed in equation (1.3.1) becomes

$$\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \psi_0(x_0) \exp\left(\frac{i}{2\hbar} \sum_{j=0}^{n-1} \left(\frac{x_{j+1} - x_j}{t_{j+1} - t_j}\right)^2 (t_{j+1} - t_j)\right) dx_0 \dots dx_{n-1},$$

with $x_n := x$. But this is still not a well-defined integral, since the integrand does not decay at infinity in the variables $x_1, \dots, x_{n-1}$. We can fix this, for example, by replacing the i in the exponent by $i - \epsilon$, where $\epsilon > 0$ is a small regularizing parameter that we will eventually send to zero. With this, we finally arrive at a well-defined regularized and discretized path integral

$$I_n(\epsilon) := \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \psi_0(x_0) \exp\left(\frac{i - \epsilon}{2\hbar} \sum_{j=0}^{n-1} \frac{(x_{j+1} - x_j)^2}{s_j}\right) dx_0 \dots dx_{n-1},$$

where $s_j := t_{j+1} - t_j$, and the subscript in I_n denotes that we are working with the partition $t_0, \dots, t_n$ of $[t_0, t]$. To evaluate this integral, let us assume that the Fourier transform $\widehat{\psi}_0$ of ψ_0 is in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.² Then by the Fourier inversion formula,

$$\psi_0(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{\psi}_0(p) e^{-ipx} dp.$$

Let $\mathcal{F}$ denote the set of all ψ_0 as above. Let us substitute this expression for $\psi_0(x)$ in the previous display. Then by Fubini's theorem³ (which applies since $\widehat{\psi}_0 \in L^1(\mathbb{R})$), we get

$$I_n(\epsilon) := \frac{1}{2\pi} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \widehat{\psi}_0(p) \exp\left(-ipx_0 + \frac{i - \epsilon}{2\hbar} \sum_{j=0}^{n-1} \frac{(x_{j+1} - x_j)^2}{s_j}\right) dx_0 \dots dx_{n-1} dp.$$

Applying the change of variable $y_j := x_{j+1} - x_j$ for $j = 0, 1, \dots, n-1$ in the above integral,

²See Section A.3 in Appendix A for an introduction to Fourier transforms.

³See Section A.1.16 in Appendix A.

we get

$$\begin{aligned}
 I_n(\epsilon) &:= \frac{1}{2\pi} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \widehat{\psi}_0(p) \exp\left(-ipx + ip \sum_{j=0}^{n-1} y_j + \frac{i-\epsilon}{2\hbar} \sum_{j=0}^{n-1} \frac{y_j^2}{s_j}\right) dy_0 \cdots dy_{n-1} dp \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{\psi}_0(p) e^{-ipx} \prod_{j=0}^{n-1} \left(\int_{-\infty}^{\infty} \exp\left(ip y + \frac{(i-\epsilon)y^2}{2\hbar s_j}\right) dy \right) dp.
 \end{aligned} \tag{1.3.2}$$

The inner integral above is a ‘Gaussian integral’ that can be evaluated by standard techniques. For example, Lemma A.3.2 in Appendix A gives

$$\int_{-\infty}^{\infty} \exp\left(ip y + \frac{(i-\epsilon)y^2}{2\hbar s_j}\right) dy = \exp\left(-\frac{p^2(\epsilon+i)}{2(\epsilon^2+1)} \hbar s_j\right).$$

Thus,

$$I_n(\epsilon) = \frac{1}{2\pi} C(\epsilon, \hbar, s_0, \dots, s_{n-1}) \int_{-\infty}^{\infty} \widehat{\psi}_0(p) e^{-ipx} \exp\left(-\frac{p^2(\epsilon+i)}{2(\epsilon^2+1)} \hbar \sum_{j=0}^{n-1} s_j\right) dp,$$

where $C(\epsilon, \hbar, s_0, \dots, s_{n-1})$ is a nonzero complex number that depends on $\epsilon, \hbar, s_0, \dots, s_{n-1}$, but not on p or x . Let us now take the discrete version of the normalization factor $Z(t)$ to be this number. Observing that $\sum_{j=0}^{n-1} s_j = t$, this shows that the discrete version of the right side in equation (1.3.1) is equal to

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{\psi}_0(p) e^{-ipx} \exp\left(-\frac{p^2(\epsilon+i)}{2(\epsilon^2+1)} \hbar t\right) dp.$$

Thus, it is now easy to take ϵ to 0 (using the dominated convergence theorem⁴) and declare that

$$\psi_t(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{\psi}_0(p) e^{-ipx - \frac{1}{2}ip^2 \hbar t} dp.$$

Note that this expression was obtained only for $\psi_0 \in \mathcal{F}$, that is, for ψ_0 such that $\widehat{\psi}_0 \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Let $U(t)$ denote the operator that takes ψ_0 to ψ_t . That is, for any $t \in \mathbb{R}$ and $\psi \in \mathcal{F}$, let

$$(U(t)\psi)(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{\psi}(p) e^{-ipx - \frac{1}{2}ip^2 \hbar t} dp.$$

By the above expression, it is clear that for each t , $U(t)$ is a linear operator from $\mathcal{F}$ into $\mathcal{F}$, because

$$\widehat{U(t)\psi}(p) = \widehat{\psi}(p) e^{-ip^2 \hbar t/2}. \tag{1.3.3}$$

We claim that $U(t)$ preserves inner products on $\mathcal{F}$ under the usual L^2 inner product. To

⁴See Section A.1.13 in Appendix A

see this, note that by the Plancherel identity⁵ (which holds on $\mathcal{F}$),

$$\begin{aligned} \int_{-\infty}^{\infty} U(t)f(x)\overline{U(t)g(x)}dx &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{U(t)f(p)}\overline{\widehat{U(t)g(p)}}dp \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{f(p)}e^{-ip^2\hbar t/4}\overline{\widehat{g(p)}e^{-ip^2\hbar t/4}}dp \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{f(p)}\overline{\widehat{g(p)}}dp = \int_{-\infty}^{\infty} f(x)\overline{g(x)}dx. \end{aligned}$$

This shows that $U(t)$ preserves inner products on $\mathcal{F}$, and therefore, is an isometry. It is not hard to show that the function class $\mathcal{F}$ is dense in $L^2(\mathbb{R})$. This implies that $U(t)$ can be extended to an isometry on the whole of $L^2(\mathbb{R})$. Now recall that a linear map from a Hilbert space into itself is called ‘unitary’ if it preserves inner products and is surjective. From the formula (1.3.3), it is easy to see that $U(t)$ is bijection from $\mathcal{F}$ on to $\mathcal{F}$, with inverse equal to $U(-t)$. This fact, together with the isometric nature of $U(-t)$ and the fact that $\mathcal{F}$ is dense in $L^2(\mathbb{R})$, shows that $U(t)$ maps $L^2(\mathbb{R})$ surjectively on to itself. Thus, $U(t)$ is a unitary operator.

One final remark is that $(U(t))_{t \in \mathbb{R}}$ is a *group* of unitary operators, in the sense that $U(s+t) = U(s)U(t)$ for all $s, t \in \mathbb{R}$. This is clear from the formula (1.3.3).

The construction of the above quantum mechanical system of a single free particle is standard fare, present in every introductory text on quantum mechanics. The presentations are, however, not always rigorously formulated as above. A classic reference for rigorous formulations of quantum mechanical systems in the spirit of the presentation in this section is Reed and Simon [26, Chapter X].

1.4 The Hamiltonian for a free particle

Let us recapitulate what we learned about a single free particle on the real line:

- There is a Hilbert space $\mathcal{H}$ of states, in this case $\mathcal{H} = L^2(\mathbb{R})$. If the particle is in a state $\psi \in \mathcal{H}$, then the probability density of its location is proportional to $|\psi(x)|^2$.
- There is a collection of unitary operators $(U(t))_{t \in \mathbb{R}}$ on $\mathcal{H}$, which form a group in the sense that $U(s+t) = U(s)U(t)$ for all $s, t \in \mathbb{R}$. In the case of the free particle, $U(t)$ is defined through the formula (1.3.3) on a dense subspace of $L^2(\mathbb{R})$, and extended to the whole of $L^2(\mathbb{R})$ by isometry.
- This group of unitary operators interacts with the quantum system in the following way: If the system is in state ψ at time s , then it is in state $U(t)\psi$ at time $s+t$.

The above framework is the setup for any quantum system, as formulated in the framework of rigorous mathematical physics [26, Chapter X]. There is a space of states $\mathcal{H}$, which is a Hilbert space. There is a group of unitary operators $(U(t))_{t \in \mathbb{R}}$. If the system is in state ψ

⁵See Section A.3 in Appendix A.

at time s , then it is in state $U(t)\psi$ at time $s + t$. Usually, if the system can be represented by a point in some set $\mathcal{S}$ at any point of time, $\mathcal{H}$ is some Hilbert space of complex-valued functions on $\mathcal{S}$. In the case of the free particle, $\mathcal{S} = \mathbb{R}$ and $\mathcal{H} = L^2(\mathbb{R})$. There is also usually some measure⁶ μ on $\mathcal{S}$, and the interpretation of a state ψ is that if the system is in a state ψ , then its probability density function with respect to μ is proportional to $|\psi(x)|^2$. In the case of the free particle, μ is Lebesgue measure on $\mathbb{R}$.

Given a group of unitary operators $(U(t))_{t \in \mathbb{R}}$ as above on a Hilbert space $\mathcal{H}$, it is called *strongly continuous* if for every $x \in \mathcal{H}$, the map $t \mapsto U(t)x$ is continuous. It is easy to show that the unitary group for the free particle is strongly continuous. A very important theorem that we will use repeatedly is the following, which we quote from Appendix D, Section D.3.

Theorem D.3.1 (Stone's theorem). *Let $(U(t))_{t \in \mathbb{R}}$ be a strongly continuous one parameter group of unitary operators on a Hilbert space $\mathcal{H}$. Let*

$$\mathcal{D} := \left\{ x \in \mathcal{H} : \lim_{t \rightarrow 0} \frac{U(t)x - x}{-it} \text{ exists} \right\}.$$

Then $\mathcal{D}$ is a dense subspace on $\mathcal{H}$. Furthermore, if we define an operator $H : \mathcal{D} \rightarrow \mathcal{H}$ as the limit displayed above, then H is self-adjoint. Conversely, given any densely defined self-adjoint operator H on $\mathcal{H}$, there is a unique strongly continuous one parameter group of unitary operators $(U(t))_{t \in \mathbb{R}}$ such that H and $\mathcal{D}$ arise from this group in the above manner, with $\mathcal{D} = \mathcal{D}(H)$.

The operator H obtained in Stone's theorem is called the Hamiltonian operator for the group $(U(t))_{t \in \mathbb{R}}$. The set $\mathcal{D}$ is called the 'domain' of H . If $x \in \mathcal{D}$, then it is easy to see that for any $t \in \mathbb{R}$, the point $x(t) := U(t)x$ is also in $\mathcal{D}$. Moreover, it satisfies the *Schrödinger equation*

$$i \frac{d}{dt} x(t) = Hx(t), \tag{1.4.1}$$

where the derivative on the left side is defined as

$$\lim_{s \rightarrow 0} \frac{1}{s} (x(t+s) - x(t)).$$

Let us work out the Hamiltonian for a free particle. Recall the class of functions $\mathcal{F}$ defined in the previous section. Take any $\psi \in \mathcal{F}$, so that $\hat{\psi} \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. For each $t \in \mathbb{R}$, let $\psi_t := U(t)\psi$, where $(U(t))_{t \in \mathbb{R}}$ is the group of unitary operators for a free particle. Recall that ψ_t is also in $\mathcal{F}$, and by equation (1.3.3),

$$\psi_t(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\psi}(p) e^{-ipx - ip^2 \hbar t / 2} dp.$$

Let $\mathcal{F}_0$ consist of all $\psi \in \mathcal{F}$ which satisfy the further condition that $\hat{\psi}$ is compactly supported.

⁶See Appendix A for an introduction to measure theory.

For $\psi \in \mathcal{F}_0$, it is easy to show using the dominated convergence theorem that

$$\frac{\psi_t(x) - \psi_0(x)}{-it} \xrightarrow{L^2} \frac{\hbar}{4\pi} \int_{-\infty}^{\infty} p^2 \widehat{\psi}(p) e^{-ipx} dp$$

as $t \rightarrow 0$. Moreover, again by the dominated convergence theorem, ψ is a smooth function and

$$\frac{d^2\psi}{dx^2} = -\frac{1}{2\pi} \int_{-\infty}^{\infty} p^2 \widehat{\psi}(p) e^{-ipx} dp.$$

Combining, we see that the Hamiltonian operator for a single free particle is given by

$$H = -\frac{\hbar}{2} \frac{d^2}{dx^2} \quad (1.4.2)$$

on the space of functions $\mathcal{F}_0$. Writing $\psi_t(x)$ as $\psi(t, x)$, it is easy to show that $(t, x) \mapsto \psi_t(x)$ is smooth, and the Schrödinger equation (1.4.1) becomes

$$i \frac{\partial \psi}{\partial t} = -\frac{\hbar}{2} \frac{\partial^2 \psi}{\partial x^2},$$

which is the familiar form of the Schrödinger equation for a particle with mass 1. Incidentally, notice that when we are working in units where Planck's constant $\hbar$ is close to 0, the above equation implies that ψ remains nearly unchanging over time. This reflects the fact that in the classical world, the state of a system does not change in the absence of external forces.

One important thing to note is that the Hamiltonian for a free particle is *not* the operator H displayed in equation (1.4.2). The operator displayed in equation (1.4.2) is the restriction of the Hamiltonian to the subspace $\mathcal{F}_0$. The actual Hamiltonian is the operator H defined on the subspace $\mathcal{D}$ consisting of all $\psi \in L^2(\mathbb{R})$ such that the limit of $(U(t)\psi - \psi)/(-it)$ exists in $L^2(\mathbb{R})$ as $t \rightarrow 0$, and this limit gives the value of $H\psi$. It just happens that $\mathcal{F}_0 \subseteq \mathcal{D}$, and the restriction of H to $\mathcal{F}_0$ takes the form given in equation (1.4.2). The full domain $\mathcal{D}$ is given by the following lemma.

Lemma 1.4.1. *The domain $\mathcal{D}$ of the generator of the group $(U(t))_{t \in \mathbb{R}}$ consists of all $\psi \in L^2(\mathbb{R})$ such that the Fourier transform $\widehat{\psi}$ satisfies*

$$\int_{-\infty}^{\infty} p^4 |\widehat{\psi}(p)|^2 dp < \infty. \quad (1.4.3)$$

For $\psi \in \mathcal{D}$, H acts on ψ as

$$\widehat{H\psi}(p) = -\frac{1}{2} p^2 \hbar \widehat{\psi}(p). \quad (1.4.4)$$

Proof. Let $\mathcal{D}$ be the set described in the statement of the lemma. Take any $\psi \in \mathcal{D}$. It is easy to see from equation (1.3.3) and isometry that $U(t)$ acts on the whole of $L^2(\mathbb{R})$ by the formula displayed in equation (1.3.3). Thus, for any $t \neq 0$, if we let

$$v_t := \frac{U(t)\psi - \psi}{-it},$$

then

$$\widehat{v}_t(p) = \widehat{\psi}(p) \frac{e^{-ip^2\hbar t/2} - 1}{-it}.$$

By the condition (1.4.3), there is a function $v \in L^2(\mathbb{R})$ such that

$$\widehat{v}(p) = -\frac{1}{2}p^2\hbar\widehat{\psi}(p).$$

Now note that

$$\begin{aligned} \|v_t - v\|_{L^2}^2 &= \frac{1}{2\pi} \|\widehat{v}_t - \widehat{v}\|_{L^2}^2 \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |\widehat{\psi}(p)|^2 \left| \frac{e^{-ip^2\hbar t/2} - 1}{-it} + \frac{1}{2}p^2\hbar \right|^2 dp. \end{aligned}$$

The integrand in the last line goes to zero as $t \rightarrow 0$ for each fixed p . Also, it is bounded by $C|\widehat{\psi}(p)|^2 p^4$ for some constant C that does not depend on p . Thus, by the condition (1.4.3) and the dominated convergence theorem, the integral goes to zero as $t \rightarrow 0$. This shows that ψ is in the domain of H and $H\psi$ is given by the formula (1.4.4). This proves one direction of the theorem.

For the converse direction, take any ψ in the domain of H . Define v_t as above. Then v_t converges to a limit in L^2 as $t \rightarrow 0$. Consequently, $\widehat{v}_t$ must also be converging to a limit in L^2 . But the formulas for $\widehat{v}_t$ and $\widehat{v}$ displayed above shows that the limit must necessarily be $\widehat{v}$. In particular, $\widehat{v} \in L^2$, which shows that (1.4.3) holds. Thus, $\psi \in \mathcal{D}$ and $H\psi$ is given by the formula (1.4.4). $\square$

1.5 The probabilistic approach to quantum mechanics

As mentioned earlier, a quantum mechanical system consists of a Hilbert space $\mathcal{H}$ and a strongly continuous one parameter group of unitary operators $(U(t))_{t \in \mathbb{R}}$. The probabilistic approach to constructing a quantum mechanical system consists of the following steps:

1. Define a strongly continuous semigroup of symmetric operators $(P(t))_{t \geq 0}$ on $\mathcal{H}$ using a stochastic process, such that its generator H is the Hamiltonian for the quantum system that we want to construct.
2. Deduce by the Hille–Yosida theorem that H is self-adjoint. This is the content of Theorem D.4.1 in Section D.4 of Appendix D. We reproduce it here for the reader's convenience:

Theorem D.4.1 (One direction of the Hille–Yosida theorem). *Let $(P(t))_{t \geq 0}$ be a strongly continuous semigroup of symmetric operators defined everywhere on a Hilbert space $\mathcal{H}$. Then the set*

$$\mathcal{D} := \left\{ x \in \mathcal{H} : \lim_{t \downarrow 0} \frac{P(t)x - x}{-t} \text{ exists} \right\}$$

is a dense subspace of $\mathcal{H}$. Moreover, if we denote the above limit by Hx for $x \in \mathcal{D}$, then H is a self-adjoint linear operator with domain $\mathcal{D}$.

3. Invoke Stone's theorem to construct the one parameter unitary group $(U(t))_{t \in \mathbb{R}}$ with Hamiltonian H .

The above process is a rigorous version of what is usually called 'Wick rotation' in physics. This is because $P(t) = e^{-tH}$, whereas $U(t) = e^{-itH}$, and the combination of Theorem D.4.1 and Stone's theorem therefore gives a way to 'replace' t by it to generate $U(t)$ using $P(t)$. For more on the probabilistic approach, see Reed and Simon [26, Chapter X].

For the simplest application, consider the case of a free particle on the real line, for which we already know the answer from previous discussion. For simplicity, let us assume that we are working in units where $\hbar = 1$. For each $t \geq 0$, define a linear operator $P(t)$ on $L^2(\mathbb{R})$ as

$$P(t)\psi(x) := \mathbb{E}(\psi(x + \sqrt{t}Z)),$$

where Z is a standard normal (Gaussian) random variable.⁷ It is easy to see that $P(t)$ is well-defined, because

$$\begin{aligned} \mathbb{E}|\psi(x + \sqrt{t}Z)| &\leq \sqrt{\mathbb{E}|\psi(x + \sqrt{t}Z)|^2} \\ &= \left(\int_{-\infty}^{\infty} |\psi(x + \sqrt{t}z)|^2 \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \right)^{1/2} \\ &\leq (2\pi)^{-1/4} \|\psi\|_{L^2} < \infty. \end{aligned}$$

Moreover, $P(t)$ maps L^2 into L^2 , since applying Fubini's theorem in the third line below, we have

$$\begin{aligned} \|P(t)\psi\|_{L^2}^2 &= \int_{-\infty}^{\infty} |P(t)\psi(x)|^2 dx = \int_{-\infty}^{\infty} |\mathbb{E}(\psi(x + \sqrt{t}Z))|^2 dx \\ &\leq \int_{-\infty}^{\infty} \mathbb{E}|\psi(x + \sqrt{t}Z)|^2 dx \\ &= \mathbb{E} \left(\int_{-\infty}^{\infty} |\psi(x + \sqrt{t}Z)|^2 dx \right) \\ &= \|\psi\|_{L^2}^2. \end{aligned} \tag{1.5.1}$$

Next, let us show that for each t , $P(t)$ is a symmetric operator. Take any $\psi, \phi \in L^2(\mathbb{R})$.

⁷See Section B.2.1 in Appendix B for an introduction to Gaussian random variables.

Then again by Fubini's theorem,

$$\begin{aligned}
 \langle P(t)\psi, \phi \rangle &= \int_{-\infty}^{\infty} P(t)\psi(x)\overline{\phi(x)}dx \\
 &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \psi(x + \sqrt{t}z) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \right) \overline{\phi(x)}dx \\
 &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \psi(x + \sqrt{t}z) \overline{\phi(x)}dx \right) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz.
 \end{aligned}$$

Now note that for any z ,

$$\int_{-\infty}^{\infty} \psi(x + \sqrt{t}z) \overline{\phi(x)}dx = \int_{-\infty}^{\infty} \psi(y) \overline{\phi(y - \sqrt{t}z)}dy.$$

Plugging this into the previous display and once again applying Fubini's theorem, we get

$$\begin{aligned}
 \langle P(t)\psi, \phi \rangle &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \psi(y) \overline{\phi(y - \sqrt{t}z)}dy \right) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \\
 &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \overline{\phi(y - \sqrt{t}z)} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \right) \psi(y)dy.
 \end{aligned}$$

But since the distribution of Z is symmetric around 0, we have

$$\begin{aligned}
 \int_{-\infty}^{\infty} \overline{\phi(y - \sqrt{t}z)} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz &= \overline{\mathbb{E}(\phi(y - \sqrt{t}Z))} \\
 &= \overline{\mathbb{E}(\psi(y + \sqrt{t}Z))} = \overline{P(t)\phi(y)}.
 \end{aligned}$$

Plugging this into the previous display, we get

$$\langle P(t)\psi, \phi \rangle = \int_{-\infty}^{\infty} \psi(y) \overline{P(t)\phi(y)}dy = \langle \psi, P(t)\phi \rangle.$$

Thus, $P(t)$ is a symmetric operator.

Clearly, $P(0) = I$. Our next claim is that $P(s+t) = P(s)P(t)$ for any $s, t \geq 0$. To see this, recall that from the basic properties of normal random variables⁸, $\sqrt{s+t}Z$ has the same distribution as $\sqrt{s}Z_1 + \sqrt{t}Z_2$, where Z_1, Z_2 are independent standard normal random variables. Thus,

$$\begin{aligned}
 P(s)P(t)\psi(x) &= \mathbb{E}(P(t)\psi(x + \sqrt{s}Z)) \\
 &= \int_{-\infty}^{\infty} P(t)\psi(x + \sqrt{s}z) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \\
 &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \psi(x + \sqrt{s}z + \sqrt{t}w) \frac{e^{-w^2/2}}{\sqrt{2\pi}} dw \right) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \\
 &= \mathbb{E}(\psi(x + \sqrt{s}Z_1 + \sqrt{t}Z_2)) = \mathbb{E}(\psi(x + \sqrt{s+t}Z)) = P(s+t)\psi(x).
 \end{aligned}$$

⁸See Section B.2.4 in Appendix B

Finally, we claim that $(P(t))_{t \geq 0}$ is a strongly continuous semigroup. To see this, recall the subspace $\mathcal{F}$ of all $\psi \in L^2(\mathbb{R})$ such that $\widehat{\psi} \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Then note that for any $\psi \in \mathcal{F}$, $p \in \mathbb{R}$ and $t \geq 0$,

$$\begin{aligned} P(t)\psi(x) &= \int_{-\infty}^{\infty} \psi(x + \sqrt{t}z) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \\ &= \frac{1}{(2\pi)^{3/2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \widehat{\psi}(p) e^{-ip(x + \sqrt{t}z) - z^2/2} dp dz \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{\psi}(p) e^{-ipx - p^2 t/2} dp. \end{aligned}$$

Thus, $P(t)\psi \in \mathcal{F}$ and

$$\widehat{P(t)\psi}(p) = \widehat{\psi}(p) e^{-p^2 t/2}. \quad (1.5.2)$$

By the Plancherel identity and the dominated convergence theorem, this shows that $t \mapsto P(t)\psi$ is a continuous map for any $\psi \in \mathcal{F}$. To show that this holds for all $\psi \in L^2(\mathbb{R})$, recall the inequality (1.5.1), which shows that $\|P(t)\| \leq 1$ for all t . Take any $\psi \in L^2(\mathbb{R})$ and any $\epsilon > 0$. Since $\mathcal{F}$ is dense in $L^2(\mathbb{R})$, we can find $\phi \in \mathcal{F}$ such that $\|\psi - \phi\|_{L^2} < \epsilon$. Then for any $s, t \geq 0$,

$$\begin{aligned} \|P(s)\psi - P(t)\psi\|_{L^2} &\leq \|P(s)\psi - P(s)\phi\|_{L^2} + \|P(s)\phi - P(t)\phi\|_{L^2} + \|P(t)\phi - P(t)\psi\|_{L^2} \\ &\leq 2\epsilon + \|P(s)\phi - P(t)\phi\|_{L^2}. \end{aligned}$$

Since $\phi \in \mathcal{F}$, this shows that for any sequence of nonnegative real numbers $t_n \rightarrow t$,

$$\limsup_{n \rightarrow \infty} \|P(t_n)\psi - P(t)\psi\|_{L^2} \leq 2\epsilon.$$

Since ϵ is arbitrary, this proves the strong continuity of the semigroup $(P(t))_{t \geq 0}$. Thus, we can now conclude that the generator H of this semigroup is self-adjoint, by Theorem D.4.1. By the identity (1.5.2), it is easy to show that for $\psi \in \mathcal{F}$,

$$H\psi = -\frac{1}{2} \frac{d^2\psi}{dx^2}.$$

By Stone's theorem, this Hamiltonian generates a one parameter group of unitary operators. Note that on $\mathcal{F}$, this H is the same as the H that we obtained in Section 1.4 as the Hamiltonian for a single free particle on the real line (with $\hbar = 1$). But are these two operators really the same, that is, do they have the same domain and agree on that domain? Indeed, this is true. The easiest way to show this is to mimic the proof of Lemma 1.4.1 and identify the domain of the H defined above, and then observe that this is the same domain as the one identified in Lemma 1.4.1.

The way we obtained H using a ‘probabilistic approach’ seems somewhat ad hoc. There is in fact a more systematic way to get such constructions, using the approach of ‘Euclidean

path integrals'. These are rigorously defined objects that we will discuss in the next chapter.

1.6 Heuristics for the Schrödinger equation with a potential

In this section, we will 'derive' the Hamiltonian for a quantum mechanical system of a single particle in a potential V on the real line, from the path integral representation of its evolution. The entire derivation will be heuristic. The purpose of this exercise is to illustrate how the Hamiltonian arises from the path integral, and then, to formulate a plan to rigorously obtain the Hamiltonian as the generator of a stochastic process, so that the analogous quantum system can be rigorously constructed via Stone's theorem.

Consider a particles on the real line with mass 1, moving under the influence of a potential $V : \mathbb{R} \rightarrow \mathbb{R}$. The action of a trajectory $\gamma : [0, t] \rightarrow \mathbb{R}$ is

$$S(\gamma) = \int_0^t \left(\frac{1}{2} \dot{\gamma}(s)^2 - V(\gamma(s)) \right) ds. \quad (1.6.1)$$

Suppose that we are working with units in which $\hbar = 1$. If this is a quantum mechanical system whose state is $\psi_0 \in L^2(\mathbb{R})$ at time 0, then its state at time t , according to the formula (1.2.1), is heuristically given by

$$\psi_t(x) = \frac{1}{Z(t)} \int_{\gamma: \gamma(t)=x} \psi_0(\gamma(0)) e^{iS(\gamma)} \mathcal{D}\gamma,$$

where $\int_{\gamma: \gamma(t)=x} \dots \mathcal{D}\gamma$ is some notion of integration over the space of trajectories that are at x at time t , and $Z(t)$ is a time-varying 'normalization factor' whose value is unimportant because two states that are nonzero scalar multiples of each other are equivalent.

Note that by reversing the direction of time when parametrizing the paths in the path integral, we may equivalently write $\psi_t(x)$ as

$$\psi_t(x) = \frac{1}{Z(t)} \int_{\gamma: \gamma(0)=x} \psi_0(\gamma(t)) e^{iS(\gamma)} \mathcal{D}\gamma, \quad (1.6.2)$$

where the integral is over all paths $\gamma : [0, t] \rightarrow \mathbb{R}$ such that $\gamma(0) = x$. We will henceforth adopt this alternate expression as the definition of our path integrals.

To compute the Hamiltonian for this evolution, we first have to choose $Z(t)$ such that the evolution is generated by a one parameter group of unitary operators. We will now show, heuristically, that this is achieved by choosing

$$Z(t) = \int_{\gamma: \gamma(0)=0} e^{iS_0(\gamma)} \mathcal{D}\gamma,$$

where, for a trajectory $\gamma : [0, t] \rightarrow \mathbb{R}$, we define

$$S_0(\gamma) := \frac{1}{2} \int_0^t \dot{\gamma}(s)^2 ds. \quad (1.6.3)$$

Let $U(t)$ be the map that sends ψ_0 to ψ_t , with the above choice of $Z(t)$. First, let us sketch why it is reasonable to believe that $U(s+t) = U(s)U(t)$. For simplicity, let us consider only the case $s, t \geq 0$. A path $\gamma : [0, t+s] \rightarrow \mathbb{R}$ that starts at 0 at time 0 can be decomposed as a concatenation of two paths $\alpha : [0, t] \rightarrow \mathbb{R}$ and $\beta : [t, t+s] \rightarrow \mathbb{R}$, where $\alpha(0) = 0$ and $\beta(t) = \alpha(t)$. Let

$$\theta(u) := \beta(t+u) - \alpha(t)$$

for $t \leq u \leq t+s$, so that $\theta : [0, s] \rightarrow \mathbb{R}$ starts at 0 at time 0. It is easy to see that $\dot{\theta}(u) = \dot{\beta}(t+u)$ for all $u \in [0, s]$, and hence, $S_0(\beta) = S_0(\theta)$. Thus,

$$\begin{aligned} Z(t+s) &= \int_{\alpha: \alpha(0)=0} \int_{\beta: \beta(t)=\alpha(t)} e^{iS_0(\alpha)+iS_0(\beta)} \mathcal{D}\alpha \mathcal{D}\beta \\ &= \int_{\alpha: \alpha(0)=0} \int_{\theta: \theta(0)=0} e^{iS_0(\alpha)+iS_0(\theta)} \mathcal{D}\alpha \mathcal{D}\theta \\ &= \left(\int_{\alpha: \alpha(0)=0} e^{iS_0(\alpha)} \mathcal{D}\alpha \right) \left(\int_{\theta: \theta(0)=0} e^{iS_0(\theta)} \mathcal{D}\theta \right) \\ &= Z(t)Z(s). \end{aligned}$$

Next, let $Q(t)$ be the map that sends a function ψ to

$$Q(t)\psi(x) := \int_{\gamma: \gamma(0)=x} \psi(\gamma(t)) e^{iS(\gamma)} \mathcal{D}\gamma,$$

where the integral is over all paths $\gamma : [0, t] \rightarrow \mathbb{R}$ with $\gamma(0) = x$. Again, by a decomposition of $\gamma : [0, t+s] \rightarrow \mathbb{R}^N$ into α and β as above, we get

$$\begin{aligned} Q(t+s)\psi(x) &= \int_{\alpha: \alpha(0)=x} \int_{\beta: \beta(t)=\alpha(t)} \psi(\beta(t+s)) e^{iS(\alpha)+iS(\beta)} \mathcal{D}\alpha \mathcal{D}\beta \\ &= \int_{\alpha: \alpha(0)=x} e^{iS(\alpha)} \left(\int_{\beta: \beta(t)=\alpha(t)} \psi(\beta(t+s)) e^{iS(\beta)} \mathcal{D}\beta \right) \mathcal{D}\alpha. \end{aligned}$$

Now, for a given α , let us make a ‘change of variable’ in the inner integral by replacing the path $\beta : [t, t+s] \rightarrow \mathbb{R}^N$ with the path $\delta : [0, s] \rightarrow \mathbb{R}^N$ defined as $\delta(u) := \beta(t+u)$. Then note that

$$\begin{aligned} \int_{\beta: \beta(t)=\alpha(t)} \psi(\beta(t+s)) e^{iS(\beta)} \mathcal{D}\beta &= \int_{\delta: \delta(0)=\alpha(t)} \psi(\delta(s)) e^{iS(\delta)} \mathcal{D}\delta \\ &= Q(s)\psi(\alpha(t)). \end{aligned}$$

Plugging this into the previous display, we get

$$\begin{aligned} Q(t+s)\psi(x) &= \int_{\alpha: \alpha(0)=x} Q(s)\psi(\alpha(t)) e^{iS(\alpha)} \mathcal{D}\alpha \\ &= Q(t)Q(s)\psi(x). \end{aligned}$$

This shows that $Q(t+s) = Q(t)Q(s)$. Now, recall that the evolution of the quantum system

under consideration is given by $\psi \mapsto U(t)\psi$, where

$$U(t)\psi(x) = \frac{1}{Z(t)}Q(t)\psi(x).$$

Thus,

$$U(t+s)\psi(x) = \frac{1}{Z(t+s)}Q(t+s)\psi(x) = \frac{1}{Z(t)Z(s)}Q(t)Q(s)\psi(x).$$

Since $Q(t)$ is a linear operator, this shows that

$$U(t+s)\psi(x) = \frac{1}{Z(t)}Q(t)U(s)\psi(x) = U(t)U(s)\psi(x).$$

This completes the heuristic argument for the claim that $(U(t))_{t \in \mathbb{R}}$ is a one parameter group of linear operators. We have not yet ‘argued’ that $U(t)$ is unitary. The easiest route to do that is to calculate the infinitesimal generator of the group, which is our main goal anyway. For that purpose, we have to understand the behavior of $U(t)\psi(x)$ for t close to zero.

First, note that if t is small, we may expect that the major contributions to the path integrals come from paths that have not ventured far from their starting points. For such a path γ that starts from a point x , we have that

$$\int_0^t V(\gamma(s))ds = V(x)t + o(t),$$

assuming that V is a continuous function. Thus,

$$Q(t)\psi(x) = e^{-iV(x)t+o(t)} \int_{\gamma:\gamma(0)=x} \psi(\gamma(t))e^{iS_0(\gamma)}\mathcal{D}\gamma$$

Also, for a path γ that stays close to its starting point x , we may heuristically expand $\psi(\gamma(t))$ in Taylor series as

$$\psi(\gamma(t)) = \psi(x) + (\gamma(t) - x)\psi'(x) + \frac{1}{2}(\gamma(t) - x)^2\psi''(x) + \dots$$

Plugging this into the previous display, we get

$$\begin{aligned} Q(t)\psi(x) = e^{-iV(x)t+o(t)} & \left[\psi(x) \int_{\gamma:\gamma(0)=x} e^{iS_0(\gamma)}\mathcal{D}\gamma + \psi'(x) \int_{\gamma:\gamma(0)=x} (\gamma(t) - x)e^{iS_0(\gamma)}\mathcal{D}\gamma \right. \\ & \left. + \frac{1}{2}\psi''(x) \int_{\gamma:\gamma(0)=x} (\gamma(t) - x)^2 e^{iS_0(\gamma)}\mathcal{D}\gamma + \dots \right]. \quad (1.6.4) \end{aligned}$$

Now, if we translate a path γ by x , $S_0(\gamma)$ remains unchanged. Thus, we may heuristically expect that for any $k \geq 0$,

$$\int_{\gamma:\gamma(0)=x} (\gamma(t) - x)^k e^{iS_0(\gamma)}\mathcal{D}\gamma = \int_{\gamma:\gamma(0)=0} \gamma(t)^k e^{iS_0(\gamma)}\mathcal{D}\gamma.$$

Since S_0 is the action of a free particle, the above path integral can be evaluated using the same techniques that we used in Section 1.3 to evaluate the path integral for a free particle. In particular, it turns out that the value is 0 for $k = 1$, and for $k = 2$, it is equal to

$$it \int_{\gamma: \gamma(0)=0} e^{iS_0(\gamma)} \mathcal{D}\gamma = itZ(t).$$

For $k = 0$, it is clear that the value is $Z(t)$. Lastly, for $k \geq 3$, the value is $o(t)Z(t)$. Using these calculations in equation (1.6.4), we get that

$$\begin{aligned} Q(t)\psi(x) &= e^{-iV(x)t+o(t)} \left(\psi(x) + \frac{i}{2}\psi''(x)t + o(t) \cdots \right) Z(t) \\ &= (1 - iV(x)t + o(t)) \left(\psi(x) + \frac{i}{2}\psi''(x)t + o(t) \cdots \right) Z(t) \\ &= \left(\psi(x) - iV(x)\psi(x)t + \frac{i}{2}\psi''(x)t + o(t) \right) Z(t). \end{aligned}$$

Consequently,

$$U(t)\psi(x) = \frac{1}{Z(t)} Q(t)\psi(x) = \psi(x) - iV(x)\psi(x)t + \frac{i}{2}\psi''(x)t + o(t).$$

Thus,

$$H\psi(x) = \lim_{t \rightarrow 0} \frac{U(t)\psi(x) - \psi(x)}{-it} = -\frac{1}{2}\psi''(x) + V(x)\psi(x). \quad (1.6.5)$$

This leads us to expect that the Hamiltonian for a single particle in a potential V on the real line should be given by the above formula, at least for its action on twice differentiable states.

Note that above Hamiltonian is heuristically a symmetric operator, because, using integration by parts, we get

$$\begin{aligned} \langle H\psi, \phi \rangle &= \int_{-\infty}^{\infty} \left(-\frac{1}{2}\psi''(x) + V(x)\psi(x) \right) \overline{\phi(x)} dx \\ &= \int_{-\infty}^{\infty} \psi(x) \overline{\left(-\frac{1}{2}\phi''(x) + V(x)\phi(x) \right)} dx \\ &= \langle \psi, H\phi \rangle. \end{aligned}$$

Since

$$\frac{d}{dt} U(t)\xi = -iH U(t)\xi$$

for any ξ , we can now use the same technique as in the proof of Lemma D.3.8 that

$$\frac{d}{dt} \langle U(t)\psi, U(t)\phi \rangle = 0.$$

This shows that each $U(t)$ is an isometry. It does not quite prove that $U(t)$ is unitary, but we will be content with this because we are not pursuing rigor in this section.

1.7 The Wick rotation heuristic for a single particle system

The Hamiltonian H in equation (1.6.5) that we derived heuristically for the evolution of a single particle in a potential V governs the evolution via the relation $U(t) = e^{-itH}$, where $(U(t))_{t \in \mathbb{R}}$ is the one parameter group of unitary operators that dictates the evolution of the state of the system. The idea of Wick rotation is to study H as the generator of the semigroup $(P(t))_{t \geq 0}$ instead, where $P(t) = e^{-tH}$, and $P(t)$ is heuristically constructed using ‘Euclidean path integrals’. The general heuristic is that since we want to replace e^{-itH} by e^{-tH} , ‘real time’ t should be replaced by ‘imaginary time’ $-it$. The prescription for doing this is to replace t by $-it$ wherever t appears in calculations, but not as arguments of functions. Explicitly, the ds in the integral defining the action S in equation (1.6.1) should be replaced by $-ids$, and the time derivative $\dot{\gamma}$ should be replaced $i\dot{\gamma}$; the latter, because

$$\dot{\gamma}(t) = \lim_{\epsilon \rightarrow 0} \frac{\gamma(t + \epsilon) - \gamma(t)}{\epsilon} \quad \text{is to be replaced by} \quad \lim_{\epsilon \rightarrow 0} \frac{\gamma(t + \epsilon) - \gamma(t)}{-i\epsilon} = i\dot{\gamma}(t).$$

With these two replacements, the action changes to

$$i \int_0^t \left(\frac{1}{2} \dot{\gamma}(s)^2 + V(\gamma(s)) \right) ds.$$

Using this in place of S in the formulas for $Q(t)\psi(x)$ and $Z(t)$ gives the following tentative expression for $P(t)\psi(x)$:

$$P(t)\psi(x) = \frac{\int_{\gamma: \gamma(0)=x} \psi(\gamma(t)) e^{-S_E(\gamma)} \mathcal{D}\gamma}{\int_{\gamma: \gamma(0)=0} e^{-S_0(\gamma)} \mathcal{D}\gamma},$$

where S_0 is as in equation (1.6.3), and S_E is the ‘Euclidean action’

$$S_E(\gamma) = \int_0^t \left(\frac{1}{2} \dot{\gamma}(s)^2 + V(\gamma(s)) \right) ds. \quad (1.7.1)$$

The path integrals defining $P(t)$ are called ‘Euclidean path integrals’. Using the same heuristics as before, we can argue that $P(t)$ is a semigroup with generator H ; that is, $P(t)$ is indeed e^{-tH} . Of course, given the shaky foundation of the heuristic, we can be sure that we are doing the right thing only if we can mathematically justify that indeed $P(t) = e^{-tH}$. We will now do this.

The advantage of Euclidean path integrals, from a mathematical point of view, is that they can sometimes be given a rigorous meaning. As before, let us start by discretizing time. Instead of continuous time, let us consider discrete time for jt/n , $j = 0, 1, \dots, n$. A path γ is then discretized into a sequence of points $x_0, x_1, \dots, x_n$ where x_j is the value of

$\gamma(jt/n)$. The time derivative of γ and time jt/n is then discretized to

$$\frac{\gamma((j+1)t/n) - \gamma(jt/n)}{t/n} = \frac{n(x_{j+1} - x_j)}{t},$$

which causes the action $S_0(\gamma)$ to be replaced by

$$\frac{1}{2} \sum_{j=0}^{n-1} \frac{t}{n} \left(\frac{\gamma((j+1)t/n) - \gamma(jt/n)}{t/n} \right)^2 = \frac{n}{2t} \sum_{j=0}^{n-1} (x_{j+1} - x_j)^2.$$

Consequently, the discretized version of the Euclidean action is

$$\frac{n}{2t} \sum_{j=0}^{n-1} (x_{j+1} - x_j)^2 + \frac{t}{n} \sum_{j=1}^n V(x_j),$$

and the discrete approximation of $P(t)\psi(x)$ is given by

$$\frac{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \psi(x_n) \exp\left(-\frac{n}{2t} \sum_{j=0}^{n-1} (x_{j+1} - x_j)^2 - \frac{t}{n} \sum_{j=1}^n V(x_j)\right) dx_1 \cdots dx_n}{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp\left(-\frac{n}{2t} \sum_{j=0}^{n-1} (x_{j+1} - x_j)^2\right) dx_1 \cdots dx_n},$$

where $x_0 := x$. Changing variables to $y_j = x_j - x_{j-1}$ for $j = 1, \dots, n$, we see that the above quantity is equal to

$$\frac{\int \cdots \int \psi(x + y_1 + \cdots + y_n) \exp\left(-\frac{n}{2t} \sum_{j=1}^n y_j^2 - \frac{t}{n} \sum_{j=1}^n V(x + y_1 + \cdots + y_j)\right) dy_1 \cdots dy_n}{\int \cdots \int \exp\left(-\frac{n}{2t} \sum_{j=1}^n y_j^2\right) dy_1 \cdots dy_n}.$$

Now, recall that the probability density function of the $N(0, \sigma^2)$ distribution at a point $y \in \mathbb{R}$ is proportional to $e^{-y^2/2\sigma^2}$. Thus, the above expression may be alternatively expressed as

$$P_n(t)\psi(x) := \mathbb{E}(\psi(x + S_n) e^{-\frac{t}{n} \sum_{j=1}^n V(x + S_j)}), \quad (1.7.2)$$

where $S_j := Y_1 + \cdots + Y_j$, and $Y_1, \dots, Y_n$ are i.i.d. $N(0, t/n)$ random variables. In the next section, we will prove the following theorem, which shows that the Wick rotation heuristic can indeed be applied to construct the quantum mechanical system of a particle in a potential V , provided that V is continuous and bounded below, and grows at most exponentially.

Theorem 1.7.1. *Suppose that $V : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and bounded below, and has at most exponential growth at infinity. For each $t > 0$, let $P_n(t)$ be the operator on $L^2(\mathbb{R})$ defined above, and let $P_n(0) := I$. Then $(P_n(t))_{t \geq 0}$ is a well-defined collection of linear operators on $L^2(\mathbb{R})$, and for each $\psi \in L^2(\mathbb{R})$ and $t \geq 0$, $P_n(t)\psi$ converges to a limit in $L^2(\mathbb{R})$ as $n \rightarrow \infty$. If we denote this limit by $P(t)\psi$, then $(P(t))_{t \geq 0}$ is a strongly continuous semigroup of symmetric operators on $L^2(\mathbb{R})$, whose generator H acts on smooth compactly supported elements of $L^2(\mathbb{R})$ according to the formula displayed in equation (1.6.5).*

In short, the above theorem shows that there is indeed a densely defined self-adjoint operator H on $L^2(\mathbb{R})$ which is a legitimate candidate for the Hamiltonian of a single free particle in the potential V . By Stone's theorem, this allows us to construct a unique solution of the Schrödinger equation

$$i\frac{\partial\psi}{\partial t} = -\frac{1}{2}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(t, x)$$

for any initial condition $\psi(0, \cdot) \in L^2(\mathbb{R})$, in the sense that $\psi(t, \cdot) = e^{-itH}\psi(0, \cdot)$. Note that here t can be any real number.

The condition that V needs to be bounded below can be relaxed. For a variety of conditions on the potential V that allow the constructions of solutions to the Schrödinger equation, see Reed and Simon [26, Chapter X].

1.8 Validity of the Wick rotation heuristic

We will prove Theorem 1.7.1 in this section, thereby proving that the validity of the Wick rotation heuristic for constructing the quantum mechanical system of a single particle in a potential V that is continuous and bounded below.

Let B be standard Brownian motion on the time interval $[0, t]$, starting from 0.⁹ It is simple to verify that if $[a, b]$ and $[c, d]$ are disjoint sub-intervals of $[0, t]$, then

$$\text{Cov}(B(b) - B(a), B(d) - B(c)) = 0.$$

Since jointly normal random variables are independent when they are uncorrelated, this implies that B has *independent increments*, meaning that for any $0 \leq s_1 \leq \dots \leq s_n \leq t$, the random variables $B(s_1), B(s_2) - B(s_1), \dots, B(s_n) - B(s_{n-1})$ are independent. Moreover, for any $0 \leq a \leq b \leq t$,

$$B(b) - B(a) \sim N(0, b - a).$$

Consequently, for any n , if we define $Y_{n,i} := B(ti/n) - B(t(i-1)/n)$, the random variables $Y_{n,1}, \dots, Y_{n,n}$ are i.i.d. $N(0, t/n)$. Thus, if we let $S_{n,j} := Y_{n,1} + \dots + Y_{n,j}$, then

$$P_n(t)\psi(x) = \mathbb{E}(\psi(x + S_{n,n})e^{-\frac{t}{n}\sum_{j=1}^n V(x+S_{n,j})}).$$

We divide the proof into a sequence of lemmas, for easier readability.

Lemma 1.8.1. *For each $n \geq 1$ and $t \geq 0$, $P_n(t)$ is a well-defined linear map from $L^2(\mathbb{R})$ into $L^2(\mathbb{R})$.*

⁹For an introduction to Brownian motion, see Section B.6 in Appendix B.

Proof. Note that since V is bounded below by a number $c \in \mathbb{R}$,

$$\begin{aligned} \int_{-\infty}^{\infty} |P_n(t)\psi(x)|^2 dx &= \int_{-\infty}^{\infty} |\mathbb{E}(\psi(x + S_{n,n})e^{-\frac{t}{n} \sum_{j=1}^n V(x+S_{n,j})})|^2 dx \\ &\leq e^{-2ct} \int_{-\infty}^{\infty} \mathbb{E}|\psi(x + S_{n,n})|^2 dx \\ &= e^{-2ct} \mathbb{E} \left(\int_{-\infty}^{\infty} |\psi(x + S_{n,n})|^2 dx \right) = e^{-2ct} \|\psi\|_{L^2}^2 < \infty, \end{aligned}$$

where the second equality holds by Fubini's theorem and the third equality holds by a change of variable within the integral. This proves the claim. $\square$

Lemma 1.8.2. *For each $\psi \in L^2(\mathbb{R})$ and $t \geq 0$, $P_n(t)\psi$ converges to a limit in L^2 as $n \rightarrow \infty$. The limit is given by*

$$P(t)\psi(x) := \mathbb{E}(\psi(x + B(t))e^{-\int_0^t V(x+B(s))ds}). \quad (1.8.1)$$

Proof. By the same argument as in the proof of Lemma 1.8.1, it is easy to show that $P(t)$ is a linear map from $L^2(\mathbb{R})$ into $L^2(\mathbb{R})$. Since $S_{n,n} = B(t)$ for each n , we see that

$$\begin{aligned} \|P_n(t)\psi - P(t)\psi\|_{L^2}^2 &= \int_{-\infty}^{\infty} |\mathbb{E}(\psi(x + B(t))(e^{-\frac{t}{n} \sum_{j=1}^n V(x+S_{n,j})} - e^{-\int_0^t V(x+B(s))ds}))|^2 dx \\ &\leq \int_{-\infty}^{\infty} \mathbb{E}|\psi(x + B(t))(e^{-\frac{t}{n} \sum_{j=1}^n V(x+S_{n,j})} - e^{-\int_0^t V(x+B(s))ds})|^2 dx. \end{aligned}$$

Since V and B are both continuous functions, we have that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{t}{n} \sum_{j=1}^n V(x + S_{n,j}) &= \lim_{n \rightarrow \infty} \frac{t}{n} \sum_{j=1}^n V(x + B(tj/n)) \\ &= \int_0^t V(B(s))ds. \end{aligned}$$

Since $V \geq c$ everywhere, an application of the dominated convergence theorem now shows that $P_n(t)\psi \rightarrow P(t)\psi$ in L^2 as $n \rightarrow \infty$. $\square$

Lemma 1.8.3. *The operator $P(t) : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ defined in equation (1.8.1) is a symmetric operator.*

Proof. Take any $\psi, \phi \in L^2(\mathbb{R})$. It is easy to justify the applications of Fubini's theorem

below, to get

$$\begin{aligned}
 \langle P(t)\psi, \phi \rangle &= \int_{-\infty}^{\infty} \mathbb{E}(\psi(x + B(t))e^{-\int_0^t V(x+B(s))ds})\overline{\phi(x)}dx \\
 &= \mathbb{E}\left(\int_{-\infty}^{\infty} \psi(x + B(t))e^{-\int_0^t V(x+B(s))ds}\overline{\phi(x)}dx\right) \\
 &= \mathbb{E}\left(\int_{-\infty}^{\infty} \psi(y)e^{-\int_0^t V(y+B(s)-B(t))ds}\overline{\phi(y-B(t))}dy\right) \\
 &= \int_{-\infty}^{\infty} \psi(y)\overline{\mathbb{E}(e^{-\int_0^t V(y+B(s)-B(t))ds}\phi(y-B(t)))}dy. \tag{1.8.2}
 \end{aligned}$$

Define $W(s) := B(t-s) - B(t)$ for $0 \leq s \leq t$. Note that W is also a random continuous function. It is easy to verify that W satisfies the conditions for being a standard Brownian motion on the time interval $[0, t]$. If we define $P(t)$ using W instead of B in equation (1.8.1), it is clear that the definition does not change, because the same argument as above shows that $P(t)\psi$ is the L^2 limit of $P_n(t)\psi$ as $n \rightarrow \infty$.

Now, note that $y - B(t) = y + W(t)$, and

$$\int_0^t V(y + B(s) - B(t))ds = \int_0^t V(y + W(t-s))ds = \int_0^t V(y + W(s))ds.$$

Thus, we get

$$\mathbb{E}(e^{-\int_0^t V(y+B(s)-B(t))ds}\phi(y-B(t))) = \mathbb{E}(e^{-\int_0^t V(y+W(s))ds}\phi(y+B(t))) = P(t)\phi(y).$$

Using this in equation (1.8.2), we see that $P(t)$ is a symmetric operator. $\square$

Lemma 1.8.4. *The collection of operators $(P(t))_{t \geq 0}$ is a semigroup. That is, for all $s, t \geq 0$, $P(s+t) = P(s)P(t)$.*

Proof. Take any $s, t \geq 0$ and any $\psi \in L^2(\mathbb{R})$. Let $\phi := P(t)\psi$. Let B be a Brownian motion on the time interval $[0, t]$ and W be a Brownian motion on the time interval $[0, s]$, independent of each other and defined on the same probability space. Then note that

$$\begin{aligned}
 P(s)\phi(x) &= \mathbb{E}(\phi(x + W(s))e^{-\int_0^s V(x+W(u))du}) \\
 &= \int_{C[0,s]} \phi(x + w(s))e^{-\int_0^s V(x+w(u))du}d\mu(w),
 \end{aligned}$$

where μ is the law of W . That is, $\mu(A) := \mathbb{P}(W \in A)$ for each $A \in \mathcal{B}(C[0, s])$. Similarly, for any $y \in \mathbb{R}$,

$$\phi(y) = P(t)\psi(y) = \int_{C[0,t]} \psi(y + b(t))e^{-\int_0^t V(y+b(u))du}d\nu(b),$$

where ν is the law of B . Plugging this into the previous display, we get

$$\begin{aligned} P(s)\phi(x) &= \int_{C[0,s]} \int_{C[0,t]} \psi(x+w(s)+b(t)) e^{-\int_0^s V(x+w(u))du - \int_0^t V(x+w(s)+b(u))du} d\nu(b) d\mu(w) \\ &= \mathbb{E}(\psi(x+W(s)+B(t)) e^{-\int_0^s V(x+W(u))du - \int_0^t V(x+W(s)+B(u))du}). \end{aligned} \quad (1.8.3)$$

Define

$$Z(u) := \begin{cases} W(u) & \text{if } 0 \leq u \leq s, \\ W(s) + B(u-s) & \text{if } s \leq u \leq s+t. \end{cases}$$

It is easy to verify that Z is a standard Brownian motion on the time interval $[0, s+t]$. Thus, by equation (1.8.3),

$$\begin{aligned} P(s)P(t)\psi(x) &= P(s)\phi(x) \\ &= \mathbb{E}(\psi(x+Z(s+t)) e^{-\int_0^{s+t} V(x+Z(u))du}) \\ &= P(s+t)\psi(x). \end{aligned}$$

This proves that $(P(t))_{t \geq 0}$ is a semigroup of operators. $\square$

Lemma 1.8.5. *The semigroup $(P(t))_{t \geq 0}$ is strongly continuous.*

Proof. Take any $\psi \in L^2(\mathbb{R})$ such that ψ is continuous and has compact support. By Lemma D.4.5, it suffices to prove that $t \mapsto P(t)\psi$ is continuous for such ψ . Let L be so large that ψ is zero outside the interval $[-L, L]$. Let B be standard Brownian motion on an interval $[0, T]$. It is easy to see that the restriction of B to $[0, t]$, for any $t \in [0, T]$, is standard Brownian motion on $[0, t]$. Thus, for any $s, t \in [0, T]$,

$$\begin{aligned} &\|P(s)\psi - P(t)\psi\|_{L^2}^2 \\ &= \int_{-\infty}^{\infty} |\mathbb{E}(\psi(x+B(s)) e^{-\int_0^s V(x+B(u))du}) - \mathbb{E}(\psi(x+B(t)) e^{-\int_0^t V(x+B(u))du})|^2 dx \\ &\leq \int_{-\infty}^{\infty} \mathbb{E} |\psi(x+B(s)) e^{-\int_0^s V(x+B(u))du} - \psi(x+B(t)) e^{-\int_0^t V(x+B(u))du}|^2 dx. \end{aligned}$$

Since ψ is continuous, the integrand converges to 0 as $s \rightarrow t$. Thus, we only have to verify the condition needed for applying the dominated convergence theorem to show that the above expression tends to zero as $s \rightarrow t$. Let $M := \max_{0 \leq u \leq T} |B(u)|$. Let c and C be constants such that $V \geq c$ and $|\psi| \leq C$ everywhere. Without loss of generality, $c \leq 0$. Since ψ vanishes outside $[-L, L]$, we see that

$$|\psi(x+B(s)) e^{-\int_0^s V(x+B(u))du} - \psi(x+B(t)) e^{-\int_0^t V(x+B(u))du}|^2 \leq 4C^2 e^{-2cT} \xi(x),$$

where

$$\xi(x) := \begin{cases} 1 & \text{if } |x+B(s)| \leq L \text{ or } |x+B(t)| \leq L, \\ 0 & \text{otherwise.} \end{cases}$$

Now, if $|x + B(s)| \leq L$, then

$$|x| \leq |x + B(s) - B(s)| \leq |x + B(s)| + |B(s)| \leq L + M.$$

Similarly, if $|x + B(t)| \leq L$, then also

$$|x| \leq L + M.$$

Thus, if $\xi(x) = 1$, then $|x| \leq L + M$. Consequently, $\xi(x) \leq \eta(x)$, where

$$\eta(x) := \begin{cases} 1 & \text{if } |x| \leq L + M, \\ 0 & \text{otherwise.} \end{cases}$$

Note that $\eta(x)$ does not depend on s or t , and by Fubini's theorem and Lemma B.6.3,

$$\int_{-\infty}^{\infty} \mathbb{E}(\eta(x)) dx = \mathbb{E} \left(\int_{-\infty}^{\infty} \eta(x) dx \right) = \mathbb{E}(2(L + M)) < \infty.$$

This allows us to apply the dominated convergence theorem and complete the proof. $\square$

Lemma 1.8.6. *For any $\psi \in L^2(\mathbb{R})$ that is smooth and compactly supported, we have*

$$\frac{P(t)\psi - \psi}{-t} \rightarrow H\psi$$

in L^2 as $t \rightarrow 0$, where H is the operator displayed in equation (1.6.5).

Proof. Let $\psi_t := P(t)\psi$ and

$$\phi_t(x) := (1 - tV(x))\mathbb{E}[\psi(x + B(t))].$$

Note that for any x ,

$$\psi_t(x) - \phi_t(x) = \mathbb{E}[\psi(x + B(t))(e^{-\int_0^t V(x+B(s))ds} - 1 + tV(x))]$$

Thus, we get

$$\begin{aligned} \|\psi_t - \phi_t\|_{L^2}^2 &\leq \int_{-\infty}^{\infty} \mathbb{E}[|\psi(x + B(t))|^2 (e^{-\int_0^t V(x+B(s))ds} - 1 + tV(x))^2] dx \\ &= \mathbb{E} \left[\int_{-\infty}^{\infty} |\psi(x + B(t))|^2 (e^{-\int_0^t V(x+B(s))ds} - 1 + tV(x))^2 dx \right] \\ &= \mathbb{E} \left[\int_{-\infty}^{\infty} |\psi(y)|^2 (e^{-\int_0^t V(y+B(s)-B(t))ds} - 1 + tV(y - B(t)))^2 dy \right]. \end{aligned} \quad (1.8.4)$$

From the continuity of V and B , it follows that

$$\left. \frac{\partial}{\partial t} e^{-\int_0^t V(y+B(s)-B(t))ds} \right|_{t=0} = -V(y - B(t)).$$

Equivalently,

$$\lim_{t \rightarrow 0} \frac{1}{t} \left(e^{-\int_0^t V(y+B(s)-B(t))ds} - 1 + tV(y-B(t)) \right) = 0.$$

Thus, from the inequality (1.8.4), we would be able to show that $(\psi_t - \phi_t)/t \rightarrow 0$ in L^2 as $t \rightarrow 0$, provided that we are able to verify the condition required for applying the dominated convergence theorem. For this, fix some $T > 0$, and let $M := \max_{0 \leq s \leq T} |B(s)|$. Take any $t \in [0, T]$. Since V has at most exponential growth, there are constants C_1 and C_2 such that $|V(x)| \leq C_1 e^{C_2|x|}$ for all $x \in \mathbb{R}$. Now, recall that V is bounded below by some $c \in \mathbb{R}$. Without loss of generality, assume that $c \leq 0$. Then

$$\begin{aligned} |e^{-\int_0^t V(y+B(s)-B(t))ds} - 1| &\leq e^{-ct} \left| \int_0^t V(y+B(s)-B(t))ds \right| \\ &\leq e^{-cT} \int_0^t |V(y+B(s)-B(t))|ds \\ &\leq C_1 e^{-cT} \int_0^t e^{C_2|y+B(s)-B(t)|} ds \leq C_1 e^{-cT} t e^{C_2(|y|+2M)}. \end{aligned}$$

Similarly, $|V(y-B(t))| \leq C_1 e^{C_2(|y|+M)}$. Combining, we get

$$\frac{1}{t^2} |\psi(y)|^2 (e^{-\int_0^t V(y+B(s)-B(t))ds} - 1 + tV(y-B(t)))^2 \leq |\psi(y)|^2 C_1^2 e^{-2cT} e^{2C_2(|y|+M)}.$$

Note that the right side does not depend on t , and if we integrate it over y and then take the expected value of the result, it is finite (by Lemma B.6.3 and the fact that ψ has compact support). This allows us to apply the dominated convergence theorem and conclude that $(\psi_t - \phi_t)/t \rightarrow 0$ in L^2 as $t \rightarrow 0$.

Next, let

$$\xi_t(x) := (1 - tV(x)) \left(\psi(x) + \frac{t}{2} \psi''(x) \right).$$

Then note that

$$\begin{aligned} \phi_t(x) - \xi_t(x) &= (1 - tV(x)) \mathbb{E} \left[\psi(x+B(t)) - \psi(x) - B(t)\psi'(x) - \frac{1}{2}B(t)^2\psi''(x) \right] \\ &= (1 - tV(x)) \mathbb{E} \left[\int_0^1 \int_0^w \int_0^y B(t)^3 \psi'''(x+zB(t)) dz dy dw \right] \end{aligned}$$

Thus, we have

$$\begin{aligned} \|\phi_t - \xi_t\|_{L^2}^2 &\leq \frac{1}{6} \mathbb{E} \left[\int_0^1 \int_0^w \int_0^y \int_{-\infty}^{\infty} |1 - tV(x)|^2 |B(t)^3 \psi'''(x+zB(t))|^2 dx dz dy dw \right] \\ &= \frac{1}{6} \mathbb{E} \left[\int_0^1 \int_0^w \int_0^y \int_{-\infty}^{\infty} |1 - tV(u-zB(t))|^2 |B(t)^3 \psi'''(u)|^2 du dz dy dw \right] \quad (1.8.5) \end{aligned}$$

Take any $t \in [0, T]$. Then

$$|1 - tV(u-zB(t))|^2 \leq 2 + 2T^2 |V(u-zB(t))|^2 \leq 2 + 2T^2 C_1^2 e^{2C_2(|u|+|z|M)}.$$

Also, note that by Lemma B.6.3,

$$\mathbb{E}(e^{2C_2(|u|+|z|M)}|B(t)|^6) \leq \sqrt{\mathbb{E}(e^{4C_2(|u|+|z|M)})\mathbb{E}|B(t)|^{12}} \leq C_5 e^{2C_2|u|} t^3,$$

where C_5 does not depend on t or u . Plugging this bound into the right side of equation (1.8.5), and invoking the assumption that ψ is compactly supported, we see that $\|\phi_t - \xi_t\|_{L^2}^2 = O(t^3)$ as $t \rightarrow 0$. Combining this with our earlier observation that $(\psi_t - \phi_t)/t \rightarrow 0$ in L^2 , we get that $(\psi_t - \xi_t)/t \rightarrow 0$ in L^2 as $t \rightarrow 0$. But this is equivalent to the statement that

$$\frac{P(t)\psi - \psi}{-t} - H\psi \rightarrow 0$$

in L^2 , where H is the operator displayed in equation (1.6.5). This completes the proof of the lemma. $\square$

Combining all of the above lemmas, we see that all the assertions of Theorem 1.7.1 have been proved. This shows that the Wick rotation heuristic is mathematically valid for the case of a single particle in a well-behaved potential on the real line.

Chapter 2

Abstracting the Probabilistic Approach

2.1 The Markov process heuristic

Suppose that we want to build a quantum system that takes value in some space $\mathcal{X}$ endowed with a σ -algebra, so that the time trajectories of the system are maps $\gamma : \mathbb{R} \rightarrow \mathcal{X}$. Let S be the action, and let S_E be its Euclidean version obtained by Wick rotation, following the method prescribed in Section 1.7. To construct the Hamiltonian for the quantum system using the ‘probabilistic approach’ outlined in Section 1.5, we need to construct a strongly continuous semigroup of symmetric operators $(P(t))_{t \geq 0}$ on some Hilbert space $\mathcal{H}$ of functions from $\mathcal{X}$ into $\mathbb{C}$, such that the generator of this semigroup is the same as the Hamiltonian for our quantum system. The path integral prescription says that we should define

$$P(t)\psi(x) := \frac{1}{Z_E(t)} \int_{\gamma: \gamma(0)=x} \psi(\gamma(t)) e^{-S_E(\gamma)} \mathcal{D}\gamma,$$

where the integration is over paths $\gamma : [0, t] \rightarrow \mathcal{X}$ satisfying $\gamma(0) = x$, $\mathcal{D}\gamma$ is formally given by

$$\mathcal{D}\gamma = \prod_{0 < s \leq t} d\lambda(\gamma(s))$$

where λ is some notion of Lebesgue measure on $\mathcal{X}$, and $Z_E(t)$ is the Wick-rotated version of the normalizing factor from the quantum mechanical path integral.

We have carried out the above construction by an ad hoc method for a single particle in a potential. A generalized approach using Markov processes is helpful for the more complex systems that arise in quantum field theory. The following is an outline of this approach.

Recall that the Hilbert space for the single particle system is $L^2(\mathbb{R})$. Generalizing this, we should take the Hilbert space for the $\mathcal{X}$ -valued system to be $L^2(\lambda)$, where λ is the analogue of Lebesgue measure on $\mathcal{X}$. But this need not be the only choice. Let μ be any measure on $\mathcal{X}$, which has density f with respect to λ , meaning that $\mu(A) = \int_A f(x) d\lambda(x)$

for any measurable set $A \subseteq \mathcal{X}$. The following map gives an embedding of $L^2(\lambda)$ into $L^2(\mu)$:

$$\psi(x) \mapsto \frac{\psi(x)}{\sqrt{f(x)}}.$$

For $\psi \in L^2(\mathbb{R})$, let $\tilde{\psi} \in L^2(\mu)$ denote the function on the right. Then we can consider $\tilde{\psi}$ to be denoting the same state of the system as ψ , provided that we take the Hilbert space to be $L^2(\mu)$ instead of $L^2(\lambda)$. This needs a re-interpretation of probabilities, as follows. If the system is in state $\phi \in L^2(\mu)$, the probability density function of the particle is proportional to $|\phi|^2$, but with respect to the measure μ instead of λ . With this re-interpretation, the state $\psi \in L^2(\lambda)$ gives the same probabilities as the state $\tilde{\psi} \in L^2(\mu)$, because for any measurable set $A \subseteq \mathcal{X}$,

$$\frac{\int_A |\psi(x)|^2 d\lambda(x)}{\int_{\mathcal{X}} |\psi(x)|^2 d\lambda(x)} = \frac{\int_A |\tilde{\psi}(x)|^2 f(x) d\lambda(x)}{\int_{\mathcal{X}} |\tilde{\psi}(x)|^2 f(x) d\lambda(x)} = \frac{\int_A |\tilde{\psi}(x)|^2 d\mu(x)}{\int_{\mathcal{X}} |\tilde{\psi}(x)|^2 d\mu(x)}.$$

What should be the time evolution semigroup $(\tilde{U}(t))_{t \geq 0}$ in $L^2(\mu)$? Clearly, if ψ_t is the state of the system at time t in $L^2(\lambda)$, then it is necessary to ensure that $\tilde{\psi}_t$ is the state at time t in $L^2(\mu)$. Thus, if $\tilde{\psi} \in L^2(\mu)$ is the state at time 0, then we should define

$$\begin{aligned} \tilde{U}(t)\tilde{\psi}(x) &= \tilde{\psi}_t(x) = \frac{\psi_t(x)}{\sqrt{f(x)}} \\ &= \frac{1}{Z(t)\sqrt{f(x)}} \int_{\gamma: \gamma(0)=x} \psi(\gamma(t)) e^{iS(\gamma)} \mathcal{D}\gamma \\ &= \frac{1}{Z(t)\sqrt{f(x)}} \int_{\gamma: \gamma(0)=x} \sqrt{f(\gamma(t))} \tilde{\psi}(\gamma(t)) e^{iS(\gamma)} \mathcal{D}\gamma. \end{aligned} \quad (2.1.1)$$

After Wick rotation, the corresponding semigroup of symmetric operators $(\tilde{P}(t))_{t \geq 0}$ should therefore be defined as

$$\tilde{P}(t)\phi(x) = \frac{1}{Z_E(t)\sqrt{f(x)}} \int_{\gamma: \gamma(0)=x} \sqrt{f(\gamma(t))} \phi(\gamma(t)) e^{-S_E(\gamma)} \mathcal{D}\gamma, \quad (2.1.2)$$

where $\phi \in L^2(\mu)$, and the integral is over all paths defined on $[0, t]$ satisfying $\gamma(0) = x$. The Markov process approach gives a generic prescription for this construction. The precondition for this construction is that the Euclidean action S_E should have the following additive property: Suppose that γ_1 is a path defined on a time interval $[t_0, t_1]$ and γ_2 is a path defined on the time interval $[t_1, t_2]$, such that $\gamma_1(t_1) = \gamma_2(t_1)$. Let γ be the path defined on the time interval $[t_0, t_2]$ obtained by concatenating γ_1 and γ_2 in the obvious way. Then we need to have

$$S_E(\gamma) = S_E(\gamma_1) + S_E(\gamma_2). \quad (2.1.3)$$

Most physical systems satisfy this condition. If this holds, then the next step is to construct a probability measure ν on the space of all paths $\gamma : \mathbb{R} \rightarrow \mathcal{X}$ that heuristically has probability

density proportional to $e^{-S_E(\gamma)}$ with respect to ‘Lebesgue measure’ on path space. Such a measure is constructed typically by first constructing it over paths $\gamma : [-T, T] \rightarrow \mathbb{R}$, and then sending T to infinity. The probability density on paths $\gamma : [-T, T] \rightarrow \mathcal{X}$ is of the form

$$\frac{1}{C(T)} e^{-S_E(\gamma)}$$

where $C(T)$ is the normalizing constant. Due to the additive nature of S_E , we have for any two intervals $[T_0, T_1]$ and $[T_1, T_2]$,

$$\int_{\gamma: [T_0, T_2] \rightarrow \mathcal{X}} e^{-S_E(\gamma)} \mathcal{D}\gamma = \left(\int_{\gamma: [T_0, T_1] \rightarrow \mathcal{X}} e^{-S_E(\gamma)} \mathcal{D}\gamma \right) \left(\int_{\gamma: [T_1, T_2] \rightarrow \mathcal{X}} e^{-S_E(\gamma)} \mathcal{D}\gamma \right).$$

Moreover, the integral on the left is clearly just a function of the difference $T_2 - T_0$. This implies that $C(T + T') = C(T)C(T')$, which, in turn, implies that there is a constant K such that $C(T) = e^{KT}$ for all $T \geq 0$. Thus, if we define, for a path $\gamma : [T_0, T_1] \rightarrow \mathcal{X}$ the modified action

$$S'_E(\gamma) := S_E(\gamma) + K(T_1 - T_0),$$

then the probability density on paths $\gamma : [-T, T] \rightarrow \mathcal{X}$ is exactly equal to (and not just proportional to) $e^{-S'_E(\gamma)}$. Thus, the same must be true for the probability density on paths $\gamma : \mathbb{R} \rightarrow \mathcal{X}$.

Suppose that such a probability measure ν on path space can be rigorously defined. Let $X = \{X_t\}_{t \in \mathbb{R}}$ be an $\mathcal{X}$ -valued stochastic process (i.e., a collection of $\mathcal{X}$ -valued random variables defined on the same probability space) with law ν . Suppose that X is stationary and reversible, meaning that the processes $\{X_{t+a}\}_{t \in \mathbb{R}}$ and $\{X_{-t}\}_{t \in \mathbb{R}}$ also have law ν , for any a . Due to stationarity, X_t has the same distribution for all t . Let μ denote this law. Assume that μ has probability density f with respect to λ . We will now construct our quantum system on the Hilbert space $L^2(\mu)$. Define¹

$$Q(t)\phi(x) := \mathbb{E}(\phi(X_t) | X_0 = x). \quad (2.1.4)$$

We claim that $Q(t)\phi(x)$ is exactly the right side of equation (2.1.2). The heuristic argument goes as follows. First, note that for any x , we have the formal identity

$$f(x) = \int_{\gamma: \gamma(0)=x} e^{-S'_E(\gamma)} \mathcal{D}\gamma, \quad (2.1.5)$$

where the integral is over all paths $\gamma : \mathbb{R} \rightarrow \mathcal{X}$ satisfying $\gamma(0) = x$. This holds because for any measurable set $A \subseteq \mathcal{X}$,

$$\int_{\mathcal{X}} \left(\int_{\gamma: \gamma(0)=x} e^{-S'_E(\gamma)} \mathcal{D}\gamma \right) d\lambda(x) = \int_{\gamma: \gamma(0) \in A} e^{-S'_E(\gamma)} \mathcal{D}\gamma = \mathbb{P}(X_0 \in A).$$

For any path γ , let γ_+ and γ_- be the restrictions of γ to the positive and negative time

¹See Section B.4.1 in Appendix B for the definition and properties of conditional expectation.

intervals. By the additive property of S_E (and hence, of S'_E), we have that $S'_E(\gamma) = S'_E(\gamma_+) + S'_E(\gamma_-)$. Thus,

$$\begin{aligned} \int_{\gamma:\gamma(0)=x} e^{-S'_E(\gamma)} \mathcal{D}\gamma &= \int_{\gamma:\gamma(0)=x} e^{-S'_E(\gamma_+)} e^{-S'_E(\gamma_-)} \mathcal{D}\gamma \\ &= \int_{\substack{\gamma_1:[0,\infty)\rightarrow\mathcal{X}, \gamma_2:(-\infty,0]\rightarrow\mathcal{X} \\ \gamma_1(0)=x, \gamma_2(0)=x}} e^{-S'_E(\gamma_1)} e^{-S'_E(\gamma_2)} \mathcal{D}\gamma_1 \mathcal{D}\gamma_2 \\ &= \left(\int_{\substack{\gamma:[0,\infty)\rightarrow\mathcal{X} \\ \gamma(0)=x}} e^{-S'_E(\gamma)} \mathcal{D}\gamma \right) \left(\int_{\substack{\gamma:(-\infty,0]\rightarrow\mathcal{X} \\ \gamma(0)=x}} e^{-S'_E(\gamma)} \mathcal{D}\gamma \right). \end{aligned}$$

But by invariance under time inversion, the two integrals in the last line are equal. Combining this with equation (2.1.5), we get

$$\begin{aligned} \int_{\substack{\gamma:[0,\infty)\rightarrow\mathcal{X} \\ \gamma(0)=x}} e^{-S'_E(\gamma)} \mathcal{D}\gamma &= \int_{\substack{\gamma:(-\infty,0]\rightarrow\mathcal{X} \\ \gamma(0)=x}} e^{-S'_E(\gamma)} \mathcal{D}\gamma \\ &= \sqrt{f(x)}. \end{aligned}$$

By time translation, the same identity holds if we replace $\gamma(0)$ by $\gamma(t)$ everywhere, for any t . Thus, we get

$$\begin{aligned} Q(t)\phi(x) &= \frac{\int_{\gamma:\gamma(0)=x} \phi(\gamma(t)) e^{-S'_E(\gamma)} \mathcal{D}(\gamma)}{\int_{\gamma:\gamma(0)=x} e^{-S'_E(\gamma)} \mathcal{D}(\gamma)} \\ &= \frac{\int_{\gamma:\gamma(0)=x} \phi(\gamma(t)) e^{-S'_E(\gamma)} \mathcal{D}(\gamma)}{\int_{\gamma:\gamma(0)=x} e^{-S'_E(\gamma)} \mathcal{D}(\gamma)} \\ &= \frac{\int_{\mathcal{X}} \int_{\gamma:\gamma(0)=x, \gamma(t)=y} \phi(y) e^{-S'_E(\gamma)} \mathcal{D}(\gamma) d\lambda(y)}{f(x)}. \end{aligned}$$

Now, the integral in the numerator can be written as

$$\begin{aligned} &\int_{\mathcal{X}} \left(\int_{\substack{\gamma:(-\infty,0]\rightarrow\mathcal{X} \\ \gamma(0)=x}} e^{-S'_E(\gamma)} \mathcal{D}\gamma \int_{\substack{\gamma:[0,t]\rightarrow\mathcal{X} \\ \gamma(0)=x, \gamma(t)=y}} \phi(y) e^{-S'_E(\gamma)} \mathcal{D}\gamma \int_{\substack{\gamma:[t,\infty)\rightarrow\mathcal{X} \\ \gamma(t)=y}} e^{-S'_E(\gamma)} \mathcal{D}\gamma \right) d\lambda(y) \\ &= \int_{\mathcal{X}} \sqrt{f(x)f(y)} \left(\int_{\substack{\gamma:[0,t]\rightarrow\mathcal{X} \\ \gamma(0)=x, \gamma(t)=y}} \phi(y) e^{-S'_E(\gamma)} \mathcal{D}\gamma \right) d\lambda(y). \end{aligned}$$

Plugging this into the previous display, we get

$$\begin{aligned} Q(t)\phi(x) &= \frac{1}{\sqrt{f(x)}} \int_{\substack{\gamma:[0,t]\rightarrow\mathcal{X} \\ \gamma(0)=x}} \sqrt{f(\gamma(t))} \phi(\gamma(t)) e^{-S'_E(\gamma)} \mathcal{D}\gamma \\ &= \frac{1}{e^{Kt} \sqrt{f(x)}} \int_{\substack{\gamma:[0,t]\rightarrow\mathcal{X} \\ \gamma(0)=x}} \sqrt{f(\gamma(t))} \phi(\gamma(t)) e^{-S_E(\gamma)} \mathcal{D}\gamma \end{aligned}$$

This gives a heuristic proof of the claim that the operator $Q(t)$ defined in equation (2.1.4) is indeed a legitimate candidate for the operator $\tilde{P}(t)$ displayed in equation (2.1.2).

To summarize the discussion, here are the steps to constructing a quantum system using Markov processes:

1. Given the action S , perform Wick rotation to obtain the Euclidean action S_E .
2. Construct a probability measure ν on the space of all infinite-time trajectories that heuristically has probability density proportional to $e^{-S_E(\gamma)}$ with respect to some notion of ‘Lebesgue measure’ on path space.
3. Let $X = \{X_t\}_{t \in \mathbb{R}}$ be a stochastic process with law ν . Let μ be the law of X_0 . Let $\mathcal{H} = L^2(\mu)$ be the Hilbert space of the quantum system, and define $Q(t) : \mathcal{H} \rightarrow \mathcal{H}$ as $Q(t)\phi(x) := \mathbb{E}(\phi(X_t) | X_0 = x)$.
4. Show that the X is a stationary reversible Markov process, so that by Theorem B.5.5, $(Q(t))_{t \geq 0}$ is a semigroup of symmetric linear operators on $\mathcal{H}$.
5. Verify that the semigroup $(Q(t))_{t \geq 0}$ is strongly continuous. By the heuristics outlined in this section, the generator of this semigroup is a valid candidate for the Hamiltonian of the quantum system with action S .

The idea of looking at Euclidean quantum field theories as Markov processes originated in the work of Symanzik [29]. The procedure for constructing the quantum field from the Markov process representing the Euclidean theory was introduced by Nelson [21]. A condition that is easier to verify than the Markov property, known as reflection positivity, was introduced by Osterwalder and Schrader [22, 23]. We will discuss reflection positivity later in this chapter.

2.2 The simple harmonic oscillator

The simple harmonic oscillator is a single particle with mass 1 on the real line moving under the quadratic potential $V(x) = \frac{1}{2}x^2$. We already know how to quantize this system using Theorem 1.7.1. We will now work out a different quantization via the Markov process heuristic. By equation (1.6.1), we know that the action of a trajectory $\gamma : \mathbb{R} \rightarrow \mathbb{R}$ is given by

$$S(\gamma) = \frac{1}{2} \int_{-\infty}^{\infty} (\dot{\gamma}(s)^2 - \gamma(s)^2) ds,$$

and by equation (1.7.1), its Euclidean version is

$$\begin{aligned} S_E(\gamma) &= \frac{1}{2} \int_{-\infty}^{\infty} (\dot{\gamma}(s)^2 + \gamma(s)^2) ds \\ &= \frac{1}{2} \int_{-\infty}^{\infty} (-\gamma(s)\ddot{\gamma}(s) + \gamma(s)^2) ds \end{aligned}$$

Thus, our first task is to construct a probability measure ν on the space of all trajectories such that ν has density proportional to $e^{-S_E(\gamma)}$ with respect to ‘Lebesgue measure’.

Now, recall that the probability density function of the Gaussian distribution² on $\mathbb{R}^n$ with mean zero and covariance matrix Σ is proportional to $\exp(-\frac{1}{2}x^T\Sigma^{-1}x)$. Since S_E is a quadratic form on paths, this suggests that ν should be a Gaussian distribution on the space of paths, such that if $X = \{X_t\}_{t \in \mathbb{R}}$ is a stochastic process with law ν , then for any n and any $t_1, \dots, t_n \in \mathbb{R}$, the random vector $(X_{t_1}, \dots, X_{t_n})$ should have a Gaussian distribution with mean zero and covariance given by

$$\mathbb{E}(X_{t_i}X_{t_j}) = K(t_i, t_j),$$

where $K : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is a kernel that acts as the inverse of the operator

$$-\frac{d^2}{dx^2} + x^2.$$

Fortunately, such a kernel can be explicitly written down:

$$K(x, y) = \int_0^\infty \frac{1}{\sqrt{4\pi t}} \exp\left(-\frac{(x-y)^2}{4t} - t\right) dt. \quad (2.2.1)$$

In Theorem 3.4.1 in Section 3.4, we will show that K is indeed the inverse of the operator $-\frac{d^2}{dx^2} + x^2$, in the following sense. Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a Schwartz function meaning that f is infinitely differentiable, and f and all its derivatives decay faster than the inverse of any polynomial at infinity. If we define

$$Kf(x) := \int_{-\infty}^\infty K(x, y)f(y)dy,$$

then Kf is also a Schwartz function, and

$$f = K\left(-\frac{d^2}{dx^2} + x^2\right)f = \left(-\frac{d^2}{dx^2} + x^2\right)Kf.$$

It turns out that $K(x, y)$ has a simpler form than the one displayed in equation (2.2.1).

Lemma 2.2.1. *We have $K(x, y) = \frac{1}{2}e^{-|x-y|}$.*

Proof. On the right side in equation (2.2.1), the change of variable $s = 2t/|x-y|$ gives

$$\begin{aligned} K(x, y) &= \frac{1}{2\sqrt{2\pi}}\sqrt{|x-y|} \int_0^\infty \frac{1}{\sqrt{s}} \exp\left(-\frac{1}{2}|x-y|\left(s + \frac{1}{s}\right)\right) ds \\ &= \sqrt{\frac{|x-y|}{2\pi}} \int_0^\infty \exp\left(-\frac{1}{2}|x-y|\left(u^2 + \frac{1}{u^2}\right)\right) du \\ &= e^{-|x-y|} \sqrt{\frac{|x-y|}{2\pi}} \int_0^\infty \exp\left(-\frac{1}{2}|x-y|\left(u - \frac{1}{u}\right)^2\right) du, \end{aligned}$$

where the second equality is obtained by the change of variable $u = \sqrt{s}$. Let I denote the

²See Sections B.2.1 and B.2.4 in Appendix B for definition and properties of Gaussian random variables.

integral on the right. By the change of variable $v = 1/u$, we get

$$I = \int_0^\infty \frac{1}{v^2} \exp\left(-\frac{1}{2}|x-y|\left(v - \frac{1}{v}\right)^2\right) dv.$$

Averaging this with the original expression for I , we get

$$I = \frac{1}{2} \int_0^\infty \left(1 + \frac{1}{u^2}\right) \exp\left(-\frac{1}{2}|x-y|\left(u - \frac{1}{u}\right)^2\right) du.$$

Making one final change of variable $w = u - \frac{1}{u}$, we get

$$I = \frac{1}{2} \int_{-\infty}^\infty e^{-\frac{1}{2}|x-y|w^2} dw = \frac{\sqrt{2\pi}}{2\sqrt{|x-y|}}.$$

This completes the proof. $\square$

Let us now construct a Gaussian Markov process with covariance kernel K . Let Z be a standard Gaussian random variable and B, B' be two independent Brownian motions³ on the time interval $[0, \infty)$, which are also independent of Z . Define X_t as

$$X_t := \begin{cases} \frac{1}{\sqrt{2}}e^{-t}Z + \frac{1}{\sqrt{2}}e^{-t}B(e^{2t} - 1) & \text{if } t \geq 0, \\ \frac{1}{\sqrt{2}}e^tZ + \frac{1}{\sqrt{2}}e^tB'(e^{-2t} - 1) & \text{if } t < 0. \end{cases}$$

This is clearly a continuous Gaussian process with $\mathbb{E}(X_t) = 0$ for each t . We claim that $\mathbb{E}(X_s X_t) = K(s, t)$ for any $s, t \in \mathbb{R}$. To see this, first take $t \geq s \geq 0$. Then

$$\begin{aligned} \mathbb{E}(X_s X_t) &= \frac{1}{2} e^{-(s+t)} (\mathbb{E}(Z^2) + \mathbb{E}(B(e^{2s} - 1)B(e^{2t} - 1))) \\ &= \frac{1}{2} e^{-(s+t)} (1 + e^{2s} - 1) = \frac{1}{2} e^{s-t} = K(s, t). \end{aligned}$$

Similarly, this holds if $t \leq s \leq 0$. Finally, take $s < 0 \leq t$. Then

$$\mathbb{E}(X_s X_t) = \frac{1}{2} e^{s-t} \mathbb{E}(Z^2) = \frac{1}{2} e^{s-t} = K(s, t).$$

This proves that X is a centered Gaussian process with covariance kernel K . It is known as the ‘Ornstein–Uhlenbeck process’.

We claim that X is a Markov process. To see this, define for each $s \leq t$,

$$X_{s,t} := e^{-(t-s)} X_s.$$

³See Section B.6 in Appendix B for the definition and basic properties of Brownian motion.

Then note that for each $u \leq s \leq t$,

$$\begin{aligned} \text{Cov}(X_u, X_t - X_{s,t}) &= \text{Cov}(X_u, X_t) - e^{-(t-s)} \text{Cov}(X_u, X_s) \\ &= \frac{1}{2}e^{-(t-u)} - e^{-(t-s)} \cdot \frac{1}{2}e^{-(s-u)} = 0. \end{aligned}$$

Since these covariances vanish for all $u \leq s \leq t$, we conclude that the Gaussian field $\{X_u\}_{u \leq s}$ is independent of the Gaussian field $\{X_t - X_{s,t}\}_{t \geq s}$. But $X_{s,t}$ is a function of X_s . Thus, $\{X_t\}_{t \geq s}$ can be expressed as a function of X_s and a field that is independent of $\{X_u\}_{u \leq s}$. By Lemma B.4.11, this proves that the stochastic process X is Markovian. The stationarity and reversibility of this Markov process are easy consequences of the translation invariance and symmetry of K and the last assertion of Proposition B.1.1.

Lastly, let us verify that the generator of Markov process X is strongly continuous. First, observe that by the form of K , the stationary distribution μ of X is the Gaussian distribution on the real line with mean zero and variance $\frac{1}{2}$. Thus, the Hilbert space for the quantum system should be $L^2(\mu)$. Explicitly, this is the set of all Borel measure $f : \mathbb{R} \rightarrow \mathbb{C}$ such that

$$\int_{-\infty}^{\infty} |f(x)|^2 e^{-x^2} dx < \infty.$$

Again, from the formula for K , it is easy to see that for any $t \in \mathbb{R}$ and any sequence $t_n \rightarrow t$,

$$\lim_{n \rightarrow \infty} \mathbb{E}[(X_{t_n} - X_t)^2] = 0.$$

Let $(Q(t))_{t \geq 0}$ be a semigroup for the Markov process X . Being a Markov semigroup, it is automatically an $L^2(\mu)$ -contraction, by Jensen's inequality for conditional expectation. It is not hard to show that the set of all Lipschitz continuous maps from $\mathbb{R}$ into $\mathbb{R}$ is dense in $L^2(\mu)$. For any such map f , the above identity implies that $t \mapsto Q(t)f$ is continuous, as a map from $\mathbb{R}$ into $L^2(\mu)$. Thus, by Lemma D.4.5, the semigroup is strongly continuous. By Stone's theorem, this allows us to construct the corresponding quantum system with Hilbert space $L^2(\mu)$ and unitary group $U(t) = e^{-itH}$, where H is the generator of the above semigroup.

2.3 Heuristic definition of vacuum expectation values

Let us now return to the abstract setting considered in Section 2.1. Assume that the semigroup $(Q(t))_{t \geq 0}$ has all the required properties. By Theorem B.5.5 from Appendix B, the semigroup for any Markov process is a contracting semigroup; in particular, $(Q(t))_{t \geq 0}$ is a contracting semigroup. By Theorem D.4.9 from Appendix D, the generator of any contracting semigroup is a positive operator. Thus, the generator H of $(Q(t))_{t \geq 0}$ is a positive operator. This means, in particular, that its spectrum is contained in $[0, \infty)$. But note that 0 is actually in the spectrum, because the function $\Omega \equiv 1$ on $\mathcal{X}$ is an eigenfunction of H with eigenvalue 0. Since μ (the law of X_0) is a probability measure on this space, $\Omega \in \mathcal{H} = L^2(\mu)$. The element Ω of $\mathcal{H}$ is called the *vacuum state* of the system.

The fact that $\Omega \equiv 1$ remains invariant under the action of $\tilde{U}(t)$ has the following important consequence. Recalling equation (2.1.1), we see that for any $x \in \mathcal{X}$,

$$\begin{aligned} 1 &= \Omega(x) = \tilde{U}(t)\Omega(x) \\ &= \frac{1}{Z(t)\sqrt{f(x)}} \int_{\substack{\gamma:[0,t] \rightarrow \mathcal{X} \\ \gamma(0)=x}} \sqrt{f(\gamma(t))} \Omega(\gamma(t)) e^{iS(\gamma)} \mathcal{D}\gamma \\ &= \frac{1}{Z(t)\sqrt{f(x)}} \int_{\substack{\gamma:[0,t] \rightarrow \mathcal{X} \\ \gamma(0)=x}} \sqrt{f(\gamma(t))} e^{iS(\gamma)} \mathcal{D}\gamma. \end{aligned}$$

Thus, for any $x \in \mathcal{X}$ and $t \geq 0$, we have the identity

$$Z(t) = \frac{1}{\sqrt{f(x)}} \int_{\substack{\gamma:[0,t] \rightarrow \mathcal{X} \\ \gamma(0)=x}} \sqrt{f(\gamma(t))} e^{iS(\gamma)} \mathcal{D}\gamma.$$

Multiplying by $f(x)$ and integrating over x on both sides, we get

$$Z(t) = \int_{\gamma:[0,t] \rightarrow \mathcal{X}} \sqrt{f(\gamma(0))f(\gamma(t))} e^{iS(\gamma)} \mathcal{D}\gamma. \quad (2.3.1)$$

A consequence of the above identities is that for any $s, t \geq 0$,

$$\begin{aligned} Z(s+t) &= \int_{\gamma:[0,s+t] \rightarrow \mathcal{X}} \sqrt{f(\gamma(0))f(\gamma(s+t))} e^{iS(\gamma)} \mathcal{D}\gamma \\ &= \int_{\gamma_1:[0,s] \rightarrow \mathcal{X}} \int_{\substack{\gamma_2:[s,s+t] \rightarrow \mathcal{X} \\ \gamma_2(s)=\gamma_1(s)}} \sqrt{f(\gamma_1(0))f(\gamma_2(s+t))} e^{iS(\gamma_1)+iS(\gamma_2)} \mathcal{D}\gamma_2 \mathcal{D}\gamma_1 \\ &= \int_{\gamma_1:[0,s] \rightarrow \mathcal{X}} \sqrt{f(\gamma_1(0))f(\gamma_1(s))} Z(t) e^{iS(\gamma_1)} \mathcal{D}\gamma_1 \\ &= Z(s)Z(t). \end{aligned}$$

Now, choose some time points $t_1 < \dots < t_k$ and some functions $F_1, \dots, F_k : \mathcal{X} \rightarrow \mathbb{C}$. Let γ denote the trajectory of our quantum system evolving over time. Suppose we are given that the system starts in a state ψ_0 at time $-T < t_1$ and ends at a state ψ_1 at time $T > t_k$, both having $L^2(\lambda)$ -norm equal to 1 (where λ is ‘Lebesgue measure’ on $\mathcal{X}$). Then the expected value of the product $F_1(\gamma(t_1))F_2(\gamma(t_2)) \dots F_k(\gamma(t_k))$, in the path integral formulation of quantum mechanics, is given by

$$\begin{aligned} &\langle \psi_0 | F_1(\gamma(t_1)) \dots F_k(\gamma(t_k)) | \psi_1 \rangle \\ &:= \frac{1}{Z(2T)} \int_{\gamma:[-T,T] \rightarrow \mathcal{X}} \psi_0(\gamma(-T)) \overline{\psi_1(\gamma(T))} F_1(\gamma(t_1)) \dots F_k(\gamma(t_k)) e^{iS(\gamma)} \mathcal{D}\gamma. \end{aligned}$$

Suppose that we now move to the Hilbert space $L^2(\mu)$, so that ψ_i is replaced by $\tilde{\psi}_i = \psi_i/\sqrt{f}$ for $i = 0, 1$, where f is the density of μ with respect to the ‘Lebesgue measure’ λ . Since the

expected value should remain unchanged in the new setting, we must define

$$\begin{aligned} & \langle \tilde{\psi}_0 | F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) | \tilde{\psi}_1 \rangle \\ &:= \frac{1}{Z(2T)} \int_{\gamma: [-T, T] \rightarrow \mathcal{X}} \sqrt{f(\gamma(-T))f(\gamma(T))} \tilde{\psi}_0(\gamma(-T)) \overline{\tilde{\psi}_1(\gamma(T))} \\ & \quad \cdot F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) e^{iS(\gamma)} \mathcal{D}\gamma. \end{aligned}$$

Now suppose we take $\tilde{\psi}_0 = \tilde{\psi}_1 = \Omega$, where Ω is the vacuum state that is identically equal to 1. Then we have

$$\begin{aligned} & \langle \Omega | F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) | \Omega \rangle_T \\ &:= \frac{1}{Z(2T)} \int \sqrt{f(\gamma(-T))f(\gamma(T))} F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) e^{iS(\gamma)} \mathcal{D}\gamma. \end{aligned} \quad (2.3.2)$$

The *vacuum expectation value* of the product $F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k))$ is formally defined to be the limit of the above quantity as $T \rightarrow \infty$. That is, it is the expected value when the system is assumed to have started in the vacuum state in the distant past, and ending in the vacuum state in the distant future. Appealing to the identity (2.3.1) and using translation invariance, we see that the above quantity can be written as the ratio

$$\frac{\int \sqrt{f(\gamma(-T))f(\gamma(T))} F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) e^{iS(\gamma)} \mathcal{D}\gamma}{\int \sqrt{f(\gamma(-T))f(\gamma(T))} e^{iS(\gamma)} \mathcal{D}\gamma}.$$

The limit of this, as $T \rightarrow \infty$, is heuristically written as

$$\langle \Omega | F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) | \Omega \rangle := \frac{\int F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) e^{iS(\gamma)} \mathcal{D}\gamma}{\int e^{iS(\gamma)} \mathcal{D}\gamma}, \quad (2.3.3)$$

with the understanding that the path integrals are over paths $\gamma : \mathbb{R} \rightarrow \mathcal{X}$, and the endpoints $\gamma(-\infty)$ and $\gamma(\infty)$ are integrated with respect to the vacuum state $\sqrt{f}$ in the Hilbert space $L^2(\lambda)$, or equivalently, the vacuum state Ω in the Hilbert space $L^2(\mu)$. The above form of the vacuum expectation value is the one that typically appears in heuristic calculations.

2.4 Rigorous definition of vacuum expectation values

Let us now try to give a rigorous meaning to vacuum expectation values. To compute the quantity displayed in equation (2.3.1), let us decompose each path $\gamma : [-T, T] \rightarrow \mathcal{X}$ as a concatenation of paths $\gamma_0 : [-T, t_1] \rightarrow \mathcal{X}, \dots, \gamma_k : [t_k, T] \rightarrow \mathcal{X}$, with $\gamma_i(t_i) = \gamma_{i-1}(t_i)$ for each i . Since Z is a multiplicative function (as shown above), the integral displayed in equation (2.3.2) becomes

$$\begin{aligned} & \frac{1}{Z(t_1 - (-T))Z(t_2 - t_1) \cdots Z(T - t_k)} \int \sqrt{f(\gamma_0(-T))f(\gamma_k(T))} F_1(\gamma_0(t_1)) \cdots \\ & \quad \cdots F_k(\gamma_{k-1}(t_k)) e^{iS(\gamma_0) + \cdots + iS(\gamma_k)} \mathcal{D}\gamma_k \cdots \mathcal{D}\gamma_0. \end{aligned}$$

Fixing $\gamma_0, \dots, \gamma_{k-1}$, note that by equation (2.3.1),

$$\frac{1}{Z(T-t_k)} \int \sqrt{f(\gamma_k(T))} e^{iS(\gamma_k)} \mathcal{D}\gamma_k = \sqrt{f(\gamma_{k-1}(t_k))}.$$

Substituting this in the main integral, we see that it simplifies to

$$\begin{aligned} & \frac{1}{Z(t_1+T)Z(t_2-t_1) \cdots Z(t_k-t_{k-1})} \int \sqrt{f(\gamma_0(-T))f(\gamma_{k-1}(t_k))} \\ & \cdot F_1(\gamma_0(t_1)) \cdots F_k(\gamma_{k-1}(t_k)) e^{iS(\gamma_0)+\cdots+iS(\gamma_{k-1})} \mathcal{D}\gamma_{k-1} \cdots \mathcal{D}\gamma_0. \end{aligned}$$

Next, let us integrate out γ_{k-1} . By equation (2.1.1), we have

$$\begin{aligned} & \frac{1}{Z(t_k-t_{k-1})} \int \sqrt{f(\gamma_{k-1}(t_k))} F_k(\gamma_{k-1}(t_k)) e^{iS(\gamma_{k-1})} \mathcal{D}\gamma_{k-1} \\ & = \sqrt{f(\gamma_{k-2}(t_{k-1}))} \tilde{U}(t_k-t_{k-1}) F_k(\gamma_{k-2}(t_{k-1})). \end{aligned}$$

Thus, the main integral simplifies to

$$\begin{aligned} & \frac{1}{Z(t_1+T) \cdots Z(t_{k-1}-t_{k-2})} \int \sqrt{f(\gamma_0(-T))f(\gamma_{k-2}(t_{k-1}))} \\ & \cdot F_1(\gamma_0(t_1)) \cdots F_{k-2}(\gamma_{k-3}(t_{k-2})) G_{k-1}(\gamma_{k-2}(t_{k-1})) \\ & \cdot e^{iS(\gamma_0)+\cdots+iS(\gamma_{k-2})} \mathcal{D}\gamma_{k-2} \cdots \mathcal{D}\gamma_0, \end{aligned}$$

where

$$G_{k-1}(x) := F_{k-1}(x) \tilde{U}(t_k-t_{k-1}) F_k(x).$$

It should now be clear how to continue like this, by backward induction: Having defined the functions $G_{k-1}, G_{k-2}, \dots, G_{j+1}$ for some $j \geq 1$, we define

$$G_j(x) := F_j(x) \tilde{U}(t_{j+1}-t_j) G_{j+1}(x).$$

Then by successively integrating out $\gamma_k, \gamma_{k-1}, \dots, \gamma_1$, we arrive at the identity

$$\begin{aligned} & \langle \Omega | F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) | \Omega \rangle_T \\ & = \frac{1}{Z(t_1+T)} \int \sqrt{f(\gamma_0(-T))f(\gamma_0(t_1))} G_1(\gamma_0(t_1)) e^{iS(\gamma_0)} \mathcal{D}\gamma_0. \end{aligned}$$

But by the identity (2.1.1),

$$\begin{aligned} & \frac{1}{Z(t_1+T)} \int_{\gamma_0: \gamma_0(-T)=x} \sqrt{f(\gamma_0(t_1))} G_1(\gamma_0(t_1)) e^{iS(\gamma_0)} \mathcal{D}\gamma_0 = \sqrt{f(x)} \tilde{U}(t_1+T) G_1(x) \\ & = \sqrt{f(x)} G_0(x), \end{aligned}$$

where $G_0 := \tilde{U}(t_1 + T)G_1$. Substituting this in the previous display, we get

$$\langle \Omega | F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) | \Omega \rangle_T = \int_{\mathcal{X}} G_0(x) f(x) d\lambda(x) = \int_{\mathcal{X}} G_0(x) d\mu(x).$$

But note that

$$\begin{aligned} \int_{\mathcal{X}} G_0(x) d\mu(x) &= \langle G_0, \Omega \rangle_{L^2(\mu)} = \langle \tilde{U}(t_1 + T)G_1, \Omega \rangle_{L^2(\mu)} \\ &= \langle G_1, \tilde{U}(t_1 + T)^* \Omega \rangle_{L^2(\mu)} = \langle G_1, \tilde{U}(-t_1 - T)\Omega \rangle_{L^2(\mu)} \\ &= \langle G_1, \Omega \rangle_{L^2(\mu)} = \int_{\mathcal{X}} G_1(x) d\mu(x). \end{aligned}$$

Remarkably, the last quantity has no dependence on T , as long as $-T < t_1$ and $T > t_k$. This is because the inductive definition of G_1 has no involvement of T . Thus, the above equation actually gives the limit as $T \rightarrow \infty$. To summarize, we have shown the following. Let

$$G_1 := F_1 \tilde{U}(t_2 - t_1) (F_2 \tilde{U}(t_3 - t_2) (\cdots \tilde{U}(t_k - t_{k-1}) F_k) \cdots),$$

assuming that this definition makes sense. Then the vacuum expectation value of the product $F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k))$ is given by

$$\langle \Omega | F_1(\gamma(t_1)) \cdots F_k(\gamma(t_k)) | \Omega \rangle := \int_{\mathcal{X}} G_1(x) d\mu(x).$$

In the next section, we will take this to be the definition of vacuum expectation value in a generalized setting, and show how to compute vacuum expectation values by first computing expected values of certain functions of random variables and then continuing analytically.

For a further discussion of the rigorous approach to defining vacuum expectation values, see [10, Chapter 6]. One thing we have not discussed here is the question of uniqueness of the vacuum state (if one defines the vacuum state as the ground state eigenvector of the Hamiltonian). Uniqueness requires an extra assumption, which is part of the Osterwalder–Schrader axioms [22, 23].

2.5 Calculating vacuum expectation values

Let $\{X_t\}_{t \in \mathbb{R}}$ be a stationary, reversible Markov process taking values in some measurable space $(\mathcal{X}, \mathcal{G})$. Let μ denote the law of X_0 , and let $\mathcal{H}$ denote the Hilbert space $L^2(\mathcal{X}, \mathcal{G}, \mu, \mathbb{C})$. For each $t \geq 0$, define $P(t) : \mathcal{H} \rightarrow \mathcal{H}$ as

$$P(t)F(x) := \mathbb{E}(F(X_t) | X_0 = x).$$

By the discussion preceding equation (B.4.2), this definition makes sense. By Theorem B.5.5, $(P(t))_{t \geq 0}$ is a semigroup of symmetric linear operators on $\mathcal{H}$. Suppose that this semigroup is strongly continuous. (In applications, strong continuity has to be proved on a case-by-case

basis.) Then by Theorem D.4.1, the semigroup $(P(t))_{t \geq 0}$ has a self-adjoint generator H defined on a dense domain $\mathcal{D}$. By Stone's theorem, H generates a strongly continuous one-parameter group of unitary operators $(U(t))_{t \in \mathbb{R}}$ on $\mathcal{H}$. This defines the time evolution group for an $\mathcal{X}$ -valued quantum field φ , whose states take values in $\mathcal{H}$. If the field is in a state $F \in \mathcal{H}$, then the probability interpretation is that for any $A \in \mathcal{G}$,

$$\mathbb{P}(\varphi \in A) = \frac{\int_A |F(x)|^2 d\mu(x)}{\int_{\mathcal{X}} |F(x)|^2 d\mu(x)}.$$

More generally, for any function $G : \mathcal{X} \rightarrow \mathbb{C}$, the expected value of $G(\varphi)$ is

$$\mathbb{E}(G(\varphi)) = \frac{\int_{\mathcal{X}} G(x) |F(x)|^2 d\mu(x)}{\int_{\mathcal{X}} |F(x)|^2 d\mu(x)},$$

provided that the integral in the numerator on the right is well-defined. If the state of the system is F at time 0, then it is $U(t)F$ at time t . The state $\Omega \equiv 1$ is a ground state of this system, by Theorem D.4.9 and the fact that $H\Omega = 0$ (since $P(t)\Omega = \Omega$ for all t).

Now, take any $k \geq 1$, $F_1, \dots, F_k \in \mathcal{H}$, and $-\infty < t_1 < t_2 < \dots < t_k < \infty$. Following the heuristics from the previous section, the vacuum expectation value of the product $F_1(\varphi_{t_1}) \cdots F_k(\varphi_{t_k})$ (where φ_t is the field at time t) is defined as

$$\langle \Omega | F_1(\varphi_{t_1}) \cdots F_k(\varphi_{t_k}) | \Omega \rangle := \int_{\mathcal{X}} G_1(x) d\mu(x),$$

where $G_1 : \mathcal{X} \rightarrow \mathbb{C}$ is the function

$$G_1 := F_1 U(t_2 - t_1) (F_2 U(t_3 - t_2) (\cdots U(t_k - t_{k-1}) F_k) \cdots), \quad (2.5.1)$$

provided that the right side is well-defined; that is, the function $G_{k-1} := F_{k-1} U(t_k - t_{k-1}) G_k$ is in $L^2(\mu)$, and inductively for each $1 \leq j \leq k-2$, the function $G_j := F_j U(t_{j+1} - t_j) G_{j+1}$ is in $L^2(\mu)$. The following theorem gives a criterion under which this vacuum expectation value is well-defined, and gives a way to calculate it.

Theorem 2.5.1. *Take any $k \geq 1$, $-\infty < t_1 \leq \dots \leq t_k < \infty$, $F_2, \dots, F_{k-1} \in L^\infty(\mu)$, and $F_1, F_k \in L^2(\mu)$. Then the vacuum expectation value $\langle \Omega | F_1(\varphi_{t_1}) \cdots F_k(\varphi_{t_k}) | \Omega \rangle$ is well-defined, and may be computed as follows. If $k = 1$, then*

$$\langle \Omega | F_1(\varphi_{t_1}) | \Omega \rangle = \int_{\mathcal{X}} F(x) d\mu(x).$$

Suppose that $k \geq 2$. Define $f : [0, \infty)^{k-1} \rightarrow \mathbb{C}$ as

$$f(s_1, \dots, s_{k-1}) := \mathbb{E}[F_1(X_0) F_2(X_{s_1}) F_3(X_{s_1+s_2}) \cdots F_k(X_{s_1+\dots+s_{k-1}})].$$

Then f can be analytically continued to $(i\mathbb{R})^{k-1}$ as follows. Fixing $s_2, \dots, s_{k-1}$, f has an analytic continuation to the open right half-plane in the first argument, which extends by continuity to the imaginary axis. This allows us to obtain $f(is_1, s_2, \dots, s_{k-1})$. Next, fixing

$s_1, s_3, \dots, s_{k-1}$, $f(is_1, \cdot, s_3, \dots, s_{k-1})$ has a similar analytic continuation that allows us to obtain the value of $f(is_1, is_2, s_3, \dots, s_{k-1})$. Progressing inductively, we get the value of $f(is_1, \dots, is_{k-1})$ for any nonnegative $s_1, \dots, s_{k-1}$. Then, for any $-\infty < t_1 \leq \dots \leq t_k < \infty$,

$$\langle \Omega | F_1(\varphi_{t_1}) \cdots F_k(\varphi_{t_k}) | \Omega \rangle = f(i(t_2 - t_1), \dots, i(t_k - t_{k-1})).$$

Proof. The case $k = 1$ follows from our definition of vacuum expectation value. So let us assume that $k \geq 2$. The conditions that $F_2, \dots, F_{k-1} \in L^\infty(\mu)$ and $F_1, F_k \in L^2(\mu)$ ensure that the function f is well-defined, and that the following functions are also well-defined and in $L^2(\mu)$. Fix some $s_1, \dots, s_{k-1} \in [0, \infty)$. Let $H_{k-1} := F_{k-1}P(s_{k-1})F_k$, and by backward induction, define $H_j := F_jP(s_j)H_{j+1}$ for $1 \leq j \leq k-2$. Then

$$f(s_1, \dots, s_{k-1}) = \mathbb{E}(H_1(X_0)) = \mathbb{E}(F_1(X_0)P(s_1)H_2(X_0)) = \langle P(s_1)H_2, \overline{F_1} \rangle_{L^2(\mu)}.$$

Moreover, note that if $F \in L^2(\mu)$ and $G \in L^\infty(\mu)$, then $FG \in L^2(\mu)$. Using this and the conditions that $F_2, \dots, F_{k-1} \in L^\infty(\mu)$ and $F_1, F_k \in L^2(\mu)$, it is easy to see that the vacuum expectation value is well-defined.

Now, if we fix $s_2, \dots, s_{k-1}$, then by Theorem D.4.6, $f(\cdot, s_2, \dots, s_{k-1})$ admits an analytic continuation to the open right half-plane in the first argument, which extends continuously to the imaginary axis, giving

$$\begin{aligned} f(is_1, s_2, \dots, s_{k-1}) &= \langle U(s_1)H_2, \overline{F_1} \rangle_{L^2(\mu)} \\ &= \langle H_2, U(-s_1)\overline{F_1} \rangle_{L^2(\mu)} \\ &= \langle H_2, \overline{U(-s_1)F_1} \rangle_{L^2(\mu)} \\ &= \langle F_2P(s_2)H_3, \overline{U(-s_1)F_1} \rangle_{L^2(\mu)} \\ &= \langle P(s_2)H_3, \overline{F_2U(-s_1)F_1} \rangle_{L^2(\mu)}, \end{aligned}$$

where the last step makes sense since $F_2U(-s_1)F_1 \in L^2(\mu)$ (since $F_1 \in L^2(\mu)$ and $F_2 \in L^\infty(\mu)$). Next, if we fix $s_1, s_3, \dots, s_{k-1}$, then Theorem D.4.6 and the above expression show that the function $f(is_1, \cdot, s_3, \dots, s_{k-1})$ can be analytically continued in the second argument to the open right half-plane, which can then be extended to the imaginary axis by continuity, giving

$$\begin{aligned} f(is_1, is_2, s_3, \dots, s_{k-1}) &= \langle U(s_2)H_3, \overline{F_2U(-s_1)F_1} \rangle_{L^2(\mu)} \\ &= \langle P(s_3)H_4, \overline{F_3U(-s_2)(F_2U(-s_1)F_1)} \rangle_{L^2(\mu)}. \end{aligned}$$

Continuing like this, we arrive at

$$f(is_1, \dots, is_{k-1}) = \langle U(s_{k-1})F_k, \overline{F_{k-1}(U(-s_{k-2})(\cdots F_2U(-s_1)F_1) \cdots)} \rangle_{L^2(\mu)}.$$

Now, we can successively bring each $U(-s_j)$ to the left side as $U(s_j)$, giving

$$\begin{aligned} f(is_1, \dots, is_{k-1}) &= \langle U(s_1)(F_2 U(s_2)(\cdots F_{k-1} U(s_{k-1}) F_k) \cdots), \overline{F_1} \rangle_{L^2(\mu)} \\ &= \int_{\mathcal{X}} G_1(x) d\mu(x), \end{aligned}$$

where G_1 is defined as in equation (2.5.1). This completes the proof. $\square$

Theorem 2.5.1, as presented here, is perhaps a new result. Calculating vacuum expectation values by analytic continuation of Euclidean expectation values in the Osterwalder–Schrader framework has been explored in [10, Chapter 19]. Theorem 2.5.1 is the analogous result for the Markov process approach, which seems not have appeared earlier in the literature. The key ingredient in the proof of this theorem is Theorem D.4.6 from Appendix D, which also appears to be a new result. We will use Theorem 2.5.1 extensively in our calculations henceforth.

2.6 The two-point function for the simple harmonic oscillator

Recall the quantum simple harmonic oscillator constructed in Section 2.2. The two-point function for this system is the function

$$D(s-t) := i \langle \Omega | \mathcal{T} \{ \varphi(s) \varphi(t) \} | \Omega \rangle,$$

where $\mathcal{T}$ is the time ordering operator, defined as

$$\mathcal{T} \{ \varphi(s) \varphi(t) \} = \begin{cases} \varphi(s) \varphi(t) & \text{if } s \leq t, \\ \varphi(t) \varphi(s) & \text{if } t \leq s. \end{cases}$$

Let us now compute this quantity using Theorem 2.5.1. Recall the Ornstein–Uhlenbeck process $\{X_t\}_{t \in \mathbb{R}}$ defined in Section 2.2. Recall that by Lemma 2.2.1,

$$f(s) := \mathbb{E}(X_0 X_s) = \frac{1}{2} e^{-s}$$

for $s \geq 0$. This function continues analytically to the entire complex plane, and in particular,

$$f(is) = \frac{1}{2} e^{-is}.$$

Thus, by Theorem 2.5.1, we get that

$$D(s-t) = \frac{i}{2} e^{-i|s-t|}.$$

The two-point function for the simple harmonic oscillator appears in any basic text on quantum mechanics.

2.7 The Osterwalder–Schrader approach

A problem with the Markov process approach is that in practice, verifying the Markov property and the strong continuity property can be difficult. The Osterwalder–Schrader approach, introduced in the papers [22, 23], replaces these two conditions with a single condition called ‘reflection positivity’, which is much easier to verify. We will not use this approach in this book, but we discuss it here for completeness. The idea goes as follows. Let $\mathcal{X}$ be a measurable space and $X = \{X_t\}_{t \in \mathbb{R}}$ be an $\mathcal{X}$ -valued stochastic process with law ν , which is stationary and reversible, in the sense that $\{X_{t+s}\}_{t \in \mathbb{R}}$ has law ν for any $s \in \mathbb{R}$, and $\{X_{-t}\}_{t \in \mathbb{R}}$ has law ν .

Let $X^+ := \{X_t\}_{t \geq 0}$ and $X^- := \{X_{-t}\}_{t \geq 0}$. We say that X is *reflection positive* if for any measurable function $F : \mathcal{X}^{[0, \infty)} \rightarrow \mathbb{C}$ such that $\mathbb{E}[|F(X^+)|^2] < \infty$, we have

$$\mathbb{E}[F(X^+) \overline{F(X^-)}] \geq 0.$$

It is easy to show that if X is Markovian (in addition to stationary and reversible), then it is reflection positive. The advantage of reflection positivity is that it is easier to establish than Markovianity and strong continuity, and it allows the construction of the semigroup without having to prove these properties. This is done as follows.

Let ν^+ denote the law of X^+ . Let $\mathcal{E}^+$ be any dense subspace of $L^2(\mathcal{X}^{[0, \infty)}, \nu^+)$. Define a form θ on $\mathcal{E}^+$ as

$$\theta(F, G) := \mathbb{E}[F(X^+) \overline{G(X^-)}],$$

which is linear in the first argument and conjugate linear in the second. By the Cauchy–Schwarz inequality and the reversibility of X , θ is a well-defined. The reversibility of X also implies that $\theta(F, G) = \overline{\theta(G, F)}$. By reflection positivity, it is positive semidefinite, meaning that $\theta(F, F) \geq 0$ for all F .

Lemma 2.7.1. *The functional θ satisfies the Cauchy–Schwarz inequality. That is, for all $F, G \in \mathcal{E}^+$,*

$$|\theta(F, G)| \leq \sqrt{\theta(F, F) \theta(G, G)}.$$

Proof. Take any F, G . For all $\lambda \in \mathbb{C}$,

$$\begin{aligned} 0 &\leq \theta(F - \lambda G, F - \lambda G) \\ &= \theta(F, F) + |\lambda|^2 \theta(G, G) - 2 \operatorname{Re}(\overline{\lambda} \theta(F, G)). \end{aligned}$$

If $\theta(G, G) = 0$, take $\lambda = a \theta(F, G)$ for some $a > 0$. The above inequality becomes

$$2|\theta(F, G)|^2 \leq \frac{\theta(F, F)}{a}.$$

Since a is arbitrary, this shows that $\theta(F, G) = 0$, which implies that the Cauchy–Schwarz inequality holds.

If $\theta(G, G) \neq 0$, taking

$$\lambda = \frac{\theta(F, G)}{\theta(G, G)}$$

completes the proof. $\square$

Now, define

$$\mathcal{N} := \{F \in \mathcal{E}^+ : \theta(F, F) = 0\}.$$

The fact that θ satisfies the Cauchy–Schwarz inequality implies that $\mathcal{N}$ is a subspace of $\mathcal{E}^+$. This allows us to define the quotient vector space $\mathcal{E}^+/\mathcal{N}$.

Lemma 2.7.2. *If $F, G \in \mathcal{E}^+$ and $N \in \mathcal{N}$, then $\theta(F + N, G) = \theta(F, G)$. Consequently, θ defines a positive definite inner product on $\mathcal{E}^+/\mathcal{N}$.*

Proof. The claimed identity is a straightforward consequence of the Cauchy–Schwarz inequality, since

$$|\theta(N, G)| \leq \sqrt{\theta(N, N)\theta(G, G)} = 0.$$

Combined with the conjugate symmetry of θ , this shows that if $F - F' \in \mathcal{N}$ and $G - G' \in \mathcal{N}$, then $\theta(F, G) = \theta(F', G')$. Thus, we can define θ on the quotient space. It retains all its properties, and moreover, becomes positive definite, by the definition of $\mathcal{N}$. $\square$

Since $\mathcal{E}^+/\mathcal{N}$ is an inner product space under the positive definite inner product θ , it can be completed to get a Hilbert space, which we call $\mathcal{H}$. This is the Hilbert space for the quantum system in the Osterwalder–Schrader approach. The following result justifies why this makes sense in the context of our previous discussions.

Proposition 2.7.3. *Suppose that X is Markovian, in addition to its other properties. Let μ be the law of X_0 . Let $\varphi : \mathcal{E}^+/\mathcal{N} \rightarrow L^2(\mu)$ be the map defined as $\varphi([F]) := H$, where H is the unique element of $L^2(\mu)$ such that $H(X_0) = \mathbb{E}(F(X^+) | X_0)$, and $[F]$ denotes the image of $F \in \mathcal{E}^+$ in the quotient space $\mathcal{E}^+/\mathcal{N}$. Then φ is an inner-product-preserving linear isomorphism, and $\varphi(\mathcal{E}^+/\mathcal{N})$ is dense in $L^2(\mu)$. Consequently, φ extends to a Hilbert space isomorphism between $\mathcal{H}$ and $L^2(\mu)$.*

Proof. First, let us show that the mapping $\varphi([F]) := H$, where $H(X_0) = \mathbb{E}(F(X^+) | X_0)$, is well-defined. Suppose that $[F] = [G]$. Then $F - G = N \in \mathcal{N}$. Thus, $\theta(N, N) = 0$. But by the Markov property and reversibility of X ,

$$\begin{aligned} \theta(N, N) &= \mathbb{E}(N(X^+) \overline{N(X^-)}) \\ &= \mathbb{E}[\mathbb{E}(N(X^+) \overline{N(X^-)} | X_0)] \\ &= \mathbb{E}[\mathbb{E}(N(X^+) | X_0) \mathbb{E}(\overline{N(X^-)} | X_0)] \\ &= \mathbb{E}[\mathbb{E}(N(X^+) | X_0)^2]. \end{aligned} \tag{2.7.1}$$

Thus, we get

$$\mathbb{E}(F(X^+) | X_0) - \mathbb{E}(G(X^+) | X_0) = \mathbb{E}(N(X^+) | X_0) = 0 \text{ a.s.}$$

This proves that φ is well-defined. Next, we claim that φ is injective. To see this, suppose that $\varphi([F]) = \varphi([G])$. Then $\mathbb{E}(N(X^+)|X_0) = 0$ a.s., where $N := F - G$. But by the identity (2.7.1), we see that this forces $\theta(N, N) = 0$. Thus, $[F] = [G]$.

Our next claim is that $\varphi(\mathcal{E}^+/\mathcal{N})$ is dense in $L^2(\mu)$. To see this, take any $H \in L^2(\mu)$. Define $F \in L^2(\mu^+)$ as $F(X^+) := H(X_0)$. Since $\mathcal{E}^+$ is dense in $L^2(\nu^+)$, there is a sequence $\{F_n\}_{n \geq 1}$ converging to F in $L^2(\nu^+)$. Let $H_n := \varphi([F_n])$, so that $H_n(X_0) = \mathbb{E}(F_n(X^+)|X_0)$ and $H_n \in \varphi(\mathcal{E}^+/\mathcal{N})$ for each n . Clearly, $H_n \rightarrow H$ in $L^2(\mu)$. This proves the claim.

Since φ is obviously linear, it only remains to show that it preserves inner products. But this, again, is a simple consequence of the Markov property and reversibility, akin to the proof of the identity (2.7.1). $\square$

Having constructed the Hilbert space $\mathcal{H}$, our next task is to define a semigroup $(P(t))_{t \geq 0}$ using reflection positivity, which is equal to the Markov semigroup when X is a Markov process. We define $P(t)$ as follows. Take any $t \geq 0$ and $[F] \in \mathcal{E}^+/\mathcal{N}$. For $x \in \mathcal{X}^{[0, \infty)}$, let $y = \tau_t x \in \mathcal{X}^{[0, \infty)}$ be the vector $y_s := x_{s+t}$, $s \geq 0$. Define $P(t)F \in \mathcal{E}^+$ as

$$P(t)F(x) := F(\tau_t x). \quad (2.7.2)$$

We make the following assumptions about the relation between $\mathcal{E}^+$ and $P(t)$:

1. For each $t \geq 0$, $P(t)$ maps $\mathcal{E}^+$ into itself.
2. For each $F \in \mathcal{E}^+$, the map $t \mapsto P(t)F$ is continuous.

We claim that if $F - G \in \mathcal{N}$, then $P(t)F - P(t)G \in \mathcal{N}$. By linearity, it suffices to show that $P(t)$ maps $\mathcal{N}$ into $\mathcal{N}$. Indeed, take any $N \in \mathcal{N}$. Then by the stationarity of X ,

$$\begin{aligned} \theta(P(t)N, P(t)N) &= \mathbb{E}(N(\tau_t X^+) \overline{N(\tau_t X^-)}) \\ &= \mathbb{E}(N(\{X_{s+t}\}_{s \geq 0}) \overline{N(\{X_{-s-t}\}_{s \geq 0})}) \\ &= \mathbb{E}(N(\{X_s\}_{s \geq 0}) \overline{N(\{X_{-s-2t}\}_{s \geq 0})}) \\ &= \mathbb{E}(N(X^+) \overline{P(2t)N(X^-)}) = \theta(N, P(2t)N). \end{aligned} \quad (2.7.3)$$

But by the Cauchy–Schwarz inequality for θ (Lemma 2.7.1),

$$\theta(N, P(2t)N) \leq \sqrt{\theta(N, N)\theta(P(2t)N, P(2t)N)} = 0.$$

Thus, we may define $P(t) : \mathcal{E}^+/\mathcal{N} \rightarrow \mathcal{E}^+/\mathcal{N}$ as

$$P(t)[F] := [P(t)F].$$

Now, note that the identity (2.7.3) holds not just for $N \in \mathcal{N}$, but for any $F \in \mathcal{E}^+$. Thus, we get

$$\theta(P(t)F, P(t)F) \leq \theta(F, F)^{1/2} \theta(P(2t)F, P(2t)F)^{1/2}.$$

But we can now again apply this inequality to the second term on the right. Iterating n times, we get

$$\theta(P(t)F, P(t)F) \leq \theta(F, F)^{1-2^{-n}} \theta(P(2^n t)F, P(2^n t)F)^{2^{-n}}. \quad (2.7.4)$$

But again by the stationarity and reversibility of X , we have that for any $u \geq 0$,

$$\begin{aligned} \theta(P(u)F, P(u)F) &= \mathbb{E}(F(\tau_u X^+) \overline{F(\tau_u X^-)}) \\ &\leq \sqrt{\mathbb{E}(|F(\tau_u X^+)|^2) \mathbb{E}(|F(\tau_u X^-)|^2)} = \mathbb{E}(|F(X^+)|^2). \end{aligned}$$

Thus, taking $n \rightarrow \infty$ in equation (2.7.4), we obtain

$$\theta(P(t)F, P(t)F) \leq \theta(F, F). \quad (2.7.5)$$

This shows that $P(t)$ is a contraction on $\mathcal{E}^+/\mathcal{N}$. Therefore, it extends uniquely by continuity to $\mathcal{H}$. This defines $P(t)$ on $\mathcal{H}$. The following theorem shows that it has the required properties; that is, $(P(t))_{t \geq 0}$ is a strongly continuous semigroup of symmetric operators.

Theorem 2.7.4. *The collection $(P(t))_{t \geq 0}$ satisfies the following properties:*

1. For all $s, t \geq 0$, $P(s+t) = P(s)P(t)$.
2. For each $x \in \mathcal{H}$, $t \mapsto P(t)x$ is a continuous map.
3. For each $t \geq 0$, $P(t)$ is a symmetric operator on $\mathcal{H}$.
4. For each $t \geq 0$, $P(t)$ is a contraction.

Proof. It is clear from the definition (2.7.2) of $P(t)$ that $P(s+t) = P(s)P(t)$ on $\mathcal{E}^+$. From this it follows that the semigroup property also holds on $\mathcal{H}$.

To prove strong continuity, recall that $P(t)$ is a contraction for each t . Thus, by Lemma D.4.5, it suffices to prove strong continuity on $\mathcal{E}^+/\mathcal{N}$. Since

$$\begin{aligned} |\theta(F, F)| &= \mathbb{E}[F(X^+) \overline{F(X^-)}] \\ &\leq \sqrt{\mathbb{E}(|F(X^+)|^2) \mathbb{E}(|F(X^-)|^2)} \\ &= \sqrt{\mathbb{E}(|F(X^+)|^2) \mathbb{E}(|F(X^+)|^2)} = \|F\|_{L^2(\nu^+)} \end{aligned}$$

for each $F \in \mathcal{E}^+$, this follows from the continuity of $t \mapsto P(t)F$ for $F \in \mathcal{E}^+$.

To see that $P(t)$ is symmetric, take any $F, G \in \mathcal{E}^+$. Then by the stationarity of X ,

$$\begin{aligned} \theta(P(t)F, G) &= \mathbb{E}[F(\tau_t X^+) \overline{G(X^-)}] \\ &= \mathbb{E}[F(\{X_{s+t}\}_{s \geq 0}) \overline{G(\{X_{-s}\}_{s \geq 0})}] \\ &= \mathbb{E}[F(\{X_s\}_{s \geq 0}) \overline{G(\{X_{-s-t}\}_{s \geq 0})}] \\ &= \mathbb{E}[F(X^+) \overline{G(\tau_t(X^-))}] = \theta(F, P(t)G). \end{aligned}$$

From our previous observations, it is easy to see that this identity carries over to $\mathcal{H}$.

Lastly, the contraction property of $P(t)$ follows from equation (2.7.5). This completes the proof of the theorem. $\square$

The following result justifies why the semigroup $(P(t))_{t \geq 0}$ is the same as the semigroup obtained in the Markov process approach.

Proposition 2.7.5. *Suppose that X is Markovian, in addition to its other properties. Let μ be the law of X_0 . Let $P(t)$ be the operator on $\mathcal{H}$ defined above, and define $Q(t) : L^2(\mu) \rightarrow L^2(\mu)$ as $Q(t) := \varphi \circ P(t) \circ \varphi^{-1}$, where φ is the isomorphism between $\mathcal{H}$ and $L^2(\mu)$ defined in Proposition 2.7.3. Then $Q(t)H(X_0) = \mathbb{E}(H(X_t)|X_0)$ for $H \in L^2(\mu)$.*

Proof. Take any $H \in L^2(\mu)$. Define $F \in L^2(\nu^+)$ as $F(X^+) := H(X_0)$. First, suppose that $F \in \mathcal{E}^+$. Then

$$\varphi([F])(X_0) = \mathbb{E}(F(X^+)|X_0) = H(X_0),$$

which implies that $[F] = \varphi^{-1}(H)$. Now note that

$$P(t)F(X^+) = F(\tau_t X^+) = F(\{X_{s+t}\}_{s \geq 0}) = H(X_t).$$

By assumption, $P(t)F \in \mathcal{E}^+$. Thus, by the definition of φ on $\mathcal{E}^+/\mathcal{N}$ and the above identity,

$$\varphi(P(t)[F]) = \varphi([P(t)F]) = \mathbb{E}(H(X_t)|X_0).$$

This proves the claim when $F \in \mathcal{E}^+$.

For the general case, let $\mathcal{G}$ be the subspace of all $H \in L^2(\mu)$ such that the corresponding F is in $\mathcal{E}^+$. We claim that $\mathcal{G}$ is dense in $L^2(\mu)$. To see this, take any $H \in L^2(\mu)$ and let $F(X^+) := H(X_0)$. Since $\mathcal{E}^+$ is dense in $L^2(\nu^+)$, there is a sequence $\{F_n\}_{n \geq 1}$ in $\mathcal{E}^+$ converging to F . Let $H_n(X_0) := \mathbb{E}(F_n(X^+)|X_0)$, so that $H_n = \varphi([F_n])$. But note that

$$H_n(X_0) = \mathbb{E}(F_n(X^+)|X_0) \rightarrow \mathbb{E}(F(X^+)|X_0) = H(X_0)$$

in $L^2(\mu)$. This proves that $\mathcal{G}$ is dense in $L^2(\mu)$. Now, the map $R(t) : L^2(\mu) \rightarrow L^2(\mu)$ defined as $R(t)H(X_0) := \mathbb{E}(H(X_t)|X_0)$ is continuous. Since φ is an isomorphism and $P(t)$ is a contraction, $Q(t)$ is continuous. Since $Q(t)$ and $R(t)$ agree on the dense subspace $\mathcal{G}$, this shows that they agree everywhere. $\square$

Chapter 3

The Free Quantum Field

3.1 Classical Klein–Gordon fields

In Chapter 1, we developed the quantum system of a single particle in a potential on the real line, using Brownian motion as a tool for justifying the Wick rotation heuristic. The same method can be used to develop many particle quantum systems, using multidimensional Brownian motion. A *field*, on the other hand, is like a system of infinitely many one-dimensional particles, indexed by points in space. That is, a field ϕ is a function from $\mathbb{R}^4$ into $\mathbb{R}$, where $\phi(t, x) \in \mathbb{R}$ denotes the location of the particle indexed by $x \in \mathbb{R}^3$ at time $t \in \mathbb{R}$. One of the simplest classical fields is the one that evolves according to the *Klein–Gordon equation*

$$\frac{\partial^2 \phi}{\partial t^2} = \Delta \phi - m^2 \phi,$$

where Δ denotes the spatial Laplacian and m is a nonnegative constant. (When $m = 0$, this is just the wave equation.) It is not hard to verify that solutions of the Klein–Gordon equation are critical points of the action

$$S(\phi) = \frac{1}{2} \int_{\mathbb{R}^4} ((\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2 - m^2 \phi(t, x)^2) dt dx, \quad (3.1.1)$$

where ∇ denotes the gradient with respect to the spatial coordinates. Indeed, if ψ is any compactly supported smooth map from $\mathbb{R}^4$ into $\mathbb{R}$, then the variational derivative of S in the direction ψ is given by

$$\begin{aligned} \left. \frac{d}{d\epsilon} S(\phi + \epsilon \psi) \right|_{\epsilon=0} &= \frac{1}{2} \int_{\mathbb{R}^4} \frac{\partial}{\partial \epsilon} ((\partial_t \phi(t, x) + \epsilon \partial_t \psi(t, x))^2 \\ &\quad - \|\nabla \phi(t, x) + \epsilon \nabla \psi(t, x)\|^2 - m^2 (\phi(t, x) + \epsilon \psi(t, x))^2) \Big|_{\epsilon=0} dt dx \\ &= \int_{\mathbb{R}^4} (\partial_t \phi(t, x) \partial_t \psi(t, x) - \nabla \phi(t, x) \cdot \nabla \psi(t, x) - m^2 \phi(t, x) \psi(t, x)) dt dx \\ &= \int_{\mathbb{R}^4} \psi(t, x) (-\partial_t^2 \phi(t, x) + \Delta \phi(t, x) - m^2 \phi(t, x)) dt dx, \end{aligned}$$

where the last step follows by integration by parts. This vanishes for all ψ if and only if ϕ satisfies the Klein–Gordon equation.

3.2 Heuristic definition of the free quantum field

The massive free quantum field with mass m is heuristically the path integral quantization of the classical Klein–Gordon field with mass m . To quantize, first note that instead of the location $\gamma(t)$ of a single particle at time t , we now have an entire field $\phi(t, \cdot)$ at time t , where $\phi(t, x)$ denotes the location on the real line of the particle indexed by $x \in \mathbb{R}^3$. When we quantized the single particle, the exact location $\gamma(t)$ was replaced by a wave function $\psi_t(\cdot)$, with the understanding that $|\psi_t(x)|^2$ is proportional to the probability density function of the particle at a point $x \in \mathbb{R}$. Analogously, we now need a wave function $\psi_t(\cdot)$, whose modulus squared gives the probability density function of ϕ . That is, we first need to settle on a space $\mathcal{S}$ of possible values of $\phi(t, \cdot)$, and then some definition of Lebesgue measure on this space, and finally declare that the probability density of $\phi(t, \cdot)$ at a point $\xi \in \mathcal{S}$, with respect to this Lebesgue measure, is proportional to $|\psi_t(\xi)|^2$. Then according to the prescription given by equation (1.6.2), ψ_t should evolve as

$$\psi_t(\xi) = \frac{1}{Z(t)} \int_{\phi: \phi(0, \cdot) = \xi(\cdot)} \psi_0(\phi(t, \cdot)) e^{iS(\phi)} \mathcal{D}\phi,$$

where S is the action displayed in equation (3.1.1) but with the integral over time going from 0 to t , $\mathcal{D}\phi$ denotes Lebesgue integration over the space of trajectories

$$\{\phi : \phi(s, \cdot) \in \mathcal{S} \text{ for all } 0 \leq s \leq t\},$$

and $Z(t)$ is some suitably chosen normalizing constant that makes the evolution $U(t)\psi_0 := \psi_t$ unitary.

Incidentally, the above is just one description of the free quantum field. There is a different description of the free quantum field where we define the Hilbert space and the time evolution group directly, via the process of *second quantization*. This produces the Fock space definition of the free quantum field. We will discuss it in Chapter 4. More detailed discussions of the Fock space construction are in Folland [6, Chapter 5] and Talagrand [30, Chapter 5].

3.3 Wick rotation for the free quantum field

First, we need to construct the Euclidean action by Wick rotating the action for the free quantum field. However, we need to be careful with this. Recall that for the one-particle system in Chapter 2, there was a given candidate for the Hamiltonian H , namely, the Schrödinger operator. For the free quantum field, the Hamiltonian is not given to us, so we cannot verify if the answer we get is ‘correct’. In this situation, the best option is to rely on the correctness of the construction that we have already established as valid, namely,

the construction of the time evolution group for a particle in a potential. The construction generalizes easily to a multi-particle system in the following way. Suppose we have N particles on the real line, each with mass 1, whose positions at time t are given by a vector $\gamma(t) = (\gamma_1(t), \dots, \gamma_N(t)) \in \mathbb{R}^N$. The particle system is moving under the influence of a potential $V : \mathbb{R}^N \rightarrow \mathbb{R}$. The action of the trajectory γ is

$$S(\gamma) = \int_{-\infty}^{\infty} \left(\frac{1}{2} \|\dot{\gamma}(t)\|^2 - V(\gamma(t)) \right) dt, \quad (3.3.1)$$

and the Euclidean action is

$$S_E(\gamma) = \int_{-\infty}^{\infty} \left(\frac{1}{2} \|\dot{\gamma}(t)\|^2 + V(\gamma(t)) \right) dt. \quad (3.3.2)$$

It is easy to generalize Theorem 1.7.1 to the above setting, thereby giving us confidence that the above Wick rotation is the correct approach.

Now, how can we connect the evolution of a particle with the evolution of a field? There is an obvious way to do it: Treat the field as a collection of particles moving on the real line, each indexed by a point in $\mathbb{R}^3$. That is, $\phi(t, x) \in \mathbb{R}$ denotes the location of the particle indexed by $x \in \mathbb{R}^3$ at time $t \in \mathbb{R}$. With this interpretation, we can view ϕ as the trajectory of this infinite particle system, with $\phi(t, \cdot)$ being its location at time t . The action $S(\phi)$ displayed in equation (3.1.1), therefore, has the same form as the action $S(\gamma)$ displayed in equation (3.3.1), with

$$V(\phi(t, \cdot)) = \frac{1}{2} \int_{\mathbb{R}^3} (\|\nabla \phi(t, x)\|^2 + \phi(t, x)^2) dx.$$

Thus, the form of the Euclidean action displayed in equation (3.3.2) suggests that we should define the Euclidean action for the free field theory as

$$S_E(\phi) = \frac{1}{2} \int_{\mathbb{R}^4} ((\partial_t \phi(t, x))^2 + \|\nabla \phi(t, x)\|^2 + m^2 \phi(t, x)^2) dt dx.$$

Note that the above action could also be obtained simply by replacing t by $-it$ everywhere, as we did for the single particle system. But the above reasoning gives more insight into the nature of the system.

It is clear that this action satisfies the additive property described in Section 2.1. So the next step is to construct a probability measure on the set of all $\phi : \mathbb{R}^4 \rightarrow \mathbb{R}$ that has probability density function proportional to $e^{-S_E(\phi)}$ with respect to ‘Lebesgue measure’ on the space of all ϕ .

We will go beyond $\mathbb{R}^4$ and define the above object, known as the Euclidean free field with mass m , on $\mathbb{R}^d$ for arbitrary $d \geq 1$. Heuristically, the Euclidean free field on $\mathbb{R}^d$ with mass m is a Gaussian field ϕ on $\mathbb{R}^d$ with probability density function (with respect to a

hypothetical ‘Lebesgue measure’ on the space of fields) proportional to

$$\exp\left(-\frac{1}{2}\int_{\mathbb{R}^d}\|\nabla\phi(x)\|^2dx - \frac{m^2}{2}\int_{\mathbb{R}^d}\phi(x)^2dx\right). \quad (3.3.3)$$

The construction of this object requires a series of steps, which we carry out in the subsequent sections.

3.4 Rigorous construction of the Euclidean free field

In this section, we will rigorously construct the Euclidean free field. This is a standard object of study in modern probability theory. For the probabilistic perspective and many more details, we refer to the surveys of Sheffield [27] and Berestycki and Powell [3].

Notice that the heuristically defined probability density function for the Euclidean free field displayed in equation (3.3.3) is of the form $\exp(-\frac{1}{2}Q(\phi, \phi))$, where Q is the positive definite inner product

$$Q(\phi, \psi) := \int_{\mathbb{R}^d} \nabla\phi(x) \cdot \nabla\psi(x)dx + m^2 \int_{\mathbb{R}^d} \phi(x)\psi(x)dx. \quad (3.4.1)$$

Integration by parts shows that

$$Q(\phi, \psi) = \int_{\mathbb{R}^d} \phi(x)(-\Delta + m^2I)\psi(x)dx.$$

Comparing this with the form of Gaussian densities¹, it seems that a rigorously defined Euclidean free field on $\mathbb{R}^d$ should be a centered Gaussian field with covariance structure

$$\text{Cov}(\phi(x), \phi(y)) = G_m(x, y),$$

where G_m is an integral kernel that is the inverse of $-\Delta + m^2I$, provided that such an object exists. Thus, we have two tasks to accomplish: (a) Prove that G_m exists, and (b) construct a centered Gaussian field with covariance kernel G_m .

Recall that a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is said to be *rapidly decaying* if for any $\alpha > 0$, there is some C such that $|f(x)| \leq C(1 + \|x\|)^{-\alpha}$ for all $x \in \mathbb{R}^d$. Let $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$ denote the space of real-valued *Schwartz functions* on $\mathbb{R}^d$, that is, infinitely differentiable functions f such that f and all its derivatives of all orders are rapidly decaying functions.

Given positive integer d and nonnegative real number m such that either $d \geq 3$ and $m \geq 0$, or $d = 1, 2$ and $m > 0$, define the function $G_m : \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{R}$ as

$$G_m(x) := \int_0^\infty \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x\|^2}{4t} - m^2t\right)dt. \quad (3.4.2)$$

It is easy to check that the integral is well-defined and finite. We extend G_m to the whole of $\mathbb{R}^d$ by defining $G_m(0) := \lim_{x \rightarrow 0} G_m(x)$. Note that $G_m(0) = \infty$ if $d \geq 2$, but finite if

¹See Sections B.2.1 and B.2.4 in Appendix B.

$d = 1$ and $m > 0$. We will view G_m as an integral operator acting on Schwartz functions as

$$G_m f(x) := \int_{\mathbb{R}^d} G_m(x-y) f(y) dy.$$

The following theorem encapsulates the essential properties of G_m .

Theorem 3.4.1. *Take any d, m such that either $d \geq 3$ and $m \geq 0$, or $d = 1, 2$ and $m > 0$. The operator G_m is well-defined on $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$, and maps into $C^\infty(\mathbb{R}^d)$. It is the inverse of $-\Delta + m^2 I$ on $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$, in the sense that for any $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$,*

$$G_m(-\Delta + m^2 I)f = (-\Delta + m^2 I)G_m f = f.$$

Lastly, when $d \geq 3$ and $m = 0$, G_m has the explicit form

$$G_m(x) = \frac{\Gamma(d/2 - 1)}{4\pi^{d/2} \|x\|^{d-2}}.$$

To prove Theorem 3.4.1, we begin with a preliminary upper bound on $G_m(x)$.

Lemma 3.4.2. *If $d \geq 3$ and $m \geq 0$, or $d = 1, 2$ and $m > 0$, the function G_m satisfies the bound*

$$G_m(x) \leq \begin{cases} C(d)e^{-\|x\|m/\sqrt{2}}\|x\|^{-(d-2)} & \text{if } d \geq 3, \\ C(d)e^{-\|x\|m/4}(1 + m^{-2}) & \text{if } d = 2 \text{ and } \|x\| \geq 2m, \\ C(d)e^{-\|x\|m/4}(m^{-2} + \log(2m/\|x\|)) & \text{if } d = 2 \text{ and } \|x\| < 2m, \\ C(d)e^{-|x|m/4}(1 + m^{-2}) & \text{if } d = 1, \end{cases}$$

for all $x \in \mathbb{R}^d \setminus \{0\}$, where $C(d)$ is a constant that depends only on d .

Proof. First, let $d \geq 3$. Making the change of variable $s = \|x\|^2/4t$ in the integral defining $G_m(x)$ in equation (3.4.2) shows that

$$G_m(x) = \frac{1}{\|x\|^{d-2}} \int_0^\infty \frac{s^{(d-4)/2}}{4\pi^{d/2}} \exp\left(-s - \frac{m^2\|x\|^2}{4s}\right) ds. \quad (3.4.3)$$

Now note that by the AM-GM inequality,

$$\frac{s}{2} + \frac{m^2\|x\|^2}{4s} \geq \left(\frac{sm^2\|x\|^2}{2s}\right)^{1/2} = \frac{\|x\|m}{\sqrt{2}}.$$

By (3.4.3), this shows that

$$G_m(x) \leq \frac{e^{-\|x\|m/\sqrt{2}}}{\|x\|^{d-2}} \int_0^\infty \frac{s^{(d-4)/2}}{4\pi^{d/2}} e^{-s/2} ds.$$

It is easy to see that the above integral is finite, since $d \geq 3$. This completes the proof when $d \geq 3$. Next, let $d = 2$. First, note that by an application of the AM-GM inequality

similar to the above, we get

$$G_m(x) \leq e^{-\|x\|^{m/4}} \int_0^\infty \frac{1}{4\pi t} \exp\left(-\frac{\|x\|^2}{8t} - \frac{m^2 t}{2}\right) dt.$$

We break up the above integral into two parts, from 0 to $\|x\|/2m$, and from $\|x\|/2m$ to infinity. For the second part, note that

$$\int_{\|x\|/2m}^\infty \frac{1}{4\pi t} \exp\left(-\frac{\|x\|^2}{8t} - \frac{m^2 t}{2}\right) dt \leq \int_{\|x\|/2m}^\infty \frac{1}{4\pi t} e^{-m^2 t/2} dt.$$

If $\|x\| \geq 2m$, then the right side is bounded above by C/m^2 for a universal constant C . If $\|x\| < 2m$, we further break up the integral into an integral from $\|x\|/2m$ to 1, and an integral from 1 to ∞ . The second integral is bounded above by C/m^2 , and the first integral is bounded above $C \log(2m/\|x\|)$. Next, note that by the change of variable $s = \|x\|^2/4t$, we get

$$\begin{aligned} \int_0^{\|x\|/2m} \frac{1}{4\pi t} \exp\left(-\frac{\|x\|^2}{8t} - \frac{m^2 t}{2}\right) dt &= \int_{\|x\|/2m}^\infty \frac{1}{4\pi s} \exp\left(-\frac{s}{2} - \frac{\lambda\|x\|^2}{8s}\right) ds \\ &\leq \int_{\|x\|/2m}^\infty \frac{1}{4\pi s} e^{-s/2} ds. \end{aligned}$$

By a similar argument as above, this is bounded by a universal constant C if $\|x\| \geq 2m$ and by $C \log(2m/\|x\|)$ if $\|x\| < 2m$. This completes the proof in the case $d = 2$.

Finally, let $d = 1$. Following exactly the same steps as for $d = 2$, we arrive at the inequality

$$G_m(x) \leq e^{-|x|^{m/4}} \int_0^\infty \frac{1}{\sqrt{4\pi t}} \exp\left(-\frac{|x|^2}{8t} - \frac{m^2 t}{2}\right) dt \leq e^{-|x|^{m/4}} \int_0^\infty \frac{e^{-m^2 t/2}}{\sqrt{4\pi t}} dt.$$

It is easy to see that the integral on the right is bounded by $C(1 + m^{-2})$. □

The next lemma proves some of the properties asserted by Theorem 3.4.1.

Lemma 3.4.3. *If $d \geq 3$ and $m \geq 0$, or $d = 1, 2$ and $m > 0$, the operator G_m sends Schwartz functions to C^∞ functions, and commutes with the derivative operators $\partial_i := \partial/\partial x_i$, $i = 1, \dots, d$.*

Proof. First, consider $d \geq 3$ and $m \geq 0$. Take any $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$. Note that $G_m f$ can be alternatively expressed as

$$G_m f(x) = \int_{\mathbb{R}^d} G_m(y) f(x - y) dy.$$

For any x, y , the bound from Lemma 3.4.2 and the fact that f is a Schwartz function show

that for any $\alpha > 1$,

$$\begin{aligned} \left| \frac{\partial}{\partial x_i} (G_m(y)f(x-y)) \right| &= G_m(y) |\partial_i f(x-y)| \\ &\leq \frac{C}{\|y\|^{d-2}(1+\|x-y\|)^\alpha} \end{aligned}$$

for some constant C that does not depend on x and y . It is easy to see from this bound (with a large enough α) and the dominated convergence theorem that $G_m f$ is a differentiable function, and for each i ,

$$\partial_i G_m f = G_m \partial_i f.$$

This proves, in particular, that G_m commutes with the derivative operators. By iteration, this also proves that $G_m f$ is infinitely differentiable for any $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$.

For $d = 1, 2$ and $m > 0$, the same argument works, observing that $G_m(x)$ is exponentially decaying in $\|x\|$ as $\|x\| \rightarrow \infty$ by Lemma 3.4.2. $\square$

We are now ready to finish the proof of Theorem 3.4.1.

Proof of Theorem 3.4.1. Take any $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$ and let $g := G_m f$. By Lemma 3.4.3, g is an infinitely differentiable function, and $\Delta g = G_m \Delta f$. Thus, by Fubini's theorem,

$$\begin{aligned} \Delta g(x) &= \int_{\mathbb{R}^d} G_m(x-y) \Delta f(y) dy \\ &= \int_{\mathbb{R}^d} \int_0^\infty \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t} - m^2 t\right) \Delta f(y) dt dy \\ &= \int_0^\infty \int_{\mathbb{R}^d} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t} - m^2 t\right) \Delta f(y) dy dt. \end{aligned}$$

Note that for any $t > 0$,

$$\begin{aligned} &\left| \int_{\mathbb{R}^d} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t} - m^2 t\right) \Delta f(y) dy \right| \\ &\leq \|\Delta f\|_\infty \int_{\mathbb{R}^d} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t}\right) dy = \|\Delta f\|_\infty, \end{aligned}$$

where $\|\Delta f\|_\infty$ denotes the supremum norm of Δf . This uniform bound shows that

$$\Delta g(x) = \lim_{\epsilon \downarrow 0} \int_\epsilon^\infty \int_{\mathbb{R}^d} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t} - m^2 t\right) \Delta f(y) dy dt.$$

Take any $t > 0$. Applying integration by parts (which is valid since the integrand is

sufficiently well-behaved and Δf is rapidly decaying), we get

$$\begin{aligned} & \int_{\mathbb{R}^d} \exp\left(-\frac{\|x-y\|^2}{4t}\right) \Delta f(y) dy \\ &= \int_{\mathbb{R}^d} f(y) \sum_{i=1}^d \frac{\partial^2}{\partial y_i^2} \exp\left(-\frac{\|x-y\|^2}{4t}\right) dy \\ &= \int_{\mathbb{R}^d} f(y) \left(-\frac{d}{2t} + \frac{\|x-y\|^2}{4t^2}\right) \exp\left(-\frac{\|x-y\|^2}{4t}\right) dy. \end{aligned}$$

Plugging this into the previous display, and again applying Fubini's theorem, we get

$$\Delta g(x) = \lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^d} \int_{\epsilon}^{\infty} \frac{1}{(4\pi t)^{d/2}} f(y) \left(-\frac{d}{2t} + \frac{\|x-y\|^2}{4t^2}\right) \exp\left(-\frac{\|x-y\|^2}{4t} - m^2 t\right) dt dy.$$

Now note that for any distinct x and y ,

$$\begin{aligned} & \frac{\partial}{\partial t} \left(\frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t}\right) \right) \\ &= \frac{1}{(4\pi t)^{d/2}} \left(-\frac{d}{2t} + \frac{\|x-y\|^2}{4t^2} \right) \exp\left(-\frac{\|x-y\|^2}{4t}\right). \end{aligned}$$

Using this in the previous display, we get

$$\Delta g(x) = \lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^d} \int_{\epsilon}^{\infty} f(y) e^{-m^2 t} \frac{\partial}{\partial t} \left(\frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t}\right) \right) dt dy.$$

Take any $\epsilon > 0$. Using integration by parts, we get

$$\begin{aligned} & \int_{\epsilon}^{\infty} e^{-m^2 t} \frac{\partial}{\partial t} \left(\frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t}\right) \right) dt \\ &= -e^{-m^2 \epsilon} \frac{1}{(4\pi \epsilon)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4\epsilon}\right) + \int_{\epsilon}^{\infty} m^2 e^{-m^2 t} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t}\right) dt. \end{aligned}$$

Combining this with the previous step gives us

$$\begin{aligned} \Delta g(x) &= -\lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^d} f(y) e^{-m^2 \epsilon} \frac{1}{(4\pi \epsilon)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4\epsilon}\right) dy \\ &\quad + \lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^d} \int_{\epsilon}^{\infty} m^2 f(y) e^{-m^2 t} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t}\right) dt dy. \end{aligned} \quad (3.4.4)$$

By the change of variable $z = \epsilon^{-1/2}(x-y)$, we have

$$\int_{\mathbb{R}^d} f(y) \frac{1}{(4\pi \epsilon)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4\epsilon}\right) dy = \int_{\mathbb{R}^d} f(x - \epsilon^{1/2} z) \frac{1}{(4\pi)^{d/2}} e^{-\|z\|^2/4} dz.$$

By the dominated convergence theorem, this shows that the first limit in (3.4.4) is equal to $f(x)$. Again, by the dominated convergence theorem and Fubini's theorem, the second limit equals $m^2 g(x)$. Thus, $\Delta g(x) = -f(x) + m^2 g(x)$. This proves that $(-\Delta + m^2)G_m = I$. Since

G_m commutes with differential operators, this also proves that $G_m(-\Delta + m^2) = I$. $\square$

Having completed the proof of Theorem 3.4.1, we are now ready to construct the Euclidean free field on $\mathbb{R}^d$ with mass m , where either $d \geq 3$ and $m \geq 0$, or $d = 2$ and $m > 0$. We leave out the case $d = 1$ since it has already been analyzed in Section 2.2. Recall that our goal is to construct a centered Gaussian field ϕ on $\mathbb{R}^d$ satisfying

$$\text{Cov}(\phi(x), \phi(y)) = G_m(x - y).$$

This means, if we consider ϕ as a linear functional acting on $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$ as

$$\phi(f) := \int_{\mathbb{R}^d} \phi(x) f(x) dx, \quad (3.4.5)$$

then $\{\phi(f) : f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})\}$ should be a centered Gaussian field indexed by $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$, satisfying

$$\text{Cov}(\phi(f), \phi(g)) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x) g(y) G_m(x - y) dy dx =: R(f, g). \quad (3.4.6)$$

We will now rigorously construct the field $\{\phi(f) : f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})\}$ as a well-defined collection of centered Gaussian random variables with the above covariance structure.

Lemma 3.4.4. *The bilinear form R defined above is a positive definite inner product on $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$. Moreover, the topology induced by this inner product is separable.*

Proof. Throughout this proof, $C_1, C_2, \dots$ will denote constants that only depend on d . First, if $d \geq 3$, then note that by Lemma 3.4.2, for any $f, g \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$,

$$\begin{aligned} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |G_m(x - y) f(x) g(y)| dy dx &\leq C_1 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \frac{|f(x) g(y)|}{\|x - y\|^{d-2}} dy dx \\ &\leq C_1 \int_{\mathbb{R}^d} |f(x)| \left(\int_{y: \|y-x\| \leq 1} \frac{\|g\|_{L^\infty}}{\|x - y\|^{d-2}} dy \right) dx \\ &\quad + C_1 \int_{\mathbb{R}^d} |f(x)| \left(\int_{y: \|y-x\| > 1} |g(y)| dy \right) dx \\ &\leq C_1 \|f\|_{L^1} \|g\|_{L^\infty} + C_2 \|f\|_{L^1} \|g\|_{L^1} < \infty. \end{aligned} \quad (3.4.7)$$

It is easy to check using Lemma 3.4.2 that this also holds if $d = 2$ and $m > 0$. This shows that R is a well-defined bilinear form. The finiteness of the above quantity, together with the formula (3.4.2) for G_m , shows that for any $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$,

$$R(f, f) = \int_0^\infty \frac{e^{-m^2 t}}{(4\pi t)^{d/2}} \left(\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x) f(y) \exp\left(-\frac{\|x - y\|^2}{4t}\right) dx dy \right) dt.$$

Now, note that if Z is a d -dimensional standard normal random vector, then

$$\exp\left(-\frac{\|x - y\|^2}{4t}\right) = \mathbb{E}(e^{i(x-y) \cdot Z / \sqrt{2t}}).$$

Thus, by Fubini's theorem,

$$\begin{aligned} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x)f(y) \exp\left(-\frac{\|x-y\|^2}{4t}\right) dx dy &= \mathbb{E}\left(\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x)f(y) e^{i(x-y)\cdot Z/\sqrt{2t}} dx dy\right) \\ &= \mathbb{E}\left|\int_{\mathbb{R}^d} f(x) e^{ix\cdot Z/\sqrt{2t}} dx\right|^2 = \mathbb{E}|\widehat{f}(Z/\sqrt{2t})|^2, \end{aligned}$$

where $\widehat{f}$ is the Fourier transform of f . This proves that $R(f, f) \geq 0$. Moreover, if $R(f, f) = 0$, then the above identity implies that for almost every $t \geq 0$, $\widehat{f}(Z/\sqrt{2t}) = 0$ with probability one. But this can happen only if $\widehat{f} = 0$ almost everywhere on the real line. Since f is a Schwartz function, it is easy to deduce from this that $f \equiv 0$. Thus, R is a positive definite inner product on $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$.

Now, it is a simple fact that $L^1(\mathbb{R}^d)$ is a separable metric space. Since any subset of a separable metric space is separable, it follows that for any positive integer N , the set

$$\{f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R}) : \|f\|_{L^\infty} \leq N\}$$

is separable in the L^1 topology. The inequality (3.4.7), therefore, shows that this set is separable in the topology induced by R . Since this holds for every N , we conclude that $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$ is separable in the R -topology. $\square$

A consequence of separability is that the space $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$ equipped with the R inner product has a countable orthonormal basis $\{f_n\}_{n \geq 1}$, meaning that the sequence $\{f_n\}_{n \geq 1}$ is orthonormal for R , and any $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$ can be written uniquely as a linear combination

$$f = \sum_{n=1}^{\infty} a_n f_n, \tag{3.4.8}$$

where $a_n = R(f, f_n)$, and the right side is the limit of finite partial sums where the order of summation is irrelevant. To see this, just take a countable dense subset and obtain an orthonormal basis by Gram-Schmidt orthogonalization, and then show that this works.

Given an orthonormal basis $\{f_n\}_{n \geq 1}$ as above, define a collection of random variables $\phi := \{\phi(f) : f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})\}$ as

$$\phi(f) := \sum_{n=1}^{\infty} a_n X_n,$$

where $\{X_n\}_{n \geq 1}$ is a sequence of i.i.d. $N(0, 1)$ random variables defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and $\{a_n\}_{n \geq 1}$ are the coefficients in the orthonormal series expansion of f as in equation (3.4.8). Note that $\phi(f)$ is a well-defined random variable since $\sum_{n=1}^{\infty} a_n^2 = R(f, f) < \infty$, which implies that the partial sums $\sum_{n=1}^N a_n X_n$ converge in $L^2(\Omega)$ as $N \rightarrow \infty$. The collection ϕ is a random linear functional on $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$, which we will call Euclidean free field on $\mathbb{R}^d$ with mass m . To see that this nomenclature is justified, note that if

$$f = \sum_{n=1}^{\infty} a_n f_n, \quad g = \sum_{n=1}^{\infty} b_n f_n,$$

then

$$\text{Cov}(\phi(f), \phi(g)) = \sum_{n=1}^{\infty} a_n b_n = R(f, g), \quad (3.4.9)$$

thereby satisfying equation (3.4.6).

Note that by the logic that we used to arrive at equation (3.4.5), $\phi(f)$ can be heuristically viewed as the L^2 inner product of ϕ and f , and ϕ itself can be viewed as a centered Gaussian field on $\mathbb{R}^d$ with $\text{Cov}(\phi(x), \phi(y)) = G_m(x - y)$. But for $d \geq 2$, since $G_m(0) = \infty$, it is impossible to rigorously define this field pointwise.

3.5 Time slices of the Euclidean free field

Take any d, m such that either $d \geq 3$ and $m \geq 0$ or $d = 2$ and $m > 0$. In this section, we will consider $\mathbb{R}^d$ as spacetime, with the first coordinate denoting time. We will denote points in $\mathbb{R}^d$ as (t, x) , where $t \in \mathbb{R}$ and $x = (x_1, \dots, x_{d-1}) \in \mathbb{R}^{d-1}$. In the previous section, we constructed the Euclidean free field ϕ on $\mathbb{R}^d$ as a random linear functional on $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$. To construct the free quantum field following the prescription outlined in Chapter 2, it is necessary to view ϕ as a stochastic process $\{\phi(t, \cdot)\}_{t \in \mathbb{R}}$ of fields evolving over time.

Fortunately, this is doable. Recall how we defined $\phi(f)$ for $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$: We first defined a positive definite inner product on $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$ in equation (3.4.6). Then we showed that $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$ is separable under the topology induced by this inner product. This allowed us to extract a countable orthonormal basis $\{f_n\}_{n \geq 1}$. Finally, for $f = \sum_{n=1}^{\infty} a_n f_n \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$, we define $\phi(f) := \sum_{n=1}^{\infty} a_n X_n$, where $\{X_n\}_{n \geq 1}$ is a sequence of i.i.d. $N(0, 1)$ random variables defined on some probability space.

Now, note that the above construction need not be limited to $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$. In fact, we can take the completion of $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$ under the inner product R . The elements of this Hilbert spaces are the formal sums $\sum_{n=1}^{\infty} a_n f_n$, where $\{a_n\}_{n \geq 1}$ is any sequence of real numbers such that $\sum_{n=1}^{\infty} a_n^2 < \infty$. For such f , we define $\phi(f)$ as before, that is, $\phi(f) := \sum_{n=1}^{\infty} a_n X_n$.

It turns out that that for any $f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$ and $t \in \mathbb{R}$, the function $\tilde{f} : \mathbb{R}^d \rightarrow \mathbb{R}$ formally defined as $\tilde{f}(s, x) := \delta_t(s) f(x)$, where δ_t is the Dirac delta at t , is an element of the above completion. This allows us to define inner products like

$$\phi_t(f) = \int_{\mathbb{R}^{d-1}} \phi(t, x) f(x) dx,$$

which define the time slice $\phi_t(\cdot) = \phi(t, \cdot)$ of the Euclidean free field ϕ . The main reason why this is possible is that if we formally define $\phi_t(f)$ as above, then

$$\text{Cov}(\phi_s(f), \phi_t(g)) = \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} f(x) g(y) G_m(s - t, x - y) dy dx,$$

which is finite, since $G_m(0, x - y)$ blows up sufficiently slowly as $x - y \rightarrow 0$. The following theorem gives a rigorous construction of ϕ_t .

Theorem 3.5.1. *Take any $w \in \mathcal{S}(\mathbb{R}, \mathbb{R})$ which is nonnegative everywhere and integrates to 1. Take any $f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$ and $t \in \mathbb{R}$. For each $n \geq 1$, define $f_n : \mathbb{R}^d \rightarrow \mathbb{R}$ as*

$$f_n(s, x) := nw(n(s - t))f(x).$$

Then $f_n \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$. Let ϕ be the Euclidean free field with mass m on $\mathbb{R}^d$, as defined in the previous section. Then $\{\phi(f_n)\}_{n \geq 1}$ is a Cauchy sequence of random variables in the L^2 metric, which therefore has a limit in L^2 , which we will call $\phi_t(f)$. Then $\{\phi_t(f) : f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R}), t \in \mathbb{R}\}$ is a Gaussian field indexed by $\mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R}) \times \mathbb{R}$, with

$$\text{Cov}(\phi_s(f), \phi_t(g)) = \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} f(x)g(y)G_m(s - t, x - y)dydx. \quad (3.5.1)$$

Proof. It is easy to verify that $f_n \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$ for each n . To show that $\{\phi(f_n)\}_{n \geq 1}$ is Cauchy in L^2 , it suffices to prove that $\{f_n\}_{n \geq 1}$ is a Cauchy sequence in the R -topology, by the identity (3.4.9). Without loss of generality, let $t = 0$. Then note that for any $k, n \geq 1$,

$$\begin{aligned} R(f_k - f_n, f_k - f_n) &= \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}} \int_{\mathbb{R}} (f_k(t, x) - f_n(t, x))(f_k(s, y) - f_n(s, y))G_m(t - s, x - y)dsdtdydx \\ &= \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}} \int_{\mathbb{R}} (kw(kt) - nw(nt))(kw(ks) - nw(ns)) \\ &\quad \cdot f(x)f(y)G_m(t - s, x - y)dsdtdydx \end{aligned}$$

Let us now expand the product of the two terms in parentheses into a sum of four terms. Consider the first term:

$$\int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}} \int_{\mathbb{R}} kw(kt)kw(ks)f(x)f(y)G_m(t - s, x - y)dsdtdydx.$$

Changing variables to $t' = kt$ and $s' = ks$, this becomes

$$\int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}} \int_{\mathbb{R}} w(t')w(s')f(x)f(y)G_m(k^{-1}t' - k^{-1}s', x - y)ds'dt'dydx.$$

Using the bound on G_m from Lemma 3.4.2 and the dominated convergence theorem, it is now easy to deduce that the above expression tends, as $k \rightarrow \infty$, to

$$\int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}} \int_{\mathbb{R}} w(t')w(s')f(x)f(y)G_m(0, x - y)ds'dt'dydx.$$

Since w integrates to 1, this is equal to

$$\int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} f(x)f(y)G_m(0, x - y)dydx.$$

By the same procedure, we see that the other three terms also tend to the same limit (with alternating signs) as k and n simultaneously tend to infinity. This proves that $\{f_n\}_{n \geq 1}$ is a

Cauchy sequence in the R -topology, thereby proving the first claim. Similar calculations prove the second claim. $\square$

3.6 Markov property of the Euclidean free field

As before, take d, m such that either $d \geq 3$ and $m \geq 0$ or $d = 2$ and $m > 0$. Let $\mathcal{A} = \mathbb{R}^{\mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})}$ be the space of all functions from $\mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$ into $\mathbb{R}$. Let $\mathcal{C}$ denote the cylinder σ -algebra on $\mathcal{A}$. Let $\{\phi_t(f) : f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})\}$ denote the time t slice of the Euclidean free field on $\mathbb{R}^d$ with mass m . By Proposition B.1.1, ϕ_t can be viewed as an $\mathcal{A}$ -valued random variable. In this section, we prove the following result. (See Section B.5.3 in Appendix B for an introduction to Markov processes.)

Theorem 3.6.1. *The $\mathcal{A}$ -valued stochastic process $\{\phi_t\}_{t \in \mathbb{R}}$ defined above is a Markov process, according to the definition of Markov process in Section B.5.3. Moreover, this Markov process is stationary and reversible.*

We need some preparation. For $t > 0$ and $x \in \mathbb{R}^{d-1}$, define

$$K_m(t, x) := \int_0^\infty \frac{t}{\sqrt{4\pi s^3}(4\pi s)^{(d-1)/2}} \exp\left(-\frac{t^2 + \|x\|^2}{4s} - m^2 s\right) ds.$$

Lemma 3.6.2. *For any $t > 0$ and $x \in \mathbb{R}^{d-1}$,*

$$G_m(t, x) = \int_{\mathbb{R}^{d-1}} G_m(0, y) K_m(t, x - y) dy.$$

Proof. Note that

$$\begin{aligned} & \int_{\mathbb{R}^{d-1}} G_m(0, y) K_m(t, x - y) dy \\ &= \int_{\mathbb{R}^{d-1}} \int_0^\infty \int_0^\infty \frac{t}{(4\pi u)^{d/2} \sqrt{4\pi s^3}(4\pi s)^{(d-1)/2}} \\ & \quad \cdot \exp\left(-\frac{\|y\|^2}{4u} - \frac{t^2 + \|x - y\|^2}{4s} - m^2(s + u)\right) ds du dy. \end{aligned}$$

Now, by the convolution properties of Gaussian densities, we have

$$\begin{aligned} & \int_{\mathbb{R}^{d-1}} \frac{1}{(4\pi s)^{(d-1)/2} (4\pi u)^{(d-1)/2}} \exp\left(-\frac{\|y\|^2}{4u} - \frac{\|x - y\|^2}{4s}\right) dy \\ &= \frac{1}{(4\pi(s + u))^{(d-1)/2}} \exp\left(-\frac{\|x\|^2}{4(s + u)}\right). \end{aligned}$$

Combining, we get

$$\begin{aligned}
 & \int_{\mathbb{R}^{d-1}} G_m(0, y) K_m(t, x - y) dy \\
 &= \int_0^\infty \int_0^\infty \frac{t}{\sqrt{4\pi s^3} (4\pi u)^{1/2} (4\pi(s+u))^{(d-1)/2}} \exp\left(-\frac{t^2}{4s} - \frac{\|x\|^2}{4(s+u)} - m^2(s+u)\right) ds du \\
 &= \int_0^\infty \int_0^r \frac{t}{\sqrt{4\pi s^3} (4\pi(r-s))^{1/2} (4\pi r)^{(d-1)/2}} \exp\left(-\frac{t^2}{4s} - \frac{\|x\|^2}{4r} - m^2 r\right) ds dr.
 \end{aligned}$$

Take any $r > 0$ and let

$$I := \int_0^r \frac{t}{\sqrt{4\pi s^3} (4\pi(r-s))^{1/2}} e^{-t^2/4s} ds.$$

Let us make the substitution $u = r/s - 1$. Then $s = r/(u+1)$, $r-s = ru/(u+1)$, and $ds = -rdu/(u+1)^2$, which gives

$$\begin{aligned}
 I &= \int_0^\infty \frac{t}{4\pi \sqrt{r^3/(u+1)^3} \sqrt{ru/(u+1)}} e^{-\frac{t^2}{4r}(u+1)} \frac{r}{(u+1)^2} du \\
 &= \frac{te^{-t^2/4r}}{4\pi r} \int_0^\infty u^{-1/2} e^{-\frac{t^2}{4r}u} du.
 \end{aligned}$$

But the last integral is a Gamma integral, which evaluates to

$$\frac{\Gamma(1/2)}{\sqrt{t^2/4r}} = \frac{\sqrt{4\pi r}}{t}.$$

Thus, we get

$$I = \frac{e^{-t^2/4r}}{\sqrt{4\pi r}}.$$

Using this in our previous computation completes the proof. $\square$

Lemma 3.6.3. *For any $s, t > 0$,*

$$K_m(s+t, x) = \int_{\mathbb{R}^{d-1}} K_m(s, y) K_m(t, x-y) dy.$$

Proof. Proceeding as in the proof of Lemma 3.6.2, we get

$$\begin{aligned}
 & \int_{\mathbb{R}^{d-1}} K_m(s, y) K_m(t, x-y) dy \\
 &= \int_0^\infty \int_0^\infty \frac{ste^{-s^2/4u-t^2/4v}}{4\pi u^{3/2} v^{3/2} (4\pi(u+v))^{(d-1)/2}} e^{-\|x\|^2/4(u+v)-m^2(u+v)} du dv \\
 &= \int_0^\infty \left(\int_0^r \frac{ste^{-s^2/4u-t^2/4(r-u)}}{4\pi u^{3/2} (r-u)^{3/2}} du \right) \frac{e^{-\|x\|^2/4r-m^2r}}{(4\pi r)^{(d-1)/2}} dr.
 \end{aligned}$$

Now, the function

$$p_t(x) := \frac{t}{\sqrt{4\pi x^3}} e^{-t^2/4x}$$

is the probability density function p_t defined in Section B.3.3. From Section B.3.3, we have that p_{s+t} is the convolution of p_s and p_t . The claim follows from these observations. $\square$

As a corollary of the above lemmas, we obtain the following.

Corollary 3.6.4. *For any $s, t > 0$,*

$$G_m(s+t, x) = \int_{\mathbb{R}^{d-1}} G_m(s, y) K_m(t, x-y) dy.$$

Proof. Using Lemma 3.6.2 to write $G_m(s, y)$ as an integral, we get

$$\int_{\mathbb{R}^{d-1}} G_m(s, y) K_m(t, x-y) dy = \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} G_m(0, z) K_m(s, y-z) K_m(t, x-y) dz dy.$$

But by Lemma 3.6.3,

$$\int_{\mathbb{R}^{d-1}} K_m(s, y-z) K_m(t, x-y) dy = K_m(s+t, x-z).$$

Substituting this in the previous display and again applying Lemma 3.6.2, we get the desired result. $\square$

Take any $-\infty < s < t < \infty$. For $f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$, define

$$K_m^{t-s} f(x) := \int_{\mathbb{R}^{d-1}} K_m(t-s, x-y) f(y) dy,$$

and then define

$$\psi_{s,t}(f) := \phi_s(K_m^{t-s} f).$$

It is not immediately clear that this makes sense, because $K_m^{t-s} f$ may not be a Schwartz function on $\mathbb{R}^{d-1}$. The following lemma shows that it does.

Lemma 3.6.5. *The mapping $(u, x) \mapsto K_m^{t-s} f(x) \delta_s(u)$ belongs to the closure $\mathcal{S}(\mathbb{R}^d, \mathbb{R})$ under the R inner product defined earlier, where δ_s denotes the Dirac delta at s . Thus, $\phi_s(K_m^{t-s} f)$ is well-defined.*

Proof. By the same argument as the one used to derive the inequality (3.4.7), one can show that for any $f, g \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$,

$$\int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} |G_m(0, x-y) f(x) g(y)| dy dx \leq C_1 \|f\|_{L^1} \|g\|_{L^p} + C_2 \|f\|_{L^1} \|g\|_{L^1}$$

for some $p \in (1, \infty)$ depending on d . Now, by the same argument as in the proof of Theorem A.2.11, one can show that Schwartz functions are dense in $L^1(\mathbb{R}^{d-1}) \cap L^p(\mathbb{R}^{d-1})$

under the norm $\|f\| := \|f\|_{L^1} + \|f\|_{L^p}$. Combining these two observations, one can now show that for any $f \in L^1(\mathbb{R}^{d-1}) \cap L^p(\mathbb{R}^{d-1})$ that there is a sequence $\{f_n\}_{n \geq 1}$ in $\mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$ such that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} G_m(0, x-y)(f(x) - f_n(x))(f(y) - f_n(y)) dy dx = 0.$$

It is not hard to check that the function $K_m^{t-s} f$, as defined above, is in $L^1(\mathbb{R}^{d-1}) \cap L^p(\mathbb{R}^{d-1})$ when $f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$. Combining these observations and mimicking the proof of Theorem 3.5.1, one can now complete the proof of this lemma. $\square$

One can formally write

$$\psi_{s,t}(f) = \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} K_m(t-s, x-y) f(y) \phi_s(x) dy dx.$$

From this, we get the following identity. For any $f, g \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$, and any $u \leq s$,

$$\begin{aligned} & \text{Cov}(\phi_u(g), \phi_t(f) - \psi_{s,t}(f)) \\ &= \text{Cov}\left(\int_{\mathbb{R}^{d-1}} \phi_u(x) g(x) dx, \int_{\mathbb{R}^{d-1}} \left(\phi_t(x) - \int_{\mathbb{R}^{d-1}} K_m(t-s, x-y) \phi_s(y) dy\right) f(x) dx\right) \\ &= \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} G_m(t-u, x-y) g(x) f(y) dx dy \\ &\quad - \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} K_m(t-s, y-z) G_m(s-u, z-x) g(x) f(y) dx dy dz. \end{aligned}$$

But by Corollary 3.6.4,

$$\int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} K_m(t-s, y-z) G_m(s-u, z-x) dz = G_m(t-u, y-x) = G_m(t-u, x-y).$$

Plugging this into the previous display, we see that

$$\text{Cov}(\phi_u(g), \phi_t(f) - \psi_{s,t}(f)) = 0.$$

Since these covariances vanish for all $f, g \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$ and all $u \leq s$, we conclude that the Gaussian field $\{\phi_u(f) : u \leq s, f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})\}$ is independent of the Gaussian field $\{\phi_t(f) - \psi_{s,t}(f) : f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})\}$. But $\psi_{s,t}(f)$ is a function of the field ϕ_s . Thus, ϕ_t can be expressed as a function of ϕ_s and a field that is independent of $\{\phi_u\}_{u \leq s}$. By Lemma B.4.11, this proves that the $\mathcal{A}$ -valued stochastic process $\{\phi_t\}_{t \in \mathbb{R}}$ is Markovian. The stationarity and reversibility of this Markov process are easy consequences of the properties of G_m and the last assertion of Proposition B.1.1.

3.7 Rigorous construction of the free quantum field

Take d, m such that either $d \geq 3$ and $m \geq 0$ or $d = 2$ and $m > 0$. As in the previous section, let $\mathcal{A} = \mathbb{R}^{\mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})}$ be the space of all functions from $\mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$ into $\mathbb{R}$, let $\mathcal{C}$ denote the

cylinder σ -algebra on $\mathcal{A}$, and let $\{\phi_t(f) : f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})\}$ denote the time t slice of the Euclidean free field on $\mathbb{R}^d$ with mass m . Since $\phi_0 : \Omega \rightarrow \mathcal{A}$ is a measurable, it follows that the law of ϕ_0 is a probability measure on $(\mathcal{A}, \mathcal{C})$. Explicitly, if μ denotes the law of ϕ_0 , then for any $C \in \mathcal{C}$,

$$\mu(C) = \mathbb{P}(\phi_0^{-1}(C)).$$

By the prescription from Chapter 2, the appropriate choice for the Hilbert space $\mathcal{H}$ for the free quantum field with mass m is the L^2 space of the probability space $(\mathcal{A}, \mathcal{C}, \mu)$. That is,

$$\begin{aligned} \mathcal{H} &= L^2(\mathcal{A}, \mathcal{C}, \mu) \\ &= \left\{ F : F \text{ is a measurable map from } \mathcal{A} \text{ into } \mathbb{C}, \text{ satisfying } \int_{\mathcal{A}} |F(\psi)|^2 d\mu(\psi) < \infty \right\}, \end{aligned}$$

where μ is the law of ϕ_0 .

Following the prescription from Chapter 2, we define $P(t) : \mathcal{H} \rightarrow \mathcal{H}$ as

$$P(t)F(\phi) := \mathbb{E}(F(\phi_t) | \phi_0 = \phi),$$

for $t \geq 0$. For the rigorous meaning of the above definition, see the discussion preceding equation (B.4.2). By Theorem 3.6.1 and Theorem B.5.5, we already know that $(P(t))_{t \geq 0}$ is a semigroup of symmetric operators on $\mathcal{H}$. It is obvious that $P(0) = I$. So, to fulfill the criteria outlined in Chapter 2, it only remains to show that it is strongly continuous. By Theorem B.5.5, $\|P(t)\| \leq 1$ for each t . Thus, by Lemma D.4.5, it suffices to show that $t \mapsto P(t)F$ is strongly continuous for a dense set of F 's. By Corollary A.2.10, square-integrable functions of the form $F(\phi(f_1), \dots, \phi(f_n))$, ranging over all $n \geq 1$ and $f_1, \dots, f_n \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$, form a dense subspace of $\mathcal{H}$. But by Theorem B.2.4, this subspace is contained in the closure of the span of all functions of the form $e^{it_1\phi(f_1) + \dots + it_n\phi(f_n)}$. By linearity of ϕ , we have

$$t_1\phi(f_1) + \dots + t_n\phi(f_n) = \phi(t_1f_1 + \dots + t_nf_n).$$

Thus, it suffices to prove that for any $f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$, the map $t \mapsto P(t)F_f$ is continuous, where $F_f(\phi) := e^{i\phi(f)}$. To show this, first observe that for any $s, t \geq 0$,

$$\begin{aligned} \mathbb{E}|P(s)F_f(\phi_0) - P(t)F_f(\phi_0)|^2 &= \mathbb{E}|\mathbb{E}(F_f(\phi_s) | \phi_0) - \mathbb{E}(F_f(\phi_t) | \phi_0)|^2 \\ &\leq \mathbb{E}|F_f(\phi_s) - F_f(\phi_t)|^2 \\ &= \mathbb{E}|e^{i(\phi_s(f) - \phi_t(f))} - 1|^2 \\ &= 2 - \mathbb{E}(e^{i(\phi_s(f) - \phi_t(f))}) - \mathbb{E}(e^{-i(\phi_s(f) - \phi_t(f))}) \\ &= 2 - 2e^{-\frac{1}{2} \text{Var}(\phi_s(f) - \phi_t(f))}, \end{aligned}$$

where the last identity holds because $\phi_s(f) - \phi_t(f)$ is a centered Gaussian random variable. From the definition of the time slices of the Euclidean free field, it is easy to deduce that

$$\text{Var}(\phi_s(f) - \phi_t(f)) = \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} f(x)f(y)(2G_m(0, x-y) - 2G_m(t-s, x-y))dxdy.$$

Again, a simple argument using the dominated convergence theorem and the properties of Schwartz functions and G_m , that the above integral converges to zero as $t - s \rightarrow 0$. This proves that $(P(t))_{t \geq 0}$ is a strongly continuous semigroup. By the Hille–Yosida theorem, this semigroup has a self-adjoint generator H . By Stone’s theorem, H generates a strongly continuous one-parameter group of unitary operators on $\mathcal{H}$. By the argument outlined in Chapter 2, this group can be considered to be the time evolution group for the free quantum field.

The construction of the free quantum field given above is slightly different than the construction in Glimm and Jaffe [10, Chapter 6], which uses the Osterwalder–Schrader axioms. It is also different than the construction via second quantization, as carried out rigorously in Folland [6, Chapter 5] and Talagrand [30, Chapter 5]. The constructed Hilbert spaces and time evolution groups are the same in all cases, nonetheless.

3.8 The Feynman propagator

Recall the definition of vacuum expectation values in a quantum system from Section 2.5. In this section, we will calculate the two-point correlation function of the free quantum field, which is the vacuum expectation value

$$\Delta_F(x - y) := i\langle \Omega | \mathcal{T}\{\varphi(x)\varphi(y)\} | \Omega \rangle, \quad (3.8.1)$$

where φ denotes a free quantum field in $\mathbb{R}^4$ with mass m , x and y are points in $\mathbb{R}^4$, and $\mathcal{T}$ denotes the time-ordering operator, defined as follows. If $x = (x_0, x_1, x_2, x_3)$ and $y = (y_0, y_1, y_2, y_3)$, with x_0 and y_0 being the time coordinates, then

$$\mathcal{T}\{\varphi(x)\varphi(y)\} = \begin{cases} \varphi(x)\varphi(y) & \text{if } x_0 \leq y_0, \\ \varphi(y)\varphi(x) & \text{if } y_0 \leq x_0. \end{cases}$$

The function (or really, the distribution) Δ_F is known as the *Feynman propagator*. Reverting back to our convention of denoting points in $\mathbb{R}^4$ by (t, x) where $t \in \mathbb{R}$ and $x \in \mathbb{R}^3$, the Feynman propagator is formally given by

$$\Delta_F(t, x) = \frac{i}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i|t|\sqrt{\|p\|^2 + m^2} + ip \cdot x}}{2\sqrt{\|p\|^2 + m^2}} dp.$$

Rigorously, this means that for each $t \in \mathbb{R}$, $\Delta_F(t, \cdot, \cdot)$ is a bilinear form acting on $\mathcal{S}(\mathbb{R}^3, \mathbb{R}) \times \mathcal{S}(\mathbb{R}^3, \mathbb{R})$ as

$$\begin{aligned} \Delta_F(t, f, g) &= \frac{i}{(2\pi)^3} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \frac{e^{-i|t|\sqrt{\|p\|^2 + m^2} + ip \cdot (x-y)} f(x)g(y)}{2\sqrt{\|p\|^2 + m^2}} dx dy dp \\ &:= \frac{i}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i|t|\sqrt{\|p\|^2 + m^2}} \widehat{f}(p) \widehat{g}(\overline{p})}{2\sqrt{\|p\|^2 + m^2}} dp. \end{aligned}$$

Note that the last expression is mathematically well-defined, and is taken as the definition of $\Delta_F(t, f, g)$. The above is not the standard form of the Feynman propagator that is displayed in physics textbooks (e.g., in [25, Chapter 4]), but it is equivalent. We will encounter the standard form in the next section.

The following is a rigorous statement of the claim made in equation (3.8.1).

Theorem 3.8.1. *Let φ denote the free quantum field on $\mathbb{R}^4$ with mass $m \geq 0$, as defined in the previous section. Let φ_t denote the value of the field at time t . Then for any $f, g \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$ and $-\infty < s \leq t < \infty$,*

$$\langle \Omega | \varphi_s(g) \varphi_t(f) | \Omega \rangle = -i \Delta_F(t - s, f, g),$$

where Δ_F is the Feynman propagator defined above. The same result holds for the free quantum field in any dimension $d \geq 3$, with the definition of Δ_F modified by changing the integration domain to $\mathbb{R}^{d-1}$, and replaced $(2\pi)^3$ to $(2\pi)^{d-1}$. For $d = 2$, the results hold if $m > 0$.

We need several lemmas for the proof of this theorem. Since the result holds in any dimension $d \geq 3$, that is what we will prove.

Lemma 3.8.2. *Take any d, m such that either $d \geq 3$ and $m \geq 0$, or $d = 2$ and $m > 0$. For any $f, g \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$, we have*

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x) g(y) G_m(x - y) dy dx = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{\widehat{f}(p) \overline{\widehat{g}(p)}}{\|p\|^2 + m^2} dp.$$

Proof. In the following, all integrals are over $\mathbb{R}^d$, except the integral over t , which is from 0 to ∞ . Take any $\epsilon > 0$. Since $\widehat{f}$ and $\widehat{g}$ are also Schwartz functions (by Theorem A.3.1),

$$\begin{aligned} & \iiint \iiint |\widehat{f}(p) \overline{\widehat{g}(q)}| e^{-\epsilon \|x\|^2} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x - y\|^2}{4t} - (m + \epsilon)^2 t\right) dt dq dp dy dx \\ &= \iiint \iiint |\widehat{f}(p) \overline{\widehat{g}(q)}| e^{-\epsilon \|x\|^2 - (m + \epsilon)^2 t} dt dq dp dx < \infty. \end{aligned}$$

Thus, by the formula for G_m from equation (3.4.2) and Fubini's theorem,

$$\begin{aligned}
 & \iint f(x)g(y)e^{-\epsilon\|x\|^2}G_{m+\epsilon}(x-y)dydx \\
 &= \frac{1}{(2\pi)^{2d}} \iiint \widehat{f}(p)\widehat{g}(q)e^{-ip\cdot x+iq\cdot y-\epsilon\|x\|^2} \\
 & \quad \cdot \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{\|x-y\|^2}{4t} - (m+\epsilon)^2 t\right) dt dq dp dy dx \\
 &= \frac{1}{(2\pi)^{2d}} \iiint \widehat{f}(p)\widehat{g}(q)e^{-i(p-q)\cdot x-\epsilon\|x\|^2-((m+\epsilon)^2+\|q\|^2)t} dt dq dp dx \\
 &= \frac{1}{(2\pi)^{2d}} \iiint \frac{\widehat{f}(p)\widehat{g}(q)}{\|q\|^2 + (m+\epsilon)^2} e^{-i(p-q)\cdot x-\epsilon\|x\|^2} dq dp dx \\
 &= \frac{(\pi/\epsilon)^{d/2}}{(2\pi)^{2d}} \iint \frac{\widehat{f}(p)\widehat{g}(q)}{\|q\|^2 + (m+\epsilon)^2} e^{-\frac{1}{4\epsilon}\|p-q\|^2} dq dp.
 \end{aligned}$$

Applying the change of variable $r = \frac{1}{\sqrt{2\epsilon}}(q-p)$, we get

$$\iint f(x)g(y)e^{-\epsilon\|x\|^2}G_{m+\epsilon}(x-y)dydx = \frac{1}{(2\pi)^{3d/2}} \iint \frac{\widehat{f}(q-\sqrt{2\epsilon}r)\widehat{g}(q)}{\|q\|^2 + (m+\epsilon)^2} e^{-\frac{1}{2}\|r\|^2} dq dr.$$

Taking $\epsilon \rightarrow 0$ on both sides and applying the dominated convergence theorem (using the bound on G_m from Lemma 3.4.2), we get the desired result. $\square$

Lemma 3.8.3. *Take any d, m such that either $d \geq 3$ and $m \geq 0$, or $d = 2$ and $m > 0$. Take any $t \geq 0$. Let G_m denote the inverse of $-\Delta + m^2 I$ on $\mathbb{R}^d$, as defined in equation (3.4.2). For any $f, g \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$, we have*

$$\int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} f(x)g(y)G_m(t, x-y)dydx = \frac{1}{(2\pi)^{d-1}} \int_{\mathbb{R}^{d-1}} \frac{e^{-t\sqrt{\|p\|^2+m^2}}\widehat{f}(p)\widehat{g}(p)}{2\sqrt{\|p\|^2+m^2}} dp.$$

Proof. In the following, integrals over s, r and q are over $\mathbb{R}$, and the rest are over $\mathbb{R}^{d-1}$. Take any $\epsilon > 0$. Define $f_\epsilon, g_\epsilon \in \mathcal{S}(\mathbb{R}^d, \mathbb{R})$ as

$$f_\epsilon(s, x) := f(x) \frac{e^{-\frac{1}{2\epsilon}(s-t)^2}}{\sqrt{2\pi\epsilon}}, \quad g_\epsilon(s, x) := g(x) \frac{e^{-\frac{1}{2\epsilon}s^2}}{\sqrt{2\pi\epsilon}}.$$

Then for $q \in \mathbb{R}$ and $p \in \mathbb{R}^{d-1}$,

$$\widehat{f}_\epsilon(q, p) = e^{itq - \frac{1}{2}\epsilon q^2} \widehat{f}(p), \quad \widehat{g}_\epsilon(q, p) = e^{-\frac{1}{2}\epsilon q^2} \widehat{g}(p).$$

Thus, applying Lemma 3.8.2 to the functions f_ϵ and g_ϵ , we get

$$\iiint f_\epsilon(s, x)g_\epsilon(r, y)G_m(s-r, x-y)dydx ds dr = \frac{1}{(2\pi)^d} \iint \frac{e^{itq - \epsilon q^2} \widehat{f}(p)\widehat{g}(p)}{q^2 + \|p\|^2 + m^2} dp dq.$$

Now, by change of variable,

$$\begin{aligned} & \iiint f_\epsilon(s, x) g_\epsilon(r, y) G_m(s - r, x - y) dy dx ds dr \\ &= \iiint f(x) g(y) \frac{e^{-\frac{1}{2}(s^2 + r^2)}}{2\pi} G_m(t + \epsilon(s - r), x - y) dy dx ds dr. \end{aligned}$$

Using the bound from Lemma 3.4.2, it is now easy to apply the dominated convergence theorem to get that

$$\lim_{\epsilon \rightarrow 0} \iiint f_\epsilon(s, x) g_\epsilon(r, y) G_m(s - r, x - y) dy dx ds dr = \iint f(x) g(y) G_m(t, x - y) dy dx.$$

On the other hand, again by the dominated convergence theorem

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \frac{1}{(2\pi)^d} \iint \frac{e^{itq - \epsilon q^2} \widehat{f}(p) \overline{\widehat{g}(p)}}{q^2 + \|p\|^2 + m^2} dp dq &= \frac{1}{(2\pi)^d} \iint \frac{e^{itq} \widehat{f}(p) \overline{\widehat{g}(p)}}{q^2 + \|p\|^2 + m^2} dp dq \\ &= \frac{1}{(2\pi)^{d-1}} \int \frac{e^{-t\sqrt{\|p\|^2 + m^2}} \widehat{f}(p) \overline{\widehat{g}(p)}}{2\sqrt{\|p\|^2 + m^2}} dp, \end{aligned}$$

where the last step follows from the well-known identity

$$\int_{-\infty}^{\infty} \frac{e^{isx}}{x^2 + a^2} dx = \frac{\pi}{|a|} e^{-|sa|},$$

which holds for $s \in \mathbb{R}$ and $a \in \mathbb{R} \setminus \{0\}$. (This follows from the computation of the characteristic function of the standard Cauchy distribution in Section B.3.2.) This completes the proof. $\square$

Lemma 3.8.4. *Take any $f, g \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$. Let $\Omega := \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$ denote the open right half-plane and let $\overline{\Omega}$ denote its closure. Define a function $h : \overline{\Omega} \rightarrow \mathbb{C}$ as*

$$h(z) := \frac{1}{(2\pi)^{d-1}} \int_{\mathbb{R}^{d-1}} \frac{e^{-z\sqrt{\|p\|^2 + m^2}} \widehat{f}(p) \overline{\widehat{g}(p)}}{2\sqrt{\|p\|^2 + m^2}} dp.$$

Then h is continuous on $\overline{\Omega}$ and analytic in Ω .

Proof. By Theorem A.3.1, $\widehat{f}$ and $\widehat{g}$ are Schwartz functions. The proof follows easily from this observation and the dominated convergence theorem, and the fact that $\operatorname{Re}(z) > 0$ for all $z \in \Omega$. $\square$

We are now ready to prove Theorem 3.8.1.

Proof of Theorem 3.8.1. By translation invariance, it suffices to consider $s = 0$ and $t \geq 0$. Let ϕ denote the Euclidean free field on $\mathbb{R}^d$ with mass m , and let $(P(t))_{t \geq 0}$ be the semigroup associated with it. Let $(U(t))_{t \in \mathbb{R}}$ be the time evolution group for the free quantum field.

Let μ denote the law of ϕ_0 , as before. Then by Theorem D.4.6, the function

$$h(t) := \langle g, P(t)f \rangle_{L^2(\mu)}$$

has an analytic continuation to the open right half-plane, which extends continuously to the imaginary axis. Let h denote this continuation. Then by Theorem D.4.6 and the definition of vacuum expectation value from Section 2.5,

$$h(it) = \langle \Omega | \varphi_0(g) \varphi_t(f) | \Omega \rangle.$$

Thus, we only need to compute h on the closed right half-plane. By Theorem 3.5.1 and Lemma 3.8.3,

$$\begin{aligned} h(t) &= \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} f(x)g(y)G_m(t, x-y)dx dy \\ &= \frac{1}{(2\pi)^{d-1}} \int_{\mathbb{R}^{d-1}} \frac{e^{-t\sqrt{\|p\|^2+m^2}} \widehat{f}(p) \overline{\widehat{g}(p)}}{2\sqrt{\|p\|^2+m^2}} dp. \end{aligned}$$

Lemma 3.8.4 shows that $h(it)$ can be obtained simply by replacing t by it in the above expression. This completes the proof. $\square$

3.9 An alternate form of the Feynman propagator

An alternative way to view the Feynman propagator is to see it as a linear functional on $(\mathcal{S}(\mathbb{R}^4, \mathbb{R}))^2$, given by

$$\begin{aligned} \Delta_F(f, g) &= \frac{i}{(2\pi)^3} \iiint \iiint \frac{e^{-i|t-s|\sqrt{\|p\|^2+m^2}} e^{ip \cdot (x-y)} f(t, x) g(s, y)}{2\sqrt{\|p\|^2+m^2}} dx dy dp dt ds \\ &:= \frac{i}{(2\pi)^3} \iiint \frac{e^{-i|t-s|\sqrt{\|q\|^2+m^2}} \widehat{f}_t(q) \overline{\widehat{g}_s(q)}}{2\sqrt{\|q\|^2+m^2}} dq dt ds, \end{aligned}$$

where $f_t := f(t, \cdot)$ and $g_s := g(s, \cdot)$, and the expression in the second line is the rigorous definition. Under this viewpoint, the Feynman propagator can be expressed in a different way, that is usually more convenient for calculations. First, for $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$, recall that its Fourier transform in Minkowski signature² is defined as

$$\widetilde{f}(p) := \int_{\mathbb{R}^4} e^{i(x,p)} f(x) dx,$$

where $p \in \mathbb{R}^4$, and for $x, p \in \mathbb{R}^4$, the Minkowski inner product of x and p is given by

$$(x, p) := p_0 x_0 - p_1 x_1 - p_2 x_2 - p_3 x_3.$$

²See Section A.3.3 in Appendix A for the basics of Fourier transforms in Minkowski signature.

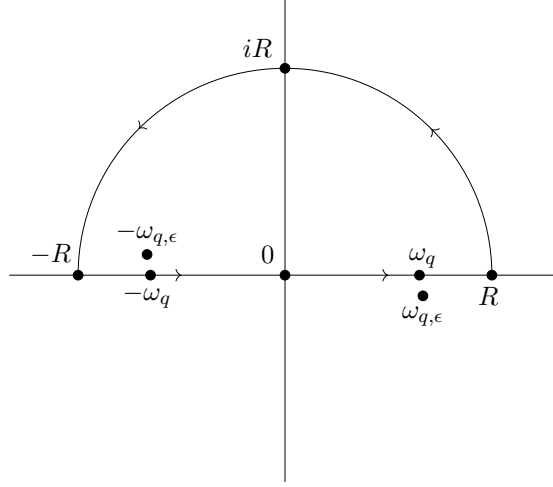


Figure 3.1: The contour used in the proof of Theorem 3.9.1 when $t - s \geq 0$.

By Theorem A.3.6, if f is a Schwartz function, then $\tilde{f}$ is also a Schwartz function.

The following theorem gives the promised alternate form of the Feynman propagator in terms of the Fourier transform in Minkowski spacetime. This is actually the standard form of the Feynman propagator as it appears in physics textbooks [25, Chapter 4]. We adopt the relativistic notation that for a vector $p = (p_0, p_1, p_2, p_3) \in \mathbb{R}^4$,

$$p^2 := p_0^2 - p_1^2 - p_2^2 - p_3^2.$$

Note this is the ‘squared norm’ induced by for the Minkowski inner product defined above.

Theorem 3.9.1. *The Feynman propagator can be alternately expressed as*

$$\Delta_F(x) = \lim_{\epsilon \downarrow 0} \frac{-1}{(2\pi)^4} \int_{\mathbb{R}^4} \frac{e^{-i(x,p)}}{p^2 - m^2 + i\epsilon} dp,$$

in the sense that for any $f, g \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$,

$$\Delta_F(f, g) = \lim_{\epsilon \downarrow 0} \frac{-1}{(2\pi)^4} \int_{\mathbb{R}^4} \frac{\tilde{f}(p) \overline{\tilde{g}(p)}}{p^2 - m^2 + i\epsilon} dp.$$

Proof. Take any $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$. Note that for any $p = (p_0, q) \in \mathbb{R}^4$, where $q = (p_1, p_2, p_3)$, we have

$$\begin{aligned} \tilde{f}(-p) &= \int_{\mathbb{R}^4} e^{-i(x,p)} f(x) dx \\ &= \int_{-\infty}^{\infty} \int_{\mathbb{R}^3} e^{-itp_0} e^{iy \cdot q} f_t(y) dy dt \\ &= \int_{-\infty}^{\infty} e^{-itp_0} \hat{f}_t(q) dt, \end{aligned} \tag{3.9.1}$$

and a similar identity holds for $\tilde{g}(p)$. Now, it is easy to show using properties of Schwartz functions that $|\hat{f}_t(q)|$ and $|\hat{g}_t(q)|$ decay fast enough as $\|q\|, |t| \rightarrow \infty$ to ensure that for any given $\epsilon > 0$,

$$\iiint \frac{|\hat{f}_t(q)\overline{\hat{g}_t(q)}|}{|p_0^2 - \|q\|^2 - m^2 + i\epsilon|} dt dp_0 dq < \infty.$$

Thus, we can apply Fubini's theorem and equation (3.9.1) to get

$$\int_{\mathbb{R}^4} \frac{\tilde{f}(-p)\overline{\tilde{g}(-p)}}{p^2 - m^2 + i\epsilon} dp = \iiint \frac{e^{-i(t-s)p_0} \hat{f}_t(q)\overline{\hat{g}_s(q)}}{p_0^2 - \|q\|^2 - m^2 + i\epsilon} dp_0 dt ds dq. \quad (3.9.2)$$

First, let us integrate over p_0 . Let $\omega_q := \sqrt{\|q\|^2 + m^2}$. It is easy to see that the equation

$$p_0^2 - \|q\|^2 - m^2 + i\epsilon = 0$$

in p_0 has exactly two roots $\pm\omega_{q,\epsilon}$, with $-\omega_{q,\epsilon}$ slightly above $-\omega_q$ in the upper half-plane, and $\omega_{q,\epsilon}$ slightly below ω_q in the lower half-plane. These points are the poles of the meromorphic function

$$h(p_0) := \frac{e^{-i(t-s)p_0}}{p_0^2 - \|q\|^2 - m^2 + i\epsilon}.$$

A simple computation shows that the residues of h at these two poles are

$$\text{Res}(h, \pm\omega_{q,\epsilon}) = \pm \frac{e^{\mp i(t-s)\omega_{q,\epsilon}}}{2\omega_{q,\epsilon}}.$$

If $t - s \geq 0$, then using a clockwise semicircular contour in the lower half-plane with vertices at $-R$ and R (see Figure 3.1), applying the residue theorem, and taking $R \rightarrow \infty$, we get

$$\frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{-i(t-s)p_0}}{p_0^2 - \|q\|^2 - m^2 + i\epsilon} dp_0 = \frac{-ie^{-i(t-s)\omega_{q,\epsilon}}}{2\omega_{q,\epsilon}}.$$

Similarly, when $t - s < 0$, using a counterclockwise semicircular contour in the upper half-plane with vertices at $-R$ and R , applying the residue theorem, and taking $R \rightarrow \infty$, we get

$$\frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{-i(t-s)p_0}}{p_0^2 - \|q\|^2 - m^2 + i\epsilon} dp_0 = \frac{-ie^{i(t-s)\omega_{q,\epsilon}}}{2\omega_{q,\epsilon}}.$$

Plugging these into equation (3.9.2), we get

$$\frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} \frac{\tilde{f}(-p)\overline{\tilde{g}(-p)}}{p^2 - m^2 + i\epsilon} dp = \frac{-i}{(2\pi)^3} \iiint \frac{e^{-i|t-s|\omega_{q,\epsilon}} \hat{f}_t(q)\overline{\hat{g}_s(q)}}{2\omega_{q,\epsilon}} dt ds dq.$$

Since $\omega_{q,\epsilon} \rightarrow \omega_q$ as $\epsilon \rightarrow 0$, and making the change of variable $p \rightarrow -p$ on the left, it is now easy to complete the proof using the dominated convergence theorem. $\square$

3.10 The Feynman propagator in momentum coordinates

In this section, we will heuristically derive the value of the Feynman propagator in momentum coordinates. This will help us define the Minkowski space Fourier transform of the free quantum field as a well-defined probabilistic object in a later section. Imagining the free quantum field as a function on $\mathbb{R}^4$, we can heuristically define the vacuum expectation value of any function $F(\varphi)$ of φ as

$$\langle F(\varphi) \rangle := \frac{\int F(\phi) e^{iS(\phi)} \mathcal{D}\phi}{\int e^{iS(\phi)} \mathcal{D}\phi}.$$

With this interpretation, the Feynman propagator is given by

$$\Delta_F(x - y) = i \langle \varphi(x) \varphi(y) \rangle.$$

It is not clear at this point how this makes sense from a rigorous point of view. We will see later that it does. Continuing with this line of thinking for the time being, we can define Minkowski space Fourier transform of φ as

$$\tilde{\varphi}(p) := \int_{\mathbb{R}^4} e^{i(x,p)} \varphi(x) dx$$

for $p \in \mathbb{R}^4$. The Feynman propagator in momentum coordinates is heuristically defined as

$$\begin{aligned} \tilde{\Delta}_F(p, q) &:= \frac{i}{(2\pi)^4} \langle \tilde{\varphi}(p) \tilde{\varphi}(q) \rangle \\ &= \frac{1}{(2\pi)^4} \iint e^{i(x,p) + i(y,q)} i \langle \varphi(x) \varphi(y) \rangle dx dy \\ &= \frac{1}{(2\pi)^4} \iint e^{i(x,p) + i(y,q)} \Delta_F(x - y) dx dy \\ &= \frac{1}{(2\pi)^4} \iint e^{i(x,p) + i(x-z,q)} \Delta_F(z) dx dz. \end{aligned}$$

Using the formal identity

$$\int_{\mathbb{R}^4} e^{i(x,p+q)} dx = \delta^{(4)}(p + q),$$

where $\delta^{(4)}$ is the Dirac delta at the origin in $\mathbb{R}^4$, we get

$$\tilde{\Delta}_F(p, q) = \delta^{(4)}(p + q) \int_{\mathbb{R}^4} e^{-i(z,q)} \Delta_F(z) dz.$$

Plugging in the value of $\Delta_F(z)$ from Theorem 3.9.1, we arrive at

$$\begin{aligned} \tilde{\Delta}_F(p, q) &= \delta^{(4)}(p + q) \lim_{\epsilon \downarrow 0} \frac{-1}{(2\pi)^4} \iint \frac{e^{i(z,r-q)}}{r^2 - m^2 + i\epsilon} dz dr \\ &= \delta^{(4)}(p + q) \lim_{\epsilon \downarrow 0} \frac{-1}{q^2 - m^2 + i\epsilon}. \end{aligned} \tag{3.10.1}$$

Although the above derivation was based in heuristics, we can actually make sense of $\tilde{\Delta}_F$ as a linear functional on $(\mathcal{S}(\mathbb{R}^4, \mathbb{R}))^2$ as follows.

Lemma 3.10.1. *The map*

$$\tilde{\Delta}_F(f, g) := -\lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^4} \frac{f(p)g(-p)}{p^2 - m^2 + i\epsilon} dp$$

is a well-defined linear functional on $(\mathcal{S}(\mathbb{R}^4, \mathbb{R}))^2$.

Proof. Let $\tilde{f}$ and $\tilde{g}$ be the Minkowski space Fourier transforms of f and g . By Theorem A.3.6, f and g are the Minkowski space Fourier transforms of the functions $(2\pi)^4 \tilde{f}(-p)$ and $(2\pi)^4 \tilde{g}(-p)$. It is now easy to see that the claim follows from Theorem 3.9.1. $\square$

3.11 Wightman functions of the free quantum field

The n -point Wightman function of the free quantum field is defined as

$$W(x_1, \dots, x_n) := \langle \Omega | \mathcal{T} \{ \varphi(x_1) \cdots \varphi(x_n) \} | \Omega \rangle, \quad (3.11.1)$$

for $x_1, \dots, x_n$. Rigorously, this is a linear functional on $(\mathcal{S}(\mathbb{R}^4, \mathbb{C}))^n$ defined as

$$W(f_1, \dots, f_n) := \int_{\mathbb{R}} \cdots \int_{\mathbb{R}} \langle \Omega | \mathcal{T} \{ \varphi_{t_1}(f_1(t_1, \cdot)) \cdots \varphi_{t_n}(f_n(t_n, \cdot)) \} | \Omega \rangle dt_1 \cdots dt_n. \quad (3.11.2)$$

To see that this correspondence makes sense, first note that

$$\begin{aligned} & \int \cdots \int W(x_1, \dots, x_n) f_1(x_1) \cdots f_n(x_n) dx_1 \cdots dx_n \\ &= \int \cdots \int W((t_1, y_1), \dots, (t_n, y_n)) f_1(t_1, y_1) \cdots f_n(t_n, y_n) dy_1 \cdots dy_n dt_1 \cdots dt_n, \end{aligned}$$

where $t_1, \dots, t_n \in \mathbb{R}$ and $y_1, \dots, y_n \in \mathbb{R}^3$. Fixing $t_1 \leq \cdots \leq t_n$, note that by the heuristic expression (3.11.1),

$$\begin{aligned} & \int \cdots \int W((t_1, y_1), \dots, (t_n, y_n)) f_1(t_1, y_1) \cdots f_n(t_n, y_n) dy_1 \cdots dy_n \\ &= \left\langle \Omega \left| \int \cdots \int \varphi(t_1, y_1) f_1(y_1) \cdots \varphi(t_n, y_n) f_n(y_n) \right| \Omega \right\rangle \\ &= \left\langle \Omega \left| \left(\int \varphi(t_1, y_1) f_1(y_1) dy_1 \right) \cdots \left(\int \varphi(t_n, y_n) f_n(y_n) dy_n \right) \right| \Omega \right\rangle \\ &= \langle \Omega | \varphi_{t_1}(f_1(t_1, \cdot)) \cdots \varphi_{t_n}(f_n(t_n, \cdot)) | \Omega \rangle. \end{aligned}$$

More generally, for any $t_1, \dots, t_n \in \mathbb{R}$, the same argument shows that

$$\begin{aligned} & \int \cdots \int W((t_1, y_1), \dots, (t_n, y_n)) f_1(t_1, y_1) \cdots f_n(t_n, y_n) dy_1 \cdots dy_n \\ &= \langle \Omega | \mathcal{T} \{ \varphi_{t_1}(f_1(t_1, \cdot)) \cdots \varphi_{t_n}(f_n(t_n, \cdot)) \} | \Omega \rangle. \end{aligned}$$

This shows why it makes sense to write the rigorous definition (3.11.2) heuristically as in equation (3.11.1).

Recall that a perfect matching of a vertex set V is a set of undirected edges E on V such that every vertex has degree exactly equal to 1. Let $\mathcal{P}_n$ be the set of perfect matchings of $1, \dots, n$. The following theorem gives the formula for the Wightman functions.

Theorem 3.11.1. *The Wightman function W is zero if n is odd, and given by*

$$W(x_1, \dots, x_n) = \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} (-i\Delta_F(x_i - x_j))$$

when n is even, in the sense that for any $f_1, \dots, f_n \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$,

$$W(f_1, \dots, f_n) = \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} (-i\Delta_F(f_i, f_j)).$$

Proof. Take any $-\infty < t_1 \leq \dots \leq t_n < \infty$ and $g_1, \dots, g_n \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$. For each j , let $F_j(\psi) := \psi(g_j)$ for a function $\psi : \mathcal{S}(\mathbb{R}^3, \mathbb{R}) \rightarrow \mathbb{R}$. Let ϕ be the Euclidean free field with mass m on $\mathbb{R}^4$, and let ϕ_t be its time slice at time t . Let $s_j := t_{j+1} - t_j$ for $j = 1, \dots, n-1$, and let

$$\begin{aligned} f(s_1, \dots, s_{n-1}) &:= \mathbb{E}(F_1(\phi_0)F_2(\phi_{s_1})F_3(\phi_{s_1+s_2}) \cdots F_n(\phi_{s_1+\dots+s_{n-1}})) \\ &= \mathbb{E}(F_1(\phi_0)F_2(\phi_{t_2-t_1})F_3(\phi_{t_3-t_1}) \cdots F_n(\phi_{t_n-t_1})) \\ &= \mathbb{E}(F_1(\phi_{t_1})F_2(\phi_{t_2})F_3(\phi_{t_3}) \cdots F_n(\phi_{t_n})), \end{aligned}$$

where the last step follows by the translation invariance of the Euclidean free field. Since ϕ is a Gaussian field and $F_1, \dots, F_n$ are linear functions, the above expected value is zero if n is odd. From this, it is easy to see by Theorem 2.5.1 that the Wightman functions are zero for odd n . For even n , the Wick product formula (Theorem B.2.7) implies that

$$\begin{aligned} f(s_1, \dots, s_{n-1}) &= \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} \mathbb{E}(F_i(\phi_{t_i})F_j(\phi_{t_j})) \\ &= \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} \mathbb{E}(\phi_{t_i}(g_i)\phi_{t_j}(g_j)). \end{aligned}$$

But by Lemma 3.8.3,

$$\mathbb{E}(\phi_{t_i}(g_i)\phi_{t_j}(g_j)) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-|t_i-t_j|\sqrt{\|p\|^2+m^2}} \widehat{g_i}(p) \overline{\widehat{g_j}(p)}}{2\sqrt{\|p\|^2+m^2}} dp.$$

Thus, we have

$$f(s_1, \dots, s_{n-1}) = \sum_{E \in \mathcal{P}_n} \prod_{\substack{\{i,j\} \in E, \\ i < j}} \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-(s_i+\dots+s_{j-1})\sqrt{\|p\|^2+m^2}} \widehat{g_i}(p) \overline{\widehat{g_j}(p)}}{2\sqrt{\|p\|^2+m^2}} dp.$$

If we now apply the method described in Theorem 2.5.1 to calculate $f(is_1, \dots, is_{n-1})$, it is easy to check that we get

$$\begin{aligned}
 \langle \Omega | F_1(\varphi_{t_1}) \cdots F_n(\varphi_{t_n}) | \Omega \rangle &= f(is_1, \dots, is_{n-1}) \\
 &= \sum_{E \in \mathcal{P}_n} \prod_{\substack{\{i,j\} \in E, \\ i < j}} \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i(s_i + \dots + s_{j-1})\sqrt{\|p\|^2 + m^2}} \widehat{g}_i(p) \overline{\widehat{g}_j(p)}}{2\sqrt{\|p\|^2 + m^2}} dp \\
 &= \sum_{E \in \mathcal{P}_n} \prod_{\substack{\{i,j\} \in E, \\ i < j}} \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i(t_j - t_i)\sqrt{\|p\|^2 + m^2}} \widehat{g}_i(p) \overline{\widehat{g}_j(p)}}{2\sqrt{\|p\|^2 + m^2}} dp \\
 &= \sum_{E \in \mathcal{P}_n} \prod_{\substack{\{i,j\} \in E, \\ i < j}} (-i\Delta_F(t_j - t_i, g_i, g_j)).
 \end{aligned}$$

Since $\Delta_F(t, g, h) = \Delta_F(t, h, g)$ for any t, g, h (which is easy to derive from the definition), this shows that for any $t_1, \dots, t_n \in \mathbb{R}$,

$$\langle \Omega | \mathcal{T}\{F_1(\varphi_{t_1}) \cdots F_n(\varphi_{t_n})\} | \Omega \rangle = \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} (-i\Delta_F(|t_i - t_j|, g_i, g_j)).$$

Thus, by equation (3.11.2), for any $f_1, \dots, f_n \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$,

$$W(f_1, \dots, f_n) = \sum_{E \in \mathcal{P}_n} \int \cdots \int \prod_{\{i,j\} \in E} (-i\Delta_F(|t_i - t_j|, f_i(t_i, \cdot), f_j(t_j, \cdot))) dt_1 \cdots dt_n.$$

It is now easy to see how the claim follows. $\square$

We also have the momentum space Wightman functions, heuristically defined as

$$\widetilde{W}(p_1, \dots, p_n) := i^{n/2} (2\pi)^{-2n} \langle \widetilde{\varphi}(p_1) \cdots \widetilde{\varphi}(p_n) \rangle,$$

where $\widetilde{\varphi}$ is the Minkowski space Fourier transform of φ . The value of this function can be calculated by taking $f_j(x) := e^{i(x, p_j)}$, $j = 1, \dots, n$, in Theorem 3.11.1. Then, by Theorem 3.11.1 and the derivation in Section 3.10, it follows that

$$\widetilde{W}(p_1, \dots, p_n) := \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} \delta^{(4)}(p_i + p_j) \lim_{\epsilon \downarrow 0} \frac{-1}{p_i^2 - m^2 + i\epsilon}, \quad (3.11.3)$$

where both sides have to be interpreted in a distributional sense.

Chapter 4

From Fields to Particles

4.1 Lorentz transforms

The Minkowski inner product on $\mathbb{R}^4$ is defined as

$$(x, y) := x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3.$$

Following usual convention, we will denote the ‘squared norm’ induced by this inner product by x^2 . That is,

$$x^2 := x_0^2 - x_1^2 - x_2^2 - x_3^2.$$

A Lorentz transformation L on $\mathbb{R}^4$ is a linear map such that $(Lx, Ly) = (x, y)$ for all $x, y \in \mathbb{R}^4$. In the language of matrices, this means that:

$$L^T \eta L = \eta,$$

where η is the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This is very similar to the orthogonality condition under the usual inner product. Like orthogonal matrices, Lorentz transformations form a Lie group. This group is denoted by $O(1, 3)$.

In analogy with the regular orthogonal groups, we can consider the map $L \mapsto \det(L)$ where the determinant serves as a homomorphism $\det : O(1, 3) \rightarrow \mathbb{Z}_2 = \{-1, 1\}$. In the case of the orthogonal groups, quotienting out by the action of this homomorphism is sufficient to reduce $O(4)$ to $SO(4)$. However, the group $O(1, 3)$ has another homomorphism:

Lemma 4.1.1. *The map $L \mapsto \text{sign}(L_{00})$, where L_{00} is the top left entry of the matrix, is a homomorphism of $O(1, 3)$ into $\mathbb{Z}_2$.*

Proof. First of all, note that the condition $L^T \eta L = \eta$ implies that for any $L \in \text{O}(1, 3)$,

$$L_{00}^2 - L_{01}^2 - L_{02}^2 - L_{03}^2 = 1. \quad (4.1.1)$$

This means that L_{00} cannot be zero. Thus, $\text{sign}(L_{00}) \in \mathbb{Z}_2$. Next, let L and M be two elements of $\text{O}(1, 3)$, and let $N := LM$. Then

$$N_{00} = \sum_{j=0}^3 L_{0j} M_{j0}$$

Suppose that L_{00} and M_{00} are both positive. The identity (4.1.1) implies that they are both ≥ 1 . Thus, by the above identity and the Cauchy–Schwarz inequality,

$$\begin{aligned} N_{00} &\geq L_{00} M_{00} - \sqrt{(L_{01}^2 + L_{02}^2 + L_{03}^2)(M_{01}^2 + M_{02}^2 + M_{03}^2)} \\ &= L_{00} M_{00} - \sqrt{(L_{00}^2 - 1)(M_{00}^2 - 1)} \geq 0. \end{aligned}$$

The same series of inequalities hold if L_{00} and M_{00} are both negative. Finally, suppose that one of L_{00}, M_{00} is negative and the other is positive. Then again by the Cauchy–Schwarz inequality,

$$\begin{aligned} N_{00} &\leq L_{00} M_{00} + \sqrt{(L_{01}^2 + L_{02}^2 + L_{03}^2)(M_{01}^2 + M_{02}^2 + M_{03}^2)} \\ &= L_{00} M_{00} + \sqrt{(L_{00}^2 - 1)(M_{00}^2 - 1)} \\ &\leq L_{00} M_{00} + |L_{00} M_{00}| = 0. \end{aligned}$$

This completes the proof. $\square$

The *restricted Lorentz group* $\text{SO}^+(1, 3)$ (also known as the *orthochronous Lorentz group*) is the set of all $L \in \text{O}(1, 3)$ such that $\det(L) = 1$ and $\text{sign}(L_{00}) = 1$. This is a subgroup of $\text{O}(1, 3)$. For a detailed mathematical treatment of the Lorentz group and associated topics, we refer to the textbook by Naber [20].

4.2 Special relativity

Classical physics is invariant under isometries of $\mathbb{R}^3$, that is, translations and rotations. What does that mean? Suppose you have a computer program simulating physics. You input the state of the system at time 0, then the program gives you the state at time t . Suppose a trickster enters the state of the system at time 0 but in a different coordinate system. If your program is properly built, the result it returns for the trickster will (after changing back coordinates) be the exact same as the result you saw earlier. This a property of classical physics.

Special relativity claims that the laws of physics remain invariant under restricted Lorentz transformations (under the assumption that we are working in units where the

speed of light is 1) and spacetime translations of $\mathbb{R}^4$, in the same sense as above. Now space and time become mixed into a single entity called spacetime, coordinates of which are changed by Lorentz transformations and spacetime translations. The same analogy as before holds, except now the trickster can change the time coordinates under a Lorentz transformation with no effect on the final results.

The parametrization of a curve in Euclidean space by arc length has the property that it is invariant under Euclidean symmetries. That is, if we apply a rotation or translation to the coordinate system, the parametrization remains unchanged. In a similar way, a curve in $\mathbb{R}^4$ can sometimes be parametrized in a way that is invariant under any change of the coordinate system by an action of $\text{SO}^+(1, 3)$ or a spacetime translation. The parametrization $x(\tau)$ of a curve by arc length requires that $\|\dot{x}(\tau)\| \equiv 1$, where $\|\cdot\|$ is Euclidean norm and $\dot{x}$ denote the derivative of x with respect to τ . In a similar vein, a Lorentz-invariant parametrization requires that the squared Minkowski norm $\dot{x}(\tau)^2$ is always equal to 1. This is known as parametrization by ‘proper time’. The following result shows that a curve can be parametrized by proper time if in some coordinate system, the curve represents the trajectory of a particle whose speed is always strictly less than 1 (which is the speed of light in our units).

Proposition 4.2.1. *A sufficient condition for a curve in $\mathbb{R}^4$ to be parametrizable by proper time is that in some coordinate system, the curve has a time parametrization $x(t) = (t, y(t))$ that satisfies $\|\dot{y}(t)\| < 1$ for all t . Moreover, the parametrization by proper time is unique up to time translations.*

Proof. Define

$$\tau(t) := \int_0^t \sqrt{1 - \|\dot{y}(s)\|^2} ds,$$

where the right side is interpreted as the negative of the integral from $-|t|$ to 0 if $t < 0$. By the given condition, τ is a strictly increasing, and hence invertible, function of t . The range of τ is a subinterval of $\mathbb{R}$ (which may be bounded or unbounded). Reparametrize the curve x as

$$\gamma(s) := x(\tau^{-1}(s))$$

for s in the range of τ . Denoting the coordinates of γ by γ_j , $0 \leq j \leq 3$, we see that

$$\dot{\gamma}_0(s) = \frac{d}{ds} \tau^{-1}(s) = \frac{1}{\tau'(\tau^{-1}(s))} = \frac{1}{\sqrt{1 - \|\dot{y}(\tau^{-1}(s))\|^2}},$$

and for $1 \leq j \leq 3$,

$$\dot{\gamma}_j(s) = \frac{d}{ds} y_j(\tau^{-1}(s)) = \frac{\dot{y}_j(\tau^{-1}(s))}{\sqrt{1 - \|\dot{y}(\tau^{-1}(s))\|^2}}.$$

This shows that $\dot{\gamma}(s)^2 = 1$. This proves the existence of a parametrization by proper time. To prove uniqueness, suppose that $\beta(s) = x(\theta(s))$ is another parametrization by proper

time, with $\theta(0) = \tau^{-1}(0)$. Then note that

$$1 = \dot{\beta}(s)^2 = \theta'(s)^2 \dot{x}(\theta(s))^2,$$

which implies that

$$\theta'(s) = \frac{1}{\sqrt{\dot{x}(\theta(s))^2}} = \frac{1}{\sqrt{1 - \|\dot{y}(\theta(s))\|^2}}.$$

Thus,

$$\frac{d}{ds} \tau(\theta(s)) = \tau'(\theta(s)) \theta'(s) = 1.$$

Since τ and θ are both strictly increasing functions which coincide at $s = 0$, this shows that $\theta(s) = \tau^{-1}(s)$ for all s . $\square$

Suppose that we are given the trajectory of a particle of mass m that always moves at a speed less than 1. Then by the above result, its trajectory can be uniquely parametrized by proper time. In analogy with Newtonian momentum, the ‘four-momentum’ of the particle at proper time τ is defined to be the 4-vector $m\dot{\gamma}(\tau)$. The following result shows why the four-momentum is a natural notion of momentum in a relativistic setting.

Proposition 4.2.2. *If the coordinate system is changed by the action of $L \in \text{SO}^+(1, 3)$, then the four-momentum of a particle also changes by a multiplication by L .*

Proof. Let $x(\tau)$ be the parametrization of the trajectory by proper time in the original coordinate system. Define $y(\tau) := Lx(\tau)$, so that y is the trajectory in the new coordinate system. Then note that

$$y(\tau)^2 = y(\tau)^T \eta y(\tau) = x(\tau)^T L^T \eta L x(\tau) = x(\tau)^T \eta x(\tau) = x(\tau)^2 = 1.$$

Thus, y is a parametrization by proper time of the trajectory in the new coordinate system. By the uniqueness of parametrization by proper time, this shows that the four-momentum in the new coordinate system is $m\dot{y}(\tau) = L(m\dot{x}(\tau))$. $\square$

Let $x(t) = (t, y(t))$ be the trajectory of a particle of mass m in some coordinate system, so that the velocity at time t is $v(t) := \dot{y}(t)$. From the calculations in the proof of Proposition 4.2.1, we see that its four-momentum at time t is given by the formula

$$p = \left(\frac{m}{\sqrt{1 - \|v(t)\|^2}}, \frac{mv(t)}{\sqrt{1 - \|v(t)\|^2}} \right).$$

Let us denote this four-momentum as

$$p = (p_0, p_1, p_2, p_3) = (p_0, q).$$

The four-momentum is also called the ‘energy-momentum’, as we can identify the first component of p_0 as the relativistic energy E . The remaining part, q , is called the relativistic

three-momentum. When the particle is at rest, $q = 0$, and we get $E = m$, the famous Einstein relation (with $c = 1$). Further, note that in the non-relativistic limit $\|v(t)\| \ll 1$, we can expand the energy p_0 as:

$$p_0 = \frac{m}{\sqrt{1 - \|v(t)\|^2}} \approx m + \frac{1}{2}m\|v(t)\|^2,$$

which is the sum of the rest energy and the classical kinetic energy. One thing we can observe from the definition of p is that

$$p^2 := p_0^2 - \|q\|^2 = m^2,$$

which in turn implies that

$$p_0 = E = \sqrt{m^2 + \|q\|^2} = \omega_q,$$

where the latter will become our conventional notation, with the subscript denoting the dependence on the relativistic three-momentum.

The topics discussed in this section are standard fare in any introductory textbook on special relativity. Again, Naber's book [20] gives a careful mathematical introduction. For the physical perspective, a classic reference is the popular science book by Einstein himself [5].

4.3 The mass shell

Notice that the four-momentum p of a particle of mass $m \geq 0$ is always an element of the manifold

$$\mathcal{X}_m := \{p \in \mathbb{R}^4 : p^2 = m^2, p_0 \geq 0\},$$

where p^2 is the squared Minkowski norm. This manifold is called the ‘mass shell’ of mass m . The mass shells are Minkowski analogues of spheres in Euclidean space. For example, just as a rotation gives a bijection of a sphere onto itself, a restricted Lorentz transform bijectively maps a mass shell onto itself, as shown by the following lemma.

Lemma 4.3.1. *Any restricted Lorentz transform bijectively maps any mass shell onto itself.*

Proof. Take $p \in \mathcal{X}_m$ and $L \in \text{SO}^+(1, 3)$. Let $q := Lp$. Then viewing p and q as column vectors, we have

$$q^2 = q^T \eta q = p^T L^T \eta L p = p^T \eta p = p^2 = m^2.$$

Now, note that

$$\eta L^T \eta L = \eta^2 = I.$$

Thus, ηL is an invertible matrix and $(\eta L)^{-1} = \eta L^T$. Consequently,

$$\eta L \eta L^T = I,$$

which gives

$$\eta = \eta^2 L \eta L^T = L \eta L^T.$$

Comparing the top-leftmost elements on both sides, we get

$$L_{00}^2 - L_{01}^2 - L_{02}^2 - L_{03}^2 = 1.$$

In particular,

$$L_{01}^2 + L_{02}^2 + L_{03}^2 = L_{00}^2 - 1 < L_{00}^2.$$

Also, the equation $p^2 = m^2$ gives

$$p_1^2 + p_2^2 + p_3^2 = p_0^2 - m^2 \leq p_0^2.$$

From these two inequalities and the facts that L_{00} and p_0 are nonnegative, we deduce that

$$\begin{aligned} q_0 &= L_{00}p_0 + L_{01}p_1 + L_{02}p_2 + L_{03}p_3 \\ &\geq L_{00}p_0 - \sqrt{(L_{01}^2 + L_{02}^2 + L_{03}^2)(p_1^2 + p_2^2 + p_3^2)} \\ &\geq L_{00}p_0 - \sqrt{L_{00}^2 p_0^2} = 0. \end{aligned}$$

Thus, $q \in \mathcal{X}_m$. This proves that L maps $\mathcal{X}_m$ into itself. This map is injective since, as we have argued above, ηL is invertible and hence L is invertible. Finally, to show that L is surjective, simply recall that since $\text{SO}^+(1, 3)$ is a group, L^{-1} is also in it. Therefore, L^{-1} maps $\mathcal{X}_m$ into $\mathcal{X}_m$, by what we have already shown. This proves that L maps $\mathcal{X}_m$ onto $\mathcal{X}_m$. $\square$

4.4 Integration on mass shells

A mass shell $\mathcal{X}_m$ inherits a topology from $\mathbb{R}^4$, and hence, a Borel σ -algebra. Just like the volume measure on a sphere in Euclidean space is the unique (up to constant multiples) measure that is invariant under rotations, there is a unique (up to constant multiples) measure λ_m on $\mathcal{X}_m$ that is invariant under restricted Lorentz transformations. It is defined as follows. Let $\pi : \mathcal{X}_m \rightarrow \mathbb{R}^3$ be the projection map

$$\pi(p_0, p_1, p_2, p_3) = (p_1, p_2, p_3). \tag{4.4.1}$$

Since p_0 is the nonnegative square-root of $p_1^2 + p_2^2 + p_3^2 + m^2$ on $\mathcal{X}_m$, the map π is a measurable bijection between $\mathcal{X}_m$ and $\mathbb{R}^3$ whose inverse is also measurable. For a Borel set $A \subseteq \mathcal{X}_m$,

we define

$$\lambda_m(A) := \frac{1}{(2\pi)^3} \int_{\pi(A)} \frac{dq}{2\omega_q}, \quad (4.4.2)$$

where we use the notation that for a point $q \in \mathbb{R}^3$,

$$\omega_q := \sqrt{\|q\|^2 + m^2}.$$

It is easy to see that λ_m is a measure on $\mathcal{X}_m$. Moreover, for any measurable function $f : \mathcal{X}_m \rightarrow \mathbb{R}$,

$$\int_{\mathcal{X}_m} f(p) d\lambda_m(p) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{f(\omega_q, q)}{2\omega_q} dq,$$

provided that the integrals are well-defined. This can be seen by starting with indicator functions (for which, it is simply the identity (4.4.2)), and then approximating general measurable functions by linear combinations of indicator functions via Theorem A.1.23.

It turns out that λ_m is the unique measure on $\mathcal{X}_m$ (up to constant multiples) that is invariant under restricted Lorentz transforms.

Theorem 4.4.1. *For any Borel set $A \subseteq \mathcal{X}_m$ and any $L \in \text{SO}^+(1, 3)$, $\lambda_m(L^{-1}(A)) = \lambda_m(A)$. Moreover, if a measure μ on $\mathcal{X}_m$ has this property, and $0 < \mu(A) < \infty$ for some Borel set A , then μ must be a nonnegative constant multiple of λ_m .*

Proof. For each $\epsilon > 0$, define

$$\mathcal{X}_{m,\epsilon} := \bigcup_{m' \in [m, \sqrt{m^2 + \epsilon}]} \mathcal{X}_{m'} = \{p \in \mathbb{R}^4 : p^2 \in [m^2, m^2 + \epsilon]\}.$$

Let $\lambda_{m,\epsilon}$ be the restriction of Lebesgue measure to $\mathcal{X}_{m,\epsilon}$, divided by $(2\pi)^3\epsilon$. Since restricted Lorentz transformations are linear maps with determinant 1, and they map $\mathcal{X}_{m,\epsilon}$ bijectively onto itself (by Lemma 4.3.1, it follows that the measure $\lambda_{m,\epsilon}$ is invariant under restricted Lorentz transformations.

Take any Schwartz function $f : \mathbb{R}^4 \rightarrow \mathbb{R}$. We claim that

$$\lim_{\epsilon \downarrow 0} \int_{\mathcal{X}_{m,\epsilon}} f(p) d\lambda_{m,\epsilon}(p) = \int_{\mathcal{X}_m} f(p) d\lambda_m(p). \quad (4.4.3)$$

To see this, let

$$\omega_{q,\epsilon} := \sqrt{\omega_q^2 + \epsilon},$$

and note that by Fubini's theorem,

$$\begin{aligned} \int_{\mathcal{X}_{m,\epsilon}} f(p) d\lambda_{m,\epsilon}(p) &= \frac{1}{(2\pi)^3\epsilon} \int_{\mathbb{R}^3} \int_{p_0 : (p_0, q) \in \mathcal{X}_{m,\epsilon}} f(p_0, q) dp_0 dq \\ &= \frac{1}{(2\pi)^3\epsilon} \int_{\mathbb{R}^3} \int_{\omega_q}^{\omega_{q,\epsilon}} f(p_0, q) dp_0 dq \end{aligned}$$

Now, for any $q \in \mathbb{R}^3$,

$$\begin{aligned} \frac{1}{\epsilon} \int_{\omega_q}^{\omega_{q,\epsilon}} f(p_0, q) dp_0 &= \frac{1}{\epsilon} \int_{\omega_q}^{\omega_{q,\epsilon}} (f(\omega_q, q) + O(\epsilon)) dp_0 \\ &= \frac{1}{\epsilon} (\omega_{q,\epsilon} - \omega_q) (f(\omega_q, q) + O(\epsilon)) dp_0. \end{aligned}$$

Since

$$\omega_{q,\epsilon} = \sqrt{\omega_q^2 + \epsilon} = \begin{cases} \omega_q + \frac{\epsilon}{2\omega_q} + O(\epsilon^2) & \text{if } \omega_q \neq 0, \\ \sqrt{\epsilon} & \text{if } \omega_q = 0, \end{cases}$$

this shows that for almost all $q \in \mathbb{R}^3$,

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \int_{\omega_q}^{\omega_{q,\epsilon}} f(p_0, q) dp_0 = \frac{f(\omega_q, q)}{2\omega_q}.$$

By an application of the dominated convergence theorem, this proves the claim (4.4.3). Next, take any restricted Lorentz transformation L and any $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$. The identity (4.4.3) and the Lorentz invariance of $\lambda_{m,\epsilon}$ implies that

$$\begin{aligned} \int_{\mathcal{X}_m} f(L(p)) d\lambda_m(p) &= \lim_{\epsilon \downarrow 0} \int_{\mathcal{X}_{m,\epsilon}} f(L(p)) d\lambda_{m,\epsilon}(p) \\ &= \lim_{\epsilon \downarrow 0} \int_{\mathcal{X}_{m,\epsilon}} f(p) d\lambda_{m,\epsilon}(p) = \int_{\mathcal{X}_m} f(p) d\lambda_m(p). \end{aligned} \quad (4.4.4)$$

Now, take any $g \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$. Define $h : \mathcal{X}_m \rightarrow \mathbb{R}$ as $h := g\pi$ and $f : \mathbb{R}^4 \rightarrow \mathbb{R}$ as

$$f(p_0, q) := g(q) e^{-(p_0^2 - \omega_q^2)^2}.$$

Then $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$ and $f = h$ on $\mathcal{X}_m$. Thus, by the identity (4.4.4),

$$\begin{aligned} \int_{\mathcal{X}_m} h(L(p)) d\lambda_m(p) &= \int_{\mathcal{X}_m} f(L(p)) d\lambda_m(p) \\ &= \int_{\mathcal{X}_m} f(p) d\lambda_m(p) = \int_{\mathcal{X}_m} h(p) d\lambda_m(p). \end{aligned}$$

Finally, following similar steps as in the proof of Theorem A.2.11, we can replace the function h in the above identity by the indicator of any Borel set $A \subseteq \mathcal{X}_m$. This proves that λ_m is invariant under any restricted Lorentz transformation.

It remains to show that λ_m is the unique measure, up to nonnegative constant multiples, that remains invariant under restricted Lorentz transformations. Take any $L \in \text{SO}^+(1, 3)$ and $f \in L^1(\lambda_m)$. Then, on the one hand, we know that

$$\int_{\mathcal{X}_m} f(L(p)) d\lambda_m(p) = \int_{\mathcal{X}_m} f(p) d\lambda_m(p).$$

On the other hand, we also know that

$$\begin{aligned}\int_{\mathcal{X}_m} f(p) d\lambda_m(p) &= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} f(\pi^{-1}q) \frac{dq}{2\omega_q}, \\ \int_{\mathcal{X}_m} f(L(p)) d\lambda_m(p) &= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} f(L\pi^{-1}q) \frac{dq}{2\omega_q}.\end{aligned}$$

Let $\psi := \pi L \pi^{-1}$. Then by the change of variable formula for integration on $\mathbb{R}^3$, we have

$$\begin{aligned}\int_{\mathbb{R}^3} f(L\pi^{-1}q) \frac{dq}{2\omega_q} &= \int_{\mathbb{R}^3} f(\pi^{-1}\psi(q)) \frac{dq}{2\omega_q} \\ &= \int_{\mathbb{R}^3} f(\pi^{-1}r) |\det J_{\psi^{-1}}(r)| \frac{dr}{2\omega_{\psi^{-1}(r)}},\end{aligned}$$

where $J_{\psi^{-1}}(r)$ is the Jacobian matrix of ψ^{-1} at r . Combining this with the previous identities, we get

$$\int_{\mathbb{R}^3} f(\pi^{-1}q) \frac{dq}{2\omega_q} = \int_{\mathbb{R}^3} f(\pi^{-1}q) |\det J_{\psi^{-1}}(q)| \frac{dq}{2\omega_{\psi^{-1}(q)}}.$$

Since this holds for any f , it is now easy to show that

$$|\det J_{\psi^{-1}}(q)| = \frac{\omega_{\psi^{-1}(q)}}{\omega_q}. \quad (4.4.5)$$

Now suppose that μ is a measure on $\mathcal{X}_m$ that is invariant under restricted Lorentz transforms. Let ν be its pushforward to $\mathbb{R}^3$ via π . That is, $\nu(A) = \mu(\pi^{-1}(A))$. Suppose that ν has a density ρ with respect to Lebesgue measure on $\mathbb{R}^3$. Then exactly the same argument as above shows that for any $f \in L^1(\mu)$,

$$\int_{\mathbb{R}^3} f(\pi^{-1}q) \rho(q) dq = \int_{\mathbb{R}^3} f(\pi^{-1}q) |\det J_{\psi^{-1}}(q)| \rho(\psi^{-1}(q)) dq.$$

This lets us conclude that

$$\rho(q) = |\det J_{\psi^{-1}}(q)| \rho(\psi^{-1}(q)).$$

Matching this with the value of $|\det J_{\psi^{-1}}(q)| \rho(\psi^{-1}(q))$ obtained in equation (4.4.5) above, we see that $\rho(q)$ must be a constant multiple of $(2\omega_q)^{-1}$.

The absolute continuity of ν with respect to Lebesgue measure is a consequence of the fact that the restricted Lorentz group acts transitively on $\mathcal{X}_m$ and the regularity of Borel measures. A small ball around a point in $\mathbb{R}^3$ can be sent a small region around any other given point by a map of the form $\pi L \pi^{-1}$ for some $L \in \text{SO}^+(1, 3)$. This allows us to show that the measure of a small ball must be small, because many disjoint sets with the same measure can fit into a bounded region. We leave the details to the reader. $\square$

4.5 Fourier transform of the Euclidean free field

Take any $d \geq 3$. Let ϕ be the Euclidean free field on $\mathbb{R}^d$ with mass m , and let ϕ_0 be its restriction to time 0, as defined in Chapter 3. Then, formally, the Fourier transform of ϕ_0 is given by

$$\widehat{\phi}_0(q) = \int_{\mathbb{R}^{d-1}} \phi_0(x) e^{iq \cdot x} dx. \quad (4.5.1)$$

Unlike ϕ_0 , which is ‘real-valued’, $\widehat{\phi}_0$ is ‘complex-valued’, in the sense that when integrated against Schwartz functions, we expect to get complex numbers. Accordingly, we can view $\widehat{\phi}_0$ as a pair of random distributions $\operatorname{Re} \widehat{\phi}_0$ and $\operatorname{Im} \widehat{\phi}_0$. These fields are defined as follows. For $f \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$, we define

$$\operatorname{Re} \widehat{\phi}_0(f) := \phi_0(\operatorname{Re} \widehat{f}), \quad \operatorname{Im} \widehat{\phi}_0(f) := -\phi_0(\operatorname{Im} \widehat{f}). \quad (4.5.2)$$

To see that this definition makes sense, let $g(x) := \widehat{f}(-x)$. Then note that by the Fourier inversion formula,

$$\begin{aligned} f(q) &= \frac{1}{(2\pi)^{d-1}} \int_{\mathbb{R}^{d-1}} e^{-ix \cdot q} \widehat{f}(x) dx \\ &= \frac{1}{(2\pi)^{d-1}} \int_{\mathbb{R}^{d-1}} e^{ix \cdot q} \widehat{f}(-x) dx \\ &= \frac{1}{(2\pi)^{d-1}} \int_{\mathbb{R}^{d-1}} e^{ix \cdot q} g(x) dx. \end{aligned}$$

Thus, $f = (2\pi)^{-(d-1)} \widehat{g}$. Since f is real-valued, we can now heuristically apply the Plancherel identity to conclude that

$$\begin{aligned} \widehat{\phi}_0(f) &= \int_{\mathbb{R}^{d-1}} \widehat{\phi}_0(q) f(q) dq \\ &= \int_{\mathbb{R}^{d-1}} \widehat{\phi}_0(q) \overline{\widehat{f}(q)} dq \\ &= \frac{1}{(2\pi)^{d-1}} \int_{\mathbb{R}^{d-1}} \widehat{\phi}_0(q) \overline{\widehat{g}(q)} dq \\ &= \int_{\mathbb{R}^{d-1}} \phi_0(x) g(x) dx = \phi_0(\operatorname{Re} g) + i \phi_0(\operatorname{Im} g). \end{aligned} \quad (4.5.3)$$

But again since f is real-valued,

$$g(x) = \widehat{f}(-x) = \int_{\mathbb{R}^{d-1}} e^{-ix \cdot q} f(q) dq = \overline{\widehat{f}(x)} = \operatorname{Re} \widehat{f}(x) - i \operatorname{Im} \widehat{f}(x), \quad (4.5.4)$$

which means that $\operatorname{Re} g = \operatorname{Re} \widehat{f}$ and $\operatorname{Im} g = -\operatorname{Im} \widehat{f}$. Equations (4.5.3) and (4.5.4) justify the definition of $\widehat{\phi}_0$ in equation (4.5.2).

Having defined the centered Gaussian field $\widehat{\phi}_0$ (or rather, the pair of fields $(\operatorname{Re} \widehat{\phi}_0, \operatorname{Im} \widehat{\phi}_0)$), let us now calculate its covariance structure.

Lemma 4.5.1. *Let $\omega_q := \sqrt{\|q\|^2 + m^2}$. For any $f, g \in \mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$,*

$$\begin{aligned} \text{Cov}(\text{Re } \widehat{\phi}_0(f), \text{Im } \widehat{\phi}_0(g)) &= 0, \\ \text{Cov}(\text{Re } \widehat{\phi}_0(f), \text{Re } \widehat{\phi}_0(g)) &= \frac{(2\pi)^{d-1}}{4} \int_{\mathbb{R}^{d-1}} \frac{1}{2\omega_q} (f(q) + f(-q))(g(q) + g(-q)) dq, \\ \text{Cov}(\text{Im } \widehat{\phi}_0(f), \text{Im } \widehat{\phi}_0(g)) &= \frac{(2\pi)^{d-1}}{4} \int_{\mathbb{R}^{d-1}} \frac{1}{2\omega_q} (f(q) - f(-q))(g(q) - g(-q)) dq. \end{aligned}$$

Proof. Note that by equation (4.5.2) and Lemma 3.8.3,

$$\begin{aligned} \text{Cov}(\text{Re } \widehat{\phi}_0(f), \text{Im } \widehat{\phi}_0(g)) &= -\text{Cov}(\phi_0(\text{Re } \widehat{f}), \phi_0(\text{Im } \widehat{g})) \\ &= -\frac{1}{(2\pi)^{d-1}} \int_{\mathbb{R}^{d-1}} \frac{u(q)\overline{v(q)}}{2\omega_q} dq, \end{aligned}$$

where u is the Fourier transform of $\text{Re } \widehat{f}$, and v is the Fourier transform of $\text{Im } \widehat{g}$. Note that by the inversion formula,

$$\begin{aligned} u(q) &= \frac{1}{2} \int_{\mathbb{R}^{d-1}} e^{iq \cdot x} (\widehat{f}(x) + \overline{\widehat{f}(x)}) dx \\ &= \frac{1}{2} \int_{\mathbb{R}^{d-1}} e^{iq \cdot x} \widehat{f}(x) dx + \frac{1}{2} \int_{\mathbb{R}^{d-1}} \overline{e^{-iq \cdot x} \widehat{f}(x)} dx \\ &= \frac{(2\pi)^{d-1}}{2} (f(-q) + \overline{f(q)}) = \frac{(2\pi)^{d-1}}{2} (f(-q) + f(q)). \end{aligned}$$

Similarly,

$$\begin{aligned} v(q) &= \frac{1}{2i} \int_{\mathbb{R}^{d-1}} e^{iq \cdot x} (\widehat{g}(x) - \overline{\widehat{g}(x)}) dx \\ &= \frac{1}{2i} \int_{\mathbb{R}^{d-1}} e^{iq \cdot x} \widehat{g}(x) dx - \frac{1}{2i} \int_{\mathbb{R}^{d-1}} \overline{e^{-iq \cdot x} \widehat{g}(x)} dx \\ &= \frac{(2\pi)^{d-1}}{2i} (g(-q) - \overline{g(q)}) = \frac{(2\pi)^{d-1}}{2i} (g(-q) - g(q)). \end{aligned}$$

Combining the last three displays, we get

$$\text{Cov}(\text{Re } \widehat{\phi}_0(f), \text{Im } \widehat{\phi}_0(g)) = \frac{(2\pi)^{d-1}}{4i} \int_{\mathbb{R}^{d-1}} \frac{1}{2\omega_q} (f(-q) + f(q))(g(-q) - g(q)) dq.$$

Since $\omega_q = \omega_{-q}$, it is easy to see that the right side is equal to zero. The other two claims follow by similar calculations. $\square$

Let us now see what the formulas from the above lemma mean intuitively. Taking

$f = \delta_q$ and $g = \delta_{q'}$ for some nonzero $q, q' \in \mathbb{R}^{d-1}$, we get the formal expressions

$$\begin{aligned} \text{Cov}(\text{Re } \widehat{\phi}_0(q), \text{Im } \widehat{\phi}_0(q')) &= 0, \\ \text{Cov}(\text{Re } \widehat{\phi}_0(q), \text{Re } \widehat{\phi}_0(q')) &= \begin{cases} \frac{(2\pi)^{d-1}}{2\omega_q} \delta_q(q) & \text{if } q = \pm q', \\ 0 & \text{otherwise,} \end{cases} \\ \text{Cov}(\text{Im } \widehat{\phi}_0(q), \text{Im } \widehat{\phi}_0(q')) &= \begin{cases} \frac{(2\pi)^{d-1}}{2\omega_q} \delta_q(q) & \text{if } q = q', \\ -\frac{(2\pi)^{d-1}}{2\omega_q} \delta_q(q) & \text{if } q = -q', \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

The first identity shows that the real and imaginary parts of $\widehat{\phi}_0$ are independent Gaussian fields. The other identities indicate that they are sort of like white noise fields, except for the weighting factor $(2\omega_q)^{-1}$. Indeed, specializing to $d = 4$, one gets a homogeneous white noise field if one ‘lifts’ $\widehat{\phi}_0$ to a field $\psi = \psi_1 + i\psi_2$ on the mass shell $\mathcal{X}_m$ as follows.

Let $\pi : \mathcal{X}_m \rightarrow \mathbb{R}^3$ be the projection map defined in equation (4.4.1). We will say that a function $f : \mathcal{X}_m \rightarrow \mathbb{R}$ is a real-valued Schwartz function if f is of the form $(2\pi)^{-3}g\pi$ for some $g \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$. For such f , define

$$\psi(f) := \widehat{\phi}_0(g) = \text{Re } \widehat{\phi}_0(g) + i \text{Im } \widehat{\phi}_0(g), \quad (4.5.5)$$

and let ψ_1 and ψ_2 be the real and imaginary parts of ψ . Then the collection $\{\psi_1(f), \psi_2(f) : f \in \mathcal{S}(\mathcal{X}_m, \mathbb{R})\}$ is jointly centered Gaussian. Then the identities from Lemma 4.5.1 can be re-expressed in terms of the field ψ as follows.

Corollary 4.5.2. *For $p = (\omega_q, q) \in \mathcal{X}_m$, let $\tilde{p}$ denote the point $(\omega_q, -q)$. For any $f, g \in \mathcal{S}(\mathcal{X}_m, \mathbb{R})$,*

$$\begin{aligned} \text{Cov}(\psi_1(f), \psi_2(g)) &= 0, \\ \text{Cov}(\psi_1(f), \psi_1(g)) &= \frac{1}{4} \int_{\mathcal{X}_m} (f(p) + f(\tilde{p}))(g(p) + g(\tilde{p})) d\lambda_m(p), \\ \text{Cov}(\psi_2(f), \psi_2(g)) &= \frac{1}{4} \int_{\mathcal{X}_m} (f(p) - f(\tilde{p}))(g(p) - g(\tilde{p})) d\lambda_m(p). \end{aligned}$$

Consequently, for any $f, g \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$,

$$\mathbb{E}(\psi(f) \overline{\psi(g)}) = \langle f, g \rangle_{L^2(\lambda_m)}, \quad \mathbb{E}(\psi(f) \psi(g)) = \int_{\mathcal{X}_m} f(p) g(\tilde{p}) d\lambda_m(p).$$

Proof. The first three identities are just restatements of the identities from Lemma 4.5.1.

Next, note that if $f, g \in \mathcal{S}(\mathcal{X}_m, \mathbb{R})$, then the three identities show that

$$\begin{aligned}
 \mathbb{E}(\psi(f)\overline{\psi(g)}) &= \mathbb{E}[(\psi_1(f) + i\psi_2(f))(\psi_1(g) - i\psi_2(g))] \\
 &= \mathbb{E}(\psi_1(f)\psi_1(g) + \psi_2(f)\psi_2(g)) + i\mathbb{E}(\psi_2(f)\psi_1(g) - \psi_1(f)\psi_2(g)) \\
 &= \mathbb{E}(\psi_1(f)\psi_1(g) + \psi_2(f)\psi_2(g)) \\
 &= \frac{1}{4} \int_{\mathcal{X}_m} (f(p) + f(\tilde{p}))(g(p) + g(\tilde{p}))d\lambda_m(p) \\
 &\quad + \frac{1}{4} \int_{\mathcal{X}_m} (f(p) - f(\tilde{p}))(g(p) - g(\tilde{p}))d\lambda_m(p).
 \end{aligned}$$

Expanding, we see that the $f(p)g(\tilde{p})$ and $f(\tilde{p})g(p)$ terms cancel out. Moreover, the change of variable $\tilde{p} \rightarrow p$ shows that the $f(\tilde{p})g(\tilde{p})$ term is equal to the $f(p)g(p)$ term. Thus, we get

$$\mathbb{E}(\psi(f)\overline{\psi(g)}) = \int_{\mathcal{X}_m} f(p)g(p)d\lambda_m(p) = \langle f, g \rangle_{L^2(\lambda_m)}.$$

Finally, take $f = f_1 + if_2 \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$ and $g = g_1 + ig_2 \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$. Then the above formula shows that

$$\begin{aligned}
 \mathbb{E}(\psi(f)\overline{\psi(g)}) &= \mathbb{E}[(\psi(f_1) + i\psi(f_2))(\overline{\psi(g_1)} - i\overline{\psi(g_2)})] \\
 &= \langle f_1, g_1 \rangle_{L^2(\lambda_m)} + i\langle f_2, g_1 \rangle_{L^2(\lambda_m)} - i\langle f_1, g_2 \rangle_{L^2(\lambda_m)} + \langle f_2, g_2 \rangle_{L^2(\lambda_m)} \\
 &= \int_{\mathcal{X}_m} (f_1(p) + if_2(p))(g_1(p) - ig_2(p))d\lambda_m(p) \\
 &= \langle f, g \rangle_{L^2(\lambda_m)}.
 \end{aligned}$$

The claimed identity for $\mathbb{E}(\psi(f)\psi(g))$ follows similarly. This completes the proof. $\square$

4.6 Bosonic Fock space

The space $L^2(\mathcal{X}_m, \lambda_m)$ is the Hilbert space of momentum states for a single scalar boson of mass m . That is, if the boson is in state f , then its four-momentum has probability density proportional to $|f(p)|^2$ with respect to the measure λ_m on $\mathcal{X}_m$. Similarly, for a system of n scalar bosons of mass m , the states are elements of $L^2(\mathcal{X}_m^n, \lambda_m^n)$, where λ_m^n is the n -fold product of λ_m . However, there is a very important constraint, coming from physics. It says that the state f of a system of n bosons must necessarily be symmetric in the arguments, that we must have that

$$f(p_1, \dots, p_n) = f(p_{\sigma(1)}, \dots, p_{\sigma(n)})$$

for any permutation σ of $1, \dots, n$ and for almost every $p_1, \dots, p_n$ (with respect to λ_m). Note that we can at most demand that this holds for almost every $p_1, \dots, p_n$ because f is an element of an L^2 space, and therefore not really a function but an equivalence class of functions. Let

$$\mathcal{B}_n := L^2_{\text{sym}}(\mathcal{X}_m^n, \lambda_m^n)$$

denote the set of all $f \in L^2(\mathcal{X}_m^n, \lambda_m^n)$ that satisfy this constraint. This is the Hilbert space of momentum states for a system of n scalar bosons of mass m . For $n = 0$, we define this to be equal to $\mathbb{C}$.

Now suppose that the number of bosons is also indeterminate, along with their momenta. Then the state of the system is an element of the Hilbert space

$$\mathcal{B} := \widehat{\bigoplus_{n \geq 0} \mathcal{B}_n},$$

where $\widehat{\bigoplus}$ is the notation for completed direct sum. Without the completion, we get the algebraic direct sum of the Hilbert spaces on the right. A typical element of this space is a formal sum like

$$\sum_{n=0}^{\infty} f_n$$

where $f_n \in \mathcal{B}_n$, and all but finitely many of the f_n 's are zero. The inner product between two such formal sums is defined as

$$\left\langle \sum_{n=0}^{\infty} f_n, \sum_{n=0}^{\infty} g_n \right\rangle := \sum_{n=0}^{\infty} \langle f_n, g_n \rangle_{L^2(\lambda_m^n)}.$$

It is easy to see that this is a positive definite inner product. The completion of the direct sum under this inner product is the Hilbert space $\mathcal{B}$. Formally, an element of $\mathcal{B}$ is again a sum like $\sum_{n=0}^{\infty} f_n$ where

$$\sum_{n=0}^{\infty} \|f_n\|_{L^2(\lambda_m^n)}^2 < \infty.$$

The probability interpretation is as follows. If the system is in the state $\sum_{n=0}^{\infty} f_n$, then the probability that the number of particles is equal to n is

$$\frac{\|f_n\|_{L^2(\lambda_m^n)}^2}{\sum_{j=0}^{\infty} \|f_j\|_{L^2(\lambda_m^j)}^2},$$

and given that the number of particles is n , the probability density of the n four-momenta is proportional to $|f_n(p_1, \dots, p_n)|^2$.

4.7 From free field to Fock space

The discussion in Section 4.5 shows that the field ϕ_0 can be alternatively viewed as the field ψ on $\mathcal{X}_m$ defined in equation (4.5.5). Recall that the field ψ is simply the collection

$$\{\psi_1(f) : f \in \mathcal{S}(\mathcal{X}_m, \mathbb{R})\} \cup \{\psi_2(f) : f \in \mathcal{S}(\mathcal{X}_m, \mathbb{R})\},$$

where ψ_1 and ψ_2 are the real and imaginary parts of ψ . By the identities (4.5.5) and (4.5.2), we see that this is exactly the collection $\{\phi_0(f) : f \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})\}$. This is because, on the

one hand we have for any $f = (2\pi)^{-3}g\pi \in \mathcal{S}(\mathcal{X}_m, \mathbb{R})$,

$$\psi_1(f) = \operatorname{Re} \widehat{\phi}_0(g) = \phi_0(\operatorname{Re} \widehat{g}), \quad \psi_2(f) = \operatorname{Im} \widehat{\phi}_0(g) = \phi_0(-\operatorname{Im} \widehat{g}),$$

and on the other hand, for any $g \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$, the Fourier inversion formula implies that $g = \widehat{h}$ where $h(q) := (2\pi)^{-3}\widehat{g}(-q)$, and hence,

$$\phi_0(g) = \phi_0(\operatorname{Re} g) = \phi_0(\operatorname{Re} \widehat{h}) = \operatorname{Re} \widehat{\phi}_0(h) = \psi_1((2\pi)^{-3}h\pi).$$

Now recall the Hilbert space $\mathcal{H} = L^2(\mathcal{A}, \mathcal{C}, \mu)$ for the free quantum field with mass m that was defined in Section 3.7. The above observation shows that $\mathcal{H}$ can be alternately expressed as $L^2(\mathcal{A}', \mathcal{C}', \mu')$, where $\mathcal{A}' := (\mathbb{R}^2)^{\mathcal{S}(\mathcal{X}_m, \mathbb{R})}$, $\mathcal{C}'$ is the cylinder σ -algebra on $\mathcal{A}'$, and μ' is the law of $\{(\psi_1(f), \psi_2(f)) : f \in \mathcal{S}(\mathcal{X}_m, \mathbb{R})\}$. Now, by Corollary A.2.10 and Corollary B.2.5, the set of polynomials (with complex coefficients) in these random variables form a dense subspace of $\mathcal{H}$. The following lemma gives an equivalent representation.

Lemma 4.7.1. *Any polynomial with complex coefficients in the real-valued random variables $\psi_1(f)$ and $\psi_2(f)$, as f ranges over $\mathcal{S}(\mathcal{X}_m, \mathbb{R})$, is the same as some polynomial with complex coefficients in the complex-valued random variables $\{\psi(f) : f \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})\}$, and vice versa.*

Proof. Take any $f \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$. Let $g(p) := \overline{f(\widetilde{p})}$, where $\widetilde{p}$ is obtained by changing the signs of the last three coordinates of p , as before. Then by Corollary 4.5.2,

$$\begin{aligned} \mathbb{E}|\overline{\psi(f)} - \psi(g)|^2 &= \mathbb{E}|\overline{\psi(f)}|^2 + \mathbb{E}|\psi(g)|^2 - \mathbb{E}(\overline{\psi(f)}\psi(g)) - \mathbb{E}(\psi(g)\overline{\psi(f)}) \\ &= \langle f, f \rangle_{L^2(\lambda_m)} + \langle g, g \rangle_{L^2(\lambda_m)} - \int_{\mathcal{X}_m} (f(p)g(\widetilde{p}) + \overline{f(p)}\overline{g(\widetilde{p})})d\lambda_m(p) \\ &= 2\|f\|_{L^2(\lambda_m)}^2 - \int_{\mathcal{X}_m} 2|f(p)|^2d\lambda_m(p) = 0. \end{aligned}$$

Thus, $\overline{\psi(f)} = \psi(g)$ almost surely, which means they are equal in the L^2 sense. Therefore, we can express

$$\begin{aligned} \psi_1(f) &= \frac{1}{2}(\psi(f) + \overline{\psi(f)}) = \frac{1}{2}(\psi(f) + \psi(g)), \\ \psi_2(f) &= \frac{1}{2i}(\psi(f) - \overline{\psi(f)}) = \frac{1}{2i}(\psi(f) - \psi(g)). \end{aligned}$$

On the other hand, for $f = f_1 + if_2 \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$, where $f_1, f_2 \in \mathcal{S}(\mathcal{X}_m, \mathbb{R})$, we have

$$\begin{aligned} \psi(f) &= \psi_1(f) + i\psi_2(f) \\ &= \psi_1(f_1) + i\psi_1(f_2) + i\psi_2(f_1) - \psi_2(f_2). \end{aligned}$$

Clearly, the above identities suffice to prove the claim. $\square$

By Lemma 4.7.1 and the preceding discussion, the set of polynomials in $\{\psi(f) : f \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})\}$ is dense in $\mathcal{H}$. We will now use this set to obtain a certain orthogonal

decomposition of $\mathcal{H}$. Let $\mathcal{H}_0$ the set of constant random variables, and for each $n \geq 1$, let $\mathcal{H}_n$ be the closure of the span of Wick ordered monomials¹ of degree n in $\{\psi(f) : f \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})\}$.

Lemma 4.7.2. *The subspaces $\{\mathcal{H}_n\}_{n \geq 0}$ defined above are mutually orthogonal, and the closure of their direct sum is equal to $\mathcal{H}$.*

Proof. The mutual orthogonality follows from the orthogonality of Wick ordered polynomials of different degrees, as proved in Theorem B.2.9. A simple argument by induction shows that any degree n polynomial is in the span of Wick ordered monomials of degree $\leq n$. As observed above, the set of polynomials is dense in $\mathcal{H}$. This completes the proof. $\square$

Now recall the space $\mathcal{B}$ defined in Section 4.6. Define a map $\tau : \oplus_{n=1}^{\infty} \mathcal{H}_n \rightarrow \mathcal{B}$ as

$$\tau(:\psi(f_1) \cdots \psi(f_n):)(p_1, \dots, p_n) := \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} f_1(p_{\sigma(1)}) \cdots f_n(p_{\sigma(n)}), \quad (4.7.1)$$

and extended by linearity. It is not obvious that this map is well-defined. The following lemma shows that it is.

Lemma 4.7.3. *The map τ defined above is well-defined, and preserves inner products.*

Proof. Take and $k, n \geq 0$, and $f_1, \dots, f_n, g_1, \dots, g_k \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$. Let $X := :\psi(f_1) \cdots \psi(f_n):$ and $Y := :\psi(g_1) \cdots \psi(g_k):$, and define

$$\begin{aligned} f(p_1, \dots, p_n) &:= \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} f_1(p_{\sigma(1)}) \cdots f_n(p_{\sigma(n)}), \\ g(p_1, \dots, p_k) &:= \frac{1}{\sqrt{k!}} \sum_{\sigma \in S_k} g_1(p_{\sigma(1)}) \cdots g_k(p_{\sigma(k)}). \end{aligned}$$

Let $h_j(p) := \overline{g_j(\widehat{p})}$ for $j = 1, \dots, k$. From the proof of Lemma 4.7.1, we know that $\overline{\psi(g_j)} = \psi(h_j)$. From this, and Theorem B.2.9, we see that if $n \neq k$, then

$$\mathbb{E}(X\overline{Y}) = 0 = \langle f, g \rangle_{\mathcal{B}}.$$

So, let us assume that $n = k$. Then again by Theorem B.2.9,

$$\begin{aligned} \mathbb{E}(X\overline{Y}) &= \mathbb{E}(:\psi(f_1) \cdots \psi(f_n)::\psi(h_1) \cdots \psi(h_n):) \\ &= \sum_{\sigma \in S_n} \prod_{j=1}^n \mathbb{E}(\psi(f_j)\psi(h_{\sigma(j)})) \\ &= \sum_{\sigma \in S_n} \prod_{j=1}^n \mathbb{E}(\psi(f_j)\overline{\psi(g_{\sigma(j)})}) \end{aligned}$$

¹See Section B.2.6 in Appendix B for the definition and properties of Wick ordered monomials.

By Corollary 4.5.2, this implies that

$$\begin{aligned}
 \mathbb{E}(X\bar{Y}) &= \sum_{\sigma \in S_n} \prod_{j=1}^n \langle f_j, g_{\sigma(j)} \rangle_{L^2(\lambda_m)} \\
 &= \frac{1}{n!} \sum_{\kappa, \sigma \in S_n} \prod_{j=1}^n \langle f_{\kappa(j)}, g_{\sigma(j)} \rangle_{L^2(\lambda_m)} \\
 &= \langle f, g \rangle_{L^2(\lambda_m^n)} = \langle f, g \rangle_{\mathcal{B}}.
 \end{aligned}$$

This proves that τ is an isometry. As a consequence, we also see that τ is well-defined, because if $X = Y$ almost surely, then

$$\begin{aligned}
 \|f - g\|_{\mathcal{B}}^2 &= \langle f, f \rangle_{\mathcal{B}} + \langle g, g \rangle_{\mathcal{B}} - 2 \operatorname{Re} \langle f, g \rangle_{\mathcal{B}} \\
 &= \mathbb{E}|X|^2 + \mathbb{E}|Y|^2 - 2 \operatorname{Re} \mathbb{E}(X\bar{Y}) = \mathbb{E}|X - Y|^2 = 0.
 \end{aligned}$$

This completes the proof. $\square$

The isometric nature of τ allows us to extend the definition of τ to the whole of $\mathcal{H}$. Our final lemma in this section, stated below, shows that τ is surjective from $\mathcal{H}$ on to $\mathcal{B}$. In other words, τ gives an isomorphism of the Hilbert spaces $\mathcal{H}$ and $\mathcal{B}$.

Lemma 4.7.4. *The image of $\oplus_{n=0}^{\infty} \mathcal{H}_n$ under τ is dense in $\mathcal{B}$. Consequently, τ is a Hilbert space isomorphism between $\mathcal{H}$ and $\mathcal{B}$.*

Proof. By the isometric nature of τ , it is easy to see that we only need to show that for any n , functions of the form

$$\sum_{\sigma \in S_n} f_1(p_{\sigma(1)}) \cdots f_n(p_{\sigma(n)}),$$

where $f_1, \dots, f_n \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$, span a dense subspace of $L_{\text{sym}}^2(\lambda_m^n)$. To show this, we claim that it suffices to prove that functions of the form $f_1(p_1) \cdots f_n(p_n)$ span a dense subspace of $L^2(\lambda_m^n)$. Indeed, suppose that the latter is true. Take any $f \in L_{\text{sym}}^2(\lambda_m^n)$ and $\epsilon > 0$. Find $f_1, \dots, f_n$ such that the function $h(p_1, \dots, p_n) := f_1(p_1) \cdots f_n(p_n)$ is within distance ϵ from f . Then for any $\sigma \in S_n$,

$$\begin{aligned}
 &\int_{\mathcal{X}_m^n} |f(p_1, \dots, p_n) - h(p_{\sigma(1)}, \dots, p_{\sigma(n)})|^2 d\lambda_m^n(p_1, \dots, p_n) \\
 &= \int_{\mathcal{X}_m^n} |f(p_{\sigma(1)}, \dots, p_{\sigma(n)}) - h(p_{\sigma(1)}, \dots, p_{\sigma(n)})|^2 d\lambda_m^n(p_1, \dots, p_n) \\
 &= \int_{\mathcal{X}_m^n} |f(p_1, \dots, p_n) - h(p_1, \dots, p_n)|^2 d\lambda_m^n(p_1, \dots, p_n) \leq \epsilon^2.
 \end{aligned}$$

Averaging over σ and applying the triangle inequality for the L^2 norm, we see that the function

$$g(p_1, \dots, p_n) := \frac{1}{n!} \sum_{\sigma \in S_n} h(p_{\sigma(1)}, \dots, p_{\sigma(n)})$$

is within distance ϵ from f . This proves the above claim.

It only remains to prove that functions of the form $f_1(p_1) \cdots f_n(p_n)$, where $f_1, \dots, f_n \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$, span a dense subspace of $L^2(\lambda_m^n)$. Take any $f \in L^2(\lambda_m^n)$, and let $h := f\pi^{-1}$, where π is the projection map from $\mathcal{X}_m$ to $\mathbb{R}^3$. Then by the definition of λ_m , we have

$$\int_{(\mathbb{R}^3)^n} \frac{|h(q_1, \dots, q_n)|^2}{\omega_{q_1} \cdots \omega_{q_n}} dq_1 \cdots dq_n < \infty,$$

where $\omega_q = \sqrt{\|q\|^2 + m^2}$, as before. In other words, the function h is in $L^2(\mu_m^n)$, where μ_m is the measure on $\mathbb{R}^3$ that has density $(2\omega_q)^{-1}$ with respect to Lebesgue measure. By the dominated convergence theorem, $L^2(\mu_m^n)$ functions on $(\mathbb{R}^3)^n$ that vanish outside a bounded region are dense in $L^2(\mu_m^n)$. By Theorem A.2.9, linear combinations of indicator functions of $(\mathbb{R}^3)^n$ -rectangles are $L^2(\mu_m^n)$ -dense in the space of such functions. But indicator functions of $(\mathbb{R}^3)^n$ -rectangles are of the form $u_1(q_1) \cdots u_n(q_n)$, where $u_1, \dots, u_n$ are indicator functions of $\mathbb{R}^3$ -rectangles. It is not hard to show that indicator functions of $\mathbb{R}^3$ -rectangles can be arbitrarily well-approximated in $L^2(\mu_m)$ by Schwartz functions. This completes the proof. $\square$

The isomorphism τ between $\mathcal{H}$ and $\mathcal{B}$ constructed above gives us a way to identify any state of the free quantum field of mass m in $\mathbb{R}^4$ with a unique state in the bosonic Fock space $\mathcal{B}$, and vice versa. In this way, the free quantum field can be seen as representing an indeterminate system of scalar bosons of mass m , and conversely.

4.8 Time evolution on Fock space

The gist of what we have accomplished in this chapter until now is that the states of the free quantum field are in a one-to-one correspondence with the states of a system of bosons, where the number of bosons is indeterminate. The correspondence is given by the map τ defined in equation (4.7.1). Therefore, the evolution of a state of the free quantum field over time should correspond to the time evolution of the corresponding state for the system of bosons. The following theorem shows that on the boson side, this evolution is particularly simple: the bosons just evolve as free non-interacting particles, and the number of bosons remains constant.

Theorem 4.8.1. *Take any $n \geq 1$ and $f_1, \dots, f_n \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$. In the Fock space representation of the free quantum field, the state*

$$\psi(p_1, \dots, p_n) = \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} f_1(p_{\sigma(1)}) \cdots f_n(p_{\sigma(n)}) \in \mathcal{B}_n$$

evolves as

$$\begin{aligned} U(t)\psi(p_1, \dots, p_n) &= \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} (e^{-itp_{\sigma(1),0}} f_1(p_{\sigma(1)})) \cdots (e^{-itp_{\sigma(n),0}} f_n(p_{\sigma(n)})) \\ &= e^{-it \sum_{j=1}^n p_{j,0}} \psi(p_1, \dots, p_n), \end{aligned}$$

where $p_{j,0}$ denotes the first coordinate of p_j . The time evolution operator $U(t)$ extends to the full Hilbert space $\mathcal{B}$ by linearity and continuity.

Proof. Let $\mathcal{H} = L^2(\mathcal{A}, \mathcal{C}, \mu)$ be the Hilbert space for the free quantum field, defined in Section 3.7, where $\mathcal{A} = \mathbb{R}^{S(\mathbb{R}^3, \mathbb{R})}$ and μ is the law of the time slice ϕ_0 of the Euclidean free field ϕ . Define the random field ψ using ϕ_0 as in Section 4.7, and instead of ψ , let us call this field ψ_0 . Take any $n \geq 1$ and $f_1, \dots, f_n, g_1, \dots, g_n \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$. Let $F, G \in \mathcal{H}$ be the states

$$F(\phi_0) := :\psi_0(f_1) \cdots \psi_0(f_n):, \quad G(\phi_0) := :\psi_0(g_1) \cdots \psi_0(g_n):.$$

Take any $t \geq 0$. Let ψ_t be defined using ϕ_t using the same prescription as was used for defining ψ_0 using ϕ_0 . Let $(P(t))_{t \geq 0}$ be the semigroup for the Euclidean free field. Then by Theorem B.2.9,

$$\begin{aligned} \mathbb{E}(G(\phi_0)P(t)F(\phi_0)) &= \mathbb{E}(G(\phi_0)F(\phi_t)) \\ &= \mathbb{E}(:\psi_0(g_1) \cdots \psi_0(g_n)::\psi_t(f_1) \cdots \psi_t(f_n):) \\ &= \sum_{\sigma \in S_n} \prod_{j=1}^n \mathbb{E}(\psi_0(g_j)\psi_t(f_{\sigma(j)})). \end{aligned} \tag{4.8.1}$$

Take any $f, g \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$. Proceeding exactly as in the proof of Lemma 4.5.1, one can get the identities

$$\begin{aligned} \text{Cov}(\text{Re } \widehat{\phi}_0(f), \text{Im } \widehat{\phi}_t(g)) &= 0, \\ \text{Cov}(\text{Re } \widehat{\phi}_0(f), \text{Re } \widehat{\phi}_t(g)) &= \frac{(2\pi)^3}{4} \int_{\mathbb{R}^3} \frac{e^{-t\omega_q}}{2\omega_q} (f(q) + f(-q))(g(q) + g(-q))dq, \\ \text{Cov}(\text{Im } \widehat{\phi}_0(f), \text{Im } \widehat{\phi}_t(g)) &= \frac{(2\pi)^3}{4} \int_{\mathbb{R}^3} \frac{e^{-t\omega_q}}{2\omega_q} (f(q) - f(-q))(g(q) - g(-q))dq, \end{aligned}$$

where $\omega_q = \sqrt{\|q\|^2 + m^2}$, as usual. Now recall that for $q \in \mathbb{R}^3$, the point $p = \pi^{-1}q \in \mathcal{X}_m$ is given by $p = (\omega_q, q)$. Thus, using the above identities and proceeding as in the proof of Corollary 4.5.2, we obtain, for any $f, g \in \mathcal{S}(\mathcal{X}_m, \mathbb{C})$,

$$\mathbb{E}(\psi_0(f)\psi_t(g)) = \int_{\mathcal{X}_m} e^{-tp_0} f(p)g(\tilde{p})d\lambda_m(p),$$

where p_0 is the first coordinate of p and $\tilde{p}$ is obtained by flipping the signs of the last three

coordinates of p . Combining this with equation (4.8.1), we get

$$\mathbb{E}(G(\phi_0)P(t)F(\phi_0)) = \sum_{\sigma \in S_n} \prod_{j=1}^n \int_{\mathcal{X}_m} e^{-itp_0} g_j(p) f_{\sigma(j)}(\tilde{p}) d\lambda_m(p).$$

Let $(U(t))_{t \in \mathbb{R}}$ be the time evolution group for the free quantum field. Then by the above formula and Theorem 2.5.1, it follows that

$$\mathbb{E}(G(\phi_0)U(t)F(\phi_0)) = \sum_{\sigma \in S_n} \prod_{j=1}^n \int_{\mathcal{X}_m} e^{-itp_0} g_j(p) f_{\sigma(j)}(\tilde{p}) d\lambda_m(p). \quad (4.8.2)$$

Now, given $f_1, \dots, f_n$, let us choose $g_1, \dots, g_n$ as

$$g_j(p) := \overline{e^{-itp_0} f_j(p)} = e^{itp_0} \overline{f_j(p)}.$$

We claim that $U(t)F = \overline{G}$. To see this, note that since $U(t)$ is a unitary operator,

$$\begin{aligned} \|U(t)F - \overline{G}\|_{\mathcal{H}}^2 &= \|U(t)F\|_{\mathcal{H}}^2 + \|\overline{G}\|_{\mathcal{H}}^2 - 2 \operatorname{Re}(\langle U(t)F, \overline{G} \rangle) \\ &= \|F\|_{\mathcal{H}}^2 + \|\overline{G}\|_{\mathcal{H}}^2 - 2 \operatorname{Re}(\langle U(t)F, \overline{G} \rangle) \\ &= \mathbb{E}(|F(\phi_0)|^2) + \mathbb{E}(|G(\phi_0)|^2) - 2 \operatorname{Re}(\mathbb{E}(G(\phi_0)U(t)F(\phi_0))). \end{aligned}$$

Applying the identity (4.8.2), we get

$$\begin{aligned} \mathbb{E}(|F(\phi_0)|^2) &= \mathbb{E}(\overline{F(\phi_0)}U(0)F(\phi_0)) = \sum_{\sigma \in S_n} \prod_{j=1}^n \int_{\mathcal{X}_m} \overline{f_j(p)} f_{\sigma(j)}(\tilde{p}) d\lambda_m(p), \\ \mathbb{E}(|G(\phi_0)|^2) &= \mathbb{E}(\overline{G(\phi_0)}U(0)G(\phi_0)) = \sum_{\sigma \in S_n} \prod_{j=1}^n \int_{\mathcal{X}_m} \overline{g_j(p)} g_{\sigma(j)}(\tilde{p}) d\lambda_m(p) \\ &= \sum_{\sigma \in S_n} \prod_{j=1}^n \int_{\mathcal{X}_m} \overline{f_j(p)} f_{\sigma(j)}(\tilde{p}) d\lambda_m(p), \\ \mathbb{E}(G(\phi_0)U(t)F(\phi_0)) &= \sum_{\sigma \in S_n} \prod_{j=1}^n \int_{\mathcal{X}_m} e^{-itp_0} g_j(p) f_{\sigma(j)}(\tilde{p}) d\lambda_m(p) \\ &= \sum_{\sigma \in S_n} \prod_{j=1}^n \int_{\mathcal{X}_m} \overline{f_j(p)} f_{\sigma(j)}(\tilde{p}) d\lambda_m(p). \end{aligned}$$

Thus, the quantities $\mathbb{E}(|F(\phi_0)|^2)$, $\mathbb{E}(|G(\phi_0)|^2)$, and $\mathbb{E}(G(\phi_0)U(t)F(\phi_0))$ are equal. This shows, in particular, that $\mathbb{E}(G(\phi_0)U(t)F(\phi_0))$ is real. This completes the proof of the claim that $U(t)F = \overline{G}$. It is easy to see that this proves the theorem. $\square$

Chapter 5

Perturbative Renormalization

5.1 The heuristic algorithm

The idea of perturbative renormalization can be mathematically formulated as follows. Suppose we have a family of real-valued functions $\{f_\Lambda\}_{\Lambda>0}$ on $\mathbb{R}^k$. For $\theta \in \mathbb{R}^k$, $f_\Lambda(\theta)$ represents the expected value of a measurement in an experiment, with θ denoting the vector of parameter values in a model and Λ denoting a ‘cutoff’ parameter. The idealized model that we use has $\Lambda = \infty$, but the model that reflects reality has an unknown (but large) value of Λ . Our theory lets us compute $f_\Lambda(\theta)$ as a formal power series in θ :

$$f_\Lambda(\theta) = \sum_{m \in \mathbb{Z}_{\geq 0}^k} a_{\Lambda,m} \theta^m, \quad (5.1.1)$$

where $\mathbb{Z}_{\geq 0}$ is the set of nonnegative integers and for $m = (m_1, \dots, m_k) \in \mathbb{Z}_{\geq 0}^k$ and $\theta = (\theta_1, \dots, \theta_k) \in \mathbb{R}^k$, θ^m denotes the product $\theta_1^{m_1} \cdots \theta_k^{m_k}$.

The problem is that if we substitute $\Lambda = \infty$ as in the idealized model, then some or most of the coefficients in the above power series turn out to be infinity. On the other hand, the true value of Λ , while large, is unknown. We cannot hope to estimate θ from the value of $f_\Lambda(\theta)$ without knowing Λ . But we can still hope to meet the following kind of objective. Suppose we want to predict the expected outcome $h_\Lambda(\theta)$ of some other experiment involving the same system, again known as a power series

$$h_\Lambda(\theta) = \sum_{m \in \mathbb{Z}_{\geq 0}^k} b_{\Lambda,m} \theta^m.$$

It turns out that in some situations, it may be possible to approximately calculate $h_\Lambda(\theta)$ using only the observed value of $f_\Lambda(\theta)$, *solely under the assumption that Λ is large, without knowing the value of Λ* . The procedure for doing this, described below, is called perturbative renormalization.

There are various approaches to perturbative renormalization. We will focus on the most popular approach, known as the *counterterm method*. In the counterterm method, we

replace each θ_j by a perturbation series in a *single* parameter g :

$$\theta_j = \sum_{r=1}^{\infty} c_{\Lambda,j,r} g^r.$$

A very important fact is that the above sum starts from $m = 1$ instead of $m = 0$. This allows the following. Substituting the above value of θ_j in equation (5.1.1), we get

$$f_{\Lambda}(\theta) = \sum_{m \in \mathbb{Z}_{\geq 0}^k} a_{\Lambda,m} \left[\prod_{j=1}^k \left(\sum_{r=1}^{\infty} c_{\Lambda,j,r} g^r \right)^{m_j} \right] = \sum_{n=1}^{\infty} d_{\Lambda,n} g^n,$$

where $d_{\Lambda,0} = a_{\Lambda,0}$, and for $n \geq 1$, $d_{\Lambda,n}$ is the sum of products of the form

$$a_{\Lambda,m} \prod_{j=1}^k \prod_{q=1}^{m_j} c_{\Lambda,j,r_{j,q}},$$

over all $m \in \mathbb{Z}_{\geq 0}^k$ and positive integers $r_{j,q}$ such that $\sum_{j=1}^k \sum_{q=1}^{m_j} r_{j,q} = n$. The goal is to choose the c 's in such a way that $d_{\Lambda,n}$ does not depend on the cutoff parameter Λ . Suppose we fix a collection of coefficients $d_1, d_2, \dots$ and try to find c 's so that

$$d_{\Lambda,n} = d_n \quad \text{for each } n \geq 1. \quad (5.1.2)$$

Now, any m that appears in the expression for $d_{\Lambda,n}$ above must satisfy $\sum_{j=1}^k m_j \leq n$, because $r_{j,q} \geq 1$ for each j, q , and $\sum_{j=1}^k \sum_{q=1}^{m_j} r_{j,q} = n$. This shows that each equation in the system of equations (5.1.2) is a linear equation involving a finite number of variables, which we can hope to solve inductively. Once that is done, we end up with the expression

$$f_{\Lambda}(\theta) = f_{\Lambda}(0) + \sum_{n=1}^{\infty} d_n g^n. \quad (5.1.3)$$

Suppose that, following the same steps, and with the same set of c 's, we have a similar expression for $h_{\Lambda}(\theta)$:

$$h_{\Lambda}(\theta) = h_{\Lambda}(0) + \sum_{n=1}^{\infty} e_{\Lambda,n} g^n. \quad (5.1.4)$$

Suppose that $f_{\Lambda}(0)$ and $h_{\Lambda}(0)$ approach known finite limits as $\Lambda \rightarrow \infty$, and for each n , $e_{\Lambda,n}$ converges to a finite limit e_n . Then, given the experimental value of $f_{\Lambda}(\theta)$, our prediction procedure for $h_{\Lambda}(\theta)$ will go as follows. First, estimate g from the observed value of $f_{\Lambda}(\theta)$ and equation (5.1.3). This may be possible to do without knowing Λ because we know the d_n 's and the limiting value of $f_{\Lambda}(0)$ as $\Lambda \rightarrow \infty$. Once we have recovered the value of g , we can substitute that in equation (5.1.4) to estimate $h_{\Lambda}(\theta)$ (again, because we know the limiting values of $h_{\Lambda}(\theta)$ and $e_{\Lambda,n}$ as $\Lambda \rightarrow \infty$).

5.2 A toy example

Before proceeding further, let us work out the perturbative renormalization heuristic in a toy example, just to see it in action. Take $k = 1$. Suppose that for $\Lambda > 0$ and $\theta \geq -\frac{1}{4\Lambda}$,

$$f_\Lambda(\theta) = \frac{\sqrt{1 + 4\Lambda\theta} - 1}{2\Lambda},$$

where the square root denotes the positive square root. It is simple to check that the Taylor series expansion of f_Λ at 0 is given by

$$f_\Lambda(\theta) = \sum_{m=1}^{\infty} \binom{\frac{1}{2}}{m} 2^{2m-1} \Lambda^{m-1} \theta^m, \quad (5.2.1)$$

with radius of convergence $-\frac{1}{4\Lambda}$. Note that all but the first coefficient blow up as $\Lambda \rightarrow \infty$. Also, let

$$h_\Lambda(\theta) := \frac{\theta}{\Lambda}.$$

Suppose we have an experiment where the theory predicts that the outcome has expected value $f_\Lambda(\theta)$, and another experiment where the outcome is $h_\Lambda(\theta)$. Here Λ and θ are both unknown, and Λ is large. The goal is to estimate the value of $h_\Lambda(\theta)$, having observed the value of $f_\Lambda(\theta)$.

The perturbative renormalization approach via counterterms would go as follows. First, let us replace θ by a power series in a new parameter g :

$$\theta = \sum_{r=1}^{\infty} c_{\Lambda,r} g^r.$$

Substituting this in equation (5.2.1), we get

$$\begin{aligned} f_\Lambda(\theta) &= \sum_{m=1}^{\infty} \binom{\frac{1}{2}}{m} 2^{2m-1} \Lambda^{m-1} \left(\sum_{r=1}^{\infty} c_{\Lambda,r} g^r \right)^m \\ &= \sum_{m=1}^{\infty} d_{\Lambda,m} g^m, \end{aligned}$$

where

$$d_{\Lambda,m} = \sum_{n=1}^m \binom{\frac{1}{2}}{n} 2^{2n-1} \Lambda^{n-1} \left(\sum_{\substack{1 \leq r_1, \dots, r_n \leq m, \\ r_1 + \dots + r_n = m}} \prod_{l=1}^n c_{\Lambda, r_l} \right).$$

We want to choose the c 's such that the d 's do not depend on Λ . To make a guess, let us

compute the first few d 's. By the above formula,

$$\begin{aligned} d_{\Lambda,1} &= c_{\Lambda,1}, \\ d_{\Lambda,2} &= c_{\Lambda,2} - \Lambda c_{\Lambda,1}^2, \\ d_{\Lambda,3} &= c_{\Lambda,3} - 2\Lambda c_{\Lambda,1} c_{\Lambda,2} + 2\Lambda^2 c_{\Lambda,1}^3, \\ d_{\Lambda,4} &= c_{\Lambda,4} - \Lambda(2c_{\Lambda,1} c_{\Lambda,3} + c_{\Lambda,2}^2) + 6\Lambda^2 c_{\Lambda,1}^2 c_{\Lambda,2} - 5\Lambda^3 c_{\Lambda,1}^4. \end{aligned}$$

Suppose we choose $c_{\Lambda,1} = 1$, so that $d_{\Lambda,1} = 1$. To make $d_{\Lambda,2}$ free of Λ , let us try to make it equal to zero. Since $c_{\Lambda,1} = 1$, this forces $c_{\Lambda,2} = \Lambda$. Next, to make $d_{\Lambda,3} = 0$, we see that it suffices to set $c_{\Lambda,3} = 0$. Finally, with $c_{\Lambda,1} = 1$, $c_{\Lambda,2} = \Lambda$, and $c_{\Lambda,3} = c_{\Lambda,4} = 0$, we get $d_{\Lambda,4} = 0$. The emerging pattern suggests that if we choose $c_{\Lambda,1} = 1$, $c_{\Lambda,2} = \Lambda$, and $c_{\Lambda,r} = 0$ for all $r \geq 3$, we will get $d_{\Lambda,m} = 0$ for all $m \geq 3$. Indeed, this is true. To see this, take any $m \geq 3$. By our choice of the c 's, the product $\prod_{l=1}^n c_{\Lambda,r_l}$ is zero if at least one $r_j \geq 3$. Now, for the existence of $r_1, \dots, r_n \in \{1, 2\}$ such that $r_1 + \dots + r_n = m$, we need $n \geq \lceil \frac{1}{2}m \rceil$. For any such n and $r_1, \dots, r_n$, exactly $m - n$ of the r_j 's are 2 and the rest are 1. Thus,

$$\sum_{\substack{1 \leq r_1, \dots, r_n \leq m, \\ r_1 + \dots + r_n = m}} \prod_{l=1}^n c_{\Lambda,r_l} = \binom{n}{m-n} \Lambda^{m-n}.$$

This gives

$$d_{\Lambda,m} = \sum_{n=\lceil \frac{1}{2}m \rceil}^m \binom{\frac{1}{2}}{n} 2^{2n-1} \Lambda^{m-1} \binom{n}{m-n}.$$

To see that this is zero, note that for $|x| < \frac{1}{4}$,

$$\sqrt{1+4x} = \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} (4x)^n, \quad (5.2.2)$$

where the left side denotes the positive square root of $1+4x$. Now, if $|x| < \frac{1}{5}$, then $|x+x^2| < \frac{1}{4}$. Thus, for $|x| < \frac{1}{5}$, substituting $x+x^2$ in place of x on both sides above, we get

$$\begin{aligned} \sqrt{1+4(x+x^2)} &= \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} 4^n x^n (1+x)^n \\ &= \sum_{n=0}^{\infty} \sum_{j=0}^n \binom{\frac{1}{2}}{n} 4^n x^n \binom{n}{j} x^j \\ &= \sum_{m=0}^{\infty} x^m \left[\sum_{n=\lceil \frac{1}{2}m \rceil}^m \binom{\frac{1}{2}}{n} \binom{n}{m-n} 4^n \right]. \end{aligned}$$

Combining, we get

$$1 + 2x = \sqrt{1 + 4(x + x^2)} = 1 + \sum_{m=1}^{\infty} 2d_{\Lambda,m}x^m.$$

Equating coefficients, we see that $d_{\Lambda,m} = 0$ for all $m \geq 3$.

To summarize, we have shown the following. Reparametrizing $\theta = g + \Lambda g^2$, we get

$$\sum_{m=1}^{\infty} \binom{\frac{1}{2}}{m} 2^{2m-1} \Lambda^{m-1} \theta^m = g. \quad (5.2.3)$$

Thus, the above renormalization has resulted in the ‘cancellation of infinities’ of the original power series (5.2.1) for f_{Λ} , because there are no coefficients that blow up as $\Lambda \rightarrow \infty$. Now, if we apply the same renormalization to h_{Λ} , we get

$$h_{\Lambda}(\theta) = \frac{g + \Lambda g^2}{\Lambda} = \frac{g}{\Lambda} + g^2. \quad (5.2.4)$$

Physicists would say that by equations (5.2.3) and (5.2.4), if Λ is large, then we can estimate $h_{\Lambda}(\theta)$ as approximately equal to the square of the observed value of $f_{\Lambda}(\theta)$ (as long as this observed value is not huge), *without knowing the values of Λ and θ* .

5.3 The paradoxical nature of perturbative renormalization

There is a well-known reason to be suspicious of the validity of perturbative renormalization. Consider the toy example from the previous section. As noted earlier, the radius of convergence of the power series (5.2.1) for f_{Λ} is of order $1/\Lambda$. Now recall that the new parameter g is related to θ as $\theta = g + \Lambda g^2$. Also, a simple calculation (given below) shows that $f_{\Lambda}(\theta) = g$. So if the observed value of $f_{\Lambda}(\theta)$ is not tiny, the value of θ is huge — of order Λ instead of $1/\Lambda$. Thus, there is no way that the power series expansion (5.2.1) equals the actual value of $f_{\Lambda}(\theta)$ for this value of θ ; in fact, the power series diverges. This is a paradox at the heart of perturbative renormalization: Taylor expansion is used to justify calculations at parameter values that are far away from the point around which the Taylor expansion is taken. Often, the original series is vigorously divergent at the location where it is evaluated. So, does it give any meaningful result at all?

We will now see why the result that $h_{\Lambda}(\theta) \approx f_{\Lambda}(\theta)^2$ that we obtained in the previous section is actually sensible for the toy example. The key observation is that for $g \geq -\frac{1}{2\Lambda}$, the reparametrization $\theta = g + \Lambda g^2$ gives

$$\begin{aligned} f_{\Lambda}(\theta) &= \frac{\sqrt{1 + 4\Lambda(g + \Lambda g^2)} - 1}{2\Lambda} \\ &= \frac{\sqrt{(1 + 2\Lambda g)^2} - 1}{2\Lambda} = g. \end{aligned}$$

Since

$$h_\Lambda(\theta) = \frac{g}{\Lambda} + g^2,$$

this shows that indeed, $h_\Lambda(\theta) \approx f_\Lambda(\theta)^2$ when $f_\Lambda(\theta)$ is nonnegative and neither tiny nor huge.

Therefore, for this toy example, the heuristic algorithm for perturbative renormalization led us to the correct conclusion, in spite of our cavalier application of the Taylor series expansion of f_Λ far away from the region where it is valid. Is this an accident? In the next section, we will show that it is not. There is a general principle that may lend validity to the perturbative renormalization heuristic in various situations.

5.4 A rigorous criterion

To summarize the preceding discussion, the paradox at the center of perturbative renormalization consists of the following:

- There is a function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ whose power series expansion at the origin has a very small radius of convergence.
- We find a renormalization map $R : \mathbb{R} \rightarrow \mathbb{R}^k$, usually with $R(0) = 0$, such that the power series expansion of $f \circ R$ at the origin has a large radius of convergence. This is done by inductively determining the coefficients in the power series expansion of R at the origin, such that the coefficients of $f \circ R$ are ‘not large’. This is an abstraction of the notion of ‘canceling infinities’ from the physics literature.
- Then, we compute $f(R(x))$ using the power series expansion of $f \circ R$ at the origin, for arbitrary x within the radius of convergence of this series. Since $R(x)$ may not necessarily be within the radius of convergence of the power series of f at the origin, this casts doubt on the validity of this approach.

We have already seen an example where this works. We will now present a theorem which gives a general criterion for the validity of this approach. We will then see a counterexample where the criterion does not hold, and the approach does not give the correct result.

Theorem 5.4.1. *Let $f : \Omega \rightarrow \mathbb{C}$ and $R : \Gamma \rightarrow \mathbb{R}^k$ be analytic functions, where $\Omega \subseteq \mathbb{R}^k$ and $\Gamma \subseteq \mathbb{R}$ are open connected sets that contains the origin and $R(0) = 0$. Define $u := f \circ R$ on $R^{-1}(\Omega)$. Let $\tilde{u}$ be the function defined by the power series of u at the origin on its domain of convergence $\mathcal{D}$. Let U be the connected component of $\mathcal{D} \cap R^{-1}(\Omega)$ that contains the origin. Then $\tilde{u} = u$ on U .*

Proof. First, note that $R^{-1}(\Omega)$ is an open set, since R continuous. Moreover, it contains the origin, since $R(0) = 0 \in \Omega$. Since f and R are both analytic on their respective domains, and compositions of analytic functions are analytic [15, Proposition 2.2.8], we conclude that u does indeed have a power series expansion at the origin with a strictly positive radius of convergence. Since $R^{-1}(\Omega)$ and $\mathcal{D}$ are open sets containing the origin, there is a

small enough open ball W centered at the origin that is contained in both $R^{-1}(\Omega)$ and $\mathcal{D}$. Clearly, $\tilde{u} = u$ on W . Now, U is an open connected set, $\tilde{u}$ and u are analytic on U , and $\tilde{u} = u$ on W , which is a nonempty open subset of U . By the identity theorem for analytic functions [15, Corollary 1.2.7], we conclude that $\tilde{u} = u$ everywhere on U . $\square$

5.5 An example and a counterexample

Recall that in our toy example from Sections 5.2 and 5.3,

$$f_\Lambda(\theta) = \frac{\sqrt{1 + 4\Lambda\theta} - 1}{2\Lambda}$$

for $\Lambda > 0$ and $\theta \geq -\frac{1}{4\Lambda}$. We claim that f_Λ is analytic on $(-\frac{1}{4\Lambda}, \infty)$. To see this, consider the analytic branch of square root on $\mathbb{C} \setminus (-\infty, 0]$, given by

$$\sqrt{re^{i\theta}} := \sqrt{r}e^{\frac{1}{2}i\theta}, \quad r > 0, \quad -\pi < \theta < \pi.$$

If we define

$$f_\Lambda(z) := \frac{\sqrt{1 + 4\Lambda z} - 1}{2\Lambda}$$

with this choice of square root, then f_Λ is analytic on $\mathbb{C} \setminus (-\infty, -\frac{1}{4\Lambda}]$. In particular, its restriction to $\Omega_\Lambda := (-\frac{1}{4\Lambda}, \infty)$ is analytic.

Recall that the renormalization map $R_\Lambda : \mathbb{R} \rightarrow \mathbb{R}$ was chosen to be $R_\Lambda(g) := g + \Lambda g^2$ for $g \in \mathbb{R}$. We claim that $R_\Lambda^{-1}(\Omega_\Lambda) = \mathbb{R} \setminus \{-\frac{1}{2\Lambda}\}$. To see this, simply note that $R_\Lambda(x) \notin \Omega_\Lambda$ if and only if $x + \Lambda x^2 \leq -\frac{1}{4\Lambda}$, which is equivalent to $(2\Lambda x + 1)^2 \leq 0$. This happens if and only if $x = -\frac{1}{2\Lambda}$.

Finally, recall that the power series expansion of $u_\Lambda := f_\Lambda \circ R_\Lambda$ at the origin is simply the function $\tilde{u}_\Lambda(x) = x$, as shown by the calculations in Section 5.2. Thus, the domain $\mathcal{D}$ in Theorem 5.4.1 is the entire real line in this example. Therefore, the connected component of $R_\Lambda^{-1}(\Omega_\Lambda) \cap \mathcal{D}$ that contains the origin is the open interval $(-\frac{1}{2\Lambda}, \infty)$. Thus, Theorem 5.4.1 tells us that if $g > -\frac{1}{2\Lambda}$, then $f_\Lambda(R_\Lambda(g)) = g$. Note that this could also be determined by direct calculation in this example.

What about g 's that do not satisfy the above criterion, but for which $f_\Lambda(R_\Lambda(g))$ is well-defined since $R_\Lambda(g)$ is in the domain of f_Λ ? Note that this consists precisely of all g strictly less than $-\frac{1}{2\Lambda}$. Take any such g . Then $1 + 2\Lambda g < 0$, which means that

$$\sqrt{(1 + 2\Lambda g)^2} = -1 - 2\Lambda g.$$

Thus, if $g < -\frac{1}{2\Lambda}$, we get

$$f_\Lambda(R_\Lambda(g)) = \frac{\sqrt{(1 + 2\Lambda g)^2} - 1}{2\Lambda} = -\frac{1}{\Lambda} - g,$$

which is not equal to g , even asymptotically as $\Lambda \rightarrow \infty$.

Chapter 6

Heuristics for φ_4^4 Theory

6.1 Perturbative expansion of φ_4^4 theory

We saw in Section 4.8 that the free quantum field corresponds to noninteracting bosons. This is an ‘uninteresting’ quantum field theory, because particles do not interact, nor does it admit creation and annihilation of particles. The simplest toy model for an interacting quantum field theory is φ_4^4 theory. The action for φ_4^4 is obtained by adding a quartic term to the action for the free theory, as follows:

$$S(\phi) = \int_{\mathbb{R}^4} \left(\frac{1}{2} (\partial_t \phi(t, x))^2 - \frac{1}{2} \|\nabla \phi(t, x)\|^2 - \frac{m^2}{2} \phi(t, x)^2 - \frac{\lambda}{4!} \phi(t, x)^4 \right) dt dx.$$

(The subscript 4 in φ_4^4 stands for the dimension, which in this case is 4. The superscript denotes the quartic term in the action.) In this chapter, we will carry out the (heuristic) perturbative renormalization of φ_4^4 theory. Recall that by equation (2.3.3), the vacuum expectation value of a function $L(\phi)$ of this quantum field is given by

$$\langle \Omega | L(\phi) | \Omega \rangle = \frac{\int L(\phi) e^{iS(\phi)} \mathcal{D}\phi}{\int e^{iS(\phi)} \mathcal{D}\phi}.$$

For notational simplicity, we will denote the above quantity by $\langle L \rangle$. Our goal is to apply the perturbative renormalization heuristic to calculate $\langle L \rangle$ for various kinds of L .

First of all, note that by rescaling ϕ in the path integrals, we can add a parameter in front of the squared gradient terms in the action. To put things in the framework introduced in the previous chapter, let us write the resulting action as

$$S(\phi) = \int_{\mathbb{R}^4} \left(\frac{1}{2} (1 + \theta_1) [(\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2] - \frac{1}{2} (m^2 + \theta_2) \phi(t, x)^2 - \frac{\theta_3}{4!} \phi(t, x)^4 \right) dt dx,$$

where $\theta_1, \theta_2, \theta_3$ are parameters of the model, to be determined from data, and m is a known

constant. Next, let us decompose S as

$$S(\phi) = S_0(\phi) + \theta_1 S_1(\phi) + \theta_2 S_2(\phi) + \theta_3 S_3(\phi),$$

where

$$\begin{aligned} S_0(\phi) &:= \int_{\mathbb{R}^4} \left(\frac{1}{2} [(\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2] - \frac{1}{2} m^2 \phi(t, x)^2 \right) dt dx, \\ S_1(\phi) &:= \frac{1}{2} \int_{\mathbb{R}^4} [(\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2] dt dx, \\ S_2(\phi) &:= -\frac{1}{2} \int_{\mathbb{R}^4} \phi(t, x)^2 dt dx, \\ S_3(\phi) &:= -\frac{1}{4!} \int_{\mathbb{R}^4} \phi(t, x)^4 dt dx. \end{aligned}$$

Let $\langle \cdot \rangle_0$ denote the vacuum expectation value with respect to the action S_0 (i.e., the free field). Note that for any L , we can heuristically express $\langle L \rangle$ as

$$\langle L \rangle = \frac{\langle L(\varphi) e^{i\theta_1 S_1(\varphi) + i\theta_2 S_2(\varphi) + i\theta_3 S_3(\varphi)} \rangle_0}{\langle e^{i\theta_1 S_1(\varphi) + i\theta_2 S_2(\varphi) + i\theta_3 S_3(\varphi)} \rangle_0}.$$

The first step towards renormalization is to expand the above as a power series in $\theta_1, \theta_2, \theta_3$. Formally, we have

$$\langle L \rangle = \sum_{m \in \mathbb{Z}_{\geq 0}^3} a_m \theta^m, \quad (6.1.1)$$

where $\theta^m := \theta_1^{m_1} \theta_2^{m_2} \theta_3^{m_3}$, and

$$a_m = \frac{1}{m_1! m_2! m_3!} \frac{\partial^{m_1+m_2+m_3} \langle L \rangle}{\partial \theta_1^{m_1} \partial \theta_2^{m_2} \partial \theta_3^{m_3}} \Big|_{\theta_1=\theta_2=\theta_3=0}. \quad (6.1.2)$$

It is not hard to see that the above coefficients can be expressed as expectations with respect to the free field. To illustrate, let us express the coefficients up to order two. First, note that

$$a_{0,0,0} = \langle L \rangle|_{\theta_1=\theta_2=\theta_3=0} = \langle L \rangle_0.$$

Next, note that

$$a_{1,0,0} = \frac{\partial \langle L \rangle}{\partial \theta_1} \Big|_{\theta_1=\theta_2=\theta_3=0} = i(\langle L S_1 \rangle_0 - \langle L \rangle_0 \langle S_1 \rangle_0).$$

Similarly,

$$\begin{aligned} a_{0,1,0} &= i(\langle L S_2 \rangle_0 - \langle L \rangle_0 \langle S_2 \rangle_0), \\ a_{0,0,1} &= i(\langle L S_3 \rangle_0 - \langle L \rangle_0 \langle S_3 \rangle_0). \end{aligned}$$

Next, we have

$$\begin{aligned} a_{2,0,0} &= \frac{1}{2} \frac{\partial^2 \langle L \rangle}{\partial \theta_1^2} \Big|_{\theta_1=\theta_2=\theta_3=0} \\ &= -\frac{1}{2} (\langle LS_1^2 \rangle_0 - \langle L \rangle_0 \langle S_1^2 \rangle_0) + \langle LS_1 \rangle_0 \langle S_1 \rangle_0 - \langle L \rangle_0 \langle S_1 \rangle_0^2. \end{aligned}$$

Similarly,

$$\begin{aligned} a_{0,2,0} &= -\frac{1}{2} (\langle LS_2^2 \rangle_0 - \langle L \rangle_0 \langle S_2^2 \rangle_0) + \langle LS_2 \rangle_0 \langle S_2 \rangle_0 - \langle L \rangle_0 \langle S_2 \rangle_0^2, \\ a_{0,0,2} &= -\frac{1}{2} (\langle LS_3^2 \rangle_0 - \langle L \rangle_0 \langle S_3^2 \rangle_0) + \langle LS_3 \rangle_0 \langle S_3 \rangle_0 - \langle L \rangle_0 \langle S_3 \rangle_0^2. \end{aligned}$$

Finally, note that

$$\begin{aligned} a_{1,1,0} &= \frac{1}{2} \frac{\partial^2 \langle L \rangle}{\partial \theta_1 \partial \theta_2} \Big|_{\theta_1=\theta_2=\theta_3=0} \\ &= -\frac{1}{2} (\langle LS_1 S_2 \rangle_0 - \langle L \rangle_0 \langle S_1 S_2 \rangle_0) + \frac{1}{2} (\langle LS_1 \rangle_0 - \langle L \rangle_0 \langle S_1 \rangle_0) \langle S_2 \rangle_0 \\ &\quad + \frac{1}{2} (\langle LS_2 \rangle_0 - \langle L \rangle_0 \langle S_2 \rangle_0) \langle S_1 \rangle_0. \end{aligned}$$

The expressions for $a_{1,0,1}$ and $a_{0,1,1}$ are similar, obtained by replacing S_1, S_2 with S_1, S_3 and S_2, S_3 , respectively.

6.2 First order of perturbation theory

Let us now choose L to be the two-point correlation function in momentum coordinates. That is,

$$L(\phi) := \frac{i}{(2\pi)^4} \tilde{\phi}(p) \tilde{\phi}(q)$$

for some given $p, q \in \mathbb{R}^4$, where $\tilde{\phi}$ denotes the Minkowski space Fourier transform of ϕ . Now, by Corollary A.3.8,

$$S_1(\tilde{\phi}) = \frac{1}{2(2\pi)^4} \int_{\mathbb{R}^4} k^2 |\tilde{\phi}(k)|^2 dk,$$

assuming that $\phi \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$. Since ϕ is real-valued, we have $\overline{\tilde{\phi}(k)} = \tilde{\phi}(-k)$. Thus,

$$S_1(\tilde{\phi}) = \frac{1}{2(2\pi)^4} \int_{\mathbb{R}^4} k^2 \tilde{\phi}(k) \tilde{\phi}(-k) dk. \quad (6.2.1)$$

Similarly, by Theorem A.3.7,

$$S_2(\tilde{\phi}) = -\frac{1}{2(2\pi)^4} \int_{\mathbb{R}^4} \tilde{\phi}(k) \tilde{\phi}(-k) dk. \quad (6.2.2)$$

Lastly, by Theorem A.3.6,

$$\begin{aligned} S_3(\tilde{\phi}) &= -\frac{1}{4!(2\pi)^{16}} \int_{\mathbb{R}^4} \left(\int_{\mathbb{R}^4} e^{-i(x,k)} \tilde{\phi}(k) dk \right)^4 dx \\ &= -\frac{1}{4!(2\pi)^{16}} \iiint\limits_{\mathbb{R}^4} e^{-i(x,k+l+m+n)} \tilde{\phi}(k) \tilde{\phi}(l) \tilde{\phi}(m) \tilde{\phi}(n) dx dk dl dm dn. \end{aligned}$$

Applying the heuristic identity

$$\frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} e^{-i(x,k+l+m+n)} dx = \delta^{(4)}(k+l+m+n)$$

to the above integral (which is easy to justify rigorously when ϕ is a Schwartz function), we get

$$S_3(\tilde{\phi}) = -\frac{1}{4!(2\pi)^{12}} \iiint \tilde{\phi}(k) \tilde{\phi}(l) \tilde{\phi}(n) \tilde{\phi}(-k-l-n) dk dl dn. \quad (6.2.3)$$

Let us now recall the identity (3.10.1):

$$\langle \tilde{\varphi}(p) \tilde{\varphi}(q) \rangle_0 = -i(2\pi)^4 \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{-1}{q^2 - m^2 + i\epsilon}. \quad (6.2.4)$$

First, this implies that

$$a_{0,0,0} = \langle L \rangle_0 = -\delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{p^2 - m^2 + i\epsilon}.$$

Next, applying equation (6.2.1) and Corollary B.2.8, we get

$$\begin{aligned} a_{1,0,0} &= i(\langle LS_1 \rangle_0 - \langle L \rangle_0 \langle S_1 \rangle_0) \\ &= -\frac{1}{2(2\pi)^8} \int k^2 (\langle \tilde{\varphi}(p) \tilde{\varphi}(q) \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0 - \langle \tilde{\varphi}(p) \tilde{\varphi}(q) \rangle_0 \langle \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0) dk \\ &= -\frac{1}{2(2\pi)^8} \int k^2 (\langle \tilde{\varphi}(p) \tilde{\varphi}(k) \rangle_0 \langle \tilde{\varphi}(q) \tilde{\varphi}(-k) \rangle_0 + \langle \tilde{\varphi}(p) \tilde{\varphi}(-k) \rangle_0 \langle \tilde{\varphi}(q) \tilde{\varphi}(k) \rangle_0) dk. \end{aligned}$$

Applying equation (6.2.4), this gives

$$a_{1,0,0} = p^2 \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 - m^2 + i\epsilon)^2}.$$

A similar calculation using equation (6.2.2) gives

$$a_{0,1,0} = -\delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 - m^2 + i\epsilon)^2}.$$

Finally, using equation (6.2.3) and Corollary B.2.8, we get

$$\begin{aligned}
 a_{0,0,1} &= \frac{1}{4!(2\pi)^{16}} \iiint (\langle \tilde{\varphi}(p) \tilde{\varphi}(q) \tilde{\varphi}(k) \tilde{\varphi}(l) \tilde{\varphi}(n) \tilde{\varphi}(-k-l-n) \rangle_0 \\
 &\quad - \langle \tilde{\varphi}(p) \tilde{\varphi}(q) \rangle_0 \langle \tilde{\varphi}(k) \tilde{\varphi}(l) \tilde{\varphi}(n) \tilde{\varphi}(-k-l-n) \rangle_0) dk dl dn \\
 &= \frac{1}{4!(2\pi)^{16}} \iiint (\langle \tilde{\varphi}(p) \tilde{\varphi}(k) \rangle_0 \langle \tilde{\varphi}(q) \tilde{\varphi}(l) \rangle_0 \langle \tilde{\varphi}(n) \tilde{\varphi}(-k-l-n) \rangle_0 \\
 &\quad + 11 \text{ similar pairings}) dk dl dn,
 \end{aligned}$$

where the ‘similar pairings’ refer to p, q being paired to two of $k, l, n, -k-l-n$ (and the other two being paired to each other). By symmetry, the six integrals are equal. Thus,

$$\begin{aligned}
 a_{0,0,1} &= -\frac{i}{2(2\pi)^4} \lim_{\epsilon \downarrow 0} \iiint \frac{\delta^{(4)}(p+k) \delta^{(4)}(q-k) \delta^{(4)}(-k-l)}{(p^2 - m^2 + i\epsilon)(q^2 - m^2 + i\epsilon)(n^2 - m^2 + i\epsilon)} dk dl dn \\
 &= -\lim_{\epsilon \downarrow 0} \frac{i \delta^{(4)}(p+q)}{2(2\pi)^4 (p^2 - m^2 + i\epsilon)^2} \int \frac{1}{n^2 - m^2 + i\epsilon} dn. \tag{6.2.5}
 \end{aligned}$$

But note that the integral in the last line is divergent. This is an instance of an ‘infinity’ appearing in the perturbative expansion of φ_4^4 theory, which makes it necessary to apply renormalization. We will now work towards ‘canceling’ this infinity.

6.3 Wick rotation

The following simple Wick rotation argument is very useful for converting integrands from Minkowski to Euclidean signature.

Lemma 6.3.1. *Suppose that $f : \mathbb{C} \rightarrow \mathbb{C}$ is a holomorphic function which has no poles in the region $\{x + iy : xy \geq 0\}$. Furthermore, suppose that $|f(z)| \leq C|z|^{-1-\epsilon}$ in this region for some $C < \infty$ and $\epsilon > 0$. Then*

$$\int_{-\infty}^{\infty} dx f(x) = i \int_{-\infty}^{\infty} dx f(ix).$$

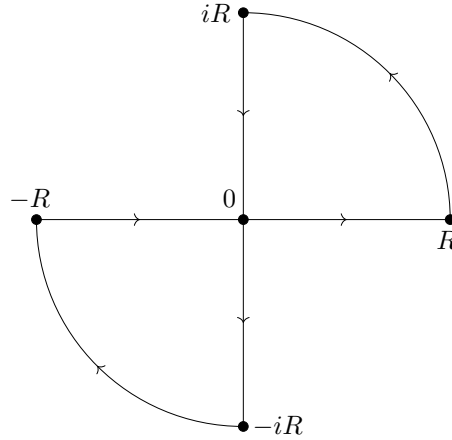
Proof. To see this, take some $R > 0$ and let γ denote the closed contour that goes from $-R$ to R in a straight line, then anticlockwise to iR in a circular arc, then to $-iR$ in a straight line, and finally to $-R$ clockwise in a circular arc (see Figure 6.1).

Note that f has no poles enclosed by this contour. The decay condition implies that the integral contribution on the circular arcs tends to 0 as $R \rightarrow \infty$. Thus, the result follows by Cauchy’s theorem and taking $R \rightarrow \infty$. $\square$

As a corollary, we obtain the following.

Corollary 6.3.2. *Take any $\alpha > 0$, $E \in \mathbb{R}$, and $\epsilon > 0$. Let p be a polynomial of degree $d \geq 0$ with complex coefficients. Let l be a positive integer such that $d - 2l < -1$. Then*

$$\int_{-\infty}^{\infty} \frac{p(x)}{(\alpha x^2 - E + i\epsilon)^l} dx = i(-1)^l \int_{-\infty}^{\infty} \frac{p(ix)}{(\alpha x^2 + E - i\epsilon)^l} dx.$$


 Figure 6.1: The contour γ in the proof of Lemma 6.3.1.

Consequently, if $E > 0$, then

$$\lim_{\epsilon \downarrow 0} \int_{-\infty}^{\infty} \frac{p(x)}{(\alpha x^2 - E + i\epsilon)^l} dx = i(-1)^l \int_{-\infty}^{\infty} \frac{p(ix)}{(\alpha x^2 + E)^l} dx.$$

Proof. Since $\epsilon > 0$, it is easy to see that the integrals are both well-defined and finite. Let

$$f(z) := \frac{p(z)}{(\alpha z^2 - E + i\epsilon)^l}.$$

Then there is some constant C (depending on p , E , α and ϵ) such that $|f(z)| \leq C|z|^{d-2l}$ for all $|z|$. Moreover, observe that for $z = x + iy$,

$$\alpha z^2 = \alpha(x^2 - y^2 + 2ixy) = \alpha(x^2 - y^2) + 2i\alpha xy.$$

So if $xy \geq 0$, then $\text{Im}(\alpha z^2) \geq 0$, and therefore $\alpha z^2 - E + i\epsilon \neq 0$. Thus we can apply Wick rotation (Lemma 6.3.1) and obtain the desired result. The second assertion follows by a straightforward application of the dominated convergence theorem. $\square$

6.4 Regularization by hard cutoff

Recall from Section 6.2 that we ran into trouble while evaluating the coefficient $a_{0,0,1}$, which had the divergent integral

$$\int_{\mathbb{R}^4} \frac{1}{k^2 - m^2 + i\epsilon} dk$$

appearing in the formula (6.2.5). Before trying to give a meaning to this integral, let us convert it into something more manageable via Wick rotation. Fixing $k_1, k_2, k_3 \in \mathbb{R}$, note

that by Corollary 6.3.2,

$$\lim_{\epsilon \downarrow 0} \int_{\mathbb{R}} \frac{1}{k^2 - m^2 + i\epsilon} dk_0 = -i \int_{\mathbb{R}} \frac{1}{\|k\|^2 + m^2} dk_0,$$

where k denotes the vector (k_0, k_1, k_2, k_3) . Substituting this in equation (6.2.5), we get the formal identity

$$a_{0,0,1} = -\frac{1}{2(2\pi)^4} \left(\int_{\mathbb{R}^4} \frac{1}{\|k\|^2 + m^2} dk \right) \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 - m^2 + i\epsilon)^2}.$$

The limit on the right is to be interpreted as a distribution. However, the integral is still divergent. To ‘tame’ this infinity, we choose a cutoff parameter Λ , and define

$$a_{\Lambda,0,0,1} := -\frac{1}{2(2\pi)^4} \left(\int_{\|k\| \leq \Lambda} \frac{1}{\|k\|^2 + m^2} dk \right) \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 - m^2 + i\epsilon)^2}.$$

Since $a_{0,0,0}$, $a_{1,0,0}$, and $a_{0,1,0}$ do not have the problem of infinities, we define

$$a_{\Lambda,0,0,0} := a_{0,0,0}, \quad a_{\Lambda,1,0,0} := a_{1,0,0}, \quad a_{\Lambda,0,1,0} := a_{0,1,0}.$$

Now, the integral in $a_{\Lambda,0,0,1}$ is finite, and evaluates to

$$\begin{aligned} 2\pi^2 \int_0^\Lambda \frac{r^3}{r^2 + m^2} dr &= 2\pi^2 \int_0^\Lambda r dr - 2\pi^2 m^2 \int_0^\Lambda \frac{r}{r^2 + m^2} dr \\ &= \pi^2 \Lambda^2 - \pi^2 m^2 \log \left(1 + \frac{\Lambda^2}{m^2} \right). \end{aligned}$$

Thus,

$$a_{\Lambda,0,0,1} = -\frac{1}{32\pi^2} \left(\Lambda^2 - m^2 \log \left(1 + \frac{\Lambda^2}{m^2} \right) \right) \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 - m^2 + i\epsilon)^2}. \quad (6.4.1)$$

6.5 Renormalizing to first order

Following the method described in the previous chapter, we reparametrize

$$\theta_j = \sum_{r=1}^{\infty} c_{\Lambda,j,r} g^r, \quad j = 1, 2, 3.$$

Our goal is to choose the c ’s such that after this substitution, the coefficients in the perturbation series of $\langle L \rangle$ (which is now a series in g of the form $\sum_{n=0}^{\infty} d_n g^n$) do not depend on the cutoff parameter Λ . We will now ensure that this true for the constant term and the coefficient of g . We already know that $d_0 = a_{\Lambda,0,0,0}$ has no dependence on Λ . Next, note

that

$$\begin{aligned} d_1 &= a_{\Lambda,1,0,0}c_{\Lambda,1,1} + a_{\Lambda,0,1,0}c_{\Lambda,2,1} + a_{\Lambda,0,0,1}c_{\Lambda,3,1} \\ &= \left[p^2 c_{\Lambda,1,1} - c_{\Lambda,2,1} - \frac{1}{32\pi^2} \left(\Lambda^2 - m^2 \log \left(1 + \frac{\Lambda^2}{m^2} \right) \right) c_{\Lambda,3,1} \right] \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 - m^2 + i\epsilon)^2}. \end{aligned}$$

Choosing $c_{\Lambda,1,1} = 0$, $c_{\Lambda,3,1} = 1$, and

$$c_{\Lambda,2,1} = -\frac{1}{32\pi^2} \left(\Lambda^2 - m^2 \log \left(1 + \frac{\Lambda^2}{m^2} \right) \right),$$

we get $d_1 = 0$.

To summarize, we have heuristically shown the following. Suppose we restrict attention to all fields ϕ such that $\hat{\phi}(p) = 0$ whenever $\|p\| > \Lambda$. One can argue that this is the ‘meaning’ of the cutoff Λ that we imposed in our calculation of $a_{\Lambda,0,0,1}$. Having done that, suppose we consider the model with action

$$\begin{aligned} S(\phi) &= \int_{\mathbb{R}^4} \left(\frac{1}{2} [(\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2] \right. \\ &\quad \left. - \frac{1}{2} (m^2 + c_{\Lambda} g + O(g^2)) \phi(t, x)^2 - \frac{g + O(g^2)}{4!} \phi(t, x)^4 \right) dt dx, \end{aligned} \quad (6.5.1)$$

where g is a small parameter and

$$c_{\Lambda} = -\frac{1}{32\pi^2} \left(\Lambda^2 - m^2 \log \left(1 + \frac{\Lambda^2}{m^2} \right) \right).$$

Then the two-point correlation function for this theory, to first order of perturbation theory, is given by

$$\langle \tilde{\varphi}(p) \tilde{\varphi}(q) \rangle = -i(2\pi)^4 \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{q^2 - m^2 + i\epsilon} + O(g^2).$$

6.6 Renormalized Wightman functions

In the previous section, we saw that with a specific choice of coefficients, the first order term in the perturbative expansion of the two-point function is no longer dependent on the cutoff parameter. But this is just the two-point function; we cannot claim to have successfully carried out renormalization to first order unless this holds for a very large class of observables. Ideally, we would like for it to hold for n -point functions for all n . Fortunately, this works out. We will now carry out this computation.

Take any positive integer n . If n is odd, then by the fact that $S(\phi) = S(-\phi)$, it follows that any n -point correlation is zero. So, let us assume that n is even. Since we already know the answer for $n = 2$, let us take $n \geq 4$. Take any $p_1, \dots, p_n \in \mathbb{R}^4$, and let

$$L(\phi) := i^{n/2} (2\pi)^{-n} \tilde{\phi}(p_1) \cdots \tilde{\phi}(p_n).$$

Let $a_m, m \in \mathbb{Z}_{\geq 0}^3$ denote the coefficients in the Taylor series expansion of $\langle L \rangle$, as before. Then by equation (3.11.3),

$$a_{0,0,0} = \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} \delta^{(4)}(p_i + p_j) \lim_{\epsilon \downarrow 0} \frac{-1}{p_i^2 - m^2 + i\epsilon},$$

where $\mathcal{P}_n$ is the set of perfect matchings of $1, \dots, n$.

By a similar calculation as for the two-point function,

$$\begin{aligned} a_{0,1,0} &= i(\langle LS_2 \rangle_0 - \langle L \rangle_0 \langle S_2 \rangle_0) \\ &= -\frac{i^{n/2+1}}{2(2\pi)^{2n+4}} \int (\langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0 - \langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \rangle_0 \langle \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0) dk. \end{aligned}$$

To evaluate the integrand, we use Corollary B.2.8 as before. We have to consider all perfect matchings of $p_1, \dots, p_n, k, -k$ such that k and $-k$ are not paired to each other. For each distinct $i, j \in \{1, \dots, n\}$, let $\mathcal{P}_{n,i,j}$ be the set of all perfect matchings of $\{1, \dots, n\} \setminus \{i, j\}$. Then

$$\begin{aligned} &\frac{i^{n/2+1}}{(2\pi)^{2n+4}} \langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0 - \langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \rangle_0 \langle \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0 \\ &= \frac{i^{n/2+1}}{(2\pi)^{2n+4}} \sum_{1 \leq i \neq j \leq n} \sum_{E \in \mathcal{P}_{n,i,j}} \langle \tilde{\varphi}(p_i) \tilde{\varphi}(k) \rangle_0 \langle \tilde{\varphi}(p_j) \tilde{\varphi}(-k) \rangle_0 \prod_{\{a,b\} \in E} \langle \tilde{\varphi}(p_a) \tilde{\varphi}(p_b) \rangle_0 \\ &= \sum_{1 \leq i \neq j \leq n} \sum_{E \in \mathcal{P}_{n,i,j}} \left(\delta^{(4)}(p_i + k) \lim_{\epsilon \downarrow 0} \frac{-1}{p_i^2 - m^2 + i\epsilon} \right) \left(\delta^{(4)}(p_j - k) \lim_{\epsilon \downarrow 0} \frac{-1}{p_j^2 - m^2 + i\epsilon} \right) \\ &\quad \cdot \prod_{\{a,b\} \in E} \left(\delta^{(4)}(p_a + p_b) \lim_{\epsilon \downarrow 0} \frac{-1}{p_a^2 - m^2 + i\epsilon} \right). \end{aligned}$$

Integrating this expression over k gives

$$\begin{aligned} a_{0,1,0} &= \frac{1}{2} \sum_{1 \leq i \neq j \leq n} \sum_{E \in \mathcal{P}_{n,i,j}} \left(\delta^{(4)}(p_i + p_j) \lim_{\epsilon \downarrow 0} \frac{1}{(p_i^2 - m^2 + i\epsilon)^2} \right) \\ &\quad \cdot \prod_{\{a,b\} \in E} \left(\delta^{(4)}(p_a + p_b) \lim_{\epsilon \downarrow 0} \frac{-1}{p_a^2 - m^2 + i\epsilon} \right). \end{aligned}$$

Similarly, we have

$$\begin{aligned} a_{0,0,1} &= i(\langle LS_3 \rangle_0 - \langle L \rangle_0 \langle S_3 \rangle_0) \\ &= -\frac{i^{n/2+1}}{4!(2\pi)^{2n+12}} \iiint (\langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \tilde{\varphi}(k) \tilde{\varphi}(l) \tilde{\varphi}(r) \tilde{\varphi}(-k-l-r) \rangle_0 \\ &\quad - \langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \rangle_0 \langle \tilde{\varphi}(k) \tilde{\varphi}(l) \tilde{\varphi}(r) \tilde{\varphi}(-k-l-r) \rangle_0) dk dl dr. \end{aligned}$$

When considering pairings for evaluating the integrand, we have to consider two types of pairings. First, when two of $k, l, r, -k-l-r$ are paired to two of p 's, and the two others

are matched to each other. Second, when all four of $k, l, r, -k - l - r$ are paired to four different p 's.

The total contribution from the first set of pairings comes to

$$\sum_{1 \leq i \neq j \leq n} \sum_{E \in \mathcal{P}_{n,i,j}} \langle \tilde{\varphi}(p_i) \tilde{\varphi}(k) \rangle_0 \langle \tilde{\varphi}(p_j) \tilde{\varphi}(l) \rangle_0 \langle \tilde{\varphi}(r) \tilde{\varphi}(-k - l - r) \rangle_0 \prod_{\{a,b\} \in E} \langle \tilde{\varphi}(p_a) \tilde{\varphi}(p_b) \rangle_0$$

+ 11 similar terms.

Multiplying this by $i^{n/2+2}(2\pi)^{-2n-8}$ and integrating over k, l, r , equation (3.11.3) gives

$$12 \lim_{\epsilon \downarrow 0} \sum_{1 \leq i \neq j \leq n} \sum_{E \in \mathcal{P}_{n,i,j}} \left(\prod_{\{a,b\} \in E} \frac{-\delta^{(4)}(p_a + p_b)}{p_a^2 - m^2 + i\epsilon} \right) \frac{\delta^{(4)}(p_i + p_j)}{(p_i^2 - m^2 + i\epsilon)^2} \int \frac{1}{r^2 - m^2 + i\epsilon} dr.$$

To tame the infinity arising from the integral on the right, we first Wick rotate and then introduce a hard cutoff as before, by replacing the above integral with

$$i \int_{\|r\| \leq \Lambda} \frac{1}{\|r\|^2 + m^2 - i\epsilon} dr.$$

For distinct $i, j, i', j' \in \{1, \dots, n\}$, let $\mathcal{P}_{n,i,j,i',j'}$ denote the set of all perfect matchings of $\{1, \dots, n\} \setminus \{i, j, i', j'\}$. Then, the total contribution from the second set of pairings amounts to

$$\sum_{\substack{1 \leq i,j,i',j' \leq n, \\ i,j,i',j' \text{ distinct}}} \sum_{E \in \mathcal{P}_{n,i,j,i',j'}} \langle \tilde{\varphi}(p_i) \tilde{\varphi}(k) \rangle_0 \langle \tilde{\varphi}(p_j) \tilde{\varphi}(l) \rangle_0$$

$$\cdot \langle \tilde{\varphi}(p_{i'}) \tilde{\varphi}(p_r) \rangle_0 \langle \tilde{\varphi}(p_{j'}) \tilde{\varphi}(-k - l - r) \rangle_0 \prod_{\{a,b\} \in E} \langle \tilde{\varphi}(p_a) \tilde{\varphi}(p_b) \rangle_0$$

When this is multiplied by $i^{n/2+2}(2\pi)^{-2n-8}$ and integrated over k, l, r , equation (3.11.3) gives

$$\lim_{\epsilon \downarrow 0} \sum_{\substack{1 \leq i,j,i',j' \leq n, \\ i,j,i',j' \text{ distinct}}} \sum_{E \in \mathcal{P}_{n,i,j,i',j'}} \left(\frac{\delta^{(4)}(p_i + p_j + p_{i'} + p_{j'})}{(p_i^2 - m^2 + i\epsilon)(p_j^2 - m^2 + i\epsilon)(p_{i'}^2 - m^2 + i\epsilon)(p_{j'}^2 - m^2 + i\epsilon)} \right)$$

$$\cdot \prod_{\{a,b\} \in E} \frac{-\delta^{(4)}(p_a + p_b)}{p_a^2 - m^2 + i\epsilon}.$$

If $a_{\Lambda,0,0,1}$ denotes the coefficient obtained from the above calculations (after the hard cutoff), then the coefficient of g in the expansion of $\langle L \rangle$ is equal to $a_{0,1,0}c_\Lambda + a_{\Lambda,0,0,1}$, since we are working with the same c 's as before. An easy verification shows that by our choice of c_Λ , the contribution from the first set of pairings in the calculation of $a_{\Lambda,0,0,1}$ exactly cancels our $a_{0,1,0}c_\Lambda$. Thus, we are left with the contributions from the second set of pairings, which have no infinities.

To summarize, we get the following result. With the renormalized action displayed in

equation (6.5.1), the momentum space Wightman function has the first order expansion

$$\begin{aligned}
 \widetilde{W}(p_1, \dots, p_n) &= \frac{i^{n/2}}{(2\pi)^{2n}} \langle \widetilde{\varphi}(p_1) \cdots \widetilde{\varphi}(p_n) \rangle \\
 &= \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} \delta^{(4)}(p_i + p_j) \lim_{\epsilon \downarrow 0} \frac{-1}{p_i^2 - m^2 + i\epsilon} \\
 &\quad + \frac{ig}{(2\pi)^4} \lim_{\epsilon \downarrow 0} \sum_{1 \leq i < j < i' < j' \leq n} \sum_{E \in \mathcal{P}_{n,i,j,i',j'}} \left(\prod_{\{a,b\} \in E} \frac{-\delta^{(4)}(p_a + p_b)}{p_a^2 - m^2 + i\epsilon} \right) \\
 &\quad \cdot \left(\frac{\delta^{(4)}(p_i + p_j + p_{i'} + p_{j'})}{(p_i^2 - m^2 + i\epsilon)(p_j^2 - m^2 + i\epsilon)(p_{i'}^2 - m^2 + i\epsilon)(p_{j'}^2 - m^2 + i\epsilon)} \right) \\
 &\quad + O(g^2).
 \end{aligned}$$

Thus, our renormalization scheme successfully renormalizes (to first order) the n -point correlation functions for all n .

Chapter 7

Heuristics for φ_3^4 Theory

7.1 Perturbative expansion of φ_3^4 theory

We will now investigate the analogue of φ_4^4 when the dimension of space is two, instead of three. This is known as φ_3^4 theory. It is an example of a ‘super-renormalizable’ theory — a term that will be explained later in the chapter. The action is now

$$S(\phi) = \int_{\mathbb{R}^3} \left(\frac{1}{2} (\partial_t \phi(t, x))^2 - \frac{1}{2} \|\nabla \phi(t, x)\|^2 - \frac{m^2}{2} \phi(t, x)^2 - \frac{\lambda}{4!} \phi(t, x)^4 \right) dt dx,$$

where now x is integrated over $\mathbb{R}^2$ and ∇ denotes the spatial gradient in $\mathbb{R}^2$. As in Chapter 6, we work formally with vacuum expectation values

$$\langle L \rangle = \frac{\int L(\phi) e^{iS(\phi)} d\phi}{\int e^{iS(\phi)} d\phi},$$

and we denote by $\langle \cdot \rangle_0$ the corresponding expectation for the free action. Exactly as before, by rescaling ϕ in the path integrals we may insert parameters in front of the kinetic, mass, and quartic terms. Write

$$S(\phi) = \int_{\mathbb{R}^3} \left(\frac{1}{2} (1 + \theta_1) [(\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2] - \frac{1}{2} (m^2 + \theta_2) \phi(t, x)^2 - \frac{\theta_3}{4!} \phi(t, x)^4 \right) dt dx.$$

Decompose

$$S(\phi) = S_0(\phi) + \theta_1 S_1(\phi) + \theta_2 S_2(\phi) + \theta_3 S_3(\phi),$$

where

$$\begin{aligned} S_0(\phi) &:= \int_{\mathbb{R}^3} \left(\frac{1}{2} [(\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2] - \frac{m^2}{2} \phi(t, x)^2 \right) dt dx, \\ S_1(\phi) &:= \frac{1}{2} \int_{\mathbb{R}^3} [(\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2] dt dx, \\ S_2(\phi) &:= -\frac{1}{2} \int_{\mathbb{R}^3} \phi(t, x)^2 dt dx, \\ S_3(\phi) &:= -\frac{1}{4!} \int_{\mathbb{R}^3} \phi(t, x)^4 dt dx. \end{aligned}$$

Then, for any observable L ,

$$\langle L \rangle = \frac{\langle L(\varphi) e^{i\theta_1 S_1(\varphi) + i\theta_2 S_2(\varphi) + i\theta_3 S_3(\varphi)} \rangle_0}{\langle e^{i\theta_1 S_1(\varphi) + i\theta_2 S_2(\varphi) + i\theta_3 S_3(\varphi)} \rangle_0}. \quad (7.1.1)$$

Expanding formally in powers of $\theta_1, \theta_2, \theta_3$ gives

$$\langle L \rangle = \sum_{m \in \mathbb{Z}_{\geq 0}^3} a_m \theta^m, \quad \theta^m := \theta_1^{m_1} \theta_2^{m_2} \theta_3^{m_3},$$

with

$$a_m = \frac{1}{m_1! m_2! m_3!} \frac{\partial^{m_1+m_2+m_3} \langle L \rangle}{\partial \theta_1^{m_1} \partial \theta_2^{m_2} \partial \theta_3^{m_3}} \Big|_{\theta_1=\theta_2=\theta_3=0}.$$

As in the φ_4^4 case, these coefficients can be expressed in terms of free-field correlators. For example,

$$\begin{aligned} a_{0,0,0} &= \langle L \rangle_0, \\ a_{1,0,0} &= i(\langle L S_1 \rangle_0 - \langle L \rangle_0 \langle S_1 \rangle_0), \\ a_{0,1,0} &= i(\langle L S_2 \rangle_0 - \langle L \rangle_0 \langle S_2 \rangle_0), \\ a_{0,0,1} &= i(\langle L S_3 \rangle_0 - \langle L \rangle_0 \langle S_3 \rangle_0), \end{aligned}$$

and similarly for the second order coefficients.

7.2 First order of perturbation theory

We now take L to be the two-point function in momentum coordinates. Let

$$L(\phi) := \frac{i}{(2\pi)^3} \tilde{\phi}(p) \tilde{\phi}(q), \quad p, q \in \mathbb{R}^3,$$

where $\tilde{\phi}$ denotes the Minkowski Fourier transform on $\mathbb{R}^3$. As in Chapter 6, we express S_1, S_2, S_3 in momentum variables. For a real field ϕ we have $\overline{\tilde{\phi}(k)} = \tilde{\phi}(-k)$. Thus,

$$S_1(\tilde{\phi}) = \frac{1}{2(2\pi)^3} \int_{\mathbb{R}^3} k^2 \tilde{\phi}(k) \tilde{\phi}(-k) dk, \quad (7.2.1)$$

where k^2 now denotes the Minkowski norm in $\mathbb{R}^{1,2}$:

$$k^2 := k_0^2 - k_1^2 - k_2^2.$$

Similarly

$$S_2(\tilde{\phi}) = -\frac{1}{2(2\pi)^3} \int_{\mathbb{R}^3} \tilde{\phi}(k) \tilde{\phi}(-k) dk. \quad (7.2.2)$$

For the quartic term, we write

$$\begin{aligned} S_3(\phi) &= -\frac{1}{4!(2\pi)^{12}} \int_{\mathbb{R}^3} \left(\int_{\mathbb{R}^3} e^{-i(x,k)} \tilde{\phi}(k) dk \right)^4 dx \\ &= -\frac{1}{4!(2\pi)^{12}} \iiint e^{-i(x,k+l+m+n)} \tilde{\phi}(k) \tilde{\phi}(l) \tilde{\phi}(m) \tilde{\phi}(n) dx dk dl dm dn. \end{aligned}$$

Using

$$\frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{-i(x,k+l+m+n)} dx = \delta^{(3)}(k+l+m+n),$$

we obtain

$$S_3(\phi) = -\frac{1}{4!(2\pi)^9} \iiint \tilde{\phi}(k) \tilde{\phi}(l) \tilde{\phi}(n) \tilde{\phi}(-k-l-n) dk dl dn. \quad (7.2.3)$$

It is not hard to show that the free two-point function in momentum space is the 3D analogue of the 4D free two-point function:

$$\langle \tilde{\varphi}(p) \tilde{\varphi}(q) \rangle_0 = -i(2\pi)^3 \delta^{(3)}(p+q) \lim_{\epsilon \downarrow 0} \frac{-1}{q^2 - m^2 + i\epsilon}. \quad (7.2.4)$$

Hence

$$a_{0,0,0} = \langle L \rangle_0 = -\delta^{(3)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{p^2 - m^2 + i\epsilon}.$$

Next, using (7.2.1) and Wick's rule exactly as in Chapter 6,

$$\begin{aligned} a_{1,0,0} &= i(\langle LS_1 \rangle_0 - \langle L \rangle_0 \langle S_1 \rangle_0) \\ &= -\frac{1}{2(2\pi)^6} \int_{\mathbb{R}^3} k^2 \left(\langle \tilde{\varphi}(p) \tilde{\varphi}(k) \rangle_0 \langle \tilde{\varphi}(q) \tilde{\varphi}(-k) \rangle_0 + \langle \tilde{\varphi}(p) \tilde{\varphi}(-k) \rangle_0 \langle \tilde{\varphi}(q) \tilde{\varphi}(k) \rangle_0 \right) dk, \end{aligned}$$

which, upon using (7.2.4), yields

$$a_{1,0,0} = p^2 \delta^{(3)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 - m^2 + i\epsilon)^2}.$$

Similarly, from (7.2.2),

$$a_{0,1,0} = -\delta^{(3)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 - m^2 + i\epsilon)^2}.$$

Finally, using (7.2.3) and Wick's rule, one finds (with the same combinatorics as before: 12 pairings contribute equally) that

$$a_{0,0,1} = -\lim_{\epsilon \downarrow 0} \frac{i \delta^{(3)}(p+q)}{2(2\pi)^3(p^2 - m^2 + i\epsilon)^2} \int_{\mathbb{R}^3} \frac{1}{n^2 - m^2 + i\epsilon} dn. \quad (7.2.5)$$

The integral diverges. Thus, even in φ_3^4 theory, perturbation theory produces an infinity at first order in the two-point function (a “tadpole” divergence). We now tame this divergence by Wick rotation and a hard ultraviolet cutoff.

7.3 Regularization by hard cutoff

In (7.2.5) the problematic factor is

$$\int_{\mathbb{R}^3} \frac{1}{n^2 - m^2 + i\epsilon} dn.$$

We Wick rotate in the time component. Write $n = (n_0, n_1, n_2) \in \mathbb{R}^3$ and fix n_1, n_2 . Applying Corollary 6.3.2 with $p \equiv 1$, $l = 1$, and $E = n_1^2 + n_2^2 + m^2$ gives

$$\lim_{\epsilon \downarrow 0} \int_{\mathbb{R}} \frac{1}{n^2 - m^2 + i\epsilon} dn_0 = -i \int_{\mathbb{R}^2} \frac{1}{\|n\|^2 + m^2} dn_0,$$

where now $\|n\|^2 := n_0^2 + n_1^2 + n_2^2$ is the Euclidean square. Substituting into (7.2.5), we obtain the formal identity

$$a_{0,0,1} = -\frac{1}{2(2\pi)^3} \left(\int_{\mathbb{R}^3} \frac{1}{\|n\|^2 + m^2} dn \right) \lim_{\epsilon \downarrow 0} \frac{\delta^{(3)}(p+q)}{(p^2 - m^2 + i\epsilon)^2}.$$

The Euclidean integral still diverges, so we introduce a hard cutoff Λ and define

$$a_{\Lambda,0,0,1} := -\frac{1}{2(2\pi)^3} \left(\int_{\|n\| \leq \Lambda} \frac{1}{\|n\|^2 + m^2} dn \right) \lim_{\epsilon \downarrow 0} \frac{\delta^{(3)}(p+q)}{(p^2 - m^2 + i\epsilon)^2},$$

and we set $a_{\Lambda,0,0,0} := a_{0,0,0}$, $a_{\Lambda,1,0,0} := a_{1,0,0}$, $a_{\Lambda,0,1,0} := a_{0,1,0}$. The cutoff integral is now finite and can be evaluated explicitly:

$$\begin{aligned} \int_{\|n\| \leq \Lambda} \frac{1}{\|n\|^2 + m^2} dn &= 4\pi \int_0^\Lambda \frac{r^2}{r^2 + m^2} dr \\ &= 4\pi \int_0^\Lambda \left(1 - \frac{m^2}{r^2 + m^2} \right) dr \\ &= 4\pi(\Lambda - m \arctan(\Lambda/m)). \end{aligned}$$

Therefore

$$a_{\Lambda,0,0,1} = -\frac{1}{4\pi^2} (\Lambda - m \arctan(\Lambda/m)) \lim_{\epsilon \downarrow 0} \frac{\delta^{(3)}(p+q)}{(p^2 - m^2 + i\epsilon)^2}. \quad (7.3.1)$$

This is the first time we see something different than in dimension four: notice that the above formula is quite different than the formula (6.4.1).

7.4 Renormalizing the two-point function to all terms

As before, we reparametrize

$$\theta_j = \sum_{r=1}^{\infty} c_{\Lambda,j,r} g^r, \quad j = 1, 2, 3,$$

and we choose the coefficients so that the resulting g -expansion $\langle L \rangle = \sum_{n=0}^{\infty} d_n g^n$ has coefficients independent of Λ . The constant term $d_0 = a_{\Lambda,0,0,0}$ has no cutoff dependence. The coefficient of g is

$$\begin{aligned} d_1 &= a_{\Lambda,1,0,0} c_{\Lambda,1,1} + a_{\Lambda,0,1,0} c_{\Lambda,2,1} + a_{\Lambda,0,0,1} c_{\Lambda,3,1} \\ &= \left[p^2 c_{\Lambda,1,1} - c_{\Lambda,2,1} - \frac{1}{4\pi^2} (\Lambda - m \arctan(\Lambda/m)) c_{\Lambda,3,1} \right] \lim_{\epsilon \downarrow 0} \frac{\delta^{(3)}(p+q)}{(p^2 - m^2 + i\epsilon)^2}, \end{aligned}$$

where we used (7.3.1). Choosing

$$c_{\Lambda,1,1} = 0, \quad c_{\Lambda,3,1} = 1, \quad c_{\Lambda,2,1} = -\frac{1}{4\pi^2} (\Lambda - m \arctan(\Lambda/m)),$$

we obtain $d_1 = 0$. In other words, if we impose an ultraviolet cutoff by restricting to fields ϕ with $\tilde{\phi}(p) = 0$ when $\|p\| > \Lambda$, and consider the action

$$\begin{aligned} S(\phi) &= \int_{\mathbb{R}^3} \left(\frac{1}{2} [(\partial_t \phi(t, x))^2 - \|\nabla \phi(t, x)\|^2] \right. \\ &\quad \left. - \frac{1}{2} (m^2 + c_{\Lambda} g + O(g^2)) \phi(t, x)^2 - \frac{g + O(g^2)}{4!} \phi(t, x)^4 \right) dt dx, \end{aligned}$$

with

$$c_{\Lambda} = -\frac{1}{4\pi^2} (\Lambda - m \arctan(\Lambda/m)),$$

then the first-order correction to the two-point function is cutoff-independent:

$$\langle \tilde{\varphi}(p) \tilde{\varphi}(q) \rangle = -i(2\pi)^3 \delta^{(3)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{q^2 - m^2 + i\epsilon} + O(g^2).$$

This completes renormalization to first order in φ_3^4 theory. However, unlike in Chapter 6, we will not stop at this. We will now continue defining the c 's such that $d_n = 0$ for every $n \geq 1$. Right away, we declare that $c_{\Lambda,1,r} = 0$ for all $r \geq 1$, $c_{\Lambda,3,1} = 1$, and $c_{\Lambda,3,r} = 0$ for all $r \geq 2$. In other words, $\theta_1 = 0$ and $\theta_3 = g$. We have also determined $c_{\Lambda,2,1}$ above. Thus, it only remains to determine $c_{\Lambda,2,r}$ for $r \geq 2$. We will do this by induction, ensuring that $d_n = 0$ for all $n \geq 2$. First, we have to understand what $a_{\Lambda,m}$ looks like.

Lemma 7.4.1. *For any multi-index $m \neq (0, 0, 0)$,*

$$a_{\Lambda, m} = \alpha_{\Lambda, m} \lim_{\epsilon \downarrow 0} \frac{\delta^{(3)}(p + q)}{(p^2 - m^2 + i\epsilon)^2}$$

for some complex coefficient $\alpha_{\Lambda, m}$.

Proof. First, we claim that for any m , the mixed partial derivative

$$\frac{\partial^{m_1+m_2+m_3} \langle L \rangle}{\partial \theta_1^{m_1} \partial \theta_2^{m_2} \partial \theta_3^{m_3}}$$

is a linear combination of terms like

$$\langle LS_1^{a_0} S_2^{b_0} S_3^{c_0} \rangle \prod_{j=1}^l \langle S_1^{a_j} S_2^{b_j} S_3^{c_j} \rangle \quad (7.4.1)$$

where l is arbitrary, the a, b, c 's are nonnegative integers, and $a_0 + \dots + a_l = m_1$, $b_0 + \dots + b_l = m_2$, and $c_0 + \dots + c_l = m_3$. This is easily seen by induction on $m_1 + m_2 + m_3$; for example, the derivative of any term as above with respect to θ_1 is

$$\begin{aligned} & (i \langle LS_1^{a_0+1} S_2^{b_0} S_3^{c_0} \rangle - i \langle LS_1^{a_0} S_2^{b_0} S_3^{c_0} \rangle \langle S_1 \rangle) \prod_{j=1}^l \langle S_1^{a_j} S_2^{b_j} S_3^{c_j} \rangle \\ & + \sum_{q=1}^l \langle LS_1^{a_0} S_2^{b_0} S_3^{c_0} \rangle \left(\prod_{\substack{1 \leq j \leq l \\ j \neq q}} \langle S_1^{a_j} S_2^{b_j} S_3^{c_j} \rangle \right) (i \langle S_1^{a_q+1} S_2^{b_q} S_3^{c_q} \rangle - i \langle S_1^{a_q} S_2^{b_q} S_3^{c_q} \rangle \langle S_1 \rangle). \end{aligned}$$

Expanding the above gives a linear combination of terms of the form (7.4.1), but with the a 's adding up to $m_1 + 1$. This proves the first claim. $\square$

Lemma 7.4.2. *After setting $\theta_1 = 0$ and $\theta_3 = g$, it is possible to choose $c_{\Lambda, 2, r}$, $r \geq 1$, such that $d_n = 0$ for all $n \geq 1$.*

Proof. For each multi-index $m = (m_1, m_2, m_3)$, the coefficient $a_{\Lambda, m}$ is a finite linear combination of expressions of the form

$$\begin{aligned} & \langle \tilde{\varphi}(p) \tilde{\varphi}(q) \cdot (\text{a product of } m_1 \text{ factors from } S_1) \\ & \cdot (\text{a product of } m_2 \text{ factors from } S_2) \cdot (\text{a product of } m_3 \text{ factors from } S_3) \rangle_0, \end{aligned}$$

and similar subtraction terms coming from the denominator. Each factor of S_2 contributes two fields φ integrated over x (equivalently, a momentum-conserving delta that forces equality of two momenta); each factor of S_3 contributes four fields integrated over x (equivalently, a delta enforcing that the sum of the four momenta is 0). After applying Wick's theorem to the resulting Gaussian expectation, every term becomes a product of free propagators

$$\langle \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0 = (2\pi)^3 C_{\Lambda}(k),$$

together with delta functions enforcing momentum conservation at each integrated vertex.

Because $\theta_1 = 0$, $\theta_2 = \sum_{r=1}^{\infty} c_{\Lambda,2,r} g^r$, and $\theta_3 = g$, we have

$$\begin{aligned} \langle L \rangle &= \sum_{m_2, m_3 \geq 0} a_{\Lambda,0,m_2,m_3} \theta_2^{m_2} \theta_3^{m_3} \\ &= \sum_{m_2, m_3 \geq 0} a_{\Lambda,0,m_2,m_3} \left(\sum_{r=1}^{\infty} c_{\Lambda,2,r} g^r \right)^{m_2} g^{m_3} \\ &= \sum_{m_2, m_3 \geq 0} \sum_{r_1, \dots, r_{m_2} \geq 1} \left(\prod_{j=1}^{m_2} c_{\Lambda,2,r_j} \right) a_{\Lambda,0,m_2,m_3} g^{r_1 + \dots + r_{m_2} + m_3}. \end{aligned}$$

Equating this with the expansion $\langle L \rangle = \sum_{n=0}^{\infty} d_n g^n$, we get

$$d_n = \sum_{m_2, m_3 \geq 0} \sum_{\substack{r_1, \dots, r_{m_2} \geq 1 \\ r_1 + \dots + r_{m_2} + m_3 = n}} \left(\prod_{j=1}^{m_2} c_{\Lambda,2,r_j} \right) a_{\Lambda,0,m_2,m_3}.$$

For each $n \geq 1$, we will ensure that $d_n = 0$ by suitably choosing $c_{\Lambda,2,r}$, $r = 1, \dots, n$. We have already done this previously for $n = 1$. Suppose this has been done up to $n - 1$. Note that on the right side of the above expression, we must have $m_2, m_3 \leq n$ (since the r 's are all ≥ 1), and each $r_j \leq n$. Moreover, if some $r_j = n$, then we must have $m_3 = 0$ and $m_2 = 1$. This shows that

$$d_n = A + c_{\Lambda,2,n} a_{\Lambda,0,1,0},$$

where A depends only on $c_{\Lambda,2,1}, \dots, c_{\Lambda,2,n-1}$. Since these c 's have already been chosen, we can now ensure that $d_n = 0$ by choosing

$$c_{\Lambda,2,n} = \frac{d_n - A}{a_{\Lambda,0,1,0}}.$$

□

$$d_n = \sum_{\substack{m_2, m_3 \geq 0 \\ m_3 + \sum_{r \geq 1} r k_r = n \\ m_2 = \sum_{r \geq 1} k_r}} a_{\Lambda,0,m_2,m_3} \cdot \prod_{r \geq 1} \frac{c_{\Lambda,2,r}^{k_r}}{k_r!}, \quad (7.4.2)$$

where the sum ranges over all ways of writing total g -order n as

$$n = m_3 + \sum_{r \geq 1} r k_r,$$

with m_3 counting how many ϕ^4 insertions (S_3 factors) appear and k_r counting how many times we pick $c_{\Lambda,2,r} g^r$ from θ_2 . (The factorials are the usual Taylor series factorials.)

7.5 Renormalization to all orders

Fix an even integer $n \geq 2$ and consider the n -point observable

$$L(\phi) = i^{n/2} (2\pi)^{-3n/2} \tilde{\phi}(p_1) \cdots \tilde{\phi}(p_n), \quad p_i \in \mathbb{R}^3,$$

with $\sum_{i=1}^n p_i = 0$ (overall momentum conservation). We make the same simplifying choices as in the first-order computation:

$$c_{\Lambda,1,r} = 0 \text{ for all } r, \quad c_{\Lambda,3,1} = 1, \quad \text{and } c_{\Lambda,3,r} = 0 \text{ for all } r \geq 2. \quad (7.5.1)$$

Thus $\theta_1 \equiv 0$ and $\theta_3 = g$. The only freedom left is

$$\theta_2 = \sum_{r=1}^{\infty} c_{\Lambda,2,r} g^r,$$

which is a (cutoff-dependent) mass counterterm series. As before, we will now try to determine the above coefficients so that $\langle L \rangle$ is formally equal to $\sum_{n=0}^{\infty} d_n g^n$, where the d_n 's do not depend on Λ .

Because $\theta_3 = g$ and $\theta_2 = \sum_{r=1}^{\infty} c_{\Lambda,2,r} g^r$, expanding $\langle L \rangle_{\Lambda}$ in θ and then substituting produces:

$$d_n = \sum_{\substack{m_2, m_3 \geq 0 \\ m_3 + \sum_{r \geq 1} r k_r = n \\ m_2 = \sum_{r \geq 1} k_r}} a_{\Lambda,0,m_2,m_3} \cdot \prod_{r \geq 1} \frac{c_{\Lambda,2,r}^{k_r}}{k_r!}, \quad (7.5.2)$$

where the sum ranges over all ways of writing total g -order n as

$$n = m_3 + \sum_{r \geq 1} r k_r,$$

with m_3 counting how many ϕ^4 insertions (S_3 factors) appear and k_r counting how many times we pick $c_{\Lambda,2,r} g^r$ from θ_2 . (The factorials are the usual Taylor series factorials.) Thus, to control d_n we must control the large- Λ behavior of the coefficients $a_{\Lambda,0,m_2,m_3}$.

Now, for each multi-index $m = (m_1, m_2, m_3)$, the coefficient $a_{\Lambda,m}$ is a finite linear combination of expressions of the form

$$\begin{aligned} & \langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \cdot (\text{a product of } m_1 \text{ factors from } S_1) \\ & \cdot (\text{a product of } m_2 \text{ factors from } S_2) \cdot (\text{a product of } m_3 \text{ factors from } S_3) \rangle_0, \end{aligned}$$

and similar subtraction terms coming from the denominator (exactly as in Chapter 6). Each factor of S_2 contributes two fields φ integrated over x (equivalently, a momentum-conserving delta that forces equality of two momenta); each factor of S_3 contributes four fields integrated over x (equivalently, a delta enforcing that the sum of the four momenta is 0). After applying Wick's theorem to the resulting Gaussian expectation, every term

becomes a product of free propagators

$$\langle \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0 = (2\pi)^3 C_\Lambda(k),$$

together with delta functions enforcing momentum conservation at each integrated vertex. After integrating out the delta functions, each term reduces to a multiple integral over some number of independent “internal” momenta, with integrand a product of factors $C_\Lambda(\cdot)$. Therefore, every coefficient $a_{\Lambda,m}$ is a finite sum of integrals of the form

$$\int_{\|q_1\| \leq \Lambda} \cdots \int_{\|q_L\| \leq \Lambda} \prod_{r=1}^I \frac{1}{\|\ell_r(q_1, \dots, q_L; p)\|^2 + m^2} dq_1 \cdots dq_L, \quad (7.5.3)$$

where:

- L is the number of independent internal momenta after using all delta constraints;
- I is the number of propagator factors;
- each ℓ_r is an *affine-linear* function of $(q_1, \dots, q_L)$ with coefficients in $\{-1, 0, 1\}$ and with a linear dependence on the external momenta p .

We need the following ‘power counting’ lemma to handle the above integrals.

Lemma 7.5.1. *Consider an integral of the form (7.5.3), and write $Q = (q_1, \dots, q_L) \in \mathbb{R}^{3L}$. Assume that $\|\ell_r(Q; p)\| \geq c\|Q\| - C$ for all Q , for some constants $c, C > 0$. Then, if $3L - 2I < 0$, the integral converges to a finite limit as $\Lambda \rightarrow \infty$.*

Proof. By the given hypotheses, we have that when $\|Q\|$ is large enough, then

$$\prod_{r=1}^I \frac{1}{\|\ell_r(Q; p)\|^2 + m^2} \leq \frac{C_0}{\|Q\|^{2I}},$$

where C_0 is a constant that does not depend on Q . Integrating in polar coordinates on $\mathbb{R}^{3L}$, we see that the integral is bounded by

$$C_1 + C_2 \int_1^\infty r^{3L-1} \cdot r^{-2I} dr,$$

for some finite constants C_1 and C_2 . This is finite if $3L - 2I < 0$. $\square$

7.6 Renormalized Wightman functions

We now check that the same first-order renormalization works for all n -point functions.

Let n be a positive integer. If n is odd, then by the symmetry $S(\phi) = S(-\phi)$ we have $\langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \rangle = 0$. Hence assume n is even and $n \geq 4$.

Fix $p_1, \dots, p_n \in \mathbb{R}^3$ and define

$$L(\phi) := i^{n/2} (2\pi)^{-n} \tilde{\phi}(p_1) \cdots \tilde{\phi}(p_n).$$

Let a_m be the Taylor coefficients of $\langle L \rangle$ in $(\theta_1, \theta_2, \theta_3)$. Then the free term is the usual sum over perfect matchings:

$$a_{0,0,0} = \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} \delta^{(3)}(p_i + p_j) \lim_{\epsilon \downarrow 0} \frac{-1}{p_i^2 - m^2 + i\epsilon},$$

where $\mathcal{P}_n$ is the set of perfect matchings of $\{1, \dots, n\}$.

Next,

$$a_{0,1,0} = i(\langle LS_2 \rangle_0 - \langle L \rangle_0 \langle S_2 \rangle_0) = -\frac{i^{n/2+1}}{2(2\pi)^{2n+3}} \int_{\mathbb{R}^3} \left(\langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0 - \langle \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \rangle_0 \langle \tilde{\varphi}(k) \tilde{\varphi}(-k) \rangle_0 \right)$$

Evaluating by Wick's rule, we must pair k and $-k$ to two of the external momenta and forbid pairing k with $-k$. For distinct $i, j \in \{1, \dots, n\}$ let $\mathcal{P}_{n,i,j}$ denote perfect matchings of $\{1, \dots, n\} \setminus \{i, j\}$. Then integrating over k yields

$$a_{0,1,0} = \frac{1}{2} \sum_{1 \leq i \neq j \leq n} \sum_{E \in \mathcal{P}_{n,i,j}} \left(\delta^{(3)}(p_i + p_j) \lim_{\epsilon \downarrow 0} \frac{1}{(p_i^2 - m^2 + i\epsilon)^2} \right) \prod_{\{a,b\} \in E} \left(\delta^{(3)}(p_a + p_b) \lim_{\epsilon \downarrow 0} \frac{-1}{p_a^2 - m^2 + i\epsilon} \right).$$

Similarly,

$$a_{0,0,1} = i(\langle LS_3 \rangle_0 - \langle L \rangle_0 \langle S_3 \rangle_0)$$

is obtained by considering Wick pairings among the n external momenta and the four internal legs $(k, l, r, -k - l - r)$ from S_3 . As in the φ_4^4 chapter there are two types of contributions:

- (I) Two of the four internal legs are paired to two of the external momenta, and the remaining two internal legs are paired to each other. This produces a tadpole integral $\int (r^2 - m^2 + i\epsilon)^{-1} dr$, hence an ultraviolet divergence.
- (II) All four internal legs are paired to four distinct external momenta. This produces a finite contribution (in φ_3^4 , this first-order “contact” correction has no ultraviolet divergence).

After Wick rotation and the same hard cutoff Λ on the tadpole integral, the total contribution of type (I) is exactly proportional to

$$\left(\int_{\|r\| \leq \Lambda} \frac{1}{\|r\|^2 + m^2} dr \right)$$

times the same external-momentum structure that appears in $a_{0,1,0}$. Therefore, when we work with the renormalized action (??), the term $a_{0,1,0} c_\Lambda$ cancels *exactly* the type-(I) part of $a_{\Lambda,0,0,1}$, just as in the φ_4^4 chapter.

Hence, to first order in g , the renormalized n -point function receives only the type-(II) contributions, which are finite and do not depend on Λ . Concretely, one obtains the

first-order expansion

$$\begin{aligned} \widetilde{W}(p_1, \dots, p_n) &:= \frac{i^{n/2}}{(2\pi)^{2n}} \langle \widetilde{\varphi}(p_1) \cdots \widetilde{\varphi}(p_n) \rangle = \sum_{E \in \mathcal{P}_n} \prod_{\{i,j\} \in E} \delta^{(3)}(p_i + p_j) \lim_{\epsilon \downarrow 0} \frac{-1}{p_i^2 - m^2 + i\epsilon} \\ &\quad + \frac{ig}{(2\pi)^3} \lim_{\epsilon \downarrow 0} \sum_{1 \leq i < j < i' < j' \leq n} \sum_{E \in \mathcal{P}_{n, i, j, i', j'}} \left(\prod_{\{a,b\} \in E} \frac{-\delta^{(3)}(p_a + p_b)}{p_a^2 - m^2 + i\epsilon} \right) \\ &\quad \cdot \left(\frac{\delta^{(3)}(p_i + p_j + p_{i'} + p_{j'})}{(p_i^2 - m^2 + i\epsilon)(p_j^2 - m^2 + i\epsilon)(p_{i'}^2 - m^2 + i\epsilon)(p_{j'}^2 - m^2 + i\epsilon)} \right) \end{aligned}$$

where $\mathcal{P}_{n, i, j, i', j'}$ denotes perfect matchings of $\{1, \dots, n\} \setminus \{i, j, i', j'\}$.

Thus, the same renormalization scheme (a mass counterterm fixed by the tadpole) renormalizes the first-order perturbation theory for *all* n -point functions in φ_3^4 . In contrast to φ_4^4 , the divergence is only linear in the hard cutoff:

$$c_\Lambda = -\frac{1}{4\pi^2} \left(\Lambda - m \arctan(\Lambda/m) \right),$$

reflecting the fact that φ_3^4 is super-renormalizable.

Chapter 8

The Yang–Mills Action

8.1 Yang–Mills action of a connection

In this chapter, we are going to heavily use notations, terminologies and results from differential geometry. It may be helpful to review the materials from Appendix C if the reader is not thoroughly familiar with those.

Throughout this chapter, let $P = M \times G$ be a trivial principal bundle, where M is a Riemannian or pseudo-Riemannian manifold with metric g and G is a matrix Lie group. Let $\mathfrak{g}$ be the Lie algebra of G . Let n be the dimension of M . Let A be a principal G -connection on P , viewed as a $\mathfrak{g}$ -valued 1-form on M . Let F be its curvature form. Recall that $F \wedge *F$ is a $\mathfrak{g}$ -valued n -form. Then in local coordinates on a chart (U, φ) ,

$$F \wedge *F = H dx_1 \wedge \cdots \wedge dx_n$$

for some smooth map $H : U \rightarrow \mathfrak{g}$. We denote by $\text{Tr}(F \wedge *F)$ the real-valued n -form

$$\text{Tr}(H) dx_1 \wedge \cdots \wedge dx_n,$$

where $\text{Tr}(M)$ denotes the trace of a linear operator M on a finite-dimensional vector space (which makes sense because $\mathfrak{g}$ is a matrix Lie algebra). It is not hard to verify that this is indeed a well-defined n -form, that is, it does not depend on the choice of coordinates. The Yang–Mills action of A with bare coupling constant g_0 is defined as

$$S(A) := -\frac{1}{2g_0^2} \int_M \text{Tr}(F \wedge *F),$$

provided that the integral on the right converges. In this chapter, we will take $g_0 = 1$ throughout, but in general g_0 may not be equal to 1.

Let us now explicitly work out the formula for $S(A)$ in local coordinates. Note that by Lemma C.6.3 (which easily generalizes to Lie algebra-valued differential forms), we get

$$F \wedge *F = \langle F, F \rangle \omega,$$

where ω is the volume form of M , and

$$\langle F, F \rangle = \sum_{1 \leq i < j \leq n} F_{ij} F^{ij},$$

where $F = \sum_{1 \leq i < j \leq n} F_{ij} dx_i \wedge dx_j$, $F_{ji} = -F_{ij}$, $F_{ii} = 0$, and

$$F^{ij} = \sum_{1 \leq k, l \leq n} g^{ik} g^{jl} F_{kl}.$$

Thus, on a chart (U, φ) , we have

$$\begin{aligned} & \int_U \text{Tr}(F \wedge *F) \\ &= \frac{1}{2} \int_{\varphi(U)} \text{Tr} \left(\sum_{1 \leq i, j, k, l \leq n} g^{ik} g^{jl} F_{ij}(\varphi^{-1}(x)) F_{kl}(\varphi^{-1}(x)) \right) \sqrt{|\det(g(\varphi^{-1}(x)))|} dx. \end{aligned} \quad (8.1.1)$$

In particular, if $M = \mathbb{R}^n$ with the standard metric, then

$$S(A) = -\frac{1}{4g_0^2} \int_{\mathbb{R}^n} \left(\sum_{1 \leq i, j \leq n} \text{Tr}(F_{ij}(x)^2) \right) dx.$$

Let us now further assume that G is a subgroup of $U(m)$. Then $\mathfrak{g}$ is a subspace of $\mathfrak{u}(m)$. In particular, the elements of $\mathfrak{g}$ are skew-Hermitian matrices. The formula (C.5.1) and the fact that $\mathfrak{g}$ is closed under the Lie product show that for each x and i, j , $F_{ij}(x) \in \mathfrak{g}$. In particular, $F_{ij}(x)$ is skew-Hermitian. Since the eigenvalues of a skew-Hermitian matrix are purely imaginary, it follows that $\text{Tr}(F_{ij}(x)^2)$ is a nonpositive real number. Thus, we conclude that when G is a subgroup of $U(m)$ and $M = \mathbb{R}^n$, $S(A) \geq 0$ for every principal G -connection A . One can similarly show that $S(A) \geq 0$ when G is a subgroup of $U(m)$ and M is any Riemannian manifold.

8.2 Covariant derivative associated with a connection

Henceforth, we will often refer to a principal G -connection as a ‘gauge field’ on M . Let A be a gauge field on M . The covariant derivative d_A associated with A is an operator acting on $\mathfrak{g}$ -valued differential forms as follows:

$$d_A \omega := d\omega + [A \wedge \omega],$$

where dA is the exterior derivative of A and $[A \wedge \omega]$ is the wedge product (or commutator) A and ω , both defined in Section C.4.3. The following identity holds universally.

Lemma 8.2.1 (Bianchi identity). *For any gauge field A with curvature form F , we have $d_A F = 0$.*

Proof. Recall from Section C.4.3 that the exterior derivative of Lie algebra-valued differential

forms satisfies the Leibniz rule and the Poincaré lemma. Thus,

$$d[A \wedge A] = [dA \wedge A] - [A \wedge dA],$$

by the Leibniz rule for the exterior derivative. Consequently, by an application of the Poincaré lemma and the identity (C.4.3),

$$dF = ddA + \frac{1}{2}d[A \wedge A] = \frac{1}{2}([dA \wedge A] - [A \wedge dA]) = -[A \wedge dA].$$

Next, note that by Lemma C.4.4,

$$\begin{aligned} [A \wedge F] &= \left[A \wedge \left(dA + \frac{1}{2}[A \wedge A] \right) \right] \\ &= [A \wedge dA] + \frac{1}{2}[A \wedge [A \wedge A]] \\ &= [A \wedge dA]. \end{aligned}$$

Combining, we get $d_A F = dF + [A \wedge F] = 0$. $\square$

Note the inner product on differential forms defined in equation (C.6.3). The following result identifies the adjoint of d_A with respect to this inner product integrated over the manifold.

Lemma 8.2.2. *Let $*$ denote the Hodge star operator defined in Section C.6.2. Given a gauge field A , define the operator d_A^* acting on $\mathfrak{g}$ -valued k -forms (for $1 \leq k \leq n$) as*

$$d_A^* \eta := (-1)^{n(k-1)+1} * d_A * \eta.$$

*Let $\langle \cdot, \cdot \rangle$ denote the inner product defined in equation (C.6.3), and let ω denote the volume form on M . Then for any $\mathfrak{g}$ -valued k -form η and $\mathfrak{g}$ -valued $(k+1)$ -form and γ (for $0 \leq k \leq n-1$) such that the $(n-1)$ -form $\eta \wedge * \gamma$ satisfies the criteria for Stokes' theorem (Theorem C.4.5), we have*

$$\int_M \text{Tr}(\langle d_A \eta, \gamma \rangle) \omega = \int_M \text{Tr}(\langle \eta, d_A^* \gamma \rangle) \omega.$$

Proof. Note that

$$\begin{aligned} \int_M \text{Tr}(\langle d_A \eta, \gamma \rangle) \omega &= \int_M \text{Tr}(d_A \eta \wedge * \gamma) \\ &= \int_M \text{Tr}((d\eta + [A \wedge \eta]) \wedge * \gamma) \\ &= \int_M \text{Tr}(d\eta \wedge * \gamma) + \int_M \text{Tr}([A \wedge \eta] \wedge * \gamma). \end{aligned} \tag{8.2.1}$$

By the Leibniz rule for the exterior derivative,

$$d(\eta \wedge * \gamma) = d\eta \wedge * \gamma + (-1)^k \eta \wedge d* \gamma.$$

But by Stokes' theorem (Theorem C.4.5), using the fact that B is compactly supported (which ensures that $d_A B \wedge * F_A$ satisfies the criteria of Stokes' theorem),

$$\int_M \text{Tr}(d(\eta \wedge * \gamma)) = 0.$$

Thus,

$$\int_M \text{Tr}(d\eta \wedge * \gamma) = -(-1)^k \int_M \text{Tr}(\eta \wedge d* \gamma). \quad (8.2.2)$$

Next, by Lemma C.4.3,

$$[A \wedge \eta] = A \wedge \eta - (-1)^k \eta \wedge A.$$

Since $\text{Tr}(MN) = \text{Tr}(NM)$ for any two square matrices M and N of the same order, and $\eta \wedge * \gamma$ is an $(n-1)$ -form, the identity (C.1.5) implies that

$$\text{Tr}(A \wedge \eta \wedge * \gamma) = (-1)^{n-1} \text{Tr}(\eta \wedge * \gamma \wedge A).$$

Combining the last two displays gives

$$\begin{aligned} \text{Tr}([A \wedge \eta] \wedge * \gamma) &= \text{Tr}(\eta \wedge ((-1)^{n-1} * \gamma \wedge A - (-1)^k A \wedge * \gamma)) \\ &= -(-1)^k \text{Tr}(\eta \wedge (A \wedge * \gamma - (-1)^{n-k-1} * \gamma \wedge A)) \\ &= -(-1)^k \text{Tr}(\eta \wedge [A \wedge * \gamma]), \end{aligned} \quad (8.2.3)$$

where the last line follows by Lemma C.4.3. Using equations (8.2.3) and (8.2.2) in equation (8.2.1), we get

$$\begin{aligned} \int_M \text{Tr}(\langle d_A \eta, \gamma \rangle) \omega &= -(-1)^k \int_M \text{Tr}(\eta \wedge d* \gamma) - (-1)^k \int_M \text{Tr}(\eta \wedge [A \wedge * \gamma]) \\ &= -(-1)^k \int_M \text{Tr}(\eta \wedge d_A * \gamma). \end{aligned}$$

But by Lemma C.6.2, we get

$$d_A * \gamma = (-1)^{k(n-k)} ** d_A * \gamma.$$

Plugging this into the previous display, we get

$$\int_M \text{Tr}(\langle d_A \eta, \gamma \rangle) \omega = -(-1)^{k+k(n-k)} \int_M \text{Tr}(\eta \wedge ** d_A * \gamma).$$

The proof is completed by noting that $k(n - k) + k \equiv nk \pmod{2}$, and that γ is a $(k + 1)$ -form. $\square$

8.3 The Yang–Mills equation

Throughout this section, we will follow the convention that F_A denotes the curvature form of a $\mathfrak{g}$ -valued 1-form A . The Yang–Mills equation identifies the critical points of the Yang–Mills action, defined below. The connections that satisfy the Yang–Mills equation are called *Yang–Mills connections*.

The notion of critical points is defined as follows. Let A be any gauge field on M and B be a compactly supported gauge field on M (meaning that $B = 0$ outside a compact set). Note that although $S(A)$ may not be finite, the difference $S(A + B) - S(A)$, interpreted as the integral

$$- \int_M (\text{Tr}(F_{A+B} \wedge *F_{A+B}) - \text{Tr}(F_A \wedge *F_A))$$

is well-defined, because the above integrand is smooth and vanishes outside a compact set. The difference being interpreted this way, the *directional derivative* of A in the direction B is defined as

$$\lim_{t \rightarrow 0} \frac{S(A + tB) - S(A)}{t}. \quad (8.3.1)$$

(We will show below that the limit always exists.) A gauge field A is said to be a critical point of the Yang–Mills action S if the directional derivative in the direction B is zero for any compactly supported B .

Theorem 8.3.1. *The directional derivative displayed in equation (8.3.1) evaluates to*

$$- \int_M \text{Tr}(B \wedge *d_A^* F_A).$$

Consequently, a connection A is a Yang–Mills connection if and only if $d_A^ F_A \equiv 0$. (This is known as the Yang–Mills equation.)*

Proof. Take any gauge field A such that $S(A)$ is finite, and a compactly supported gauge field B . Note that for any t ,

$$\begin{aligned} F_{A+tB} &= d(A + tB) + \frac{1}{2}[(A + tB) \wedge (A + tB)] \\ &= dA + tdB + \frac{1}{2}[A \wedge A] + \frac{t}{2}[A \wedge B] + \frac{t}{2}[B \wedge A] + \frac{t^2}{2}[B \wedge B]. \end{aligned}$$

Now, by the identity (C.4.3), $[A \wedge B] = [B \wedge A]$. Thus, we get

$$\begin{aligned} F_{A+tB} &= F_A + tdB + t[A \wedge B] + \frac{t^2}{2}[B \wedge B] \\ &= F_A + td_A B + \frac{t^2}{2}[B \wedge B]. \end{aligned}$$

Consequently,

$$\langle F_{A+tB}, F_{A+tB} \rangle = \langle F_A, F_A \rangle + t\langle F_A, d_AB \rangle + t\langle d_AB, F_A \rangle + O(t^2),$$

where $O(t^2)$ denotes a collection of terms with t^2 , t^3 or t^4 as coefficients. It is not hard to see that

$$\text{Tr}(\langle F_A, d_AB \rangle) = \text{Tr}(\langle d_AB, F_A \rangle).$$

Thus,

$$S(A + tB) - S(A) = \int_M (-t \text{Tr}(\langle d_AB, F_A \rangle) + O(t^2)),$$

where we are using the convention that the integral of a smooth real-valued function on a Riemannian manifold is defined as the integral of that function times the volume form. From this equation, using the finiteness of $S(A)$ and the fact that B is compactly supported, it is not hard to see that the dominated convergence theorem can be applied, letting us conclude that

$$\lim_{t \rightarrow 0} \frac{S(A + tB) - S(A)}{t} = - \int_M \text{Tr}(\langle d_AB, F_A \rangle).$$

But by Lemma 8.2.2,

$$\int_M \text{Tr}(\langle d_AB, F_A \rangle) = \int_M \text{Tr}(\langle B, d_A^* F_A \rangle).$$

Thus, we arrive at the conclusion that A is a critical point of S if and only if

$$\int_M \text{Tr}(\langle B, d_A^* F_A \rangle) = 0 \tag{8.3.2}$$

for every compactly supported connection B . Using the smoothness of A , and taking a sequence of B 's approaching $d_A^* F_A(x) \delta_x$ for a given x (where δ_x is the Dirac delta at x), it is easy to show that this holds if and only if $d_A^* F_A \equiv 0$. $\square$

Note that $A \equiv 0$ is always a solution to the Yang–Mills equation. Next, suppose that G is Abelian, so that the Lie bracket vanishes identically on $\mathfrak{g}$. Therefore, if we take $A = df$ for some smooth $\mathfrak{g}$ -valued 1-form f , then $F_A = dA =ddf = 0$, which means that A is a Yang–Mills connection. We will see later that such connections are also trivial in a certain sense. A connection is said to be a nontrivial solution of the Yang–Mills equation if its curvature form is not constant.

It is instructive to work out the details of the following nontrivial solution. Let $M = \mathbb{R}^4$ with the usual Euclidean metric, and $G = \text{U}(1)$. Since G is Abelian, the Yang–Mills equation is simply $d^*F = 0$, where F is the curvature form of A . The formula (C.6.2) for the Hodge star operator gives, for any $1 \leq i < j \leq 4$,

$$*F_{ij} = \frac{1}{2} \sum_{1 \leq k, l \leq 4} \epsilon_{k,l,i,j} F_{kl}.$$

Consequently, $*F_{ij} = \text{sign}(\pi)F_{kl}$, where $k < l$ are the two indices such that k, l, i, j is a permutation π of $1, 2, 3, 4$. Explicitly,

$$\begin{aligned} *F_{12} &= F_{34}, & *F_{13} &= -F_{24}, & *F_{14} &= F_{23} \\ *F_{23} &= F_{14}, & *F_{24} &= -F_{13}, & *F_{34} &= F_{12}. \end{aligned}$$

(Note that we have the above relations only because $n = 4$.) Thus,

$$d*F = H_1 dx_2 \wedge dx_3 \wedge dx_4 + H_2 dx_1 \wedge dx_3 \wedge dx_4 + H_3 dx_1 \wedge dx_2 \wedge dx_4 + H_4 dx_1 \wedge dx_2 \wedge dx_3,$$

where

$$\begin{aligned} H_1 &= \frac{\partial *F_{34}}{\partial x_2} - \frac{\partial *F_{24}}{\partial x_3} + \frac{\partial *F_{23}}{\partial x_4} = \frac{\partial F_{12}}{\partial x_2} + \frac{\partial F_{13}}{\partial x_3} + \frac{\partial F_{14}}{\partial x_4}, \\ H_2 &= \frac{\partial *F_{34}}{\partial x_1} - \frac{\partial *F_{14}}{\partial x_3} + \frac{\partial *F_{13}}{\partial x_4} = \frac{\partial F_{12}}{\partial x_2} - \frac{\partial F_{23}}{\partial x_3} - \frac{\partial F_{24}}{\partial x_4}, \\ H_3 &= \frac{\partial *F_{24}}{\partial x_1} - \frac{\partial *F_{14}}{\partial x_2} + \frac{\partial *F_{12}}{\partial x_4} = -\frac{\partial F_{13}}{\partial x_1} - \frac{\partial F_{23}}{\partial x_2} + \frac{\partial F_{34}}{\partial x_4}, \\ H_4 &= \frac{\partial *F_{23}}{\partial x_1} - \frac{\partial *F_{13}}{\partial x_2} + \frac{\partial *F_{12}}{\partial x_3} = \frac{\partial F_{14}}{\partial x_1} + \frac{\partial F_{24}}{\partial x_2} + \frac{\partial F_{34}}{\partial x_3}. \end{aligned}$$

Thus, A satisfying the Yang–Mills equations if and only if $H_i = 0$ for each i . Using the conventions $F_{ji} = -F_{ij}$ and $F_{ii} = 0$, these equations can be written more succinctly as

$$\sum_{j=1}^4 \frac{\partial F_{ij}}{\partial x_j} = 0, \quad i = 1, 2, 3, 4, \quad (8.3.3)$$

where, from equation (C.5.1), we have

$$F_{ij} = \frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j}.$$

Take any A such that $A_2 = A_3 = A_4 = 0$. Then $F_{ij} = 0$ for $2 \leq i < j \leq 4$, and

$$F_{1j} = -\frac{\partial A_1}{\partial x_j}, \quad j = 2, 3, 4.$$

Thus, the Yang–Mills equation (8.3.3) reduces to

$$\sum_{j=2}^4 \frac{\partial^2 A_1}{\partial x_j^2} = 0.$$

To summarize, if we take $A_2 = A_3 = A_4 = 0$ and A_1 to be harmonic and nonlinear in (x_2, x_3, x_4) , then A is a nontrivial Yang–Mills connection.

8.4 Pseudo-Riemannian manifolds

Let M be a smooth n -dimensional manifold. Let p and q be two nonnegative integers such that $p + q = n$. Let S be the set of all triple (x, u, v) such that $x \in M$ and $u, v \in T_x M$. Let us endow S with a manifold structure in the usual way. A smooth map $g : S \rightarrow \mathbb{R}$ is called a pseudo-Riemannian metric on M with signature (p, q) if for each x , $g(x, \cdot, \cdot) : T_x M \times T_x M \rightarrow \mathbb{R}$ is a symmetric bilinear form with p positive eigenvalues and q negative eigenvalues. (Meaning that the matrix of $g(x, \cdot, \cdot)$ with respect to any local coordinate system has p positive eigenvalues and q negative eigenvalues.) Pseudo-Riemannian manifolds occur in relativistic physics. The most basic example is the metric

$$g \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (8.4.1)$$

on $\mathbb{R}^4$. This is known as the Lorentz metric on $\mathbb{R}^4$. When $\mathbb{R}^4$ is equipped with this metric, it is usually denoted by $\mathbb{R}^{1,3}$. The signature $(1, 3)$ is called the Lorentzian signature. Everything that we have discussed until now about Riemannian manifolds remain valid for pseudo-Riemannian manifolds with signature (p, q) , with the following differences:

- In the definitions of the volume form and the Hodge star operator, $\det(g)$ is replaced by $|\det(g)|$, since $\det(g)$ may not be positive.
- A consequence of the above replacement is that the Hodge star operator now satisfies

$$**\eta = (-1)^{k(n-k)+q}\eta.$$

- Due to the above, the formula for d_A^* becomes

$$d_A^* \eta := (-1)^{n(k-1)+1+q} * d_A * \eta.$$

- Theorem 8.3.1 remains valid. To see this, note that by following exactly the same steps, we arrive at equation (8.3.2). Now, note that since all the eigenvalues of $g(x)$ are nonzero, $g(x)$ is a nonsingular matrix at every $x \in M$. This implies that if

$$u = \sum_{1 \leq i_1 < \dots < i_k \leq n} u_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k} \in \Lambda^k T_x^* M$$

is a vector such that for any $v \in \Lambda^k T_x^* M$ we have

$$\sum_{1 \leq i_1, \dots, i_k \leq n} g^{i_1 j_1} \dots g^{i_k j_k} u_{i_1, \dots, i_k} v_{j_1, \dots, j_k} = 0,$$

then $u = 0$. From this, it is easy to show that $d_A^*F = 0$ if (8.3.2) holds for any B .

- The Yang–Mills action is not guaranteed to be nonnegative on pseudo-Riemannian manifolds.

8.5 Maxwell’s equations from $U(1)$ Yang–Mills theory

Recall that $\mathbb{R}^{1,3}$ denotes $\mathbb{R}^4$ endowed with the Minkowski metric g defined in equation (8.4.1). Let $\mathbf{B}$ be a magnetic field and $\mathbf{E}$ be an electric field, meaning that at each point (t, x) in spacetime $\mathbb{R}^{1,3}$, we have a vector $\mathbf{B}(t, x) \in \mathbb{R}^3$ and vector $\mathbf{E}(t, x) \in \mathbb{R}^3$. In the absence of anything else, and in units where the speed of light is 1, Maxwell’s equations for a free electromagnetic field govern the behavior of $(\mathbf{E}, \mathbf{B})$:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0 && \text{(Gauss’s law for electricity),} \\ \nabla \cdot \mathbf{B} &= 0 && \text{(Gauss’s law for magnetism),} \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} && \text{(Faraday’s law of induction),} \\ \nabla \times \mathbf{B} &= \frac{\partial \mathbf{E}}{\partial t} && \text{(Ampère’s law in vacuum).} \end{aligned}$$

Here $\nabla \cdot$ denotes the divergence operator and $\nabla \times$ denotes the curl operator, both with respect to the spatial coordinates. That is, for a function $f = (f_1, f_2, f_3) : \mathbb{R}^{1,3} \rightarrow \mathbb{R}^3$, $\nabla \cdot f$ is a function from $\mathbb{R}^{1,3}$ into $\mathbb{R}$, defined as

$$\nabla \cdot f = \frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2} + \frac{\partial f_3}{\partial x_3},$$

and $\nabla \times f$ is a function from $\mathbb{R} \times \mathbb{R}^3$ into $\mathbb{R}^3$, defined as

$$\nabla \times f = \left(\frac{\partial f_3}{\partial x_2} - \frac{\partial f_2}{\partial x_3}, \frac{\partial f_1}{\partial x_3} - \frac{\partial f_3}{\partial x_1}, \frac{\partial f_2}{\partial x_1} - \frac{\partial f_1}{\partial x_2} \right).$$

It turns out that Maxwell’s equations can be obtained from Bianchi’s identity and the Yang–Mills equation for Yang–Mills theory corresponding to the Lie group $U(1)$ on the manifold $\mathbb{R}^4$. Let $M = \mathbb{R} \times \mathbb{R}^3$ and $G = U(1)$. Recall that the Lie algebra of G is the space of all 1×1 skew-Hermitian matrices, which is simply the imaginary axis $i\mathbb{R} \subseteq \mathbb{C}$. Thus, a gauge field $A = A_0 dt + \sum_{j=1}^3 A_j dx_j$ is simply a map $A : M \rightarrow (i\mathbb{R})^4$. Given a gauge field (also called gauge potential or the electromagnetic four-potential) A , it is the convention to denote A_0 by $-\phi$ and call ϕ the ‘scalar potential’, and the triple (A_1, A_2, A_3) by $\mathbf{A}$ and call $\mathbf{A}$ the ‘vector potential’. Note that ϕ and the components of $\mathbf{A}$ are real-valued.

Given a gauge potential $A = (-\phi, \mathbf{A})$, it gives rise to an electromagnetic field $(\mathbf{E}, \mathbf{B})$ as:

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (8.5.1)$$

As an application of the techniques developed so far, we now prove the following theorem, which shows how Maxwell's equations are the same as the combination of the Bianchi identity and the Yang–Mills equations for $U(1)$ Yang–Mills theory on $\mathbb{R}^{1,3}$.

Theorem 8.5.1. *Let A be a $U(1)$ Yang–Mills connection on $\mathbb{R}^{1,3}$ and let $(\mathbf{E}, \mathbf{B})$ be defined through equation (8.5.1). Then $(\mathbf{E}, \mathbf{B})$ satisfies Maxwell's equations. Conversely, if $(\mathbf{E}, \mathbf{B})$ is a smooth electromagnetic field that satisfies Maxwell's equations, then there exists at least one Yang–Mills connection A such that $(\mathbf{E}, \mathbf{B})$ arises from A via equation (8.5.1).*

For the proof of the converse direction, we need the following lemma about differential forms on $\mathbb{R}^n$, known as Poincaré's lemma.

Lemma 8.5.2 (Poincaré's lemma for $\mathbb{R}^n$). *Let ω be a smooth k -form on $\mathbb{R}^n$, for some $1 \leq k \leq n$, such that $d\omega = 0$. Then there is a $(k-1)$ -form η such that $d\eta = \omega$.*

Proof. Let us write in coordinates

$$\omega(x) = \sum_{1 \leq i_1 < \dots < i_k \leq n} \omega_{i_1, \dots, i_k}(x) dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

As usual, extend the definition of $\omega_{i_1, \dots, i_k}$ for arbitrary $1 \leq i_1, \dots, i_k \leq n$ by defining $\omega_{i_{\pi(1)}, \dots, i_{\pi(k)}} = \text{sign}(\pi) \omega_{i_1, \dots, i_k}$ for any $\pi \in S_k$ and letting $\omega_{i_1, \dots, i_k} = 0$ if $i_1, \dots, i_k$ are not distinct. Define a $(k-1)$ -form η as

$$\eta(x) = \sum_{1 \leq j_1 < \dots < j_{k-1} \leq n} \sum_{i=1}^n \left(\int_0^1 t^{k-1} x_i \omega_{i, j_1, \dots, j_{k-1}}(tx) dt \right) dx_{j_1} \wedge \dots \wedge dx_{j_{k-1}}.$$

Now, moving the derivative inside the integral is allowed because we are dealing with smooth functions on $\mathbb{R}^n$. Thus,

$$\frac{\partial}{\partial x_{i'}} \left(\int_0^1 t^{k-1} x_i \omega_{i, j_1, \dots, j_{k-1}}(tx) dt \right) = \int_0^1 t^{k-1} (\delta_{ii'} \omega_{i, j_1, \dots, j_{k-1}}(tx) + t x_i \partial_{i'} \omega_{i, j_1, \dots, j_{k-1}}(tx)) dt,$$

where $\partial_{i'}$ is shorthand for $\frac{\partial}{\partial x_{i'}}$ on the right side, and $\delta_{ii'} = 1$ if $i = i'$ and 0 if not. Thus,

$$\begin{aligned} d\eta &= \sum_{1 \leq j_1 < \dots < j_{k-1} \leq n} \sum_{i, i'=1}^n \frac{\partial}{\partial x_{i'}} \left(\int_0^1 t^{k-1} x_i \omega_{i, j_1, \dots, j_{k-1}}(tx) dt \right) dx_{i'} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_{k-1}} \\ &= \sum_{1 \leq j_1 < \dots < j_{k-1} \leq n} \sum_{i, i'=1}^n \left(\int_0^1 t^{k-1} (\delta_{ii'} \omega_{i, j_1, \dots, j_{k-1}}(tx) + t x_i \partial_{i'} \omega_{i, j_1, \dots, j_{k-1}}(tx)) dt \right) \\ &\quad \cdot dx_{i'} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_{k-1}}. \end{aligned}$$

Rearranging $i', j_1, \dots, j_{k-1}$ as an increasing sequence $i_1 < \dots < i_k$, it is easy to see that

this can be rewritten as

$$\begin{aligned}
 d\eta = \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{i=1}^n \sum_{j=1}^k (-1)^{j-1} \left(\int_0^1 t^{k-1} (\delta_{ii_j} \omega_{i, i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_k}(tx) \right. \\
 \left. + tx_i \partial_{i_j} \omega_{i, i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_k}(tx)) dt \right) dx_{i_1} \wedge \dots \wedge dx_{i_k}.
 \end{aligned} \tag{8.5.2}$$

Now take any $1 \leq i_1 < \dots < i_k \leq n$ and $1 \leq i \leq n$. If $i = i_l$ for some l , then

$$\begin{aligned}
 \sum_{j=1}^k (-1)^{j-1} \partial_{i_j} \omega_{i, i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_k} &= (-1)^{l-1} \partial_{i_l} \omega_{i, i_1, \dots, i_{l-1}, i_{l+1}, \dots, i_k} \\
 &= \partial_i \omega_{i_1, \dots, i_k}.
 \end{aligned} \tag{8.5.3}$$

If $i_{l-1} < i < i_l$ for some j (which includes $i < i_0$ if $l = 0$ and $i_k < i$ if $l = k$), then rename the sequence $i_1 < \dots < i_{l-1} < i < i_{l+1} < \dots < i_k$ and $j_1 < \dots < j_{k+1}$. Since $d\omega = 0$, we have

$$\begin{aligned}
 0 &= \sum_{m=1}^{k+1} (-1)^{m-1} \partial_{j_m} \omega_{j_1, \dots, j_{m-1}, j_{m+1}, \dots, j_{k+1}} \\
 &= \sum_{j=1}^{l-1} (-1)^{j-1} (-1)^{l-2} \partial_{i_j} \omega_{i, i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_k} + (-1)^{l-1} \partial_i \omega_{i_1, \dots, i_k} \\
 &\quad + \sum_{j=l}^k (-1)^j (-1)^{l-1} \partial_{i_j} \omega_{i, i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_k}.
 \end{aligned}$$

This shows that

$$\sum_{j=1}^k (-1)^{j-1} \partial_{i_j} \omega_{i, i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_k} = \partial_i \omega_{i_1, \dots, i_k}. \tag{8.5.4}$$

Lastly, note that

$$\begin{aligned}
 \sum_{i=1}^n \sum_{j=1}^k (-1)^{j-1} \delta_{ii_j} \omega_{i, i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_k} &= \sum_{j=1}^k (-1)^{j-1} \omega_{i_j, i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_k} \\
 &= k \omega_{i_1, \dots, i_k}.
 \end{aligned} \tag{8.5.5}$$

Using equations (8.5.3), (8.5.4) and (8.5.5) in equation (8.5.2), we get

$$d\eta = \sum_{1 \leq i_1 < \dots < i_k \leq n} \left(\int_0^1 \left(kt^{k-1} \omega_{i_1, \dots, i_k}(tx) + t^k \sum_{i=1}^n x_i \partial_i \omega_{i_1, \dots, i_k}(tx) \right) dt \right) dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

But note that

$$\begin{aligned} \int_0^1 \left(k t^{k-1} \omega_{i_1, \dots, i_k}(tx) + t^k \sum_{i=1}^n x_i \partial_i \omega_{i_1, \dots, i_k}(tx) \right) dt &= \int_0^1 \frac{\partial}{\partial t} (t^k \omega_{i_1, \dots, i_k}(tx)) dt \\ &= \omega_{i_1, \dots, i_k}(x). \end{aligned}$$

This proves that $d\eta = \omega$. $\square$

We are now ready to prove Theorem 8.5.1.

Proof of Theorem 8.5.1. First, let A be a Yang–Mills connection and $(\mathbf{E}, \mathbf{B})$ be defined via equation (8.5.1). Let F denote the curvature form of A . Then we know that F satisfies the Bianchi identity $d_A F = 0$, and since it is a Yang–Mills connection, Theorem 8.3.1 implies that it satisfies the Yang–Mills equation $d_A^* F = 0$. We claim that the Bianchi identity is equivalent to the combination of Gauss’s law for magnetism and Faraday’s law of induction (collectively called the homogeneous Maxwell equations), and the Yang–Mills equation is equivalent to the combination of Gauss’s law for electricity and Ampère’s law in the vacuum (collectively called the inhomogeneous Maxwell equations).

To prove these claims, first recall that

$$F = \sum_{i=1}^3 F_{0i} dt \wedge dx_i + \sum_{1 \leq i < j \leq 3} F_{ij} dx_i \wedge dx_j,$$

where

$$F_{0i} = \frac{\partial A_i}{\partial t} - \frac{\partial A_0}{\partial x_i} = -iE_i, \quad (8.5.6)$$

for $i = 1, 2, 3$, and for $1 \leq i < j \leq 3$,

$$F_{ij} = \frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j}. \quad (8.5.7)$$

In particular,

$$\mathbf{B} = \nabla \times \mathbf{A} = (-iF_{23}, iF_{13}, -iF_{12}). \quad (8.5.8)$$

Since $U(1)$ is an Abelian group, the Lie product vanishes on its algebra. Thus,

$$\begin{aligned} d_A F = dF &= \sum_{1 \leq i < j \leq 3} \left(\frac{\partial F_{ij}}{\partial t} - \frac{\partial F_{0j}}{\partial x_i} + \frac{\partial F_{0i}}{\partial x_j} \right) dt \wedge dx_i \wedge dx_j \\ &\quad + \left(\frac{\partial F_{23}}{\partial x_1} - \frac{\partial F_{13}}{\partial x_2} + \frac{\partial F_{12}}{\partial x_3} \right) dx_1 \wedge dx_2 \wedge dx_3. \end{aligned}$$

Now, by equation (8.5.8), the coefficient of $dx_1 \wedge dx_2 \wedge dx_3$ in the above display vanishes if and only if $\nabla \cdot \mathbf{B} = 0$ (i.e., Gauss’s law for magnetism). The vanishing of the other three

coordinates, together with the identity (8.5.6), gives rise to the following three equations:

$$\begin{aligned} -\frac{\partial B_1}{\partial t} &= i \frac{\partial F_{23}}{\partial t} = i \frac{\partial F_{03}}{\partial x_2} - i \frac{\partial F_{02}}{\partial x_3} = \frac{\partial E_3}{\partial x_2} - \frac{\partial E_2}{\partial x_3}, \\ -\frac{\partial B_2}{\partial t} &= -i \frac{\partial F_{13}}{\partial t} = -i \frac{\partial F_{03}}{\partial x_1} + i \frac{\partial F_{01}}{\partial x_3} = \frac{\partial E_1}{\partial x_3} - \frac{\partial E_3}{\partial x_1}, \\ -\frac{\partial B_3}{\partial t} &= i \frac{\partial F_{12}}{\partial t} = i \frac{\partial F_{02}}{\partial x_1} - i \frac{\partial F_{01}}{\partial x_2} = \frac{\partial E_2}{\partial x_1} - \frac{\partial E_1}{\partial x_2}. \end{aligned}$$

Collectively, the above equations are the same as

$$-\frac{\partial \mathbf{B}}{\partial t} = \nabla \times \mathbf{E}.$$

Note that this is Faraday's law of induction. This proves the claim that the Bianchi identity is equivalent to the combination of Gauss's law for magnetism and Faraday's law of induction.

Next, note that by Lemma 8.2.2, the Yang–Mills equation $d_A^* F = 0$ in this problem reduces to the simpler equation $d^* F = 0$. Note that

$$*F = \sum_{i=1}^3 *F_{0i} dt \wedge dx_i + \sum_{1 \leq i < j \leq 3} *F_{ij} dx_i \wedge dx_j,$$

where, by the formula (C.6.2),

$$*F_{ij} = \frac{1}{2} \sum_{0 \leq k, l \leq 3} \sum_{0 \leq k', l' \leq 3} \epsilon_{k, l, i, j} g^{kk'} g^{ll'} F_{kl} \quad (8.5.9)$$

for all $0 \leq i < j \leq 3$ (where $F_{lk} = F_{kl}$ and $F_{kk} = 0$, by our usual convention). By equation (8.5.8) and the definition of g , this gives

$$*F_{01} = F_{23} = iB_1, \quad *F_{02} = -F_{13} = iB_2, \quad *F_{03} = F_{12} = iB_3. \quad (8.5.10)$$

On the other hand, by equations (8.5.6) and (8.5.9) and the definition of g , we get

$$*F_{12} = -F_{03} = iE_3, \quad *F_{13} = F_{02} = -iE_2, \quad *F_{23} = -F_{01} = iE_1. \quad (8.5.11)$$

A simple inspection shows that equations (8.5.10) and (8.5.11) are the same as equations (8.5.6) and (8.5.8), after replacing F with $*F$, $\mathbf{E}$ with $-\mathbf{B}$, and $\mathbf{B}$ with $\mathbf{E}$. Thus, following the same steps as before, we get the equations $\nabla \cdot \mathbf{E} = 0$ and

$$\frac{\partial \mathbf{E}}{\partial t} = \nabla \times \mathbf{B},$$

which are Gauss's law for electricity and Ampère's law in the vacuum. This completes the proof of one direction of the theorem.

For the converse direction, let $(\mathbf{E}, \mathbf{B})$ be an electromagnetic field that satisfies Maxwell's

equations. Let us define a real-valued 2-form $\omega = \sum_{0 \leq i < j \leq 3} \omega_{ij} dx_i \wedge dx_j$ on $\mathbb{R}^4$ as

$$\begin{aligned}\omega_{0i} &:= -E_i, \quad i = 1, 2, 3, \\ \omega_{23} &:= -B_1, \quad \omega_{13} := B_2, \quad \omega_{12} := -B_3.\end{aligned}$$

In other words, ω mimics the role of iF in equations (8.5.6) and (8.5.8). Thus, following the same calculation as the one that follows those equations (using that $(\mathbf{E}, \mathbf{B})$ satisfies the homogeneous Maxwell equations), we conclude that $d\omega = 0$. Therefore by Poincaré’s lemma for $\mathbb{R}^n$ (Lemma 8.5.2), there exists a real-valued 1-form $\eta = \sum_{j=0}^3 \eta_j dx_j$ such that $d\eta = \omega$. Let $A_j := i\eta_j$ for $0 \leq j \leq 3$. Then $A := \sum_{j=0}^3 A_j dx_j$ is a $u(1)$ -valued 1-form, and a simple verification shows that $(\mathbf{E}, \mathbf{B})$ arise from A via equation (8.5.1). The fact that $(\mathbf{E}, \mathbf{B})$ satisfies the inhomogeneous Maxwell equations implies that A is a Yang–Mills connection, by the same argument as before. $\square$

Going from Euclidean signature to Lorentzian signature changes the nature of the nontrivial solutions to the Yang–Mills equations. An example of a nontrivial solution to Maxwell’s equations is given by the gauge potential

$$A_0 = A_1 = A_2 = 0, \quad A_3(t, x) = i \sin(x_2 - t).$$

To see that this is a solution, note that by equation (8.5.1),

$$\begin{aligned}\mathbf{E} &= -\frac{\partial \mathbf{A}}{\partial t} = (0, 0, -\cos(x_2 - t)), \\ \mathbf{B} &= \nabla \times \mathbf{A} = \left(-i \frac{\partial A_3}{\partial x_2}, i \frac{\partial A_3}{\partial x_1}, 0 \right) = (\cos(x_2 - t), 0, 0).\end{aligned}$$

Then clearly, $\nabla \cdot \mathbf{E} = 0$, and

$$\nabla \times \mathbf{B} = \left(0, \frac{\partial B_1}{\partial x_3}, -\frac{\partial B_1}{\partial x_2} \right) = (0, 0, \sin(x_2 - t)),$$

which proves that Ampère’s law holds. Since the homogeneous Maxwell equations are always valid when $(\mathbf{E}, \mathbf{B})$ is defined via equation (8.5.1), this proves that the above example works. In this example, the electric and magnetic fields are waves traveling in mutually perpendicular directions at the speed of light.

8.6 Gauge transformations

We have seen in the previous section how an electromagnetic field can arise from a $U(1)$ gauge field. Is this mapping one-to-one? It turns out that it is not. In fact, while an electromagnetic field is a physical object that can be measured in experiments, a gauge field is not. It is only when we take the quotient of the space of gauge fields with respect to the action of a certain group known as the group of ‘gauge transformations’ that we obtain physically measurable objects (which are points of the quotient space). In the case

of $U(1)$ Yang–Mills theory, the electromagnetic field is the image of the gauge field under this quotient map.

Let us return to the setting introduced at the beginning of this chapter, where we have a trivial principal bundle $M \times G$, where M is an n -dimensional smooth manifold and G is a matrix Lie group with Lie algebra $\mathfrak{g}$. Henceforth, we will work under the assumption that M is endowed with a pseudo-Riemannian (instead of Riemannian) metric.

Definition 8.6.1. *In the above setting, a gauge transformation is a smooth map from M into G .*

Let $\mathcal{G}$ denote the set of all gauge transformations $C^\infty(M, G)$ and $\mathcal{A}$ denote the space of gauge fields $\Omega^1(M, \mathfrak{g})$. One can define a group structure on $\mathcal{G}$ by pointwise multiplication:

$$(U_1 U_2)(x) := U_1(x) U_2(x).$$

The identity element of this group is the map $I(x) \equiv I$.

In the following, we will frequently use the convention that if η and γ are two matrix-valued differential forms such that at least one is a 0-form, then $\eta\gamma$ denotes $\eta \wedge \gamma$. We will also use the observation that if η, γ, θ are three matrix-valued differential forms, and γ is a 0-form, then

$$\eta \wedge \gamma \theta = \eta \gamma \wedge \theta. \quad (8.6.1)$$

Now, since G is a matrix Lie group, we can view gauge transformations as matrix-valued 0-forms. With that interpretation, for any $U \in \mathcal{G}$, dU is a matrix-valued 1-form. In local coordinates within a chart (W, φ) ,

$$dU = \sum_{i=1}^n \frac{\partial U}{\partial x_i} dx_i,$$

where, by equation (C.2.2),

$$\frac{\partial U}{\partial x_i} = \left. \frac{d}{dt} U(\varphi^{-1}(\varphi(x) + te_i)) \right|_{t=0},$$

where $e_1, \dots, e_n$ are the standard basis vectors of $\mathbb{R}^n$.

We claim that the matrix-valued 1-form $(dU)U^{-1}$ is actually a $\mathfrak{g}$ -valued 1-form (where, as mentioned above, $(dU)U^{-1}$ is shorthand for $dU \wedge U^{-1}$). To see this, take any i and x , and let γ be the curve $\gamma(t) := U(\varphi^{-1}(\varphi(x) + te_i))U(x)^{-1}$. This is a smooth curve in G , and $\gamma(0) = I$. Thus, $\gamma'(0)$ is an element of $\mathfrak{g}$. But

$$\gamma'(0) = \frac{\partial U}{\partial x_i} U(x)^{-1}.$$

This shows that $(dU)U^{-1}$ is a $\mathfrak{g}$ -valued 1-form. By a similar argument, the matrix-valued 1-form UAU^{-1} is actually $\mathfrak{g}$ -valued.

Definition 8.6.2. *The action of a gauge transformation U on a gauge field A is defined as*

$$A^U := UAU^{-1} - (dU)U^{-1}.$$

Usually, we refer to both U and the action of U as gauge transformations.

Lemma 8.6.3. *The action defined above is a group action, in the sense that $(A^U)^V = A^{VU}$.*

Proof. First note that

$$\begin{aligned} (A^U)^V &= VA^UV^{-1} - (dV)V^{-1} \\ &= V(UAU^{-1} - (dU)U^{-1})V^{-1} - (dV)V^{-1} \\ &= (VU)A(VU)^{-1} - V(dU)(VU)^{-1} - (dV)V^{-1}. \end{aligned}$$

On the other hand, by the Leibniz rule for the exterior derivative,

$$\begin{aligned} (d(VU))(VU)^{-1} &= ((dV)U + VdU)(VU)^{-1} \\ &= (dV)V^{-1} + V(dU)(VU)^{-1}. \end{aligned}$$

This proves that $(A^U)^V = A^{VU}$. Thus, $A \mapsto A^U$ is a group action of $\mathcal{G}$ on $\mathcal{A}$. $\square$

Two gauge fields A and B are said to be *gauge equivalent* if $B = A^U$ for some gauge transformation U . It is easy to see that this is an equivalence relation. The set of equivalence classes is denoted by $\mathcal{A}/\mathcal{G}$.

A function f on $\mathcal{A}$ is said to be gauge invariant if $f(A^U) = f(A)$ for any $A \in \mathcal{A}$ and $U \in \mathcal{G}$. A key property of the Yang–Mills action is that it is gauge invariant.

Theorem 8.6.4. *Take any $A \in \mathcal{A}$ and $U \in \mathcal{G}$, and let $B := A^U$. Then $F_B = UF_AU^{-1}$. As a consequence, if $S(A)$ is well-defined and finite, then so is $S(B)$, and $S(A) = S(B)$.*

Proof. Take any $A \in \mathcal{A}$ and $U \in \mathcal{G}$. Let $B := A^U$. Then note that

$$\begin{aligned} F_B &= dB + B \wedge B \\ &= d(UAU^{-1} - (dU)U^{-1}) + (UAU^{-1} - (dU)U^{-1}) \wedge (UAU^{-1} - (dU)U^{-1}). \end{aligned}$$

By the Leibniz rule for the exterior derivative,

$$\begin{aligned} d(UAU^{-1}) &= dU \wedge AU^{-1} + U d(AU^{-1}) \\ &= dU \wedge AU^{-1} + U((dA)U^{-1} - A(dU^{-1})) \\ &= dU \wedge AU^{-1} + U(dA)U^{-1} - UA(dU^{-1}). \end{aligned}$$

Since $UU^{-1} \equiv I$, we have (again by the Leibniz rule)

$$0 = d(UU^{-1}) = (dU)U^{-1} + U d(U^{-1}).$$

Thus, $d(U^{-1}) = -U^{-1}(dU)U^{-1}$. Substituting this in the previous display, we get

$$d(UAU^{-1}) = dU \wedge AU^{-1} + U(dA)U^{-1} + UAU^{-1} \wedge (dU)U^{-1}.$$

Similarly,

$$\begin{aligned} -d((dU)U^{-1}) &= -ddU \wedge U^{-1} + dU \wedge dU^{-1} \\ &= -dU \wedge U^{-1}(dU)U^{-1}. \end{aligned}$$

Lastly, by generous applications of equation (8.6.1),

$$\begin{aligned} &(UAU^{-1} - (dU)U^{-1}) \wedge (UAU^{-1} - (dU)U^{-1}) \\ &= UAU^{-1} \wedge UAU^{-1} - (dU)U^{-1} \wedge UAU^{-1} \\ &\quad - UAU^{-1} \wedge (dU)U^{-1} + (dU)U^{-1} \wedge (dU)U^{-1} \\ &= UA \wedge AU^{-1} - dU \wedge AU^{-1} - UAU^{-1} \wedge (dU)U^{-1} + (dU)U^{-1} \wedge (dU)U^{-1}. \end{aligned}$$

Adding up the right sides of the last three displays and plugging into the expression for F_B obtained above, we get that

$$F_B = U(dA)U^{-1} + UA \wedge AU^{-1} = UF_AU^{-1}.$$

Now, for notational convenience, let $F := F_A$ and $F' := F_B$. Then note that in local coordinates,

$$\begin{aligned} \text{Tr}(\langle F', F' \rangle) &= \sum_{1 \leq i, j, k, l \leq n} g^{ik} g^{jl} \text{Tr}(F'_{ij} F'_{kl}) \\ &= \sum_{1 \leq i, j, k, l \leq n} g^{ik} g^{jl} \text{Tr}(U^{-1} F_{ij} U U^{-1} F_{kl} U) \\ &= \sum_{1 \leq i, j, k, l \leq n} g^{ik} g^{jl} \text{Tr}(U^{-1} F_{ij} F_{kl} U) \\ &= \sum_{1 \leq i, j, k, l \leq n} g^{ik} g^{jl} \text{Tr}(F_{ij} F_{kl}) \\ &= \text{Tr}(\langle F, F \rangle). \end{aligned}$$

This proves that $S(A) = S(B)$ if $S(A)$ is well-defined and finite. $\square$

The physics of gauge theories says that the image of a connection A under the quotient map from $\mathcal{A}$ into $\mathcal{A}/\mathcal{G}$ is the physically measurable object, and not A itself. Another way to say this is that only gauge invariant functions of a gauge field are physically observable. For example, the following result shows that this is indeed the case for electromagnetism (in vacuum). The gauge field is only a mathematical construct. The physically measurable quantities are only those that depend on the gauge field through the electromagnetic field generated by it.

Theorem 8.6.5. *Consider $U(1)$ connections on $\mathbb{R}^{1,3}$. Two connections A and B yield the same electromagnetic field $(\mathbf{E}, \mathbf{B})$ via equation (8.5.1) if and only if they are gauge equivalent.*

Proof. For $U(1)$ theory on $\mathbb{R}^{1,3}$, a gauge transformation is a smooth map $U : \mathbb{R}^{1,3} \rightarrow U(1)$. It is not hard to show that U can be written as $U(x) = e^{i\psi(x)}$ for some smooth map $\psi : \mathbb{R}^{1,3} \rightarrow \mathbb{R}$ (for example, by taking $\psi(x)$ to be the line integral of $U'(x)/U(x)$ along any piecewise smooth curve joining 0 and x , and showing that value does not depend on the choice of the curve). Thus,

$$(dU)U^{-1} = i \sum_{i=0}^3 \frac{\partial \psi}{\partial x_i} dx_i = i d\psi,$$

where x_0 is the same as t . Also, due to the Abelian nature of $U(1)$, $U^{-1}AU = A$. Thus, a gauge transformation in this setting is simply a transformation like

$$A^\psi := A - i d\psi,$$

where $\psi \in C^\infty(\mathbb{R}^{1,3})$. In particular,

$$\begin{aligned} A_0^\psi &= A_0 - i \frac{\partial \psi}{\partial t}, \\ A_j^\psi &= A_j - i \frac{\partial \psi}{\partial x_j}, \quad j = 1, 2, 3. \end{aligned}$$

Thus, if $(\mathbf{E}, \mathbf{B})$ is obtained from A and $(\mathbf{E}^\psi, \mathbf{B}^\psi)$ is obtained from A^ψ via equation (8.5.1), then for $j = 1, 2, 3$,

$$\begin{aligned} E_j^\psi &= -i \frac{\partial A_0^\psi}{\partial x_j} + i \frac{\partial A_j^\psi}{\partial t} \\ &= -i \frac{\partial A_0}{\partial x_j} + \frac{\partial^2 \psi}{\partial x_j \partial t} + i \frac{\partial A_j}{\partial t} - \frac{\partial^2 \psi}{\partial t \partial x_j} \\ &= -i \frac{\partial A_0}{\partial x_j} + i \frac{\partial A_j}{\partial t} = E_j. \end{aligned}$$

Similarly, it is easy to check that $\mathbf{B}^\psi = \mathbf{B}$. This completes the proof of one direction of the claim. For the opposite direction, let A and B be two gauge field that generate the same electromagnetic field $(\mathbf{E}, \mathbf{B})$ via equation (8.5.1). By equations (8.5.6) and (8.5.8), this implies that $F_A = F_B$. But for $U(1)$ connections, the explicit form given in equation (8.5.7) implies that $F_A - F_B = F_{A-B}$. Thus, $F_C = 0$, where $C := A - B$. But $F_C = dC$ and C is an $i\mathbb{R}$ -valued 1-form. Thus, if we let $C' := -iC$, then C' is a real-valued 1-form with $dC' = 0$. Thus, by the Poincaré lemma for $\mathbb{R}^n$ (Lemma 8.5.2), there is a real-valued 0-form ψ such that $C' = d\psi$. Thus, for each j ,

$$B_j(x) = A_j(x) - C_j(x) = A_j(x) - iC'_j(x) = A_j(x) - i \frac{\partial \psi}{\partial x_j}.$$

This shows that $B = A^\psi$.

□

Chapter 9

The Free Electromagnetic Field

9.1 The Lorenz gauge condition

Recall from Section 8.1 and Section 8.5 that a gauge field in the theory of electromagnetism is a smooth map $A : \mathbb{R}^4 \rightarrow (\mathbb{i}\mathbb{R})^4$, with coordinates denoted by A_0, A_1, A_2, A_3 , and the action of such a field is formally given by

$$S(A) = \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}^3} \left(\sum_{j=1}^3 \left(\frac{\partial A_j}{\partial t} - \frac{\partial A_0}{\partial x_j} \right)^2 - \sum_{1 \leq j < k \leq 3} \left(\frac{\partial A_k}{\partial x_j} - \frac{\partial A_j}{\partial x_k} \right)^2 \right) dx dt. \quad (9.1.1)$$

(As in the previous chapter, we are denoting points in $\mathbb{R}^4$ as (t, x_1, x_2, x_3) , and taking the bare coupling constant g_0 to be 1.) Recall that a gauge field A is an abstract object that gives rise to a physically observable electromagnetic field $(\mathbf{E}, \mathbf{B})$ as

$$E_j = \mathbf{i} \frac{\partial A_j}{\partial t} - \mathbf{i} \frac{\partial A_0}{\partial x_j} \quad (9.1.2)$$

for $j = 1, 2, 3$, and

$$B_1 = \mathbf{i} \frac{\partial A_3}{\partial x_2} - \mathbf{i} \frac{\partial A_2}{\partial x_3}, \quad B_2 = \mathbf{i} \frac{\partial A_1}{\partial x_3} - \mathbf{i} \frac{\partial A_3}{\partial x_1}, \quad B_3 = \mathbf{i} \frac{\partial A_2}{\partial x_1} - \mathbf{i} \frac{\partial A_1}{\partial x_2}. \quad (9.1.3)$$

Moreover, two gauge fields A and A' gives rise to the same electromagnetic field if and only if A' is gauge equivalent to A , meaning that there is some $\psi : \mathbb{R}^4 \rightarrow \mathbb{R}$ such that

$$\begin{aligned} A'_0 &= A_0 - \mathbf{i} \frac{\partial \psi}{\partial t}, \quad \text{and} \\ A'_j &= A_j - \mathbf{i} \frac{\partial \psi}{\partial x_j}, \quad \text{for } j = 1, 2, 3. \end{aligned}$$

Let $\mathcal{A}_s$ denote the set of all Schwartzian $\mathrm{U}(1)$ connections on $\mathbb{R}^4$, that is, all $\mathrm{U}(1)$ connections A such that $A_j \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$ for $j = 0, 1, 2, 3$. Similarly, let $\mathcal{G}_s$ be the set of all Schwartzian gauge transforms. It is easy to see that $\mathcal{G}_s$ is a group that acts on $\mathcal{A}_s$ in the above manner. It turns out that there are simple ways to choose a unique element from each equivalence

class in $\mathcal{A}_s/\mathcal{G}_s$. One such method is *Lorenz gauge fixing*. A connection $A \in \mathcal{A}_s$ is said to satisfy the Lorenz gauge condition if

$$-\frac{\partial A_0}{\partial t} + \sum_{j=1}^3 \frac{\partial A_j}{\partial x_j} = 0. \quad (9.1.4)$$

Let $\mathcal{A}_s^L$ be the set of all $A \in \mathcal{A}_s$ that satisfy the Lorenz gauge condition. The following result shows that each element of $\mathcal{A}_s$ can be uniquely represented as A^ψ for some $A \in \mathcal{A}_s^L$ and $\psi \in \mathcal{G}_s$.

Proposition 9.1.1. *The mapping $\Phi(A, \psi) := A^\psi$ is an isomorphism of the vector spaces $\mathcal{A}_s^L \times \mathcal{G}_s$ and $\mathcal{A}_s$.*

Proof. It is clear that Φ is linear. To see that Φ is injective, take any (A, ψ) in the kernel of Φ . Then

$$\begin{aligned} i \frac{\partial \psi}{\partial t} &= A_0, \quad \text{and} \\ i \frac{\partial \psi}{\partial x_j} &= A_j, \quad \text{for } j = 1, 2, 3. \end{aligned}$$

Since $A \in \mathcal{A}_s^L$, this shows that

$$\square \psi := \frac{\partial^2 \psi}{\partial t^2} - \sum_{j=1}^3 \frac{\partial^2 \psi}{\partial x_j^2} = -i \frac{\partial A_0}{\partial t} + i \sum_{j=1}^3 \frac{\partial A_j}{\partial x_j} = 0.$$

Since $\psi \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$, an easy calculation shows that

$$\widehat{\square \psi}(p) = (p_0^2 - p_1^2 - p_2^2 - p_3^2) \widehat{\psi}(p)$$

for $p = (p_0, p_1, p_2, p_3) \in \mathbb{R}^4$. But since $\square \psi = 0$, its Fourier transform is also zero. Thus, the above equation shows that $\widehat{\psi} = 0$ almost everywhere on $\mathbb{R}^4$. We conclude that $\psi \equiv 0$. Since we started with the assumption that $A^\psi = 0$, this proves that A is also zero. Thus, Φ is injective.

Next, take any $A \in \mathcal{A}_s$. Define

$$\phi := -\frac{\partial A_0}{\partial t} + \sum_{j=1}^3 \frac{\partial A_j}{\partial x_j}.$$

First, suppose that the map

$$g(p) := \frac{i \widehat{\phi}(p)}{p_0^2 - p_1^2 - p_2^2 - p_3^2}$$

defined on the domain $\{p \in \mathbb{R}^4 : p_0^2 - p_1^2 - p_2^2 - p_3^2 \neq 0\}$ extends continuously to a Schwartz

function on $\mathbb{R}^4$. Then define

$$\psi(t, x) := \frac{1}{(2\pi)^4} \int_{\mathbb{R}^3} \int_{\mathbb{R}} g(p) e^{-i(p_0 t + q \cdot x)} dp_0 dq, \quad (9.1.5)$$

so that $\psi \in \mathcal{G}_s$ and $g = \widehat{\psi}$. Let $B := A^{-\psi}$. Then note that

$$-\frac{\partial B_0}{\partial t} + \sum_{j=1}^3 \frac{\partial B_j}{\partial x_j} = \phi - i\Box\psi$$

But, we have

$$\widehat{\Box\psi}(p) = (p_0^2 - p_1^2 - p_2^2 - p_3^2)\widehat{\psi}(p) = (p_0^2 - p_1^2 - p_2^2 - p_3^2)g(p) = i\widehat{\phi}(p). \quad (9.1.6)$$

This shows that $B \in \mathcal{A}_s^L$. Thus, any A satisfying the above condition is in the image of Φ .

If g does not extend continuously to a Schwartz function, we have to work harder. We still want to define ψ via equation (9.1.5), but since the integral is not absolutely convergent, we have to define it as a principal value:

$$\psi(t, x) := \lim_{\epsilon \rightarrow 0} \frac{1}{(2\pi)^4} \int_{\mathbb{R}^3} \int_{p_0: |p_0 - \|q\|| > \epsilon} g(p_0, q) e^{-i(p_0 t + q \cdot x)} dp_0 dq.$$

We will not bother to show that the above limit exists. Instead, we will just write down $\psi(t, x)$ as the explicit form of the limit and verify that it works (that is, satisfies equation (9.1.6)). To do this, we define

$$\begin{aligned} \psi(t, x) = & \int_{\mathbb{R}^3} \left(\int_0^{\|q\|} (g(\|q\| + r, q) e^{-i(\|q\| + r)t + q \cdot x} + g(\|q\| - r, q) e^{-i(\|q\| - r)t + q \cdot x}) dr \right. \\ & \int_0^{\|q\|} (g(-\|q\| + r, q) e^{-i(-\|q\| + r)t + q \cdot x} + g(-\|q\| - r, q) e^{-i(-\|q\| - r)t + q \cdot x}) dr \\ & \left. + \int_{2\|q\|}^{\infty} g(r, q) e^{-i(rt + q \cdot x)} dr + \int_{-\infty}^{-2\|q\|} g(r, q) e^{-i(rt + q \cdot x)} dr \right) dq. \end{aligned}$$

We first claim that the above integrals over r are absolutely convergent. The third and fourth integrals converge because g has rapid decay at infinity. For the first integral, note that

$$\begin{aligned} & g(\|q\| + r, q) e^{-i(\|q\| + r)t + q \cdot x} + g(\|q\| - r, q) e^{-i(\|q\| - r)t + q \cdot x} \\ &= \frac{i\widehat{\phi}(\|q\| + r, q) e^{-i(\|q\| + r)t + q \cdot x}}{2\|q\|r + r^2} - \frac{i\widehat{\phi}(\|q\| - r, q) e^{-i(\|q\| - r)t + q \cdot x}}{2\|q\|r - r^2} \end{aligned}$$

Now, there is a constant C , not depending on q or r , such that

$$\begin{aligned} |\widehat{\phi}(\|q\| + r, q) - \widehat{\phi}(\|q\| - r, q)| &\leq Cr, \\ |e^{-i(\|q\| + r)t + q \cdot x} - e^{-i(\|q\| - r)t + q \cdot x}| &\leq C|t|r, \end{aligned}$$

and since $0 \leq r \leq \|q\|$,

$$\left| \frac{1}{2\|q\|r + r^2} - \frac{1}{2\|q\|r - r^2} \right| \leq C\|q\|^{-1}r.$$

Moreover, $|\widehat{\phi}|$ is uniformly bounded, and

$$\frac{1}{|2\|q\|r \pm r^2|} \leq C\|q\|^{-1}r.$$

Using these bounds, it is now easy to deduce that

$$|g(\|q\| + r, q)e^{-i(\|q\|+r)t+q \cdot x} + g(\|q\| - r, q)e^{-i(\|q\|-r)t+q \cdot x}| \leq C(1 + |t| + \|q\|^{-1}).$$

This allows us to prove that ψ is well-defined and satisfies equation (9.1.6). $\square$

9.2 Rigorous construction of the free electromagnetic field

By Proposition 9.1.1, we write each Schwartzian connection A uniquely as B^ψ where B is a Schwartzian connection that satisfies the Lorenz gauge condition and ψ is a Schwartzian gauge transform. We claim that the vacuum expectation value of any gauge invariant observable does not change if we replace the action S by the modified action

$$\widetilde{S}(A) := S(B) + F(\psi)$$

for *any* function F . The heuristic justification goes as follows. Recall from equation (2.3.3) that given a function $L(A)$ of the field, the vacuum expectation value of $L(A)$ is heuristically given by

$$\langle \Omega | L(A) | \Omega \rangle = \frac{\int L(A) e^{iS(A)} \mathcal{D}A}{\int e^{iS(A)} \mathcal{D}A}.$$

Writing $A = B^\psi$ as above, and noting that the map $(B, \psi) \mapsto B^\psi$ is linear, we get

$$\langle \Omega | L(A) | \Omega \rangle = \frac{\iint L(B^\psi) e^{iS(B^\psi)} \mathcal{D}B \mathcal{D}\psi}{\iint e^{iS(B^\psi)} \mathcal{D}B \mathcal{D}\psi}.$$

Now, $S(B^\psi) = S(B)$, and if L is gauge invariant, then $L(B^\psi) = L(B)$. Thus,

$$\langle \Omega | L(A) | \Omega \rangle = \frac{\iint L(B) e^{iS(B)} \mathcal{D}B \mathcal{D}\psi}{\iint e^{iS(B)} \mathcal{D}B \mathcal{D}\psi} = \frac{\int L(B) e^{iS(B)} \mathcal{D}B \int \mathcal{D}\psi}{\int e^{iS(B)} \mathcal{D}B \int \mathcal{D}\psi} = \frac{\int L(B) e^{iS(B)} \mathcal{D}B}{\int e^{iS(B)} \mathcal{D}B}.$$

But on the other hand,

$$\begin{aligned}
 \frac{\int L(A) e^{\tilde{S}(A)} \mathcal{D}A}{\int e^{\tilde{S}(A)} \mathcal{D}A} &= \frac{\iint L(B^\psi) e^{iS(B) + iF(\psi)} \mathcal{D}B \mathcal{D}\psi}{\iint e^{iS(B) + iF(\psi)} \mathcal{D}B \mathcal{D}\psi} \\
 &= \frac{\iint L(B) e^{iS(B) + iF(\psi)} \mathcal{D}B \mathcal{D}\psi}{\iint e^{iS(B) + iF(\psi)} \mathcal{D}B \mathcal{D}\psi} \\
 &= \frac{\int L(B) e^{iS(B)} \mathcal{D}B \int e^{iF(\psi)} \mathcal{D}\psi}{\int e^{iS(B)} \mathcal{D}B \int e^{iF(\psi)} \mathcal{D}\psi} \\
 &= \frac{\int L(B) e^{iS(B)} \mathcal{D}B}{\int e^{iS(B)} \mathcal{D}B}.
 \end{aligned}$$

This completes the heuristic justification of the claim that replacing S by $\tilde{S}$ leaves expectation values of gauge invariant observables unchanged. Let us now make the specific choice

$$F(\psi) := \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}^3} (\Box \psi(t, x))^2 dx dt. \quad (9.2.1)$$

This is known as the *Feynman gauge fixing term*. Then note that since B satisfies the Lorenz gauge condition and $S(A) = S(B)$, we have

$$\begin{aligned}
 \tilde{S}(A) &= S(B) + \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}^3} (\Box \psi(t, x))^2 dx dt \\
 &= S(A) - \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}^3} \left(\frac{\partial B_0}{\partial t} - \sum_{j=1}^3 \frac{\partial B_j}{\partial x_j} + i \Box \psi(t, x) \right)^2 dx dt \\
 &= S(A) - \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}^3} \left(\frac{\partial A_0}{\partial t} - \sum_{j=1}^3 \frac{\partial A_j}{\partial x_j} \right)^2 dx dt.
 \end{aligned} \quad (9.2.2)$$

Integration by parts gives

$$\int_{\mathbb{R}^4} \frac{\partial A_k}{\partial x_j} \frac{\partial A_j}{\partial x_k} dx = \int_{\mathbb{R}^4} \frac{\partial A_k}{\partial x_k} \frac{\partial A_j}{\partial x_j} dx$$

for any $0 \leq j, k \leq 3$, where x_0 stands for t . Applying this in the expression (9.1.1) for $S(A)$, and letting $\eta_0 = 1$, $\eta_1 = \eta_2 = \eta_3 = -1$, we get

$$\begin{aligned}
 S(A) &= -\frac{1}{2} \int_{\mathbb{R}^4} \sum_{0 \leq j < k \leq 3} \eta_j \eta_k \left(\frac{\partial A_k}{\partial x_j} - \frac{\partial A_j}{\partial x_k} \right)^2 dx \\
 &= -\frac{1}{4} \int_{\mathbb{R}^4} \sum_{j, k=0}^3 \eta_j \eta_k \left(\frac{\partial A_k}{\partial x_j} - \frac{\partial A_j}{\partial x_k} \right)^2 dx \\
 &= -\frac{1}{2} \int_{\mathbb{R}^4} \left[\sum_{j, k=0}^3 \eta_j \eta_k \left(\frac{\partial A_k}{\partial x_j} \right)^2 - \left(\sum_{j=0}^3 \eta_j \frac{\partial A_j}{\partial x_j} \right)^2 \right] dx.
 \end{aligned} \quad (9.2.3)$$

Combining equations (9.2.2) and (9.2.3) and again applying integration by parts, we get

$$\begin{aligned}\tilde{S}(A) &= -\frac{1}{2} \int_{\mathbb{R}^4} \sum_{j,k=0}^3 \eta_j \eta_k \left(\frac{\partial A_k}{\partial x_j} \right)^2 dx \\ &= -\frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}^3} ((\partial_t A_0(t, x))^2 - \|\nabla A_0(t, x)\|^2) dx dt \\ &\quad + \frac{1}{2} \sum_{j=1}^3 \int_{\mathbb{R}} \int_{\mathbb{R}^3} ((\partial_t A_j(t, x))^2 - \|\nabla A_j(t, x)\|^2) dx dt,\end{aligned}$$

where ∇ denotes the gradient in the spatial coordinates. To construct a quantum field theory with this action, let us attempt to follow the Markov process heuristic from Chapter 2. Doing Wick rotation by replacing t by $-it$ everywhere and then multiplying the resulting action by i (as we did for the one particle systems of Chapter 1), we get the Euclidean action

$$\begin{aligned}\tilde{S}_E(A) &= -\frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}^3} ((\partial_t A_0(t, x))^2 + \|\nabla A_0(t, x)\|^2) dx dt \\ &\quad + \frac{1}{2} \sum_{j=1}^3 \int_{\mathbb{R}} \int_{\mathbb{R}^3} ((\partial_t A_j(t, x))^2 + \|\nabla A_j(t, x)\|^2) dx dt.\end{aligned}$$

Thus, an Euclidean free electromagnetic field $A^E = (A_0^E, A_1^E, A_2^E, A_3^E)$ should be a random field on $\mathbb{R}^4$ with probability density proportional to $e^{-\tilde{S}_E(A)}$. This is achieved by taking four independent massless Euclidean free fields $\phi_0, \phi_1, \phi_2, \phi_3$ on $\mathbb{R}^4$, and defining

$$A_0^E := i\phi_0, \quad A_1^E := \phi_1, \quad A_2^E := \phi_2, \quad A_3^E := \phi_3. \quad (9.2.4)$$

Note that the i is included in the definition of A_0^E to account for the fact that the term involving A_0 in $\tilde{S}_E(A)$ has the ‘wrong sign’. With this definition, it is easy to see from the properties of the Euclidean free field that $\{A^E(t, \cdot)\}_{t \in \mathbb{R}}$ is a stationary, reversible Markov process on the state space

$$\mathcal{U} := (i\mathbb{R})^{\mathcal{S}(\mathbb{R}^3, \mathbb{R})} \times \mathbb{R}^{\mathcal{S}(\mathbb{R}^3, \mathbb{R})} \times \mathbb{R}^{\mathcal{S}(\mathbb{R}^3, \mathbb{R})} \times \mathbb{R}^{\mathcal{S}(\mathbb{R}^3, \mathbb{R})},$$

where $\mathcal{U}$ is equipped with its cylinder σ -algebra. The law μ of $A^E(0, \cdot)$ is the product of the laws of $i\phi_0(0, \cdot)$, $\phi_1(0, \cdot)$, $\phi_2(0, \cdot)$ and $\phi_3(0, \cdot)$, and the coordinates evolve independently according to the transition probabilities of the Euclidean free field (and multiplied by i for the first coordinate). Thus, the Hilbert space for the (quantum) free electromagnetic field in the Feynman gauge should be $L^2(\mu)$, and its Hamiltonian H should be the generator of the semigroup $(P(t))_{t \geq 0}$ on $L^2(\mu)$ given by

$$P(t)F(X) := \mathbb{E}(F(A^E(t, \cdot)) | A^E(0, \cdot) = X).$$

It is easy to verify that this is a strongly continuous semigroup of symmetric operators, and that the function $\Omega \equiv 1$ in $L^2(\mu)$ satisfies $H\Omega = 0$. This allows us to complete the construction and define vacuum expectation values as before.

9.3 The photon propagator in the Feynman gauge

Let A be a free electromagnetic field in the Feynman gauge, as defined in the previous section. The photon propagator $\Pi = (\Pi_{jk})_{0 \leq j, k \leq 3}$ for this field is defined as

$$\Pi_{jk}(x - y) := i\langle \Omega | \mathcal{T} \{ A_j(x) A_k(y) \} | \Omega \rangle.$$

Since we constructed the free electromagnetic field in the Feynman gauge using four independent Euclidean free fields, and we already know the form of the Feynman propagator, it is relatively easy to compute the photon propagator rigorously.

Reverting back to our convention of denoting points in $\mathbb{R}^4$ by (t, x) where $t \in \mathbb{R}$ and $x \in \mathbb{R}^3$, the photon propagator is formally given by

$$\Pi_{jk}(t, x) = -\frac{i\eta_{jk}}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i|t||p| + ip \cdot x}}{2||p||} dp, \quad (9.3.1)$$

where η is the Minkowski metric, given by

$$\eta_{jk} := \begin{cases} 1 & \text{if } j = k = 0, \\ -1 & \text{if } j = k \in \{1, 2, 3\}, \\ 0 & \text{if } j \neq k. \end{cases}$$

Rigorously, this means that for each $t \in \mathbb{R}$, $\Pi_{jk}(t, \cdot, \cdot)$ is a bilinear form acting on $\mathcal{S}(\mathbb{R}^3, \mathbb{R}) \times \mathcal{S}(\mathbb{R}^3, \mathbb{R})$ as

$$\Pi_{jk}(t, f, g) := -\frac{i\eta_{jk}}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i|t||p|} \widehat{f}(p) \overline{\widehat{g}(p)}}{2||p||} dp.$$

Note that the expression on the right is mathematically well-defined, and is taken as the definition of $\Pi_{jk}(t, f, g)$. The following is a rigorous statement of the claim made above in equation (9.3.1).

Theorem 9.3.1. *Let A denote the free electromagnetic field in the Feynman gauge, as defined in the previous section. Let $A(t, \cdot)$ denote the value of the field at time t . Then for any $f, g \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$ and $-\infty < s \leq t < \infty$,*

$$\langle \Omega | A_j(s, g) A_k(t, f) | \Omega \rangle = -i\Pi_{jk}(t - s, f, g),$$

where Π_{jk} is the photon propagator defined above.

Proof. Take any $t > 0$, $f, g \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$, and $0 \leq j, k \leq 3$. Let A^E be the Euclidean free

electromagnetic field defined in equation (9.2.4) in the previous section. Let $\theta_0 := i$ and $\theta_j := 1$ for $j = 1, 2, 3$. Let $\delta_{jk} = 1$ if $j = k$ and 0 otherwise. Then note that by Lemma 3.8.3 and the independence of the fields $\phi_0, \phi_1, \phi_2, \phi_3$, we have

$$\begin{aligned} \mathbb{E}[g(A_j^E(0, \cdot))f(A_k^E(t, \cdot))] &= \theta_j \theta_k \mathbb{E}[g(\phi_j(0, \cdot))f(\phi_k(t, \cdot))] \\ &= \theta_j \theta_k \delta_{jk} \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-t\|p\|} \widehat{f}(p) \overline{\widehat{g}(p)}}{2\|p\|} dp. \end{aligned}$$

But $\theta_j \theta_k \delta_{jk} = -\eta_{jk}$. By Lemma 3.8.4, the right side can be analytically continued to the imaginary axis (as a function of t) simply by replacing t by it in the integral. This completes the proof. $\square$

9.4 Expected value of a gauge invariant observable

Let us now work out the two-point correlation function for the electric field in a free electromagnetic field. Since the electric field is a gauge-invariant object, the answer should not depend on our specific choice of the Feynman gauge. That is, if we chose the gauge fixing term $F(\psi)$ in equation (9.2.1) to be something different, then although the photon propagator would change, the object that we will calculate below would remain the same.

By equation (9.1.2), a formal computation gives, for any $1 \leq j, k \leq 3$, $-\infty < s \leq t < \infty$ and $x, y \in \mathbb{R}^3$,

$$\begin{aligned} &\langle \Omega | E_j(s, x) E_k(t, y) | \Omega \rangle \\ &= \left\langle \Omega \left| \left(i \frac{\partial A_j}{\partial s}(s, x) - i \frac{\partial A_0}{\partial x_j}(s, x) \right) \left(i \frac{\partial A_k}{\partial t}(t, y) - i \frac{\partial A_0}{\partial y_k}(t, y) \right) \right| \Omega \right\rangle \\ &= -\frac{\partial^2}{\partial s \partial t} \langle \Omega | A_j(s, x) A_k(t, y) | \Omega \rangle + \frac{\partial^2}{\partial s \partial y_k} \langle \Omega | A_j(s, x) A_0(t, y) | \Omega \rangle \\ &\quad + \frac{\partial^2}{\partial x_j \partial t} \langle \Omega | A_0(s, x) A_k(t, y) | \Omega \rangle - \frac{\partial^2}{\partial x_j \partial y_k} \langle \Omega | A_0(s, x) A_0(t, y) | \Omega \rangle. \end{aligned}$$

By Theorem 9.3.1, the second and third terms on the right are zero,

$$\begin{aligned} \frac{\partial^2}{\partial s \partial t} \langle \Omega | A_j(s, x) A_k(t, y) | \Omega \rangle &= -\frac{\partial^2}{\partial s \partial t} \frac{\eta_{jk}}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i(t-s)\|p\| + ip \cdot (x-y)}}{2\|p\|} dp \\ &= -\frac{\eta_{jk}}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i(t-s)\|p\| + ip \cdot (x-y)}}{2} \|p\| dp, \end{aligned}$$

and

$$\begin{aligned} \frac{\partial^2}{\partial x_j \partial y_k} \langle \Omega | A_0(s, x) A_0(t, y) | \Omega \rangle &= \frac{\partial^2}{\partial x_j \partial y_k} \frac{-1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-i(t-s)\|p\| + ip \cdot (x-y)}}{2\|p\|} dp \\ &= \frac{-1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{p_j p_k e^{-i(t-s)\|p\| + ip \cdot (x-y)}}{2\|p\|} dp. \end{aligned}$$

Combining, we get the formal expression

$$\langle \Omega | E_j(s, x) E_k(t, y) | \Omega \rangle = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{1}{2} \left(\frac{p_j p_k}{\|p\|} - \eta_{jk} \|p\| \right) e^{-i(t-s)\|p\| + i p \cdot (x-y)} dp.$$

One can make this formal computation precise by treating the photon propagator and the above two-point correlation function as distributions rather than functions.

9.5 The photon propagator in momentum coordinates

By Theorem 3.9.1, the photon propagator defined in equation (9.3.1) can be alternatively expressed as

$$\Pi_{jk}(x) = \lim_{\epsilon \downarrow 0} \frac{\eta_{jk}}{(2\pi)^4} \int_{\mathbb{R}^4} \frac{e^{-i(x,p)}}{p^2 + i\epsilon} dp$$

for $0 \leq j, k \leq 3$ and $x \in \mathbb{R}^4$, where (x, p) denotes the inner product in Minkowski signature. Now, suppose we view the free electromagnetic field as an actual field define on $\mathbb{R}^4$, and consider the Minkowski space Fourier transforms of its four components:

$$\tilde{A}_j(p) := \int_{\mathbb{R}^4} e^{i(x,p)} A_j(x) dx, \quad j = 0, 1, 2, 3.$$

Then, just as we did for the free quantum field in Section 3.10, we can define the photon propagator in momentum coordinates as

$$\tilde{\Pi}_{jk}(p, q) := \frac{i}{(2\pi)^4} \langle \tilde{A}_j(p) \tilde{A}_k(q) \rangle.$$

Then, from the calculation in Section 3.10, it is clear that

$$\tilde{\Pi}_{jk}(p, q) = \eta_{jk} \delta^{(4)}(p + q) \lim_{\epsilon \downarrow 0} \frac{1}{q^2 + i\epsilon}. \quad (9.5.1)$$

We will use this formula later, in our calculations for non-Abelian Yang–Mills theories.

Chapter 10

Grassmann Fields

10.1 Grassmann variables

Let Θ be any set. Let V be the vector space over $\mathbb{C}$ with basis Θ . That is, each element of V is either zero, or uniquely represented as

$$c_1\theta_1 + \cdots + c_n\theta_n$$

for some $n \geq 1$, $c_1, \dots, c_n \in \mathbb{C}$, and distinct $\theta_1, \dots, \theta_n \in \Theta$. Recall the exterior algebra $\Lambda(V)$ of V , defined in Section C.1.3 of Appendix C. We will henceforth refer to this algebra as the ‘Grassmann algebra’ generated by the set of ‘Grassman variables’ Θ . For ease of notation, we will omit the wedge product symbol and simply write fg instead of $f \wedge g$ for $f, g \in \Lambda(V)$.

In a nutshell, the Grassmann algebra can also be defined as follows. Let F be the free algebra generated by Θ , that is, all formal (finite) noncommutative products of elements of Θ , together with linear combinations of these products with complex coefficients. Let I be the two-sided ideal¹ of F generated by elements of the form $\theta_1\theta_2 + \theta_2\theta_1$. Then the Grassmann algebra generated by Θ is the quotient algebra R/I .

An element of the Grassman algebra is can be seen as a function of the Grassmann variables. So, for example, the element $1 + \theta_1 + \theta_2 + \theta_1\theta_2$ is seen as a function $f(\theta_1, \theta_2)$ of the two Grassmann variables $\theta_1, \theta_2 \in \Theta$, just as we would see it as a function if θ_1 and θ_2 were real variables. In general, any Grassmann function is a polynomial in finitely many of the Grassmann variables with complex coefficients, where no term has an element raised to a power greater than one (because raising to a higher power yields zero). To multiply two such functions, we simply have to follow the usual rule of multiplication (without commutativity) and then simplify using the rules that $\theta^2 = 0$ for any θ and $\theta_1\theta_2 = -\theta_1\theta_2$ for any θ_1, θ_2 . For example,

$$(\theta_1 + \theta_2)(\theta_1 + \theta_3) = \theta_1^2 + \theta_2\theta_1 + \theta_1\theta_3 + \theta_2\theta_3 = -\theta_1\theta_2 + \theta_1\theta_3 + \theta_2\theta_3.$$

¹That is, all elements of the form $a_1b_1c_1 + \cdots + a_nb_nc_n$ where $b_1, \dots, b_n$ are elements of F of the form $\theta_1\theta_2 + \theta_2\theta_1$, n is an arbitrary nonnegative integer, and $a_1, \dots, a_n, c_1, \dots, c_n$ are arbitrary elements of F .

We will refer to these functions as ‘Grassmann functions’. Once we start seeing these elements as functions, the advantage is that we can start talking about differentiating and integrating these functions. We do this in the subsequent sections. Throughout, we will work with the Grassmann algebra defined above.

10.2 Differentiating Grassmann functions

Differentiation of a function of Grassmann variables yields another function of Grassmann variables, as follows. For distinct $\theta_1, \dots, \theta_n \in \Theta$, we define

$$\frac{\partial}{\partial \theta_1} \theta_1 \cdots \theta_n = \theta_2 \cdots \theta_n.$$

As usual, when $n = 1$, the empty product on the right is taken to be 1. The derivative of a constant is defined to be zero, and we also define

$$\frac{\partial}{\partial \theta} \theta_1 \cdots \theta_n = 0$$

for any $\theta \notin \{\theta_1, \dots, \theta_n\}$. To be consistent with the above definition and be linear, we must have for any $1 \leq m \leq n$,

$$\begin{aligned} \frac{\partial}{\partial \theta_m} \theta_1 \cdots \theta_m &= \frac{\partial}{\partial \theta_m} ((-1)^{m-1} \theta_m \theta_1 \cdots \theta_{m-1} \theta_{m+1} \cdots \theta_n) \\ &= (-1)^m \theta_1 \cdots \theta_{m-1} \theta_{m+1} \cdots \theta_n. \end{aligned}$$

Once we have the above definition, it is now clear how to extend it by linearity to define $\frac{\partial f}{\partial \theta}$ for any Grassmann function f and any θ . For example,

$$\frac{\partial}{\partial \theta_2} (1 + \theta_1 + \theta_2 + \theta_1 \theta_2) = 1 - \theta_1.$$

We can also differentiate repeatedly in the obvious way. Note, however, that the order of differentiation matters, because the definition implies that

$$\frac{\partial}{\partial \theta} \frac{\partial}{\partial \theta'} = - \frac{\partial}{\partial \theta'} \frac{\partial}{\partial \theta} \tag{10.2.1}$$

for any $\theta, \theta' \in \Theta$ (not necessarily distinct). This proves, in particular, that repeated differentiation in any single variable yields zero.

10.3 Berezin integrals

Given a function of these Grassmann variables, if we integrate out one of the variables (as a definite integral), we should get a function of the remaining variables. In particular, the integral of a single variable should be constant. We *define*, for each $\theta \in \Theta$, the definite

integrals

$$\int \theta d\theta := 1, \quad \int d\theta = 0.$$

With these definitions, and treating the Grassmann variables as independent variables, note that for any distinct $\theta_1, \dots, \theta_n \in \Theta$, we must have

$$\int \theta_1 \cdots \theta_n d\theta_1 = \theta_2 \cdots \theta_n.$$

But then note that the right side is in fact equal to

$$\frac{\partial}{\partial \theta_1} \theta_1 \cdots \theta_n.$$

Extending by linearity, we see that for any function Grassman function f and any θ , we must have

$$\int f d\theta := \frac{\partial f}{\partial \theta}.$$

Note that this also holds for constant functions. These are known as ‘Berezin integrals’. The above definition allows us to integrate over several Grassmann variables, by differentiating repeatedly. As before, the order of integration matters.

An important observation that we will use later is that for distinct Grassmann variables $\theta_1, \dots, \theta_n$ and a permutation $\sigma \in S_n$,

$$\int \cdots \int \theta_{\sigma(1)} \cdots \theta_{\sigma(n)} d\theta_1 \cdots d\theta_n = \int \cdots \int \text{sign}(\sigma) \theta_1 \cdots \theta_n d\theta_1 \cdots d\theta_n = \text{sign}(\sigma).$$

10.4 Exponentials of Grassmann functions

Let f be a Grassmann function that has no constant term. Let n be the number of Grassmann variables that appear in f . Then $f^k = 0$ for any $k > n$, because each monomial in the expansion of f^k has at least one Grassmann variable appearing more than once. Thus, it makes sense to define the exponential of f as

$$e^f := \sum_{k=0}^{\infty} \frac{f^k}{k!}.$$

If f has a constant term c , then $f - c$ has no constant term, and we define

$$e^f := e^c e^{f-c}.$$

For example,

$$\begin{aligned}
 e^{\theta_1\theta_2+\theta_3\theta_4} &= 1 + (\theta_1\theta_2 + \theta_3\theta_4) + \frac{1}{2}(\theta_1\theta_2 + \theta_3\theta_4)^2 \\
 &= 1 + (\theta_1\theta_2 + \theta_3\theta_4) + \frac{1}{2}((\theta_1\theta_2)^2 + (\theta_3\theta_4)^2 + (\theta_1\theta_2\theta_3\theta_4 + \theta_3\theta_4\theta_1\theta_2)) \\
 &= 1 + \theta_1\theta_2 + \theta_3\theta_4 + \theta_1\theta_2\theta_3\theta_4.
 \end{aligned}$$

Now suppose that the Grassmann functions f and g are homogeneous polynomials of degrees k and l , respectively. In the vector space notation, this means $f \in \Lambda^k(V)$ and $g \in \Lambda^l(V)$. Then it is easy to see, and already noted in equation (C.1.5) in Appendix C that

$$fg = (-1)^{kl}gf.$$

In particular, if at least one of k and l is even, then f and g commute. This implies that if $f_1, \dots, f_n$ are homogeneous polynomials of even degrees, then

$$e^{f_1+\dots+f_n} = e^{f_1} \dots e^{f_n}.$$

(Note that being homogeneous of even degree automatically ensures that there is no constant term.)

10.5 Gaussian integration with Grassmann variables

The following formula is known as the Gaussian integral with Grassmann variables.

Theorem 10.5.1. *Take any $n \geq 1$ and let $\theta_1, \dots, \theta_n, \bar{\theta}_1, \dots, \bar{\theta}_n \in \Theta$ be distinct Grassmann variables. Let*

$$d\theta d\bar{\theta} := d\theta_1 d\bar{\theta}_1 \dots d\theta_n d\bar{\theta}_n.$$

Then for any $n \times n$ complex matrix $M = (m_{ij})_{1 \leq i,j \leq n}$,

$$\int \exp\left(-\sum_{i,j=1}^n m_{ij} \bar{\theta}_i \theta_j\right) d\theta d\bar{\theta} = \det M.$$

Proof. From the discussion in the previous section, we know that

$$\exp\left(-\sum_{i,j=1}^n m_{ij} \bar{\theta}_i \theta_j\right) = \prod_{i,j=1}^n \exp(-m_{ij} \bar{\theta}_i \theta_j),$$

where the terms on the right commute with each other, so the order in which the product is taken does not matter. But again, from the same discussion,

$$\begin{aligned}
 \exp(-m_{ij} \bar{\theta}_i \theta_j) &= 1 + (-m_{ij} \bar{\theta}_i \theta_j) + \frac{1}{2}(-m_{ij} \bar{\theta}_i \theta_j)^2 \\
 &= 1 - m_{ij} \bar{\theta}_i \theta_j.
 \end{aligned}$$

Thus, we get

$$\int \exp\left(-\sum_{i,j=1}^n m_{ij} \bar{\theta}_i \theta_j\right) d\theta d\bar{\theta} = \int \prod_{i,j=1}^n (1 - m_{ij} \bar{\theta}_i \theta_j) d\theta d\bar{\theta}.$$

Expanding the product, we see that the integral on the right is the sum of the integrals

$$\int (-1)^k (m_{i_1 j_1} \bar{\theta}_{i_1} \theta_{j_1}) \cdots (m_{i_k j_k} \bar{\theta}_{i_k} \theta_{j_k}) d\theta d\bar{\theta}$$

over all $0 \leq k \leq n^2$ and all distinct $(i_1, j_1), \dots, (i_k, j_k)$ that are lexicographically increasing in $\{1, \dots, n\}^2$. It is easy to see that the above integral is zero unless $k = n$, $i_1, \dots, i_n$ is a permutation of $1, \dots, n$, and $j_1, \dots, j_n$ is a permutation of $1, \dots, n$. But then, the condition that $(i_1, j_1), \dots, (i_n, j_n)$ are lexicographically increasing implies that $i_l = l$ for $l = 1, \dots, n$. Thus,

$$\int \exp\left(-\sum_{i,j=1}^n m_{ij} \bar{\theta}_i \theta_j\right) d\theta d\bar{\theta} = (-1)^n \sum_{\sigma \in S_n} m_{1\sigma(1)} \cdots m_{n\sigma(n)} \int \bar{\theta}_1 \theta_{\sigma(1)} \cdots \bar{\theta}_n \theta_{\sigma(n)} d\theta d\bar{\theta}.$$

Take any $\sigma \in S_n$. Since moving from $\bar{\theta}_1 \theta_{\sigma(1)} \cdots \bar{\theta}_n \theta_{\sigma(n)}$ to $\bar{\theta}_1 \theta_1 \cdots \bar{\theta}_n \theta_n$ requires the same number of transpositions as needed to move σ to the identity permutation, we see that

$$\begin{aligned} \int \bar{\theta}_1 \theta_{\sigma(1)} \cdots \bar{\theta}_n \theta_{\sigma(n)} d\theta d\bar{\theta} &= \int \text{sign}(\sigma) \bar{\theta}_1 \theta_1 \cdots \bar{\theta}_n \theta_n d\theta d\bar{\theta} \\ &= (-1)^n \text{sign}(\sigma) \int \theta_1 \bar{\theta}_1 \cdots \theta_n \bar{\theta}_n d\theta d\bar{\theta} \\ &= (-1)^n \text{sign}(\sigma). \end{aligned}$$

Combining the above observations, we get

$$\int \exp\left(-\sum_{i,j=1}^n m_{ij} \bar{\theta}_i \theta_j\right) d\bar{\theta} d\theta = \sum_{\sigma \in S_n} \text{sign}(\sigma) m_{1\sigma(1)} \cdots m_{n\sigma(n)} = \det M.$$

This completes the proof. $\square$

10.6 Grassmann-valued Gaussian random variables

Let n be a positive integer and $M = (m_{ij})_{1 \leq i,j \leq n}$ be an $n \times n$ invertible complex matrix. We define Grassmann-valued Gaussian random variables $(X, \bar{X}) = (X_1, \dots, X_n, \bar{X}_1, \dots, \bar{X}_n)$ with mean zero and inverse covariance matrix M as follows. For any function F that can be expressed as a polynomial with complex coefficients, the expected value of $F(X, \bar{X})$ is defined as

$$\mathbb{E}(F(X, \bar{X})) := \frac{\int F(\theta, \bar{\theta}) \exp(-\sum_{i,j=1}^n m_{ij} \bar{\theta}_i \theta_j) d\theta d\bar{\theta}}{\int \exp(-\sum_{i,j=1}^n m_{ij} \bar{\theta}_i \theta_j) d\theta d\bar{\theta}},$$

whenever the right side makes sense in the framework of Berezin integrals. The following theorem shows why it makes sense to call M the inverse covariance matrix.

Theorem 10.6.1. *Let $(X, \bar{X})$ be as above. Let $\Sigma = (\sigma_{ij})_{1 \leq i, j \leq n} := M^{-1}$. Then for any $1 \leq i, j \leq n$, $\mathbb{E}(X_i \bar{X}_j) = \sigma_{ij}$ and $\mathbb{E}(X_i X_j) = \mathbb{E}(\bar{X}_i \bar{X}_j) = 0$.*

Proof. Proceeding as in the proof of Theorem 10.5.1, we see that the integral in the numerator of $\mathbb{E}(X_i \bar{X}_j)$ can be expressed as the sum of the integrals

$$\int (-1)^k \theta_i \bar{\theta}_j (m_{i_1 j_1} \bar{\theta}_{i_1} \theta_{j_1}) \cdots (m_{i_k j_k} \bar{\theta}_{i_k} \theta_{j_k}) d\theta d\bar{\theta} \quad (10.6.1)$$

over all $0 \leq k \leq n^2$ and all distinct $(i_1, j_1), \dots, (i_k, j_k)$ that are lexicographically increasing in $\{1, \dots, n\}^2$. The above integral is zero unless $k = n - 1$, $i_1, \dots, i_k$ is a permutation of $\{1, \dots, n\} \setminus \{j\}$, and $j_1, \dots, j_k$ is a permutation of $\{1, \dots, n\} \setminus \{i\}$. But then, the condition that $(i_1, j_1), \dots, (i_k, j_k)$ are lexicographically increasing implies that

$$(i_1, \dots, i_k) = (1, 2, \dots, j-1, j+1, \dots, n).$$

Thus, the above sum can be written as

$$\begin{aligned} & (-1)^{n-1} \sum_{\sigma \in S'} m_{1\sigma(1)} \cdots m_{(j-1)\sigma(j-1)} m_{(j+1)\sigma(j+1)} \cdots m_{n\sigma(n)} \\ & \cdot \int \theta_i \bar{\theta}_j \bar{\theta}_1 \theta_{\sigma(1)} \cdots \bar{\theta}_{j-1} \theta_{\sigma(j-1)} \cdots \bar{\theta}_{j+1} \theta_{\sigma(j+1)} \bar{\theta}_n \theta_{\sigma(n)} d\theta d\bar{\theta}, \end{aligned}$$

where S' is the set of all bijections from $\{1, \dots, n\} \setminus \{j\}$ into $\{1, \dots, n\} \setminus \{i\}$. For $\sigma \in S'$, let $\text{sign}(\sigma)$ denote the sign of σ , viewing σ as a permutation of $\{1, \dots, n\} \setminus \{i\}$. Then, again proceeding as in the proof of Theorem 10.5.1, we see that the above sum can be written as

$$\begin{aligned} & \sum_{\sigma \in S'} \text{sign}(\sigma) m_{1\sigma(1)} \cdots m_{(j-1)\sigma(j-1)} m_{(j+1)\sigma(j+1)} \cdots m_{n\sigma(n)} \int (-1)^{i+j} \bar{\theta}_1 \theta_1 \cdots \bar{\theta}_n \theta_n d\theta d\bar{\theta} \\ & = (-1)^{i+j} (\det M_{ji}) \int \bar{\theta}_1 \theta_1 \cdots \bar{\theta}_n \theta_n d\theta d\bar{\theta}, \end{aligned}$$

where M_{ji} is the matrix obtained by omitting row j and column i from M . But recall that

$$\frac{(-1)^{i+j} \det M_{ij}}{\det M}$$

is the $(i, j)^{\text{th}}$ element of M^{-1} . This completes the proof of the first identity. The other two follow easily from the observation that the integral in equation (10.6.1) is zero if $\theta_i \bar{\theta}_j$ is replaced by either $\theta_i \theta_j$ or $\bar{\theta}_i \bar{\theta}_j$. $\square$

Theorem 10.6.1 generalizes to arbitrary functions of $(X, \bar{X})$, as follows. It suffices to compute the expected values of monomials. Looking at the expression in equation (10.6.1), it should be clear that the expected value is zero unless the monomial is of the form

$X_{a_1}\overline{X}_{b_1}\cdots X_{a_m}\overline{X}_{b_m}$ for some distinct $a_1, \dots, a_m$ and distinct $b_1, \dots, b_m$, or some permuted version of it. The following theorem gives the expected value of such a monomial.

Theorem 10.6.2. *Let $(X, \overline{X})$ and Σ be as in Theorem 10.6.1. Then for distinct $a_1, \dots, a_m$ and distinct $b_1, \dots, b_m$ in $\{1, \dots, n\}$, we have*

$$\mathbb{E}(X_{a_1}\overline{X}_{b_1}\cdots X_{a_m}\overline{X}_{b_m}) = \det K,$$

where K is the $m \times m$ matrix $(\sigma_{a_i b_j})_{1 \leq i, j \leq m}$.

Proof. Let π_1 be the permutation of $1, \dots, n$ that moves $a_1, \dots, a_m$ to $1, \dots, m$, respectively, and keeps the relative ordering of the remaining indices intact. Let π_2 be the permutation of $1, \dots, n$ that moves $b_1, \dots, b_m$ to $1, \dots, m$, respectively, and keeps the relative ordering of the remaining indices intact. Let N be the matrix obtained by removing rows $a_1, \dots, a_m$ and columns $b_1, \dots, b_m$ of M . Then proceeding exactly as in the proof of Theorem 10.6.1, one can show that

$$\mathbb{E}(X_{a_1}\overline{X}_{b_1}\cdots X_{a_m}\overline{X}_{b_m}) = \frac{\text{sign}(\pi_1) \text{sign}(\pi_2) \det N}{\det M}. \quad (10.6.2)$$

Next, let $\widetilde{M}$ be the matrix obtained by applying the permutation π_1 to the rows of M and the permutation π_2 to the columns of M . Note that $\widetilde{M}$ can be written as the block matrix

$$\widetilde{M} = \begin{pmatrix} L & A \\ B & N \end{pmatrix},$$

where N is the matrix from the previous step. Now, on the one hand,

$$\det \widetilde{M} = \text{sign}(\pi_1) \text{sign}(\pi_2) \det M, \quad (10.6.3)$$

and on the other hand, by the well-known formula for the determinant of a block matrix,

$$\det \widetilde{M} = \det N \det(L - AN^{-1}B). \quad (10.6.4)$$

Combining equations (10.6.2), (10.6.3) and (10.6.4), we get

$$\mathbb{E}(X_{a_1}\overline{X}_{b_1}\cdots X_{a_m}\overline{X}_{b_m}) = \frac{1}{\det(L - AN^{-1}B)} = \det((L - AN^{-1}B)^{-1}).$$

But by the formula for the inverse of a block matrix, the top-left $m \times m$ block of $\widetilde{M}^{-1}$ is $(L - AN^{-1}B)^{-1}$. This completes the proof. $\square$

10.7 Heuristic description of the Grassmann-valued free field

Heuristically, the Grassmann-valued free field on $\mathbb{R}^4$ with mass m is a pair $(\gamma, \bar{\gamma})$ of Grassmann-valued fields on $\mathbb{R}^4$ with joint ‘probability density’ proportional to

$$\exp\left(i \int_{\mathbb{R}^4} \sum_{j=0}^3 \eta_j \partial_j \bar{\gamma}(x) \partial_j \gamma(x) dx - im^2 \int_{\mathbb{R}^4} \bar{\gamma}(x) \gamma(x) dx\right),$$

where ∂_j denotes partial derivative in coordinate j , and $\eta_0 = -1$, $\eta_1 = \eta_2 = \eta_3 = 1$. This is the Grassmannian analogue of the free quantum field that we constructed in Chapter 3. Recall that by equation (2.3.3), the free quantum field φ can be heuristically described as a real-valued field on $\mathbb{R}^4$ with probability density proportional to

$$\exp\left(\frac{i}{2} \int_{\mathbb{R}^4} \sum_{j=0}^3 \eta_j (\partial_j \varphi(x))^2 dx - \frac{im^2}{2} \int_{\mathbb{R}^4} \varphi(x)^2 dx\right).$$

Now, we know from the discussion in Section 3.8 that the covariance structure of φ is given by

$$\langle \varphi(x) \varphi(y) \rangle = -i\Delta_F(x - y),$$

where Δ_F is the Feynman propagator. Thus, Theorem 10.6.1 suggests that it is reasonable to define

$$\langle \gamma(x) \bar{\gamma}(y) \rangle := -i\Delta_F(x - y)$$

and $\langle \gamma(x) \gamma(y) \rangle = \langle \bar{\gamma}(x) \bar{\gamma}(y) \rangle = 0$ for all $x, y \in \mathbb{R}^4$. Theorem 10.6.2 then shows how the expected value of any polynomial in $(\gamma, \bar{\gamma})$ can be computed. Note that this process is purely a computational prescription. We will see how it connects it to actual quantum systems later.

Let $\tilde{\gamma}$ and $\tilde{\bar{\gamma}}$ denote the Minkowski space Fourier transforms of the Grassmann fields γ and $\bar{\gamma}$. That is, heuristically,

$$\tilde{\gamma}(p) = \int_{\mathbb{R}^4} e^{i(x,p)} \gamma(x) dx, \quad \tilde{\bar{\gamma}}(p) = \int_{\mathbb{R}^4} e^{i(x,p)} \bar{\gamma}(x) dx.$$

Then repeating the heuristic calculation from Section 3.10, we see that

$$\langle \tilde{\gamma}(p) \tilde{\bar{\gamma}}(q) \rangle = -(2\pi)^4 i \tilde{\Delta}_F(p, q) = -(2\pi)^4 i \delta^{(4)}(p + q) \lim_{\epsilon \downarrow 0} \frac{-1}{q^2 - m^2 + i\epsilon}, \quad (10.7.1)$$

and $\langle \tilde{\gamma}(p) \tilde{\gamma}(q) \rangle = \langle \tilde{\bar{\gamma}}(p) \tilde{\bar{\gamma}}(q) \rangle = 0$. While this looks exactly the same as for the free quantum field, the higher correlations are different, because of the difference between Theorem 10.6.2

and Theorem B.2.7. For example, Theorem 10.6.2 suggests that

$$\begin{aligned}
 \langle \tilde{\gamma}(p) \tilde{\gamma}(q) \tilde{\gamma}(p') \tilde{\gamma}(q') \rangle &= -(2\pi)^8 \det \begin{pmatrix} \tilde{\Delta}_F(p, q) & \tilde{\Delta}_F(p, q') \\ \tilde{\Delta}_F(p', q) & \tilde{\Delta}_F(p', q') \end{pmatrix} \\
 &= -(2\pi)^8 (\tilde{\Delta}_F(p, q) \tilde{\Delta}_F(p', q') - \tilde{\Delta}_F(p', q) \tilde{\Delta}_F(p, q')) \\
 &= \lim_{\epsilon \downarrow 0} \frac{(2\pi)^8 (\delta^{(4)}(p+q) \delta^{(4)}(p'+q') - \delta^{(4)}(p'+q) \delta^{(4)}(p+q'))}{(p^2 - m^2 + i\epsilon)(p'^2 - m^2 + i\epsilon)}. \tag{10.7.2}
 \end{aligned}$$

10.8 Fermionic Fock space

10.9 Rigorous construction of the Grassmann-valued free field

Chapter 11

Faddeev–Popov Ghosts

11.1 The heuristic

The presence of gauge symmetry is one of the main obstacles in properly defining path integrals in gauge theory. The Faddeev–Popov procedure is a method of breaking gauge symmetry by introducing Grassmann-valued ‘ghost fields’. Heuristically, the idea goes as follows.

Suppose we want to compute the integral of a function $f : S \rightarrow \mathbb{C}$, where S is some abstract space endowed with some measure μ , possibly neither of which are rigorously defined. We are given that f is invariant under the action of some group G on S , and that μ is invariant under the action of any $g \in G$. That is, $f(x^g) = f(x)$ for any $x \in S$ and $g \in G$, where x^g denotes the action of g on x , and that $\mu(A^g) = \mu(A)$ for any $g \in G$ and $A \subseteq S$, where $A^g := \{x^g : x \in A\}$. Suppose we are able to choose a subset $T \subseteq S$ such that each orbit of the group action intersects T exactly once. For each $x \in S$, let $\pi(x)$ be the unique representative in T of the orbit of x . The measure μ can be pushed forward to T by π as $\nu(A) := \mu(\pi^{-1}(A))$ for $A \subseteq T$. Then it is intuitively clear, and easily provable under suitable assumptions, that

$$\int_S f(x) d\mu(x) = \int_T f(y) d\nu(y).$$

The relevance to Yang–Mills theory is clear, because the Yang–Mills action and physically relevant observables are invariant under the action of the group of gauge transformations. The challenge is to figure out the measure ν , given μ . This means the following.

Usually, the set T , being a subset of S , ‘inherits’ a nontrivial measure from S , even if $\mu(T) = 0$. For example, suppose that S is a metric space. Then for a subset $A \subseteq T$, one can take the set A_ϵ of all $x \in S$ that are within distance ϵ from some point in A , scale $\mu(A_\epsilon)$ by some suitable power of ϵ and send ϵ to zero to obtain a kind of μ -measure of A . For example, if $S = \mathbb{R}^2$, μ is Lebesgue measure on S , and $T = \{(x, 0) : x \geq 0\}$, then the nontrivial measure inherited in the above manner is simply the one-dimensional Lebesgue measure on T . In the abstract setup, let μ_0 denote the measure inherited by T . In the context of path integrals, this is usually the ‘Lebesgue measure’ used for path integrals on

T . The task of ‘figuring out’ ν is usually interpreted as the task of figuring out the density of ν with respect to μ_0 .

For a concrete example, consider S and T as in the previous paragraph. Let G be the group of rotations acting on $\mathbb{R}^2$ in the usual way. Then T intersects each orbit of G exactly once. If a function $f : \mathbb{R}^2 \rightarrow \mathbb{C}$ is invariant under the action of G , it just means that f is radially symmetric. For such a function, transforming the integral over $\mathbb{R}^2$ to polar coordinates, we get

$$\int_{\mathbb{R}^2} f(x, y) dx dy = 2\pi \int_0^\infty f(r, 0) dr.$$

Recall our previous observation that the measure μ_0 on T is just the one-dimensional Lebesgue measure. Thus, the above formula shows that

$$d\nu(r) = 2\pi r d\mu_0(r).$$

Note that it is intuitively clear where the factor $2\pi r$ comes from: It is the size of the orbit containing $(r, 0)$.

The Faddeev–Popov procedure gives a systematic prescription for calculating the density of ν with respect to μ_0 as the determinant of a certain operator. The fundamental fact that underlies this identification is the following. Let $u : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a smooth bijection and let $J_u(z)$ denote the Jacobian matrix of u at $z \in \mathbb{R}^n$. Then

$$\int_{\mathbb{R}^n} \delta^{(n)}(u(z)) |\det J_u(z)| dz = 1, \quad (11.1.1)$$

where $\delta^{(n)}$ denotes the Dirac delta at the origin in $\mathbb{R}^n$. To see this, simply note that by the change of variable formula and the fact that f is a bijection,

$$\int_{\mathbb{R}^n} h(u(z)) |\det J_u(z)| dz = \int_{\mathbb{R}^n} h(z) dz$$

for any smooth h . Now take a sequence of h ’s approaching $\delta^{(n)}$ in a suitable sense to get equation (11.1.1).

Next, assume that the group G is a Lie group¹ and let $\mathfrak{g}$ be its Lie algebra. Suppose that we can describe the set T as the set of zeros of a ‘gauge fixing function’ $\Psi : \mathfrak{g} \times S \rightarrow \mathbb{R}$, as follows. For each $x \in S$ and $X \in \mathfrak{g}$, $\Psi(X, x) = 0$ if and only if $x^{e^X} \in T$, where e^X denotes the exponential of X . Since $\mathfrak{g}$ can be identified with $\mathbb{R}^n$ for some n , equation (11.1.1) lets us conclude that for any $x \in S$,

$$\int_{\mathfrak{g}} \delta^{(n)}(\Psi(X, x)) |\det J_{\Psi(\cdot, x)}(X)| dX = 1.$$

¹See Appendix C, and specifically Sections C.3 and C.7, for background on Lie theory.

Thus, we can proceed heuristically to get

$$\begin{aligned}
 \int_S f(x) d\mu(x) &= \int_S f(x) \left(\int_{\mathfrak{g}} \delta^{(n)}(\Psi(X, x)) |\det J_{\Psi(\cdot, x)}(X)| dX \right) d\mu(x) \\
 &= \int_{S \times \mathfrak{g}} f(x) \delta^{(n)}(\Psi(X, x)) |\det J_{\Psi(\cdot, x)}(X)| dX d\mu(x) \\
 &= \int_T f(y) |\det J_{\Psi(\cdot, y)}(0)| d\mu_0(y),
 \end{aligned}$$

where the last identity holds because the delta function restricts the domain of integration of x to the subset T . The last step of the Faddeev–Popov method is to replace the determinant in the above integral by an integral over a field of Grassmann variables, via Theorem 10.5.1 from the previous chapter. These Grassmann variables are known as ‘Faddeev–Popov ghosts’.

11.2 A rigorous formulation in finite dimensions

We will now rigorously formulate and prove a finite-dimensional version of the Faddeev–Popov method. One of the main difficulties of making the heuristic rigorous is that it is generally impossible to choose exactly one representative from each gauge equivalence class in a smooth way. This is known as the problem of Gribov ambiguity [12]. We will bypass this problem in the finite-dimensional setting by assuming below that a finite constant number of representatives can be chosen from equivalence class.

Let n be a positive integer, and let G be a matrix Lie group contained in $U(N)$ for some N , such that $m := \dim(G) < n$. Consider an action of G on $\mathbb{R}^n$, denoted by $x \mapsto x^g$, for $x \in \mathbb{R}^n$ and $g \in G$. Suppose that

1. for each $g \in G$, the action $x \mapsto x^g$ preserves Lebesgue measure, and
2. the map $(g, x) \mapsto x^g$ is smooth.

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function that is invariant under these gauge transforms, meaning that $F(x^g) = F(x)$ for all $x \in \mathbb{R}^n$ and $g \in G$. For $x \in \mathbb{R}^n$, let O_x denote the orbit of x under the action of G . Let $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a surjective linear map such that

1. for each $x \in \mathbb{R}^n$, O_x intersects $\ker \Phi$, and
2. there exist distinct elements $g_1, \dots, g_k \in G$ such that for almost all $y \in \ker \Phi$, $\{g \in G : y^g \in \ker \Phi\} = \{g_1, \dots, g_k\}$.

Let $\mathfrak{g}$ denote the Lie algebra of G . Define a function $\Psi : \mathfrak{g} \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ as

$$\Psi(X, x) := \Phi(x^{e^X}). \quad (11.2.1)$$

Fix an orthonormal basis $T_1, \dots, T_m$ of the Lie algebra $\mathfrak{g}$ of G (under the inner product $\langle X, Y \rangle := \text{Tr}(XY^*)$), so that any $X \in \mathfrak{g}$ can be written uniquely as $X = \sum_{a=1}^m X_a T_a$ for

some coefficients $X_1, \dots, X_m \in \mathbb{R}$. Let $\Psi_1, \dots, \Psi_m$ denote the components of Ψ , and define the Jacobian matrix

$$\partial_X \Psi := (\partial_{X_a} \Psi_b)_{1 \leq a, b \leq m},$$

where $\partial_{X_a} \Psi_b$ denotes the partial derivative of Ψ_b with respect to X_a . We make the following two assumptions:

1. $\partial_X \Psi(0, y)$ is nonsingular for almost all $y \in \ker \Phi$.
2. For almost all $y \in \ker \Phi$, $|\det \partial_X \Psi(0, w)|$ is the same for all $w \in O_y \cap \ker \Phi$.

The following theorem shows that the size of the gauge orbit containing $z \in \mathcal{G}^+$ is proportional to the determinant of the matrix $\partial_X \Psi(0, z)$.

Theorem 11.2.1. *In the above setting, for any gauge invariant $F \in C(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$, we have*

$$\int_{\mathbb{R}^n} F(x) dx = \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} F(y) |\det(\partial_X \Psi(0, y))| dy,$$

where the integral on the right is absolutely convergent, μ is the unnormalized Haar measure² on G , and $\Delta := |\det(\Phi|_{(\ker \Phi)^\perp})|$. Consequently, if $\gamma_1, \dots, \gamma_m, \bar{\gamma}_1, \dots, \bar{\gamma}_m$ are distinct Grassmann variables, then

$$\int_{\mathbb{R}^n} F(x) dx = \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} \int F(y) \exp\left(i\pi\sigma(y) - \sum_{a,j=1}^m \partial_{X_a} \Psi_j(0, y) \bar{\gamma}_j \gamma_a\right) d\gamma d\bar{\gamma} dy,$$

where $d\gamma d\bar{\gamma} := d\gamma_1 d\bar{\gamma}_1 \cdots d\gamma_m d\bar{\gamma}_m$, and $\sigma(y) = 1$ if $\det(\partial_X \Psi(0, y)) < 0$ and 0 otherwise.

The second assertion of the theorem follows from the first, by Theorem 10.5.1. So, we only need to prove the first claim. We need some preparation for the proof. Take any $x \in \mathbb{R}^n$. For each $\epsilon > 0$, let

$$Z_\epsilon(x) := \frac{1}{(2\pi\epsilon)^{m/2}} \int_G \exp\left(-\frac{\|\Phi(x^g)\|^2}{2\epsilon}\right) d\mu(g).$$

We need several lemmas about Z_ϵ .

Lemma 11.2.2. *For any $\epsilon > 0$, Z_ϵ is a gauge invariant function.*

Proof. Note that by the invariance property of the Haar measure,

$$\begin{aligned} Z_\epsilon(x^h) &:= \frac{1}{(2\pi\epsilon)^{m/2}} \int_G \exp\left(-\frac{\|\Phi((x^h)^g)\|^2}{2\epsilon}\right) d\mu(g) \\ &= \frac{1}{(2\pi\epsilon)^{m/2}} \int_G \exp\left(-\frac{\|\Phi(x^{gh})\|^2}{2\epsilon}\right) d\mu(g) \\ &= \frac{1}{(2\pi\epsilon)^{m/2}} \int_G \exp\left(-\frac{\|\Phi(x^g)\|^2}{2\epsilon}\right) d\mu(g) = Z_\epsilon(x) \end{aligned}$$

for any $x \in \mathbb{R}^n$ and $h \in G$. □

²See Section C.7 in Appendix C for Haar measures on matrix Lie groups.

Lemma 11.2.3. *Let $M := \ker \Phi$ and let $M^\perp$ denote the orthogonal complement of M in $\mathbb{R}^n$. Then for any $y \in M$ such that $\{g \in G : y^g \in \ker \Phi\} = \{g_1, \dots, g_k\}$ and $|\det \partial_X(0, y^{g_j})|$ is nonzero and the same for each j , and any $w \in M^\perp$, we have*

$$\lim_{\epsilon \rightarrow 0} Z_\epsilon(y + \sqrt{\epsilon}w) = \frac{k}{|\det \partial_X \Psi(0, y)|}.$$

Proof. Let $y_j := y^{g_j}$. Choose $\delta > 0$ so small that the balls

$$B_j := \{g \in G : \|g - g_j\| < \delta\}, \quad j = 1, \dots, k$$

are disjoint. Then, with $D := G \setminus (B_1 \cup \dots \cup B_k)$, we have

$$\liminf_{\epsilon \rightarrow 0} \inf_{g \in D} \|\Phi((y + \sqrt{\epsilon}w)^g)\| > 0,$$

because otherwise we can produce $g \in D$ such that $\Phi(y^g) = 0$, which is impossible. This lets us conclude that

$$\lim_{\epsilon \rightarrow 0} \left(Z_\epsilon(y + \sqrt{\epsilon}w) - \sum_{j=1}^k Z_\epsilon^j \right) = 0, \quad (11.2.2)$$

where

$$Z_\epsilon^j := \frac{1}{(2\pi\epsilon)^{m/2}} \int_{B_j} \exp\left(-\frac{\|\Phi((y + \sqrt{\epsilon}w)^g)\|^2}{2\epsilon}\right) d\mu(g).$$

Without loss, let us assume that g_1 is the identity element of G . Since $G \subseteq \mathrm{U}(N)$, it follows that $B_j = g_j B_1$. Thus,

$$Z_\epsilon^j := \frac{1}{(2\pi\epsilon)^{m/2}} \int_{B_1} \exp\left(-\frac{\|\Phi((y + \sqrt{\epsilon}w)^{g_j g})\|^2}{2\epsilon}\right) d\mu(g).$$

Now recall from Theorem C.7.4 in Appendix C that if δ is small enough, then the term on the right is equal to

$$\frac{1}{(2\pi\epsilon)^{m/2}} \int_U \exp\left(-\frac{\|\Phi((y + \sqrt{\epsilon}w)^{g_j e^X})\|^2}{2\epsilon}\right) J(X) dX$$

for some bounded open neighborhood U of the origin in $\mathfrak{g}$, and some smooth function $J : \mathfrak{g} \rightarrow [0, \infty)$ satisfying $J(0) = 1$, where dX denotes integration with respect to Lebesgue measure on $\mathfrak{g}$. Moreover, the diameter of U can be made as small as we like by decreasing δ . Applying the change of variable $Y = \epsilon^{-1/2}X$, we get

$$Z_\epsilon^j = \frac{1}{(2\pi)^{m/2}} \int_{\epsilon^{-1/2}U} \exp\left(-\frac{\|\Phi((y + \sqrt{\epsilon}w)^{g_j e^{\sqrt{\epsilon}Y}})\|^2}{2\epsilon}\right) J(\sqrt{\epsilon}Y) dY. \quad (11.2.3)$$

Define the function $\Psi^j : \mathfrak{g} \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ as

$$\Psi^j(X, x) := \Phi(x^{g_j} e^X), \quad (11.2.4)$$

and let $\partial_X \Psi^j$ and $\partial_x \Psi^j$ denote the partial derivatives of Ψ^j in X and x , so that $\partial_X \Psi^j$ is an $m \times m$ matrix-valued function, and $\partial_x \Psi^j$ is an $m \times n$ matrix-valued function. Note that the matrix

$$\partial_X \Psi^j(0, y) = \partial_X \Psi(0, y^{g_j}) = \partial_X \Psi(0, y_j)$$

is nonsingular, by the assumptions that $\partial_X \Psi(0, y)$ is nonsingular and $|\det \partial_X \Psi(0, y_j)|$ is the same for each j .

Now, take any $X = \sum_{a=1}^m X_a T_a \in U$. By first-order Taylor expansion of Ψ^j , we get

$$\Psi^j(X, y + \sqrt{\epsilon}w) = \Psi^j(0, y) + \sqrt{\epsilon} \partial_x \Psi^j(0, y)w + \partial_X \Psi^j(0, y)X + R, \quad (11.2.5)$$

where the X on the right is the vector $(X_a)_{1 \leq a \leq m}$, and the remainder term $R \in \mathbb{R}^m$ satisfies

$$\|R\| \leq C(\epsilon + \|X\|^2) \quad (11.2.6)$$

for some constant C that does not depend on ϵ or X , as long as $\epsilon \in (0, 1)$ and $X \in U$. Since $\Psi^j(0, y) = 0$ and $\partial_X \Psi^j(0, y)$ is nonsingular, this implies (by the inequality $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$) that if δ is small enough, then there are positive constants C_1, C_2, C_3, C_4 such that for any $X \in U$,

$$\begin{aligned} \|\Psi^j(X, y + \sqrt{\epsilon}w)\|^2 &\geq \frac{1}{3} \|\partial_X \Psi^j(0, y)X\|^2 - \|\sqrt{\epsilon} \partial_x \Psi^j(0, y)w\|^2 - \|R\|^2 \\ &\geq C_1 \|X\|^2 - C_2 \epsilon - C_3 \|X\|^4 \geq C_4 \|X\|^2 - C_2 \epsilon. \end{aligned} \quad (11.2.7)$$

Now, equation (11.2.3) can be rewritten as

$$Z_\epsilon^j = \frac{1}{(2\pi)^{m/2}} \int_{\epsilon^{-1/2}U} \exp\left(-\frac{\|\Psi^j(\sqrt{\epsilon}Y, y + \sqrt{\epsilon}w)\|^2}{2\epsilon}\right) J(\sqrt{\epsilon}Y) dY$$

The Taylor expansion (11.2.5), the bound (11.2.6), and the fact that $J(0) = 1$ show that for any fixed $Y \in \mathfrak{g}$,

$$\begin{aligned} &\lim_{\epsilon \rightarrow 0} \exp\left(-\frac{\|\Psi^j(\sqrt{\epsilon}Y, y + \sqrt{\epsilon}w)\|^2}{2\epsilon}\right) J(\sqrt{\epsilon}Y) \\ &= \exp\left(-\frac{1}{2} \|\partial_x \Psi^j(0, y)w + \partial_X \Psi^j(0, y)Y\|^2\right). \end{aligned}$$

Additionally, by the inequality (11.2.7) and the fact that J is a bounded function (because

it is smooth), we see that for any $Y \in \epsilon^{-1/2}U$,

$$\left| \exp\left(-\frac{\|\Psi^j(\sqrt{\epsilon}Y, y + \sqrt{\epsilon}w)\|^2}{2\epsilon}\right) J(\sqrt{\epsilon}Y) \right| \leq C_5 e^{-C_6 \|Y\|^2}$$

for some positive constants C_5 and C_6 that do not depend on ϵ or Y . These two observations allow us to apply the dominated convergence theorem to conclude that

$$\lim_{\epsilon \rightarrow 0} Z_\epsilon^j = \frac{1}{(2\pi)^{m/2}} \int_{\mathfrak{g}} \exp\left(-\frac{1}{2} \|\partial_x \Psi^j(0, y)w + \partial_X \Psi^j(0, y)Y\|^2\right) dY.$$

A standard Gaussian computation shows that the integral on the right is equal to

$$\begin{aligned} (2\pi)^{m/2} [\det(\partial_X \Psi^j(0, y)^T \partial_X \Psi^j(0, y))]^{-1/2} &= (2\pi)^{m/2} |\det \partial_X \Psi^j(0, y)|^{-1} \\ &= (2\pi)^{m/2} |\det \partial_X \Psi(0, y_j)|^{-1} \\ &= (2\pi)^{m/2} |\det \partial_X \Psi(0, y)|^{-1}. \end{aligned}$$

where the second equality follows from the observation (11.2.4), and the third equality holds due to our assumption that $|\det \partial_X \Psi(0, y_j)|$ is the same for all j . Recalling equation (11.2.2), we conclude that

$$\lim_{\epsilon \rightarrow 0} Z_\epsilon(y + \sqrt{\epsilon}w) = k |\det \partial_X \Psi(0, y)|^{-1}.$$

This completes the proof of the lemma. $\square$

Lemma 11.2.4. *There is some $\epsilon_0 \in (0, 1)$ such that for any $K > 0$,*

$$\inf\{Z_\epsilon(x) : \|x\| \leq K, \epsilon \in (0, \epsilon_0)\} > 0.$$

Proof. By the smoothness of $(x, g) \mapsto x^g$ and the compactness of G , the number

$$L := \sup\{\|x^g\| : \|x\| \leq K, g \in G\}$$

is finite. Take any x with $\|x\| \leq K$ and any $\epsilon \in (0, 1)$. Since O_x intersects $\ker \Phi$, there is some $y \in \ker \Phi$ that is in the orbit of x , and by the above observation, $\|y\| \leq L$. By Lemma 11.2.2, $Z_\epsilon(x) = Z_\epsilon(y)$.

Choose an open neighborhood V of the identity in G , with diameter so small that by Theorem C.7.4, there is a bounded open neighborhood U of the origin in $\mathfrak{g}$ such that for any $\epsilon \in (0, 1)$,

$$\begin{aligned} Z_\epsilon(y) &\geq \frac{1}{(2\pi\epsilon)^{m/2}} \int_V \exp\left(-\frac{\|\Phi(y^g)\|^2}{2\epsilon}\right) d\mu(g) \\ &= \frac{1}{(2\pi\epsilon)^{m/2}} \int_U \exp\left(-\frac{\|\Phi(y^{e^X})\|^2}{2\epsilon}\right) J(X) dX. \end{aligned}$$

Since $\Phi(y) = 0$ and $\|y\| \leq L$, Taylor expansion shows that for any $X \in U$,

$$\|\Phi(y^{e^X})\| \leq C\|X\|^2$$

for some constant C that does not depend on ϵ or x . Plugging this into the previous display and applying the change of variable $Y = \epsilon^{-1/2}X$, we get

$$Z_\epsilon(y) \geq \frac{1}{(2\pi)^{m/2}} \int_{\epsilon^{-1/2}U} e^{-\frac{1}{2}C\|Y\|^2} J(\sqrt{\epsilon}Y) dY.$$

If ϵ is sufficiently small, $\epsilon^{-1/2}U$ contains the unit ball of $\mathbb{R}^n$. Thus, a lower bound on the right side above can be obtained by replacing the domain of integration by the unit ball. Recalling that $J(0) = 1$, this completes the proof. $\square$

We are now ready to complete the proof of Theorem 11.2.1.

Proof of Theorem 11.2.1. Let us first prove the theorem under the additional assumption that F vanishes outside a bounded region. Throughout, we will assume that $\epsilon \in (0, \epsilon_0)$, where ϵ_0 is the number from Lemma 11.2.4. Note that by Fubini's theorem,

$$\begin{aligned} \int_{\mathbb{R}^n} F(x) dx &= \frac{1}{(2\pi\epsilon)^{m/2}} \int_{\mathbb{R}^n} F(x) \left(\frac{1}{Z_\epsilon(x)} \int_G \exp\left(-\frac{\|\Phi(x^g)\|^2}{2\epsilon}\right) d\mu(g) \right) dx \\ &= \frac{1}{(2\pi\epsilon)^{m/2}} \int_G \int_{\mathbb{R}^n} \frac{F(x)}{Z_\epsilon(x)} \exp\left(-\frac{\|\Phi(x^g)\|^2}{2\epsilon}\right) dx d\mu(g). \end{aligned}$$

Since F and Z_ϵ are both gauge invariant (by Lemma 11.2.2) and the gauge transforms preserve Lebesgue measure, this gives

$$\begin{aligned} \int_{\mathbb{R}^n} F(x) dx &= \frac{1}{(2\pi\epsilon)^{m/2}} \int_G \int_{\mathbb{R}^n} \frac{F(x^g)}{Z_\epsilon(x^g)} \exp\left(-\frac{\|\Phi(x^g)\|^2}{2\epsilon}\right) dx d\mu(g) \\ &= \frac{1}{(2\pi\epsilon)^{m/2}} \int_G \int_{\mathbb{R}^n} \frac{F(x)}{Z_\epsilon(x)} \exp\left(-\frac{\|\Phi(x)\|^2}{2\epsilon}\right) dx d\mu(g) \\ &= \frac{\mu(G)}{(2\pi\epsilon)^{m/2}} \int_{\mathbb{R}^n} \frac{F(x)}{Z_\epsilon(x)} \exp\left(-\frac{\|\Phi(x)\|^2}{2\epsilon}\right) dx. \end{aligned}$$

Let $M := \ker \Phi$. Decomposing Lebesgue integration on $\mathbb{R}^n$ as Lebesgue integration on the product space $M \times M^\perp$, this shows that

$$\begin{aligned} \int_{\mathbb{R}^n} F(x) dx &= \frac{\mu(G)}{(2\pi\epsilon)^{m/2}} \int_{M^\perp} \int_M \frac{F(y+z)}{Z_\epsilon(y+z)} \exp\left(-\frac{\|\Phi(z)\|^2}{2\epsilon}\right) dy dz \\ &= \frac{\mu(G)}{(2\pi)^{m/2}} \int_{M^\perp} \int_M \frac{F(y+\sqrt{\epsilon}w)}{Z_\epsilon(y+\sqrt{\epsilon}w)} e^{-\frac{1}{2}\|\Phi(w)\|^2} dy dw. \end{aligned}$$

Take any $y \in M$ that satisfies the assumptions of Lemma 11.2.3 and any $w \in M^\perp$. Then

by Lemma 11.2.3 and the continuity of F ,

$$\lim_{\epsilon \rightarrow 0} \frac{F(y + \sqrt{\epsilon}w)}{Z_\epsilon(y + \sqrt{\epsilon}w)} e^{-\frac{1}{2}\|\Phi(w)\|^2} = F(y) \frac{|\det \partial_X \Psi(0, y)|}{k} e^{-\frac{1}{2}\|\Phi(w)\|^2}. \quad (11.2.8)$$

Let L be so large that $F(x) = 0$ when $\|x\| > L$. If $\|y + \sqrt{\epsilon}w\| > L$, then $F(y + \sqrt{\epsilon}w) = 0$. On the other hand, if $\|y + \sqrt{\epsilon}w\| \leq L$, then by Lemma 11.2.4, $Z_\epsilon(y + \sqrt{\epsilon}w) \geq C_1$ for some positive constant C_1 that does not depend on ϵ , y or w . Moreover, by the orthogonality of y and w , $\|y\| \leq L$. Combining these observations, we see that

$$\left| \frac{F(y + \sqrt{\epsilon}w)}{Z_\epsilon(y + \sqrt{\epsilon}w)} e^{-\frac{1}{2}\|\Phi(w)\|^2} \right| \leq C_2 \xi(y) e^{-\frac{1}{2}\|\Phi(w)\|^2},$$

where C_2 is a positive constant that does not depend on ϵ , y and w , and $\xi(y) = 1$ if $\|y\| \leq L$ and 0 otherwise. Since Φ is a linear map and is injective on $M^\perp$, there is a positive constant C_3 such that $\|\Phi(w)\| \geq C_3\|w\|$ for all $w \in M^\perp$. Thus,

$$\int_{M^\perp} \int_M \xi(y) e^{-\frac{1}{2}\|\Phi(w)\|^2} dy dw < \infty.$$

Together with equation (11.2.8), this allows us to apply the dominated convergence theorem and complete the proof for compactly supported F .

Next, take any $F \in C(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$. Let $h : \mathbb{R}^n \rightarrow [0, 1]$ be a continuous function that is 1 inside the unit ball and 0 outside the ball of radius 2 centered at the origin. For each $l \geq 1$, let

$$F_l(x) := F(x)h(l^{-1}x).$$

Note that $|F_l(x)| \leq |F(x)|$ for all l and x , and $F_l \rightarrow F$ pointwise as $l \rightarrow \infty$. Now, by what we have already shown,

$$\int_{\mathbb{R}^n} F_l(x) dx = \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} F_l(y) |\det(\partial_X \Psi(0, y))| dy \quad (11.2.9)$$

for each l . By the dominated convergence theorem and the assumption that $F \in L^1(\mathbb{R}^n)$, we get

$$\lim_{l \rightarrow \infty} \int_{\mathbb{R}^n} F_l(x) dx = \int_{\mathbb{R}^n} F(x) dx. \quad (11.2.10)$$

Next, again by what we have already shown,

$$\frac{\mu(G)}{k\Delta} \int_{\ker \Phi} |F_l(y) \det(\partial_X \Psi(0, y))| dy = \int_{\mathbb{R}^n} |F_l(x)| dx \leq \int_{\mathbb{R}^n} |F(x)| dx.$$

Therefore, by Fatou's lemma,

$$\begin{aligned} \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} |F(z) \det(\partial_X \Psi(0, y))| dy &= \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} \lim_{l \rightarrow \infty} |F_l(y) \det(\partial_X \Psi(0, y))| dy \\ &\leq \liminf_{l \rightarrow \infty} \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} |F_l(y) \det(\partial_X \Psi(0, y))| dy \\ &\leq \int_{\mathbb{R}^n} |F(x)| dx < \infty. \end{aligned}$$

This allows us to apply the dominated convergence theorem to conclude that

$$\lim_{l \rightarrow \infty} \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} |F_l(y) \det(\partial_X \Psi(0, y))| dy = \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} |F(z) \det(\partial_X \Psi(0, y))| dy. \quad (11.2.11)$$

Combining equations (11.2.9), (11.2.10) and (11.2.11) completes the proof. $\square$

11.3 A toy example

Identifying $\mathbb{R}^2$ with $\mathbb{C}$, let us consider the action of $U(1)$ by multiplication as our set of gauge transformations. Note that this group actions satisfies the required conditions; that is, it is smooth, and it preserves Lebesgue measure.

Since $\dim(U(1)) = 1$, we need a gauge fixing map $\Phi : \mathbb{C} \rightarrow \mathbb{R}$. Consider the map $\Phi(x + iy) = y$. Let us verify that this map satisfies the required conditions. First, note that Φ is linear, as required. The kernel of Φ is the real line. Thus, O_z intersects $\ker \Phi$ for any $z \in \mathbb{C}$. Moreover, it is clear that for every nonzero $x \in \ker \Phi$, $\{g \in G; x^g \in \ker \Phi\} = \{1, -1\}$.

Recall that the Lie algebra $\mathfrak{u}(1)$ of $U(1)$ is the vector space (over reals) of all imaginary numbers. An orthonormal basis of this one-dimensional vector space is given by $T_1 = i$. Let $\Psi : \mathfrak{u}(1) \times \mathbb{C} \rightarrow \mathbb{R}$ be the map

$$\Psi(X, x) := \Phi(x^{e^X}),$$

where Φ is the gauge fixing function considered above. Writing

$$X = X_1 T_1 = iX_1,$$

we see that for any $z = x_1 + i0 \in \ker \Phi$, the representative of x^{e^X} in $\mathbb{C}^2$ is

$$e^{iX_1} z = (\cos X_1 + i \sin X_1) x_1 = x_1 \cos X_1 + ix_1 \sin X_1.$$

Thus, $\Psi(X, x) = x_1 \sin X_1$, and hence, $\partial_X \Psi(X, x)$ is the 1×1 matrix $x_1 \cos X_1$. We observe that $\partial_X \Psi$ satisfies the required conditions: $\partial_X \Psi(0, x)$ is nonsingular for almost all $x \in \ker \Phi$, and for almost all $x \in \ker \Phi$, and $|\det \partial_X \Psi(0, y)|$ is the same for all $y \in O_x \cap \ker \Phi$.

Thus, Theorem 11.2.1 shows that for any gauge invariant $F \in C(\mathbb{R}^2) \cap L^1(\mathbb{R}^2)$,

$$\int_{\mathbb{R}^4} F(x) dx = \frac{\mu(\mathrm{U}(1))}{2\Delta} \int_{-\infty}^{\infty} F(x_1, 0) |x_1| dx_1, \quad (11.3.1)$$

where μ is the unnormalized Haar measure on $\mathrm{U}(1)$ and $\Delta = |\det(\Phi|_{(\ker \Phi)^\perp})|$. Now, $(\ker \Phi)^\perp$ is simply the imaginary axis in $\mathbb{C}$, and $\Phi(iy) = y$. This shows that $\Delta = 1$. Next, if we identify $\mathrm{U}(1)$ with S^1 in the usual way, then it can be shown that $\mu(\mathrm{U}(1)) = 2\pi$. Thus, we get

$$\int_{\mathbb{R}^2} F(x) dx = \pi \int_{-\infty}^{\infty} F(x_1, 0) |x_1| dx_1.$$

Note that this can be obtained straightforwardly by shifting the integral on the left to polar coordinates. The purpose of the above calculation is to provide a toy example for the Faddeev–Popov procedure.

Lastly, in terms of Grassmann variables, the identity (11.3.1) can be written as

$$\int_{\mathbb{R}^2} F(x) dx = \pi \int_{-\infty}^{\infty} \iint F(x_1, 0) e^{i\sigma(x_1) - x_1 \bar{\gamma}_1 \gamma_1} d\gamma_1 d\bar{\gamma}_1 dx_1,$$

where $\gamma_1, \bar{\gamma}_1$ are distinct Grassmann variables, and $\sigma(x) = 1$ if $x < 0$ and 0 if $x > 0$.

Chapter 12

Dimensional Regularization

12.1 The main idea

The key idea behind dimensional regularization is as follows. Suppose that we want to integrate a radially symmetric function $f : \mathbb{R}^{d_0} \rightarrow \mathbb{C}$ over the whole of $\mathbb{R}^{d_0}$, where d_0 is a given dimension (usually four), but the integral diverges. Dimensional regularization consists of the following steps:

1. Find a class of functions $g_d : [0, \infty) \rightarrow \mathbb{C}$ for d belonging to a subset of the real line containing d_0 , such that $f(x) = g_{d_0}(\|x\|)$ for all $x \in \mathbb{R}^d$.
2. Observe that by transforming to polar coordinates and the assumption that f is radially symmetric, we have the formal relation

$$\int_{\mathbb{R}^{d_0}} f(x) dx = \frac{2\pi^{d_0/2}}{\Gamma(d_0/2)} \int_0^\infty r^{d_0-1} g_{d_0}(r) dr,$$

where Γ is the classical Gamma function.

3. To give a meaning to the right side above, define

$$h(d) := \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty r^{d-1} g_d(r) dr$$

for any d where the integral converges absolutely and $\Gamma(d/2)$ has a well-defined finite value via analytic continuation.

4. Analytically continue h to d_0 to assign a value to the original integral.

Often, in practice, it turns out that h has a pole at d_0 . In that case, we have to work at a dimension $d_0 - \epsilon$ and then apply renormalization to get rid of the pole. We will see examples later.

12.2 A key integral

The following integral is key to a lot of computations in dimensional regularization. Here, we are using the convention that for $z \in \mathbb{C} \setminus (-\infty, 0]$ and $w \in \mathbb{C}$, z^w denotes $e^{w \log z}$, where $\log$ is the analytic branch of logarithm on $\mathbb{C} \setminus (-\infty, 0]$ that is real-valued on $(0, \infty)$. In other words, if $z = re^{i\theta}$ for some $\theta \in (-\pi, \pi)$, then $z^w := r^w e^{i\theta w}$.

Lemma 12.2.1. *For $b > 0$, $-1 < a < 2b - 1$ and $\Delta \in \mathbb{C} \setminus (-\infty, 0]$, the integral*

$$G(a, b, \Delta) := \int_0^\infty \frac{r^a}{(r^2 + \Delta)^b} dr$$

is convergent. If we further have that $a \geq 2b - 3$, then

$$G(a, b, \Delta) = \frac{\Delta^{\frac{1}{2}(1+a-2b)} \Gamma(\frac{a+1}{2}) \Gamma(b - \frac{a+1}{2})}{2\Gamma(b)}.$$

Proof. The condition $-1 < a < 2b - 1$ renders the integral absolutely convergent, because

$$\left| \frac{z^a}{(z^2 + \Delta)^b} \right| \sim |z|^{a-2b}$$

as $|z| \rightarrow \infty$, and

$$\left| \frac{z^a}{(z^2 + \Delta)^b} \right| \sim |\Delta|^{-b} |z|^a$$

as $|z| \rightarrow 0$. Moreover, the first estimate and the assumption that $a - 2b < -1$ also allow us to change the contour of integration from $(0, \infty)$ to $(0, e^{\frac{1}{2}i\theta}\infty)$, where $\theta \in (-\pi, \pi)$ is the argument of Δ . Thus,

$$\begin{aligned} G(a, b, \Delta) &= e^{\frac{1}{2}i\theta} \int_0^\infty \frac{(re^{\frac{1}{2}i\theta})^a}{((re^{\frac{1}{2}i\theta})^2 + |\Delta|e^{i\theta})^b} dr \\ &= e^{\frac{1}{2}i\theta(1+a-2b)} \int_0^\infty \frac{r^a}{(r^2 + |\Delta|)^b} dr. \end{aligned}$$

Let us make the change of variable

$$x := \frac{r^2}{r^2 + |\Delta|}.$$

Then x ranges from 0 to 1 monotonically as r ranges from 0 to ∞ . Also,

$$\frac{dx}{dr} = \frac{2|\Delta|r}{(r^2 + |\Delta|)^2}.$$

Thus,

$$\begin{aligned}
 G(a, b, \Delta) &= e^{\frac{1}{2}i\theta(1+a-2b)} \int_0^\infty \frac{r^{a-1}}{2|\Delta|(r^2 + |\Delta|)^{b-2}} \frac{2|\Delta|r}{(r^2 + |\Delta|)^2} dr \\
 &= e^{\frac{1}{2}i\theta(1+a-2b)} \int_0^\infty \frac{1}{2|\Delta|^{1+b-2-\frac{a-1}{2}}} \left(\frac{r^2}{r^2 + |\Delta|} \right)^{\frac{a-1}{2}} \left(\frac{|\Delta|}{r^2 + |\Delta|} \right)^{b-2-\frac{a-1}{2}} \frac{2|\Delta|r}{(r^2 + |\Delta|)^2} dr \\
 &= \frac{1}{2} |\Delta|^{\frac{1}{2}(1+a-2b)} e^{\frac{1}{2}i\theta(1+a-2b)} \int_0^1 x^{\frac{a-1}{2}} (1-x)^{b-\frac{a+3}{2}} dx.
 \end{aligned}$$

Now, the last integral is the classical Beta integral, whose value is

$$\frac{\Gamma(\frac{a+1}{2})\Gamma(b - \frac{a+1}{2})}{\Gamma(b)}.$$

Finally, if $a \geq 2b - 3$, then

$$\frac{1}{2}|1 + a - 2b| = \frac{1}{2}(2b - a - 1) \leq 1,$$

which implies that

$$|\Delta|^{\frac{1}{2}(1+a-2b)} e^{\frac{1}{2}i\theta(1+a-2b)} = \Delta^{\frac{1}{2}(1+a-2b)}.$$

This completes the proof. $\square$

12.3 Some examples

In this section, we will work out three examples of dimensional regularization. All of these will appear later in our calculations for Yang–Mills theories.

Example 12.3.1. As our first example, let us consider the integral

$$\int_{\mathbb{R}^4} \frac{1}{\|x\|^2 - i\epsilon} dx,$$

where ϵ is some given positive constant. In polar coordinates, this is formally equal to

$$2\pi^2 \int_0^\infty \frac{r^3}{r^2 - i\epsilon} dr.$$

We are interested in assigning a value to this integral as $\epsilon \rightarrow 0$. To generalize this to an arbitrary dimension $d \in \mathbb{R}$, let us define

$$h_\epsilon(d) := \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty \frac{r^{d-1}}{r^2 - i\epsilon} dr.$$

We are in the setting of Lemma 12.2.1 with $a = d - 1$, $b = 1$, and $\Delta = -i\epsilon$. Note that $\Delta \in \mathbb{C} \setminus (-\infty, 0]$, $b > 0$, and the condition $-1 < a < 2b - 1$ is equivalent to $0 < d < 2$. Moreover, the condition $a \geq 2b - 3$ is equivalent to $d \geq 0$. Thus, Lemma 12.2.1 implies

that if $d \in (0, 2)$, then the integral defining $h_\epsilon(d)$ is convergent, and

$$\begin{aligned} h_\epsilon(d) &= \frac{\pi^{\frac{d}{2}} \Delta^{\frac{1}{2}(1+a-2b)} \Gamma(\frac{a+1}{2}) \Gamma(b - \frac{a+1}{2})}{\Gamma(\frac{d}{2}) \Gamma(b)} \\ &= \pi^{\frac{d}{2}} (-i\epsilon)^{\frac{1}{2}(d-2)} \Gamma\left(1 - \frac{d}{2}\right). \end{aligned}$$

Note that the above function admits an analytic continuation to the set of all $d \in \mathbb{C}$ such that $1 - \frac{d}{2}$ is not a nonpositive integer. This is the set of all $z \in \mathbb{C}$ that are not positive even integers. Thus, the dimensional regularization heuristic suggests that we should use the above formula to define $h_\epsilon(d)$ for any d that is not a positive even integer. With this definition, we get that for any $d \in (2, 4)$,

$$h(d) := \lim_{\epsilon \rightarrow 0} h_\epsilon(d) = 0.$$

Consequently, dimensional regularization suggests that we should define

$$\lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^{4-\delta}} \frac{1}{\|x\|^2 - i\epsilon} dx := 0$$

for any $\delta \in (0, 2)$.

Example 12.3.2. Next, let us consider the integral

$$\int_{\mathbb{R}^4} \frac{1}{(\|x\|^2 + \alpha - i\epsilon)^2} dx,$$

where $\alpha \in \mathbb{R}$ and $\epsilon > 0$. Replacing $\mathbb{R}^4$ by $\mathbb{R}^d$ and converting to polar coordinates, this becomes

$$h_\epsilon(\alpha, d) := \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty \frac{r^{d-1}}{(r^2 + \alpha - i\epsilon)^2} dr.$$

This brings us to the setting of Lemma 12.2.1, with $\Delta = \alpha - i\epsilon$, $b = 2$, and $a = d - 1$. Note that $\Delta \in \mathbb{C} \setminus (-\infty, 0]$ and $b > 0$. The condition $-1 < a < 2b - 1$ is equivalent to $0 < d < 4$, and the condition $a \geq 2b - 3$ is equivalent to $d \geq 2$. Thus, for $d \in [2, 4)$, we get

$$\begin{aligned} h_\epsilon(\alpha, d) &= \frac{\pi^{\frac{d}{2}} \Delta^{\frac{1}{2}(1+a-2b)} \Gamma(\frac{a+1}{2}) \Gamma(b - \frac{a+1}{2})}{\Gamma(\frac{d}{2}) \Gamma(b)} \\ &= \pi^{\frac{d}{2}} (\alpha - i\epsilon)^{\frac{1}{2}(d-4)} \Gamma\left(2 - \frac{d}{2}\right) \end{aligned}$$

The above function admits an analytic continuation to all $d \in \mathbb{C}$ such that $2 - \frac{d}{2}$ is not a nonpositive integer. This is the set of all $d \in \mathbb{C}$ that are not positive integer multiples of 4. In particular, dimensional regularization suggests that we can use this formula for any $d \in (0, 4)$, to get

$$\int_{\mathbb{R}^{4-\delta}} \frac{1}{(\|x\|^2 + \alpha - i\epsilon)^2} dx := \pi^{2-\frac{\delta}{2}} (\alpha - i\epsilon)^{-\frac{\delta}{2}} \Gamma\left(\frac{\delta}{2}\right)$$

for any $0 < \delta < 1$. This gives

$$\lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^{4-\delta}} \frac{1}{(\|x\|^2 + \alpha - i\epsilon)^2} dx := \pi^{2-\frac{\delta}{2}} \alpha^{-\frac{\delta}{2}} \Gamma\left(\frac{\delta}{2}\right).$$

Now, it is well known that as $z \rightarrow 0$,

$$\Gamma(z) = \frac{1}{z} - \gamma + O(z),$$

where γ is the Euler–Mascheroni constant. Thus, as $\delta \rightarrow 0$,

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^{4-\delta}} \frac{1}{(\|x\|^2 + \alpha - i\epsilon)^2} dx &= \pi^2 \left(1 - \frac{\delta}{2} \log(\pi\alpha) + O(\delta^2)\right) \left(\frac{2}{\delta} - \gamma + O(\delta)\right) \\ &= \frac{2\pi^2}{\delta} - \pi^2 \log(\pi\alpha) - \pi^2 \gamma + O(\delta). \end{aligned}$$

Although α was assumed to be positive, it does not make an appearance in the leading term. So it is reasonable to assume that the leading term remains the same for any α .

Example 12.3.3. As our final example, let us consider the integral

$$\int_{\mathbb{R}^4} \frac{\|x\|^2}{(\|x\|^2 + \alpha - i\epsilon)^2} dx,$$

where $\alpha \in \mathbb{R}$ and $\epsilon > 0$. As usual, we are interested in the limit $\epsilon \rightarrow 0$. Proceeding as in the previous example, we define

$$h_\epsilon(\alpha, d) := \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty \frac{r^{d+1}}{(r^2 + \alpha - i\epsilon)^2} dr.$$

We are in the setting of Lemma 12.2.1 with $\Delta = \alpha - i\epsilon$, $b = 2$, and $a = d + 1$. Note that $\Delta \in \mathbb{C} \setminus (-\infty, 0]$, $b > 0$, and the condition $-1 < a < 2b - 1$ is equivalent to $-2 < d < 2$. Also, the condition $a \geq 2b - 3$ is equivalent to $d \geq 0$. Thus, for $d \in [0, 2)$, Lemma 12.2.1 gives

$$\begin{aligned} h_\epsilon(\alpha, d) &= \frac{\pi^{\frac{d}{2}} \Delta^{\frac{1}{2}(1+a-2b)} \Gamma(\frac{a+1}{2}) \Gamma(b - \frac{a+1}{2})}{\Gamma(\frac{d}{2}) \Gamma(b)} \\ &= \frac{\pi^{\frac{d}{2}} (\alpha - i\epsilon)^{\frac{1}{2}(d-2)} \Gamma(\frac{d+2}{2}) \Gamma(1 - \frac{d}{2})}{\Gamma(\frac{d}{2})} \\ &= \frac{1}{2} d \pi^{\frac{d}{2}} (\alpha - i\epsilon)^{\frac{1}{2}(d-2)} \Gamma\left(1 - \frac{d}{2}\right). \end{aligned}$$

The above formula continues analytically to all $d \in \mathbb{C}$ such that $1 - \frac{d}{2}$ is not a nonpositive integer. This is the set of all $d \in \mathbb{C}$ that are not positive even integers. In particular, take

$d = 4 - \delta$ for some $\delta \in (0, 2)$, we get

$$\int_{\mathbb{R}^{4-\delta}} \frac{\|x\|^2}{(\|x\|^2 + \alpha - i\epsilon)^2} dx = \frac{1}{2}(4 - \delta)\pi^{2-\frac{\delta}{2}}(\alpha - i\epsilon)^{1-\frac{\delta}{2}}\Gamma\left(-1 + \frac{\delta}{2}\right).$$

This gives

$$\lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^{4-\delta}} \frac{\|x\|^2}{(\|x\|^2 + \alpha - i\epsilon)^2} dx = \frac{1}{2}(4 - \delta)\pi^{2-\frac{\delta}{2}}\alpha^{1-\frac{\delta}{2}}\Gamma\left(-1 + \frac{\delta}{2}\right).$$

Again, by standard facts about the Gamma function,

$$\Gamma(-1 + z) = -\frac{1}{z} + \gamma - 1 + O(z)$$

as $z \rightarrow 0$. Also, as $\delta \rightarrow 0$,

$$\begin{aligned} (4 - \delta)\pi^{2-\frac{\delta}{2}}\alpha^{1-\frac{\delta}{2}} &= 4\pi^2\alpha \exp\left(\log\left(1 - \frac{\delta}{4}\right) - \frac{\delta}{2}\log(\pi\alpha)\right) \\ &= 4\pi^2\alpha\left(1 - \delta\left(\frac{1}{4} + \frac{1}{2}\log(\pi\alpha)\right) + O(\delta^2)\right). \end{aligned}$$

Thus, as $\delta \rightarrow 0$,

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^{4-\delta}} \frac{\|x\|^2}{(\|x\|^2 + \alpha - i\epsilon)^2} dx &= 2\pi^2\alpha\left(1 - \delta\left(\frac{1}{4} + \frac{1}{2}\log(\pi\alpha)\right) + O(\delta^2)\right) \\ &\quad \cdot \left(-\frac{2}{\delta} + \gamma - 1 + O(\delta)\right) \\ &= -\frac{4\pi^2\alpha}{\delta} + 2\pi^2\alpha\left(\gamma - \frac{1}{2} + \log(\pi\alpha)\right) + O(\delta). \end{aligned}$$

Chapter 13

Heuristics for Pure Yang–Mills Theories

13.1 Gauge fixing and ghost fields

Let G be a compact matrix Lie group contained in $U(n)$ for some n , and let $\mathfrak{g}$ be its Lie algebra. Recall from equation (8.1.1) in Section 8.1 that the action for pure Yang–Mills theory with gauge group G on $\mathbb{R}^{1,3}$ is given by

$$S(A) = -\frac{1}{4g_0^2} \int_{\mathbb{R}^4} \text{Tr} \left(\sum_{0 \leq i, j \leq 3} \eta_i \eta_j F_{ij}(x)^2 \right) dx,$$

where $\eta_0 = 1$ and $\eta_1 = \eta_2 = \eta_3 = -1$, A is a $\mathfrak{g}$ -valued 1-form on $\mathbb{R}^{1,3}$ and F is its curvature form, and g_0 is the bare coupling constant. Previously, when we investigated $U(1)$ theory, we took $g_0 = 1$. We did this because this does not matter for $U(1)$ theory, where g_0 can be altered simply by rescaling the gauge field A . For non-Abelian theories, we cannot simply rescale the gauge field and get $g_0 = 1$. Thus, the theory has to be analyzed with g_0 present in the equations. However, it is still convenient and conventional to rescale A , as follows. Recall from equation (C.5.1) in Appendix C that $F = \sum_{0 \leq i < j \leq 3} F_{ij} dx_i \wedge dx_j$ is explicitly given by

$$F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j],$$

where $\partial_i A_j$ is shorthand for $\frac{\partial A_j}{\partial x_i}$ and $[A_i, A_j]$ is the commutator of A_i and A_j . This shows that if we rescale A by replacing it with $g_0 A$, then the Yang–Mills action becomes

$$S(A) = -\frac{1}{4} \int_{\mathbb{R}^4} \text{Tr} \left(\sum_{0 \leq i, j \leq 3} \eta_i \eta_j (\partial_i A_j(x) - \partial_j A_i(x) + g_0 [A_i(x), A_j(x)])^2 \right) dx. \quad (13.1.1)$$

This is the form of the action that we will work with henceforth.

Now, recall from Section 8.6 that a gauge transformation U is simply a smooth map from $\mathbb{R}^4$ into G , which acts on a connection A as $A^U = UAU^{-1} - (dU)U^{-1}$. But if we are

working with the action displayed above, it means that the gauge fields are rescaled by g_0 . To be consistent with the previous definition, we now need to define gauge transformation by U as

$$A^U := UAU^{-1} - \frac{1}{g_0}(dU)U^{-1}.$$

In Theorem 8.6.4, we showed that the Yang–Mills action is invariant under gauge transformations. If L is a gauge-invariant observable on the space of connections, recall from equation (2.3.3) in Section 2.3 that the expected value of $L(A)$ is heuristically given by

$$\langle L(A) \rangle = \frac{\int L(A) e^{iS(A)} \mathcal{D}A}{\int e^{iS(A)} \mathcal{D}A}. \quad (13.1.2)$$

We will now apply the Faddeev–Popov procedure to reduce the integrals in the numerator and denominator to integrals over gauge-fixed fields and a set of Grassmann-valued fields. The process will not be rigorous, since our Theorem 11.2.1 works only for finite-dimensional integrals. First, note that gauge transformations should heuristically preserve ‘Lebesgue measure’ on the space $\mathcal{A}$ of gauge connections. This is because conjugation by a unitary matrix preserves Lebesgue measure on the space of $n \times n$ matrix, and so does translation by an arbitrary matrix. Smoothness of gauge transformations is clear from the definition.

Next, let $m := \dim(G) = \dim(\mathfrak{g})$. Fix an orthonormal basis $T_1, \dots, T_m$ of $\mathfrak{g}$. We can write $A = \sum_{i=0}^3 A_i dx_i \in \mathcal{A}$ as

$$A(x) = \sum_{i=0}^3 \left(\sum_{a=1}^m A_{i,a}(x) T_a \right) dx_i = \sum_{a=1}^m \left(\sum_{i=0}^3 A_{i,a}(x) dx_i \right) T_a.$$

Thus, A may be represented as the collection of real-valued 1-forms $\sum_{i=0}^3 A_{i,a}(x) dx_i$, for $a = 1, \dots, m$. This is the more common way to represent it in the literature on Yang–Mills theories.

Since a gauge transformation U is a map from $\mathbb{R}^4$ into G , the group $\mathcal{G}$ of gauge transformations is locally like $(\mathbb{R}^m)^{\mathbb{R}^4}$. This suggests that we need to define a linear gauge fixing map Φ from $\mathcal{A}$ into $(\mathbb{R}^m)^{\mathbb{R}^4}$. We choose the following map, that fixes the Lorenz gauge for each component of A in the above representation:

$$\Phi(A)(x) := \left(\sum_{i=0}^3 \eta_i \partial_i A_{i,a}(x) \right)_{1 \leq a \leq m}.$$

Proposition 9.1.1 shows that for $U(1)$ theory, every gauge orbit intersects the kernel of Φ exactly once, as long as we restrict ourselves to Schwartzian connections and gauge transforms. Although the connections that typically arise in quantized gauge theories are far from Schwartzian, we will pretend as if the above conclusion is true in justifying the following steps; furthermore, Proposition 9.1.1 shows this only for $U(1)$ theory, but we will pretend as if we know this for any G .

Let us now try to understand the map Ψ defined in equation (11.2.1), adapted to the

present setting. Heuristically, the ‘Lie algebra’ of the gauge group $\mathcal{G}$ is the space $\mathcal{X}$ of all $\mathfrak{g}$ -valued smooth maps on $\mathbb{R}^4$. For any such map X and any $A \in \mathcal{A}$, we should define

$$\Psi(X, A) := \Phi(A^{e^X}),$$

where e^X denotes the element of $\mathcal{G}$ given by

$$e^X(x) := e^{X(x)}.$$

To apply the Faddeev–Popov procedure, we now have to calculate the variational derivatives

$$\mathcal{M}_{a,b}(x, y) := \frac{\partial \Psi_a(X, A)(x)}{\partial X_b(y)}$$

for $x, y \in \mathbb{R}^4$ and $1 \leq a, b \leq m$, at $X = 0$ and $A \in \ker \Phi$. Here we are representing X as $\sum_{a=1}^m X_a T_a$, where $X_1, \dots, X_m$ are real-valued fields on $\mathbb{R}^4$.

To calculate these derivatives, take any $x, y \in \mathbb{R}^4$ and $1 \leq a, b \leq m$. First, note that

$$\begin{aligned} B := A^{e^X} &= e^X A e^{-X} - \frac{1}{g_0} (de^X) e^{-X} \\ &= \sum_{i=0}^3 e^X A_i e^{-X} dx_i - \frac{1}{g_0} \sum_{i=0}^3 (\partial_i e^X) e^{-X} dx_i \\ &= \sum_{i=0}^3 \left(\sum_{c=1}^m A_{i,c} e^X T_c e^{-X} \right) dx_i - \frac{1}{g_0} \sum_{i=0}^3 (\partial_i e^X) e^{-X} dx_i. \end{aligned}$$

Since $T_1, \dots, T_m$ is an orthonormal basis, this shows that $B = \sum_{a=1}^m B_{i,a} T_a$, where

$$B_{i,a} = \langle B_i, T_a \rangle = \sum_{c=1}^m A_{i,c} \operatorname{Tr}(e^X T_c e^{-X} T_a^*) - \frac{1}{g_0} \operatorname{Tr}((\partial_i e^X) e^{-X} T_a^*).$$

Thus, we have

$$\begin{aligned} \Psi_a(X, A) &= \sum_{i=0}^3 \eta_i \partial_i B_{i,a} \\ &= \sum_{c=1}^m \sum_{i=0}^3 \eta_i \partial_i (A_{i,c} \operatorname{Tr}(e^X T_c e^{-X} T_a^*)) - \frac{1}{g_0} \sum_{i=0}^3 \eta_i \partial_i (\operatorname{Tr}((\partial_i e^X) e^{-X} T_a^*)). \end{aligned}$$

By the definition of variational derivative, when $X = 0$, $\mathcal{M}_{a,b}$ is a generalized function such that for any $Y = \sum_{c=1}^m Y_c T_c \in \mathcal{A}$,

$$\left. \frac{d}{d\epsilon} \Psi_a(\epsilon Y, A)(x) \right|_{\epsilon=0} = \int_{\mathbb{R}^4} \sum_{b=1}^m \mathcal{M}_{a,b}(x, y) Y_b(y) dy. \quad (13.1.3)$$

From the previous display, we see that

$$\begin{aligned} \Psi_a(\epsilon Y, A) &= \sum_{c=1}^m \sum_{i=0}^3 \eta_i \partial_i \left\{ A_{i,c} \operatorname{Tr} \left[\left(1 + \epsilon \sum_{b=1}^m Y_b T_b + O(\epsilon^2) \right) T_c \left(1 - \epsilon \sum_{b=1}^m Y_b T_b + O(\epsilon^2) \right) T_a^* \right] \right\} \\ &\quad - \frac{1}{g_0} \sum_{i=0}^3 \eta_i \partial_i \left\{ \operatorname{Tr} \left[\left(\epsilon \sum_{b=1}^m (\partial_i Y_b) T_b + O(\epsilon^2) \right) \left(1 - \epsilon \sum_{b=1}^m Y_b T_b + O(\epsilon^2) \right) T_a^* \right] \right\}. \end{aligned}$$

This shows that

$$\left. \frac{d}{d\epsilon} \Psi_a(\epsilon Y, A) \right|_{\epsilon=0} = \sum_{b,c=1}^m \sum_{i=0}^3 \eta_i \partial_i (A_{i,c} Y_b \operatorname{Tr}([T_b, T_c] T_a^*)) - \frac{1}{g_0} \sum_{b=1}^m \sum_{i=0}^3 \eta_i (\partial_i^2 Y_b) \operatorname{Tr}(T_b T_a^*).$$

Now note that $\operatorname{Tr}([T_b, T_c] T_a^*) = f_{bc}^a$, where f_{bc}^a are the structure constants¹ of $\mathfrak{g}$. Also, $\operatorname{Tr}(T_b T_a^*) = \delta_{ab}$ (where $\delta_{ab} = 0$ if $a \neq b$ and 1 if $a = b$). Comparing with equation (13.1.3), we see that

$$\mathcal{M}_{a,b}(x, y) = \left(\sum_{c=1}^m \sum_{i=0}^3 \eta_i f_{bc}^a \partial_i (A_{i,c}(x) \cdot) - \frac{1}{g_0} \sum_{i=0}^3 \delta_{ab} \eta_i \partial_i^2(\cdot) \right) \delta^{(4)}(x - y), \quad (13.1.4)$$

where $\delta^{(4)}$ is the Dirac delta at the origin in $\mathbb{R}^4$. Thus, a heuristic extension of Theorem 11.2.1 to infinite dimensions suggests that we should consider the action

$$\begin{aligned} S^L(A, \gamma, \bar{\gamma}) &:= -\frac{1}{4} \int_{\mathbb{R}^4} \operatorname{Tr} \left(\sum_{0 \leq i, j \leq 3} \eta_i \eta_j (\partial_i A_j(x) - \partial_j A_i(x) + g_0 [A_i(x), A_j(x)])^2 \right) dx \\ &\quad + \kappa \int_{\mathbb{R}^4} \left(\sum_{a,b,c=1}^m \sum_{i=0}^3 \eta_i f_{bc}^a \bar{\gamma}_a(x) \partial_i (A_{i,c}(x) \gamma_b(x)) - \frac{1}{g_0} \sum_{a=1}^m \bar{\gamma}_a(x) \square \gamma_a(x) \right) dx \\ &\quad + i\pi \sigma(A), \end{aligned}$$

where $\gamma_a, \bar{\gamma}_a$, $a = 1, \dots, m$ are Grassmann-valued fields on $\mathbb{R}^4$, $\square = \sum_{i=0}^3 \eta_i \partial_i^2$, κ is an arbitrary nonzero constant, A is a gauge field satisfying the Lorenz gauge condition, and $\sigma(A) = 1$ if $\det(\kappa \mathcal{M}) < 0$ and 0 otherwise, where $\mathcal{M}$ is the Faddeev–Popov operator displayed in equation (13.1.4). The fields γ and $\bar{\gamma}$ are called *ghost fields*, since they are unphysical.

Heuristically, the factor κ contributes a constant multiple to the determinant arising in the Faddeev–Popov procedure, which cancels out between the numerator and denominator in equation (13.1.2). We will work with the choice $\kappa = g_0$. The superscript L indicates that this is the action when we fix the Lorenz gauge. With this action, the vacuum expected

¹See Section C.3.4 in Appendix C

value of a gauge-invariant observable A is given by

$$\langle L(A) \rangle = \frac{\iiint L(A) e^{iS^L(A, \gamma, \bar{\gamma})} \delta(\Phi(A)) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A}{\iiint e^{iS^L(A, \gamma, \bar{\gamma})} \delta(\Phi(A)) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A},$$

where the factors $\delta(\Phi(A))$ indicate that the path integrals are taken over the subspace $\ker \Phi$ of gauge configurations that satisfy the Lorenz gauge condition.

13.2 Covariant gauge fixing

An improvement due to 't Hooft allows us to obtain the action in the so-called covariant gauge. Take any arbitrary function $f : \mathbb{R}^4 \rightarrow \mathbb{R}^m$. Instead of Φ , consider the gauge-fixing function $\Phi - f$, defined as

$$(\Phi - f)(A)(x) := \Phi(A)(x) - f(x).$$

A slight modification of the proof of Proposition 9.1.1 ‘justifies’ that $\ker(\Phi - f)$ intersects every gauge orbit. Since the variational derivatives of $\Phi - f$ are unaffected by the choice of f , this shows that for a gauge invariant function L ,

$$\iiint L(A) e^{iS^L(A, \gamma, \bar{\gamma})} \delta(\Phi(A) - f) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A$$

does not depend on the choice of f . So, if we now integrate the above quantity over f , and even inserting a factor depending on f , the result will just be a constant multiple of the above. Thus, for any $\xi \in \mathbb{R}$,

$$\begin{aligned} \langle L(A) \rangle &= \frac{\iiint L(A) \exp(iS^L(A, \gamma, \bar{\gamma}) - \frac{i(1+\xi)}{2} \int_{\mathbb{R}^4} \|f(x)\|^2 dx) \delta(\Phi(A) - f) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A \mathcal{D}f}{\iiint \exp(iS^L(A, \gamma, \bar{\gamma}) - \frac{i(1+\xi)}{2} \int_{\mathbb{R}^4} \|f(x)\|^2 dx) \delta(\Phi(A) - f) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A \mathcal{D}f} \\ &= \frac{\iiint L(A) \exp(iS^L(A, \gamma, \bar{\gamma}) - \frac{i(1+\xi)}{2} \int_{\mathbb{R}^4} \|\Phi(A)(x)\|^2 dx) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A}{\iiint \exp(iS^L(A, \gamma, \bar{\gamma}) - \frac{i(1+\xi)}{2} \int_{\mathbb{R}^4} \|\Phi(A)(x)\|^2 dx) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A} \\ &= \frac{\iiint L(A) \exp(iS^C(A, \gamma, \bar{\gamma})) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A}{\iiint \exp(iS^C(A, \gamma, \bar{\gamma})) \mathcal{D}\gamma \mathcal{D}\bar{\gamma} \mathcal{D}A}, \end{aligned}$$

where S^C is the action in the covariant gauge, given by

$$\begin{aligned} S^C(A, \gamma, \bar{\gamma}) &:= -\frac{1}{4} \int_{\mathbb{R}^4} \text{Tr} \left(\sum_{0 \leq i, j \leq 3} \eta_i \eta_j (\partial_i A_j(x) - \partial_j A_i(x) + g_0 [A_i(x), A_j(x)])^2 \right) dx \\ &\quad - \frac{1+\xi}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \left(\sum_{i=0}^3 \eta_i \partial_i A_{i,a}(x) \right)^2 dx \\ &\quad + \int_{\mathbb{R}^4} \left(g_0 \sum_{a,b,c=1}^m \sum_{i=0}^3 \eta_i f_{bc}^a \bar{\gamma}_a(x) \partial_i (A_{i,c}(x) \gamma_b(x)) - \sum_{a=1}^m \bar{\gamma}_a(x) \square \gamma_a(x) \right) dx + i\pi \sigma(A). \end{aligned}$$

Note that with $\xi = 1$, this is the Feynman gauge that we encountered in our analysis of $U(1)$ theory. Here, however, we will need to keep ξ arbitrary, for a reason that will become clear later. Observe that

$$\begin{aligned}\mathrm{Tr}[(\partial_i A_j - \partial_j A_i)^2] &= \mathrm{Tr} \left[\left(\sum_{a=1}^m (\partial_i A_{j,a} - \partial_j A_{i,a}) T_a \right)^2 \right] \\ &= \sum_{a,b=1}^m (\partial_i A_{j,a} - \partial_j A_{i,a}) (\partial_i A_{j,b} - \partial_j A_{i,b}) \mathrm{Tr}(T_a T_b).\end{aligned}$$

Since G is a subgroup of $U(n)$, $\mathfrak{g}$ consists of skew-Hermitian matrices. Thus,

$$\mathrm{Tr}(T_a T_b) = -\mathrm{Tr}(T_a T_b^*) = -\langle T_a, T_b \rangle.$$

Since $T_1, \dots, T_m$ is an orthonormal basis, we conclude that

$$\mathrm{Tr}[(\partial_i A_j - \partial_j A_i)^2] = \sum_{a=1}^m (\partial_i A_{j,a} - \partial_j A_{i,a})^2.$$

Now, integration by parts gives

$$\int_{\mathbb{R}^4} \partial_i A_{j,a}(x) \partial_j A_{i,a}(x) dx = \int_{\mathbb{R}^4} \partial_i A_{i,a}(x) \partial_j A_{j,a}(x) dx.$$

Thus,

$$\begin{aligned}& -\frac{1}{4} \int_{\mathbb{R}^4} \mathrm{Tr} \left(\sum_{0 \leq i,j \leq 3} \eta_i \eta_j (\partial_i A_j(x) - \partial_j A_i(x))^2 \right) dx \\ &= -\frac{1}{4} \int_{\mathbb{R}^4} \sum_{a=1}^m \sum_{0 \leq i,j \leq 3} \eta_i \eta_j [(\partial_i A_{j,a}(x))^2 + (\partial_j A_{i,a}(x))^2 - 2\partial_i A_{i,a}(x) \partial_j A_{j,a}(x)] dx \\ &= -\frac{1}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \sum_{0 \leq i,j \leq 3} \eta_i \eta_j (\partial_i A_{j,a}(x))^2 dx + \frac{1}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \left(\sum_{i=0}^3 \eta_i \partial_i A_{i,a}(x) \right)^2 dx.\end{aligned}$$

The second term on the right partially cancels out the covariant gauge fixing term in the action, leaving us with

$$\begin{aligned}S^C(A, \gamma, \bar{\gamma}) &= -\frac{1}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \sum_{0 \leq i,j \leq 3} \eta_i \eta_j (\partial_i A_{j,a}(x))^2 dx - \frac{\xi}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \left(\sum_{i=0}^3 \eta_i \partial_i A_{i,a}(x) \right)^2 dx \\ &\quad - \frac{g_0}{2} \int_{\mathbb{R}^4} \sum_{0 \leq i,j \leq 3} \eta_i \eta_j \mathrm{Tr} \left\{ (\partial_i A_j(x) - \partial_j A_i(x)) [A_i(x), A_j(x)] + \frac{g_0}{2} [A_i(x), A_j(x)]^2 \right\} dx \\ &\quad + \int_{\mathbb{R}^4} \left(g_0 \sum_{a,b,c=1}^m \sum_{i=0}^3 \eta_i f_{bc}^a \bar{\gamma}_a(x) \partial_i (A_{i,c}(x) \gamma_b(x)) - \sum_{a=1}^m \bar{\gamma}_a(x) \square \gamma_a(x) \right) dx + i\pi \sigma(A).\end{aligned}$$

To simplify further, notice that

$$\begin{aligned}
 & \text{Tr}((\partial_i A_j(x) - \partial_j A_i(x))[A_i(x), A_j(x)]) \\
 &= \text{Tr} \left[\left(\sum_{a=1}^m (\partial_i A_{j,a}(x) - \partial_j A_{i,a}(x)) T_a \right) \sum_{a,b=1}^m A_{i,a}(x) A_{j,b}(x) [T_a, T_b] \right] \\
 &= \sum_{a,b,c=1}^m (\partial_i A_{j,a}(x) - \partial_j A_{i,a}(x)) A_{i,b}(x) A_{j,c}(x) \text{Tr}(T_a [T_b, T_c]).
 \end{aligned}$$

Now,

$$\text{Tr}(T_a [T_b, T_c]) = -\text{Tr}(T_a^* [T_b, T_c]) = -\text{Tr}([T_b, T_c] T_a^*) = -f_{bc}^a.$$

Thus,

$$\begin{aligned}
 & \text{Tr}((\partial_i A_j(x) - \partial_j A_i(x))[A_i(x), A_j(x)]) \\
 &= - \sum_{a,b,c=1}^m f_{bc}^a (\partial_i A_{j,a}(x) - \partial_j A_{i,a}(x)) A_{i,b}(x) A_{j,c}(x).
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 \text{Tr}([A_i(x), A_j(x)]^2) &= \text{Tr} \left[\left(\sum_{a,b=1}^m A_{i,a}(x) A_{j,b}(x) [T_a, T_b] \right)^2 \right] \\
 &= \text{Tr} \left[\left(\sum_{a,b,c=1}^m f_{ab}^c A_{i,a}(x) A_{j,b}(x) T_c \right)^2 \right] \\
 &= \sum_{a,b,c,a',b',c'=1}^m f_{ab}^c f_{a'b'}^{c'} A_{i,a}(x) A_{j,b}(x) A_{i,a'}(x) A_{j,b'}(x) \text{Tr}(T_c T_{c'}).
 \end{aligned}$$

As we have already observed earlier, $\text{Tr}(T_c T_{c'}) = -\text{Tr}(T_c T_{c'}^*) = \delta_{c,c'}$. Thus,

$$\text{Tr}([A_i(x), A_j(x)]^2) = - \sum_{a,b,a',b',c=1}^m f_{ab}^c f_{a'b'}^c A_{i,a}(x) A_{j,b}(x) A_{i,a'}(x) A_{j,b'}(x).$$

Combining all of the above, we arrive at the following formula:

$$\begin{aligned}
 S^C(A, \gamma, \bar{\gamma}) &= -\frac{1}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \sum_{0 \leq i,j \leq 3} \eta_i \eta_j (\partial_i A_{j,a}(x))^2 dx - \frac{\xi}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \left(\sum_{i=0}^3 \eta_i \partial_i A_{i,a}(x) \right)^2 dx \\
 &+ \frac{g_0}{2} \int_{\mathbb{R}^4} \sum_{0 \leq i,j \leq 3} \sum_{a,b,c=1}^m \eta_i \eta_j f_{bc}^a (\partial_i A_{j,a}(x) - \partial_j A_{i,a}(x)) A_{i,b}(x) A_{j,c}(x) dx \\
 &+ \frac{g_0^2}{4} \int_{\mathbb{R}^4} \sum_{0 \leq i,j \leq 3} \sum_{a,b,a',b',c=1}^m \eta_i \eta_j f_{ab}^c f_{a'b'}^c A_{i,a}(x) A_{j,b}(x) A_{i,a'}(x) A_{j,b'}(x) dx \\
 &+ \int_{\mathbb{R}^4} \left(g_0 \sum_{a,b,c=1}^m \sum_{i=0}^3 \eta_i f_{bc}^a \bar{\gamma}_a(x) \partial_i (A_{i,c}(x) \gamma_b(x)) - \sum_{a=1}^m \bar{\gamma}_a(x) \square \gamma_a(x) \right) dx + i\pi \sigma(A),
 \end{aligned}$$

where $\sigma(A) = 1$ if $\det(g_0\mathcal{M}) < 0$ and 0 otherwise, with $\mathcal{M}$ being the Faddeev–Popov operator displayed in equation (13.1.4).

Since $\sigma(A)$ is either 0 or 1, it should remain constant under small perturbations and should therefore play no role in this perturbation theory. Ignoring $\sigma(A)$, the action can be decomposed into the sum of a ‘free’ and an ‘interaction’ part, given by

$$\begin{aligned} S_{\text{free}}^{\text{C}}(A, \gamma, \bar{\gamma}) &:= -\frac{1}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \sum_{0 \leq i, j \leq 3} \eta_i \eta_j (\partial_i A_{j,a}(x))^2 dx - \int_{\mathbb{R}^4} \sum_{a=1}^m \bar{\gamma}_a(x) \square \gamma_a(x) dx \\ &\quad - \frac{\xi}{2} \int_{\mathbb{R}^4} \sum_{a=1}^m \left(\sum_{i=0}^3 \eta_i \partial_i A_{i,a}(x) \right)^2 dx \end{aligned}$$

and

$$\begin{aligned} S_{\text{int}}^{\text{C}}(A, \gamma, \bar{\gamma}) &:= \frac{g_0}{2} \int_{\mathbb{R}^4} \sum_{0 \leq i, j \leq 3} \sum_{a, b, c=1}^m \eta_i \eta_j f_{bc}^a (\partial_i A_{j,a}(x) - \partial_j A_{i,a}(x)) A_{i,b}(x) A_{j,c}(x) dx \\ &\quad + \frac{g_0^2}{4} \int_{\mathbb{R}^4} \sum_{0 \leq i, j \leq 3} \sum_{a, b, a', b', c=1}^m \eta_i \eta_j f_{ab}^c f_{a'b'}^c A_{i,a}(x) A_{j,b}(x) A_{i,a'}(x) A_{j,b'}(x) dx \\ &\quad + g_0 \int_{\mathbb{R}^4} \sum_{a, b, c=1}^m \sum_{i=0}^3 \eta_i f_{bc}^a \bar{\gamma}_a(x) \partial_i (A_{i,c}(x) \gamma_b(x)) dx. \end{aligned}$$

13.3 The action in momentum coordinates

We will now write down the action in momentum coordinates, that is, in terms of the Fourier transforms of the gauge fields and ghost fields. This will simplify calculations later. Recall that for $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$, its Fourier transform in Minkowski signature is defined as

$$\tilde{f}(p) := \int_{\mathbb{R}^4} e^{i(x,p)} f(x) dx,$$

where $p \in \mathbb{R}^4$, and for $x, p \in \mathbb{R}^4$, the Minkowski inner product of x and p is defined as

$$(x, p) := p_0 x_0 - p_1 x_1 - p_2 x_2 - p_3 x_3 = \sum_{j=0}^3 \eta_j p_j x_j.$$

Also recall that $\tilde{f}$ is also a Schwartz function, and that the following inversion formula holds:

$$f(x) = \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} e^{-i(x,p)} \tilde{f}(p) dp.$$

We will now rewrite the action in terms of the Fourier transforms $\tilde{A}_{ia}$, $\tilde{\gamma}_a$ and $\tilde{\bar{\gamma}}_a$. Note that

$$\partial_i A_{j,a}(x) = -\frac{i\eta_i}{(2\pi)^4} \int_{\mathbb{R}^4} p_i e^{-i(x,p)} \tilde{A}_{j,a}(p) dp.$$

From this and the formal identity

$$\frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} e^{-i(x,p)} dx = \delta^{(4)}(p),$$

it follows that

$$\begin{aligned} \int_{\mathbb{R}^4} (\partial_i A_{j,a}(x))^2 dx &= -\frac{1}{(2\pi)^8} \iiint e^{-i(x,p+q)} p_i q_i \tilde{A}_{j,a}(p) \tilde{A}_{j,a}(q) dx dp dq \\ &= -\frac{1}{(2\pi)^4} \int p_i^2 \tilde{A}_{j,a}(p) \tilde{A}_{j,a}(-p) dp, \end{aligned}$$

Similarly,

$$\begin{aligned} &\int_{\mathbb{R}^4} (\partial_i A_{j,a}(x)) A_{i,b}(x) A_{j,c}(x) dx \\ &= -\frac{i\eta_i}{(2\pi)^{12}} \iiint p_i e^{-i(x,p+q+r)} \tilde{A}_{j,a}(p) \tilde{A}_{i,b}(q) \tilde{A}_{j,c}(r) dx dp dq dr \\ &= -\frac{i\eta_i}{(2\pi)^8} \iint p_i \tilde{A}_{j,a}(p) \tilde{A}_{i,b}(q) \tilde{A}_{j,c}(-p-q) dp dq, \end{aligned}$$

where $\delta^{(4)}$ is the Dirac delta at zero in $\mathbb{R}^4$. This step can be easily justified (as we have done several times earlier) by introducing a regularization term $e^{-\epsilon r^2}$, evaluating the integral, and then sending ϵ to zero.

From the above calculation, it follows that

$$\begin{aligned} \int \sum_{a=1}^m \sum_{0 \leq i, j \leq 3} \eta_i \eta_j (\partial_i A_{j,a}(x))^2 dx &= -\frac{1}{(2\pi)^4} \int \sum_{a=1}^m \sum_{0 \leq i, j \leq 3} \eta_i \eta_j p_i^2 \tilde{A}_{j,a}(p) \tilde{A}_{j,a}(-p) dp \\ &= -\frac{1}{(2\pi)^4} \int p^2 \sum_{a=1}^m \sum_{j=0}^3 \eta_j \tilde{A}_{j,a}(p) \tilde{A}_{j,a}(-p) dp, \end{aligned}$$

and

$$\begin{aligned} &\int_{\mathbb{R}^4} \sum_{0 \leq i, j \leq 3} \sum_{a, b, c=1}^m \eta_i \eta_j f_{bc}^a (\partial_i A_{j,a}(x) - \partial_j A_{i,a}(x)) A_{i,b}(x) A_{j,c}(x) dx \\ &= -\frac{i}{(2\pi)^8} \iint \sum_{0 \leq i, j \leq 3} \sum_{a, b, c=1}^m \eta_i \eta_j f_{bc}^a (\eta_i p_i \tilde{A}_{j,a}(p) - \eta_j p_j \tilde{A}_{i,a}(p)) \\ &\quad \cdot \tilde{A}_{i,b}(q) \tilde{A}_{j,c}(-p-q) dp dq. \end{aligned}$$

Similarly,

$$\begin{aligned}
 & \int_{\mathbb{R}^4} \sum_{0 \leq i, j \leq 3} \sum_{a, b, a', b', c=1}^m \eta_i \eta_j f_{ab}^c f_{a'b'}^c A_{i,a}(x) A_{j,b}(x) A_{i,a'}(x) A_{j,b'}(x) dx \\
 &= \frac{1}{(2\pi)^{12}} \iiint \sum_{0 \leq i, j \leq 3} \sum_{a, b, a', b', c=1}^m \eta_i \eta_j f_{ab}^c f_{a'b'}^c \tilde{A}_{i,a}(p) \\
 & \quad \cdot \tilde{A}_{j,b}(q) \tilde{A}_{i,a'}(r) \tilde{A}_{j,b'}(-p-q-r) dp dq dr,
 \end{aligned}$$

and

$$\begin{aligned}
 & \int_{\mathbb{R}^4} \sum_{a, b, c=1}^m \sum_{i=0}^3 \eta_i f_{bc}^a \bar{\gamma}_a(x) \partial_i (A_{i,c}(x) \gamma_b(x)) dx \\
 &= -\frac{i}{(2\pi)^8} \iint \sum_{a, b, c=1}^m \sum_{i=0}^3 \eta_i f_{bc}^a (\eta_i p_i + \eta_i q_i) \tilde{\gamma}_a(-p-q) \tilde{A}_{i,c}(p) \tilde{\gamma}_b(q) dp dq.
 \end{aligned}$$

The covariant gauge fixing term has a similar expression:

$$\int_{\mathbb{R}^4} \sum_{a=1}^m \left(\sum_{i=0}^3 \eta_i \partial_i A_{i,a}(x) \right)^2 dx = -\frac{1}{(2\pi)^4} \int \sum_{a=1}^m \sum_{0 \leq i, j \leq 3} p_i p_j \tilde{A}_{i,a}(p) \tilde{A}_{j,a}(-p) dp.$$

Lastly, note that by the formal identities

$$\gamma_a(x) = \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} e^{-i(x,p)} \tilde{\gamma}_a(p) dp, \quad \bar{\gamma}_a(x) = \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} e^{-i(x,p)} \tilde{\bar{\gamma}}_a(p) dp,$$

we get

$$\begin{aligned}
 \square \gamma_a(x) &= \sum_{i=0}^3 \eta_i \partial_i^2 \gamma_a(x) = \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} \left(\sum_{i=0}^3 \eta_i (-i \eta_i p_i)^2 \right) e^{-i(x,p)} \tilde{\gamma}_a(p) dp \\
 &= -\frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} \left(\sum_{i=0}^3 \eta_i p_i^2 \right) e^{-i(x,p)} \tilde{\gamma}_a(p) dp = -\frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} p^2 e^{-i(x,p)} \tilde{\gamma}_a(p) dp,
 \end{aligned}$$

and hence,

$$\begin{aligned}
 \int \bar{\gamma}_a(x) \square \gamma_a(x) dx &= -\frac{1}{(2\pi)^8} \iiint e^{-i(x,p+q)} q^2 \tilde{\bar{\gamma}}_a(p) \tilde{\gamma}_a(q) dx dp dq \\
 &= -\frac{1}{(2\pi)^4} \int p^2 \tilde{\bar{\gamma}}_a(p) \tilde{\gamma}_a(-p) dp.
 \end{aligned}$$

Thus, we get $\tilde{S}^C(\tilde{A}, \tilde{\gamma}, \tilde{\gamma}) = \tilde{S}_{\text{free}}^C(\tilde{A}, \tilde{\gamma}, \tilde{\gamma}) + \tilde{C}_{\text{int}}^F(\tilde{A}, \tilde{\gamma}, \tilde{\gamma})$, where

$$\begin{aligned} \tilde{S}_{\text{free}}^C(\tilde{A}, \tilde{\gamma}, \tilde{\gamma}) &= \frac{1}{2(2\pi)^4} \int p^2 \sum_{a=1}^m \sum_{j=0}^3 \eta_j \tilde{A}_{j,a}(p) \tilde{A}_{j,a}(-p) dp \\ &\quad + \frac{1}{(2\pi)^4} \int p^2 \sum_{a=1}^m \tilde{\gamma}_a(p) \tilde{\gamma}_a(-p) dp \\ &\quad + \frac{\xi}{2(2\pi)^4} \int \sum_{a=1}^m \sum_{0 \leq i, j \leq 3} p_i p_j \tilde{A}_{i,a}(p) \tilde{A}_{j,a}(-p) dp \end{aligned}$$

and

$$\begin{aligned} &\tilde{S}_{\text{int}}^C(\tilde{A}, \tilde{\gamma}, \tilde{\gamma}) \\ &= -\frac{i g_0}{2(2\pi)^8} \iint \sum_{i,j=0}^3 \sum_{a,b,c=1}^m f_{bc}^a(\eta_j p_i \tilde{A}_{j,a}(p) - \eta_i p_j \tilde{A}_{i,a}(p)) \tilde{A}_{i,b}(q) \tilde{A}_{j,c}(-p-q) dp dq \\ &\quad + \frac{g_0^2}{4(2\pi)^{12}} \iiint \sum_{i,j=0}^3 \sum_{a,b,a',b',c=1}^m \eta_i \eta_j f_{ab}^c f_{a'b'}^c \tilde{A}_{i,a}(p) \tilde{A}_{j,b}(q) \tilde{A}_{i,a'}(r) \tilde{A}_{j,b'}(-p-q-r) dp dq dr \\ &\quad - \frac{i g_0}{(2\pi)^8} \iint \sum_{a,b,c=1}^m \sum_{i=0}^3 f_{bc}^a(p_i + q_i) \tilde{\gamma}_a(-p-q) \tilde{A}_{i,c}(p) \tilde{\gamma}_b(q) dp dq. \end{aligned}$$

13.4 First order of perturbation theory

In this section, we assume that G is a simple Lie group², in addition to the properties listed earlier. Heuristically, the gluon propagator in momentum coordinates is defined as the vacuum expectation value

$$D_{jk}^{ab}(p, q) := \frac{i}{(2\pi)^4} \langle \tilde{A}_{j,a}(p) \tilde{A}_{k,b}(q) \rangle = \frac{i}{(2\pi)^4} \frac{\int \tilde{A}_{j,a}(p) \tilde{A}_{k,b}(q) e^{i\tilde{S}^C(\tilde{A}, \tilde{\gamma}, \tilde{\gamma})} \mathcal{D}\tilde{\gamma} \mathcal{D}\tilde{\gamma} \mathcal{D}\tilde{A}}{\int e^{i\tilde{S}^C(\tilde{A}, \tilde{\gamma}, \tilde{\gamma})} \mathcal{D}\tilde{\gamma} \mathcal{D}\tilde{\gamma} \mathcal{D}\tilde{A}}.$$

For the perturbative expansion of the gluon propagator, we introduce three new parameters $\theta_1, \theta_2, \theta_3$. The first two parameters are used to rescale the gauge fields $\tilde{A}_{j,a}$ and the ghost fields $\tilde{\gamma}_a, \tilde{\bar{\gamma}}_a$ by replacing them with $\sqrt{1+\theta_1} \tilde{A}_{j,a}$, $\sqrt{1+\theta_2} \tilde{\gamma}_a$, and $\sqrt{1+\theta_2} \tilde{\bar{\gamma}}_a$ in the action. The third parameter is used to reparametrize ξ as

$$\xi = \frac{\theta_3}{1+\theta_1}.$$

²See Section C.3.5 in Appendix C for an introduction to simple Lie groups.

The free and interacting parts of the resulting action are then given by

$$\begin{aligned}\tilde{S}_{\text{free}}^{\text{C}}(\tilde{A}, \tilde{\gamma}, \tilde{\gamma}) &= \frac{1+\theta_1}{2(2\pi)^4} \int p^2 \sum_{a=1}^m \sum_{j=0}^3 \eta_j \tilde{A}_{j,a}(p) \tilde{A}_{j,a}(-p) dp \\ &\quad + \frac{1+\theta_2}{(2\pi)^4} \int p^2 \sum_{a=1}^m \tilde{\gamma}_a(p) \tilde{\gamma}_a(-p) dp \\ &\quad + \frac{\theta_3}{2(2\pi)^4} \int \sum_{a=1}^m \sum_{0 \leq i,j \leq 3} p_i p_j \tilde{A}_{i,a}(p) \tilde{A}_{j,a}(-p) dp\end{aligned}$$

and

$$\begin{aligned}\tilde{S}_{\text{int}}^{\text{F}}(\tilde{A}, \tilde{\gamma}, \tilde{\gamma}) &= -\frac{ig_0(1+\theta_1)^{3/2}}{2(2\pi)^8} \iint \sum_{i,j=0}^3 \sum_{a,b,c=1}^m f_{bc}^a (\eta_j p_i \tilde{A}_{j,a}(p) - \eta_i p_j \tilde{A}_{i,a}(p)) \tilde{A}_{i,b}(q) \tilde{A}_{j,c}(-p-q) dp dq \\ &\quad + \frac{g_0^2(1+\theta_1)^2}{4(2\pi)^{12}} \iiint \sum_{i,j=0}^3 \sum_{a,b,a',b',c=1}^m \eta_i \eta_j f_{ab}^c f_{a'b'}^c \tilde{A}_{i,a}(p) \\ &\quad \quad \cdot \tilde{A}_{j,b}(q) \tilde{A}_{i,a'}(r) \tilde{A}_{j,b'}(-p-q-r) dp dq dr \\ &\quad - \frac{ig_0 \sqrt{1+\theta_1}(1+\theta_2)}{(2\pi)^8} \iint \sum_{a,b,c=1}^m \sum_{i=0}^3 f_{bc}^a (p_i + q_i) \tilde{\gamma}_a(-p-q) \tilde{A}_{i,c}(p) \tilde{\gamma}_b(q) dp dq.\end{aligned}$$

Let us now expand the gluon propagator as a power series in $g_0, \theta_1, \theta_2, \theta_3$. Throughout, let $\langle \cdot \rangle_0$ denote expectations with respect to the free theory. Also, we will use the notation

$$\langle A; B \rangle_0 := \langle AB \rangle_0 - \langle A \rangle_0 \langle B \rangle_0.$$

First, note that in the free theory, the components of the gauge field are decoupled. Thus, by the formula (9.5.1) for the photon propagator, we have

$$D_{jk}^{ab}(p, q)|_{\theta_1=\theta_2=\theta_3=g_0=0} = \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{p^2 + i\epsilon}. \quad (13.4.1)$$

A simple computation shows that

$$\left. \frac{\partial}{\partial \theta_1} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=g_0=0} = \frac{1}{2(2\pi)^8} \int r^2 \sum_{c=1}^m \sum_{l=0}^3 \eta_l \langle \tilde{A}_{j,a}(p) \tilde{A}_{k,b}(q); \tilde{A}_{l,c}(r) \tilde{A}_{l,c}(-r) \rangle_0 dr.$$

To evaluate this, note that by equation (9.5.1) and Corollary B.2.8,

$$\begin{aligned}
 & \langle \tilde{A}_{j,a}(p) \tilde{A}_{k,b}(q); \tilde{A}_{l,c}(r) \tilde{A}_{l,c}(-r) \rangle_0 \\
 &= \langle \tilde{A}_{j,a}(p) \tilde{A}_{l,c}(r) \rangle \langle \tilde{A}_{k,b}(q) \tilde{A}_{l,c}(-r) \rangle_0 + \langle \tilde{A}_{j,a}(p) \tilde{A}_{l,c}(-r) \rangle_0 \langle \tilde{A}_{k,b}(q) \tilde{A}_{l,c}(r) \rangle_0 \\
 &= -(2\pi)^8 \eta_j \eta_k \delta_{jl} \delta_{kl} \delta_{ac} \delta_{bc} \delta^{(4)}(p+r) \delta^{(4)}(q-r) \lim_{\epsilon \downarrow 0} \frac{1}{(r^2 + i\epsilon)^2} \\
 &\quad - (2\pi)^8 \eta_j \eta_k \delta_{jl} \delta_{kl} \delta_{ac} \delta_{bc} \delta^{(4)}(p-r) \delta^{(4)}(q+r) \lim_{\epsilon \downarrow 0} \frac{1}{(r^2 + i\epsilon)^2}.
 \end{aligned}$$

Integrating over r , we get

$$\left. \frac{\partial}{\partial \theta_1} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=g_0=0} = -p^2 \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2}. \quad (13.4.2)$$

Next, note that since the gauge field and the ghost fields decouple in the free theory, we have

$$\langle \tilde{A}_{j,a}(p) \tilde{A}_{k,b}(q); \tilde{\gamma}_c(r) \tilde{\gamma}_c(-r) \rangle_0 = 0.$$

Thus, by a similar calculation as above, we get

$$\left. \frac{\partial}{\partial \theta_2} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=g_0=0} = 0. \quad (13.4.3)$$

Next, note that

$$\begin{aligned}
 & \left. \frac{\partial}{\partial \theta_3} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=g_0=0} \\
 &= \frac{1}{2(2\pi)^8} \int \sum_{c=1}^m \sum_{0 \leq i', j' \leq 3} r_{i'} r_{j'} \langle \tilde{A}_{j,a}(p) \tilde{A}_{k,b}(q); \tilde{A}_{i',c}(r) \tilde{A}_{j',c}(-r) \rangle_0 dr.
 \end{aligned}$$

Again, as before,

$$\begin{aligned}
 & \langle \tilde{A}_{j,a}(p) \tilde{A}_{k,b}(q); \tilde{A}_{i',c}(r) \tilde{A}_{j',c}(-r) \rangle_0 \\
 &= \langle \tilde{A}_{j,a}(p) \tilde{A}_{i',c}(r) \rangle_0 \langle \tilde{A}_{k,b}(q) \tilde{A}_{j',c}(-r) \rangle_0 + \langle \tilde{A}_{j,a}(p) \tilde{A}_{j',c}(-r) \rangle_0 \langle \tilde{A}_{k,b}(q) \tilde{A}_{i',c}(r) \rangle_0 \\
 &= -(2\pi)^8 \eta_j \eta_k \delta_{ji'} \delta_{kj'} \delta_{ac} \delta_{bc} \delta^{(4)}(p+r) \delta^{(4)}(q-r) \lim_{\epsilon \downarrow 0} \frac{1}{(r^2 + i\epsilon)^2} \\
 &\quad - (2\pi)^8 \eta_j \eta_k \delta_{jj'} \delta_{ki'} \delta_{ac} \delta_{bc} \delta^{(4)}(p-r) \delta^{(4)}(q+r) \lim_{\epsilon \downarrow 0} \frac{1}{(r^2 + i\epsilon)^2}.
 \end{aligned}$$

Integrating over r , we get

$$\left. \frac{\partial}{\partial \theta_3} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=g_0=0} = -\eta_j \eta_k p_j p_k \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2}. \quad (13.4.4)$$

Lastly, we claim that

$$\left. \frac{\partial}{\partial g_0} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=g_0=0} = 0. \quad (13.4.5)$$

To see this, note that the free field expectations that arise in the above quantity are all expectations of products of odd numbers of field variables. Therefore, they vanish.

Combining equations (13.4.1), (13.4.2), (13.4.3), (13.4.4) and (13.4.5), we get the first order of perturbation theory for the gluon propagator:

$$\begin{aligned} D_{jk}^{ab}(p, q) &= \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{p^2 + i\epsilon} \\ &\quad - \theta_1 p^2 \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ &\quad - \theta_3 \eta_j \eta_k p_j p_k \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ &\quad + O(\theta_1^2 + \theta_2^2 + \theta_3^2 + g_0^2). \end{aligned} \quad (13.4.6)$$

Note that there are no infinities in this expansion. But the expansion is not interesting, since the parameter g_0 makes no appearance, which means that we are not seeing any non-Abelian effects. To see non-Abelian effects, we have to go to the second order of perturbation theory. Infinities will arise in the second order, and we will have to apply renormalization to tame the infinities.

13.5 Second order of perturbation theory

Our next goal is to compute the second partial derivative with respect to g_0 . This computation is rather complicated. For ease of notation, we will denote $\tilde{A}_{j,a}(p)$ by $A_{j,a,p}$, $\tilde{\gamma}_a(p)$ by $\gamma_{a,p}$, and $\tilde{\bar{\gamma}}_a(p)$ by $\bar{\gamma}_{a,p}$ in the following. First, let us decompose the interacting part of the action $\tilde{S}_{\text{int}}^F(\tilde{A}, \tilde{\gamma}, \tilde{\bar{\gamma}})$, when $\theta_1 = \theta_2 = \theta_3 = 0$, as a sum $g_0 S_1 + g_0^2 S_2$. Then, heuristically,

$$\begin{aligned} \left. \frac{\partial}{\partial g_0} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=0} &= \frac{i}{(2\pi)^4} \frac{\partial}{\partial g_0} \frac{\langle A_{j,a,p} A_{k,b,q} e^{ig_0 S_1 + ig_0^2 S_2} \rangle_0}{\langle e^{ig_0 S_1 + ig_0^2 S_2} \rangle_0} \\ &= -\frac{1}{(2\pi)^4} \left(\frac{\langle A_{j,a,p} A_{k,b,q} (S_1 + 2g_0 S_2) e^{ig_0 S_1 + ig_0^2 S_2} \rangle_0}{\langle e^{ig_0 S_1 + ig_0^2 S_2} \rangle_0} \right. \\ &\quad \left. - \frac{\langle A_{j,a,p} A_{k,b,q} e^{ig_0 S_1 + ig_0^2 S_2} \rangle_0 \langle (S_1 + 2g_0 S_2) e^{ig_0 S_1 + ig_0^2 S_2} \rangle_0}{\langle e^{ig_0 S_1 + ig_0^2 S_2} \rangle_0^2} \right). \end{aligned}$$

Differentiating again, and setting $g_0 = 0$, a simple calculation gives

$$\left. \frac{\partial^2}{\partial g_0^2} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=g_0=0} = -\frac{1}{(2\pi)^4} \left(2\langle A_{j,a,p} A_{k,b,q}; S_2 \rangle_0 + i\langle A_{j,a,p} A_{k,b,q}; S_1^2 \rangle_0 - 2i\langle A_{j,a,p} A_{k,b,q}; S_1 \rangle_0 \langle S_1 \rangle_0 \right).$$

Since $\langle S_1 \rangle_0 = 0$, this simplifies to

$$\begin{aligned} & \left. \frac{\partial^2}{\partial g_0^2} D_{jk}^{ab}(p, q) \right|_{\theta_1=\theta_2=\theta_3=g_0=0} \\ &= -\frac{2}{(2\pi)^4} \langle A_{j,a,p} A_{k,b,q}; S_2 \rangle_0 - \frac{i}{(2\pi)^4} \langle A_{j,a,p} A_{k,b,q}; S_1^2 \rangle_0. \end{aligned} \quad (13.5.1)$$

Let us now evaluate the above terms. First note that

$$\begin{aligned} & \langle A_{j,a,p} A_{k,b,q}; S_2 \rangle_0 \\ &= \frac{1}{4(2\pi)^{12}} \sum_{i',j'=0}^3 \sum_{a',b',a'',b'',c=1}^m \eta_{i'} \eta_{j'} f_{a'b'}^c f_{a''b''}^c \\ & \quad \cdot \iiint \langle A_{j,a,p} A_{k,b,q}; A_{i',a',p'} A_{j',b',q'} A_{i'',a'',r'} A_{j'',b'',-p'-q'-r'} \rangle_0 dp' dq' dr'. \end{aligned}$$

Let $t' := -p' - q' - r'$. By Corollary B.2.8, we see that the integrand is equal to the sum of 12 terms which correspond to the 12 ways of pairing up $A_{j,a,p}$, $A_{k,b,q}$, $A_{i',a',p'}$, $A_{j',b',q'}$, $A_{i'',a'',r'}$, $A_{j'',b'',t'}$ where $A_{j,a,p}$ is not matched with $A_{k,b,q}$. We can also ignore terms where a' is matched to b' or a'' is matched to b'' , because of the totally antisymmetric nature of the structure constants³. This leaves us with 8 matchings to consider. Now, by equation (9.5.1), for any l, l', d, d', s, s' ,

$$\langle A_{l,d,s} A_{l',d',s'} \rangle_0 = -(2\pi)^4 i \eta_l \delta_{ll'} \delta_{dd'} \delta^{(4)}(s + s') \lim_{\epsilon \downarrow 0} \frac{1}{s^2 + i\epsilon}. \quad (13.5.2)$$

Thus, each matching forces certain identities. We will denote the matching $a \leftrightarrow a'$, $b \leftrightarrow a''$, $b' \leftrightarrow b''$ by $aa'ba''b'b''$, and so on. Table 13.1 tabulates the identities forced by the various matchings.

Since $p' + q' + r' + t' = 0$, Table 13.1 shows that $p + q = 0$ is forced by each matching. Moreover, it shows that the contribution to $\langle A_{j,a,p} A_{k,b,q}; S_2 \rangle_0$ from each matching is of the form

$$\frac{i}{4} \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p + q) C}{(p^2 + i\epsilon)^2} \int \frac{1}{r^2 + i\epsilon} dr,$$

where C is a number that depends on the matching. Table 13.2 tabulates the values of C for the various matchings. To calculate these values, we used the fact from Corollary C.3.10

³See Lemma C.3.7.

Table 13.1: Identities forced by matchings in the S_2 term

Matching	Identities
$aa'ba''b'b''$	$j = i', k = i', p = -p', q = -r', q' = -t'$
$aa'bb''b'a''$	$j = i', k = j', i' = j', p = -p', q = -t', q' = -r'$
$ab'ba''a'b''$	$j = j', k = i', i' = j', p = -q', q = -r', p' = -t'$
$ab'bb''a'a''$	$j = j', k = j', p = -q', q = -t', p' = -r'$
$ab''ba'b'a''$	$j = j', k = i', j' = i', p = -t', q = -p', q' = -r'$
$ab''bb'a'a''$	$j = j', k = j', p = -t', q = -q', p' = -r'$
$aa''ba'b'b''$	$j = i', k = i', p = -r', q = -p', q' = -t'$
$aa''bb'a'b''$	$j = i', k = j', i' = j', p = -r', q = -q', p' = -t'$

 Table 13.2: The value of C for various matchings in the S_2 term

Matching	C
$aa'ba''b'b''$	$4C_{\mathfrak{g}}\eta_j\delta_{jk}\delta_{ab}$
$aa'bb''b'a''$	$-C_{\mathfrak{g}}\eta_j\delta_{jk}\delta_{ab}$
$ab'ba''a'b''$	$-C_{\mathfrak{g}}\eta_j\delta_{jk}\delta_{ab}$
$ab'bb''a'a''$	$4C_{\mathfrak{g}}\eta_j\delta_{jk}\delta_{ab}$
$ab''ba'b'a''$	$-C_{\mathfrak{g}}\eta_j\delta_{jk}\delta_{ab}$
$ab''bb'a'a''$	$4C_{\mathfrak{g}}\eta_j\delta_{jk}\delta_{ab}$
$aa''ba'b'b''$	$4C_{\mathfrak{g}}\eta_j\delta_{jk}\delta_{ab}$
$aa''bb'b'a''$	$-C_{\mathfrak{g}}\eta_j\delta_{jk}\delta_{ab}$

that

$$\sum_{b,c=1}^m f_{bc}^a f_{bc}^{a'} = C_{\mathfrak{g}} \delta_{aa'} \quad (13.5.3)$$

for some constant $C_{\mathfrak{g}}$ that depends only on the Lie algebra $\mathfrak{g}$. Summing up the values from Table 13.2, we arrive at the heuristic identity

$$\langle A_{j,a,p} A_{k,b,q}; S_2 \rangle_0 = 3i \lim_{\epsilon \downarrow 0} \frac{C_{\mathfrak{g}} \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{1}{r^2 + i\epsilon} dr. \quad (13.5.4)$$

The integral on the right is divergent. We will take care of it later by renormalization. Next, let us calculate $\langle A_{j,a,p} A_{k,b,q}; S_1^2 \rangle_0$. First, recall that $S_1 = T_1 + T_2$, where

$$T_1 := -\frac{i}{2(2\pi)^8} \iint \sum_{i,j=0}^3 \sum_{a,b,c=1}^m f_{bc}^a (\eta_j p_i A_{j,a,p} - \eta_i p_j A_{i,a,p}) A_{i,b,q} A_{j,c,-p-q} dp dq,$$

$$T_2 := -\frac{i}{(2\pi)^8} \iint \sum_{a,b,c=1}^m \sum_{i=0}^3 f_{bc}^a (p_i + q_i) \bar{\gamma}_{a,-p-q} A_{i,c,p} \gamma_{b,q} dp dq.$$

The term T_1 can be written as $T_{11} - T_{12}$, where

$$T_{11} := -\frac{i}{2(2\pi)^8} \iint \sum_{i,j=0}^3 \sum_{a,b,c=1}^m f_{bc}^a \eta_j p_i A_{j,a,p} A_{i,b,q} A_{j,c,-p-q} dp dq,$$

$$T_{12} := -\frac{i}{2(2\pi)^8} \iint \sum_{i,j=0}^3 \sum_{a,b,c=1}^m f_{bc}^a \eta_i p_j A_{i,a,p} A_{i,b,q} A_{j,c,-p-q} dp dq.$$

Now, in T_{12} , if we swap the indices i and j , b and c , and make the change of variable $r = -p - q$, we get

$$T_{12} = \frac{i}{2(2\pi)^8} \iint \sum_{i,j=0}^3 \sum_{a,b,c=1}^m f_{cb}^a \eta_j p_i A_{j,a,p} A_{j,c,-p-r} A_{i,b,r} dp dr.$$

Since $f_{cb}^a = -f_{bc}^a$, this shows that $T_{12} = -T_{11}$. Thus, S_1 simplifies to $2T_{11} + T_2$. Therefore,

$$\begin{aligned} \langle A_{j,a,p} A_{k,b,q}; S_1^2 \rangle_0 &= 4 \langle A_{j,a,p} A_{k,b,q}; T_{11}^2 \rangle_0 + 4 \langle A_{j,a,p} A_{k,b,q}; T_{11} T_2 \rangle_0 \\ &\quad + \langle A_{j,a,p} A_{k,b,q}; T_2^2 \rangle_0. \end{aligned} \tag{13.5.5}$$

Let us now evaluate the three terms on the right. First, note that

$$\begin{aligned} \langle A_{j,a,p} A_{k,b,q}; T_{11}^2 \rangle_0 &= -\frac{1}{4(2\pi)^{16}} \iiint \sum_{i',j',i'',j''=0}^3 \sum_{a',b',c',a'',b'',c''=1}^m f_{b'c'}^{a'} f_{b''c''}^{a''} \eta_{j'} \eta_{j''} p_{i'} p_{i''} \\ &\quad \cdot \langle A_{j,a,p} A_{k,b,q}; A_{j',a',p'} A_{i',b',q'} A_{j',c',-p'-q'} A_{j'',a'',p''} A_{i'',b'',q''} A_{j'',c'',-p''-q''} \rangle_0 dp' dq' dp'' dq''. \end{aligned}$$

By Corollary B.2.8, the term within the parentheses is equal to the sum of products of all pairings except those in which p is matched to q . Since the structure constants are totally antisymmetric (Lemma C.3.7), we can ignore terms where any of a', b', c' is matched to another, or where any of a'', b'', c'' is matched to another.

This leaves us with the following options:

1. Match a to one of a', b', c' and b to one of a'', b'', c'' , or
2. match a to one of a'', b'', c'' and b to one of a', b', c' .

In both options, the remaining two indices from a', b', c' have to be matched with the remaining two indices from a'', b'', c'' . For ease of notation, let $r' := -p' - q'$ and $r'' := -p'' - q''$. Suppose we match a to a' , b to a'' , b' with b'' , and c' with c'' . Then equation (13.5.2) forces $q' = -q''$ and $r' = -r''$. Since $p' + q' + r' = 0$ and $p'' + q'' + r'' = 0$, this implies that $p' = -p''$. But equation (13.5.2) also forces $p = -p'$ and $q = -p''$. Thus, we must have $p + q = 0$. A moment's thought shows that by a similar logic, all other matchings also force the condition that $p + q = 0$, and that each term is equal to

$$-\frac{1}{4} \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{P(p, r)}{(r^2 + i\epsilon)((p-r)^2 + i\epsilon)} dr,$$

Table 13.3: Identities forced by matchings in the T_{11}^2 term

Matching	Identities
$aa'ba''b'b''c'c''$	$j = j', k = j'', i' = i'', j' = j'', p' = -p, p'' = -q, q' = -q'', r' = -r''$
$aa'ba''b'c'c'b''$	$j = j', k = j'', i' = j'', j' = i'', p' = -p, p'' = -q, q' = -r'', r' = -q''$
$aa'bb''b'a''c'c''$	$j = j', k = i'', i' = j'', j' = j'', p' = -p, q = -q'', q' = -p'', r' = -r''$
$aa'bb''b'c'c'a''$	$j = j', k = i'', i' = j'', j' = j'', p' = -p, q = -q'', q' = -r'', r' = -p''$
$aa'bc''b'a''c'b''$	$j = j', k = j'', i' = j'', j' = i'', p' = -p, q = -r'', q' = -p'', r' = -q''$
$aa'bc''b'b''c'a''$	$j = j', k = j'', i' = i'', j' = j'', p' = -p, q = -r'', q' = -q'', r' = -p''$
$ab'ba''a'b''c'c''$	$j = i', k = j'', j' = i'', j' = j'', p = -q', q = -p'', p' = -q'', r' = -r''$
$ab'ba''a'c''c'b''$	$j = i', k = j'', j' = j'', j' = i'', p = -q', q = -p'', p' = -r'', r' = -q''$
$ab'bb''a'a''c'c''$	$j = i', k = i'', j' = j'', p = -q', q = -q'', p' = -p'', r' = -r''$
$ab'bb''a'c'c'a''$	$j = i', k = i'', j' = j'', p = -q', q = -q'', p' = -r'', r' = -p''$
$ab'bc''a'a''c'b''$	$j = i', k = j'', j' = j'', j' = i'', p = -q', q = -r'', p' = -p'', r' = -q''$
$ab'bc''a'b''c'a''$	$j = i', k = j'', j' = i'', j' = j'', p = -q', q = -r'', p' = -q'', r' = -p''$
$ac'ba''a'b''b'c''$	$j = j', k = j'', j' = i'', i' = j'', p = -r', q = -p'', p' = -q'', q' = -r''$
$ac'ba''a'c''b'b''$	$j = j', k = j'', j' = j'', i' = i'', p = -r', q = -p'', p' = -r'', q' = -q''$
$ac'bb''a'a''b'c''$	$j = j', k = i'', j' = j'', i' = j'', p = -r', q = -q'', p' = -p'', q' = -r''$
$ac'bb''a'c'c'b'a''$	$j = j', k = i'', j' = j'', i' = j'', p = -r', q = -q'', p' = -r'', q' = -p''$
$ac'bc''a'a''b'b''$	$j = j', k = j'', j' = j'', i' = i'', p = -r', q = -r'', p' = -p'', q' = -q''$
$ac'bc''a'b''b'a''$	$j = j', k = j'', j' = i'', i' = j'', p = -r', q = -r'', p' = -q'', q' = -p''$

where $P(p, r)$ is a polynomial in the entries of p and r , depending on the term. Here the dummy variable r stands for one of the two variables among p', q', r' that are not matched to p , and $-p - r$ is the other one.

For ease of notation, let $C(aa'ba''b'b''c'c'')$ denote the contribution from the above matching, and so on. There are 36 matchings with potentially nonzero contributions, by the above paragraph. By the symmetry between $'$ and $''$, we can reduce the problem to 18 matchings, where a is matched to one of a', b', c' and b is matched to one of a'', b'', c'' . For example, $C(aa'ba''b'b''c'c'') = C(aa''ba'b'b''c'c'')$, and so on. We will evaluate the above quantities. For each quantity, equation (13.5.2) forces some identities. Table 13.3 tabulates the identities that are relevant for us.

Table 13.3 helps us apply equation (13.5.2) and calculate the polynomial $P(p, r)$ for every matching. For example, consider the matching $aa'ba''b'b''c'c''$. By the identities from Table 13.3, and the forced identity $p = -q$ (as observed earlier),

$$\begin{aligned}
 P(p, r) &= \sum_{i', j', i'', j''=0}^3 \sum_{a', b', c', a'', b'', c''=1}^m f_{b'c'}^{a'} f_{b''c''}^{a''} \eta_{j'} \eta_{j''} p_{i'}' p_{i''}'' \eta_{j'} \eta_{j''} \eta_{i'} \eta_{i''} \\
 &\quad \cdot \delta_{jj'} \delta_{kk''} \delta_{i'i''} \delta_{j'j''} \delta_{aa'} \delta_{ba''} \delta_{b'b''} \delta_{c'c''} \\
 &= -\eta_j \delta_{jk} \sum_{i'=0}^3 \sum_{b', c'=1}^m f_{b'c'}^a f_{b'c'}^b \eta_{i'} p_{i'}^2 \\
 &= -C_{\mathfrak{g}} \delta_{ab} \eta_j \delta_{jk} p^2.
 \end{aligned}$$

Table 13.4: $P(p, r)$ for various matchings

Matching	$P(p, r)$
$aa'ba''b'b''c'c''$	$-C_g \delta_{ab} \eta_j \delta_{jk} p^2$
$aa'ba''b'c''c'b''$	$C_g \delta_{ab} \eta_k \eta_j p_k p_j$
$aa'bb''b'a''c'c''$	$-C_g \delta_{ab} \eta_k \eta_j p_j r_k$
$aa'bb''b'c''c'a''$	$C_g \delta_{ab} \eta_k \eta_j p_j r_k$
$aa'bc''b'a''c'b''$	$C_g \delta_{ab} \eta_k \eta_j p_k r_j$
$aa'bc''b'b''c'a''$	$-C_g \delta_{ab} \eta_j \delta_{jk}(p, r)$
$ab'ba''a'b''c'c''$	$-C_g \delta_{ab} \eta_j \eta_k p_k r_j$
$ab'ba''a'c''c'b''$	$C_g \delta_{ab} \eta_j \eta_k p_k r_j$
$ab'bb''a'a''c'c''$	$-4C_g \delta_{ab} \eta_j \eta_k r_j r_k$
$ab'bb''a'c''c'a''$	$-4C_g \delta_{ab} \eta_j \eta_k r_j(r_k - p_k)$
$ab'bc''a'a''c'b''$	$C_g \delta_{ab} \eta_j \eta_k r_j r_k$
$ab'bc''a'b''c'a''$	$C_g \delta_{ab} \eta_j \eta_k r_j(r_k - p_k)$
$ac'ba''a'b''b'c''$	$C_g \delta_{ab} \eta_j \eta_k p_j r_k$
$ac'ba''a'c''b'b''$	$-C_g \delta_{ab} \eta_j \delta_{jk}(p, r)$
$ac'bb''a'a''b'c''$	$C_g \delta_{ab} \eta_j \eta_k r_j r_k$
$ac'bb''a'c''b'a''$	$C_g \delta_{ab} \eta_j \eta_k r_j(r_k - p_k)$
$ac'bc''a'a''b'b''$	$-C_g \delta_{ab} \eta_j \delta_{jk} r^2$
$ac'bc''a'b''b'a''$	$-C_g \delta_{ab} \eta_j \eta_k r_k(r_j - p_j)$

Applying similar calculations, we obtain Table 13.4. From Table 13.4, we see that the sum of $P(p, r)$ over all terms is equal to $C_g \delta_{ab}$ times

$$-\eta_j \delta_{jk}(p+r)^2 + \eta_k \eta_j p_k p_j - 3\eta_j \eta_k r_j(r_k - p_k) - 2\eta_j \eta_k r_k(r_j - p_j).$$

By the change of variable $r \rightarrow p - r$, we have

$$\int \frac{r_j(p_k - r_k)}{(r^2 + i\epsilon)((p-r)^2 + i\epsilon)} dr = \int \frac{r_k(p_j - r_j)}{(r^2 + i\epsilon)((p-r)^2 + i\epsilon)} dr. \quad (13.5.6)$$

Using this, and making the change of variable $r \rightarrow -r$, we arrive at the identity

$$\langle A_{j,a,p} A_{k,b,q}; T_{11}^2 \rangle_0 = -\frac{1}{2} C_g \delta_{ab} \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{N_{jk}(p, r)}{(r^2 + i\epsilon)((r+p)^2 + i\epsilon)} dr, \quad (13.5.7)$$

where

$$N_{jk}(p, r) := -\eta_j \delta_{jk}(p-r)^2 + \eta_k \eta_j p_k p_j - 5\eta_k \eta_j r_k(r_j + p_j).$$

Next, note that

$$\begin{aligned} \langle A_{j,a,p} A_{k,b,q}; T_{11} T_2 \rangle_0 &= \frac{1}{2(2\pi)^{16}} \iiint \sum_{i', j', i''=0}^3 \sum_{a', b', c', a'', b'', c''=1}^m f_{b'c'}^{a'} f_{b''c''}^{a''} \eta_{j'} p_{i'}' (p_{i''}'' + q_{i''}'') \\ &\cdot \langle A_{j,a,p} A_{k,b,q}; A_{j',a',p'} A_{i',b',q'} A_{j',c',-p'-q'} A_{i'',c'',p''} \bar{\gamma}_{a'',-p'',-q''} \gamma_{b'',q''} \rangle_0 dp' dq' dp'' dq''. \end{aligned}$$

For the integrand to be nonzero, we must match $\bar{\gamma}$ to γ . But then a'' is matched to b'' , which renders $f_{b''c''}^{a''} = 0$. Therefore,

$$\langle A_{j,a,p} A_{k,b,q}; T_{11} T_2 \rangle_0 = 0. \quad (13.5.8)$$

Finally, note that

$$\begin{aligned} \langle A_{j,a,p} A_{k,b,q}; T_2^2 \rangle_0 &= -\frac{1}{(2\pi)^{16}} \iiint \sum_{i', i''=0}^3 \sum_{a', b', c', a'', b'', c''=1}^m f_{b'c'}^{a'} f_{b''c''}^{a''} (p_{i'}' + q_{i'}')(p_{i''}'' + q_{i''}'') \\ &\cdot \langle A_{j,a,p} A_{k,b,q}; A_{i',c',p'} \bar{\gamma}_{a',-p'-q'} \gamma_{b',q'} A_{i'',c'',p''} \bar{\gamma}_{a'',-p''-q''} \gamma_{b'',q''} \rangle_0 dp' dq' dp'' dq''. \end{aligned}$$

The integrand is simplified by noticing that

$$\begin{aligned} &\langle A_{j,a,p} A_{k,b,q} A_{i',c',p'} \bar{\gamma}_{a',-p'-q'} \gamma_{b',q'} A_{i'',c'',p''} \bar{\gamma}_{a'',-p''-q''} \gamma_{b'',q''} \rangle_0 \\ &= \langle A_{j,a,p} A_{k,b,q}; A_{i',c',p'} A_{i'',c'',p''} \rangle_0 \langle \bar{\gamma}_{a',-p'-q'} \gamma_{b',q'} \bar{\gamma}_{a'',-p''-q''} \gamma_{b'',q''} \rangle_0. \end{aligned}$$

The terms where $a' = b'$ do not contribute, since then $f_{b'c'}^{a'} = 0$. If $a' \neq b'$, then by Theorem 10.6.2 and the formula (10.7.1),

$$\begin{aligned} &\langle \bar{\gamma}_{a',-p'-q'} \gamma_{b',q'} \bar{\gamma}_{a'',-p''-q''} \gamma_{b'',q''} \rangle_0 \\ &= \delta_{a'b''} \delta_{b'a''} \lim_{\epsilon \downarrow 0} \frac{(2\pi)^8 \delta^{(4)}(q' - p'' - q'') \delta^{(4)}(-p' - q' + q'')}{(q'^2 + i\epsilon)(q''^2 + i\epsilon)}. \end{aligned}$$

On the other hand, by Corollary B.2.8,

$$\begin{aligned} \langle A_{j,a,p} A_{k,b,q}; A_{i',c',p'} A_{i'',c'',p''} \rangle_0 &= \langle A_{j,a,p} A_{i',c',p'} \rangle_0 \langle A_{k,b,q} A_{i'',c'',p''} \rangle_0 \\ &+ \langle A_{j,a,p} A_{i'',c'',p''} \rangle_0 \langle A_{k,b,q} A_{i',c',p'} \rangle_0. \end{aligned}$$

Thus, by equation (13.5.2), we end up with two terms. The first term forces the identities

$$\begin{aligned} a &= c', \quad b = c'', \quad a' = b'', \quad b' = a'', \\ j &= i', \quad k = i'', \quad p = -p', \quad q = -p'', \quad q' = p'' + q'', \quad q'' = p' + q', \end{aligned}$$

and the second term forces

$$\begin{aligned} a &= c'', \quad b = c', \quad a' = b'', \quad b' = a'', \\ j &= i'', \quad k = i', \quad p = -p'', \quad q = -p', \quad q' = p'' + q'', \quad q'' = p' + q'. \end{aligned}$$

Therefore, both terms force

$$p + q = -p' - p'' = (q' - q'') + (q'' - q') = 0.$$

Lastly, the first term forces $q'' - q' = -p'' = q = -p$, and the second term forces

$q'' - q' = -p'' = p$. Combining the above observations, we get

$$\begin{aligned} \langle A_{j,a,p} A_{k,b,q}; T_2^2 \rangle_0 &= C_{\mathfrak{g}} \delta_{ab} \eta_j \eta_k \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \left(- \int \frac{r_j(r_k - p_k)}{(r^2 + i\epsilon)((r-p)^2 + i\epsilon)} dr \right. \\ &\quad \left. - \int \frac{r_k(r_j + p_j)}{(r^2 + i\epsilon)((r+p)^2 + i\epsilon)} dr \right) \\ &= -C_{\mathfrak{g}} \delta_{ab} \eta_j \eta_k \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{r_j(r_k + p_k) + r_k(r_j + p_j)}{(r^2 + i\epsilon)((r+p)^2 + i\epsilon)} dr. \end{aligned} \quad (13.5.9)$$

Combining equations (13.5.5), (13.5.7), (13.5.8), and (13.5.9), and using the symmetry relation (13.5.6), we get

$$\langle A_{j,a,p} A_{k,b,q}; S_1^2 \rangle_0 = C_{\mathfrak{g}} \delta_{ab} \lim_{\epsilon \downarrow 0} \frac{\delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{L_{jk}(p, r)}{(r^2 + i\epsilon)((r+p)^2 + i\epsilon)} dr,$$

where

$$\begin{aligned} L_{jk}(p, r) &= -2N_{jk}(p, r) - 2\eta_j \eta_k r_k(r_j + p_j) \\ &= 2\eta_j \delta_{jk}(p-r)^2 - 2\eta_j \eta_k p_j p_k + 8\eta_j \eta_k r_k(r_j + p_j). \end{aligned} \quad (13.5.10)$$

Combining this with equations (13.5.1) and (13.5.4), we get

$$\begin{aligned} \frac{\partial^2}{\partial g_0^2} D_{jk}^{ab}(p, q) \Big|_{\theta_1=\theta_2=\theta_3=g_0=0} &= -\frac{3i}{(2\pi)^4} \lim_{\epsilon \downarrow 0} \frac{C_{\mathfrak{g}} \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{1}{r^2 + i\epsilon} dr \\ &\quad + \frac{i}{(2\pi)^4} \lim_{\epsilon \downarrow 0} \frac{C_{\mathfrak{g}} \delta_{ab} \delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{L_{jk}(p, r)}{(r^2 + i\epsilon)((r+p)^2 + i\epsilon)} dr. \end{aligned} \quad (13.5.11)$$

Combining equations (13.4.6) and (13.5.11), we get the second order perturbative expansion

$$\begin{aligned} D_{jk}^{ab}(p, q) &= \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{q^2 + i\epsilon} - \theta_1 p^2 \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ &\quad - \theta_3 \eta_j \eta_k p_j p_k \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ &\quad - \frac{3ig_0^2}{(2\pi)^4} \lim_{\epsilon \downarrow 0} \frac{C_{\mathfrak{g}} \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{1}{r^2 + i\epsilon} dr \\ &\quad + \frac{ig_0^2}{2(2\pi)^4} \lim_{\epsilon \downarrow 0} \frac{C_{\mathfrak{g}} \delta_{ab} \delta^{(4)}(p+q)}{(p^2 + i\epsilon)^2} \int \frac{L_{jk}(p, r)}{(r^2 + i\epsilon)((r+p)^2 + i\epsilon)} dr \\ &\quad + O(\theta_1^2 + \theta_2^2 + \theta_3^2 + g_0^3), \end{aligned} \quad (13.5.12)$$

where $C_{\mathfrak{g}}$ is the constant defined in equation (13.5.3) and L_{jk} is the polynomial defined in equation (13.5.10). Our next task is to renormalize the parameters to cancel out the infinities.

13.6 The Feynman parameter

The Feynman parameter is an integration trick invented by Richard Feynman, that is widely used in computations in quantum field theory. The key lemma is the following. The parameter u introduced below is known as the Feynman parameter.

Lemma 13.6.1. *Suppose that $A, B \in \mathbb{C}$ are such that the line segment joining A and B in the complex plane does not pass through 0. Then*

$$\frac{1}{AB} = \int_0^1 \frac{1}{(Au + B(1-u))^2} du.$$

Proof. Note that

$$\frac{d}{du} \left(\frac{1}{(B-A)(Au + B(1-u))} \right) = \frac{1}{(Au + B(1-u))^2}$$

and apply the fundamental theorem of calculus. $\square$

As an example, consider the integral

$$\int_{\mathbb{R}^4} \frac{L_{jk}(p, r)}{(r^2 + i\epsilon)((r+p)^2 + i\epsilon)} dr$$

appearing in equation (13.5.12). By Lemma 13.6.1, this can be expressed as

$$\begin{aligned} & \int_{\mathbb{R}^4} \int_0^1 \frac{L_{jk}(p, r)}{((r+p)^2 u + r^2(1-u) + i\epsilon)^2} du dr \\ &= \int_{\mathbb{R}^4} \int_0^1 \frac{L_{jk}(p, r)}{(r^2 + 2(r, up) + up^2 + i\epsilon)^2} du dr = \int_{\mathbb{R}^4} \int_0^1 \frac{L_{jk}(p, r)}{((r+up)^2 + u(1-u)p^2 + i\epsilon)^2} du dr. \end{aligned}$$

By the change of variable $r \rightarrow r - up$, this becomes

$$\int_{\mathbb{R}^4} \int_0^1 \frac{L_{jk}(p, r - up)}{(r^2 + u(1-u)p^2 + i\epsilon)^2} du dr.$$

Rewriting this as

$$\int_0^1 \int_{\mathbb{R}^3} \int_{\mathbb{R}} \frac{L_{jk}(p, r - up)}{(r_0^2 - \|s\|^2 + u(1-u)p^2 + i\epsilon)^2} dr_0 ds du,$$

where $r = (r_0, s)$, and applying Corollary 6.3.2 to the integral over r_0 , we get the above integral is formally equal to

$$i \int_0^1 \int_{\mathbb{R}^4} \frac{L_{jk}(p, r' - up)}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr du,$$

where $r' := (ir_0, r_1, r_2, r_3)$. We can simplify this integral as follows. First, note that by the

symmetry $r_j \rightarrow -r_j$,

$$\int_{\mathbb{R}^4} \frac{r'_j}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr = 0,$$

and the same holds with r_k instead of r_j . By the symmetry $r \rightarrow -r$,

$$\int_{\mathbb{R}^4} \frac{(p, r')}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr = 0.$$

Next, note that if $j \neq k$, then the symmetry $r_j \rightarrow -r_j$ implies that

$$\int_{\mathbb{R}^4} \frac{r'_j r'_k}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr = 0.$$

On the other hand, if $j = k$, then the symmetry between coordinates implies that

$$\begin{aligned} \int_{\mathbb{R}^4} \frac{r'_j r'_k}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr &= \int_{\mathbb{R}^4} \frac{r_j'^2}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr \\ &= \int_{\mathbb{R}^4} \frac{-\eta_j r_j^2}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr \\ &= -\frac{\eta_j}{4} \int_{\mathbb{R}^4} \frac{\|r\|^2}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr. \end{aligned}$$

Combining all of the above observations, we get the simplification

$$\begin{aligned} &\int_{\mathbb{R}^4} \frac{L_{jk}(p, r' - up)}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr \\ &= \int_{\mathbb{R}^4} \frac{2\eta_j \delta_{jk} (p - r' + up)^2 - 2\eta_j \eta_k p_j p_k + 8\eta_j \eta_k (r'_k - up_k)(r'_j + (1-u)p_j)}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr \\ &= - \int_{\mathbb{R}^4} \frac{M_{jk}(p, r)}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr, \end{aligned}$$

where $M_{jk}(p, r) := 2\eta_j \delta_{jk} (2\|r\|^2 - (1+u)^2 p^2) + (8u(1-u) + 2)\eta_j \eta_k p_j p_k$. Similarly,

$$\int_{\mathbb{R}^4} \frac{1}{r^2 + i\epsilon} dr = -i \int_{\mathbb{R}^4} \frac{1}{\|r\|^2 - i\epsilon} dr.$$

Thus, the perturbative expansion (13.5.12) can be rewritten as

$$\begin{aligned} D_{jk}^{ab}(p, q) &= \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{q^2 + i\epsilon} - \theta_1 p^2 \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ &\quad - \theta_3 \eta_j \eta_k p_j p_k \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ &\quad - \frac{3g_0^2}{(2\pi)^4} C_{\mathfrak{g}} \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \left(\int \frac{1}{\|r\|^2 - i\epsilon} dr \right) \frac{1}{(p^2 + i\epsilon)^2} \\ &\quad - \frac{g_0^2}{2(2\pi)^4} C_{\mathfrak{g}} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \left(\int_{\mathbb{R}^4} \int_0^1 \frac{M_{jk}(p, r)}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} du dr \right) \frac{1}{(p^2 + i\epsilon)^2} \\ &\quad + O(\theta_1^2 + \theta_2^2 + \theta_3^2 + g_0^3). \end{aligned} \tag{13.6.1}$$

13.7 Dimensional regularization of Yang–Mills theories

To renormalize, we first have to regularize the theory. The regularization that works best for Yang–Mills theories is dimensional regularization. Recall that by Example 12.3.1 from Section 12.3,

$$\lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^{4-\delta}} \frac{1}{\|r\|^2 - i\epsilon} dr = 0.$$

The above identity is actually the key reason why dimensional regularization is preferred for Yang–Mills theories. If we opt for a hard cutoff as we did for φ_4^4 theory in Chapter 6, then the above integral does not vanish, and neither can it be renormalized by adjusting the coefficients of the other terms. In hard cutoff, we have to introduce an additional quadratic gauge fixing term to perform the renormalization.

Next, by Example 12.3.2, for any δ we have

$$\lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^{4-\delta}} \frac{1}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr = \frac{2\pi^2}{\delta} - \pi^2 \log(-\pi u(1-u)p^2) - \pi^2 \gamma + O(\delta),$$

where $\log$ denotes the branch of logarithm on $\mathbb{C} \setminus (-\infty, 0]$ which is real-valued on $(0, \infty)$, and γ is the Euler–Mascheroni constant. Integrating over u from 0 to 1 (and assuming that limit and integral can be interchanged), we get

$$\begin{aligned} & \lim_{\epsilon \downarrow 0} \int_0^1 \int_{\mathbb{R}^{4-\delta}} \frac{1}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr du \\ &= \frac{2\pi^2}{\delta} - \pi^2 \log(-\pi p^2) + \pi^2(1 - \gamma) + O(\delta). \end{aligned}$$

Similarly,

$$\begin{aligned} & \lim_{\epsilon \downarrow 0} \int_0^1 \int_{\mathbb{R}^{4-\delta}} \frac{(1+u)^2}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr du \\ &= \frac{14\pi^2}{3\delta} - \pi^2 \log(-\pi p^2) + \frac{85\pi^2}{18} - \pi^2 \gamma + O(\delta), \end{aligned}$$

and

$$\begin{aligned} & \lim_{\epsilon \downarrow 0} \int_0^1 \int_{\mathbb{R}^{4-\delta}} \frac{u(1-u)}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr du \\ &= \frac{\pi^2}{3\delta} - \pi^2 \log(-\pi p^2) + \frac{5\pi^2}{18} - \pi^2 \gamma + O(\delta). \end{aligned}$$

Next, by Example 12.3.3,

$$\begin{aligned} & \lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^{4-\delta}} \frac{\|r\|^2}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr \\ &= \frac{4\pi^2 u(1-u)p^2}{\delta} - 2\pi^2 u(1-u)p^2 \left(\gamma - \frac{1}{2} + \log(-\pi u(1-u)p^2) \right) + O(\delta). \end{aligned}$$

Again, integrating over u , we get

$$\begin{aligned} & \lim_{\epsilon \downarrow 0} \int_0^1 \int_{\mathbb{R}^{4-\delta}} \frac{\|r\|^2}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr du \\ &= \frac{2\pi^2 p^2}{3\delta} - \pi^2 p^2 \left(\frac{\gamma + \log(-\pi p^2)}{3} - \frac{13}{18} \right) + O(\delta). \end{aligned}$$

Combining the above calculations, we get

$$\begin{aligned} & \lim_{\epsilon \downarrow 0} \int_0^1 \int_{\mathbb{R}^{4-\delta}} \frac{M_{jk}(p, r)}{(\|r\|^2 - u(1-u)p^2 - i\epsilon)^2} dr du \\ &= \eta_j \delta_{jk} p^2 \left(-\frac{20}{3} \frac{\pi^2}{\delta} + \frac{2}{3} \pi^2 (\gamma + \log(-\pi p^2)) - \frac{59}{9} \pi^2 \right) \\ & \quad + \eta_j \eta_k p_j p_k \left(\frac{20\pi^2}{3\delta} - 10\pi^2 \log(-\pi p^2) + \frac{38\pi^2}{9} - 10\pi^2 \gamma \right) + O(\delta). \end{aligned}$$

Thus, the dimension regularized version of the gluon propagator is given by

$$\begin{aligned} & \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{q^2 + i\epsilon} - \theta_1 p^2 \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ & - \theta_3 \eta_j \eta_k p_j p_k \delta_{ab} \delta^{(4)}(p+q) \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ & - \frac{g_0^2}{2(2\pi)^4} C_{\mathfrak{g}} \delta_{ab} \delta^{(4)}(p+q) \left\{ \eta_j \delta_{jk} p^2 \left(-\frac{20}{3} \frac{\pi^2}{\delta} + \frac{2}{3} \pi^2 (\gamma + \log(-\pi p^2)) - \frac{59}{9} \pi^2 \right) \right. \\ & \quad \left. + \eta_j \eta_k p_j p_k \left(\frac{20\pi^2}{3\delta} - 10\pi^2 \log(-\pi p^2) + \frac{38\pi^2}{9} - 10\pi^2 \gamma \right) + O(\delta) \right\} \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} \\ & + O(\theta_1^2 + \theta_2^2 + \theta_3^2 + g_0^3). \end{aligned} \tag{13.7.1}$$

13.8 Renormalization to first order

To renormalize the dimension regularized version of the gluon propagator, we take a renormalization parameter g , and make the following choices for $\theta_1, \theta_2, \theta_3, g_0$ in terms of g :

$$\begin{aligned} \theta_1 &= \frac{5C_{\mathfrak{g}}}{24\pi^2\delta} g^2 + O(g^3), \\ \theta_2 &= O(g^3), \\ \theta_3 &= -\frac{5C_{\mathfrak{g}}}{24\pi^2\delta} g^2 + O(g^3), \\ g_0 &= g + O(g^2). \end{aligned}$$

These choices ensure that the $1/\delta$ terms in equation (13.7.1) cancel out exactly. Thus, we can now send δ to zero and obtain the renormalized gluon propagator:

$$\begin{aligned}
 D_{jk}^{ab}(p, q) &= \eta_j \delta_{jk} \delta_{ab} \delta^{(4)}(p + q) \lim_{\epsilon \downarrow 0} \frac{1}{q^2 + i\epsilon} \\
 &\quad + C_{\mathfrak{g}} g^2 \delta_{ab} \delta^{(4)}(p + q) \left\{ -\eta_j \delta_{jk} p^2 \left(\frac{1}{48\pi^2} (\gamma + \log(-\pi p^2)) - \frac{59}{288\pi^2} \right) \right. \\
 &\quad \left. + \eta_j \eta_k p_j p_k \left(\frac{5}{16\pi^2} (\gamma + \log(-\pi p^2)) - \frac{19}{144\pi^2} \right) \right\} \lim_{\epsilon \downarrow 0} \frac{1}{(p^2 + i\epsilon)^2} + O(g^4).
 \end{aligned}$$

Chapter 14

The Wightman Axioms

14.1 The Poincaré group

Let $\mathbb{R}^{1,3}$ denote the group of spacetime translations acting on the vector space $\mathbb{R}^{1,3}$, by the action $x(y) := x + y$. Recall the restricted Lorentz group $\mathrm{SO}^+(1, 3)$ defined in Chapter 4. The Poincaré group $\mathcal{P}$ is the semi-direct product $\mathbb{R}^{1,3} \rtimes \mathrm{SO}^+(1, 3)$, meaning that

$$\mathcal{P} = \{(a, A) : a \in \mathbb{R}^{1,3}, A \in \mathrm{SO}^+(1, 3)\},$$

with the group operation defined as

$$(a, A)(b, B) := (a + Ab, AB).$$

This is the group of isometries of Minkowski spacetime. It is simple to verify that this is a Lie group. The Poincaré group acts on $\mathbb{R}^{1,3}$ as

$$(a, A)(x) := a + Ax.$$

A representation U of the Poincaré group in a Hilbert space $\mathcal{H}$ is a map U from $\mathcal{P}$ into the space of unitary operators on $\mathcal{H}$ such that

$$U((a, A)(b, B)) = U(a, A)U(b, B)$$

for all (a, A) and (b, B) . The representation is called strongly continuous if for any $x \in \mathcal{H}$, the map $(a, A) \mapsto U(a, A)x$ is continuous.

14.2 The framework of relativistic quantum mechanics

Until now, we have treated a quantum system as a Hilbert space of states, coupled with a strongly continuous group of unitary operators which determines how a given state evolves in time. But there is a fundamental difficulty in reconciling the above framework of quantum mechanics with special relativity. Recall from our discussion in Chapter 4 that the basic

postulate of special relativity is that the laws of the universe should remain invariant under change of spacetime coordinates by restricted Lorentz transforms. But the notion of two events happening ‘at the same time’ need not remain the same under a change of coordinate system. So what does even mean to talk about time evolution groups of unitary operators?

To resolve this difficulty, we have to replace the time evolution group by a representation U of the Poincaré group. This works as follows. If an observer in a given coordinate system observes the system to be in state ψ , then an observer in a new coordinate system obtained by the action of (a, A) on $\mathbb{R}^{1,3}$ observes the state of the same system as $U(a, A)\psi$.

The correspondence of this new picture with the earlier picture of a time evolution group is as follows. Consider the transformation $(a, 1)$ acting on a point $x = (x_0, x_1, x_2, x_3)$, where $a = a(t) := (-t, 0, 0, 0)$ for some t . The result of this action gives us the point $(x_0 - t, x_1, x_2, x_3)$. This reduces the time coordinate by t , so the new reference frame obtained by the action of $(a, 1)$ is that of an observer who is t units ahead in time. The new state $U(a, 1)\psi$ is what the system will look like t units of time later, if it looks like ψ now.

Note that the family of unitary operators $(U(a(t), 1))_{t \in \mathbb{R}}$ is a strongly continuous unitary group acting on $\mathcal{H}$, so by Stone’s theorem, there exists a self-adjoint operator H such that $U(a(t), 1) = e^{-itH}$. So if the observer remains stationary, then the system evolves in time just as if we were in the original quantum mechanical setting, with H being the Hamiltonian governing time evolution of states.

14.3 The Wightman axioms

The Wightman axioms give a framework for rigorously defining interacting quantum field theories. The axioms are designed to ensure that there are no physical inconsistencies. Let us only consider the axioms for bosonic scalar fields for now. Recall the Minkowski inner product and the Minkowski norm on $\mathbb{R}^{1,3}$, defined in Chapter 4.

Axiom 1. There exists a Hilbert space $\mathcal{H}$ whose elements represent the possible states of the quantum system under consideration, and a unitary representation U of $\mathcal{P}$ in $\mathcal{H}$ which satisfies the following assumption. Note that $(U(a, 1))_{a \in \mathbb{R}^{1,3}}$ is a commuting family of unitary operators. Therefore by the spectral theorem, we have that

$$U(a, 1) = \int_{\mathbb{R}^{1,3}} e^{i(a,p)} dE(p),$$

where E is a projection-valued measure on $\mathbb{R}^{1,3}$. Let

$$L := \{p = (p_0, p_1, p_2, p_3) \in \mathbb{R}^{1,3} : p^2 \geq 0, p_0 \geq 0\}$$

be the ‘forward light cone’. We assume that

$$E(A) = 0 \text{ whenever } A \cap L = \emptyset. \quad (14.3.1)$$

This assumption is required to ensure that there are no particles with negative masses in the theory. There is a simple way to check equation (14.3.1). The condition is satisfied if and only if for any $\xi, \eta \in \mathcal{H}$ and any $f \in \mathcal{S}(\mathbb{R}^{1,3}, \mathbb{C})$, such that the support of $\hat{f}$ does not intersect L , we have

$$\int_{\mathbb{R}^4} (\xi, U(a, 1)\eta) da = 0.$$

Axiom 2. There exists a dense subspace $\mathcal{D}$ of $\mathcal{H}$ and a linear map

$$\varphi : \mathcal{S}(\mathbb{R}^4, \mathbb{C}) \rightarrow \{\text{unbounded operators on } \mathcal{H}\}$$

such that for any $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$, $\varphi(f)$ is defined on $\mathcal{D}$ and $\varphi(f)(\mathcal{D}) \subseteq \mathcal{D}$. Moreover, we assume that for any $\xi, \eta \in \mathcal{D}$,

$$(\varphi(\bar{f})\xi, \eta) = (\xi, \varphi(f)\eta).$$

In particular if f is real-valued, then $\varphi(f)$ is a symmetric operator on $\mathcal{D}$.

Axiom 3. For any $\xi, \eta \in \mathcal{D}$, $f \mapsto (\xi, \varphi(f)\eta)$ is a tempered distribution.

Axiom 4. For any $(a, A) \in \mathcal{P}$,

$$U(a, A)\varphi(f)U(a, A)^{-1} = \varphi(V(a, A)f),$$

where

$$V(a, A)f(x) = f(A^{-1}(x - a)).$$

This is the axiom of Lorentz invariance, designed to ensure compatibility with special relativity.

Axiom 5. If the supports of $f, g \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$ are ‘spacelike separated’, then

$$[\varphi(f), \varphi(g)] = 0.$$

(Two points $x, y \in \mathbb{R}^{1,3}$ are called spacelike separated if $(x - y)^2 < 0$. Two sets are spacelike separated if any point in one is spacelike separated from any point in the other.) This is the locality axiom, designed to ensure compatibility with the stipulation of special relativity that a signal cannot travel faster than the speed of light.

Axiom 6. There exists $\Omega \in \mathcal{D}$ such that $U(a, 1)\Omega = \Omega$ for any a , and it is the unique element satisfying this condition. (One can actually deduce from the axioms that $U(a, A)\Omega = \Omega$ for any $(a, A) \in \mathcal{P}$.) This is the axiom that ensures the existence and uniqueness of the vacuum state.

Axiom 7. The closed linear span of elements like $\varphi(f_1) \cdots \varphi(f_n)\Omega$ over all n and $f_1, \dots, f_n$ is equal to the Hilbert space $\mathcal{H}$. This axiom is designed to ensure that the field φ is ‘rich enough’ to generate all possible states.

14.4 Verifying the axioms for the free quantum field

Chapter 15

Introduction to Lattice Gauge Theories

15.1 Euclidean Yang–Mills theories

Let G be a compact matrix Lie group contained in $U(n)$ for some n , and let $\mathfrak{g}$ be its Lie algebra. Recall from Section C.4 in Appendix C that a $\mathfrak{g}$ -valued 1-form A on the Minkowski space $\mathbb{R}^{1,3}$ has the form $\sum_{i=0}^3 A_i dx_i$, where A_0, A_1, A_2, A_3 are $\mathfrak{g}$ -valued smooth maps on $\mathbb{R}^3$. Recall that the curvature form of A is the $\mathfrak{g}$ -valued 2-form $F = \sum_{0 \leq i < j \leq 3} F_{ij} dx_i \wedge dx_j$ given by

$$F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j],$$

where $\partial_i A_j$ is shorthand for $\frac{\partial A_j}{\partial x_i}$ and $[A_i, A_j]$ is the commutator of A_i and A_j . Recall from Section 8.1 that the action for pure Yang–Mills theory with gauge group G on $\mathbb{R}^{1,3}$ is given by

$$S(A) = -\frac{1}{2g_0^2} \int_{\mathbb{R}^4} \text{Tr} \left(\sum_{0 \leq i < j \leq 3} \eta_i \eta_j F_{ij}(x)^2 \right) dx,$$

where $\eta_0 = 1$ and $\eta_1 = \eta_2 = \eta_3 = -1$, A is a $\mathfrak{g}$ -valued 1-form on $\mathbb{R}^{1,3}$ and F is its curvature form, and g_0 is the bare coupling constant.

If one wants to construct quantum Yang–Mills theories following the approach outlined in Chapter 2, the first step would be to define an Euclidean version of the action S , following the prescription from Section 1.7. Recall that the prescription is to replace t by $-it$ wherever t appears in calculations, but not as arguments of functions (where $t = x_0$ in the above discussion). Thus, the dx_0 in the integral should be replaced by $-id x_0$, and $\partial_0 = \frac{\partial}{\partial x_0}$ should be replaced by $\frac{1}{-i} \frac{\partial}{\partial x_0} = i\partial_0$. Lastly, we need to also replace A_0 by iA_0 , for the following reason. Recall that $A = \sum_{j=0}^3 A_j dx_j$. Now, if dx_0 is replaced by $-id x_0$, then to keep A the same, we need to replace A_0 by iA_0 .

With the above replacements, notice that for $1 \leq j \leq 3$, $\partial_0 A_j$ is replaced by $i\partial_0 A_j$, $\partial_j A_0$

is replaced by $i\partial_j A_0$, and $[A_0, A_j]$ is replaced by $i[A_0, A_j]$. Lastly, the integration element $dx = dx_0 dx_1 dx_2 dx_3$ is replaced by $-i dx_0 dx_1 dx_2 dx_3 = -i dx$ in the integral. Everything else remains unaltered. Thus, F_{0j} is replaced by iF_{0j} for $1 \leq j \leq 3$, and F_{ij} remains unaltered for $1 \leq i < j \leq 3$. Putting it all together, we see $S(A)$ changes to

$$\frac{i}{2g_0^2} \int_{\mathbb{R}^4} \text{Tr} \left(\sum_{0 \leq i < j \leq 3} F_{ij}(x)^2 \right) dx.$$

Recall from Section 1.7 that the Euclidean action is obtained by multiplying the above by i . This gives the Euclidean action

$$S_E(A) := -\frac{1}{2g_0^2} \int_{\mathbb{R}^4} \text{Tr} \left(\sum_{0 \leq i < j \leq 3} F_{ij}(x)^2 \right) dx. \quad (15.1.1)$$

Heuristically, a Euclidean Yang–Mills theory is a probability measure on the space of $\mathfrak{g}$ -valued 1-forms on $\mathbb{R}^4$ that has probability density proportional to $e^{-S_E(A)}$ with respect to a ‘Lebesgue measure’ on this space. Other than $G = \text{U}(1)$, we do not know how to construct these theories at the present time.

The Euclidean action S_E defined in equation (15.1.1) can be readily generalized to d dimensions for any $d \geq 2$. The connection $A = \sum_{j=0}^{d-1} A_j dx_j$ is then a $\mathfrak{g}$ -valued 1-form on $\mathbb{R}^d$, with curvature form $F = \sum_{0 \leq i < j \leq d-1} F_{ij} dx_i \wedge dx_j$, where F_{ij} given by the same formula as before. The Euclidean action S_E is the defined just as in equation (15.1.1), with the sum going over $0 \leq i < j \leq d-1$.

Henceforth, we will run the indices from 1 to d instead of 0 to $d-1$, for ease of notation. Thus, a connection A would be written as $\sum_{j=1}^d A_j dx_j$, and so on.

15.2 Lattice gauge theories

In 1974, Wilson [32] proposed a discretization of Euclidean Yang–Mills theories. These are now known as lattice gauge theories or lattice Yang–Mills theories. Let G be as in the previous section. The lattice gauge theory with gauge group G on a finite set $\Lambda \subseteq \mathbb{Z}^d$ is defined as follows. Suppose that for any two adjacent vertices $x, y \in \Lambda$, we have a unitary matrix $U(x, y) \in G$, with the constraint that $U(y, x) = U(x, y)^{-1}$. Let us call any such assignment of matrices to edges a ‘configuration’. Let $G(\Lambda)$ denote the set of all configurations. A square bounded by four edges is called a plaquette. Let $P(\Lambda)$ denote the set of all plaquettes in Λ . For a plaquette $p \in P(\Lambda)$ with vertices x_1, x_2, x_3, x_4 in anti-clockwise order, and a configuration $U \in G(\Lambda)$, define

$$U_p := U(x_1, x_2)U(x_2, x_3)U(x_3, x_4)U(x_4, x_1). \quad (15.2.1)$$

The Wilson action of U is defined as

$$S_\Lambda(U) := \sum_{p \in P(\Lambda)} \operatorname{Re}(\operatorname{Tr}(I - U_p)), \quad (15.2.2)$$

where I is the identity matrix of order n . Let σ_Λ be the product Haar measure¹ on $G(\Lambda)$, normalized to be a probability measure. Given $\beta > 0$, let $\mu_{\Lambda,\beta}$ be the probability measure on $G(\Lambda)$ defined as

$$d\mu_{\Lambda,\beta}(U) := \frac{1}{Z} e^{-\beta S_\Lambda(U)} d\sigma_\Lambda(U),$$

where Z is the normalizing constant. This probability measure is called the lattice gauge theory on Λ for the gauge group G , with inverse coupling strength β , and free boundary condition.

The following result gives some justification why lattice gauge theories may be viewed as lattice approximations of Euclidean Yang–Mills theories.

Theorem 15.2.1. *Take any $d \geq 2$. Let $A = \sum_{j=1}^d A_j dx_j$ be a compactly supported smooth $\mathfrak{g}$ -valued 1-form on $\mathbb{R}^d$. Let $e_1, \dots, e_d$ denote the standard basis vectors of $\mathbb{R}^d$. For a directed edge $(x, x + \epsilon e_j)$ of $\epsilon\mathbb{Z}^d$, define $U(x, x + \epsilon e_j) := e^{\epsilon A_j(x)}$, and let $U(x + \epsilon e_j, x) := U(x, x + \epsilon e_j)^{-1}$. For a plaquette p in $\epsilon\mathbb{Z}^d$, let U_p be defined as in equation (15.2.1). Let $S(U)$ be defined as in equation (15.2.2), but with p running over all plaquettes of $\epsilon\mathbb{Z}^d$. Then*

$$\lim_{\epsilon \rightarrow 0} g_0^{-2} \epsilon^{d-4} S(U) = S_E(A),$$

where S_E is the Euclidean Yang–Mills action defined in the previous section.

To prove this result, we need the following four-term version of the Baker–Campbell–Hausdorff formula for the product of matrix exponentials.

Lemma 15.2.2. *Let A and B be two square matrices of order n , and let K be an upper bound on the absolute values of the entries of A and B . Then for any $t \in [0, 1]$,*

$$\begin{aligned} e^{tA} e^{tB} = \exp \bigg(& t(A + B) + \frac{t^2}{2} [A, B] + \frac{t^3}{12} ([A, [A, B]] + [B, [B, A]]) \\ & - \frac{t^4}{24} [A, [B, [A, B]]] \bigg) + O(t^5), \end{aligned}$$

where $O(t^5)$ denotes a matrix whose entries have absolute values bounded by Ct^5 , where C depends only on K and n .

Proof. Define $g(t) := e^{h(t)}$, where

$$h(t) := t(A + B) + \frac{t^2}{2} [A, B] + \frac{t^3}{12} ([A, [A, B]] + [B, [B, A]]) - \frac{t^4}{24} [A, [B, [A, B]]].$$

¹See Section C.7 in Appendix C for an introduction to Haar measures on Lie groups.

Then note that $h(t) = O(t)$, and hence,

$$g(t) = I + h(t) + \frac{1}{2}h(t)^2 + \frac{1}{6}h(t)^3 + \frac{1}{24}h(t)^4 + O(t^5).$$

But, we have

$$\begin{aligned} h(t)^2 &= t^2(A+B)^2 + \frac{1}{2}t^3((A+B)[A,B] + [A,B](A+B)) \\ &\quad + \frac{t^4}{12}((A+B)([A,[A,B]] + [B,[B,A]]) \\ &\quad + ([A,[A,B]] + [B,[B,A]])(A+B)) + \frac{t^4}{4}[A,B]^2 + O(t^5), \end{aligned}$$

where we clubbed all terms of order t^5 and smaller into the $O(t^5)$ remainder. Similarly,

$$\begin{aligned} h(t)^3 &= t^3(A+B)^3 + \frac{1}{4}t^4((A+B)^2[A,B] + (A+B)[A,B](A+B) \\ &\quad + [A,B](A+B)^2) + O(t^5), \end{aligned}$$

and

$$h(t)^4 = t^4(A+B)^4 + O(t^5).$$

Combining, we get

$$\begin{aligned} g(t) &= I + t(A+B) + t^2\left(\frac{1}{2}[A,B] + \frac{1}{2}(A+B)^2\right) \\ &\quad + t^3\left(\frac{1}{12}([A,[A,B]] + [B,[B,A]]) + \frac{1}{4}((A+B)[A,B] + [A,B](A+B))\right. \\ &\quad \left.+ \frac{1}{6}(A+B)^3\right) \\ &\quad + t^4\left(-\frac{1}{24}[A,[B,[A,B]]] + \frac{1}{24}((A+B)([A,[A,B]] + [B,[B,A]]))\right. \\ &\quad \left.+ \frac{1}{24}((A+B)^2[A,B] + (A+B)[A,B](A+B) + [A,B](A+B)^2)\right. \\ &\quad \left.+ \frac{1}{24}(A+B)^4\right) + O(t^5). \end{aligned}$$

Simplifying the above gives

$$\begin{aligned} g(t) &= I + t(A+B) + \frac{1}{2}t^2(A^2 + 2AB + B^2) + \frac{1}{6}t^3(A^3 + 3A^2B + 3AB^2 + B^3) \\ &\quad + \frac{1}{24}t^4(A^4 + 4A^3B + 6A^2B^2 + 4AB^3 + B^4) + O(t^5). \end{aligned}$$

But note that

$$e^{tA}e^{tB} = \left(I + tA + \frac{1}{2}t^2A^2 + \frac{1}{6}t^3A^3 + \frac{1}{24}t^4A^4 + O(t^5) \right) \cdot \left(I + tB + \frac{1}{2}t^2B^2 + \frac{1}{6}t^3B^3 + \frac{1}{24}t^4B^4 + O(t^5) \right).$$

Upon expanding, this gives the same terms up to $O(t^5)$ as in the above expression for $g(t)$. This completes the proof. $\square$

Specializing to skew-Hermitian matrices and iterating Lemma 15.2.2 gives the following result.

Lemma 15.2.3. *Let $B_1, \dots, B_m$ be skew-Hermitian matrices of order n , and let K be an upper bound on the absolute values of the entries of these matrices. Then for any $t \in [0, 1]$,*

$$e^{tB_1} \dots e^{tB_m} = \exp \left(t \sum_{i=1}^m B_i + \frac{t^2}{2} \sum_{1 \leq i < j \leq m} [B_i, B_j] + R \right) + O(t^5),$$

where $R = O(t^3)$ and R is skew-Hermitian, and as before, $O(t^\alpha)$ denotes a matrix whose entries have absolute values bounded by Ct^α , where C depends only on m, n , and K .

Proof. Lemma 15.2.2 proves the claim for $m = 2$. Suppose that it is true for up to $m - 1$. Take any $t \in [0, 1]$. Then by the induction hypothesis, $e^{tB_1} \dots e^{tB_{m-1}} = e^{tA} + O(t^5)$, where

$$A = \sum_{i=1}^{m-1} B_i + \frac{t}{2} \sum_{1 \leq i < j \leq m-1} [B_i, B_j] + Q,$$

where $Q = O(t^2)$ and Q is skew-Hermitian. Note that the absolute values of the entries of A are bounded by CK , where C depends only on m and n . Since the commutator of two skew-Hermitian matrices is skew-Hermitian, Lemma 15.2.2 implies that for all $s \in [0, 1]$,

$$e^{sA}e^{sB_m} = \exp \left(s(A + B_m) + \frac{s^2}{2}[A, B_m] + S \right) + O(s^5),$$

where $S = O(s^3)$ and S is skew-Hermitian. Taking $s = t$ gives and again invoking the observation that the set of skew-Hermitian matrices is closed under commutation, we get the desired result. $\square$

Lemma 15.2.3 yields the following corollary.

Corollary 15.2.4. *Let $B_1, \dots, B_m$ be skew-Hermitian matrices of order n , and let K be an*

upper bound on the absolute values of the entries of these matrices. Then for any $t \in [0, 1]$,

$$\begin{aligned} \operatorname{Re}(\operatorname{Tr}(I - e^{tB_1} \dots e^{tB_m})) &= -\frac{1}{2} \operatorname{Tr} \left[\left(t \sum_{i=1}^m B_i + \frac{t^2}{2} \sum_{1 \leq i < j \leq m} [B_i, B_j] \right)^2 \right] \\ &\quad + \operatorname{Tr} \left(O(t^4) \sum_{i=1}^m B_i \right) + O(t^5), \end{aligned}$$

where $O(t^\alpha)$ denotes a matrix whose entries have absolute values bounded by Ct^α , where C depends only on m , n , and K .

Proof. By Lemma 15.2.3, we have

$$e^{tB_1} \dots e^{tB_m} = I + \sum_{k=1}^4 \frac{1}{k!} \left(t \sum_{i=1}^m B_i + \frac{t^2}{2} \sum_{1 \leq i < j \leq m} [B_i, B_j] + R \right)^k + O(t^5),$$

where R is skew-Hermitian and $R = O(t^3)$. Note that the eigenvalues of a skew-Hermitian matrix are all purely imaginary. This implies that

$$\operatorname{Tr} \left[\left(t \sum_{i=1}^m B_i + \frac{t^2}{2} \sum_{1 \leq i < j \leq m} [B_i, B_j] + R \right)^k \right]$$

is purely imaginary for each odd k and purely real for each even k . Combining these observations, we get

$$\begin{aligned} \operatorname{Re}(\operatorname{Tr}(I - e^{tB_1} \dots e^{tB_m})) &= -\frac{1}{2} \operatorname{Tr} \left[\left(t \sum_{i=1}^m B_i + \frac{t^2}{2} \sum_{1 \leq i < j \leq m} [B_i, B_j] + R \right)^2 \right] \\ &\quad - \frac{1}{24} \operatorname{Tr} \left[\left(t \sum_{i=1}^m B_i + \frac{t^2}{2} \sum_{1 \leq i < j \leq m} [B_i, B_j] + R \right)^4 \right] + O(t^5). \end{aligned}$$

Since $R = O(t^3)$, it is easy to see that this gives the desired result. $\square$

We are now ready to prove Theorem 15.2.1.

Proof of Theorem 15.2.1. In this proof, $O(\epsilon^\alpha)$ will denote any matrix whose entries have absolute values bounded by $C\epsilon^\alpha$, where C is a constant that may depend only on the group G , the dimension d , and the connection A . Take any $x \in \epsilon\mathbb{Z}^d$ and any $1 \leq j < k \leq d$, and let

$$x_1 = x, \quad x_2 = x + \epsilon e_j, \quad x_3 = x + \epsilon e_j + \epsilon e_k, \quad x_4 = x + \epsilon e_k.$$

Let p be the plaquette formed by the vertices x_1, x_2, x_3, x_4 . Let U_p be defined as in equations (15.2.1). Then

$$U_p = e^{\epsilon A_j(x_1)} e^{\epsilon A_k(x_2)} e^{-\epsilon A_j(x_4)} e^{-\epsilon A_k(x_1)}.$$

Since G is a subgroup of $U(n)$, the elements of $\mathfrak{g}$ are skew-Hermitian. Therefore by

Corollary 15.2.4, we get

$$\begin{aligned} \operatorname{Re}(\operatorname{Tr}(I - U_p)) &= -\frac{1}{2} \operatorname{Tr} \left[\left(\epsilon(A_j(x_1) + A_k(x_2) - A_j(x_4) - A_k(x_1)) \right. \right. \\ &\quad \left. \left. + \frac{\epsilon^2}{2} \left\{ [A_j(x_1), A_k(x_2)] - [A_j(x_1), A_j(x_4)] - [A_j(x_1), A_k(x_1)] \right. \right. \right. \\ &\quad \left. \left. \left. - [A_k(x_2), A_j(x_4)] - [A_k(x_2), A_k(x_1)] + [A_j(x_4), A_k(x_1)] \right\} \right) \right]^2 \\ &\quad + \operatorname{Re}[\operatorname{Tr}(O(\epsilon^4)(A_j(x_1) + A_k(x_2) - A_j(x_4) - A_k(x_1))) + O(\epsilon^5)]. \end{aligned}$$

Writing

$$A_k(x_2) = A_k(x + \epsilon e_j) = A_k(x) + \epsilon \partial_j A_k(x) + O(\epsilon^2)$$

and using a similar Taylor expansion for $A_j(x_4)$, we get

$$A_j(x_1) + A_k(x_2) - A_j(x_4) - A_k(x_1) = \epsilon(\partial_j A_k(x) - \partial_k A_j(x)) + O(\epsilon^2).$$

Similarly, $[A_j(x_1), A_k(x_2)]$, $[A_j(x_1), A_k(x_1)]$, $-[A_k(x_2), A_j(x_4)]$, and $[A_j(x_4), A_k(x_1)]$ are equal to $[A_j(x), A_k(x)] + O(\epsilon)$, and $[A_j(x_1), A_j(x_4)]$ and $[A_k(x_2), A_k(x_1)]$ are equal to $O(\epsilon)$. Combining the above observations gives

$$\begin{aligned} \operatorname{Re}(\operatorname{Tr}(I - U_p)) &= -\frac{1}{2} \epsilon^4 \operatorname{Tr}[(\partial_j A_k(x) - \partial_k A_j(x) + [A_j(x), A_k(x)] + O(\epsilon))^2] + O(\epsilon^5) \\ &= -\frac{1}{2} \epsilon^4 \operatorname{Tr}(F_{jk}(x)^2) + O(\epsilon^5). \end{aligned}$$

Thus, we get

$$S(U) = \sum_p \operatorname{Re}(\operatorname{Tr}(I - U_p)) = -\frac{1}{2} \sum_{x \in \mathbb{Z}^d} \sum_{1 \leq j < k \leq d} (\epsilon^4 \operatorname{Tr}(F_{jk}(x)^2) + O(\epsilon^5)).$$

Since A is compactly supported, it suffices to sum over only $O(\epsilon^{-d})$ plaquettes instead of all plaquettes. The sum over x can also be reduced to a sum over $O(\epsilon^{-d})$ many x 's for the same reason. Thus, the error terms add up to $O(\epsilon^{5-d})$, and the main terms add up to

$$-\frac{\epsilon^{4-d}}{2} \int_{\mathbb{R}^d} \sum_{1 \leq j < k \leq d} \operatorname{Tr}(F_{jk}(x)^2) dx + O(\epsilon^{5-d}).$$

This completes the proof. $\square$

15.3 Gauge transformations

Let us recall the definition of a smooth gauge transformation from Section 8.6, specialized to the trivial principal bundle $\mathbb{R}^d \times G$. A gauge transformation in this setting is simply a

smooth map U from $\mathbb{R}^d$ into G , which acts on a $\mathfrak{g}$ -valued 1-form $A = \sum_{j=1}^d A_j dx_j$ as

$$A^U := UAU^{-1} - (dU)U^{-1}$$

Explicitly, this means that for $1 \leq j \leq d$,

$$A_j^U(x) = U(x)A_j(x)U(x)^{-1} - (\partial_j U(x))U(x)^{-1}.$$

As argued in Section 8.6, smooth gauge transforms form a group under pointwise multiplication, which acts on smooth $\mathfrak{g}$ -valued 1-forms via the above action.

The definition of gauge transformations for lattice gauge theories is as follows. Consider the lattice gauge theory on a finite set $\Lambda \subseteq \mathbb{Z}^d$ defined in the previous section. The group of gauge transforms for this theory is the set G^Λ of all maps from Λ into G . Again, this is a group under pointwise multiplication. It acts on a configuration $U \in G(\Lambda)$ as follows. Given two adjacent vertices $x, y \in \Lambda$ and a gauge transform $g \in G^\Lambda$, we define

$$U^g(x, y) := g(x)U(x, y)g(y)^{-1}.$$

Note that $U^g \in G(\Lambda)$, since

$$\begin{aligned} U^g(y, x) &= g(y)U(y, x)g(x)^{-1} = g(y)U(x, y)^{-1}g(x)^{-1} \\ &= (g(x)U(x, y)g(y)^{-1})^{-1} = U^g(x, y)^{-1}. \end{aligned}$$

Also, this is a group action, since

$$\begin{aligned} (U^g)^h(x, y) &= h(x)U^g(x, y)h(y)^{-1} \\ &= h(x)g(x)U(x, y)g(y)^{-1}h(y)^{-1} = U^{hg}(x, y). \end{aligned}$$

We will now see why the above definition of a gauge transformation is a reasonable discrete analogue of the definition in the continuum. In the following, the logarithm of a matrix $U \in \mathrm{U}(n)$ is defined as follows. Let VDV^* be the spectral decomposition of U , where V is unitary and $D = \mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})$ for some $\theta_1, \dots, \theta_n \in [-\pi, \pi)$, which are unique given V . We define

$$\log U := V \mathrm{diag}(i\theta_1, \dots, i\theta_n) V^*.$$

Note that with this definition, $\exp(\log U) = U$.

Theorem 15.3.1. *Let A be a smooth, compactly supported $\mathfrak{g}$ -valued 1-form on $\mathbb{R}^d$, and $g \in C^\infty(\mathbb{R}^d, G)$ that is equal to I outside a compact set. Define $U(x, x + \epsilon e_j)$ as in Theorem 15.2.1. Let A^g and U^g be defined as above. For each $x \in \epsilon\mathbb{Z}^d$ and $1 \leq j \leq d$, define $B_j(x) := \epsilon^{-1} \log U^g(x, x + \epsilon e_j)$. Then*

$$\lim_{\epsilon \rightarrow 0} \sup_{x \in \epsilon\mathbb{Z}^d, 1 \leq j \leq d} \|A_j^g(x) - B_j(x)\| = 0,$$

where $\|\cdot\|$ denotes Euclidean norm on $n \times n$ matrices.

We need the following lemma for the matrix logarithm defined above.

Lemma 15.3.2. *Let $U \in \mathrm{U}(n)$ be such that $\|U - I\| \leq \frac{1}{2}$. Then*

$$\log U = - \sum_{k=1}^{\infty} \frac{(I - U)^k}{k},$$

where $\|\cdot\|$ denotes Euclidean norm, and the series on the right is absolutely convergent with respect to this norm.

Proof. First, to see that the series is absolutely convergent, simply note that the Euclidean norm on $n \times n$ matrices is submultiplicative (meaning that $\|UV\| \leq \|U\|\|V\|$) by a simple application of the Cauchy–Schwarz inequality. Thus, $\|(U - I)^k\| \leq \|U - I\|^k \leq 2^{-k}$.

Next, let VDV^* be the spectral decomposition of U , where V is unitary and $D = \mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})$ for some $\theta_1, \dots, \theta_n \in [0, 2\pi)$. Then $\log U = V\Lambda V^*$, where $\Lambda = \mathrm{diag}(i\theta_1, \dots, i\theta_n)$. Let $\log$ denote the analytic branch of logarithm on $\mathbb{C} \setminus (-\infty, 0]$ that agrees with the natural logarithm on $(0, \infty)$. Then recall that $\log$ has the following power series expansion around 1:

$$\log z = - \sum_{k=1}^{\infty} \frac{(1 - z)^k}{k},$$

with radius of convergence equal to 1. Now, note that

$$\|U - I\|^2 = \mathrm{Tr}((U - I)^*(U - I)) = \sum_{j=1}^n |1 - e^{i\theta_j}|^2.$$

Since $\|U - I\| \leq \frac{1}{2}$, this implies that $|1 - e^{i\theta_j}| \leq \frac{1}{2}$ for each j . In particular, no θ_j is equal to $-\pi$, which gives $\log e^{i\theta_j} = i\theta_j$ for each j . On the other hand, this inequality also tells us that by the above power series expansion,

$$\log e^{i\theta_j} = - \sum_{k=1}^{\infty} \frac{(1 - e^{i\theta_j})^k}{k}.$$

Combining, we get

$$i\theta_j = - \sum_{k=1}^{\infty} \frac{(1 - e^{i\theta_j})^k}{k}.$$

By the spectral decompositions of U and V , this proves the claimed identity. $\square$

We are now ready to prove Theorem 15.3.1.

Proof of Theorem 15.3.1. Throughout this proof, we will use the convention that $O(\epsilon^\alpha)$ denotes any matrix whose entries have absolute values bounded by $C\epsilon^\alpha$, where C depends

only on A , g , G , n , and d . Take any $x \in \epsilon \mathbb{Z}^d$ and $1 \leq j \leq d$. Then

$$\begin{aligned}
 U^g(x, x + \epsilon e_j) &= g(x)U(x, x + \epsilon e_j)g(x + \epsilon e_j)^{-1} \\
 &= g(x)e^{\epsilon A_j(x)}g(x + \epsilon e_j)^{-1} \\
 &= g(x)(I + \epsilon A_j(x) + O(\epsilon^2))g(x + \epsilon e_j)^{-1} \\
 &= g(x)g(x + \epsilon e_j)^{-1} + \epsilon g(x)A_j(x)g(x + \epsilon e_j)^{-1} + O(\epsilon^2).
 \end{aligned} \tag{15.3.1}$$

Since $g(x)g(x)^{-1} = I$ for all x , we have

$$0 = \partial_j(g(x)g(x)^{-1}) = (\partial_j g(x))g(x)^{-1} + g(x)\partial_j(g(x)^{-1}),$$

which gives

$$\partial_j(g(x)^{-1}) = -g(x)^{-1}(\partial_j g(x))g(x)^{-1}.$$

Thus,

$$\begin{aligned}
 g(x + \epsilon e_j)^{-1} &= g(x)^{-1} + \epsilon \partial_j(g(x)^{-1}) + O(\epsilon^2) \\
 &= g(x)^{-1} - \epsilon g(x)^{-1}(\partial_j g(x))g(x)^{-1} + O(\epsilon^2).
 \end{aligned}$$

Substituting this in equation (15.3.1), we get

$$\begin{aligned}
 U^g(x, x + \epsilon e_j) &= g(x)(g(x)^{-1} - \epsilon g(x)^{-1}(\partial_j g(x))g(x)^{-1}) + \epsilon g(x)A_j(x)g(x)^{-1} + O(\epsilon^2) \\
 &= I + \epsilon(-(\partial_j g(x))g(x)^{-1} + g(x)A_j(x)g(x)^{-1}) + O(\epsilon^2) \\
 &= I + \epsilon A_j^g(x) + O(\epsilon^2).
 \end{aligned}$$

This implies, in particular, that if ϵ is sufficiently small, then we can apply Lemma 15.3.2 to get

$$B_j(x) = -\epsilon^{-1} \sum_{k=1}^{\infty} \frac{1}{k} (I - U^g(x, x + \epsilon e_j))^k = A_j^g(x) + O(\epsilon).$$

Since the error bound is uniform in x , this completes the proof. $\square$

15.4 Gauge invariant observables

Consider the lattice gauge theory on Λ defined in Section 15.2. For a measurable function $f : G(\Lambda) \rightarrow \mathbb{C}$, its expected value under the probability measure $\mu_{\Lambda, \beta}$ is usually denoted by $\langle f \rangle$, provided that the expected value is well-defined. Computing or estimating such expected values is one of the main objectives of lattice gauge theory, because one can then hope to use them for computing vacuum expectation values in quantum Yang–Mills theories via the route described in Chapter 2.

An observable f is called ‘gauge invariant’ if it is invariant under gauge transformations, meaning that for any $U \in G(\Lambda)$ and $g \in G^\Lambda$, $f(U^g) = f(U)$. For reasons explained

in Section 8.6, gauge invariant observables are the only observables that are physically interesting. Gauge invariant observables also suffice from a mathematical point of view, as we will see a little later.

An important class of gauge invariant observables in lattice gauge theories are the so-called *Wilson loop observables*. A Wilson loop ℓ is a sequence of points in $x_0, x_1, \dots, x_k$ in $\mathbb{Z}^d$, not necessarily distinct, such that for each $0 \leq j \leq k-1$, x_{j+1} is a neighbor of x_j , and x_0 is a neighbor of x_k . In other words, ℓ is a closed loop in $\mathbb{Z}^d$. Suppose that a Wilson loop $\ell = (x_0, x_1, \dots, x_k)$ is contained in Λ . The Wilson loop observable for ℓ is defined as

$$W_\ell(U) := \text{Tr}(U(x_0, x_1)U(x_1, x_2) \cdots U(x_{k-1}, x_k)U(x_k, x_0)).$$

To see that W_ℓ is gauge invariant, take any U and g , and let $V := U^g$. Then

$$\begin{aligned} W_\ell(V) &= \text{Tr}(V(x_0, x_1)V(x_1, x_2) \cdots V(x_{k-1}, x_k)V(x_k, x_0)) \\ &= \text{Tr}(g(x_0)U(x_0, x_1)g(x_1)^{-1}g(x_1)U(x_1, x_2)g(x_2)^{-1} \cdots \\ &\quad \cdots g(x_{k-1})U(x_{k-1}, x_k)g(x_k)^{-1}g(x_k)U(x_k, x_0)g(x_0)^{-1}) \\ &= \text{Tr}(g(x_0)U(x_0, x_1)U(x_1, x_2) \cdots U(x_{k-1}, x_k)U(x_k, x_0)g(x_0)^{-1}) \\ &= \text{Tr}(g(x_0)^{-1}g(x_0)U(x_0, x_1)U(x_1, x_2) \cdots U(x_{k-1}, x_k)U(x_k, x_0)) \\ &= \text{Tr}(g(x_0)U(x_0, x_1)U(x_1, x_2) \cdots U(x_{k-1}, x_k)U(x_k, x_0)g(x_0)^{-1}) = W_\ell(U). \end{aligned}$$

Note that for a plaquette p in Λ , the map $U \mapsto \text{Tr}(U_p)$ itself is a Wilson loop observable, with p being the loop. Thus, the Wilson action S is gauge invariant, just as we proved in Theorem 8.6.4 that the Yang–Mills action is gauge invariant.

A consequence of the gauge invariance of the Wilson action is that the probability measure $\mu_{\Lambda, \beta}$ is invariant under gauge transforms, meaning that for any measurable $f : G(\Lambda) \rightarrow \mathbb{C}$ and any $g \in G^\Lambda$,

$$\int_{G(\Lambda)} f(U) d\mu_{\Lambda, \beta}(U) = \int_{G(\Lambda)} f(U^g) d\mu_{\Lambda, \beta}(U). \quad (15.4.1)$$

To see this, first note that the product Haar measure σ_Λ is invariant under gauge transforms. This is a simple consequence of the invariance properties of the Haar measure, as follows. For any f and g ,

$$\int_{G(\Lambda)} f(U^g) d\sigma_\Lambda(U) = \int_{G(\Lambda)} f((g(x)U(x, y)g(y)^{-1})_{(x, y) \in E(\Lambda)}) \prod_{(x, y) \in E(\Lambda)} d\sigma(U(x, y)),$$

where σ is the normalized Haar measure on G and $E(\Lambda)$ is the set of positively oriented edges of Λ (i.e., (x, y) where x is lexicographically smaller than y). By the right- and left-invariance of σ , we conclude that the right side is equal to

$$\int_{G(\Lambda)} f((U(x, y))_{(x, y) \in E(\Lambda)}) \prod_{(x, y) \in E(\Lambda)} d\sigma(U(x, y)) = \int_{G(\Lambda)} f(U) d\sigma_\Lambda(U).$$

By the above observation and the gauge invariance of the Wilson action, we get

$$\begin{aligned} \int_{G(\Lambda)} f(U^g) \exp(-\beta S_\Lambda(U)) d\sigma_\Lambda(U) &= \int_{G(\Lambda)} f(U^g) \exp(-\beta S_\Lambda(U^g)) d\sigma_\Lambda(U) \\ &= \int_{G(\Lambda)} f(U) \exp(-\beta S_\Lambda(U)) d\sigma_\Lambda(U). \end{aligned}$$

Dividing throughout by the normalizing constant Z , we get equation (15.4.1).

A consequence of equation (15.4.1) is the following. Take any f such that $\langle |f| \rangle < \infty$ (and hence, $\langle f \rangle$ is well-defined). Define a function $h : G(\Lambda) \rightarrow \mathbb{C}$ as

$$h(U) := \int_{G^\Lambda} f(U^g) \prod_{x \in \Lambda} d\sigma(g(x)). \quad (15.4.2)$$

Then note that by equation (15.4.1) applied to $|f|$,

$$\begin{aligned} \int_{G^\Lambda} \int_{G(\Lambda)} |f(U^g)| d\mu_{\Lambda,\beta}(U) \prod_{x \in \Lambda} d\sigma(g(x)) &= \int_{G^\Lambda} \int_{G(\Lambda)} |f(U)| d\mu_{\Lambda,\beta}(U) \prod_{x \in \Lambda} d\sigma(g(x)) \\ &= \int_{G^\Lambda} \langle |f| \rangle \prod_{x \in \Lambda} d\sigma(g(x)) = \langle |f| \rangle < \infty. \end{aligned}$$

This allows us to apply Fubini's theorem and equation (15.4.1) to get

$$\begin{aligned} \langle h \rangle &= \int_{G(\Lambda)} \int_{G^\Lambda} f(U^g) \prod_{x \in \Lambda} d\sigma(g(x)) d\mu_{\Lambda,\beta}(U) \\ &= \int_{G^\Lambda} \int_{G(\Lambda)} f(U^g) d\mu_{\Lambda,\beta}(U) \prod_{x \in \Lambda} d\sigma(g(x)) \\ &= \int_{G^\Lambda} \int_{G(\Lambda)} f(U) d\mu_{\Lambda,\beta}(U) \prod_{x \in \Lambda} d\sigma(g(x)) \\ &= \int_{G^\Lambda} \langle f \rangle \prod_{x \in \Lambda} d\sigma(g(x)) = \langle f \rangle. \end{aligned}$$

Thus, f and h have the same expected value under $\mu_{\Lambda,\beta}$. But note that h is gauge invariant, because for any $g \in G^\Lambda$ and $U \in G(\Lambda)$,

$$\begin{aligned} h(U^g) &= \int_{G^\Lambda} f((U^g)^w) \prod_{x \in \Lambda} d\sigma(w(x)) \\ &= \int_{G^\Lambda} f(U^{wg}) \prod_{x \in \Lambda} d\sigma(w(x)) \\ &= \int_{G^\Lambda} f(U^w) \prod_{x \in \Lambda} d\sigma(w(x)) = h(U), \end{aligned}$$

where the penultimate identity follows from the invariance of Haar measure. Thus, to summarize, we have shown that for any observable f , there is a gauge invariant observable h produced using f through the formula (15.4.2), such that $\langle h \rangle = \langle f \rangle$. This shows that

even from a mathematical point of view, it suffices to focus on gauge invariant observables.

15.5 Gauge fixing

We have discussed gauge fixing in earlier chapters in the context of Yang–Mills theories in the continuum. For a lattice gauge theory on a finite set $\Lambda \subseteq \mathbb{Z}^d$ under free boundary condition, as we have considered in this chapter until now, gauge fixing takes the following form. Recall that $E(\Lambda)$ is the set of positively oriented edges in Λ . Let Γ be a subset of $E(\Lambda)$ such that there are no loops formed by the undirected versions of the edges in Γ . It turns out that for studying the lattice gauge theory on Λ , it suffices to study the theory on the subset of the configuration space where the matrices attached to the edges in Γ are all set to be equal to the identity matrix.

The idea is formalized as follows. Let $G(\Lambda, \Gamma)$ be the subset of configurations $U \in G(\Lambda)$ such that $U(x, y) = I$ for any edge $(x, y) \in \Gamma$. Let $\mu_{\Lambda, \Gamma, \beta}$ be the lattice gauge theory restricted to the configuration space $G(\Lambda, \Gamma)$. That is,

$$d\mu_{\Lambda, \Gamma, \beta}(U) := \frac{1}{Z_\Gamma} e^{-\beta S_\Lambda(U)} d\sigma_{\Lambda, \Gamma}(U),$$

where Z_Γ is the normalizing constant that makes this a probability measure, and $\sigma_{\Lambda, \Gamma}(U)$ is the product Haar measure on $G(\Lambda, \Gamma)$; that is,

$$d\sigma_{\Lambda, \Gamma}(U) = \prod_{(x, y) \in E(\Lambda) \setminus \Gamma} d\sigma(U(x, y)).$$

The following theorem shows that for evaluating the expected values of gauge invariant observables, it suffices to consider the restricted theory $\mu_{\Lambda, \Gamma, \beta}$.

Theorem 15.5.1. *Let Γ and $\mu_{\Lambda, \Gamma, \beta}$ be as above. Then for any gauge invariant observable $f : G(\Lambda) \rightarrow \mathbb{C}$ such that $\langle |f| \rangle < \infty$, we have*

$$\langle f \rangle = \int_{G(\Lambda, \Gamma)} f(U) d\mu_{\Lambda, \Gamma, \beta}(U).$$

Proof. Let $(V(x, y))_{(x, y) \in E(\Lambda) \setminus \Gamma}$ be a collection of i.i.d. random matrices from the normalized Haar measure σ on G . Extend V to a configuration in $G(\Lambda)$ by setting $V(x, y) = I$ for $(x, y) \in \Gamma$. Let $(g(x))_{x \in \Lambda}$ be another collection of i.i.d. random matrices from σ . Let $W := V^g$. We claim that the random matrices $(W(x, y))_{(x, y) \in E(\Lambda)}$ are i.i.d. from σ .

To prove this claim, the first step is to show that $(W(x, y))_{(x, y) \in \Gamma}$ are i.i.d. from σ . Note that for $(x, y) \in \Gamma$, we have $V(x, y) = I$, and therefore,

$$W(x, y) = g(x)V(x, y)g(y)^{-1} = g(x)g(y)^{-1}.$$

Since Γ has no loops, it is easy to see, as follows, that the elements of Γ can be arranged as $(x_1, y_1), (x_2, y_2), \dots$ such that for each n , at *most* one of x_n and y_n is in

$\{x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n\}$. First, suppose that the graph formed by the edges in Γ is connected. Choose (x_1, y_1) to be any element of Γ . Having chosen $(x_1, y_1), \dots, (x_{n-1}, y_{n-1})$, choose (x_n, y_n) to be any of the remaining edges of Γ such that at *least* one of x_n and y_n is in $\{x_1, y_1, \dots, x_{n-1}, y_{n-1}\}$. If there is no such edge, we must have exhausted Γ , because otherwise it will contradict the assumption that Γ is connected. If such an edge can be found, then it cannot be that both x_n and y_n are in $\{x_1, y_1, \dots, x_{n-1}, y_{n-1}\}$, because otherwise we have found a loop in Γ , since the graph formed by $(x_1, y_1), \dots, (x_{n-1}, y_{n-1})$ is connected by construction. Therefore, this construction ensures that for each n , exactly one of x_n and y_n is in $\{x_1, y_1, \dots, x_{n-1}, y_{n-1}\}$. Lastly, if the graph formed by Γ has multiple connected components, we can do the above enumeration successively, component by component.

We will now prove by induction on n that for each n , $W(x_1, y_1), \dots, W(x_n, y_n)$ are i.i.d. from σ . This is clearly true for $n = 1$. Suppose that it holds up to $n - 1$. If neither x_n nor y_n is in $\{x_1, y_1, \dots, x_{n-1}, y_{n-1}\}$, then clearly, $W(x_n, y_n) = g(x_n)g(y_n)^{-1}$ is independent of $W(x_1, y_1), \dots, W(x_{n-1}, y_{n-1})$. If $y_n \notin \{x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n\}$ but x_n belongs to this set, then $g(y_n)$ is independent of $g(x_1), g(y_1), \dots, g(x_{n-1}), g(y_{n-1})$. Since $g(y_n)$ is Haar-distributed, this implies that conditionally given $g(x_1), g(y_1), \dots, g(x_{n-1}), g(y_{n-1})$, the random matrix $W(x_n, y_n) = g(x_n)g(y_n)^{-1}$ is Haar-distributed. Since the conditional distribution does not depend on the conditioning, $W(x_n, y_n)$ is Haar-distributed and is independent of $g(x_1), g(y_1), \dots, g(x_{n-1}), g(y_{n-1})$.² This implies that $W(x_n, y_n)$ is independent of $W(x_1, y_1), \dots, W(x_{n-1}, y_{n-1})$. Similarly, the same conclusion holds if x_n is not a member of $\{x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n\}$ but y_n is in it. We have now exhausted all possibilities, because by construction, it cannot be that both x_n and y_n are in $\{x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n\}$. This completes the induction step, proving the claim that $(W(x, y))_{(x, y) \in \Gamma}$ are i.i.d. from σ .

Next, note that conditionally on g , $(W(x, y))_{(x, y) \in E(\Lambda) \setminus \Gamma}$ are i.i.d. from σ , because $W(x, y) = g(x)V(x, y)g(y)^{-1}$ and the $V(x, y)$'s are i.i.d. from σ . Since the conditional distribution does not depend on the conditioning, we conclude that $(W(x, y))_{(x, y) \in E(\Lambda) \setminus \Gamma}$ are i.i.d. from σ and are independent of g . But we have already shown that $(W(x, y))_{(x, y) \in \Gamma}$ are i.i.d. from σ , and they are functions of g only, with no dependence on V . This completes the proof of the claim that $(W(x, y))_{(x, y) \in E(\Lambda)}$ are i.i.d. from σ .

Take any gauge invariant observable $h : G(\Lambda) \rightarrow \mathbb{C}$ that is integrable with respect to the product Haar measure σ_Λ on $G(\Lambda)$. Then

$$\begin{aligned} \int_{G(\Lambda, \Gamma)} h(U) d\sigma_{\Lambda, \Gamma}(U) &= \mathbb{E}(h(V)) \\ &= \mathbb{E}(h(V^g)) \quad \text{since } h \text{ is gauge invariant,} \\ &= \mathbb{E}(h(W)) \quad \text{since } V^g = W, \\ &= \int_{G(\Lambda)} h(U) d\sigma_\Lambda(U) \quad \text{since } (W(x, y))_{(x, y) \in E(\Lambda)} \text{ are i.i.d. from } \sigma. \end{aligned} \tag{15.5.1}$$

²See Lemma B.4.12 for the justification of this step, which is a standard tool in probability theory.

Now take any gauge invariant f such that $\langle |f| \rangle < \infty$. Then

$$\int_{G(\Lambda)} |f(U)| e^{-\beta S_\Lambda(U)} d\sigma_\Lambda(U) < \infty.$$

Since both f and S_Λ are gauge invariant, so is the function

$$h(U) := f(U) e^{-\beta S_\Lambda(U)}.$$

By the previous display, h is integrable with respect to σ_Λ . Thus, by equation (15.5.1), we conclude that

$$\int_{G(\Lambda, \Gamma)} f(U) e^{-\beta S_\Lambda(U)} d\sigma_{\Lambda, \Gamma}(U) = \int_{G(\Lambda)} f(U) e^{-\beta S_\Lambda(U)} d\sigma_\Lambda(U).$$

Taking $f \equiv 1$ (which is gauge invariant), we get $Z = Z_\Gamma$. Dividing both sides by this normalizing constant completes the proof. $\square$

15.6 Mass gap and quark confinement

The two most important questions about lattice gauge theories are those of mass gap and quark confinement. We will discuss the physical origins of these questions. For now, let us only see the mathematical formulations, which are relatively simple to state.

Suppose we have a sequence of lattice gauge theories, with the N^{th} theory defined on a finite set $\Lambda_N \subseteq \mathbb{Z}^d$, with $\Lambda_N \uparrow \mathbb{Z}^d$ as $N \rightarrow \infty$. We say that the theory (or, more accurately, the sequence of theories) has a mass gap (equivalently, exponential decay of correlations) if the following holds. If f is a gauge invariant observable, we say that it is supported on a set E of edges if $f(U)$ has no dependence on $U(x, y)$ for any $(x, y) \notin E$. For example, a Wilson loop observable is supported on the set of edges in the loop. Let f_1 and f_2 be two bounded gauge invariant observables, supported on two finite sets of edges S_1 and S_2 , respectively. We say that the theory has a mass gap if there are positive constants C_1 and C_2 , where C_1 may depend on f_1, f_2 but C_2 may not, and neither C_1 nor C_2 may depend on N , such that whenever N is so large that $S_1, S_2 \subseteq \Lambda_N$, we have

$$|\langle f_1 f_2 \rangle - \langle f_1 \rangle \langle f_2 \rangle| \leq C_1 e^{-C_2 \text{dist}(S_1, S_2)},$$

where $\text{dist}(S_1, S_2)$ is the smallest graph distance between a vertex of an edge in S_1 and a vertex of an edge in S_2 .

The theory is said to confine quarks (or simply, confining), if the following holds. There is a function $V : (0, \infty) \rightarrow \mathbb{R}$ satisfying $\lim_{x \rightarrow \infty} V(x) = \infty$, such that for any rectangular loop ℓ in $\mathbb{Z}^d$ with side lengths R, T , where $R \leq T$, we have

$$|\langle W_\ell \rangle| \leq C e^{-V(R)T},$$

where C is a constant that does not depend on R or T or N , and N is so large that ℓ is contained in Λ_N . A theory that is not confining is called deconfining or unconfining. A special case of confining theories are theories that satisfy *Wilson's area law*, where $V(R) \propto R$. This means that the expected value of a Wilson loop observable, for a rectangular loop, decays at least as fast as $\exp(-c \text{area}(\ell))$, where $\text{area}(\ell)$ is the area enclosed by ℓ and c is some constant that does not depend on ℓ . Examples of deconfining theories usually satisfy the *perimeter law*, that $\langle W_\ell \rangle \sim C_1 e^{-C_2(R+T)}$.

15.7 Mass gap in 2D lattice gauge theories

It turns out 2D lattice gauge theories always have mass gap, in the sense that we defined in the previous section. This is the content of the following theorem.

Theorem 15.7.1. *Let $\{\Lambda_N\}_{N \geq 1}$ be any sequence of sets increasing to $\mathbb{Z}^2$. Consider pure lattice gauge theories with free boundary condition on this sequence, with a compact Lie group $G \subseteq \text{U}(n)$ as before, and any $\beta \geq 0$. Such a sequence has a mass gap, in the sense defined in the previous section.*

Proof. The crux of the proof is in choosing a suitable gauge fixing. Let f_1 and f_2 be bounded gauge invariant observables with finite supports S_1 and S_2 , and take any N such that S_1 and S_2 are both contained in Λ_N . Define Γ to be the set of all edges in $E(\Lambda_N)$ that are parallel to the x -axis. Then clearly, Γ has loops. Therefore by Theorem 15.5.1, it suffices to consider the gauge-fixed theory where the matrices attached to the horizontal edges are all equal to I .

If some edge e in Λ_N is not in any plaquette that is fully contained in Λ_N , then it is easy to see that $U(e)$ is independent of the rest of the system in the lattice gauge theory on Λ_N . So, we may assume without loss of generality that there is no such edge in Λ_N .

Let us now check what the probability density function of the gauge-fixed theory looks like. Take any plaquette p in Λ_N with vertices $(x, y), (x+1, y), (x+1, y+1), (x, y+1)$. Since $U((x, y), (x+1, y)) = I$ and $U((x, y+1), (x+1, y+1)) = I$, we have

$$U_p = U((x+1, y), (x+1, y+1))U((x, y), (x, y+1))^{-1}.$$

Now take any $y \in \mathbb{Z}$. Let $P(y) \subseteq \mathbb{Z}$ be the set of all x 's as above (i.e., such that the plaquette p is contained in Λ_N). Let $e_{x,y}$ denote the vertical edge $((x, y), (x, y+1))$. Then by the above observation,

$$S_{\Lambda_N}(U) = \sum_{y \in \mathbb{Z}} \sum_{x \in P(y)} \text{Re}(\text{Tr}(I - U(e_{x+1,y})U(e_{x,y})^{-1})). \quad (15.7.1)$$

Now, for each y , $P(y)$ is the union of finitely many connected components (under the usual

graph structure on $\mathbb{Z}$). Let $\mathcal{C}(y)$ be the set of components, so that

$$P(y) = \bigcup_{C \in \mathcal{C}(y)} C.$$

Now, note that if $U(e_{x,y})$ and $U(e_{x',y'})$ occur within the same term on the right in equation (15.7.1), then we must have that $y = y'$ and $|x - x'| = 1$. In other words, we must have that $y = y'$ and x, x' are in the same element of $\mathcal{C}(y)$. This implies that $\exp(-\beta S_{\Lambda_N}(U))$ factorizes as

$$\exp(-\beta S_{\Lambda_N}(U)) = \prod_{y \in \mathbb{Z}} \prod_{C \in \mathcal{C}(y)} T_C(U),$$

where, for a component $C = \{x, x+1, \dots, x+k\} \in \mathcal{C}(y)$,

$$T_C(U) := \exp\left(-\beta \sum_{j=0}^k \operatorname{Re}(\operatorname{Tr}(I - U(e_{x+j+1,y})U(e_{x+j,y})^{-1}))\right).$$

Let C be as above and let $U(C)$ be the collection $(U(e_{x+j,y}))_{0 \leq j \leq k+1}$. The above decomposition shows that if U is a random configuration from our lattice gauge theory on Λ_N , then the collections $\{U(C) : C \in \mathcal{C}(y), y \in \mathbb{Z}\}$ are independent. Moreover, by the assumption that each edge in Λ_N belongs to at least one plaquette that is fully contained in Λ_N , every $U(e)$ must be in one of the $U(C)$'s.

Continuing with C be as above, we claim that $U(C)$ is a Markov chain with transition density function

$$p(V, W) = \frac{1}{K} \exp(-\beta \operatorname{Re}(\operatorname{Tr}(I - WV^{-1}))),$$

where K is the normalizing constant. Note that K does not depend on V , because

$$K = \int_G \exp(-\beta \operatorname{Re}(\operatorname{Tr}(I - WV^{-1}))) d\sigma(W) = \int_G \exp(-\beta \operatorname{Re}(\operatorname{Tr}(I - W))) d\sigma(W),$$

by the invariance of Haar measure. The fact that K does not depend on V implies that a Markov chain with transition density p has joint density of the form T_C up to a normalizing constant. This proves our claim that $U(C)$ is a Markov chain with transition density p .

Now note that p is bounded below by a positive constant on G , since G is compact and p is never zero. Thus, there is some $\epsilon > 0$ such that $p \geq \epsilon p_0$ everywhere, where $p_0 \equiv 1$ is the density of the Haar measure with respect to itself. A simple verification shows that the Haar measure is invariant for this Markov chain. Thus, by Doeblin's condition (Theorem B.5.1 and Corollary B.5.2), this chain has exponential decay of correlations, in the sense of Corollary B.5.2, where the rate of decay depends only on ϵ , and thus, only on β and G . It is now easy to complete the proof using Corollary B.5.2, by explicitly writing down the expected values as integrals with respect to the relevant joint densities. $\square$

15.8 Area law in 2D lattice gauge theories

Let us continue working in the same setting as before. We will now show that 2D lattice gauge theories satisfy Wilson's area law, as long as a certain irreducibility condition is satisfied.

Theorem 15.8.1. *Let $\{\Lambda_N\}_{N \geq 1}$ be any sequence of sets increasing to $\mathbb{Z}^2$. Consider pure lattice gauge theories with free boundary condition on this sequence, with a compact Lie group $G \subseteq \mathrm{U}(n)$ as before, and any $\beta \geq 0$. Assume that there is no vector in $\mathbb{C}^n$ that is fixed by all elements of G . Then this sequence of theories satisfies Wilson's area law, in the sense defined in Section 15.6.*

Recall that the ℓ^2 operator norm of an $n \times n$ complex matrix M is defined as

$$\|M\| := \sup\{\|Mx\| : x \in \mathbb{C}^n, \|x\| = 1\}. \quad (15.8.1)$$

It is easy to see that this norm is submultiplicative, meaning that $\|MN\| \leq \|M\|\|N\|$. Now, if M is random, let $\mathbb{E}(M)$ denote the $n \times n$ matrix containing the expected values of the entries of M . We will use the following consequence of the irreducibility assumption.

Lemma 15.8.2. *Let G satisfy the condition stated in Theorem 15.8.1. Let U be a Haar-distributed random element of G . Then $\mathbb{E}(U) = 0$.*

Proof. First, we claim that $\|\mathbb{E}(U)\| \leq 1$. The easiest way to see this is as follows. Let $U_1, U_2, \dots$ be i.i.d. from the Haar measure. Take any $x \in \mathbb{C}^n$ with $\|x\| = 1$. Then by the law of large numbers,

$$\frac{1}{n} \sum_{j=1}^n U_j x \rightarrow \mathbb{E}(U)x$$

with probability one as $n \rightarrow \infty$. But for each n , the triangle inequality gives

$$\left\| \frac{1}{n} \sum_{j=1}^n U_j x \right\| \leq \frac{1}{n} \sum_{j=1}^n \|U_j x\| = \|x\| = 1.$$

This proves the claim that $\|\mathbb{E}(U)\| \leq 1$. Next, we claim that $\|\mathbb{E}(U)\| < 1$. Indeed, if that were not true, then $\|\mathbb{E}(U)\| = 1$; this implies that there exists $x \in \mathbb{C}^n$ such $\|x\| = 1$ and $\|\mathbb{E}(U)x\| = 1$. Fix such an x . Let U_1, U_2 be i.i.d. from the Haar measure. Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{C}^n$. Then note that by the independence of U_1, U_2 and the hypothesis that $\|\mathbb{E}(U)x\| = 1$, we have

$$\begin{aligned} \mathbb{E}(\|U_1 x - U_2 x\|^2) &= 2 - 2\mathbb{E}(\langle U_1 x, U_2 x \rangle) \\ &= 2 - 2\langle \mathbb{E}(U_1)x, \mathbb{E}(U_2)x \rangle \\ &= 2 - 2\langle \mathbb{E}(U)x, \mathbb{E}(U)x \rangle = 2 - 2\|\mathbb{E}(U)x\|^2 = 0. \end{aligned} \quad (15.8.2)$$

Thus, $U_1 x = U_2 x$ with probability one. But this means $U_2^{-1} U_1 x = x$ with probability one. Since $U_2^{-1} U_1$ is also Haar-distributed, we see that $Ux = x$ with probability one. We claim

that $Vx = x$ for all $V \in G$. To see this, note that by the nature of the Haar measure on a compact matrix Lie group (as constructed in Section C.7 of Appendix C), the Haar measure puts positive mass on arbitrarily small open neighborhoods of any given V . Thus, there is some W arbitrarily close to V that satisfies $Wx = x$. By continuity, this proves that $Vx = x$. This yields a contradiction and proves that $\|\mathbb{E}(U)\| < 1$.

Again let U_1, U_2 be i.i.d. from the Haar measure. Then $U_1 U_2$ is also Haar-distributed. Thus,

$$\begin{aligned} \|\mathbb{E}(U)\| &= \|\mathbb{E}(U_1 U_2)\| \\ &= \|\mathbb{E}(U_1) \mathbb{E}(U_2)\| \leq \|\mathbb{E}(U_1)\| \|\mathbb{E}(U_2)\| = \|\mathbb{E}(U)\|^2. \end{aligned}$$

Since $\|\mathbb{E}(U)\| < 1$, this proves that $\mathbb{E}(U) = 0$. $\square$

We are now ready to prove Theorem 15.8.1.

Proof of Theorem 15.8.1. As in the proof of Theorem 15.7.1, let us fix the gauge and assume that the matrices attached to all horizontal edges are identity. Let ℓ be the rectangular loop with vertices at $(0, 0)$, $(T, 0)$, (T, R) and $(0, R)$, where $R \leq T$ are two positive integers. Let N be so large that this loop is contained in Λ_N . Let U be a random configuration from the lattice gauge theory on Λ_N . For $0 \leq j \leq T$ and $0 \leq k \leq R - 1$, let

$$U_{j,k} := U((j, k), (j, k + 1)).$$

Then, since the matrices attached to the horizontal edges are identity, we have

$$W_\ell = \text{Tr}(U_{T,0} U_{T,1} \cdots U_{T,R-1} U_{0,R-1}^* U_{0,R-2}^* \cdots U_{0,0}^*).$$

Let $U_{j,k}^{a,b}$ denote the $(a, b)^{\text{th}}$ entry of $U_{j,k}$. Then we can expand the above identity as

$$W_\ell = \sum_{1 \leq a_0, \dots, a_{2R-1} \leq n} U_{T,0}^{a_0, a_1} U_{T,1}^{a_1, a_2} \cdots U_{T,R-1}^{a_{R-1}, a_R} \overline{U_{0,R-1}^{a_{R+1}, a_R}} \overline{U_{0,R-2}^{a_{R+2}, a_{R+1}}} \cdots \overline{U_{0,0}^{a_0, a_{2R-1}}}. \quad (15.8.3)$$

Now recall from the proof of Theorem 15.7.1 that for each j , the sequence $U_{0,j}, \dots, U_{T,j}$ is a Markov chain satisfying Doeblin's condition with an ϵ that depends only on G and β . Moreover, these Markov chains are independent. Thus,

$$\begin{aligned} &\mathbb{E}(U_{T,0}^{a_0, a_1} U_{T,1}^{a_1, a_2} \cdots U_{T,R-1}^{a_{R-1}, a_R} \overline{U_{0,R-1}^{a_{R+1}, a_R}} \overline{U_{0,R-2}^{a_{R+2}, a_{R+1}}} \cdots \overline{U_{0,0}^{a_0, a_{2R-1}}}) \\ &= \prod_{j=0}^{R-1} \mathbb{E}(U_{T,j}^{a_j, a_{j+1}} \overline{U_{0,j}^{a_{2R-j}, a_{2R-j-1}}}), \end{aligned} \quad (15.8.4)$$

where $a_{2R} = a_0$. But by Doeblin's theorem (Theorem B.5.1 and Corollary B.5.2),

$$|\mathbb{E}(U_{T,j}^{a_j, a_{j+1}} \overline{U_{0,j}^{a_{2R-j}, a_{2R-j-1}}}) - \mathbb{E}(U_{T,j}^{a_j, a_{j+1}}) \mathbb{E}(\overline{U_{0,j}^{a_{2R-j}, a_{2R-j-1}}})| \leq C_1 e^{-C_2 T}$$

where C_1, C_2 are positive constants that depend only on G and β . But by Lemma 15.8.2, $\mathbb{E}(U_{T,j}^{a_j, a_{j+1}}) = 0$. Thus,

$$|\mathbb{E}(U_{T,j}^{a_j, a_{j+1}} \overline{U_{0,j}^{a_{2R-j}, a_{2R-j-1}}})| \leq C_1 e^{-C_2 T}.$$

Using this in equation (15.8.4), we get

$$|\mathbb{E}(U_{T,0}^{a_0, a_1} U_{T,1}^{a_1, a_2} \cdots U_{T,R-1}^{a_{R-1}, a_R} \overline{U_{0,R-1}^{a_{R+1}, a_R}} \overline{U_{0,R-2}^{a_{R+2}, a_{R+1}}} \cdots \overline{U_{0,0}^{a_0, a_{2R-1}}})| \leq C_1^R e^{-C_2 RT}.$$

Using this bound in equation (15.8.3) gives

$$|\mathbb{E}(W_\ell)| \leq n^{2R} C_1^R e^{-C_2 RT} \leq C_3^R e^{-C_2 RT},$$

where $C_3 := n^2 C_1$ depends only on G and β . If $T \geq 2C_2^{-1} \log C_3$, then the above bound gives

$$|\mathbb{E}(W_\ell)| \leq e^{R \log C_3 - C_2 RT} \leq e^{-\frac{1}{2} C_2 RT}.$$

Choose C_4 so large that $|\mathbb{E}(W_\ell)| \leq C_4 e^{-\frac{1}{2} C_2 RT}$ when $T < 2C_2^{-1} \log C_3$. It is possible to find such a C_4 because $C_2 > 0$ and the condition $R \leq T$ implies that there are only finitely many pairs (R, T) satisfying $T < 2C_2^{-1} \log C_3$. This completes the proof. $\square$

Chapter 16

Lattice Gauge Theories in $d \geq 3$

16.1 The strong and weak coupling regimes

Recall from the discussion in Section 13.1 of Chapter 13 that upon rescaling the gauge field A by replacing it with $g_0 A$, the Yang–Mills action becomes

$$S(A) = -\frac{1}{4} \int_{\mathbb{R}^4} \text{Tr} \left(\sum_{0 \leq i, j \leq 3} \eta_i \eta_j (\partial_i A_j(x) - \partial_j A_i(x) + g_0 [A_i(x), A_j(x)])^2 \right) dx.$$

This rescaled version explains why g_0 is called the ‘coupling constant’. If g_0 is very small, then after gauge fixing, the coordinates of A become essentially independent, as in the case of U(1) theory discussed in Chapter 9. That is, the fields $A_0, \dots, A_3$ become ‘decoupled’.

We carry over this terminology to lattice gauge theories. From Theorem 15.2.1, we see that the parameter β plays the role of g_0^{-2} . Thus, β is small if and only if g_0 is large. This leads to the following convention. If a theorem is true when β is small enough, we say that it is valid in the ‘strong coupling regime’. Similarly, if a theorem holds for large enough β , we say that it is a weak coupling result.

The theorems about 2D lattice gauge theories that we proved in the two preceding sections are valid for all $\beta \geq 0$. We say that such a result holds in both the strong and weak coupling regimes.

It is often easier to prove theorems in the strong coupling regime. Physical interest, however, is concentrated in the weak coupling regime. This is because the actual values of β that reflect the physical world are believed to be extremely large; so, a theorem that only holds for small enough β runs the risk of being unphysical. On the other hand, a theorem that holds for all large enough β is more likely to reflect physical reality.

16.2 Correlation decay under Dobrushin’s condition

Let $(\Omega, \mathcal{F}, \mu)$ be a probability space and let ν be a probability measure on $(\Omega^n, \mathcal{F}^n, \mu^n)$ for some $n \geq 1$. Suppose that ν has a density ρ with respect to μ^n . Let X be an Ω^n -valued

random variable with law ν . Then the marginal density¹ of $(X_j)_{j \neq i}$ is

$$\rho_{-i}((x_j)_{j \neq i}) = \int \rho(x_1, \dots, x_n) d\mu(x_i),$$

and the conditional density of X_i given $(X_j)_{j \neq i}$ is

$$\rho_i(x_i | (x_j)_{j \neq i}) = \frac{\rho(x_1, \dots, x_n)}{\rho_{-i}((x_j)_{j \neq i})}$$

if $\rho_{-i}((x_j)_{j \neq i}) \neq 0$, and defined arbitrarily (for example, zero) if the denominator is zero.

Suppose that the above system has a *conditional dependency graph* G , consisting of a set of undirected edges on $\{1, \dots, n\}$, such that for any i , $\rho_i(x_i | (x_j)_{j \neq i})$ is a function only of x_i and those x_j such that j is a neighbor of i . This means that conditionally on $(X_j)_{j \in N(i)}$, where $N(i)$ is the neighborhood of i , X_i is independent of $(X_j)_{j \notin N(i) \cup \{i\}}$. Let Δ be the maximum vertex degree of G . We assume that

$$s := \sup_{1 \leq i \leq n, x \in \Omega^n} |\rho_i(x_i | (x_j)_{j \neq i}) - 1| < \frac{1}{\Delta}. \quad (16.2.1)$$

The above assumption is known as *Dobrushin's condition*. The theorem presented below is inspired by the work of Dobrushin and Shlosman [4] from 1985.

Theorem 16.2.1. *Let all notations and assumptions be as above. Take any nonempty disjoint sets $A, B \subseteq \{1, \dots, n\}$. Let $X_A := (X_i)_{i \in A}$ and $X_B := (X_i)_{i \in B}$. Let r be the minimum graph distance between a vertex in A and a vertex in B . Then for any measurable $f : \Omega^{|A|} \rightarrow [-1, 1]$ and $g : \Omega^{|B|} \rightarrow [-1, 1]$,*

$$|\mathbb{E}(f(X_A)g(X_B)) - \mathbb{E}(f(X_A))\mathbb{E}(g(X_B))| \leq \frac{2s^r \Delta^r |A||B|}{1 - s\Delta}.$$

Proof. Take any $x, x' \in \Omega^{|B|}$. We claim that²

$$|\mathbb{E}(f(X_A) | X_B = x) - \mathbb{E}(f(X_A) | X_B = x')| \leq \frac{2s^r \Delta^r |A||B|}{1 - s\Delta}. \quad (16.2.2)$$

where the conditional expectations are calculated using the respective conditional densities.

To prove this claim, fix x, x' . Let τ and τ' be the conditional distributions of X given $X_B = x$ and $X_B = x'$, respectively. Let X_0 and Y_0 be independent Ω^n -valued random variables with laws τ and τ' . We will now inductively define a Markov chain $(X_k, Y_k)_{k \geq 0}$.

Given $(X_k, Y_k) = (x, y)$ for some k , generate (X_{k+1}, Y_{k+1}) as follows. Choose a coordinate I uniformly at random from $\{1, \dots, n\} \setminus B$. Generate a pair (U, V) of Ω -valued random variables such that U has density $\rho_I(\cdot | (X_{k,j})_{j \neq I})$ and V has density $\rho_I(\cdot | (Y_{k,j})_{j \neq I})$, but their joint distribution is such that $\mathbb{P}(U \neq V)$ is equal to the total variation distance between these two laws. By Lemma B.1.7, it is possible to generate such a pair (U, V) .

¹See Subsection B.4.4 in Appendix B for the definitions of marginal density and conditional density.

²See Subsection B.4.1 in Appendix B for the definition of $\mathbb{E}(Y | X = x)$, and Subsection B.4.4 for how it can be defined in terms of conditional densities.

Having obtained U and V , produce X_{k+1} by replacing the I^{th} coordinate of X_k by U (keeping all other coordinates the same) and produce Y_{k+1} similarly using V . Then note that separately, the transition from X_k to X_{k+1} and from Y_k to Y_{k+1} happen according to the respective Gibbs sampling algorithms. By Proposition B.5.3, this shows that for each k , $X_k \sim \tau$ and $Y_k \sim \tau'$.

Now, from the definition of s in equation (16.2.1) and the formula for total variation distance in terms of densities (Lemma B.1.7), it is easy to see that for the pair (U, V) generated above, $\mathbb{P}(U \neq V) \leq s$ if $(X_{k,j})_{j \in N(I)} \neq (Y_{k,j})_{j \in N(I)}$, and $\mathbb{P}(U \neq V) = 0$ otherwise. This implies that

$$\begin{aligned} \mathbb{P}(X_{k+1,i} \neq Y_{k+1,i} | X_k, Y_k) &\leq \left(1 - \frac{1}{m}\right) 1_{\{X_{k,i} \neq Y_{k,i}\}} \\ &\quad + \frac{s}{m} 1_{\{(X_{k,j})_{j \in N(i)} \neq (Y_{k,j})_{j \in N(i)}\}}, \end{aligned}$$

where $m := n - |B|$. This implies that

$$\begin{aligned} \mathbb{P}(X_{k+1,i} \neq Y_{k+1,i} | X_k, Y_k) &\leq \left(1 - \frac{1}{m}\right) 1_{\{X_{k,i} \neq Y_{k,i}\}} + \frac{s}{m} \sum_{j \in N(i) \setminus B} 1_{\{X_{k,j} \neq Y_{k,j}\}} \\ &\quad + \frac{s}{m} 1_{\{(X_{k,j})_{j \in N(i) \cap B} \neq (Y_{k,j})_{j \in N(i) \cap B}\}} \\ &\leq \left(1 - \frac{1}{m}\right) 1_{\{X_{k,i} \neq Y_{k,i}\}} + \frac{s}{m} \sum_{j \in N(i) \setminus B} 1_{\{X_{k,j} \neq Y_{k,j}\}} + \frac{s}{m} h_i, \end{aligned}$$

where $h_i := 1_{\{N(i) \cap B \neq \emptyset\}}$. Now let $\ell_k = (\ell_{k,i})_{i \notin B}$ denote the vector in $\mathbb{R}^m$ whose i^{th} coordinate is $\mathbb{P}(X_{k,i} \neq Y_{k,i})$, and let $h := (h_i)_{i \notin B}$. Let Q denote the adjacency matrix of G restricted to $\{1, \dots, n\} \setminus B$. Taking expectation on both sides of the above inequality, we get

$$\ell_{k+1} \leq \left(1 - \frac{1}{m}\right) \ell_k + \frac{s}{m} Q \ell_k + \frac{s}{m} h,$$

where the inequality means that each coordinate of the vector on the left is dominated by the corresponding coordinate on the right. Let

$$\ell := \limsup_{k \rightarrow \infty} \ell_k,$$

where the lim sup is taken coordinatewise. By the above inequality, it follows that

$$\ell \leq \left(1 - \frac{1}{m}\right) \ell + \frac{s}{m} Q \ell + \frac{s}{m} h,$$

which simplifies to

$$(I - sQ)\ell \leq sh, \tag{16.2.3}$$

where I is the $n \times n$ identity matrix.

Let $q_{i,j}^{(k)}$ denote the $(i, j)^{\text{th}}$ element of Q^k . Note that $q_{i,j}^{(k)}$ is simply the number of B -avoiding paths of length k from i to j in the graph G . Clearly, this is bounded above

by Δ^k . Since $s\Delta < 1$, this shows that $s^k Q^k$ goes to zero exponentially fast as $k \rightarrow \infty$. In particular, the series $\sum_{k=0}^{\infty} s^k Q^k$ converges, and

$$\left(\sum_{k=0}^{\infty} s^k Q^k \right) (I - sQ) = \sum_{k=0}^{\infty} s^k Q^k - \sum_{k=0}^{\infty} s^{k+1} Q^{k+1} = I.$$

Thus, multiplying both sides of equation (16.2.3) by $\sum_{k=0}^{\infty} s^k Q^k$, and noting that all entries of this matrix are nonnegative, we get

$$\ell \leq \sum_{k=0}^{\infty} s^k Q^k h.$$

Let $D := \{i \notin B : N(i) \cap B \neq \emptyset\}$. Then the above inequality shows that for each $i \notin B$,

$$\ell_i \leq \sum_{k=0}^{\infty} \sum_{j \notin B} s^{k+1} q_{i,j}^{(k)} h_j = \sum_{k=0}^{\infty} \sum_{j \in D} s^{k+1} q_{i,j}^{(k)} = \sum_{j \in D} \sum_{k=0}^{\infty} s^{k+1} q_{i,j}^{(k)}.$$

Take any $i \in A$ and $j \in D$. Then $q_{i,j}^{(k)}$, being the number of paths of length k from i to j , is zero if $k < r - 1$. Also, we have already noted that $q_{i,j}^{(k)} \leq \Delta^k$ in general. Thus,

$$\sum_{k=0}^{\infty} s^{k+1} q_{i,j}^{(k)} \leq \sum_{k=r-1}^{\infty} s^{k+1} \Delta^k = \frac{s^r \Delta^{r-1}}{1 - s\Delta}.$$

Plugging this into the previous display, we get that for $i \in A$,

$$\ell_i \leq \frac{s^r \Delta^{r-1} |D|}{1 - s\Delta} \leq \frac{s^r \Delta^r |B|}{1 - s\Delta}.$$

Take any k . Since $X_k \sim \tau$, $Y_k \sim \tau'$, and $|f| \leq 1$, we have

$$\begin{aligned} |\mathbb{E}(f(X_A) | X_B = x) - \mathbb{E}(f(X_A) | X_B = x')| &= |\mathbb{E}(f((X_{k,i})_{i \in A})) - \mathbb{E}(f((Y_{k,i})_{i \in A}))| \\ &\leq 2\mathbb{P}((X_{k,i})_{i \in A} \neq (Y_{k,i})_{i \in A}) \\ &\leq 2 \sum_{i \in A} \mathbb{P}(X_{k,i} \neq Y_{k,i}) = 2 \sum_{i \in A} \ell_{k,i}. \end{aligned}$$

Taking the lim sup of the right side as $k \rightarrow \infty$ and applying the bound from the previous display completes the proof of the claim (16.2.2).

Let τ_B be the law of X_B . Then note that

$$\mathbb{E}(f(X_A)) = \int \mathbb{E}(f(X_A) | X_B = x) d\tau_B(x).$$

Thus, the inequality (16.2.2) implies that for any x ,

$$\begin{aligned}
 & |\mathbb{E}(f(X_A)|X_B = x) - \mathbb{E}(f(X_A))| \\
 &= \left| \int (\mathbb{E}(f(X_A)|X_B = x) - \mathbb{E}(f(X_A)|X_B = x')) d\tau_B(x') \right| \\
 &\leq \int |\mathbb{E}(f(X_A)|X_B = x) - \mathbb{E}(f(X_A)|X_B = x')| d\tau_B(x') \\
 &\leq \frac{2s^r \Delta^r |A||B|}{1 - s\Delta}.
 \end{aligned}$$

Since

$$\mathbb{E}(f(X_A)g(X_B)) - \mathbb{E}(f(X_A))\mathbb{E}(g(X_B)) = \mathbb{E}[(\mathbb{E}(f(X_A)|X_B) - \mathbb{E}(f(X_A)))g(X_B)]$$

and $|g(X_B)| \leq 1$, this completes the proof. $\square$

16.3 Mass gap at strong coupling

As in the previous chapter, let G be a compact Lie subgroup of $U(n)$ for some n . Take any $d \geq 2$, and let $\{\Lambda_N\}_{N \geq 1}$ be a sequence of finite subsets of $\mathbb{Z}^d$ that increase to $\mathbb{Z}^d$. Let β be a nonnegative real number. Consider lattice gauge theories on Λ_N with this G and β , under free boundary condition. The main result is the following. This theorem was proved by Osterwalder and Seiler [24] in 1978 using the technique of cluster expansions. We will take a different route, using Theorem 16.2.1 from the previous section.

Theorem 16.3.1. *There is some $\beta_0 > 0$, depending on G and d , such that if $\beta \leq \beta_0$, then the above theory (or rather, sequence of theories) has a mass gap in the sense defined in Section 15.6.*

Proof. Let f_1, f_2 be two gauge invariant observables supported on two disjoint finite sets of positively oriented edges S_1, S_2 , and let N be so large that $S_1, S_2 \subseteq \Lambda_N$. Let U be a random configuration from our lattice gauge theory on Λ_N . Define a graph structure on $E(\Lambda_N)$ by decreeing that e, e' are neighbors if and only if e, e' are both part of a common plaquette that is fully contained in Λ_N . It is easy to see from the form of the density of lattice gauge theory that this is a conditional dependency graph, in the sense defined in the previous section.

Let $\mathcal{P}(e)$ denote the set of plaquettes that contain e and are fully contained in Λ_N . Then note that the conditional density of $U(e)$ given $(U(f))_{f \neq e}$ is equal to

$$\rho(U(e)) = \frac{1}{Z(e)} \exp\left(-\beta \sum_{p \in \mathcal{P}(e)} \operatorname{Re}(\operatorname{Tr}(I - U_p))\right),$$

where $Z(e)$ is the normalizing constant, given by

$$Z(e) = \int_G \exp\left(-\beta \sum_{p \in \mathcal{P}(e)} \operatorname{Re}(\operatorname{Tr}(I - U_p))\right) d\sigma(U(e)).$$

From this, it is easy to see that

$$|\rho(U(e)) - 1| \leq C\beta,$$

where C is a constant that depends only on G and d . This implies that the Dobrushin condition (16.2.1) is satisfied in β is small enough. By Theorem 16.2.1, this completes the proof. $\square$

16.4 Area law at strong coupling

Just like mass gap, quark confinement also holds at strong coupling in any dimension under fairly general conditions. Again, this was proved by Osterwalder and Seiler [24] using cluster expansions in the same paper where they proved mass gap under strong coupling. We will use Theorem 16.2.1. As before, let G be a compact Lie subgroup of $U(n)$ for some n . Take any $d \geq 2$, and let $\{\Lambda_N\}_{N \geq 1}$ be a sequence of finite subsets of $\mathbb{Z}^d$ that increase to $\mathbb{Z}^d$. Let β be a nonnegative real number. Consider lattice gauge theories on Λ_N with this G and β , under free boundary condition.

Theorem 16.4.1. *Assume that there is no vector in $\mathbb{C}^n$ that is fixed by all elements of G . Then there is some $\beta_0 > 0$, depending on G and d , such that if $\beta \leq \beta_0$, then the above theory (or rather, sequence of theories) exhibits quark confinement in the sense defined in Section 15.6.*

Proof. Let $e_1, \dots, e_d$ be the standard basis vectors of $\mathbb{R}^d$. Without loss of generality, let ℓ be the rectangular loop with vertices $0, Te_1, Re_2, Te_1 + Re_2$, where $R \leq T$. Let N be so large that ℓ is contained in Λ_N . Let U be a random configuration drawn from the lattice gauge theory on Λ_N . For $0 \leq j \leq T-1$ and $0 \leq k \leq R-1$, let

$$\begin{aligned} H_{j,0} &:= U((j, 0, \dots, 0), (j+1, 0, \dots, 0)), \\ H_{j,R} &:= U((j, R, 0, \dots, 0), (j+1, R, 0, \dots, 0)), \\ V_{0,k} &:= U((0, k, 0, \dots, 0), (0, k+1, 0, \dots, 0)), \\ V_{T,k} &:= U((T, k, 0, \dots, 0), (T, k+1, 0, \dots, 0)). \end{aligned}$$

(That is, the H 's are the matrices attached to the horizontal edges in ℓ and the V 's are the matrices attached to the vertical edges in ℓ , where ‘horizontal’ means the direction of e_1 and ‘vertical’ means the direction of e_2 .) Let $H_{j,k}^{a,b}$ denote the $(a, b)^{\text{th}}$ element of $H_{j,k}$, etc.

Then, proceeding as the proof of Theorem 15.8.1, we get

$$\begin{aligned}
 W_\ell = & \sum_{1 \leq a_0, \dots, a_{2R+2T-1} \leq n} H_{0,0}^{a_0, a_1} H_{1,0}^{a_1, a_2} \dots H_{T-1,0}^{a_{T-1}, a_T} \\
 & \cdot V_{T,0}^{a_T, a_{T+1}} V_{T,1}^{a_{T+1}, a_{T+2}} \dots V_{T,R-1}^{a_{T+R-1}, a_{T+R}} \\
 & \cdot \overline{H_{T-1,R}^{a_{T+R+1}, a_{T+R}} H_{T-2,R}^{a_{T+R+2}, a_{T+R+1}} \dots H_{0,R}^{a_{2T+R}, a_{2T+R-1}}} \\
 & \cdot \overline{V_{0,R-1}^{a_{2T+R+1}, a_{2T+R}} V_{0,R-2}^{a_{2T+R+2}, a_{2T+R+1}} \dots V_{0,0}^{a_0, a_{2T+2R-1}}}. \tag{16.4.1}
 \end{aligned}$$

For each $j \in \mathbb{Z}$, let E_j be the set of edges in $E(\Lambda_N)$ that are of the form $(x, x + e_2)$ for some $x = (x_1, \dots, x_d) \in \mathbb{Z}^d$ with $x_2 = j$. Define

$$E' := \bigcup_{j \in \mathbb{Z}} E_j.$$

That is, E' consists of all edges of the form $(x, x + e_2)$ in $E(\Lambda_N)$. In the following, $\mathbb{E}'$ and $\mathbb{P}'$ will denote conditional probability and conditional expectation given $(U(e))_{e \in E(\Lambda_N) \setminus E'}$. Also, whenever we say ‘conditional distribution’, it will be conditional distribution given this this collection.

Note that $e \in E_j$ and $e' \in E_{j'}$ for some $j \neq j'$, then e, e' cannot be in the same plaquette. This shows that conditionally, the collections $\{(U(e))_{e \in E_j} : j \in \mathbb{Z}^d\}$ are independent, because the conditional density factorizes. By the expansion (16.4.1), this shows that

$$\begin{aligned}
 \mathbb{E}'(W_\ell) = & \sum_{1 \leq a_0, \dots, a_{2R+2T-1} \leq n} H_{0,0}^{a_0, a_1} H_{1,0}^{a_1, a_2} \dots H_{T-1,0}^{a_{T-1}, a_T} \\
 & \cdot \mathbb{E}'(V_{T,0}^{a_T, a_{T+1}} \overline{V_{0,0}^{a_0, a_{2T+2R-1}}}) \mathbb{E}'(V_{T,1}^{a_{T+1}, a_{T+2}} \overline{V_{0,1}^{a_{2T+2R-1}, a_{2T+2R-2}}}) \dots \\
 & \dots \mathbb{E}'(V_{T,R-1}^{a_{T+R-1}, a_{T+R}} \overline{V_{0,R-1}^{a_{2T+R}, a_{2T+R-1}}}) \\
 & \cdot \overline{H_{T-1,R}^{a_{T+R+1}, a_{T+R}} H_{T-2,R}^{a_{T+R+2}, a_{T+R+1}} \dots H_{0,R}^{a_{2T+R}, a_{2T+R-1}}}.
 \end{aligned}$$

Now, for any $j \in \mathbb{Z}$, the form of the conditional density of $(U(e))_{e \in E_j}$ shows that this collection has exponentially decaying correlations if β is smaller than some positive constant that depends only on G and d . Thus, each $\mathbb{E}'(\dots)$ term in the above expression has absolute value bounded above by $C_1 e^{-C_2 T}$, where C_1, C_2 are positive constants that depend only on G and d . The proof is now completed by proceeding as in the last part of the proof of Theorem 15.8.1. $\square$

16.5 Confinement in 3D lattice gauge theories

Using an inequality of Messager, Miracle-Sole, and Pfister [19], Fröhlich [7] proved that $U(n)$ lattice gauge theory confines whenever the corresponding $U(1)$ theory does. Earlier, Glimm and Jaffe [9] established confinement for three-dimensional $U(1)$ gauge theory with a logarithmically growing potential, using ideas originating in the McBryan–Spencer approach to the Mermin–Wagner theorem [16]. Combining these results yields the theorem stated

below; indeed it appears as Corollary 2 in [7]. We will give a relatively short, self-contained proof of this result in this section.

Theorem 16.5.1. *Let G be a compact Lie subgroup of $U(n)$ for some n . Assume that $zI \in G$ for all $z \in \mathbb{C}$ with $|z| = 1$, where I denotes the $n \times n$ identity matrix. Let Λ be a finite subset of $\mathbb{Z}^3$ and β be a positive real number. Consider the lattice gauge theory on Λ with this G and β , under free boundary condition. For any rectangular loop ℓ contained in Λ with side lengths $R \leq T$, we have*

$$|\langle W_\ell \rangle| \leq ne^{-C(1+n\beta)^{-1}T \log(R+1)},$$

where C is a positive universal constant.

For $U(1)$ gauge theory in $d = 3$, confinement is known to be substantially stronger: G6pfert and Mack [11] proved positivity of the string tension for all couplings in a Villain-type formulation (see Section 16.6), i.e., an area law and at least linear growth of the quark–antiquark potential. It is natural to ask whether an appropriate version of Fr6hlich’s comparison inequality could be combined with such inputs to yield an area law for three-dimensional $U(n)$ gauge theory with Wilson action. We do not pursue this here, since doing so carefully would require additional technology and a careful comparison between formulations and limits.

Our proof of Theorem 16.5.1 requires the following ‘anti-concentration’ lemma about a certain class of probability measures on $U(1)$.

Lemma 16.5.2. *Take any $w \in \mathbb{C}$ and let μ be the probability measure on $U(1)$ that has density proportional to $\exp\{\operatorname{Re}(z\bar{w})\}$ with respect to the Haar measure on $U(1)$. Then*

$$\iint |z_1 - z_2|^2 d\mu(z_1) d\mu(z_2) \geq C \min\{1, |w|^{-1}\},$$

where C is a positive universal constant.

Proof. By Taylor expansion, we have that for any $x \in \mathbb{R}$,

$$|e^{ix} - 1 - ix| \leq \frac{x^2}{2}. \tag{16.5.1}$$

Thus, for any $z \in U(1)$ and $x \in \mathbb{R}$,

$$\begin{aligned} |z - ze^{ix}| &\geq |izx| - |z(1 - e^{ix} + ix)| \\ &\geq |x| - \frac{1}{2}x^2. \end{aligned}$$

Choosing

$$\epsilon := \min\{1, |w|^{-1/2}\} \leq 1,$$

we get that for any $z \in \mathbf{U}(1)$,

$$|z - ze^{\pm i\epsilon}| \geq \epsilon - \frac{1}{2}\epsilon^2 \geq \frac{\epsilon}{2}. \quad (16.5.2)$$

Define $\rho : \mathbf{U}(1) \rightarrow [0, \infty)$ as

$$\rho(z) := \exp\{\operatorname{Re}(z\bar{w})\},$$

and let

$$Z := \int_{\mathbf{U}(1)} \rho(z) d\sigma_0(z),$$

where σ_0 is the normalized Haar measure on $\mathbf{U}(1)$. Then note that μ has density $f = Z^{-1}\rho$ with respect to σ_0 . Define $\rho^+(z) := \rho(ze^{i\epsilon})$ and $\rho^-(z) := \rho(ze^{-i\epsilon})$. Let f^+ and f^- be the probability densities proportional to ρ^+ and ρ^- , respectively. By the invariance of Haar measure, the respective normalizing constants are both equal to Z . Let μ^+ and μ^- denote the respective probability measures. Then note that for any measurable set $A \subseteq \mathbf{U}(1)$, the invariance of Haar measure implies that

$$\begin{aligned} \mu^\pm(A) &= \frac{1}{Z} \int_A \rho(ze^{\pm i\epsilon}) d\sigma_0(z) \\ &= \frac{1}{Z} \int_{Ae^{\pm i\epsilon}} \rho(z') d\sigma_0(z') = \mu(Ae^{\pm i\epsilon}). \end{aligned} \quad (16.5.3)$$

Next, note that for any $z \in \mathbf{U}(1)$,

$$\begin{aligned} \sqrt{\rho^+(z)\rho^-(z)} &= \exp\left\{\frac{1}{2} \operatorname{Re}(\bar{w}ze^{i\epsilon} + \bar{w}ze^{-i\epsilon})\right\} \\ &= \rho(z) \exp\left\{\frac{1}{2} \operatorname{Re}(\bar{w}z(e^{i\epsilon} + e^{-i\epsilon} - 2))\right\} \\ &\geq \rho(z) \exp\left\{-\frac{1}{2}|w||e^{i\epsilon} + e^{-i\epsilon} - 2|\right\}. \end{aligned}$$

But, by the inequality (16.5.1),

$$|e^{i\epsilon} + e^{-i\epsilon} - 2| \leq |e^{i\epsilon} - 1 - i\epsilon| + |e^{-i\epsilon} - 1 + i\epsilon| \leq \epsilon^2.$$

Thus, we get

$$\sqrt{\rho^+(z)\rho^-(z)} \geq \rho(z)e^{-\frac{1}{2}|w|\epsilon^2} \geq \rho(z)e^{-1/2},$$

where the last inequality holds because $\epsilon \leq |w|^{-1/2}$. By the Cauchy–Schwarz inequality,

this gives for any measurable set $A \subseteq \mathbf{U}(1)$,

$$\begin{aligned} \sqrt{\mu^+(A)\mu^-(A)} &= \left(\int_A f^+(z) d\sigma_0(z) \int_A f^-(z) d\sigma_0(z) \right)^{1/2} \\ &\geq \int_A \sqrt{f^+(z)f^-(z)} d\sigma_0(z) \\ &= \frac{1}{Z} \int_A \sqrt{\rho^+(z)\rho^-(z)} d\sigma_0(z) \\ &\geq \frac{1}{Z} \int_A \rho(z) e^{-1/2} d\sigma_0(z) = \mu(A) e^{-1/2}. \end{aligned}$$

Take any $z \in \mathbf{U}(1)$ and let $A := \{z' \in \mathbf{U}(1) : |z - z'| < \frac{1}{4}\epsilon\}$. The above inequality shows that at least one $\mu^+(A)$ and $\mu^-(A)$ is $\geq \mu(A)e^{-1/2}$. First, suppose that $\mu^+(A) \geq \mu(A)e^{-1/2}$. Then by equation (16.5.3),

$$\begin{aligned} \mu(A \cap Ae^{i\epsilon}) &\geq \mu(A) + \mu(Ae^{i\epsilon}) - 1 \\ &= \mu(A) + \mu^+(A) - 1 \geq \mu(A)(1 + e^{-1/2}) - 1. \end{aligned}$$

We claim that $A \cap Ae^{i\epsilon}$ is empty. To see this, notice that if $z' \in A \cap Ae^{i\epsilon}$, then $|z' - z| < \frac{1}{4}\epsilon$ and $|z'e^{-i\epsilon} - z| < \frac{1}{4}\epsilon$. Thus, $|z' - z'e^{-i\epsilon}| < \frac{1}{2}\epsilon$. But this contradicts the inequality (16.5.2). So, $A \cap Ae^{i\epsilon}$ must be empty. Thus, the above inequality implies that

$$\mu(A) \leq \frac{1}{1 + e^{-1/2}}.$$

The same argument shows that this inequality also holds if $\mu^-(A) \geq \mu(A)e^{-1/2}$. This leads to the conclusion that for any $z \in \mathbf{U}(1)$,

$$\begin{aligned} \int |z - z'|^2 d\mu(z') &\geq \frac{1}{16}\epsilon^2 \mu\left(\left\{z' \in \mathbf{U}(1) : |z - z'| \geq \frac{1}{4}\epsilon\right\}\right) \\ &\geq \frac{1}{16}\epsilon^2 \left(1 - \frac{1}{1 + e^{-1/2}}\right) \\ &= \frac{e^{-1/2}}{16(1 + e^{-1/2})} \min\{1, |w|^{-1}\}. \end{aligned}$$

Integrating over z completes the proof. $\square$

Lemma 16.5.2 yields the following corollary.

Corollary 16.5.3. *Let ξ be a $\mathbf{U}(1)$ -valued random variable following the probability density f defined in Lemma 16.5.2. Then*

$$|\mathbb{E}(\xi)| \leq 1 - C \min\{1, |w|^{-1}\},$$

where C is a positive universal constant.

Proof. Let ξ_1, ξ_2 be i.i.d. copies of ξ . Then

$$|\xi_1 - \xi_2|^2 = 2 - 2\operatorname{Re}(\xi_1 \bar{\xi}_2).$$

This gives

$$\mathbb{E}(|\xi_1 - \xi_2|^2) = 2 - 2\operatorname{Re}(\mathbb{E}(\xi_1)\mathbb{E}(\bar{\xi}_2)) = 2 - 2|\mathbb{E}(\xi)|^2.$$

Rearranging and applying Lemma 16.5.2, we get

$$|\mathbb{E}(\xi)|^2 = 1 - \frac{1}{2}\mathbb{E}(|\xi_1 - \xi_2|^2) \leq 1 - C \min\{1, |w|^{-1}\},$$

where C is a positive universal constant. Without loss, $C \leq 1$. Taking square-root on both sides and applying the inequality $\sqrt{1-a} \leq 1 - \frac{1}{2}a$ (which holds for $a \in [0, 1]$) completes the proof. $\square$

We will now use Corollary 16.5.3 to prove the following generalization of the classical Mermin–Wagner theorem [17, 18].

Lemma 16.5.4. *Let Λ be a finite subset of $\mathbb{Z}^2$ and let E be the set positively oriented nearest-neighbor edges with both endpoints in Λ . Suppose that we are given a collection of complex numbers $(w_e)_{e \in E}$. Consider the probability measure γ on $\mathrm{U}(1)^\Lambda$ that has density $\propto \exp(\sum_{e=(x,y) \in E} \operatorname{Re}(w_e \xi_x \bar{\xi}_y))$ with respect to the product of normalized Haar measures on $\mathrm{U}(1)^\Lambda$, at each $\xi \in \mathrm{U}(1)^\Lambda$. Let ϕ be a random configuration drawn from γ . Then for any $x \in \Lambda$ and $R > 0$ such that $y := x + (R, 0) \in \Lambda$, we have*

$$|\mathbb{E}(\phi_x \bar{\phi}_y)| \leq e^{-CL^{-1} \log(R+1)},$$

where C is a positive universal constant and $L := 1 + \max_{e \in E} |w_e|$.

Proof. Fix x and R . First, we note that there is no loss of generality in assuming that $\Lambda = x + \{-N, \dots, N\}^2$ for some sufficiently large N . To see this, note that the original system can be ‘expanded’ to a system of this form by defining $w_e = 0$ for the new edges. This just couples the old system with a collection of independent uniformly distributed ϕ_x ’s. Proving the required bound for the new system proves it for the old system.

For each $0 \leq k \leq N$, let $y_k = x + (k, 0)$ and let L_k be the set of vertices in Λ that are at ℓ^∞ distance equal to k from x . Define a map $\tau : \mathrm{U}(1)^\Lambda \rightarrow \mathrm{U}(1)^\Lambda$ as follows. For $\xi \in \mathrm{U}(1)^\Lambda$, let $\tau(\xi)_x := \xi_x$, and for each $1 \leq k \leq N$, let $\tau(\xi)_{y_k} := \xi_{y_k} \bar{\xi}_{y_{k-1}}$. Next, for each $1 \leq k \leq N$ and $y \in L_k \setminus \{y_k\}$, let $\tau(\xi)_y := \xi_y \bar{\xi}_{y_k}$.

Note that each $e \in E$ is either an edge connecting two neighboring vertices in some L_k , or connecting a vertex in some L_k with a neighboring vertex in L_{k-1} . For each $1 \leq k \leq N$, let E_k denote the set of edges of the first kind, and let F_k denote the set of edges of the second kind, except the edge (y_{k-1}, y_k) . Further decompose F_k as $F_k^1 \cup F_k^2$, where F_k^1 is the set of all $(y, y') \in F_k$ with $y \in L_{k-1}$ and $y' \in L_k$, and F_k^2 is the set of all $(y, y') \in F_k$ with $y \in L_k$ and $y' \in L_{k-1}$.

Take any $\xi \in U(1)^\Lambda$ and let $\chi := \tau(\xi)$. Then by the above observation,

$$\begin{aligned} f(\xi) := \sum_{e=(y,y') \in E} w_e \xi_y \bar{\xi}_{y'} &= \sum_{k=1}^N w_{(y_{k-1}, y_k)} \xi_{y_{k-1}} \bar{\xi}_{y_k} + \sum_{k=1}^N \sum_{e=(y,y') \in E_k} w_e \xi_y \bar{\xi}_{y'} \\ &\quad + \sum_{k=1}^N \sum_{e=(y,y') \in F_k} w_e \xi_y \bar{\xi}_{y'}. \end{aligned}$$

Take any $2 \leq k \leq N$. Note that for $(y, y') \in F_k^1$,

$$\xi_y \bar{\xi}_{y'} = \xi_y \bar{\xi}_{y_{k-1}} \bar{\xi}_{y'} \xi_{y_k} \bar{\xi}_{y_k} \xi_{y_{k-1}} = \chi_y \bar{\chi}_{y'} \bar{\chi}_{y_k}.$$

Similarly, for any $(y, y') \in F_k^2$,

$$\xi_y \bar{\xi}_{y'} = \xi_y \bar{\xi}_{y_k} \bar{\xi}_{y'} \xi_{y_{k-1}} \bar{\xi}_{y_{k-1}} \xi_{y_k} = \chi_y \bar{\chi}_{y'} \chi_{y_k}.$$

When $k = 1$ and $(y, y') \in F_k^1$, we have $y = x = y_0$ and $y' \in L_1$, which gives

$$\xi_y \bar{\xi}_{y'} = \xi_{y_0} \bar{\xi}_{y_1} \bar{\xi}_{y'} \xi_{y_1} = \bar{\chi}_{y_1} \bar{\chi}_{y'}.$$

Similarly, if $k = 1$ and $(y, y') \in F_k^2$, then $y' = x = y_0$ and $y \in L_1$, which gives

$$\xi_y \bar{\xi}_{y'} = \xi_y \bar{\xi}_{y_1} \xi_{y_1} \bar{\xi}_{y_0} = \chi_y \chi_{y_1}.$$

Next, take any $1 \leq k \leq N$ and note that for $(y, y') \in E_k$ with $y, y' \neq y_k$,

$$\xi_y \bar{\xi}_{y'} = \xi_y \bar{\xi}_{y_k} \bar{\xi}_{y'} \xi_{y_k} = \chi_y \bar{\chi}_{y'}.$$

On the other hand, if $(y, y') \in E_k$ and $y = y_k$, then

$$\xi_y \bar{\xi}_{y'} = \bar{\xi}_{y'} \xi_{y_k} = \bar{\chi}_{y'},$$

and if $(y, y') \in E_k$ and $y' = y_k$, then

$$\xi_y \bar{\xi}_{y'} = \xi_y \bar{\xi}_{y_k} = \chi_y.$$

Combining the above observations, we get

$$\begin{aligned}
 f(\xi) &= \sum_{k=1}^N w_{(y_{k-1}, y_k)} \bar{\chi}_{y_k} + \sum_{k=1}^N \sum_{\substack{e=(y, y') \in E_k, \\ y, y' \neq y_k}} w_e \chi_y \bar{\chi}_{y'} \\
 &\quad + \sum_{k=1}^N w_{(y_k, y_k + (0, 1))} \bar{\chi}_{y_k + (0, 1)} + \sum_{k=1}^N w_{(y_k - (0, 1), y_k)} \chi_{y_k - (0, 1)} \\
 &\quad + \sum_{k=2}^N \left(\sum_{e=(y, y') \in F_k^1} w_e \chi_y \bar{\chi}_{y'} \right) \bar{\chi}_{y_k} + \sum_{k=2}^N \left(\sum_{e=(y, y') \in F_k^2} w_e \chi_y \bar{\chi}_{y'} \right) \chi_{y_k} \\
 &\quad + \sum_{e=(y, y') \in F_1^1} w_e \bar{\chi}_{y'} \bar{\chi}_{y_1} + \sum_{e=(y, y') \in F_1^2} w_e \chi_y \chi_{y_1} =: g(\chi).
 \end{aligned}$$

Let ϕ be a random configuration with density proportional to $e^{\text{Re}(f)}$, and let $\psi := \tau(\phi)$. We claim that ψ has density proportional to $e^{\text{Re}(g)}$. To see this, let us first show that for any bounded measurable $F : \text{U}(1)^\Lambda \rightarrow \mathbb{C}$,

$$\int F(\tau(\xi)) \prod_{y \in \Lambda} d\sigma_0(\xi_y) = \int F(\xi) \prod_{y \in \Lambda} d\sigma_0(\xi_y), \quad (16.5.4)$$

where σ_0 denotes the normalized Haar measure on $\text{U}(1)$. To prove this claim, we first integrate over ξ_y for all $y \in \Lambda \setminus \{y_0, \dots, y_N\}$. Take any $y \in L_k \setminus \{y_k\}$. In the integrand, ξ_y appears only through $\tau(\xi)_y = \xi_y \bar{\xi}_{y_k}$. By the invariance of Haar measure, we can replace this simply by ξ_y . Next, we successively integrate over $\xi_{y_N}, \xi_{y_{N-1}}, \dots, \xi_{y_0}$, in this order. Note that ξ_{y_N} appears in the integrand only through $\tau(\xi)_{y_N} = \xi_{y_N} \bar{\xi}_{y_{N-1}}$. Again, by the invariance of Haar measure, we can replace this by ξ_{y_N} . After this replacement, the integrand now depends on $\xi_{y_{N-1}}$ only through $\tau(\xi)_{y_{N-1}} = \xi_{y_{N-1}} \bar{\xi}_{y_{N-2}}$. Thus, we can replace this term by $\xi_{y_{N-1}}$, and so on. This proves the claim (16.5.4).

Let us now prove that ψ has density proportional to $e^{\text{Re}(g)}$. Let F be as above. Then by equation (16.5.4) and the definition of g ,

$$\begin{aligned}
 \int F(\tau(\xi)) e^{\text{Re}(f(\xi))} \prod_{y \in \Lambda} d\sigma_0(\xi_y) &= \int F(\tau(\xi)) e^{\text{Re}(g(\tau(\xi)))} \prod_{y \in \Lambda} d\sigma_0(\xi_y) \\
 &= \int F(\xi) e^{\text{Re}(g(\xi))} \prod_{y \in \Lambda} d\sigma_0(\xi_y).
 \end{aligned}$$

Since this holds also for $F \equiv 1$, we get

$$\begin{aligned}
 \mathbb{E}(F(\psi)) &= \frac{\int F(\tau(\xi)) e^{\text{Re}(f(\xi))} \prod_{y \in \Lambda} d\sigma_0(\xi_y)}{\int e^{\text{Re}(f(\xi))} \prod_{y \in \Lambda} d\sigma_0(\xi_y)} \\
 &= \frac{\int F(\xi) e^{\text{Re}(g(\xi))} \prod_{y \in \Lambda} d\sigma_0(\xi_y)}{\int e^{\text{Re}(g(\xi))} \prod_{y \in \Lambda} d\sigma_0(\xi_y)}.
 \end{aligned}$$

This proves that ψ has density proportional to $e^{\text{Re}(g)}$.

Let $\mathbb{E}'$ denote conditional expectation given $\{\psi(y) : y \notin \{y_1, \dots, y_N\}\}$. From the form of g , we see that under this conditioning, the random variables $\psi_{y_1}, \dots, \psi_{y_N}$ are independent, with ψ_{y_k} having conditional density proportional to $\exp(\text{Re}(\bar{u}_k \chi))$ at $\chi \in \text{U}(1)$, where

$$u_k := w_{(y_{k-1}, y_k)} + \sum_{e=(y, y') \in F_k^1} w_e \psi_y \bar{\psi}_{y'} + \sum_{e=(y, y') \in F_k^2} \bar{w}_e \bar{\psi}_y \psi_{y'}$$

if $2 \leq k \leq N$, and

$$u_1 := w_{(y_0, y_1)} + \sum_{e=(y, y') \in F_1^1} w_e \bar{\psi}_{y'} + \sum_{e=(y, y') \in F_1^2} \bar{w}_e \bar{\psi}_y.$$

Note that $|u_k| \leq CLk$, where C is a universal constant and $L = 1 + \max_{e \in E} |w_e|$. Thus, by Corollary 16.5.3,

$$|\mathbb{E}'(\psi_{y_k})| \leq 1 - C_0 \min\{1, (C_1 Lk)^{-1}\}$$

for some positive universal constants C_0 and C_1 . Thus, for any $1 \leq R \leq N$,

$$\begin{aligned} |\mathbb{E}'(\psi_{y_1} \cdots \psi_{y_R})| &= \prod_{k=1}^R |\mathbb{E}'(\psi_{y_k})| \leq \prod_{k=1}^R (1 - C_0 \min\{1, (C_1 Lk)^{-1}\}) \\ &\leq \prod_{k=1}^R e^{-C_0 \min\{1, (C_1 Lk)^{-1}\}} = \exp\left(-C_0 \sum_{k=1}^R \min\{1, (C_1 Lk)^{-1}\}\right). \end{aligned}$$

Now, note that since $L \geq 1$, we have

$$\min\{1, (C_1 Lk)^{-1}\} \geq \min\{(Lk)^{-1}, (C_1 Lk)^{-1}\} = C_2 (Lk)^{-1},$$

where $C_2 := \min\{1, C_1^{-1}\}$. Thus, we get

$$|\mathbb{E}'(\psi_{y_1} \cdots \psi_{y_R})| \leq \exp\left(-\frac{C_0 C_2}{L} \sum_{k=1}^R \frac{1}{k}\right) \leq \exp\left(-\frac{C_0 C_2}{L} \log(R+1)\right).$$

Thus, the unconditional expected value $|\mathbb{E}(\psi_{y_1} \cdots \psi_{y_R})|$ is also bounded above by the same quantity. But $\psi_{y_1} \cdots \psi_{y_R} = \bar{\phi}_x \phi_{y_R}$. This completes the proof. $\square$

We are now ready to prove Theorem 16.5.1.

Proof of Theorem 16.5.1. First, we construct an ‘expanded’ model that is equivalent to our lattice gauge theory on Λ in a certain sense. In the expanded model, configurations consist of maps from $E(\Lambda)$ into $G \times \text{U}(1)$. Let (U, ξ) be such a configuration. For a negatively oriented edge (x, y) , we let $U(y, x) := U(x, y)^{-1}$ and $\xi(y, x) := \xi(x, y)^{-1}$. For a plaquette p with successive oriented edges e_1, e_2, e_3, e_4 , we define U_p as before and analogously let

$\xi_p := \xi(e_1)\xi(e_2)\xi(e_3)\xi(e_4)$. Finally, we consider the probability measure

$$d\nu(U, \xi) = \frac{1}{\tilde{Z}} \exp\left(-\beta \sum_{p \in P(\Lambda)} \operatorname{Re}(\operatorname{Tr}(I - \xi_p U_p))\right) \prod_{e \in E(\Lambda)} d\sigma(U(e)) d\sigma_0(\xi(e)),$$

where σ and σ_0 are the normalized Haar measures on G and $U(1)$, respectively, and $\tilde{Z}$ is the normalizing constant.

We claim that $\tilde{Z} = Z$, where Z is the normalizing constant of our original lattice gauge theory on Λ . To see this, recall that $zI \in G$ for all $z \in \mathbb{C}$ with $|z| = 1$. Take any ξ . Since $|\xi(e)| = 1$ for all e , and

$$\xi_p U_p = \xi(e_1)U(e_1)\xi(e_2)U(e_2)\xi(e_3)U(e_3)\xi(e_4)U(e_4)$$

for a plaquette p with successive oriented edges e_1, e_2, e_3, e_4 , the invariance of σ implies that

$$\begin{aligned} \tilde{Z} &= \int \exp\left(-\beta \sum_{p \in P(\Lambda)} \operatorname{Re}(\operatorname{Tr}(I - \xi_p U_p))\right) \prod_{e \in E(\Lambda)} d\sigma(U(e)) \\ &= \int \exp\left(-\beta \sum_{p \in P(\Lambda)} \operatorname{Re}(\operatorname{Tr}(I - U_p))\right) \prod_{e \in E(\Lambda)} d\sigma(U(e)) = Z. \end{aligned}$$

Since this holds for any ξ and σ_0 is a probability measure, we can now integrate over ξ and conclude that $\tilde{Z} = Z$.

Next, for any (U, ξ) , let $\xi U \in G^{E(\Lambda)}$ denote the configuration $(\xi U)(e) := \xi(e)U(e)$. Then the same argument as above implies that for any bounded measurable function $f : G^{E(\Lambda)} \rightarrow \mathbb{C}$,

$$\begin{aligned} &\int f(\xi U) \exp\left(-\beta \sum_{p \in P(\Lambda)} \operatorname{Re}(\operatorname{Tr}(I - \xi_p U_p))\right) \prod_{e \in E(\Lambda)} d\sigma(U(e)) \\ &= \int f(U) \exp\left(-\beta \sum_{p \in P(\Lambda)} \operatorname{Re}(\operatorname{Tr}(I - U_p))\right) \prod_{e \in E(\Lambda)} d\sigma(U(e)). \end{aligned}$$

Combining, we get that

$$\int f(\xi U) d\nu(U, \xi) = \int f(U) d\mu(U), \quad (16.5.5)$$

where μ is the probability measure for the original lattice gauge theory.

Henceforth, we will let (U, ξ) denote a random configuration drawn from ν , and V be a random configuration drawn from μ . Let ℓ be the rectangular loop with vertices $(0, 0, 0)$, $(R, 0, 0)$, $(0, T, 0)$ and $(R, T, 0)$. Let $e_1, \dots, e_k$ be the successive oriented edges of ℓ (with

some arbitrary e_1), and define

$$\begin{aligned} W_\ell &:= \text{Tr}(V(e_1) \cdots V(e_k)), \\ Q_\ell &:= \text{Tr}(U(e_1) \cdots U(e_k)), \\ \chi_\ell &:= \xi(e_1) \cdots \xi(e_k). \end{aligned}$$

Then by equation (16.5.5),

$$\mathbb{E}(W_\ell) = \mathbb{E}[\text{Tr}(\xi(e_1)U(e_1) \cdots \xi(e_k)U(e_k))] = \mathbb{E}(\chi_\ell Q_\ell).$$

Let $\mathbb{E}'$ denote conditional expectation given U . Then by the above equation,

$$\mathbb{E}(W_\ell) = \mathbb{E}(\mathbb{E}'(\chi_\ell)Q_\ell),$$

which implies that

$$|\mathbb{E}(W_\ell)| \leq \mathbb{E}(|\mathbb{E}'(\chi_\ell)Q_\ell|) \leq n\mathbb{E}(|\mathbb{E}'(\chi_\ell)|).$$

Thus, it suffices to show that

$$|\mathbb{E}'(\chi_\ell)| \leq e^{-C(1+n\beta)^{-1}T \log(R+1)} \quad (16.5.6)$$

for some positive universal constants C .

Let $E' \subseteq E(\Lambda)$ denote the set of edges of the form $(x, x + e_2)$, where $e_2 = (0, 1, 0)$. Let $E'' := E(\Lambda) \setminus E'$, and let $\mathbb{E}''$ denote conditional expectation given U and $(\xi(e))_{e \in E''}$. We will prove that

$$|\mathbb{E}''(\chi_\ell)| \leq e^{-C(1+n\beta)^{-1}T \log(R+1)} \quad (16.5.7)$$

for some positive universal constant C . Clearly, this implies the bound (16.5.6) and completes the proof.

For $0 \leq j < T$, let p_j be the edge $(je_2, (j+1)e_2)$ and let q_j be the edge $(Re_1 + je_2, Re_1 + (j+1)e_2)$. Also, let $E_j \subseteq E(\Lambda)$ be the set of edges of the form $(x, x + e_2)$ where the second coordinate of x is j . Repeating the observation that we have already made several times earlier, note that conditional on U and $(\xi(e))_{e \in E''}$, the collections $(\xi(e))_{e \in E_j}$, $j = 0, 1, \dots, T-1$ are independent. Consequently,

$$\mathbb{E}''(\chi_\ell) = \chi'_\ell \prod_{j=0}^{T-1} \mathbb{E}''(\overline{\xi(p_j)} \xi(q_j)),$$

where χ'_ℓ is the product of $\xi(e)$ over all e in the lower boundary of ℓ and $\overline{\xi(e)}$ over all e in

the upper boundary of ℓ . In particular,

$$|\mathbb{E}''(\chi_\ell)| = \prod_{j=0}^{T-1} |\mathbb{E}''(\overline{\xi(p_j)}\xi(q_j))|. \quad (16.5.8)$$

Now, it is easy to see that conditional on U and $(\xi(e))_{e \in E''}$, the collection $(\xi(e))_{e \in E_j}$ is a 2D spin system, with $U(1)$ -valued spins, and conditional density of the form considered in Lemma 16.5.4. The corresponding w 's are uniformly bounded by $C_0 n\beta$, where C_0 is a positive universal constant. Thus, for any j ,

$$|\mathbb{E}''(\overline{\xi(p_j)}\xi(q_j))| \leq e^{-C_1(1+n\beta)^{-1} \log(R+1)},$$

where, again, C_1 is a positive universal constant. By equation (16.5.8), this proves the claim (16.5.7) and therefore, completes the proof of the theorem. $\square$

16.6 The Villain action

The Villain action was introduced by Jacques Villain in [31] in the context of 2D magnets with continuous spins. Adapted to $U(1)$ lattice gauge theory, the action is as follows. Let Λ be a finite subset of $\mathbb{Z}^d$ for some $d \geq 2$, and let E be the set of positively oriented nearest-neighbor edges with both endpoints in Λ . Recall that the space of configurations for $U(1)$ lattice gauge theory on Λ is $U(1)^E$. Thus, we can alternatively view the configuration space as $[-\pi, \pi)^E$, with the interpretation that $\theta \in [-\pi, \pi)$ represents the element $e^{i\theta} \in U(1)$.

Let $\theta = (\theta(e))_{e \in E}$ be a configuration. For an edge $(x, y) \in E$, we define $\theta(y, x) := -\theta(x, y)$, so that $e^{i\theta(y, x)} = e^{-i\theta(x, y)} = (e^{i\theta(x, y)})^{-1}$, as required. For a plaquette p with vertices x_1, x_2, x_3, x_4 in anticlockwise order, we define

$$\theta_p := \theta(x_1, x_2) + \theta(x_2, x_3) + \theta(x_3, x_4) + \theta(x_4, x_1). \quad (16.6.1)$$

Again, this ensures that

$$e^{i\theta_p} = e^{i\theta(x_1, x_2)} e^{i\theta(x_2, x_3)} e^{i\theta(x_3, x_4)} e^{i\theta(x_4, x_1)}.$$

Let P denote the set of plaquettes with all vertices in Λ . The Villain action of θ at inverse coupling parameter β is defined as

$$S_\beta(\theta) := - \sum_{p \in P} \log \left[\sum_{n \in \mathbb{Z}} \exp \left(-\frac{\beta}{2} (\theta_p - 2\pi n)^2 \right) \right]. \quad (16.6.2)$$

Lattice gauge theory with Villain action has density proportional to $e^{-S_\beta(\theta)}$ with respect to Lebesgue measure on $[-\pi, \pi)^E$. The base measure is chosen to be Lebesgue because that is pushed forward to Haar measure on $U(1)$ under the map $x \mapsto e^{ix}$.

Let us first check that the Villain action is indeed a reasonable discrete approximation

of the $U(1)$ Yang–Mills action in the continuum. For this, we have to prove two things. First, we have to prove the analogue of Theorem 15.2.1 for the Villain action. Second, we have to show that it is gauge invariant. For the first property, we cannot work with the action S_β defined in equation (16.6.2), since it has nonzero contributions from plaquettes where $\theta_p = 0$. We modify S_β as

$$\begin{aligned} \tilde{S}_\beta(\theta) := & - \sum_{p \in P} \left\{ \log \left[\sum_{n \in \mathbb{Z}} \exp \left(-\frac{\beta}{2} (\theta_p - 2\pi n)^2 \right) \right] \right. \\ & \left. - \log \left[\sum_{n \in \mathbb{Z}} \exp \left(-\frac{\beta}{2} (-2\pi n)^2 \right) \right] \right\}. \end{aligned} \quad (16.6.3)$$

This gives the same theory as S_β , but has the property that plaquettes where $\theta_p = 0$ do not contribute to the action. With this action, we have the following result.

Theorem 16.6.1. *Take any $d \geq 2$ and $g_0 > 0$. Let $A = \sum_{j=1}^d A_j dx_j$ be a compactly supported smooth $u(1)$ -valued 1-form on $\mathbb{R}^d$. Let $e_1, \dots, e_d$ denote the standard basis vectors of $\mathbb{R}^d$. For a directed edge $(x, x + \epsilon e_j)$ of $\epsilon \mathbb{Z}^d$, define $\theta(x, x + \epsilon e_j) := -i \log(e^{\epsilon A_j(x)})$, where we define $\log$ on the unit circle as $\log(e^{ia}) := ia$ for $-\pi \leq a < \pi$. Let $\theta(x + \epsilon e_j, x) := -\theta(x, x + \epsilon e_j)$. For a plaquette p in $\epsilon \mathbb{Z}^d$, let θ_p be defined as in equation (16.6.1). Let $\beta = \beta(\epsilon) := g_0^{-2} \epsilon^{d-4}$, and let $\tilde{S}_\beta(\theta)$ be defined as in equation (16.6.3), but with p running over all plaquettes of $\epsilon \mathbb{Z}^d$. Then, for $d \geq 5$, $\tilde{S}_{\beta(\epsilon)}(\theta) \rightarrow 0$ as $\epsilon \rightarrow 0$. For $d = 2, 3$, we have*

$$\lim_{\epsilon \rightarrow 0} \tilde{S}_{\beta(\epsilon)}(\theta) = S_E(A),$$

where S_E is the Euclidean Yang–Mills action defined in equation (15.1.1). For $d = 4$,

$$\lim_{\epsilon \rightarrow 0} \tilde{S}_{\beta(\epsilon)}(\theta) = \kappa S_E(A),$$

where

$$\kappa = 1 - \frac{4\pi^2}{g_0^2} \frac{\sum_{n \in \mathbb{Z}} n^2 e^{-2\pi^2 g_0^{-2} n^2}}{\sum_{n \in \mathbb{Z}} e^{-2\pi^2 g_0^{-2} n^2}}.$$

Moreover in $d = 4$, given any $g_0 > 0$, one can find $\tilde{g}_0 > 0$ such that if we define $\beta := \tilde{g}_0^{-2}$, then $\tilde{S}_\beta(\theta)$ converges to $S_E(A)$ as $\epsilon \rightarrow 0$.

Proof. In this proof, $O(\epsilon^\alpha)$ will denote any quantity whose absolute value is bounded by $C\epsilon^\alpha$, where C is a constant that may depend only on the dimension d , the constant g_0 , and the connection A . Take any $x \in \epsilon \mathbb{Z}^d$ and any $1 \leq j < k \leq d$, and let

$$x_1 = x, \quad x_2 = x + \epsilon e_j, \quad x_3 = x + \epsilon e_j + \epsilon e_k, \quad x_4 = x + \epsilon e_k.$$

Let p be the plaquette formed by the vertices x_1, x_2, x_3, x_4 . Now, note that for any edge $(x, x + \epsilon e_j)$,

$$\theta(x, x + \epsilon e_j) \equiv -i\epsilon A_j(x) \pmod{2\pi}.$$

Thus, we have

$$\begin{aligned}\theta_p &= \theta(x_1, x_2) + \theta(x_2, x_3) + \theta(x_3, x_4) + \theta(x_4, x_1) \\ &\equiv -i\epsilon(A_j(x_1) + A_k(x_2) - A_j(x_4) - A_k(x_1)) \pmod{2\pi}.\end{aligned}$$

Consequently,

$$\begin{aligned}&\sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) \\ &= \sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(-i\epsilon(A_j(x_1) + A_k(x_2) - A_j(x_4) - A_k(x_1)) - 2\pi n)^2\right) \\ &= \sum_{n \in \mathbb{Z}} \exp\left(\frac{\epsilon^{d-2}}{2g_0^2}(A_j(x_1) + A_k(x_2) - A_j(x_4) - A_k(x_1) - 2\pi i\epsilon^{-1}n)^2\right).\end{aligned}\tag{16.6.4}$$

Writing

$$A_k(x_2) = A_k(x + \epsilon e_j) = A_k(x) + \epsilon \partial_j A_k(x) + O(\epsilon^2)$$

and using a similar Taylor expansion for $A_j(x_4)$, we get

$$A_j(x_1) + A_k(x_2) - A_j(x_4) - A_k(x_1) = \epsilon(\partial_j A_k(x) - \partial_k A_j(x)) + O(\epsilon^2).$$

Plugging this into equation (16.6.4) gives

$$\begin{aligned}&\sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) \\ &= \sum_{n \in \mathbb{Z}} \exp\left(\frac{\epsilon^{d-2}}{2g_0^2}(\epsilon(\partial_j A_k(x) - \partial_k A_j(x)) + O(\epsilon^2) - 2\pi i\epsilon^{-1}n)^2\right) \\ &= \sum_{n \in \mathbb{Z}} \exp\left(\frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon) - 2\pi i\epsilon^{-2}n)^2\right).\end{aligned}$$

Take any $n \in \mathbb{Z}$, and note that

$$\begin{aligned}&\exp\left(\frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon) - 2\pi i\epsilon^{-2}n)^2\right) \\ &= \exp\left(-\frac{2\epsilon^{d-4}\pi^2 n^2}{g_0^2}\right) \exp\left(\frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))^2\right. \\ &\quad \left.- \frac{2\pi i\epsilon^{d-2}n}{g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))\right),\end{aligned}$$

where the $O(\epsilon)$ inside the cosine term has no dependence on n . This shows that for any

nonnegative integer n ,

$$\begin{aligned} & \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) + \exp\left(-\frac{\beta}{2}(\theta_p + 2\pi n)^2\right) \\ &= 2 \exp\left(-\frac{2\epsilon^{d-4}\pi^2 n^2}{g_0^2}\right) \exp\left(\frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))^2\right) \\ & \quad \cdot \cos\left(\frac{2\pi\epsilon^{d-2}n}{g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))\right). \end{aligned} \quad (16.6.5)$$

We will treat the cases $d = 2, 3$, $d = 4$, and $d \geq 5$ separately. First, suppose that $d = 4$. Then note that

$$\exp\left(\frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))^2\right) = 1 + \frac{\epsilon^4}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^5),$$

and

$$\begin{aligned} \cos\left(\frac{2\pi\epsilon^{d-2}n}{g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))\right) &= 1 - \frac{2\pi^2\epsilon^4 n^2}{g_0^4}(\partial_j A_k(x) - \partial_k A_j(x))^2 \\ & \quad + O(\epsilon^5)n^2 + O(\epsilon^8)n^4. \end{aligned}$$

Combining the last three displays and simplifying, we get that for $d = 4$ and any nonnegative integer n ,

$$\begin{aligned} & \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) + \exp\left(-\frac{\beta}{2}(\theta_p + 2\pi n)^2\right) \\ &= 2 \exp\left(-\frac{2\pi^2 n^2}{g_0^2}\right) \left(1 + \epsilon^4 \left(\frac{1}{2g_0^2} - \frac{2\pi^2 n^2}{g_0^4}\right) (\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^5)n^4\right). \end{aligned}$$

From this, we get

$$\sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) = K_1 + K_2 \epsilon^4 (\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^5),$$

where

$$\begin{aligned} K_1 &:= \sum_{n \in \mathbb{Z}} \exp\left(-\frac{2\pi^2 n^2}{g_0^2}\right), \\ K_2 &:= \sum_{n \in \mathbb{Z}} \exp\left(-\frac{2\pi^2 n^2}{g_0^2}\right) \left(\frac{1}{2g_0^2} - \frac{2\pi^2 n^2}{g_0^4}\right). \end{aligned}$$

Note that $\log K_1$ is the quantity that we subtract from each term to get $\tilde{S}_\beta$ from S_β . Thus,

for $d = 4$,

$$\begin{aligned}\tilde{S}_\beta(\theta) &= - \sum_{p \in P} \{\log(K_1 + K_2 \epsilon^4 (\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^5)) - \log K_1\} \\ &= - \frac{K_2}{K_1} \sum_{p \in P} (\epsilon^4 (\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^5)),\end{aligned}$$

where P denotes the set of plaquettes with nonzero contributions. Since $|P| = O(\epsilon^{-4})$ and

$$\frac{K_2}{K_1} = \frac{1}{2g_0^2} \left(1 - \frac{4\pi^2 \sum_{n \in \mathbb{Z}} n^2 e^{-2\pi^2 g_0^{-2} n^2}}{\sum_{n \in \mathbb{Z}} e^{-2\pi^2 g_0^{-2} n^2}} \right) = \frac{\kappa}{2g_0^2},$$

we can now proceed as in the proof of Theorem 15.2.1 to complete the proof when $d = 4$. We will establish the claim about the existence of $\tilde{g}_0$ at the end of the proof.

Next, suppose that d is 2 or 3. Starting with equation (16.6.5), we get that for any positive integer n ,

$$\exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) + \exp\left(-\frac{\beta}{2}(\theta_p + 2\pi n)^2\right) \leq C_1 e^{-C_2 \epsilon^{d-4} n^2},$$

where C_1, C_2 are positive constants that depend only on d, g_0 , and A . On the other hand, for $n = 0$, equation (16.6.5) gives

$$\exp\left(-\frac{\beta}{2}\theta_p^2\right) = \exp\left(\frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))^2\right).$$

Combining, we get that for any $\alpha > 0$,

$$\begin{aligned}\sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) &= \exp\left(\frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))^2\right) + O(\epsilon^\alpha) \\ &= 1 + \frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^{d+1}) + O(\epsilon^\alpha).\end{aligned}$$

Choosing $\alpha = d + 1$, we get

$$\log \left[\sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) \right] = \frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^{d+1}).$$

On the other hand,

$$\begin{aligned}\sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(-2\pi n)^2\right) &= \sum_{n \in \mathbb{Z}} \exp\left(-\frac{\pi^2 \epsilon^{d-4} n^2}{g_0^2}\right) \\ &= 1 + O(\epsilon^\alpha)\end{aligned}$$

for any $\alpha > 0$. It is now easy to complete the proof for $d = 2, 3$.

Finally, suppose that $d \geq 5$. Then note that

$$\exp\left(\frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))^2\right) = 1 + \frac{\epsilon^d}{2g_0^2}(\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^{d+1}),$$

and for any nonnegative integer n ,

$$\begin{aligned} & \cos\left(\frac{2\pi\epsilon^{d-2}n}{g_0^2}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))\right) \\ &= 1 - \frac{2\pi^2\epsilon^{2d-4}n^2}{g_0^4}(\partial_j A_k(x) - \partial_k A_j(x) + O(\epsilon))^2 + O(\epsilon^{4d-8})n^4 \\ &= 1 - \frac{2\pi^2\epsilon^{2d-4}n^2}{g_0^4}(\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^{2d-3})n^2 + O(\epsilon^{4d-8})n^4. \end{aligned}$$

Plugging the above estimates into equation (16.6.5), we get that for any nonnegative integer n ,

$$\begin{aligned} & \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) + \exp\left(-\frac{\beta}{2}(\theta_p + 2\pi n)^2\right) \\ &= 2 \exp\left(-\frac{2\epsilon^{d-4}\pi^2 n^2}{g_0^2}\right) \left(1 + \left(\frac{\epsilon^d}{2g_0^2} - \frac{2\pi^2\epsilon^{2d-4}n^2}{g_0^4}\right)(\partial_j A_k(x) - \partial_k A_j(x))^2\right. \\ & \quad \left.+ O(\epsilon^{d+1}) + O(\epsilon^{2d-3})n^2 + O(\epsilon^{4d-8})n^4\right). \end{aligned}$$

Define

$$\begin{aligned} K_1(\epsilon) &:= \sum_{n \in \mathbb{Z}} \exp\left(-\frac{2\epsilon^{d-4}\pi^2 n^2}{g_0^2}\right), \\ K_2(\epsilon) &:= \sum_{n \in \mathbb{Z}} n^2 \exp\left(-\frac{2\epsilon^{d-4}\pi^2 n^2}{g_0^2}\right), \\ K_3(\epsilon) &:= \sum_{n \in \mathbb{Z}} n^4 \exp\left(-\frac{2\epsilon^{d-4}\pi^2 n^2}{g_0^2}\right). \end{aligned}$$

Then by the previous display, we get

$$\begin{aligned} & \sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) \\ &= K_1(\epsilon) \left(1 + \left(\frac{\epsilon^d}{2g_0^2} - \frac{2\pi^2\epsilon^{2d-4}K_2(\epsilon)}{g_0^4 K_1(\epsilon)}\right)(\partial_j A_k(x) - \partial_k A_j(x))^2 + O(\epsilon^{d+1})\right) \\ & \quad + O(\epsilon^{2d-3})K_2(\epsilon) + O(\epsilon^{4d-8})K_3(\epsilon). \end{aligned}$$

Thus,

$$\begin{aligned}\tilde{S}_\beta(\theta) &= - \sum_{p \in P} \left\{ \log \left[\sum_{n \in \mathbb{Z}} \exp \left(-\frac{\beta}{2} (\theta_p - 2\pi n)^2 \right) \right] - \log K_1(\epsilon) \right\} \\ &= - \sum_{p \in P} \log \left[1 + \left(\frac{\epsilon^d}{2g_0^2} - \frac{2\pi^2 \epsilon^{2d-4} K_2(\epsilon)}{g_0^4 K_1(\epsilon)} \right) (\partial_j A_k(x) - \partial_k A_j(x))^2 \right. \\ &\quad \left. + O(\epsilon^{d+1}) + O(\epsilon^{2d-3}) \frac{K_2(\epsilon)}{K_1(\epsilon)} + O(\epsilon^{4d-8}) \frac{K_3(\epsilon)}{K_1(\epsilon)} \right].\end{aligned}$$

Now, note that by Riemann sum approximation,

$$\begin{aligned}\lim_{\epsilon \rightarrow 0} \epsilon^{(d-4)/2} K_1(\epsilon) &= \int_{-\infty}^{\infty} \exp \left(-\frac{2\pi^2 x^2}{g_0^2} \right) dx, \\ \lim_{\epsilon \rightarrow 0} \epsilon^{3(d-4)/2} K_2(\epsilon) &= \int_{-\infty}^{\infty} x^2 \exp \left(-\frac{2\pi^2 x^2}{g_0^2} \right) dx, \\ \lim_{\epsilon \rightarrow 0} \epsilon^{5(d-4)/2} K_3(\epsilon) &= \int_{-\infty}^{\infty} x^4 \exp \left(-\frac{2\pi^2 x^2}{g_0^2} \right) dx.\end{aligned}$$

Consequently, as $\epsilon \rightarrow 0$,

$$\frac{K_2(\epsilon)}{K_1(\epsilon)} \sim \epsilon^{-(d-4)} \frac{g_0^2}{4\pi^2}, \quad \frac{K_3(\epsilon)}{K_1(\epsilon)} = O(\epsilon^{-2(d-4)}).$$

This gives

$$\begin{aligned}\tilde{S}_\beta(\theta) &= - \sum_{p \in P} \left\{ \log \left[\sum_{n \in \mathbb{Z}} \exp \left(-\frac{\beta}{2} (\theta_p - 2\pi n)^2 \right) \right] - \log K_1(\epsilon) \right\} \\ &= - \sum_{p \in P} \log(1 + \epsilon^d o(1) + O(\epsilon^{d+1}) + O(\epsilon^{2d})).\end{aligned}$$

The proof can now be completed as before.

Finally, to prove the existence of $\tilde{g}_0$ when $d = 4$, define a function $f : (0, \infty) \rightarrow \mathbb{R}$ as

$$f(x) := \frac{\sum_{n \in \mathbb{Z}} n^2 x e^{-n^2 x}}{\sum_{n \in \mathbb{Z}} e^{-n^2 x}}.$$

It is easy to see that f is continuous and $f(x) \rightarrow 0$ as $x \rightarrow \infty$. On the other hand, as $x \rightarrow 0$, Riemann sum approximation gives

$$\lim_{x \rightarrow 0} \sqrt{x} \sum_{n \in \mathbb{Z}} e^{-n^2 x} = \int_{-\infty}^{\infty} e^{-y^2} dy, \quad \lim_{x \rightarrow 0} x^{3/2} \sum_{n \in \mathbb{Z}} n^2 e^{-n^2 x} = \int_{-\infty}^{\infty} y^2 e^{-y^2} dy.$$

Thus,

$$\lim_{x \rightarrow 0} f(x) = \frac{\int_{-\infty}^{\infty} y^2 e^{-y^2} dy}{\int_{-\infty}^{\infty} e^{-y^2} dy} = \frac{1}{2}.$$

Take any $g_0 > 0$. To make $\tilde{S}_\beta(\theta)$ converge to $S_E(A)$ as $\epsilon \rightarrow 0$, we need to choose $\beta = \tilde{g}_0^{-2}$, where $\tilde{g}_0 > 0$ solves

$$\frac{1}{2\tilde{g}_0^2}(1 - 2f(2\pi^2\tilde{g}_0^{-2})) = \frac{1}{2g_0^2}.$$

But the left side, as a function of $\tilde{g}_0$, is continuous, tends to ∞ as $\tilde{g}_0 \rightarrow 0$, and tends to 0 as $\tilde{g}_0 \rightarrow \infty$. Thus, a solution to the above equation exists. $\square$

Next, let us show that the Villain action is gauge invariant.

Lemma 16.6.2. *Consider $U(1)$ lattice gauge theory on a finite set $\Lambda \subseteq \mathbb{Z}^d$ (for any $d \geq 2$) with Villain action and free boundary condition. Then, the Villain action is gauge invariant in the sense defined in Section 15.4.*

Proof. Consider a configuration $\theta \in [-\pi, \pi)^E$. This corresponds to the configuration $U \in U(1)^E$ given by $U(e) := e^{i\theta(e)}$. Let $g \in U(1)^\Lambda$ be a gauge transformation and let $V := U^g$. That is, for any $(x, y) \in E$, $V(x, y) = g(x)U(x, y)g(y)^{-1}$. Let $\xi \in [-\pi, \pi)^E$ be the configuration corresponding to V . That is, for any $e \in E$, $\xi(e)$ is the unique element of $[-\pi, \pi)$ such that $e^{i\xi(e)} = V(e)$. Also, let $\eta \in [-\pi, \pi)^\Lambda$ be the unique element satisfying $e^{i\eta(x)} = g(x)$ for each x . Then we have that for any $(x, y) \in E$,

$$\xi(x, y) \equiv \eta(x) + \theta(x, y) - \eta(y) \pmod{2\pi}.$$

This shows that for any plaquette p ,

$$\xi_p \equiv \theta_p \pmod{2\pi}.$$

Consequently,

$$\sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) = \sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\xi_p - 2\pi n)^2\right).$$

This completes the proof. $\square$

16.7 Deconfinement in 4D $U(1)$ theory

Our goal in this section is to prove that $U(1)$ lattice gauge theory in dimension four, with Villain action, undergoes a *deconfinement transition* at weak coupling. This result was established in the theoretical physics literature by Guth [13], and a complete rigorous proof was given by Fröhlich and Spencer [8]. The result is as follows.

Theorem 16.7.1. *There exists $\beta_0 > 0$ such that the following holds. Let $\Lambda \subseteq \mathbb{Z}^4$ be a hypercube. Consider $U(1)$ lattice gauge theory on Λ under free boundary condition and inverse coupling parameter $\beta \geq \beta_0$. Let ℓ be a rectangular loop contained in Λ , with side lengths R and T . Then there is some constant $C(\beta) > 0$ depending only on β (and not on Λ or ℓ), such that $\langle W_\ell \rangle \geq e^{-C(\beta)(R+T)}$.*

We present below the proof due to Fröhlich and Spencer [8]. The proof is divided into several subsections for readability.

16.7.1 Dual of 4D U(1) theory

It turns out that four-dimensional U(1) lattice gauge theory with the Villain action has a dual theory which can help with various calculations. To arrive at the dual theory, we begin by fixing some $\beta > 0$ and defining a function $\varphi_\beta : [-\pi, \pi) \rightarrow \mathbb{R}$ as

$$\varphi_\beta(x) := \sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(x - 2\pi n)^2\right).$$

Clearly, this is a well-defined function, and an easy application of the dominated convergence theorem shows that it is infinitely differentiable in $(-\pi, \pi)$. Let $\varphi_\beta^{(k)}$ denote the k^{th} derivative of φ_β , with $\varphi_\beta^{(0)} := \varphi_\beta$. Note that

$$\varphi_\beta^{(k)}(x) := \sum_{n \in \mathbb{Z}} p_{k,\beta}(x - 2\pi n) \exp\left(-\frac{\beta}{2}(x - 2\pi n)^2\right),$$

where $p_{k,\beta}$ is a polynomial depending on k and β . Again by a simple application of the dominated convergence theorem, this shows that

$$\begin{aligned} \lim_{x \uparrow \pi} \varphi_\beta^{(k)}(x) &= \sum_{n \in \mathbb{Z}} p_{k,\beta}(\pi - 2\pi n) \exp\left(-\frac{\beta}{2}(\pi - 2\pi n)^2\right) \\ &= \sum_{n \in \mathbb{Z}} p_{k,\beta}(\pi - 2\pi(n+1)) \exp\left(-\frac{\beta}{2}(\pi - 2\pi(n+1))^2\right) \\ &= \sum_{n \in \mathbb{Z}} p_{k,\beta}(-\pi - 2\pi n) \exp\left(-\frac{\beta}{2}(-\pi - 2\pi n)^2\right) \\ &= \lim_{x \downarrow -\pi} \varphi_\beta^{(k)}(x). \end{aligned}$$

This shows that $\varphi_\beta \in C^\infty(S^1)$, in the sense defined in Subsection A.3.4 of Appendix C. Thus, the Fourier coefficients³ $\widehat{\varphi}_\beta(n)$, $n \in \mathbb{Z}$ are well-defined and rapidly decaying⁴, and we have the pointwise identity⁵

$$\varphi_\beta(x) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \widehat{\varphi}_\beta(n) e^{-inx}. \quad (16.7.1)$$

It turns out that the Fourier coefficients can be explicitly calculated:

Lemma 16.7.2. *For each $n \in \mathbb{Z}$, we have*

$$\widehat{\varphi}_\beta(n) = \sqrt{\frac{2\pi}{\beta}} \exp\left(-\frac{n^2}{2\beta}\right).$$

³See Subsection A.3.4 in Appendix C for the definition of these Fourier coefficients.

⁴By Lemma A.3.9.

⁵By Lemma A.3.10.

Proof. By the dominated convergence theorem, we get

$$\begin{aligned}\widehat{\varphi}_\beta(n) &= \int_{-\pi}^{\pi} \varphi_\beta(x) e^{inx} dx \\ &= \sum_{m \in \mathbb{Z}} \int_{-\pi}^{\pi} \exp\left(-\frac{\beta}{2}(x - 2\pi m)^2 + inx\right) dx.\end{aligned}$$

Take any $m \in \mathbb{Z}$, and note that

$$\begin{aligned}\int_{-\pi}^{\pi} \exp\left(-\frac{\beta}{2}(x - 2\pi m)^2 + inx\right) dx &= \int_{-\pi}^{\pi} \exp\left(-\frac{\beta}{2}(x - 2\pi m)^2 + in(x - 2\pi m)\right) dx \\ &= \int_{-(2m+1)\pi}^{-(2m-1)\pi} \exp\left(-\frac{\beta}{2}x^2 + inx\right) dx.\end{aligned}$$

Summing over $m \in \mathbb{Z}$, we get

$$\widehat{\varphi}_\beta(n) = \int_{-\infty}^{\infty} \exp\left(-\frac{\beta}{2}x^2 + inx\right) dx = \sqrt{\frac{2\pi}{\beta}} \exp\left(-\frac{n^2}{2\beta}\right).$$

This completes the proof. $\square$

Let $\Lambda \subseteq \mathbb{Z}^4$ be a finite hypercube. Consider U(1) lattice gauge theory on Λ with Villain action at inverse coupling parameter β , under free boundary condition. Let E be the set of positively oriented edges with both endpoints in Λ and P be the set of plaquettes with all four vertices in Λ . Given any $\theta \in [-\pi, \pi)^E$, note that by equation (16.6.2),

$$\exp(-S_\beta(\theta)) = \prod_{p \in P} \left[\sum_{n \in \mathbb{Z}} \exp\left(-\frac{\beta}{2}(\theta_p - 2\pi n)^2\right) \right] = \prod_{p \in P} \varphi_\beta(\theta_p).$$

Thus, by equation (16.7.1), the normalizing constant for our theory is given by

$$\begin{aligned}Z &= \int_{[-\pi, \pi)^E} \exp(-S_\beta(\theta)) d\theta = \int_{[-\pi, \pi)^E} \prod_{p \in P} \varphi_\beta(\theta_p) d\theta \\ &= \int_{[-\pi, \pi)^E} \prod_{p \in P} \left(\frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \widehat{\varphi}_\beta(n) e^{-in\theta_p} \right) d\theta \\ &= \frac{1}{(2\pi)^{|P|}} \int_{[-\pi, \pi)^E} \sum_{n \in \mathbb{Z}^P} \left(\prod_{p \in P} \widehat{\varphi}_\beta(n_p) e^{-in_p \theta_p} \right) d\theta.\end{aligned}\tag{16.7.2}$$

As noted before, the Fourier coefficients of φ_β are rapidly decaying. Since P is finite, this allows us to apply the dominated convergence theorem and bring the sum outside the

integral, to get

$$\begin{aligned} Z &= \frac{1}{(2\pi)^{|P|}} \sum_{n \in \mathbb{Z}^P} \int_{[-\pi, \pi]^E} \left(\prod_{p \in P} \widehat{\varphi}_\beta(n_p) e^{-i n_p \theta_p} \right) d\theta \\ &= \frac{1}{(2\pi)^{|P|}} \sum_{n \in \mathbb{Z}^P} \left(\prod_{p \in P} \widehat{\varphi}_\beta(n_p) \right) \int_{[-\pi, \pi]^E} \exp\left(-i \sum_{p \in P} n_p \theta_p\right) d\theta. \end{aligned} \quad (16.7.3)$$

Now take any $\theta \in [-\pi, \pi]^E$. We can consider θ as a real-valued 1-form $\theta = \sum_{j=1}^4 \theta_j dx_j$ on $\mathbb{Z}^d$, defined as:

$$\theta_j(x) := \begin{cases} \theta(x, x + e_j) & \text{if } (x, x + e_j) \in E, \\ 0 & \text{otherwise.} \end{cases} \quad (16.7.4)$$

With this definition, note that if $p \in P$ has vertices $x, x + e_j, x + e_j + e_k, x + e_k$ for some $1 \leq j < k \leq 4$, and θ_p is defined as in equation (16.6.1), then by equation (C.8.2),

$$\begin{aligned} d\theta_{jk}(x) &= \partial_j \theta_k(x) - \partial_k \theta_j(x) \\ &= \theta_k(x + e_j) - \theta_k(x) - \theta_j(x + e_k) + \theta_j(x) \\ &= \theta(x + e_j, x + e_j + e_k) - \theta(x, x + e_k) - \theta(x + e_k, x + e_j + e_k) + \theta(x, x + e_j) \\ &= \theta_p. \end{aligned} \quad (16.7.5)$$

Next, take any $n \in \mathbb{Z}^P$. Define a real-valued 2-form n on $\mathbb{Z}^4$ as

$$n_{jk}(x) := \begin{cases} n_p & \text{if } p \in P, \\ 0 & \text{otherwise,} \end{cases}$$

where $1 \leq j < k \leq 4$, and p is the plaquette with vertices $x, x + e_j, x + e_j + e_k, x + e_k$. Then note that by equation (16.7.5) and the fact that the 2-form n vanishes on plaquettes outside P , we have

$$\sum_{p \in P} n_p \theta_p = \sum_{x \in \mathbb{Z}^4} \sum_{1 \leq j < k \leq 4} n_{jk}(x) d\theta_{jk}(x) = (n, d\theta),$$

where the inner product on the right is defined as in Subsection C.8.8 of Appendix C. But by Lemma C.8.5, we have the summation by parts identity $(n, d\theta) = (\delta n, \theta)$. Since θ vanishes on edges outside E , this gives

$$\sum_{p \in P} n_p \theta_p = \sum_{e \in E} \delta n(e) \theta(e),$$

where $\delta n(e) := \delta n_j(x)$ if e is the edge $(x, x + e_j)$. Plugging this into equation (16.7.2), we

get

$$\begin{aligned} Z &= \frac{1}{(2\pi)^{|P|}} \sum_{n \in \mathbb{Z}^P} \left(\prod_{p \in P} \widehat{\varphi}_\beta(n_p) \right) \int_{[-\pi, \pi]^E} \exp \left(-i \sum_{e \in E} \delta n(e) \theta(e) \right) d\theta \\ &= \frac{1}{(2\pi)^{|P|}} \sum_{n \in \mathbb{Z}^P} \left(\prod_{p \in P} \widehat{\varphi}_\beta(n_p) \right) \left(\prod_{e \in E} \int_{-\pi}^{\pi} e^{-i\delta n(e)\theta(e)} d\theta(e) \right). \end{aligned}$$

Now, note that for any $k \in \mathbb{Z}$,

$$\int_{-\pi}^{\pi} e^{-ikx} dx = \begin{cases} 2\pi & \text{if } k = 0, \\ 0 & \text{if } k \neq 0. \end{cases}$$

This shows that

$$\prod_{e \in E} \int_{-\pi}^{\pi} e^{-i\delta n(e)\theta(e)} d\theta(e) = \begin{cases} (2\pi)^{|P|} & \text{if } \delta n(e) = 0 \text{ for each } e \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Now, by equation (C.8.9),

$$\delta n_j(x) = - \sum_{i=1}^4 \bar{\partial}_i n_{ij}(x) = \sum_{i=1}^4 (n_{ij}(x - e_i) - n_{ij}(x)). \quad (16.7.6)$$

This shows that for an edge e in $\mathbb{Z}^d$, $\delta n(e)$ is a linear combination of $n(p)$ over a subset of plaquettes p that contain e . Since n vanishes on plaquettes that are not in P , this shows that for any $n \in \mathbb{Z}^P$, $\delta n(e) = 0$ for all $e \notin E$. Thus, if $\delta n(e) = 0$ for all $e \in E$, then the 1-form δn vanishes everywhere. This brings us to the identity

$$\begin{aligned} Z &= \sum_{n \in \mathbb{Z}^P, \delta n=0} \left(\prod_{p \in P} \widehat{\varphi}_\beta(n_p) \right) \\ &= (2\pi)^{|E|-|P|} \left(\frac{2\pi}{\beta} \right)^{\frac{1}{2}|P|} \sum_{n \in \mathbb{Z}^P, \delta n=0} \exp \left(-\frac{1}{2\beta} \sum_{p \in P} n_p^2 \right), \end{aligned} \quad (16.7.7)$$

where the second identity follows by Lemma 16.7.2.

Let $\mathcal{N}$ denote the set of all $n \in \mathbb{Z}^P$ such that the integer-valued 2-form n on $\mathbb{Z}^4$ satisfies $\delta n = 0$. Let $\mathcal{A}$ denote the set of all integer-valued 1-forms supported on the dual hypercube⁶ $*\Lambda$ such that the 2-form $*d\alpha$ is vanishes on plaquettes outside P . The set $\mathcal{A}$ has the following simpler characterization. Let Λ^* be the hypercube in $*\mathbb{Z}^4$ consisting of the centers of all 4-cells in $\mathbb{Z}^4$ which have at least one vertex in Λ . Explicitly, if

$$\Lambda = \times_{i=1}^4 \{a_i, a_i + 1, \dots, b_i - 1, b_i\},$$

⁶See Subsection C.8.6 in Appendix C.

then

$$\Lambda^* = \times_{i=1}^4 \left\{ a_i - \frac{1}{2}, a_i + \frac{1}{2}, \dots, b_i - \frac{1}{2}, b_i + \frac{1}{2} \right\}.$$

Note that $*\Lambda$ is a proper subset of Λ^* , by equation (C.8.10). Let E^* be the set of all negatively oriented edges in $*\mathbb{Z}^4$ with both endpoints in Λ^* . The boundary $\partial\Lambda^*$ of Λ^* is the set of all elements of Λ^* which have at least one neighbor outside Λ^* . The boundary ∂E^* of E^* is the set of elements of E^* whose endpoints are both in $\partial\Lambda^*$. Let $E_0^* := E^* \setminus \partial E^*$ and $\Lambda_0^* := \Lambda^* \setminus \partial\Lambda^*$.

Lemma 16.7.3. *The set $\mathcal{A}$ defined above is equal to the set of all $\mathbb{Z}$ -valued 1-forms on $*\mathbb{Z}^4$ that vanish outside E_0^* .*

Proof. Let $\mathcal{B}$ be the set of all $\mathbb{Z}$ -valued 1-forms on $*\mathbb{Z}^4$ that vanish outside E^* and also on ∂E^* . Take any $\alpha \in \mathcal{A}$. Since α is supported on $*\Lambda$, and each negatively oriented edge coming out of a vertex in $*\Lambda$ is a member of E^* , we see that α vanishes outside E^* . Next, take any $e \in \partial E^*$. Then both endpoints of e are boundary vertices of Λ^* . This implies that e is contained in a plaquette p whose remaining two vertices are outside Λ^* . Thus, the three other edges of p are outside E^* . This means that $d\alpha(p)$ cannot be zero unless $\alpha(e) = 0$. We will now show that $d\alpha(p) = 0$.

Let q be the dual plaquette of p . We claim that $q \notin P$. Indeed, if $q \in P$, then all four vertices of q are in Λ . This implies that all four vertices of p are in Λ^* , giving a contradiction to the observation (made above) that at least two vertices of p are outside Λ^* . Thus, $q \notin P$, and so, $d\alpha(p) = *\alpha(q) = 0$. This shows that $\mathcal{A} \subseteq \mathcal{B}$.

For the opposite inclusion, take any $\alpha \in \mathcal{B}$. If e is a negatively oriented edge in $*\mathbb{Z}^4$ coming out of a vertex $v \notin *\Lambda$, it is easy to see that e is either outside E^* , or in ∂E^* . This shows that α is supported on $*\Lambda$. Next, take any plaquette $p \notin P$. Let q be the dual plaquette of p . We have to show that $*d\alpha(p) = 0$, which is equivalent to showing that $d\alpha(q) = 0$. So, it suffices to show that all edges of p are either outside E^* or in ∂E^* .

Without loss, suppose that the four vertices of p are $x, x + e_1, x + e_2$, and $x + e_1 + e_2$. Since $p \in P$, at least one of these vertices is not in Λ . If x_1, x_2, x_3, x_4 denote the coordinates of x , and $\Lambda = ([a_1, b_1] \times [a_2, b_2] \times [a_3, b_3] \times [a_4, b_4]) \cap \mathbb{Z}^4$ for some integers $a_1, b_1, \dots, a_4, b_4$, this means that at least one of the following must be true: $x_1 \notin [a_1, b_1 - 1]$, or $x_2 \notin [a_2, b_2 - 1]$, or $x_3 \notin [a_3, b_3]$, or $x_4 \notin [a_4, b_4]$. Let $y := x + \frac{1}{2}(e_1 + e_2 + e_3 + e_4)$ be the dual vertex of x . Then the above condition implies that at least one of the following must be true: $y_1 \notin [a_1 + \frac{1}{2}, b_1 - \frac{1}{2}]$, or $y_2 \notin [a_2 + \frac{1}{2}, b_2 - \frac{1}{2}]$, or $y_3 \notin [a_3 + \frac{1}{2}, b_3 + \frac{1}{2}]$, or $y_4 \notin [a_4 + \frac{1}{2}, b_4 + \frac{1}{2}]$. The vertices of q are $y, y - e_3, y - e_4$ and $y - e_3 - e_4$. From the deduction about y in the previous sentence, it is now easy to verify that none of the vertices of q can be a non-boundary vertex of Λ^* . This completes the proof. $\square$

We will say that $\alpha, \alpha' \in \mathcal{A}$ are equivalent if $d(\alpha - \alpha') = 0$. We will denote the equivalence relation by $\sim$ and denote the quotient space $\mathcal{A}/\sim$ by $\mathcal{A}_0$. For $\alpha \in \mathcal{A}$, we will denote its equivalence class by $[\alpha]$. Note that by Lemma C.8.1 and the Poincaré lemma for the discrete exterior derivative (Lemma C.8.2), $\alpha \sim \alpha'$ if and only if there is an integer-valued 0-form γ

supported on $*\Lambda$ such that $\alpha' = \alpha + d\gamma$. The following lemma is the key to the construction of the dual model.

Lemma 16.7.4. *Let $\mathcal{N}$ and $\mathcal{A}_0$ be as above. Then the map $[\alpha] \mapsto *d\alpha$ is a bijection between $\mathcal{A}_0$ and $\mathcal{N}$.*

Proof. First, note that for any $\alpha \in \mathcal{A}$, $*d\alpha(p) = 0$ for $p \notin P$. Moreover, by Lemma C.8.3, equation (C.8.11), and Lemma C.8.1,

$$\delta*d\alpha = (-1)^{4(2+1)+1}*d**d\alpha = (-1)^{4(2+1)+1+2(4-2)}*dd\alpha = 0.$$

Thus, the map $[\alpha] \mapsto *d\alpha$ maps $\mathcal{A}_0$ into $\mathcal{N}$.

Next, take any $n \in \mathcal{N}$. Then by the Poincaré lemma for the discrete coderivative (Lemma C.8.4), there exists an integer-valued 3-form m supported on Λ (since Λ is a hypercube), such that $n = \delta m$. Let $\alpha := *(-m)$, so that α is an integer-valued 1-form supported on $*\Lambda$. By Lemma C.8.3 and equation (C.8.11),

$$n = \delta m = (-1)^{4(3+1)+1}*d*m(y) = *d\alpha.$$

Thus, we have arrived at the conclusion that for any $n \in \mathcal{N}$, there is an integer-valued 1-form α supported on $*\Lambda$ such that $n = *d\alpha$. In other words, the map $[\alpha] \mapsto *d\alpha$ is surjective from $\mathcal{A}_0$ onto $\mathcal{N}$.

Finally, suppose that α and α' are two integer-valued 1-forms supported on $*\Lambda$ such that $*d\alpha = *d\alpha'$. Then $*d(\alpha - \alpha') = 0$, and hence $d(\alpha - \alpha') = 0$. Thus, $[\alpha] = [\alpha']$. This shows that the map $[\alpha] \mapsto *d\alpha$ is injective. $\square$

Now take any $[\alpha] \in \mathcal{A}_0$ and let $n := *d\alpha$. Then for a plaquette p with vertices x , $x + e_j$, $x + e_j + e_k$, and $x + e_k$,

$$n_p = n_{jk}(x) = sd\alpha_{j'k'}(y),$$

where $\{j', k'\} = \{1, 2, 3, 4\} \setminus \{j, k\}$, $j' < k'$, s is the sign of the permutation (j, k, j', k') , and y is the dual vertex of x (i.e., the center of the 4-cell $dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4$). Since n_p vanishes for plaquettes outside P , this shows that

$$\begin{aligned} \sum_{p \in P} n_p^2 &= \sum_{x \in \mathbb{Z}^4} \sum_{1 \leq j < k \leq 4} n_{jk}(x)^2 \\ &= \sum_{y \in *\mathbb{Z}^4} \sum_{1 \leq j' < k' \leq 4} d\alpha_{j'k'}(y)^2 = \|d\alpha\|^2. \end{aligned}$$

Note that if $\alpha \sim \alpha'$, then $d\alpha = d\alpha'$. Thus, the right side is actually a function of the equivalence class $[\alpha]$. Combining these observations with Lemma 16.7.4 and equation (16.7.7), we arrive at the following result.

Lemma 16.7.5. *Let $\mathcal{A}_0$ be as above. Then*

$$Z = (2\pi)^{|E|-|P|} \left(\frac{2\pi}{\beta} \right)^{\frac{1}{2}|P|} \sum_{[\alpha] \in \mathcal{A}_0} \exp\left(-\frac{\|d\alpha\|^2}{2\beta}\right).$$

The dual model is defined to be the probability measure μ_0 on $\mathcal{A}_0$ that has probability mass function proportional to

$$\exp\left(-\frac{\|d\alpha\|^2}{2\beta}\right).$$

(Note that we do not have to specify a base measure, since $\mathcal{A}_0$ is a discrete set; by default, the counting measure is the base measure.)

16.7.2 Wilson loop expectations in the dual model

Let us now see how to express Wilson loop expectations as expected values of certain observables in the dual theory. For simplicity, we will only consider rectangular loops. Let ℓ be a rectangular Wilson loop with successive vertices $x_0, x_1, \dots, x_n \in \Lambda \setminus \partial\Lambda$ (where $\partial\Lambda$ is the boundary of Λ), such that $x_1 = x_0 + e_j$ and $x_n = x_0 + e_k$ for some $1 \leq j < k \leq 4$. Then for any $\theta \in [-\pi, \pi)^E$, the value of the Wilson loop observable W_ℓ for the configuration θ is given by

$$W_\ell(\theta) = \prod_{j=0}^n e^{i\theta(x_j, x_{j+1})},$$

where $x_{n+1} := x_0$. Let Σ denote the set of plaquettes enclosed by this rectangle. The following lemma is a special case of the discrete Stokes' theorem.

Lemma 16.7.6. *For any $\theta \in [-\pi, \pi)^E$,*

$$W_\ell(\theta) = \exp\left(-i \sum_{p \in \Sigma} \theta_p\right).$$

Proof. Given $\theta \in [-\pi, \pi)^E$, let θ also denote the real-valued 1-form on $\mathbb{Z}^4$ defined via equation (16.7.4). Then by equation (16.7.5), $\theta_p = d\theta_{jk}(x)$ for any plaquette p with vertices $x, x + e_j, x + e_j + e_k$, and $x + e_k$. Thus, if we let η be the real-valued 2-form given by $\eta(p) = 1$ if $p \in \Sigma$ and 0 otherwise, then by Lemma C.8.5,

$$\sum_{p \in \Sigma} \theta_p = (d\theta, \eta) = (\theta, \delta\eta). \quad (16.7.8)$$

Recalling equation (16.7.6), we see that for $\delta\eta(e)$ to be nonzero, e has to belong to at least one plaquette in Σ . If $e = (x, x + e_i)$ is such an edge, then i is either j or k . Suppose $i = j$. Then by equation (16.7.6),

$$\delta\eta(e) = \delta\eta_j(x) = \sum_{l=1}^4 (\eta_l(x - e_l) - \eta_l(x)) = \eta_{jk}(x) - \eta_{jk}(x - e_k) = \eta(p) - \eta(q),$$

where p is the plaquette vertices x , $x + e_j$, $x + e_j + e_k$, and $x + e_k$, and q is p shifted by $-e_k$. Thus, the right side above is 1 if e is one of the positively oriented edges on the left boundary of ℓ , and -1 if e is one of the positively oriented edges on the right boundary of ℓ . (Here we are imagining the j direction as the horizontal direction and the k direction as the vertical direction.) If e is not on the boundary, the above formula shows that $\delta\eta(e) = 0$.

A similar argument shows that if $i = k$, then $\delta\eta(e) = 1$ if e is a positively oriented edge on the upper boundary, $\delta\eta(e) = -1$ if e is a positively oriented edge on the lower boundary, and $\delta\eta(e) = 0$ if e is not on the boundary. These observations, together with equation (16.7.8), show that the sum of θ_p over all $p \in \Sigma$ is equal to the sum of $\theta(e)$ over all positively oriented edges on the left and upper boundaries of ℓ , minus the sum of $\theta(e)$ over all positively oriented edges on the right and bottom boundaries. This completes the proof. $\square$

To calculate $\langle W_\ell \rangle$, we have to now express

$$Z_\ell := \int_{[-\pi, \pi]^E} W_\ell(\theta) \exp(-S_\beta(\theta)) d\theta$$

in terms of the dual model. Proceeding exactly as in the proof of equation (16.7.3) and using Lemma 16.7.6, we arrive at the identity

$$Z_\ell = \frac{1}{(2\pi)^{|P|}} \sum_{n \in \mathbb{Z}^P} \left(\prod_{p \in P} \widehat{\varphi}_\beta(n_p) \right) \int_{[-\pi, \pi]^E} \exp\left(-i \sum_{p \in P} n_p \theta_p - i \sum_{p \in \Sigma} \theta_p\right) d\theta.$$

Let us now make the change of index in the above summation by replacing n with m , defined as

$$m_p := \begin{cases} n_p + 1 & \text{if } p \in \Sigma, \\ n_p & \text{if } p \in P \setminus \Sigma. \end{cases}$$

Since the sum remains unchanged, we get

$$Z_\ell = \frac{1}{(2\pi)^{|P|}} \sum_{m \in \mathbb{Z}^P} \left(\prod_{p \in P \setminus \Sigma} \widehat{\varphi}_\beta(m_p) \prod_{p \in \Sigma} \widehat{\varphi}_\beta(m_p - 1) \right) \int_{[-\pi, \pi]^E} \exp\left(-i \sum_{p \in P} m_p \theta_p\right) d\theta.$$

Now proceeding as in the proof of equation (16.7.7), we get

$$\begin{aligned} Z_\ell &= (2\pi)^{|E|-|P|} \sum_{m \in \mathbb{Z}^P, \delta m=0} \left(\prod_{p \in P \setminus \Sigma} \widehat{\varphi}_\beta(m_p) \prod_{p \in \Sigma} \widehat{\varphi}_\beta(m_p - 1) \right) \\ &= (2\pi)^{|E|-|P|} \left(\frac{2\pi}{\beta} \right)^{\frac{1}{2}|P|} \sum_{m \in \mathbb{Z}^P, \delta m=0} \exp\left(-\frac{1}{2\beta} \sum_{p \in P \setminus \Sigma} m_p^2 - \frac{1}{2\beta} \sum_{p \in \Sigma} (m_p - 1)^2\right) \\ &= (2\pi)^{|E|-|P|} \left(\frac{2\pi}{\beta} \right)^{\frac{1}{2}|P|} \sum_{m \in \mathbb{Z}^P, \delta m=0} H_\ell(m) \exp\left(-\frac{1}{2\beta} \sum_{p \in P} m_p^2\right), \end{aligned}$$

where

$$H_\ell(m) := \exp\left(\frac{1}{\beta} \sum_{p \in \Sigma} m_p - \frac{|\Sigma|}{2\beta}\right).$$

This yields the following analogue of Lemma 16.7.5.

Lemma 16.7.7. *Let $\mathcal{A}_0$ be as before, and H_ℓ be defined as above. Define a map $D_\ell : \mathcal{A}_0 \rightarrow [0, \infty)$ as $D_\ell(\alpha) := H_\ell(*d\alpha)$, noting that D_ℓ is actually a function of $[\alpha]$. Then*

$$Z_\ell = (2\pi)^{|E|-|P|} \left(\frac{2\pi}{\beta}\right)^{\frac{1}{2}|P|} \sum_{[\alpha] \in \mathcal{A}_0} D_\ell(\alpha) \exp\left(-\frac{1}{2\beta} \|d\alpha\|^2\right).$$

Combining Lemma 16.7.5 and Lemma 16.7.7, we get the following theorem.

Theorem 16.7.8. *Let $\langle \cdot \rangle^*$ denote the expectation with respect to the dual model. Then we have $\langle W_\ell \rangle = \langle D_\ell \rangle^*$.*

16.7.3 The Gaussian $U(1)$ model

We will approximate the dual model from the previous section by a Gaussian model. Let $\epsilon > 0$ be a parameter that will later be taken to zero. Let $\mathcal{U}$ denote the finite-dimensional vector space of all real-valued 1-forms on $*\mathbb{Z}^4$ which are supported on $*\Lambda$ and satisfy the condition that $*d\alpha = 0$ on plaquettes outside P . By the same argument as in Lemma 16.7.3, we see that $\mathcal{U}$ is simply the set of real-valued 1-forms on $*\mathbb{Z}^4$ that vanish outside E^* and on ∂E^* . In other words, we can think of $\mathcal{U}$ as $\mathbb{R}^{E_0^*}$. Note that $\mathcal{U}$ is simply the range of the orthogonal projection operator Π_{Λ^*} defined in Subsection C.8.10 of Appendix C, acting on real-valued 1-forms.

We define γ_ϵ to be the centered Gaussian measure on $\mathcal{U}$ with probability density function proportional to

$$\exp\left(-\frac{1}{2\beta} (\|d\alpha\|^2 + \epsilon \|\alpha\|^2)\right).$$

We will now compute the covariance matrix of this Gaussian density.

Lemma 16.7.9. *The restriction of the operator $\Pi_{\Lambda^*}(\delta d + \epsilon)$ to $\mathcal{U}$ maps $\mathcal{U}$ into $\mathcal{U}$, is self-adjoint, positive semidefinite, and invertible. The covariance matrix V_ϵ of γ_ϵ , viewed as a linear map from $\mathcal{U}$ into $\mathcal{U}$, is equal to β times the inverse of this map.*

Proof. It is clear that $\Pi_{\Lambda^*}(\delta d + \epsilon)$ maps into $\mathcal{U}$, since Π_{Λ^*} maps into $\mathcal{U}$. Let S denote the restriction of this map to $\mathcal{U}$. We claim that for any $\alpha, \alpha' \in \mathcal{U}$,

$$(\alpha, S\alpha') = (d\alpha, d\alpha') + \epsilon(\alpha, \alpha'). \quad (16.7.9)$$

Indeed, note that since $\alpha \in \mathcal{U}$ and Π_{Λ^*} is the orthogonal projection on to $\mathcal{U}$, we have

$$(\alpha, S\alpha') = (\alpha, \Pi_{\Lambda^*}(\delta d + \epsilon)\alpha') = (\alpha, (\delta d + \epsilon)\alpha').$$

Applying summation by parts (Lemma C.8.5) to the right side yields equation (16.7.9). Equation (16.7.9) shows that S is self-adjoint and positive semidefinite. Moreover, it shows that if $S\alpha = 0$, then $(\alpha, S\alpha) = 0$ and hence, $(\alpha, \alpha) = 0$. Thus, $\ker S = \{0\}$. Since $\mathcal{U}$ is finite-dimensional, we conclude that S is invertible. The proof is completed by noticing that by equation (16.7.9), the density of γ_ϵ at $\alpha \in \mathcal{U}$ is proportional to $\exp(-\frac{1}{2\beta}(\alpha, S\alpha))$, and that $\mathcal{U}$ is simply the Euclidean space $\mathbb{R}^{E^* \setminus \partial E^*}$. $\square$

Let $\mathcal{V}_0$ be the subspace of $\mathcal{U}$ consisting of all $\alpha \in \mathcal{U}$ such that $d\alpha = 0$. Let $\mathcal{U}_0$ denote the orthogonal complement of $\mathcal{V}_0$ in $\mathcal{U}$. The following lemma gives the explicit descriptions of these two subspaces.

Lemma 16.7.10. *The subspace $\mathcal{V}_0$ is the space of all $d\theta$ where θ is a real-valued 0-form supported on Λ_0^* . The subspace $\mathcal{U}_0$ is the space of all real valued 1-forms α such that $E(\alpha) \subseteq E_0^*$ and $\delta\alpha(x) = 0$ for all $x \in \Lambda_0^*$. Alternatively, $\mathcal{U}_0$ is the set of all real-valued 1-forms α such that $\alpha = \Pi_{\Lambda^*}\delta\tau$ for some real-valued 2-form τ .*

Proof. Take any $\alpha \in \mathcal{V}_0$. Since $d\alpha = 0$ and α is supported on Λ^* , there is some 0-form θ supported on Λ^* such that $\alpha = d\theta$. Since α vanishes on ∂E^* and outside E^* , we deduce that θ must be zero on $\partial\Lambda^*$. Thus, θ is supported on Λ_0^* . Conversely, if θ is a 0-form supported on Λ_0^* , then $dd\theta = 0$, and it is easy to see that $E(d\theta) \subseteq E_0^*$. Thus, $d\theta \in \mathcal{V}_0$. This proves the first claim of the lemma.

For the second claim, take any $\alpha \in \mathcal{U}_0$. By definition, this means that $E(\alpha) \subseteq E_0^*$ and $(\alpha, \eta) = 0$ for all $\eta \in \mathcal{V}_0$. By the first part, this is equivalent to saying that $E(\alpha) \subseteq E_0^*$ and $(\alpha, d\theta) = 0$ for any 0-form θ supported on Λ_0^* . But $(\alpha, d\theta) = (\delta\alpha, \theta)$. Thus, clearly, the condition that $(\alpha, d\theta) = 0$ for any 0-form θ supported on Λ_0^* is equivalent to the condition that $\delta\alpha(x) = 0$ for all $x \in \Lambda_0^*$.

Finally, let us prove the alternative characterization of $\mathcal{U}_0$. If $\alpha = \Pi_{\Lambda^*}\delta\tau$ for some 2-form τ , then for any $\eta \in \mathcal{V}_0$,

$$(\eta, \alpha) = (\eta, \Pi_{\Lambda^*}\delta\tau) = (\eta, \delta\tau) = (d\eta, \tau) = 0.$$

Conversely, take any $\alpha \in \mathcal{U}_0$. For each $x \in \partial\Lambda^*$ that is in exactly one of the faces of $\partial\Lambda^*$, there is exactly one directed edge $e = (x, y)$, with $y \in \Lambda_0^*$, that is incident to x . Define $\eta(x) := \alpha(e)$. Let η be zero outside this set of x 's. We claim that

$$\sum_{x \in \mathbb{Z}^4} \eta(x) = 0. \tag{16.7.10}$$

To see this, note that if h denotes the 0-form that is 1 on Λ_0^* and 0 elsewhere. Then it is easy to see that

$$\sum_{x \in \mathbb{Z}^4} \eta(x) = (\alpha, dh).$$

Since $(\alpha, dh) = (\delta\alpha, h)$, h is supported on Λ_0^* , and $\delta\alpha$ vanishes on Λ_0^* , this proves the claim (16.7.10).

Now, by equation (16.7.10) and the Poincaré lemma for 0-forms (Lemma C.8.6), there is a 1-form κ supported on boundary edges of Λ^* , such that $\eta = \delta\kappa$. Let $\xi := \alpha - \kappa$. Then, for $x \in \Lambda_0^*$,

$$\delta\xi(x) = \delta\alpha(x) - \delta\kappa(x) = 0,$$

since $\delta\alpha(x) = 0$ and $\delta\kappa(x) = \eta(x) = 0$. Similarly, $\delta\xi(x) = 0$ for $x \notin \Lambda^*$. Finally, for $x \in \partial\Lambda^*$, the definition of η shows that $\eta(x) = \delta\alpha(x)$, and therefore, $\delta\xi(x) = 0$. Thus, $\delta\xi = 0$ everywhere. Consequently, there is a 2-form τ such that $\xi = \delta\tau$. We claim that $\alpha = \Pi_{\Lambda^*}\delta\tau$. Indeed, for $e \in E_0^*$, $\delta\tau(e) = \xi(e) = \alpha(e)$, since $\kappa(e) = 0$. This completes the proof. $\square$

The next lemma shows that a certain operator, which will turn out to be the limit (in a certain sense) of V_ϵ^{-1} as $\epsilon \rightarrow 0$, is invertible on $\mathcal{U}_0$.

Lemma 16.7.11. *The operator $\Pi_{\Lambda^*}\delta d$ maps $\mathcal{U}$ into $\mathcal{U}_0$, and its restriction to $\mathcal{U}_0$ is self-adjoint, positive semidefinite, and invertible.*

Proof. First, we claim that $\Pi_{\Lambda^*}\delta d$ maps $\mathcal{U}$ into $\mathcal{U}_0$. To see this, take any $\alpha \in \mathcal{U}$ and $\eta \in \mathcal{V}_0$. Then $(\eta, \Pi_{\Lambda^*}\delta d\alpha) = (\eta, \delta d\alpha)$, since $\eta \in \mathcal{U}$. By Lemma C.8.5, this is equal to $(d\eta, d\alpha)$, which is zero since $d\eta = 0$. Thus, $\Pi_{\Lambda^*}\delta d\alpha \in \mathcal{V}_0^\perp = \mathcal{U}_0$.

Next, let T denote the restriction of $\Pi_{\Lambda^*}\delta d$ to $\mathcal{U}_0$. We need to show that T is invertible. Since $\mathcal{U}_0$ is finite-dimensional, this is equivalent to showing that $\ker T = \{0\}$. Take any $\alpha \in \ker T$. Then $(\alpha, T\alpha) = 0$. But, since $\alpha \in \mathcal{U}$,

$$(\alpha, T\alpha) = (\alpha, \delta d\alpha) = (d\alpha, d\alpha).$$

Thus, $d\alpha = 0$, which means that $\alpha \in \mathcal{V}_0$. But α is also in $\mathcal{U}_0 = \mathcal{V}_0^\perp$. Therefore, $\alpha = 0$. This shows that T is invertible. Moreover, the identity $(\alpha, T\alpha') = (d\alpha, d\alpha')$ shows that T is self-adjoint and positive semidefinite. $\square$

Lemma 16.7.11 yields the following corollary.

Corollary 16.7.12. *For any $\epsilon > 0$, the operator $\Pi_{\Lambda^*}(\delta d + \epsilon)$ maps $\mathcal{U}_0$ into $\mathcal{U}_0$, and its restriction to $\mathcal{U}_0$ is invertible. Consequently, the same assertions are also true for V_ϵ .*

Proof. Take any $\alpha \in \mathcal{U}_0$. By Lemma 16.7.11, $\Pi_{\Lambda^*}\delta d\alpha \in \mathcal{U}_0$. Since $\mathcal{U}_0$ is a subspace of $\mathcal{U}$, this implies that

$$\Pi_{\Lambda^*}(\delta d + \epsilon)\alpha = \Pi_{\Lambda^*}\delta d\alpha + \epsilon\Pi_{\Lambda^*}\alpha = \Pi_{\Lambda^*}\delta d\alpha + \epsilon\alpha \in \mathcal{U}_0.$$

Thus, $\Pi_{\Lambda^*}(\delta d + \epsilon)$ maps $\mathcal{U}_0$ into $\mathcal{U}_0$. Since this map is injective on $\mathcal{U}$, it is also injective on $\mathcal{U}_0$. Since $\mathcal{U}_0$ is finite-dimensional, we deduce that this map is invertible on $\mathcal{U}_0$. Clearly, its inverse is the restriction of V_ϵ to $\mathcal{U}_0$, and therefore, the same conclusions hold for V_ϵ too. $\square$

Let $\|\cdot\|$ denote the norm induced by the inner product $(\cdot, \cdot)$ on $\mathcal{U}$. We need the following lemma.

Lemma 16.7.13. *There is a positive constant c such that $\|d\alpha\| \geq c\|\alpha\|$ for all $\alpha \in \mathcal{U}_0$.*

Proof. By linearity of the exterior derivative operator, this is equivalent to showing that there is some $c > 0$ such that $\|d\alpha\| \geq c$ for all $\alpha \in \mathcal{U}_0$ with $\|\alpha\| = 1$. Suppose that this is not true. Then by a compactness argument, we deduce that there is some $\alpha \in \mathcal{U}_0$ with $\|\alpha\| = 1$ and $d\alpha = 0$. But this means $\alpha \in \mathcal{U}_0 \cap \mathcal{U}_0^\perp = \{0\}$, giving a contradiction. $\square$

Lemma 16.7.13 and Corollary 16.7.12 yield the following corollary.

Corollary 16.7.14. *There is some $C > 0$ such that for any $\epsilon > 0$ and $\alpha \in \mathcal{U}_0$, $\|V_\epsilon \alpha\| \leq C\|\alpha\|$.*

Proof. Take any $\alpha \in \mathcal{U}_0$ and $\epsilon > 0$. Let c be as in Lemma 16.7.13. Let $\eta := V_\epsilon \alpha$. By Corollary 16.7.12, $\eta \in \mathcal{U}_0$. Thus, by equation (16.7.9) and Lemma 16.7.13,

$$\begin{aligned} (\alpha, V_\epsilon \alpha) &= (\eta, V_\epsilon^{-1} \eta) \\ &= \|d\eta\|^2 + \epsilon \|\eta\|^2 \geq \|d\eta\|^2 \geq c^2 \|\eta\|^2. \end{aligned}$$

By the Cauchy–Schwarz inequality, this gives

$$c^2 \|\eta\|^2 \leq \|\alpha\| \|\eta\|,$$

which implies that $\|V_\epsilon \alpha\| = \|\eta\| \leq c^{-2} \|\alpha\|$. $\square$

Let V denote the inverse of the restriction of $\Pi_\Lambda^* \delta d$ to $\mathcal{U}_0$. By Lemma 16.7.11, this is a positive definite operator. Let γ_0 denote the centered Gaussian measure on $\mathcal{U}_0$ with covariance V . The next lemma identifies the behavior of a random 1-form drawn from the probability measure γ_ϵ and its behavior as $\epsilon \rightarrow 0$.

Lemma 16.7.15. *Let η_ϵ be a random 1-form with law γ_ϵ . Let $\eta_{0,\epsilon}$ and $\eta_{1,\epsilon}$ be its orthogonal projections on to $\mathcal{U}_0$ and $\mathcal{V}_0$. Then $\eta_{0,\epsilon}$ and $\eta_{1,\epsilon}$ are independent centered Gaussian 1-forms on $\mathcal{U}_0$ and $\mathcal{V}_0$, with covariance matrices βV_ϵ and $\beta \epsilon^{-1} I$, respectively. Moreover, as $\epsilon \rightarrow 0$, the law of $\eta_{0,\epsilon}$ converges weakly to γ_0 .*

Proof. Since η_ϵ is a centered Gaussian 1-form and $\eta_{0,\epsilon}, \eta_{1,\epsilon}$ are obtained by applying linear transformations to η_ϵ , they are jointly centered Gaussian. Let Π_0 denote the orthogonal projection from $\mathcal{U}$ on to $\mathcal{U}_0$, so that $\eta_{0,\epsilon} = \Pi_0 \eta_\epsilon$ and $\eta_{1,\epsilon} = (I - \Pi_0) \eta_\epsilon$. Take any $\alpha, \alpha' \in \mathcal{U}$. Let $\alpha_0 := \Pi_0 \alpha$, $\alpha_1 := (I - \Pi_0) \alpha$, $\alpha'_0 := \Pi_0 \alpha'$ and $\alpha'_1 := (I - \Pi_0) \alpha'$. Then note that

$$\begin{aligned} \mathbb{E}[(\langle \alpha, \eta_{0,\epsilon} \rangle + \langle \alpha', \eta_{1,\epsilon} \rangle)^2] &= \mathbb{E}[(\langle \alpha, \Pi_0 \eta_\epsilon \rangle + \langle \alpha', (I - \Pi_0) \eta_\epsilon \rangle)^2] \\ &= \mathbb{E}[(\langle \Pi_0 \alpha, \eta_\epsilon \rangle + \langle (I - \Pi_0) \alpha', \eta_\epsilon \rangle)^2] \\ &= \mathbb{E}[(\langle \alpha_0 + \alpha'_1, \eta_\epsilon \rangle)^2] = \beta(\alpha_0 + \alpha'_1, V_\epsilon(\alpha_0 + \alpha'_1)). \end{aligned}$$

But since $\alpha'_1 \in \mathcal{V}_0$, we have $d\alpha'_1 = 0$. Thus,

$$\Pi_{\Lambda^*}(\delta d + \epsilon)\alpha'_1 = \epsilon\alpha'_1.$$

This implies that

$$V_\epsilon\alpha'_1 = (\Pi_{\Lambda^*}(\delta d + \epsilon))^{-1}\alpha'_1 = \epsilon^{-1}\alpha'_1.$$

Since V_ϵ is self-adjoint, this gives

$$\begin{aligned} \frac{1}{\beta}\mathbb{E}[(\langle\alpha, \eta_{0,\epsilon}\rangle + \langle\alpha', \eta_{1,\epsilon}\rangle)^2] &= (\alpha_0, V_\epsilon\alpha_0) + (\alpha_0, V_\epsilon\alpha'_1) + (\alpha'_1, V_\epsilon\alpha_0) + (\alpha'_1, V_\epsilon\alpha'_1) \\ &= (\alpha_0, V_\epsilon\alpha_0) + 2(\alpha_0, V_\epsilon\alpha'_1) + (\alpha'_1, V_\epsilon\alpha'_1) \\ &= (\alpha_0, V_\epsilon\alpha_0) + \epsilon^{-1}\|\alpha'_1\|^2. \end{aligned} \tag{16.7.11}$$

This proves the first assertion of the lemma. Next, for each $\epsilon > 0$, let $\xi_\epsilon := V_\epsilon\alpha_0$, so that

$$\alpha_0 = V_\epsilon^{-1}\xi_\epsilon = \Pi_{\Lambda^*}(\delta d + \epsilon)\xi_\epsilon.$$

By Corollary 16.7.14, there is some constant C such that $\|\xi_\epsilon\| \leq C\|\alpha_0\|$ for any $\epsilon > 0$. Thus, ξ_ϵ has at least one subsequential limit as $\epsilon \rightarrow 0$. Let ξ be any such limit. Then by the above equation, $\alpha_0 = \Pi_{\Lambda^*}\delta d\xi$. But this means that $\xi = V\alpha_0$. Thus, $\xi_\epsilon \rightarrow \xi$ as $\epsilon \rightarrow 0$. Consequently, $(\alpha_0, V_\epsilon\alpha_0) = (\alpha_0, \xi_\epsilon) \rightarrow (\alpha_0, \xi) = (\alpha_0, V\alpha_0)$ as $\epsilon \rightarrow 0$. By equation (16.7.11), this completes the proof. $\square$

16.7.4 Relating the discrete and continuous models

Recall the countable sets $\mathcal{A}$ and $\mathcal{A}_0$, and the probability measure μ_0 on $\mathcal{A}_0$ defined in Subsection 16.7.1: $\mathcal{A}$ is the set of all $\mathbb{Z}$ -valued 1-forms α on $*\mathbb{Z}^4$ such that α vanishes outside E^* and $*d\alpha$ vanishes outside P , $\mathcal{A}_0 = \mathcal{A}/\sim$ where $\sim$ is the equivalence relation $\alpha \sim \alpha'$ if $d(\alpha - \alpha') = 0$, and μ_0 is the probability measure on $\mathcal{A}_0$ with probability mass function proportional to $\exp(-\frac{1}{2\beta}\|d\alpha\|^2)$.

We will now relate μ_0 with the Gaussian measure γ_0 defined in the previous subsection. The first step is to show that μ_0 can be obtained as the limit of a sequence of probability measures on $\mathcal{A}$, analogous to γ_ϵ from the previous subsection. For each $\epsilon > 0$, let μ_ϵ be the probability measure on $\mathcal{A}_0$ with probability mass function proportional to

$$\exp\left(-\frac{1}{2\beta}(\|d\alpha\|^2 + \epsilon\|\alpha\|^2)\right).$$

Let $\langle \cdot \rangle_\epsilon$ denote expectation with respect to μ_ϵ .

Instead of viewing $\mathcal{A}_0$ as the set of equivalence classes, we will henceforth take $\mathcal{A}_0$ to be a subset of $\mathcal{A}$ comprised of one element from each equivalence class. Specifically, we will take an element with minimum norm from each class. This makes sense, because the squared norms of the elements are nonnegative integers. With this convention, μ_0 can be viewed as a probability measure on $\mathcal{A}$ which is supported on $\mathcal{A}_0$.

Lemma 16.7.16. *There is a positive constant c_0 , possibly depending on Λ , such that $\|d\alpha\| \geq c_0\|\alpha\|$ for all $\alpha \in \mathcal{A}_0$.*

Proof. Take any $\alpha \in \mathcal{A}_0$. Let α_0 be the orthogonal projection of α on the subspace $\mathcal{U}_0$ defined in the previous section, and let α_1 be its projection on the orthogonal subspace $\mathcal{V}_0$. Since $d\alpha_1 = 0$ and α_1 is supported on $*\Lambda$, there is a 0-form κ supported on $*\Lambda$ such that $\alpha_1 = d\kappa$. Construct an ‘integer approximation’ η of κ by defining $\eta(x) := \lfloor \kappa(x) \rfloor$ for each $x \in *\mathbb{Z}^4$. Let $\theta := \alpha - d\eta$. Then $\theta \in \mathcal{A}$ and $\theta \sim \alpha$. Thus, $\|\alpha\| \leq \|\theta\|$. Now,

$$\|\theta - \alpha_0\| = \|\alpha_1 - d\theta\| = \|d\kappa - d\theta\|.$$

But from the above construction, it is clear that $\|d\kappa - d\theta\| \leq C$ for some constant C that depends only on the size of Λ . Combining this with Lemma 16.7.13, we see that

$$\|\theta\| \leq \|\alpha_0\| + C \leq c^{-1}\|d\alpha_0\| + C.$$

But $d\alpha_0 = d\alpha$. This gives $\|\alpha\| \leq \|\theta\| \leq c^{-1}\|d\alpha\| + C$. But $\|\alpha\|^2$ and $\|d\alpha\|^2$ are integer-valued, and $\|\alpha\| = 0$ when $\|d\alpha\| = 0$, because $d\alpha = 0$ implies that $\alpha \sim 0$. This completes the proof. $\square$

We will say that a function $f : \mathcal{A} \rightarrow \mathbb{C}$ has sub-exponential growth if there are constants C_1 and C_2 such that $|f(\alpha)| \leq C_1 e^{C_2\|\alpha\|}$ for all $\alpha \in \mathcal{A}_0$. We are now ready to show that μ_ϵ converges to μ_0 as ϵ tends to zero.

Lemma 16.7.17. *Let $f : \mathcal{A} \rightarrow \mathbb{C}$ be a function of sub-exponential growth which is constant on equivalence classes. Then*

$$\langle f \rangle^* = \lim_{\epsilon \rightarrow 0} \langle f \rangle_\epsilon^*.$$

Proof. In this proof, $C, C_0, C_1, \dots$ will denote positive constants that may depend only on β and the size of Λ , whose values may change from line to line. For $\alpha \in \mathcal{A}_0$, the equivalence class of α is given by

$$[\alpha] = \{\alpha + \eta : \eta \in \mathcal{A}, d\eta = 0\} = \alpha + \mathcal{B},$$

where $\mathcal{B} := \{\eta \in \mathcal{A} : d\eta = 0\}$. Note that for any $\eta \in \mathcal{B}$ and $\alpha' = \alpha + \eta$,

$$\|d\alpha'\|^2 + \epsilon\|\alpha'\|^2 = \|d\alpha\|^2 + \epsilon\|\alpha\|^2 + \epsilon\|\eta\|^2 + 2\epsilon(\alpha, \eta).$$

Since f is constant on equivalence classes, this shows that

$$\begin{aligned} & \sum_{\alpha \in \mathcal{A}} f(\alpha) \exp\left(-\frac{1}{2\beta}(\|d\alpha\|^2 + \epsilon\|\alpha\|^2)\right) \\ &= \sum_{\alpha \in \mathcal{A}_0} \sum_{\eta \in \mathcal{B}} f(\alpha) \exp\left(-\frac{1}{2\beta}(\|d\alpha\|^2 + \epsilon\|\alpha\|^2 + \epsilon\|\eta\|^2 + 2\epsilon(\alpha, \eta))\right). \end{aligned} \quad (16.7.12)$$

Recall that $\Lambda_0^* := \Lambda^* \setminus \partial\Lambda^*$ denotes the set of interior vertices of Λ^* . For $\alpha \in \mathcal{A}_0$, let

$$g_\epsilon(\alpha) := \epsilon^{\frac{1}{2}|\Lambda_0^*|} \sum_{\eta \in \mathcal{B}} \exp\left(-\frac{\epsilon}{2\beta}(\|\alpha\|^2 + \|\eta\|^2 + 2(\alpha, \eta))\right).$$

Equation (16.7.12) shows that

$$\langle f \rangle_\epsilon^* = \frac{\langle f g_\epsilon \rangle^*}{\langle g_\epsilon \rangle^*}.$$

We claim that $\eta \in \mathcal{B}$ if and only if $\eta = d\theta$ for some $\mathbb{Z}$ -valued 0-form θ supported on Λ_0^* . To see this, first take any θ as above. Then it is easy to see that $d\theta$ is a 1-form that is zero outside E_0^* . Also, by Lemma C.8.1, $dd\theta = 0$. Conversely, take any $\eta \in \mathcal{B}$. Since $d\eta = 0$ and η is supported in Λ_0^* , there is a $\mathbb{Z}$ -valued 0-form supported on Λ_0^* such that $\eta = d\theta$. By the Poincaré lemma for 0-forms, θ is unique. This shows that for any $\alpha \in \mathcal{A}_0$,

$$\begin{aligned} g_\epsilon(\alpha) &= \epsilon^{\frac{1}{2}|\Lambda_0^*|} \sum_{\theta \in \mathbb{Z}^{\Lambda_0^*}} \exp\left(-\frac{\epsilon}{2\beta}(\|\alpha\|^2 + \|d\theta\|^2 + 2(\alpha, d\theta))\right) \\ &= \epsilon^{\frac{1}{2}|\Lambda_0^*|} \sum_{\theta \in \sqrt{\epsilon}\mathbb{Z}^{\Lambda_0^*}} \exp\left(-\frac{1}{2\beta}(\epsilon\|\alpha\|^2 + \|d\theta\|^2 + 2\sqrt{\epsilon}(\alpha, d\theta))\right). \end{aligned} \quad (16.7.13)$$

For each $\theta \in \sqrt{\epsilon}\mathbb{Z}^{\Lambda_0^*}$, let B_θ denote the open cube of side-length $\sqrt{\epsilon}$ and center θ in $\mathbb{R}^{\Lambda_0^*}$. Define a function $h_\epsilon : \mathcal{A}_0 \times \mathbb{R}^{\Lambda_0^*} \rightarrow [0, \infty)$ as follows. For each $\alpha \in \mathcal{A}_0$, $\theta_0 \in \sqrt{\epsilon}\mathbb{Z}^{\Lambda_0^*}$, and $\theta \in B_{\theta_0}$, let

$$h_\epsilon(\alpha, \theta) := \exp\left(-\frac{1}{2\beta}(\epsilon\|\alpha\|^2 + \|d\theta_0\|^2 + 2\sqrt{\epsilon}(\alpha, d\theta_0))\right).$$

We define $h_\epsilon = 0$ on the boundaries of the cubes. Then equation (16.7.13) can be rewritten as

$$g_\epsilon(\alpha) = \int_{\mathbb{R}^{\Lambda_0^*}} h_\epsilon(\alpha, \theta) d\theta.$$

Thus,

$$\langle f g_\epsilon \rangle^* = \int_{\mathcal{A}_0} \int_{\mathbb{R}^{\Lambda_0^*}} f(\alpha) h_\epsilon(\alpha, \theta) d\theta d\mu_0(\alpha).$$

By the Poincaré lemma (Lemma C.8.2), $d\theta = 0$ for a real-valued 0-form θ if and only if $\theta = 0$. By linearity and compactness, this implies the existence of a positive constant C_0 such that $\|d\theta\| \geq C_0\|\theta\|$ for all $\theta \in \mathbb{R}^{\Lambda_0^*}$.

For any $\epsilon > 0$, $\theta_0 \in \sqrt{\epsilon}\mathbb{Z}^{\Lambda_0^*}$, and $\theta \in B_{\theta_0}$,

$$\begin{aligned} h_\epsilon(\alpha, \theta) &\leq \exp\left(-\frac{1}{2\beta}(\epsilon\|\alpha\|^2 + C_0^2\|\theta_0\|^2 + 2\sqrt{\epsilon}(\alpha, d\theta_0))\right) \\ &\leq \exp\left(-\frac{1}{2\beta}C_0^2\|\theta_0\|^2 - \frac{\sqrt{\epsilon}}{\beta}(\delta\alpha, \theta_0)\right). \end{aligned}$$

Now, $\|\theta - \theta_0\| \leq C\sqrt{\epsilon}$, which implies that $\|\theta\|^2 \leq 2\|\theta_0\|^2 + 2C^2\epsilon$. Similarly,

$$|(\delta\alpha, \theta_0) - (\delta\alpha, \theta)| \leq C\sqrt{\epsilon}\|\alpha\|.$$

Combining the above observations, we get that for any $\epsilon_0 > 0$ and $\epsilon \in (0, \epsilon_0)$,

$$\begin{aligned} h_\epsilon(\alpha, \theta) &\leq C_1 e^{C_2\epsilon\|\alpha\| + C_3\epsilon} \exp\left(-\frac{1}{4\beta}C_0^2\|\theta\|^2 - \frac{\sqrt{\epsilon}}{\beta}(\delta\alpha, \theta)\right) \\ &\leq 2C_1 e^{C_2\epsilon\|\alpha\| + C_3\epsilon} \exp\left(-\frac{1}{4\beta}C_0^2\|\theta\|^2\right) \cosh\left(\frac{\sqrt{\epsilon}}{\beta}(\delta\alpha, \theta)\right) \\ &\leq 2C_1 e^{C_2\epsilon_0\|\alpha\| + C_3\epsilon_0} \exp\left(-\frac{1}{4\beta}C_0^2\|\theta\|^2\right) \cosh\left(\frac{\sqrt{\epsilon_0}}{\beta}(\delta\alpha, \theta)\right) =: H(\alpha, \theta). \end{aligned}$$

Note that

$$\int_{\mathbb{R}^{\Lambda_0^*}} H(\alpha, \theta) d\theta = C_1 e^{C_2\epsilon_0\|\alpha\| + C_3\epsilon_0 + C_4\epsilon_0\|\delta\alpha\|^2}.$$

By Lemma 16.7.16 and the assumption that f has sub-exponential growth, this shows that

$$\begin{aligned} \int_{\mathcal{A}_0} \int_{\mathbb{R}^{\Lambda_0^*}} |f(\alpha)| H(\alpha, \theta) d\theta d\mu_0(\alpha) &\leq \int_{\mathcal{A}_0} C_1 e^{C_2\epsilon_0\|\alpha\| + C_3\epsilon_0 + C_4\epsilon_0\|\alpha\|^2} d\mu_0(\alpha) \\ &\leq C_1 \sum_{\alpha \in \mathcal{A}_0} e^{C_2\epsilon_0\|\alpha\| + C_3\epsilon_0 + C_4\epsilon_0\|\alpha\|^2 - C_5\|\alpha\|^2}. \end{aligned}$$

The right side is finite if ϵ_0 is chosen small enough. Thus, by the dominated convergence theorem,

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} g_\epsilon(\alpha) &= \lim_{\epsilon \rightarrow 0} \int_{\mathcal{A}_0} \int_{\mathbb{R}^{\Lambda_0^*}} f(\alpha) h_\epsilon(\alpha, \theta) d\theta d\mu_0(\alpha) \\ &= \int_{\mathcal{A}_0} \int_{\mathbb{R}^{\Lambda_0^*}} f(\alpha) \lim_{\epsilon \rightarrow 0} h_\epsilon(\alpha, \theta) d\theta d\mu_0(\alpha). \end{aligned}$$

But $\lim_{\epsilon \rightarrow 0} h_\epsilon(\alpha, \theta) = \exp(-\frac{1}{2\beta}\|d\theta\|^2)$, which has no dependence on α . This completes the proof. $\square$

Our next goal is to show how certain expectations with respect to μ_0 can be computed using integrals with respect to γ_0 . Heuristically, the following result says that

$$d\mu_0(\eta) \propto \left(\sum_{\alpha \in \mathcal{A} \cap \mathcal{U}_0} e^{2\pi i(\alpha, \eta)} \right) d\gamma_0(\eta). \quad (16.7.14)$$

Of course, this doesn't make sense rigorously as written, since μ_0 is not absolutely continuous with respect to γ_0 , and the above sum is not convergent. But it can still have a rigorous meaning as follows.

Lemma 16.7.18. *Take any $\tau \in \mathcal{U}_0$. Let $f(\alpha) := e^{(\tau, \alpha)}$ for $\alpha \in \mathcal{U}$. Let η_0 be an $\mathcal{U}_0$ -valued*

centered Gaussian 1-form with law γ_0 . Then

$$\langle f \rangle^* = \frac{\sum_{\alpha \in \mathcal{A} \cap \mathcal{U}_0} \mathbb{E}[f(\eta_0) e^{2\pi i(\alpha, \eta_0)}]}{\sum_{\alpha \in \mathcal{A} \cap \mathcal{U}_0} \mathbb{E}[e^{2\pi i(\alpha, \eta_0)}]},$$

where the sums on the right are absolutely convergent.

Proof. Fix $\epsilon \in (0, 1)$. Let m_ϵ denote the probability mass function of the measure μ_ϵ and Z_ϵ denote the normalizing constant

$$Z_\epsilon := \sum_{\alpha \in \mathcal{A}_0} \exp\left(-\frac{1}{2\beta}(\|d\alpha\|^2 + \epsilon\|\alpha\|^2)\right).$$

From the given conditions on f , it is easy to see that $f m_\epsilon$ is a Schwartz function. Thus, by the Poisson summation formula⁷,

$$\begin{aligned} \langle f \rangle_\epsilon^* &= \sum_{\alpha \in \mathcal{A}} \widehat{f m_\epsilon}(2\pi\alpha) \\ &= \frac{1}{Z_\epsilon} \sum_{\alpha \in \mathcal{A}} \int_{\mathcal{U}} f(\kappa) e^{2\pi i(\alpha, \kappa)} \exp\left(-\frac{1}{2\beta}(\|d\kappa\|^2 + \epsilon\|\kappa\|^2)\right) d\kappa, \end{aligned}$$

and the sum on the right is absolutely convergent. Using this with $f \equiv 1$ and recalling that μ_ϵ is a probability measure, we get

$$Z_\epsilon = \sum_{\alpha \in \mathcal{A}} \int_{\mathcal{U}} e^{2\pi i(\alpha, \kappa)} \exp\left(-\frac{1}{2\beta}(\|d\kappa\|^2 + \epsilon\|\kappa\|^2)\right) d\kappa.$$

Let η_ϵ be a random 1-form with law γ_ϵ . Combining the above observations, we get

$$\begin{aligned} \langle f \rangle_\epsilon^* &= \frac{\sum_{\alpha \in \mathcal{A}} \int_{\mathcal{U}} f(\eta) e^{2\pi i(\alpha, \eta)} \exp(-\frac{1}{2\beta}(\|d\eta\|^2 + \epsilon\|\eta\|^2)) d\eta}{\sum_{\alpha \in \mathcal{A}} \int_{\mathcal{U}} e^{2\pi i(\alpha, \eta)} \exp(-\frac{1}{2\beta}(\|d\eta\|^2 + \epsilon\|\eta\|^2)) d\eta} \\ &= \frac{\sum_{\alpha \in \mathcal{A}} \mathbb{E}[f(\eta_\epsilon) e^{2\pi i(\alpha, \eta_\epsilon)}]}{\sum_{\alpha \in \mathcal{A}} \mathbb{E}[e^{2\pi i(\alpha, \eta_\epsilon)}]}. \end{aligned}$$

Let $\eta_{0,\epsilon}$ and $\eta_{1,\epsilon}$ be the projections of η_ϵ on to $\mathcal{U}_0$ and $\mathcal{V}_0$ respectively. By Lemma 16.7.15, and the fact that $f(\eta_\epsilon) = e^{(\tau, \eta_\epsilon)} = e^{(\tau, \eta_{0,\epsilon})} = f(\eta_{0,\epsilon})$ (since $\tau \in \mathcal{U}_0$), we get

$$\begin{aligned} \mathbb{E}[f(\eta_\epsilon) e^{2\pi i(\alpha, \eta_\epsilon)}] &= \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha, \eta_{0,\epsilon} + \eta_{1,\epsilon})}] \\ &= \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha_0, \eta_{0,\epsilon})}] e^{-2\pi^2 \beta \epsilon^{-1} \|\alpha_{1,\epsilon}\|^2}, \end{aligned}$$

where α_0, α_1 denote the orthogonal projections of α on $\mathcal{U}_0$ and $\mathcal{V}_0$, respectively. Now, as shown in the proof of Lemma 16.7.17, any $\xi \in \mathcal{A}$ can be decomposed as $\xi = \alpha + d\theta$, where $\alpha \in \mathcal{A}_0$ and θ is an integer-valued 0-form supported on Λ_0^* . Moreover, note that this decomposition is unique, since $d(\theta - \theta') = 0$ implies $\theta = \theta'$, by the Poincaré lemma. Thus, if Θ denotes the set of $\mathbb{Z}$ -valued 0-form supported on Λ_0^* , and α_0, α_1 denote the projections

⁷See Subsection A.3.5 in Chapter A.

of $\alpha \in \mathcal{A}_0$ on to $\mathcal{U}_0$ and $\mathcal{V}_0$, then

$$\begin{aligned} \sum_{\alpha \in \mathcal{A}} \mathbb{E}[f(\eta_\epsilon) e^{2\pi i(\alpha, \eta_\epsilon)}] &= \sum_{\alpha \in \mathcal{A}_0, \theta \in \Theta} \mathbb{E}[f(\eta_\epsilon) e^{2\pi i(\alpha + d\theta, \eta_\epsilon)}] \\ &= \sum_{\alpha \in \mathcal{A}_0, \theta \in \Theta} \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha_0, \eta_{0,\epsilon})} e^{2\pi i(\alpha_1 + d\theta, \eta_{1,\epsilon})}] \\ &= \sum_{\alpha \in \mathcal{A}_0, \theta \in \Theta} \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha_0, \eta_{0,\epsilon})}] e^{-2\pi^2 \beta \epsilon^{-1} \|\alpha_1 + d\theta\|^2}. \end{aligned}$$

We claim that there is some finite constant C , not depending on α or ϵ (recall that $0 < \epsilon < 1$), such that

$$\sum_{\theta \in \Theta} e^{-2\pi^2 \beta \epsilon^{-1} \|\alpha_1 + d\theta\|^2} \leq C. \quad (16.7.15)$$

To see this, take any α . Since $d\alpha_1 = 0$, there is some real-valued 0-form ξ supported on $*\Lambda$ such that $\alpha_1 = d\xi$. Take an integer approximation κ of ξ by defining $\kappa(x) := \lfloor \xi(x) \rfloor$ for each x . Then $\|\alpha_1 - d\kappa\| = \|d\xi - d\kappa\| \leq C$ for some C that does not depend on α or ϵ . But since κ is $\mathbb{Z}$ -valued,

$$\begin{aligned} \sum_{\theta \in \Theta} e^{-2\pi^2 \beta \epsilon^{-1} \|\alpha_1 + d\theta\|^2} &= \sum_{\theta \in \Theta} e^{-2\pi^2 \beta \epsilon^{-1} \|(\alpha_1 - d\kappa) + d\theta\|^2} \\ &\leq \sum_{\theta \in \Theta} e^{-2\pi^2 \beta \|(\alpha_1 - d\kappa) + d\theta\|^2}. \end{aligned}$$

Since $\|\alpha_1 - d\kappa\| \leq C$, a simple argument using compactness now proves the claim (16.7.15).

Moreover, note that if $\min_{\theta \in \Theta} \|\alpha_1 + d\theta\| = 0$, then $\alpha_1 = -d\theta$ for some integer-valued 0-form θ supported on Λ_0^* . In particular, α_1 is integer-valued. But this is impossible unless $\alpha_1 = 0$, since we chose α_0 to minimize the norm in the equivalence of α . Thus, we conclude that

$$\lim_{\epsilon \rightarrow 0} \sum_{\theta \in \Theta} e^{-2\pi^2 \beta \epsilon^{-1} \|\alpha_1 + d\theta\|^2} = \begin{cases} 1 & \text{if } \alpha \in \mathcal{A}_0 \cap \mathcal{U}_0, \\ 0 & \text{if } \alpha \in \mathcal{A}_0 \setminus \mathcal{U}_0. \end{cases}$$

By Lemma 16.7.15,

$$\begin{aligned} \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha_0, \eta_{0,\epsilon})}] &= \exp\left(\frac{\beta}{2}(\tau + 2\pi i\alpha_0, V_\epsilon \tau + 2\pi i\alpha_0)\right) \\ &= \exp\left(\frac{\beta}{2}((\tau, V_\epsilon \tau) + 4\pi i(\alpha_0, V_\epsilon \tau) - 4\pi^2(\alpha_0, V_\epsilon \alpha_0))\right). \end{aligned}$$

From the spectral decomposition of V_ϵ , it follows that if $\epsilon \leq \epsilon'$, then $(\xi, V_\epsilon \xi) \geq (\xi, V_{\epsilon'} \xi)$ for any $\xi \in \mathcal{U}_0$. Thus, for any $\epsilon \in (0, 1)$,

$$|\mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha_0, \eta_{0,\epsilon})}]| \leq \exp\left(\frac{\beta}{2}((\tau, V \tau) - 4\pi^2(\alpha_0, V_1 \alpha_0))\right).$$

Note that the bound has no dependence on ϵ , and is summable in α_0 . The above observations

allow us to apply the dominated convergence theorem and conclude that

$$\lim_{\epsilon \rightarrow 0} \langle f \rangle_{\epsilon}^* = \frac{\sum_{\alpha_0 \in \mathcal{A}_0 \cap \mathcal{U}_0} \mathbb{E}[f(\eta_0) e^{2\pi i(\alpha_0, \eta_0)}]}{\sum_{\alpha_0 \in \mathcal{A}_0 \cap \mathcal{U}_0} \mathbb{E}[e^{2\pi i(\alpha_0, \eta_0)}]},$$

where η_0 has law γ_0 . Moreover, it also shows that the sums on the right are absolutely convergent.

To complete the proof, we now show that $\mathcal{A}_0 \cap \mathcal{U}_0 = \mathcal{A} \cap \mathcal{U}_0$. Indeed, take any $\alpha \in \mathcal{A} \cap \mathcal{U}_0$. We claim that α is the unique norm minimizing element in its equivalence class. To see this, take any $\alpha' \sim \alpha$. Then $d(\alpha' - \alpha) = 0$, which means that $\alpha' - \alpha$ is in $\mathcal{V}_0$ and hence, orthogonal to elements of $\mathcal{U}_0$. In particular, it is orthogonal to α . Thus,

$$\begin{aligned} \|\alpha'\|^2 &= \|\alpha' - \alpha\|^2 + 2(\alpha' - \alpha, \alpha) + \|\alpha\|^2 \\ &= \|\alpha' - \alpha\|^2 + \|\alpha\|^2 \geq \|\alpha\|^2. \end{aligned}$$

To show uniqueness, just observe by the above calculation, if α' is another norm minimizing element in the equivalence class of α , then $\|\alpha' - \alpha\| = 0$. Thus, we conclude that $\alpha \in \mathcal{A}_0$. This completes the proof. $\square$

16.7.5 Towards renormalization

Although Lemma 16.7.18 does give a rigorous meaning to the heuristic identity (16.7.14), it does not give it in a way that can be exploited gainfully. We will now do a further ‘processing’ of this identity to bring it to form that is more suitable for applying a renormalization step later. First, we rewrite the conclusion of Lemma 16.7.18 in a more convenient form. Let $\{z_q\}_{q=1}^{\infty}$ be a sequence of positive numbers such that

$$\sum_{q=1}^{\infty} \frac{2}{z_q} = 1.$$

The result that we are going to state below is valid for any choice of this sequence. Let $\mathcal{Q} := \{1, 2, \dots\}^{E_0^*}$. For each $q \in \mathcal{Q}$, let

$$c(q) := \prod_{e \in E_0^*} \frac{2}{z_{q(e)}}.$$

For each $N \geq 1$, let $\mathcal{Q}_N := \{1, \dots, N\}^{E_0^*}$. Let

$$a_N := \sum_{q=1}^N \frac{2}{z_q},$$

so that $a_N \uparrow 1$ as $N \rightarrow \infty$. For $\alpha \in \mathcal{U}$, define

$$\rho_N(\alpha) := \sum_{q \in \mathcal{Q}_N} c(q) \prod_{e \in E_0^*} (1 + a_N z_{q(e)} \cos(2\pi i q(e) \alpha(e))).$$

The following lemma is a restatement of Lemma 16.7.18.

Lemma 16.7.19. *Take any $\tau \in \mathcal{U}_0$ and define $f : \mathcal{U} \rightarrow \mathbb{C}$ as $f(\alpha) := e^{(\alpha, \tau)}$. For each $\epsilon > 0$, let η_ϵ be a random 1-form with law γ_ϵ . Let $\eta_{0,\epsilon}$ be the projection of η_ϵ on $\mathcal{U}_0$. Then*

$$\langle f \rangle^* = \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \frac{\mathbb{E}[f(\eta_{0,\epsilon}) \rho_N(\eta_\epsilon)]}{\mathbb{E}[\rho_N(\eta_\epsilon)]}.$$

Proof. Let $\mathcal{A}_N := \{-N, \dots, N\}^{E_0^*}$, so that $\mathcal{A}$ is the increasing limit of $\mathcal{A}_N$ as $N \rightarrow \infty$. Since the sums in Lemma 16.7.18 are absolutely convergent, we have

$$\langle f \rangle^* = \lim_{N \rightarrow \infty} \frac{\sum_{\alpha \in \mathcal{A}_N \cap \mathcal{U}_0} \mathbb{E}[f(\eta_0) e^{2\pi i(\alpha, \eta_0)}]}{\sum_{\alpha \in \mathcal{A}_N \cap \mathcal{U}_0} \mathbb{E}[e^{2\pi i(\alpha, \eta_0)}]}.$$

Take any N and $\alpha \in \mathcal{A}_N$. Let α_0 and α_1 be the projections of α on $\mathcal{U}_0$ and $\mathcal{V}_0$. Let $\eta_{0,\epsilon}$ and $\eta_{1,\epsilon}$ denote the projections of η_ϵ on $\mathcal{U}_0$ and $\mathcal{V}_0$. Since f is constant on equivalence classes, Lemma 16.7.15 gives

$$\begin{aligned} \mathbb{E}[f(\eta_\epsilon) e^{2\pi i(\alpha, \eta_\epsilon)}] &= \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha_0, \eta_{0,\epsilon})} e^{2\pi i(\alpha_1, \eta_{1,\epsilon})}] \\ &= \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha_0, \eta_{0,\epsilon})}] \mathbb{E}[e^{2\pi i(\alpha_1, \eta_{1,\epsilon})}] \\ &= \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha_0, \eta_{0,\epsilon})}] e^{-2\pi^2 \beta \epsilon^{-1} \|\alpha_1\|^2} \\ &= \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha, \eta_{0,\epsilon})}] e^{-2\pi^2 \beta \epsilon^{-1} \|\alpha_1\|^2} \end{aligned}$$

Thus, we have (again by Lemma 16.7.15),

$$\lim_{\epsilon \rightarrow 0} \mathbb{E}[f(\eta_\epsilon) e^{2\pi i(\alpha, \eta_\epsilon)}] = \begin{cases} \mathbb{E}[f(\eta_0) e^{2\pi i(\alpha, \eta_0)}] & \text{if } \alpha \notin \mathcal{U}_0, \\ 0 & \text{otherwise.} \end{cases}$$

Since $\mathcal{A}_N$ is a finite set, this gives

$$\sum_{\alpha \in \mathcal{A}_N \cap \mathcal{U}_0} \mathbb{E}[f(\eta_0) e^{2\pi i(\alpha, \eta_0)}] = \lim_{\epsilon \rightarrow 0} \sum_{\alpha \in \mathcal{A}_N} \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha, \eta_\epsilon)}].$$

Clearly, this holds also with $f \equiv 1$. Thus, we conclude that

$$\begin{aligned} \langle f \rangle^* &= \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \frac{\sum_{\alpha \in \mathcal{A}_N} \mathbb{E}[f(\eta_{0,\epsilon}) e^{2\pi i(\alpha, \eta_\epsilon)}]}{\sum_{\alpha \in \mathcal{A}_N} \mathbb{E}[e^{2\pi i(\alpha, \eta_\epsilon)}]} \\ &= \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \frac{\mathbb{E}[f(\eta_{0,\epsilon}) (\sum_{\alpha \in \mathcal{A}_N} e^{2\pi i(\alpha, \eta_\epsilon)})]}{\mathbb{E}[\sum_{\alpha \in \mathcal{A}_N} e^{2\pi i(\alpha, \eta_\epsilon)}]}. \end{aligned}$$

Now note that

$$\begin{aligned} \sum_{\alpha \in \mathcal{A}_N} e^{2\pi i(\alpha, \eta_\epsilon)} &= \sum_{\alpha \in \mathcal{A}_N} \prod_{e \in E_0^*} e^{2\pi i \alpha(e) \eta_\epsilon(e)} \\ &= \prod_{e \in E_0^*} \sum_{q(e)=-N}^N e^{2\pi i q(e) \eta_\epsilon(e)} = \prod_{e \in E_0^*} \left(1 + 2 \sum_{q(e)=1}^N \cos(2\pi i q(e) \eta_\epsilon(e)) \right). \end{aligned}$$

Rewriting

$$1 + 2 \sum_{q(e)=1}^N \cos(2\pi i q(e) \eta_\epsilon(e)) = \sum_{q(e)=1}^N \frac{2}{z_{q(e)}} (a_N^{-1} + z_{q(e)} \cos(2\pi i q(e) \eta_\epsilon(e))),$$

we get

$$\prod_{e \in E_0^*} \left(1 + 2 \sum_{q(e)=1}^N \cos(2\pi i q(e) \eta_\epsilon(e)) \right) = a_N^{-|E_0^*|} \rho_N(\eta_\epsilon).$$

Since the factor $a_N^{-|E_0^*|}$ cancels out between the numerator and the denominator, this completes the proof. $\square$

We will now do a further ‘processing’ to get an even more convenient version of the ‘density’ of μ_0 with respect to γ_0 . For an element $\alpha \in \mathcal{A}$, we define the *edge-support* of α , denoted by $E(\alpha)$, as the set of edges $e \in E_0$ where $\alpha(e) \neq 0$. We define the distance $\text{dist}(e, e')$ between two edges $e, e' \in E_0$ as the minimum of the Euclidean distance between a vertex of e and a vertex of e' . For $E, E' \subseteq E_0$, we define $\text{dist}(E, E') := \min\{\text{dist}(e, e') : e \in E, e' \in E'\}$. We will say that a finite set $\mathcal{E} \subseteq \mathcal{A}$ is an *ensemble* if for any distinct $\alpha, \alpha' \in \mathcal{E}$, $E(\alpha)$ and $E(\alpha')$ are disjoint. For $k = 0, 1, 2, \dots$, will say that $\mathcal{E}$ is a k -ensemble if for any distinct $\alpha, \alpha' \in \mathcal{E}$, $\text{dist}(E(\alpha), E(\alpha')) \geq 2^{k/2}$.

Lemma 16.7.20. *Given any $q \in \mathcal{Q}$, there is a finite set $\Gamma(q)$, strictly positive coefficients $\{c_\gamma\}_{\gamma \in \Gamma(q)}$, and 1-ensembles $\{\mathcal{E}_\gamma\}_{\gamma \in \Gamma(q)}$, such that the following is true. For any $N \geq 1$, there are strictly positive coefficients $\{K_\gamma^N(\xi)\}_{\gamma \in \Gamma(q), \xi \in \mathcal{E}_\gamma}$, such that for any $\alpha \in \mathcal{U}$,*

$$\prod_{e \in E_0^*} (1 + a_N z_{q(e)} \cos(2\pi q(e) \alpha(e))) = \sum_{\gamma \in \Gamma(q)} c_\gamma \prod_{\xi \in \mathcal{E}_\gamma} (1 + K_\gamma^N(\xi) \cos(2\pi(\alpha, \xi))),$$

and moreover, for each $\gamma \in \Gamma(q)$ and $\xi \in \mathcal{E}_\gamma$,

$$K_\gamma^N(\xi) \leq 3^{N_1(\xi)} \prod_{e \in E(\xi)} z_{|\xi(e)|},$$

where $N_1(\xi)$ is the number of edges in E_0^* which at distance ≤ 1 from $E(\xi)$.

Proof. Note that for any $K_1, K_2, \theta_1, \theta_2 \in \mathbb{R}$, we have the identity

$$\begin{aligned}
 & \frac{1}{3}(1 + 3K_1 \cos \theta_1) + \frac{1}{3}(1 + 3K_2 \cos \theta_2) \\
 & + \frac{1}{6}(1 + 3K_1 K_2 \cos(\theta_1 - \theta_2)) + \frac{1}{6}(1 + 3K_1 K_2 \cos(\theta_1 + \theta_2)) \\
 & = 1 + K_1 \cos \theta_1 + K_2 \cos \theta_2 + K_1 K_2 \cos \theta_1 \cos \theta_2 \\
 & = (1 + K_1 \cos \theta_1)(1 + K_2 \cos \theta_2).
 \end{aligned} \tag{16.7.16}$$

Now, let

$$I := \prod_{e \in E_0} (1 + a_N z_{q(e)} \cos(2\pi q(e) \alpha(e))).$$

Note that I can be written as

$$\sum_{\gamma \in \Gamma(q)} c_\gamma \prod_{\xi \in \mathcal{E}_\gamma} (1 + K_\gamma^N(\xi) \cos(2\pi(\alpha, \xi))), \tag{16.7.17}$$

where $\Gamma(q) = \{0\}$, $c_0 = 1$, $\mathcal{E}_0 \subseteq \mathcal{A}$ is the set of 1-forms $\{\xi_e\}_{e \in E_0}$, defined as

$$\xi_e(e') = \begin{cases} q(e) & \text{if } e' = e, \\ 0 & \text{otherwise,} \end{cases}$$

and $K_0^N(\xi_e) = a_N z_{q(e)}$ for each e . This is an expression of the form that we see in the statement of the lemma, except that $\mathcal{E}_0$ is not a 1-ensemble (although it is an ensemble).

The strategy, now, is to inductively modify $\Gamma(q)$ until it becomes a collection of 1-ensembles. In the first step, take two distinct edges $a, b \in E_0^*$ such that $\text{dist}(a, b) < \sqrt{2}$. By the identity (16.7.16), we get

$$\begin{aligned}
 & (1 + a_N z_{q(a)} \cos(2\pi q(a) \alpha(a)))(1 + a_N z_{q(b)} \cos(2\pi q(b) \alpha(b))) \\
 & = \frac{1}{3}(1 + 3a_N z_{q(a)} \cos(2\pi q(a) \alpha(a))) + \frac{1}{3}(1 + 3a_N z_{q(b)} \cos(2\pi q(b) \alpha(b))) \\
 & \quad + \frac{1}{6}(1 + 3a_N^2 z_{q(a)} z_{q(b)} \cos(2\pi(q(a) \alpha(a) - q(b) \alpha(b)))) \\
 & \quad + \frac{1}{6}(1 + 3a_N^2 z_{q(a)} z_{q(b)} \cos(2\pi(q(a) \alpha(a) + q(b) \alpha(b)))).
 \end{aligned}$$

Substituting this expression back into the product defining I and expanding, we again get an expression of the form (16.7.17), where now $\Gamma(q) = \{1, 2, 3, 4\}$, with $\mathcal{E}_1 = \{\xi_e\}_{e \in E_0^* \setminus \{b\}}$, $\mathcal{E}_2 = \{\xi_e\}_{e \in E_0^* \setminus \{a\}}$, $\mathcal{E}_3 = \{\xi_e\}_{e \in E_0^* \setminus \{a, b\}} \cup \{\xi_1\}$, and $\mathcal{E}_4 = \{\xi_e\}_{e \in E_0^* \setminus \{a, b\}} \cup \{\xi_2\}$, where

$$\xi_1(e') = \begin{cases} q(a) & \text{if } e' = a, \\ -q(b) & \text{if } e' = b, \\ 0 & \text{otherwise,} \end{cases} \quad \xi_2(e') = \begin{cases} q(a) & \text{if } e' = a, \\ q(b) & \text{if } e' = b, \\ 0 & \text{otherwise.} \end{cases}$$

If $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_4$ are 1-ensembles, there is nothing more to do. If not, we proceed by induction,

as described in the following paragraph. Note that $|\mathcal{E}_i| < |\mathcal{E}|$ for $i = 1, 2, 3, 4$ in this base case of the induction.

Suppose we have that I is equal to the expression (16.7.17) for some $\Gamma(q)$, where $\mathcal{E}_\gamma$ is not a 1-ensemble for at least one $\gamma \in \Gamma(q)$. This means that there are distinct $\xi, \xi' \in \mathcal{E}_\gamma$ such that $\text{dist}(E(\xi), E(\xi')) < \sqrt{2}$. Then there is a pair of edges a, b such that $\text{dist}(a, b) < \sqrt{2}$ and both $\xi(a)$ and $\xi'(b)$ are nonzero. By the identity (16.7.16), we get

$$\begin{aligned} & (1 + K_\gamma^N(\xi) \cos(2\pi(\alpha, \xi)))(1 + K_\gamma^N(\xi') \cos(2\pi(\alpha, \xi'))) \\ &= \frac{1}{3}(1 + 3K_\gamma^N(\xi) \cos(2\pi(\alpha, \xi))) + \frac{1}{3}(1 + 3K_\gamma^N(\xi') \cos(2\pi(\alpha, \xi'))) \\ & \quad + \frac{1}{6}(1 + 3K_\gamma^N(\xi)K_\gamma^N(\xi') \cos(2\pi(\alpha, \xi - \xi'))) \\ & \quad + \frac{1}{6}(1 + 3K_\gamma^N(\xi)K_\gamma^N(\xi') \cos(2\pi(\alpha, \xi + \xi'))). \end{aligned}$$

We use this to decompose the product coming from $\mathcal{E}_\gamma$ into a linear combination of four products, corresponding to the collections $\mathcal{E}_{\gamma_1}, \mathcal{E}_{\gamma_2}, \mathcal{E}_{\gamma_3}, \mathcal{E}_{\gamma_4}$, given by

$$\begin{aligned} \mathcal{E}_{\gamma_1} &= \mathcal{E}_\gamma \setminus \{\xi'\}, \quad \mathcal{E}_{\gamma_2} = \mathcal{E}_\gamma \setminus \{\xi\}, \\ \mathcal{E}_{\gamma_3} &= (\mathcal{E}_\gamma \setminus \{\xi, \xi'\}) \cup \{\xi_1\}, \quad \mathcal{E}_{\gamma_4} = (\mathcal{E}_\gamma \setminus \{\xi, \xi'\}) \cup \{\xi_2\}, \end{aligned}$$

where $\xi_1 = \xi - \xi'$ and $\xi_2 = \xi + \xi'$. In particular, $|\mathcal{E}_{\gamma_i}| < |\mathcal{E}_\gamma|$ for $i = 1, 2, 3, 4$.

Thus, we have shown that if at least one $\mathcal{E}_\gamma$ that is not a 1-ensemble, we can replace this element by four new elements, each having size strictly less than $|\mathcal{E}_\gamma|$. Since this process cannot go on indefinitely, this completes the proof of the formula claimed in the lemma. Also, the construction makes it clear that c_γ and $K_\gamma^N(\xi)$ are strictly positive, and $\Gamma(q)$, the c_γ 's, and the $\mathcal{E}_\gamma$'s have no dependence on N .

It remains to prove the upper bound on $K_\gamma^N(\xi)$. Note that in the above construction, at any step, we are replacing an ensemble $\mathcal{E}_\gamma$ by four ‘children’ $\mathcal{E}_{\gamma_1}, \mathcal{E}_{\gamma_2}, \mathcal{E}_{\gamma_3}, \mathcal{E}_{\gamma_4}$. Thus, the ensembles evolve like a tree. We will talk about ancestors and descendants in the usual sense below.

Now take any $\kappa \in \mathcal{E}_{\gamma_i}$ for some i . If $\kappa \in \mathcal{E}_\gamma \setminus \{\xi, \xi'\}$, then $K_{\gamma_i}^N(\kappa) = K_\gamma^N(\kappa)$. If $i = 1$ and $\kappa = \xi$, then $K_{\gamma_1}^N(\kappa) = 3K_\gamma^N(\kappa)$. Similarly, $i = 2$ and $\kappa = \xi'$, then $K_{\gamma_2}^N(\kappa) = 3K_\gamma^N(\kappa)$. If $i = 3$ and $\kappa = \xi_1$ or $i = 4$ and $\kappa = \xi_2$, then $K_{\gamma_i}^N(\kappa) = 3K_\gamma^N(\xi)K_\gamma^N(\xi')$. Also, if $i = 3$ or 4 , we will say that ξ, ξ' are the parents of κ , and if $i = 1$ or 2 , we will say that κ is its own parent.

We will say that the pair $\{a, b\}$ was ‘used’ in the above birthing process. For any ensemble $\mathcal{E}_\gamma$ obtained at any stage of the algorithm, and any $\xi \in \mathcal{E}_\gamma$, we will say a pair $\{a, b\}$ of edges (with $\text{dist}(a, b) < \sqrt{2}$) was ‘used’ in the construction ξ if it was used in the birthing process of an ancestor ξ_0 of ξ . Note that ξ_0 must have been an element of an ancestor $\mathcal{E}_\delta$ of $\mathcal{E}_\gamma$.

A pair $\{a, b\}$ cannot be used more than once in the construction of ξ , for the following reason. When $\{a, b\}$ is used, one of the following happens. Either a is thrown out of the

child ensemble of $\mathcal{E}_\delta$ that is an ancestor of $\mathcal{E}_\gamma$, or b is thrown out of it, or a, b are both included in the support of a single element of the child ensemble that is an ancestor of $\mathcal{E}_\gamma$. In none of these cases can we encounter the situation that $\{a, b\}$ is used again at some later stage.

From the above observations, it is now easy to see by induction that $K_\gamma^N(\xi)$ can be at most $3^n \prod_{e \in E(\xi)} z_{|\xi(e)|}$, where n is the number of pairs $\{a, b\}$ used in the construction of ξ . But if $\{a, b\}$ is used, then either a or b or both must be present in $E(\xi)$. This is because at least one of them must be present in $E(\xi_0)$ (where ξ_0 is the ancestor of ξ which was produced using $\{a, b\}$), and $E(\xi) \supseteq E(\xi_0)$. Thus, $n \leq N_1(\xi)$. This completes the proof. $\square$

Since the $\Gamma(q)$'s in Lemma 16.7.20 are indexing sets, we can assume without loss that they are disjoint. Then, letting

$$\Gamma := \bigcup_{q \in \mathcal{Q}} \Gamma(q), \quad \Gamma_N := \bigcup_{q \in \mathcal{Q}_N} \Gamma(q),$$

we obtain the following corollary.

Corollary 16.7.21. *Let Γ and Γ_N be as above. Then, there are strictly positive coefficients $\{c_\gamma\}_{\gamma \in \Gamma}$ and $\{K_\gamma^N(\xi)\}_{N \geq 1, \gamma \in \Gamma_N, \xi \in \mathcal{E}_\gamma}$, and 1-ensembles $\{\mathcal{E}_\gamma\}_{\gamma \in \Gamma}$, such that for any $\alpha \in \mathcal{U}$ and $N \geq 1$,*

$$\rho_N(\alpha) = \sum_{\gamma \in \Gamma_N} c_\gamma \prod_{\xi \in \mathcal{E}_\gamma} (1 + K_\gamma^N(\xi) \cos(2\pi(\alpha, \xi))),$$

and moreover, for each $N \geq 1$, $\gamma \in \Gamma_N$, and $\xi \in \mathcal{E}_\gamma$,

$$K_\gamma^N(\xi) \leq 3^{N_1(\xi)} \prod_{e \in E(\xi)} z_{|\xi(e)|},$$

where $N_1(\xi)$ is the number of edges in E_0^* which at distance ≤ 1 from $E(\xi)$.

The next step is to further simplify the form of ρ_N . We define $\mathcal{E}_\gamma^0 := \mathcal{E}_\gamma \cap \mathcal{U}_0$, and then define $\rho_N^0 : \mathcal{U} \rightarrow \mathbb{R}$ as

$$\rho_N^0(\alpha) := \sum_{\gamma \in \Gamma_N} c_\gamma \prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\alpha, \xi))). \quad (16.7.18)$$

The following lemma shows that ρ_N can be replaced by ρ_N^0 in Lemma 16.7.19. This is the version that will assist us in our renormalization step.

Lemma 16.7.22. *Take any $\tau \in \mathcal{U}_0$ and define $f : \mathcal{U} \rightarrow \mathbb{C}$ as $f(\alpha) := e^{(\alpha, \tau)}$. Let η_0 be a random 1-form with law γ_0 . Then*

$$\langle f \rangle^* = \lim_{N \rightarrow \infty} \frac{\mathbb{E}[f(\eta_0) \rho_N^0(\eta_0)]}{\mathbb{E}[\rho_N^0(\eta_0)]}.$$

Proof. Let η_ϵ be a random 1-form with law γ_ϵ and let $\eta_{0,\epsilon}$ and $\eta_{1,\epsilon}$ be its projections on $\mathcal{U}_0$

and $\mathcal{V}_0$. Then note that by Corollary 16.7.21,

$$\mathbb{E}(f(\eta_{0,\epsilon})\rho_N(\eta_\epsilon)) = \sum_{\gamma \in \Gamma_N} c_\gamma \mathbb{E}\left(f(\eta_{0,\epsilon}) \prod_{\xi \in \mathcal{E}_\gamma} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_\epsilon, \xi)))\right).$$

The expectation on the right can be expanded as

$$\sum_{\mathcal{E} \subseteq \mathcal{E}_\gamma} \left(\prod_{\xi \in \mathcal{E}} K_\gamma^N(\xi) \right) \mathbb{E}\left(f(\eta_{0,\epsilon}) \prod_{\xi \in \mathcal{E}} \cos(2\pi(\eta_\epsilon, \xi))\right).$$

Take any nonempty $\mathcal{E} \subseteq \mathcal{E}_\gamma$. Then

$$\begin{aligned} \mathbb{E}\left(f(\eta_{0,\epsilon}) \prod_{\xi \in \mathcal{E}} \cos(2\pi(\eta_\epsilon, \xi))\right) &= 2^{-|\mathcal{E}|} \sum_{\sigma \in \{-1,1\}^\mathcal{E}} \mathbb{E}\left(f(\eta_{0,\epsilon}) \prod_{\xi \in \mathcal{E}} e^{2\pi i \sigma_\xi(\eta_\epsilon, \xi)}\right) \\ &= 2^{-|\mathcal{E}|} \sum_{\sigma \in \{-1,1\}^\mathcal{E}} \mathbb{E}(f(\eta_{0,\epsilon}) e^{2\pi i(\eta_\epsilon, \alpha_\sigma)}), \end{aligned}$$

where

$$\alpha_\sigma := \sum_{\xi \in \mathcal{E}} \sigma_\xi \xi.$$

We claim that if $\mathcal{E} \setminus \mathcal{U}_0$ is nonempty, then $\alpha_\sigma \notin \mathcal{U}_0$ for each σ . Indeed, suppose that $\kappa \in \mathcal{E} \setminus \mathcal{U}_0$. Then by Lemma 16.7.10, there is some $x \in \Lambda_0^*$ such that $\delta\kappa(x) \neq 0$. This implies that there is some $e \in E_0^*$ with x being one of the vertices of e , such that $\kappa(e) \neq 0$. That is, $e \in E(\kappa)$. Since $\mathcal{E}_\gamma$ is a 1-ensemble, this implies that no other edge incident to x can be in $E(\xi)$ for any $\xi \in \mathcal{E} \setminus \{\kappa\}$. Consequently, $\delta\alpha_\sigma(x) = \delta\kappa(x) \neq 0$. So, again by Lemma 16.7.10, $\alpha_\sigma \notin \mathcal{U}_0$. This proves the claim. (Note that in this step, we do not need the full power of the 1-ensemble property; it would be enough to have that each $\mathcal{E}_\gamma$ is a 0-ensemble. We will need the 1-ensemble property later.)

Thus, by Lemma 16.7.15, it follows that if $\mathcal{E} \setminus \mathcal{U}_0$ is nonempty, then

$$\lim_{\epsilon \rightarrow 0} \mathbb{E}\left(f(\eta_{0,\epsilon}) \prod_{\xi \in \mathcal{E}} \cos(2\pi(\eta_\epsilon, \xi))\right) = 0.$$

Consequently,

$$\begin{aligned} &\lim_{\epsilon \rightarrow 0} \mathbb{E}\left(f(\eta_{0,\epsilon}) \prod_{\xi \in \mathcal{E}_\gamma} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_\epsilon, \xi)))\right) \\ &= \lim_{\epsilon \rightarrow 0} \sum_{\mathcal{E} \subseteq \mathcal{E}_\gamma^0} \left(\prod_{\xi \in \mathcal{E}} K_\gamma^N(\xi) \right) \mathbb{E}\left(f(\eta_{0,\epsilon}) \prod_{\xi \in \mathcal{E}} \cos(2\pi(\eta_\epsilon, \xi))\right) \\ &= \lim_{\epsilon \rightarrow 0} \mathbb{E}\left(f(\eta_{0,\epsilon}) \prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_\epsilon, \xi)))\right). \end{aligned}$$

But for $\xi \in \mathcal{E}_\gamma^0$, we have $(\eta_\epsilon, \xi) = (\eta_{0,\epsilon}, \xi)$. It is now clear how the proof can be completed using Lemma 16.7.15. $\square$

16.7.6 Completing the square

Recall the function $D_\ell : \mathcal{A}_0 \rightarrow [0, \infty)$ defined in Lemma 16.7.7. Note that for $\alpha \in \mathcal{A}_0$,

$$\begin{aligned} D_\ell(\alpha) &= \exp\left(\frac{1}{\beta} \sum_{p \in \Sigma} *d\alpha(p) - \frac{|\Sigma|}{2\beta}\right) \\ &= \exp\left(\frac{1}{\beta} \sum_{p \in \Sigma} d\alpha(*p) - \frac{|\Sigma|}{2\beta}\right), \end{aligned}$$

where $*p$ is the dual plaquette of p . Let $*\Sigma$ be the set of plaquettes that are duals of plaquettes in Σ . Let σ be the 2-form on $*\mathbb{Z}^4$ defined as

$$\sigma(p) := \begin{cases} 1 & \text{if } p \in *\Sigma, \\ 0 & \text{otherwise.} \end{cases}$$

Then note that

$$\sum_{p \in \Sigma} d\alpha(*p) - \frac{|\Sigma|}{2} = (d\alpha, \sigma) - \frac{1}{2}\|\sigma\|^2 = (\alpha, \delta\sigma) - \frac{1}{2}\|\sigma\|^2. \quad (16.7.19)$$

We claim that $\delta\sigma \in \mathcal{U}_0$. Indeed, for any $\xi \in \mathcal{V}_0$, $(\xi, \delta\sigma) = (d\xi, \sigma) = 0$, and $\delta\sigma(e) = 0$ for $e \notin E_0^*$ since ℓ does not touch the boundary of Λ . Thus, we can apply Lemma 16.7.22 to conclude that

$$\langle D_\ell \rangle^* = \lim_{N \rightarrow \infty} \frac{\mathbb{E}[D_\ell(\eta_0)\rho_N^0(\eta_0)]}{\mathbb{E}[\rho_N^0(\eta_0)]},$$

where η_0 is a random 1-form with law γ_0 , and D_ℓ is extended to $\mathcal{U}$ in the natural way.

We will now employ a standard ‘completing the square’ trick to re-express the above formula differently. Recall the discrete Hodge Laplacian operator $\Delta = d\delta + \delta d$ discussed in Subsection C.8.10 of Appendix C, and the restrictions of Δ to hypercubes. Specifically, we will need the restriction of Δ to the hypercube Λ^* . Since ℓ does not touch $\partial\Lambda$, it follows that $\sigma(p) = 0$ unless all vertices of p are in Λ^* . Thus, $\sigma \in \text{range } \Pi_{\Lambda^*}$ (where Π_{Λ^*} is now acting on 2-forms), which means that $\Delta_{\Lambda^*}^{-1}\sigma$ is well-defined.

Lemma 16.7.23. *Let η_0 be a random 1-form with law γ_0 , as before. Let $\tau := -\delta\Delta_{\Lambda^*}^{-1}\sigma$, $\tau_0 := \Pi_{\Lambda^*}\tau$, and $\kappa := -\Pi_{\Lambda^*}\delta d\Delta_{\Lambda^*}^{-1}\sigma$. Then $\tau_0 \in \mathcal{U}_0$, and*

$$\langle D_\ell \rangle^* = e^{\frac{1}{2\beta}\|d\tau - \Pi_{\Lambda^*}d\tau\|^2 - \frac{1}{2\beta}\|\kappa\|^2} \lim_{N \rightarrow \infty} \frac{\mathbb{E}[\rho_N^0(\eta_0 + \tau_0)]}{\mathbb{E}[\rho_N^0(\eta_0)]}.$$

Proof. We begin with the calculation

$$\begin{aligned}
 \Pi_{\Lambda^*} d\tau &= -\Pi_{\Lambda^*} d\delta \Delta_{\Lambda^*}^{-1} \sigma \\
 &= -\Pi_{\Lambda^*} (d\delta + \delta d) \Delta_{\Lambda^*}^{-1} \sigma + \Pi_{\Lambda^*} \delta d \Delta_{\Lambda^*}^{-1} \sigma \\
 &= -\Pi_{\Lambda^*} \sigma - \kappa = \sigma - \kappa,
 \end{aligned} \tag{16.7.20}$$

Next, recall that γ_0 has density proportional to $e^{-\frac{1}{2\beta} \|d\alpha\|^2}$. Thus,

$$\begin{aligned}
 \mathbb{E}[D_\ell(\eta_0) \rho_N^0(\eta_0)] &= \mathbb{E}[e^{\frac{1}{\beta}(d\eta_0, \sigma) - \frac{1}{2\beta} \|\sigma\|^2} \rho_N^0(\eta_0)] \\
 &= \frac{1}{Z_0} \int_{\mathcal{U}_0} e^{\frac{1}{\beta}(d\alpha, \sigma) - \frac{1}{2\beta} \|\sigma\|^2 - \frac{1}{2\beta} \|d\alpha\|^2} \rho_N^0(\alpha) d\lambda(\alpha) \\
 &= \frac{1}{Z_0} \int_{\mathcal{U}_0} e^{-\frac{1}{2\beta} \|d\alpha - \sigma\|^2} \rho_N^0(\alpha) d\lambda(\alpha),
 \end{aligned} \tag{16.7.21}$$

where λ denotes Lebesgue measure on $\mathcal{U}_0$ and Z_0 is the normalizing constant of γ_0 . Now, by equation (16.7.20), we have that for any $\alpha \in \mathcal{U}_0$,

$$\begin{aligned}
 \|d\alpha - \kappa\|^2 &= \|d\alpha + \Pi_{\Lambda^*} d\tau - \sigma\|^2 \\
 &= \|d\alpha\|^2 + 2(d\alpha, \Pi_{\Lambda^*} d\tau - \sigma) + \|\Pi_{\Lambda^*} d\tau - \sigma\|^2 \\
 &= \|d\alpha\|^2 + 2(d\alpha, d\tau - \sigma) + \|d\tau - \sigma\|^2 + \|d\tau - \Pi_{\Lambda^*} d\tau\|^2 \\
 &= \|d(\alpha + \tau) - \sigma\|^2 + \|d\tau - \Pi_{\Lambda^*} d\tau\|^2,
 \end{aligned}$$

where the third equality follows from the fact that $d\alpha$ and σ are in the range of Π_{Λ^*} , and that Π_{Λ^*} is an orthogonal projection. By the translation invariance of Lebesgue measure on $\mathcal{U}_0$, this shows that

$$\begin{aligned}
 \int_{\mathcal{U}_0} e^{-\frac{1}{2\beta} \|d\alpha - \sigma\|^2} \rho_N^0(\alpha) d\lambda(\alpha) &= \int_{\mathcal{U}_0} e^{-\frac{1}{2\beta} \|d(\alpha + \tau) - \sigma\|^2} \rho_N^0(\alpha + \tau) d\lambda(\alpha) \\
 &= e^{\frac{1}{2\beta} \|d\tau - \Pi_{\Lambda^*} d\tau\|^2} \int_{\mathcal{U}_0} e^{-\frac{1}{2\beta} \|d\alpha - \kappa\|^2} \rho_N^0(\alpha + \tau) d\lambda(\alpha).
 \end{aligned} \tag{16.7.22}$$

But note that

$$\begin{aligned}
 \|d\alpha - \kappa\|^2 &= \|d\alpha\|^2 - 2(d\alpha, \kappa) + \|\kappa\|^2 = \|d\alpha\|^2 + 2(d\alpha, \Pi_{\Lambda^*} \delta d \Delta_{\Lambda^*}^{-1} \sigma) + \|\kappa\|^2 \\
 &= \|d\alpha\|^2 + 2(d\alpha, \delta d \Delta_{\Lambda^*}^{-1} \sigma) + \|\kappa\|^2 = \|d\alpha\|^2 + 2(dd\alpha, d\Delta_{\Lambda^*}^{-1} \sigma) + \|\kappa\|^2 \\
 &= \|d\alpha\|^2 + \|\kappa\|^2.
 \end{aligned}$$

Combining this with equations (16.7.21) and (16.7.22), we get

$$\begin{aligned}
 \mathbb{E}[D_\ell(\eta_0) \rho_N^0(\eta_0)] &= \frac{1}{Z_0} e^{\frac{1}{2\beta} \|d\tau - \Pi_{\Lambda^*} d\tau\|^2 - \frac{1}{2\beta} \|\kappa\|^2} \int_{\mathcal{U}_0} e^{-\frac{1}{2\beta} \|d\alpha\|^2} \rho_N^0(\alpha + \tau) d\lambda(\alpha) \\
 &= e^{\frac{1}{2\beta} \|d\tau - \Pi_{\Lambda^*} d\tau\|^2 - \frac{1}{2\beta} \|\kappa\|^2} \mathbb{E}[\rho_N^0(\eta_0 + \tau)].
 \end{aligned}$$

Note that by the definition (16.7.18) of ρ_N^0 , we have $\rho_N^0(\eta_0 + \tau) = \rho_N^0(\eta_0 + \tau_0)$. Also, by

Lemma 16.7.10, it follows that $\tau_0 \in \mathcal{U}_0$. This completes the proof. $\square$

16.7.7 The renormalization transform

Let η_0 be a random 1-form with law γ_0 . Fix some N . We will now introduce certain ‘renormalization transforms’ $\xi \rightarrow \bar{\xi}$ and $K_\gamma^N(\xi) \rightarrow R_\gamma^N(\bar{\xi})$ such that for any $\gamma \in \Gamma_N$ and real numbers $(\theta_\xi)_{\xi \in \mathcal{E}_\gamma^0}$, we have

$$\mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_0, \xi) - \theta_\xi)) \right] = \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_\gamma^N(\xi) \cos(2\pi(\eta_0, \bar{\xi}) - \theta_\xi)) \right],$$

with the idea being that $R_\gamma^N(\xi)$ is ‘small’ (whereas $K_\gamma^N(\xi)$ can be very large).

Defining the transforms requires some work, which we now commence. For $\epsilon > 0$, let η_ϵ be a random 1-form with law γ_ϵ . Recall that P^* denotes the set of plaquettes with all vertices in Λ^* . Note that each $e \in E_0^*$ is contained in exactly 6 elements of P^* . For $e \in E_0^*$, define the random variable

$$\bar{\eta}_\epsilon(e) := \frac{1}{6} \sum_{p \in P^*, p \ni e} \sum_{e' \in p, e' \neq e} s_{e,e'} \eta_\epsilon(e'),$$

where $s_{e,e'} = -1$ if e and e' are both oriented positively or both oriented negatively when traversing p as a loop, and $s_{e,e'} = 1$ otherwise. Notice that $\bar{\eta}_\epsilon(e)$ is a function of $\{\eta_\epsilon(e') : e' \neq e, e' \text{ shares a plaquette with } e\}$. We extend the definition of $s_{e,e'}$ by setting $s_{e,e'} = 0$ if e, e' do not share a plaquette.

Lemma 16.7.24. *Take any $\epsilon > 0$ and $e \in E_0^*$. Conditionally on $\{\eta_\epsilon(e')\}_{e' \in E_0^* \setminus \{e\}}$, $\eta_0(e)$ is distributed as a normal random variable with mean $6\bar{\eta}_\epsilon(e)/(6 + \epsilon)$ and variance $\beta/(6 + \epsilon)$.*

Proof. From the form of the density of η_ϵ , we see that the conditional density of $\eta_\epsilon(e)$ given $\{\eta_\epsilon(e')\}_{e' \in E_0^* \setminus \{e\}}$ at a point $x \in \mathbb{R}$ is proportional to

$$\begin{aligned} & \exp \left(-\frac{1}{2\beta} \left(6x^2 - 2 \sum_{p \in P^*, p \ni e} \sum_{e' \in p, e' \neq e} x s_{e,e'} \eta_\epsilon(e') \right) - \frac{\epsilon x^2}{2\beta} \right) \\ &= \exp \left(-\frac{1}{2\beta} (6x^2 - 12x\bar{\eta}_\epsilon(e)) - \frac{\epsilon x^2}{2\beta} \right) \\ &= \exp \left(-\frac{6 + \epsilon}{2\beta} \left(x - \frac{6\bar{\eta}_\epsilon(e)}{6 + \epsilon} \right)^2 + \frac{(6\bar{\eta}_\epsilon(e))^2}{6 + \epsilon} \right). \end{aligned}$$

Clearly, this is proportional to the density of the normal distribution with mean $6\bar{\eta}_\epsilon(e)/(6 + \epsilon)$ and variance $\beta/(6 + \epsilon)$. $\square$

Take any $\gamma \in \Gamma_N$ and $\xi \in \mathcal{E}_\gamma^0$. Consider the following undirected graph structure on $E(\xi)$. We will say that two distinct edges $e, e' \in E(\xi)$ are neighbors if $\text{dist}(e, e') < \sqrt{2}$. The maximum degree of this graph is at most a universal constant c . From this, it follows easily (by the greedy algorithm) that this graph can be colored with $c + 1$ colors, such that

no two neighboring nodes receive the same color. Thus, we can write $E(\xi)$ as the disjoint union of $c + 1$ subsets, such that no two edges within the same subset share a plaquette. Consequently, we can find $\mathcal{B}_\xi \subseteq E(\xi)$ such that $\text{dist}(e, e') \geq \sqrt{2}$ for any distinct $e, e' \in \mathcal{B}_\xi$, and

$$\sum_{e \in \mathcal{B}_\xi} \xi(e)^2 \geq \frac{1}{c+1} \|\xi\|^2. \quad (16.7.23)$$

Fix $\mathcal{B}_\xi$ as above, and let $\mathcal{B}_\xi^c := E(\xi) \setminus \mathcal{B}_\xi$. We are now ready to define the renormalization transforms. For $\xi \in \mathcal{E}_\gamma^0$, we define

$$\bar{\xi}(e) := \xi(e) + \frac{1}{6} \sum_{p \in P^*, p \ni e} \sum_{e' \in p, e' \in \mathcal{B}_\xi} s_{e', e} \xi(e').$$

if $e \in \mathcal{B}_\xi^c$, and $\bar{\xi}(e) = 0$ if $e \in \mathcal{B}_\xi$; and we define

$$R_\gamma^N(\xi) := K_\gamma^N(\xi) \exp\left(-\frac{\pi^2 \beta}{3} \sum_{e \in \mathcal{B}_\xi} \xi(e)^2\right). \quad (16.7.24)$$

We will now prove the following lemma.

Lemma 16.7.25. *If $\bar{\xi}$ and $R_\gamma^N(\xi)$ are defined as above, then we have for each $\gamma \in \Gamma_N$,*

$$\mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_0, \xi) - \theta_\xi)) \right] = \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_\gamma^N(\xi) \cos(2\pi(\eta_0, \bar{\xi}) - \theta_\xi)) \right].$$

The first step towards the proof of Lemma 16.7.25 is the following.

Lemma 16.7.26. *If $\xi \in \mathcal{U}_0$, then $\bar{\xi} \in \mathcal{U}_0$.*

Proof. By Lemma 16.7.10, we have to show that $\delta \bar{\xi}(x) = 0$ for all $x \in \Lambda_0^*$. Take any such x . If none of the edges incident to x are in $\mathcal{B}_\xi$, then $\delta \bar{\xi}(x) = \delta \xi(x) = 0$, since $\xi \in \mathcal{U}_0$. Since $\text{dist}(e, e') \geq \sqrt{2}$ for any distinct $e, e' \in \mathcal{B}_\xi$, the only other possibility is that exactly one edge $e \in \mathcal{B}_\xi$ is incident to x , and for any other edge e' incident to x , either e' does not share a plaquette with any element of $\mathcal{B}_\xi$, or e is the only element of $\mathcal{B}_\xi$ with which e' shares a plaquette. Thus, for any such e' ,

$$\bar{\xi}(e') := \xi(e') + \frac{1}{6} s_{e, e'} \xi(e).$$

It is easy to see from the definition of the coderivative operator that

$$\delta \bar{\xi}(x) = \sum_{e' \ni x} t_{e', x} \bar{\xi}(e'),$$

where $t_{e', x} = 1$ if e' is entering into x and -1 if e' is exiting from x . Thus, by the previous

display, we get

$$\begin{aligned}\delta\bar{\xi}(x) &= \sum_{e' \ni x, e' \neq e} t_{e',x} \left(\xi(e') + \frac{1}{6} s_{e,e'} \xi(e) \right) \\ &= \sum_{e' \ni x, e' \neq e} t_{e',x} \xi(e') + \frac{1}{6} \sum_{e' \ni x, e' \neq e} t_{e',x} s_{e,e'} \xi(e).\end{aligned}$$

But $s_{e,e'} = t_{e,x} t_{e',x}$ if e, e' share a plaquette, and zero otherwise. Therefore, we get

$$\begin{aligned}\delta\bar{\xi}(x) &= \sum_{e' \ni x, e' \neq e} t_{e',x} \xi(e') + \frac{1}{6} \sum_{e' \ni x, e' \neq e} t_{e,x} s_{e,e'} \xi(e) \\ &= \sum_{e' \ni x} t_{e',x} \xi(e') = \delta\xi(x) = 0.\end{aligned}$$

This proves the claim. $\square$

The second step is to prove an ‘ ϵ -version’ of Lemma 16.7.25. Define ‘ ϵ -versions’ of $\bar{\xi}$ and $R_\gamma^N(\xi)$ as:

$$\bar{\xi}_\epsilon(e) := \xi(e) + \frac{1}{6 + \epsilon} \sum_{p \in P^*, p \ni e} \sum_{e' \in p, e' \in \mathcal{B}_\xi} s_{e',e} \xi(e').$$

if $e \in \mathcal{B}_\xi^c$, and $\bar{\xi}_\epsilon(e) := 0$ if $e \in \mathcal{B}_\xi$; and

$$R_{\gamma,\epsilon}^N(\xi) := K_\gamma^N(\xi) \exp\left(-\frac{2\pi^2\beta}{6 + \epsilon} \sum_{e \in \mathcal{B}_\xi} \xi(e)^2\right).$$

Lemma 16.7.27. *With $\bar{\xi}_\epsilon$ and $K_{\gamma,\epsilon}^N(\xi)$ defined as above, we have*

$$\mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_\epsilon, \xi) - \theta_\xi)) \right] = \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_{\gamma,\epsilon}^N(\xi) \cos(2\pi(\eta_\epsilon, \bar{\xi}_\epsilon) - \theta_\xi)) \right].$$

Proof. Let us begin by expanding

$$\begin{aligned}& \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_\epsilon, \xi) - \theta_\xi)) \right] \\ &= \sum_{\mathcal{E} \subseteq \mathcal{E}_\gamma^0} \mathbb{E} \left[\prod_{\xi \in \mathcal{E}} K_\gamma^N(\xi) \cos(2\pi(\eta_\epsilon, \xi) - \theta_\xi) \right] \\ &= \sum_{\mathcal{E} \subseteq \mathcal{E}_\gamma^0} \left(\prod_{\xi \in \mathcal{E}} K_\gamma^N(\xi) \right) \left\{ 2^{-|\mathcal{E}|} \sum_{\sigma \in \{-1,1\}^\mathcal{E}} \mathbb{E} \left[\exp \left(i \sum_{\xi \in \mathcal{E}} (2\pi\sigma_\xi(\eta_\epsilon, \xi) - \sigma_\xi\theta_\xi) \right) \right] \right\}.\end{aligned}$$

Take any nonempty $\mathcal{E} \subseteq \mathcal{E}_\gamma^0$. Let

$$\mathcal{B}_\mathcal{E} := \bigcup_{\xi \in \mathcal{E}} \mathcal{B}_\xi,$$

and let $\mathcal{B}_\mathcal{E}^c := E_0^* \setminus \mathcal{B}_\mathcal{E}$. Note that since $\mathcal{E}_\gamma^0$ is an ensemble, the right side above is a disjoint union. Next, note that by construction, no two elements of $\mathcal{B}_\mathcal{E}$ can share a plaquette. Also, since $\mathcal{E}_\gamma^0$ is a 1-ensemble, no two elements from two different $\mathcal{B}_\mathcal{E}$ and $\mathcal{B}_{\mathcal{E}'}$ can share a plaquette. Thus, no two elements from $\mathcal{B}_\mathcal{E}$ can share a plaquette. Therefore by Lemma 16.7.24,

$$\begin{aligned} & \mathbb{E} \left[\exp \left(2\pi i \sum_{\xi \in \mathcal{E}} \sum_{e \in \mathcal{B}_\xi} \sigma_\xi \eta_e(e) \xi(e) \right) \middle| \{ \eta_e(e) \}_{e \in \mathcal{B}_\mathcal{E}^c} \right] \\ &= \prod_{\xi \in \mathcal{E}} \prod_{e \in \mathcal{B}_\xi} \mathbb{E} [e^{2\pi i \sigma_\xi \eta_e(e) \xi(e)} | \{ \eta_e(e) \}_{e \in \mathcal{B}_\mathcal{E}^c}] \\ &= \prod_{\xi \in \mathcal{E}} \prod_{e \in \mathcal{B}_\xi} \exp \left(\frac{12\pi i \sigma_\xi \bar{\eta}_e(e) \xi(e)}{6 + \epsilon} - \frac{2\pi^2 \beta \xi(e)^2}{6 + \epsilon} \right). \end{aligned}$$

This implies that

$$\begin{aligned} & \mathbb{E} \left[\exp \left(i \sum_{\xi \in \mathcal{E}} (2\pi \sigma_\xi (\eta_\epsilon, \xi) - \sigma_\xi \theta_\xi) \right) \right] \\ &= \mathbb{E} \left[\exp \left\{ i \sum_{\xi \in \mathcal{E}} \left(2\pi \sigma_\xi (\tilde{\eta}_{\epsilon, \xi}, \xi) - \frac{2\pi^2 \beta}{6 + \epsilon} \sum_{e \in \mathcal{B}_\xi} \xi(e)^2 - \sigma_\xi \theta_\xi \right) \right\} \right], \end{aligned}$$

where

$$\tilde{\eta}_{\epsilon, \xi}(e) := \begin{cases} \eta_e(e) & \text{if } e \in \mathcal{B}_\xi^c, \\ \frac{6}{6 + \epsilon} \bar{\eta}_e(e) & \text{if } e \in \mathcal{B}_\xi. \end{cases}$$

Now note that

$$\begin{aligned} (\tilde{\eta}_{\epsilon, \xi}, \xi) &= \sum_{e \in \mathcal{B}_\xi^c} \eta_e(e) \xi(e) + \frac{6}{6 + \epsilon} \sum_{e \in \mathcal{B}_\xi} \bar{\eta}_e(e) \xi(e) \\ &= \sum_{e \in \mathcal{B}_\xi^c} \eta_e(e) \xi(e) + \frac{1}{6 + \epsilon} \sum_{e \in \mathcal{B}_\xi} \sum_{p \in P^*, p \ni e} \sum_{e' \in p, e' \neq e} s_{e, e'} \eta_e(e') \xi(e) \\ &= \sum_{e \in \mathcal{B}_\xi^c} \eta_e(e) \xi(e) + \frac{1}{6 + \epsilon} \sum_{e' \in \mathcal{B}_\xi^c} \sum_{p \in P^*, p \ni e'} \sum_{e \in p, e \in \mathcal{B}_\xi} s_{e, e'} \eta_e(e') \xi(e) \\ &= (\eta_\epsilon, \bar{\xi}_\epsilon), \end{aligned}$$

where Combining all of the above, we get

$$\begin{aligned} & 2^{-|\mathcal{E}|} \sum_{\sigma \in \{-1, 1\}^\mathcal{E}} \mathbb{E} \left[\exp \left(i \sum_{\xi \in \mathcal{E}} (2\pi \sigma_\xi (\eta_\epsilon, \xi) - \sigma_\xi \theta_\xi) \right) \right] \\ &= \exp \left(-\frac{2\pi^2 \beta}{6 + \epsilon} \sum_{\xi \in \mathcal{E}} \sum_{e \in \mathcal{B}_\xi} \xi(e)^2 \right) \mathbb{E} \left[\prod_{\xi \in \mathcal{E}} \cos(2\pi (\eta_\epsilon, \bar{\xi}_\epsilon) - \theta_\xi) \right]. \end{aligned}$$

Tracing back the steps, it is clear that this completes the proof. $\square$

We are now ready to prove Lemma 16.7.25.

Proof of Lemma 16.7.25. Let $\eta_{0,\epsilon}$ and $\eta_{1,\epsilon}$ be the orthogonal projections of η_0 on $\mathcal{U}_0$ and $\mathcal{V}_0$. Now, any $\xi \in \mathcal{E}_\gamma^0$ is in $\mathcal{U}_0$. This implies that $(\eta_{0,\epsilon}, \xi) = (\eta_\epsilon, \xi)$. Thus, by Lemma 16.7.15, we get

$$\begin{aligned} & \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_0, \xi) - \theta_\xi)) \right] \\ &= \lim_{\epsilon \rightarrow 0} \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_{0,\epsilon}, \xi) - \theta_\xi)) \right] \\ &= \lim_{\epsilon \rightarrow 0} \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_\epsilon, \xi) - \theta_\xi)) \right]. \end{aligned}$$

Applying Lemma 16.7.27 to the last line, we get

$$\begin{aligned} & \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + K_\gamma^N(\xi) \cos(2\pi(\eta_0, \xi) - \theta_\xi)) \right] \\ &= \lim_{\epsilon \rightarrow 0} \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_{\gamma,\epsilon}^N(\xi) \cos(2\pi(\eta_\epsilon, \bar{\xi}_\epsilon) - \theta_\xi)) \right]. \end{aligned}$$

From the definitions of $\bar{\xi}$ and $\bar{\xi}_\epsilon$, it is easy to see that $\|\bar{\xi} - \bar{\xi}_\epsilon\| \leq C\epsilon\|\xi\|$, where C has no dependence on ϵ or ξ . Also, by Lemma 16.7.26,

$$\begin{aligned} (\eta_\epsilon, \bar{\xi}_\epsilon) &= (\eta_\epsilon, \bar{\xi}) + (\eta_\epsilon, \bar{\xi}_\epsilon - \bar{\xi}) \\ &= (\eta_{0,\epsilon}, \bar{\xi}) + (\eta_\epsilon, \bar{\xi}_\epsilon - \bar{\xi}). \end{aligned}$$

This gives

$$\begin{aligned} |(\eta_\epsilon, \bar{\xi}_\epsilon) - (\eta_{0,\epsilon}, \bar{\xi})| &= |(\eta_\epsilon, \bar{\xi}_\epsilon - \bar{\xi})| \\ &\leq \|\eta_\epsilon\| \|\bar{\xi}_\epsilon - \bar{\xi}\| \leq C\epsilon\|\xi\| \|\eta_\epsilon\| \leq C\epsilon\|\xi\| (\|\eta_{0,\epsilon}\| + \|\eta_{1,\epsilon}\|). \end{aligned}$$

By Lemma 16.7.15, we see that the upper bound on the right tends to zero in probability as $\epsilon \rightarrow 0$. From this, and another application of Lemma 16.7.15, we get

$$\begin{aligned} & \lim_{\epsilon \rightarrow 0} \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_{\gamma,\epsilon}^N(\xi) \cos(2\pi(\eta_\epsilon, \bar{\xi}_\epsilon) - \theta_\xi)) \right] \\ &= \lim_{\epsilon \rightarrow 0} \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_{\gamma,\epsilon}^N(\xi) \cos(2\pi(\eta_{0,\epsilon}, \bar{\xi}) - \theta_\xi)) \right] \\ &= \mathbb{E} \left[\prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_\gamma^N(\xi) \cos(2\pi(\eta_0, \bar{\xi}) - \theta_\xi)) \right]. \end{aligned}$$

This completes the proof. $\square$

16.7.8 The key step

Recall Theorem 16.7.8, which allowed us to express $\langle W_\ell \rangle$ as the expectation $\langle D_\ell \rangle^*$ under the dual model, and Lemma 16.7.23, which let us express $\langle D_\ell \rangle^*$ as a limit involving the function ρ_N^0 defined in equation (16.7.18). Lemma 16.7.25 gives an alternate expression for ρ_N^0 , which yields the following corollary.

Corollary 16.7.28. *Let $\bar{\xi}$ and R_γ^N be defined as in the previous subsection, and define*

$$\rho_N^1(\alpha) := \sum_{\gamma \in \Gamma_N} c_\gamma \prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_\gamma^N(\xi) \cos(2\pi(\alpha, \bar{\xi}))). \quad (16.7.25)$$

Let η_0 , τ , τ_0 , and κ be as in Lemma 16.7.23. Then

$$\langle W_\ell \rangle = e^{\frac{1}{2\beta} \|d\tau - \Pi_{\Lambda^*} d\tau\|^2 - \frac{1}{2\beta} \|\kappa\|^2} \lim_{N \rightarrow \infty} \frac{\mathbb{E}[\rho_N^1(\eta_0 + \tau)]}{\mathbb{E}[\rho_N^1(\eta_0)]}.$$

Proof. Taking $\theta_\xi = 0$ for all ξ in Lemma 16.7.25, we get $\mathbb{E}[\rho_N^0(\eta_0)] = \mathbb{E}[\rho_N^1(\eta_0)]$. Similarly, taking $\theta_\xi = 2\pi(\tau, \xi)$ in Lemma 16.7.25, we get $\mathbb{E}[\rho_N^0(\eta_0 + \tau)] = \mathbb{E}[\rho_N^1(\eta_0 + \tau)]$. By Lemma 16.7.25 and Theorem 16.7.8, this completes the proof. $\square$

The advantage of Corollary 16.7.28 over Lemma 16.7.25 is that the function ρ_N^1 does not become more and more singular as $N \rightarrow \infty$. We will now work towards establishing this claim. We now make a specific choice for the sequence $\{z_q\}_{q \geq 1}$ introduced in Subsection 16.7.5. Recall that $\{z_q\}_{q \geq 1}$ can be any sequence of positive real numbers satisfying the condition

$$\sum_{q=1}^{\infty} \frac{2}{z_q} = 1.$$

Now consider the form $z_q = e^{\lambda q^2}$. Then the above sum is a strictly decreasing continuous function of λ , which goes to infinity as $\lambda \rightarrow 0$ and to 0 as $\lambda \rightarrow \infty$. Thus, there is a unique λ such that the above sum is equal to one. We define $z_q := e^{\lambda q^2}$ with this λ . Note that λ is a universal constant.

The first step towards showing the stability of ρ_N^1 is to show that the coefficients $R_\gamma^N(\xi)$ are not allowed to blow up as N grows, unlike the coefficients $K_\gamma^N(\xi)$. The following lemma shows that this happens if β is greater than a universal constant.

Lemma 16.7.29. *For any $N \geq 1$, $\gamma \in \Gamma_N$, and $\xi \in \mathcal{E}_\gamma^0$, we have*

$$0 < R_\gamma^N(\xi) \leq e^{(C_0 - C_1\beta)\|\xi\|^2},$$

where C_0, C_1 are positive universal constants.

Proof. From Corollary 16.7.21, recall that

$$0 < K_\gamma^N(\xi) \leq 3^{N_1(\xi)} \prod_{e \in E(\xi)} z_{|\xi(e)|},$$

where $N_1(\xi)$ is the number of edges in E_0^* which at distance ≤ 1 from $E(\xi)$. With our choice of z_q , this gives

$$0 < K_\gamma^N(\xi) \leq 3^{N_1(\xi)} e^{\lambda \|\xi\|^2}.$$

Now, $N_1(\xi) \leq C|E(\xi)|$, where C a universal constant. Also, $|E(\xi)| \leq \|\xi\|^2$, since $|\xi(e)| \geq 1$ for each $e \in E(\xi)$. Since λ is a universal constant, these observations show that there is a universal constant C_0 such that

$$0 < K_\gamma^N(\xi) \leq e^{C_0 \|\xi\|^2}.$$

Now, by the definition (16.7.24) of $R_\gamma^N(\xi)$ and the inequality (16.7.23), we get

$$0 < R_\gamma^N(\xi) \leq K_\gamma^N(\xi) e^{-C_1 \beta \|\xi\|^2},$$

where C_1 is a positive universal constant. The proof is now completed by combining the last two displays. $\square$

The main utility of Lemma 16.7.29 is that it shows that each term in the sum defining ρ_N^1 is a positive function if β is large enough. This allows us to apply Jensen's inequality in a useful way. First, we need the following lemma.

Lemma 16.7.30. *For any $z \in [0, 1)$ and $\alpha, \theta \in \mathbb{R}$, we have*

$$\frac{1 + z \cos(\alpha - \theta)}{1 + z \cos \alpha} \geq \exp\left(E(\alpha, \theta) + O(\alpha, \theta) - \frac{2z^2\theta^2}{(1-z)^2}\right),$$

where

$$E(\alpha, \theta) := \frac{z(\cos \alpha)(\cos \theta - 1)}{1 + z \cos \alpha}, \quad O(\alpha, \theta) := \frac{z \sin \alpha \sin \theta}{1 + z \cos \alpha}.$$

Moreover, we have the bound

$$|E(\alpha, \theta)| \leq \frac{z\theta^2}{2(1-z)}.$$

Proof. Fix $z \in [0, 1)$ and $\alpha, \theta \in \mathbb{R}$. Set

$$R := \frac{1 + z \cos(\alpha - \theta)}{1 + z \cos \alpha}.$$

Since $z \in [0, 1)$ and $\cos x \geq -1$, we have $1 + z \cos x \geq 1 - z > 0$ for all $x \in \mathbb{R}$, hence $R > 0$. Using $\cos(\alpha - \theta) = \cos \alpha \cos \theta + \sin \alpha \sin \theta$, we obtain

$$1 + z \cos(\alpha - \theta) = 1 + z \cos \alpha + z \cos \alpha (\cos \theta - 1) + z \sin \alpha \sin \theta.$$

Dividing by $1 + z \cos \alpha$ gives

$$R = 1 + \frac{z \cos \alpha (\cos \theta - 1)}{1 + z \cos \alpha} + \frac{z \sin \alpha \sin \theta}{1 + z \cos \alpha} = 1 + E(\alpha, \theta) + O(\alpha, \theta),$$

where E and O are as in the statement of the lemma. Next, recall the elementary inequality

$$\log x \geq 1 - \frac{1}{x} \quad (x > 0). \quad (16.7.26)$$

(Indeed, for $f(x) := \log x + \frac{1}{x} - 1$ we have $f'(x) = \frac{x-1}{x^2}$, so f is minimized at $x = 1$ and $f(1) = 0$.) Applying (16.7.26) with $x = R$ yields

$$\log R \geq 1 - \frac{1}{R} = \frac{R-1}{R} = (R-1) - \frac{(R-1)^2}{R}.$$

Since $R-1 = E + O$, this becomes

$$\log R \geq E(\alpha, \theta) + O(\alpha, \theta) - \frac{(R-1)^2}{R}. \quad (16.7.27)$$

Now, observe that

$$R-1 = \frac{z(\cos(\alpha-\theta) - \cos \alpha)}{1 + z \cos \alpha}.$$

Therefore,

$$\begin{aligned} \frac{(R-1)^2}{R} &= \frac{z^2(\cos(\alpha-\theta) - \cos \alpha)^2}{(1 + z \cos \alpha)^2} \cdot \frac{1 + z \cos \alpha}{1 + z \cos(\alpha-\theta)} \\ &= \frac{z^2(\cos(\alpha-\theta) - \cos \alpha)^2}{(1 + z \cos \alpha)(1 + z \cos(\alpha-\theta))}. \end{aligned}$$

Since $1 + z \cos x \geq 1 - z$ for all $x \in \mathbb{R}$, we get

$$\frac{(R-1)^2}{R} \leq \frac{z^2(\cos(\alpha-\theta) - \cos \alpha)^2}{(1-z)^2}. \quad (16.7.28)$$

Next, using the trigonometric identity

$$\begin{aligned} \cos(\alpha-\theta) - \cos \alpha &= \cos\left(\alpha - \frac{\theta}{2} - \frac{\theta}{2}\right) - \cos\left(\alpha - \frac{\theta}{2} + \frac{\theta}{2}\right) \\ &= \cos\left(\alpha - \frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) + \sin\left(\alpha - \frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right) \\ &\quad - \left[\cos\left(\alpha - \frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) - \sin\left(\alpha - \frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right) \right] \\ &= 2 \sin\left(\alpha - \frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right), \end{aligned}$$

we obtain

$$(\cos(\alpha-\theta) - \cos \alpha)^2 \leq 4 \sin^2\left(\frac{\theta}{2}\right) \leq 4 \left(\frac{\theta}{2}\right)^2 = \theta^2,$$

where we used $|\sin u| \leq |u|$. Combining this with (16.7.28) gives

$$\frac{(R-1)^2}{R} \leq \frac{z^2 \theta^2}{(1-z)^2} \leq \frac{2z^2 \theta^2}{(1-z)^2}. \quad (16.7.29)$$

Substituting (16.7.29) into (16.7.27) yields

$$\log R \geq E(\alpha, \theta) + O(\alpha, \theta) - \frac{2z^2\theta^2}{(1-z)^2}.$$

Exponentiating both sides proves the first claim of the lemma. For the second, recall that

$$E(\alpha, \theta) = \frac{z \cos \alpha (\cos \theta - 1)}{1 + z \cos \alpha} = z(1 - \cos \theta) \frac{-\cos \alpha}{1 + z \cos \alpha}.$$

Since $z \in [0, 1)$ and $\cos \alpha \geq -1$, we have $1 + z \cos \alpha \geq 1 - z > 0$. Therefore, the above identity shows that

$$|E(\alpha, \theta)| \leq \frac{z(1 - \cos \theta)}{1 - z}.$$

Finally,

$$1 - \cos \theta = 2 \sin^2 \left(\frac{\theta}{2} \right) \leq 2 \left(\frac{\theta}{2} \right)^2 = \frac{\theta^2}{2},$$

using $|\sin x| \leq |x|$. Combining the last two displays yields

$$|E(\alpha, \theta)| \leq \frac{z}{1-z} \cdot \frac{\theta^2}{2} = \frac{z\theta^2}{2(1-z)}.$$

This completes the proof. $\square$

Lemma 16.7.31. *Let C_0 and C_1 be as in Lemma 16.7.29, and let $\beta_0 := C_0/C_1$. Suppose that $\beta > \beta_0$. Let η_0 and τ_0 be as in Lemma 16.7.23. Let*

$$h(x) := \frac{x}{2(1-x)} + \frac{2x^2}{(1-x)^2}.$$

Then for any N , there are positive weights $(w_{N,\gamma})_{\gamma \in \Gamma_N}$ which add up to 1, and

$$\frac{\mathbb{E}[\rho_N^1(\eta_0 + \tau_0)]}{\mathbb{E}[\rho_N^1(\eta_0)]} \geq \sum_{\gamma \in \Gamma_N} w_{N,\gamma} \prod_{\xi \in \mathcal{E}_\gamma^0} e^{-h(R_\gamma^N(\xi))\theta_\xi^2},$$

where $\theta_\xi := 2\pi(\tau_0, \bar{\xi})$.

Proof. Take any N and any $\gamma \in \Gamma_N$. For $\alpha \in \mathcal{U}_0$, let

$$\Upsilon_{N,\gamma}(\alpha) := \prod_{\xi \in \mathcal{E}_\gamma^0} (1 + R_\gamma^N(\xi) \cos(2\pi(\alpha, \bar{\xi}))). \quad (16.7.30)$$

Since $\beta > \beta_0$, Lemma 16.7.29 shows that $\Upsilon_{N,\gamma}$ is a positive function. Also, clearly, it is a bounded function (although the bound may depend on N and γ). Thus, there is a probability measure $\nu_{N,\gamma}$ on $\mathcal{U}_0$ that has density proportional to $\Upsilon_{N,\gamma}$ with respect to the

law of η_0 . Let Θ be a random variable with law $\nu_{N,\gamma}$. Then note that

$$\mathbb{E}[\Upsilon_{N,\gamma}(\eta_0 + \tau_0)] = Z_{N,\gamma} \mathbb{E} \left[\frac{\Upsilon_{N,\gamma}(\Theta + \tau_0)}{\Upsilon_{N,\gamma}(\Theta)} \right],$$

where $Z_{N,\gamma} := \mathbb{E}[\Upsilon_{N,\gamma}(\eta_0)]$. By Jensen's inequality, this shows that

$$\begin{aligned} \mathbb{E}[\Upsilon_{N,\gamma}(\eta_0 + \tau_0)] &\geq Z_{N,\gamma} \exp \left(\mathbb{E} \left[\log \frac{\Upsilon_{N,\gamma}(\Theta + \tau_0)}{\Upsilon_{N,\gamma}(\Theta)} \right] \right) \\ &= Z_{N,\gamma} \prod_{\xi \in \mathcal{E}_\gamma^0} \exp \left(\mathbb{E} \left[\log \frac{1 + R_\gamma^N(\xi) \cos(2\pi(\Theta + \tau_0, \bar{\xi}))}{1 + R_\gamma^N(\xi) \cos(2\pi(\Theta, \bar{\xi}))} \right] \right). \end{aligned}$$

Define

$$\alpha_\xi := 2\pi(\Theta, \bar{\xi}), \quad \theta_\xi := 2\pi(\tau_0, \bar{\xi}),$$

noting that α_ξ is a random variable and θ_ξ is deterministic. Then by Lemma 16.7.30,

$$\log \frac{1 + R_\gamma^N(\xi) \cos(2\pi(\Theta + \tau_0, \bar{\xi}))}{1 + R_\gamma^N(\xi) \cos(2\pi(\Theta, \bar{\xi}))} \geq E(\alpha_\xi, \theta_\xi) + O(\alpha_\xi, \theta_\xi) - \frac{2R_\gamma^N(\xi)^2 \theta_\xi^2}{(1 - R_\gamma^N(\xi))^2}.$$

Thus, we arrive at the lower bound

$$\mathbb{E}[\Upsilon_{N,\gamma}(\eta_0 + \tau_0)] \geq Z_{N,\gamma} \prod_{\xi \in \mathcal{E}_\gamma^0} \exp \left(\mathbb{E}[E(\alpha_\xi, \theta_\xi)] + \mathbb{E}[O(\alpha_\xi, \theta_\xi)] - \frac{2R_\gamma^N(\xi)^2 \theta_\xi^2}{(1 - R_\gamma^N(\xi))^2} \right).$$

Since η_0 has the same law as $-\eta_0$, and $\Upsilon_{N,\gamma}(\alpha) = \Upsilon_{N,\gamma}(-\alpha)$, we see that Θ has the same law as $-\Theta$. From this it follows that $O(\alpha_\xi, \theta_\xi)$ has the same law as $-O(\alpha_\xi, \theta_\xi)$. Consequently, $\mathbb{E}(O(\alpha_\xi, \theta_\xi)) = 0$. Next, by the bound on $|E(\alpha, \theta)|$ from Lemma 16.7.30, we have

$$|E(\alpha_\xi, \theta_\xi)| \leq \frac{R_\gamma^N(\xi) \theta_\xi^2}{2(1 - R_\gamma^N(\xi))}.$$

Combining, we get

$$\mathbb{E}[\Upsilon_{N,\gamma}(\eta_0 + \tau_0)] \geq Z_{N,\gamma} \prod_{\xi \in \mathcal{E}_\gamma^0} e^{-h(R_\gamma^N(\xi)) \theta_\xi^2},$$

where h is as in the statement of the lemma. Recalling the formula (16.7.25) for ρ_N^1 and the formula (16.7.30) for $\Upsilon_{N,\gamma}$, and applying the above inequality, we see that

$$\begin{aligned} \mathbb{E}[\rho_N^1(\eta_0 + \tau_0)] &= \sum_{\gamma \in \Gamma_N} c_\gamma \mathbb{E}[\Upsilon_{N,\gamma}(\eta_0 + \tau_0)] \\ &= \sum_{\gamma \in \Gamma_N} c_\gamma Z_{N,\gamma} \prod_{\xi \in \mathcal{E}_\gamma^0} e^{-h(R_\gamma^N(\xi)) \theta_\xi^2}. \end{aligned}$$

But again, by the formulas (16.7.25) and (16.7.30),

$$\mathbb{E}[\rho_N^1(\eta_0)] = \sum_{\gamma \in \Gamma_N} c_\gamma \mathbb{E}[\Upsilon_{N,\gamma}(\eta_0)] = \sum_{\gamma \in \Gamma_N} c_\gamma Z_{N,\gamma}.$$

Clearly, this completes the proof. □

Appendix A

Measure Theory and Analysis

A.1 Basics of measure theory

A.1.1 Measure spaces

Let Ω be a set. A σ -algebra $\mathcal{F}$ on Ω is a collection of subsets such that

- $\emptyset \in \mathcal{F}$ (where $\emptyset$ denotes the empty set),
- if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (where A^c denotes the complement of A in Ω), and
- if $A_1, A_2, \dots$ is a countable collection of sets in $\mathcal{F}$, then

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

A pair $(\Omega, \mathcal{F})$, where Ω is a set and $\mathcal{F}$ is a σ -algebra on Ω , is called a measurable space. If the countable union condition is replaced by a finite union condition, then $\mathcal{F}$ is called an algebra instead of a σ -algebra. Note that σ -algebras are also closed under finite unions, since we can append empty sets to convert a finite collection into a countable collection.

Let Ω be a set and $\mathcal{A}$ be a collection of subsets of Ω . It is easy to show that the intersection of any arbitrary collection of σ -algebras is a σ -algebra. The intersection of all σ -algebras containing $\mathcal{A}$ is therefore a σ -algebra (because there is at least one, namely, the power set of Ω), which is the smallest σ -algebra containing $\mathcal{A}$. The smallest σ -algebra containing $\mathcal{A}$ is called the σ -algebra generated by $\mathcal{A}$, and is denoted by $\sigma(\mathcal{A})$. The following is an important class of σ -algebras. Let Ω be a set endowed with a topology. The Borel σ -algebra on Ω is the σ -algebra generated by the collection of open subsets of Ω . It is sometimes denoted by $\mathcal{B}(\Omega)$. In particular, the Borel σ -algebra of the real line $\mathbb{R}$ is the σ -algebra generated by all open subsets of $\mathbb{R}$.

Let $(\Omega, \mathcal{F})$ be a measurable space and take any $\Omega' \in \mathcal{F}$. Consider the set $\mathcal{F}' := \{A \cap \Omega' : A \in \mathcal{F}\}$. It is easy to see that this is a σ -algebra. The σ -algebra $\mathcal{F}'$ is called the restriction of $\mathcal{F}$ to Ω' . The following lemma is sometimes useful.

Lemma A.1.1. *Let $\mathcal{F}'$ be as above. If $\mathcal{F}$ is generated by a collection of sets $\mathcal{A}$, then $\mathcal{F}'$ is generated by the collection $\mathcal{A}' := \{A \cap \Omega' : A \in \mathcal{A}\}$.*

Proof. Since $\mathcal{F}'$ is a σ -algebra and it contains $\mathcal{A}'$, we have that $\sigma(\mathcal{A}') \subseteq \mathcal{F}'$. For the reverse inclusion, take any σ -algebra $\mathcal{G}'$ on Ω' that contains $\mathcal{A}'$. Define

$$\mathcal{G} := \{B \subseteq \Omega : B \cap \Omega' \in \mathcal{G}'\}.$$

An easy check shows that $\mathcal{G}$ is a σ -algebra on Ω . Moreover, $\mathcal{G} \supseteq \mathcal{A}$ since $A \cap \Omega' \in \mathcal{A}' \subseteq \mathcal{G}'$ for all $A \in \mathcal{A}$. Thus, $\mathcal{G} \supseteq \mathcal{F}$. This means that for all $B \in \mathcal{F}$, $B \cap \Omega' \in \mathcal{G}'$. This implies that $\mathcal{F}' \subseteq \mathcal{G}'$. Thus, $\mathcal{F}' \subseteq \sigma(\mathcal{A}')$. $\square$

Let $(\Omega, \mathcal{F})$ be a measurable space. A measure μ on this space is a function from $\mathcal{F}$ into $[0, \infty]$ such that

- $\mu(\emptyset) = 0$, and
- if $A_1, A_2, \dots$ is a countable sequence of disjoint sets in $\mathcal{F}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

The triple $(\Omega, \mathcal{F}, \mu)$ is called a measure space. The second condition above is known as the countable additivity condition. Note that finite additivity is also valid, since we can append empty sets to a finite collection to make it countable. If a measure μ on a measurable space $(\Omega, \mathcal{F})$ satisfies $\mu(\Omega) = 1$, then it is called a probability measure, and the triple $(\Omega, \mathcal{F}, \mu)$ is called a probability space. In this case, elements of $\mathcal{F}$ are often called ‘events’.

A.1.2 Dynkin’s π - λ theorem

Let Ω be a set. A collection $\mathcal{P}$ of subsets of Ω is called a π -system if it is closed under finite intersections. A collection $\mathcal{L}$ of subsets of Ω is called a λ -system (or Dynkin system) if $\Omega \in \mathcal{L}$ and $\mathcal{L}$ is closed under taking complements and countable disjoint unions. The following theorem is a cornerstone of measure theory.

Theorem A.1.2 (Dynkin’s π - λ theorem). *Let Ω be a set. Let $\mathcal{P}$ be a π -system of subsets of Ω , and let $\mathcal{L} \supseteq \mathcal{P}$ be a λ -system of subsets of Ω . Then $\mathcal{L} \supseteq \sigma(\mathcal{P})$.*

We need a small lemma for the proof.

Lemma A.1.3 (Dynkin’s π - λ theorem). *If a λ -system is also a π -system, then it is a σ -algebra.*

Proof. Let $\mathcal{L}$ be a λ -system which is also a π -system. Then $\mathcal{L}$ is closed under complements by definition, and clearly, $\emptyset \in \mathcal{L}$. Suppose that $A_1, A_2, \dots \in \mathcal{L}$. Let $B_1 = A_1$, and for each $i \geq 2$, let

$$B_i = A_i \cap A_1^c \cap A_2^c \cap \dots \cap A_{i-1}^c.$$

Since $\mathcal{L}$ is a λ -system, each A_i^c is in $\mathcal{L}$. Therefore, since $\mathcal{L}$ is also a π -system, each $B_i \in \mathcal{L}$. By construction, $B_1, B_2, \dots$ are disjoint sets and

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i.$$

Thus $\cup A_i \in \mathcal{L}$. This shows that $\mathcal{L}$ is a σ -algebra. $\square$

We are now ready to prove Theorem A.1.2.

Proof of Theorem A.1.2. Since the intersection of all λ -systems containing $\mathcal{P}$ is again a λ -system, we may assume that $\mathcal{L}$ is the smallest λ -system containing $\mathcal{P}$.

Take any $A \in \mathcal{P}$. Let

$$\mathcal{G}_A := \{B \in \mathcal{L} : A \cap B \in \mathcal{L}\}. \quad (\text{A.1.1})$$

Then $\mathcal{G}_A \supseteq \mathcal{P}$ since $\mathcal{P}$ is a π -system and $\mathcal{P} \subseteq \mathcal{L}$. Clearly, $\Omega \in \mathcal{G}_A$. If $B \in \mathcal{G}_A$, then $B^c \in \mathcal{L}$ and

$$A \cap B^c = (A^c \cup (A \cap B))^c \in \mathcal{L},$$

since A^c and $A \cap B$ are disjoint elements of $\mathcal{L}$ and $\mathcal{L}$ is a λ -system. Thus, $B^c \in \mathcal{G}_A$. If $B_1, B_2, \dots$ are disjoint sets in $\mathcal{G}_A$, then

$$A \cap (B_1 \cup B_2 \cup \dots) = (A \cap B_1) \cup (A \cap B_2) \cup \dots \in \mathcal{L},$$

again since $\mathcal{L}$ is a λ -system. Thus, $\mathcal{G}_A$ is a λ -system containing $\mathcal{P}$. By the minimality of $\mathcal{L}$, this shows that $\mathcal{G}_A = \mathcal{L}$. In particular, if $A \in \mathcal{P}$ and $B \in \mathcal{L}$, then $A \cap B \in \mathcal{L}$.

Next, for $A \in \mathcal{L}$, let $\mathcal{G}_A$ be defined as in (A.1.1). By the deduction in the previous paragraph, $\mathcal{G}_A \supseteq \mathcal{P}$. As before, $\mathcal{G}_A$ is a λ -system. Thus, $\mathcal{G}_A = \mathcal{L}$. In particular, $\mathcal{L}$ is a π -system. By Lemma A.1.3, this completes the proof. $\square$

A.1.3 Uniqueness of measures

An important corollary of the π - λ theorem is the following result about uniqueness of measures.

Theorem A.1.4. *Let $\mathcal{P}$ be a π -system. If μ_1 and μ_2 are measures on $\sigma(\mathcal{P})$ that agree on $\mathcal{P}$, and there is a sequence $A_1, A_2, \dots \in \mathcal{P}$ such that A_n increases to Ω and $\mu_1(A_n)$ and $\mu_2(A_n)$ are finite for every n , then $\mu_1 = \mu_2$ on $\sigma(\mathcal{P})$.*

Proof. Take any $A \in \mathcal{P}$ such that $\mu_1(A) = \mu_2(A) < \infty$. Let

$$\mathcal{L} := \{B \in \sigma(\mathcal{P}) : \mu_1(A \cap B) = \mu_2(A \cap B)\}.$$

Clearly, $\Omega \in \mathcal{L}$. If $B \in \mathcal{L}$, then

$$\begin{aligned}\mu_1(A \cap B^c) &= \mu_1(A) - \mu_1(A \cap B) \\ &= \mu_2(A) - \mu_2(A \cap B) = \mu_2(A \cap B^c),\end{aligned}$$

and hence $B^c \in \mathcal{L}$. If $B_1, B_2, \dots \in \mathcal{L}$ are disjoint and B is their union, then

$$\begin{aligned}\mu_1(A \cap B) &= \sum_{i=1}^{\infty} \mu_1(A \cap B_i) \\ &= \sum_{i=1}^{\infty} \mu_2(A \cap B_i) = \mu_2(A \cap B),\end{aligned}$$

and therefore $B \in \mathcal{L}$. This shows that $\mathcal{L}$ is a λ -system. Therefore by the π - λ theorem, $\mathcal{L} = \sigma(\mathcal{P})$. In other words, for every $B \in \sigma(\mathcal{P})$ and $A \in \mathcal{P}$ such that $\mu_1(A) < \infty$, $\mu_1(A \cap B) = \mu_2(A \cap B)$. By the given condition, there is a sequence $A_1, A_2, \dots \in \mathcal{P}$ such that $\mu_1(A_n) < \infty$ for every n and $A_n \uparrow \Omega$. Thus, for any $B \in \sigma(\mathcal{P})$,

$$\mu_1(B) = \lim_{n \rightarrow \infty} \mu_1(A_n \cap B) = \lim_{n \rightarrow \infty} \mu_2(A_n \cap B) = \mu_2(B).$$

This completes the proof of the theorem. $\square$

A.1.4 An approximation theorem for measurable sets

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. In measure theory, $\mu(A)$ is thought of as a measure of the ‘size’ of A . When $\mu(A)$ is small, we think of A as a ‘small’ set. Following this line of thought, we think of $\mu(A \Delta B)$ as a kind of distance between two sets A and B (where $A \Delta B$ is the symmetric difference of A and B). If this is small, then A and B are ‘almost the same set’. The following useful result shows that if μ is a finite measure space (meaning that $\mu(\Omega) < \infty$), then any element of $\mathcal{F}$ can be arbitrarily well-approximated by elements of any generating algebra.

Theorem A.1.5. *Let $(\Omega, \mathcal{F}, \mu)$ be a finite measure space, and let $\mathcal{A}$ be an algebra of sets generating $\mathcal{F}$. Then for any $A \in \mathcal{F}$ and any $\epsilon > 0$, there is some $B \in \mathcal{A}$ such that $\mu(A \Delta B) < \epsilon$, where $A \Delta B$ is the symmetric difference of A and B .*

Proof. Let $\mathcal{G}$ be the collection of all $A \in \mathcal{F}$ for which the stated property holds. Clearly, $\mathcal{A} \subseteq \mathcal{G}$. In particular, $\Omega \in \mathcal{G}$. Take $A \in \mathcal{G}$ and any $\epsilon > 0$. Find $B \in \mathcal{A}$ such that $\mu(A \Delta B) < \epsilon$. Since $A^c \Delta B^c = A \Delta B$, we get $\mu(A^c \Delta B^c) < \epsilon$. Thus, $A^c \in \mathcal{G}$. Finally, take any $A_1, A_2, \dots \in \mathcal{G}$. Let $A := \bigcup_{i=1}^{\infty} A_i$ and take any $\epsilon > 0$. Since μ is a probability measure, there is some n large enough such that

$$\mu\left(A \setminus \bigcup_{i=1}^n A_i\right) < \frac{\epsilon}{2}.$$

For each i , find $B_i \in \mathcal{A}$ such that $\mu(A_i \Delta B_i) < 2^{-i-1}\epsilon$. Let $A' := \bigcup_{i=1}^n A_i$ and $B' := \bigcup_{i=1}^n B_i$.

It is not hard to see that

$$A' \Delta B' \subseteq \bigcup_{i=1}^n (A_i \Delta B_i).$$

Thus,

$$\mu(A' \Delta B') \leq \sum_{i=1}^n \mu(A_i \Delta B_i) \leq \frac{\epsilon}{2}.$$

Again, it is not hard to check that $A \Delta B' \subseteq (A \setminus A') \cup (A' \Delta B')$. Therefore

$$\mu(A \Delta B') \leq \mu(A \setminus A') + \mu(A' \Delta B') < \epsilon.$$

Thus, $A \in \mathcal{G}$. This proves that $\mathcal{G}$ is a σ -algebra, and hence contains $\sigma(\mathcal{A}) = \mathcal{F}$. $\square$

A.1.5 Outer measures

Let Ω be any set and let 2^Ω denote its power set. A function $\phi : 2^\Omega \rightarrow [0, \infty]$ is called an outer measure if it satisfies the following conditions:

- $\phi(\emptyset) = 0$.
- $\phi(A) \leq \phi(B)$ whenever $A \subseteq B$.
- For any $A_1, A_2, \dots \subseteq \Omega$,

$$\phi\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \phi(A_i).$$

Note that there is no σ -algebra in the definition of an outer measure. In fact, we will show below that an outer measure generates its own σ -algebra. If ϕ is an outer measure on a set Ω , a subset $A \subseteq \Omega$ is called ϕ -measurable if for all $B \subseteq \Omega$,

$$\phi(B) = \phi(B \cap A) + \phi(B \cap A^c).$$

Note that A is ϕ -measurable if and only if

$$\phi(B) \geq \phi(B \cap A) + \phi(B \cap A^c)$$

for every B , since the opposite inequality follows by subadditivity. The following result is the most important fact about outer measures.

Theorem A.1.6. *Let Ω be a set and ϕ be an outer measure on Ω . Let $\mathcal{F}$ be the collection of all ϕ -measurable subsets of Ω . Then $\mathcal{F}$ is a σ -algebra and ϕ is a measure on $\mathcal{F}$.*

The proof of Theorem A.1.6 is divided into a sequence of lemmas.

Lemma A.1.7. *The collection $\mathcal{F}$ is an algebra.*

Proof. Clearly, $\emptyset \in \mathcal{F}$, since $\phi(\emptyset) = 0$. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ by the definition of $\mathcal{F}$. If $A, B \in \mathcal{F}$, let $D := A \cup B$ and note that by subadditivity of ϕ , for any E we have

$$\begin{aligned} & \phi(E \cap D) + \phi(E \cap D^c) \\ &= \phi((E \cap A) \cup (E \cap B \cap A^c)) + \phi(E \cap A^c \cap B^c) \\ &\leq \phi(E \cap A) + \phi(E \cap B \cap A^c) + \phi(E \cap A^c \cap B^c). \end{aligned}$$

But since $B \in \mathcal{F}$,

$$\phi(E \cap B \cap A^c) + \phi(E \cap A^c \cap B^c) = \phi(E \cap A^c).$$

Thus,

$$\phi(E \cap D) + \phi(E \cap D^c) \leq \phi(E \cap A) + \phi(E \cap A^c).$$

But since $A \in \mathcal{F}$, the right side equals $\phi(E)$. This completes the proof. $\square$

Lemma A.1.8. *If $A_1, \dots, A_n \in \mathcal{F}$ are disjoint and $E \subseteq \Omega$, then*

$$\phi(E \cap (A_1 \cup \dots \cup A_n)) = \sum_{i=1}^n \phi(E \cap A_i).$$

Proof. For each j , let $B_j := A_1 \cup \dots \cup A_j$. Since $A_n \in \mathcal{F}$,

$$\phi(E \cap B_n) = \phi(E \cap B_n \cap A_n) + \phi(E \cap B_n \cap A_n^c).$$

But since $A_1, \dots, A_n$ are disjoint, $B_n \cap A_n^c = B_{n-1}$ and $B_n \cap A_n = A_n$. Thus,

$$\phi(E \cap B_n) = \phi(E \cap A_n) + \phi(E \cap B_{n-1}).$$

The proof is now completed by induction. $\square$

A consequence of the last two lemmas is the following.

Lemma A.1.9. *If $A_1, A_2, \dots$ is a sequence of sets in $\mathcal{F}$ increasing to a set $A \subseteq \Omega$, then for any $E \subseteq \Omega$,*

$$\phi(E \cap A) = \lim_{n \rightarrow \infty} \phi(E \cap A_n).$$

Proof. By monotonicity of ϕ , the left side dominates the right. For the opposite inequality, let $B_n := A_n \cap (A_1 \cup \dots \cup A_{n-1})^c$ for each n , so that the sets $B_1, B_2, \dots$ are disjoint and $A_n = B_1 \cup \dots \cup B_n$ for each n . By Lemma A.1.7, $B_n \in \mathcal{F}$ for each n . Thus, by Lemma A.1.8,

$$\phi(E \cap A_n) = \sum_{i=1}^n \phi(E \cap B_i).$$

Consequently,

$$\lim_{n \rightarrow \infty} \phi(E \cap A_n) = \sum_{i=1}^{\infty} \phi(E \cap B_i).$$

Since ϕ is countably subadditive, this completes the proof of the lemma. $\square$

We are now ready to prove Theorem A.1.6.

Proof of Theorem A.1.6. Let $A_1, A_2, \dots \in \mathcal{F}$ and let $A := \cup A_i$. For each n , let

$$B_n := \bigcup_{i=1}^n A_i.$$

Take any $E \subseteq \Omega$ and any n . By Lemma A.1.7, $B_n \in \mathcal{F}$ and hence

$$\phi(E) = \phi(E \cap B_n) + \phi(E \cap B_n^c).$$

By monotonicity of ϕ , $\phi(E \cap B_n^c) \geq \phi(E \cap A^c)$. Thus,

$$\phi(E) \geq \phi(E \cap B_n) + \phi(E \cap A^c).$$

By Lemma A.1.9,

$$\lim_{n \rightarrow \infty} \phi(E \cap B_n) = \phi(E \cap A).$$

Thus, $A \in \mathcal{F}$. Together with Lemma A.1.7, this shows that $\mathcal{F}$ is a σ -algebra. To show that ϕ is a measure on $\mathcal{F}$, take any disjoint collection of sets $A_1, A_2, \dots \in \mathcal{F}$. Let B be the union of these sets, and let $B_n = A_1 \cup \dots \cup A_n$. By the monotonicity of ϕ and Lemma A.1.8,

$$\phi(B) \geq \phi(B_n) = \sum_{i=1}^n \phi(A_i).$$

Letting $n \rightarrow \infty$, we get $\phi(B) \geq \sum \phi(A_i)$. On the other hand, subadditivity of ϕ gives the opposite inequality. This shows that ϕ is a measure on $\mathcal{F}$ and completes the proof. $\square$

A.1.6 Carathéodory's extension theorem

Let Ω be a set and $\mathcal{A}$ be an algebra of subsets of Ω . A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is called a measure on $\mathcal{A}$ if

- $\mu(\emptyset) = 0$, and
- if $A_1, A_2, \dots$ is a countable collection of disjoint elements of $\mathcal{A}$ such that their union is also in $\mathcal{A}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

A measure μ on an algebra $\mathcal{A}$ is called σ -finite if there is a countable family of sets $A_1, A_2, \dots \in \mathcal{A}$ such that $\mu(A_i) < \infty$ for each i , and $\Omega = \cup_{i=1}^{\infty} A_i$.

Theorem A.1.10 (Carathéodory's extension theorem). *If $\mathcal{A}$ is an algebra of subsets of a set Ω and μ is a measure on $\mathcal{A}$, then μ has an extension to $\sigma(\mathcal{A})$. Moreover, if μ is σ -finite on $\mathcal{A}$, then the extension is unique.*

The plan of the proof is to construct an outer measure on Ω that agrees with μ on $\mathcal{A}$. Then Theorem A.1.6 will give the required extension. The outer measure is defined as follows. For each $A \subseteq \Omega$, let

$$\mu^*(A) := \inf \left\{ \sum_{i=1}^{\infty} \mu(A_i) : A_1, A_2, \dots \in \mathcal{A}, A \subseteq \bigcup_{i=1}^{\infty} A_i \right\}. \quad (\text{A.1.2})$$

The next lemma establishes that μ^* is an outer measure on Ω .

Lemma A.1.11. *The functional μ^* is an outer measure on Ω .*

Proof. It is obvious from the definition of μ^* that $\mu^*(\emptyset) = 0$ and μ^* is monotone. For proving subadditivity, take any sequence of sets $\{A_i\}_{i \geq 1}$ and let $A := \bigcup A_i$. Fix some $\epsilon > 0$, and for each i , let $\{A_{ij}\}_{j=1}^{\infty}$ be a collection of elements of $\mathcal{A}$ such that $A_i \subseteq \bigcup_j A_{ij}$ and

$$\sum_{j=1}^{\infty} \mu(A_{ij}) \leq \mu^*(A_i) + 2^{-i}\epsilon.$$

Then $\{A_{ij}\}_{i,j=1}^{\infty}$ is a countable cover for A , and so

$$\begin{aligned} \mu^*(A) &\leq \sum_{i,j=1}^{\infty} \mu(A_{ij}) \\ &\leq \sum_{i=1}^{\infty} (\mu^*(A_i) + 2^{-i}\epsilon) = \epsilon + \sum_{i=1}^{\infty} \mu^*(A_i). \end{aligned}$$

Since ϵ is arbitrary, this completes the proof of the lemma. $\square$

The next lemma shows that μ^* is a viable candidate for an extension.

Lemma A.1.12. *For $A \in \mathcal{A}$, $\mu^*(A) = \mu(A)$.*

Proof. Take any $A \in \mathcal{A}$. By definition, $\mu^*(A) \leq \mu(A)$. Conversely, take any $A_1, A_2, \dots \in \mathcal{A}$ such that $A \subseteq \bigcup A_i$. Then $A = \bigcup (A \cap A_i)$. Note that each $A \cap A_i \in \mathcal{A}$, and their union is A , which is also in $\mathcal{A}$. It is easy to check that countable subadditivity continues to be valid for measures on algebras, provided that the union belongs to the algebra. Thus, we get

$$\mu(A) \leq \sum_{i=1}^{\infty} \mu(A \cap A_i) \leq \sum_{i=1}^{\infty} \mu(A_i).$$

This shows that $\mu(A) \leq \mu^*(A)$. $\square$

We are now ready to prove Carathéodory's extension theorem.

Proof of Theorem A.1.10. Let $\mathcal{A}^*$ be the set of all μ^* -measurable sets. By Theorem A.1.6, we know that $\mathcal{A}^*$ is a σ -algebra and that μ^* is a measure on $\mathcal{A}^*$. We now claim that $\mathcal{A} \subseteq \mathcal{A}^*$. To prove this, take any $A \in \mathcal{A}$ and $E \subseteq \Omega$. Let $A_1, A_2, \dots$ be any sequence of elements of

$\mathcal{A}$ that cover E . Then $\{A \cap A_i\}_{i=1}^\infty$ is a cover for $E \cap A$ and $\{A^c \cap A_i\}_{i=1}^\infty$ is a cover for $E \cap A^c$. Consequently,

$$\begin{aligned} \mu^*(E \cap A) + \mu^*(E \cap A^c) &\leq \sum_{i=1}^\infty (\mu(A \cap A_i) + \mu(A^c \cap A_i)) \\ &= \sum_{i=1}^\infty \mu(A_i). \end{aligned}$$

Taking infimum over all choices of $\{A_i\}_{i=1}^\infty$, this shows that $\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(E)$, which means that $A \in \mathcal{A}^*$. Thus, $\mathcal{A} \subseteq \mathcal{A}^*$. This proves the existence part of the theorem. If μ is σ -finite on $\mathcal{A}$, then the uniqueness of the extension follows from Theorem A.1.4. $\square$

A.1.7 Construction of Lebesgue measure

Let $\mathcal{A}$ be the set of all subsets of $\mathbb{R}$ that are finite disjoint unions of half-open intervals of the form $(a, b] \cap \mathbb{R}$, where $-\infty \leq a \leq b \leq \infty$. Here we write $(a, b] \cap \mathbb{R}$ to ensure that the interval is (a, ∞) if $b = \infty$. If $a = b$, the interval is empty. It is easy to show that $\mathcal{A}$ is an algebra of sets, and that it generates the Borel σ -algebra of $\mathbb{R}$. Define a functional $\lambda : \mathcal{A} \rightarrow \mathbb{R}$ as:

$$\lambda\left(\bigcup_{i=1}^n (a_i, b_i] \cap \mathbb{R}\right) := \sum_{i=1}^n (b_i - a_i),$$

where remember that $(a_1, b_1], \dots, (a_n, b_n]$ are disjoint. In other words, λ measures the length of an element of $\mathcal{A}$, as understood in the traditional sense. It is obvious that λ is finitely additive on $\mathcal{A}$ (that is, $\lambda(A_1 \cup \dots \cup A_n) = \lambda(A_1) + \dots + \lambda(A_n)$ when $A_1, \dots, A_n$ are disjoint elements of $\mathcal{A}$). It is also obvious that λ is monotone, that is, $\lambda(A) \leq \lambda(B)$ when $A \subseteq B$ (just observe that B is the disjoint union of A and $B \setminus A$, and apply finite additivity).

Lemma A.1.13. *For any $A_1, \dots, A_n \in \mathcal{A}$ and any $A \subseteq A_1 \cup \dots \cup A_n$, $\lambda(A) \leq \sum \lambda(A_i)$.*

Proof. Let $B_1 = A_1$ and $B_i = A_i \setminus (A_1 \cup \dots \cup A_{i-1})$ for $2 \leq i \leq n$. Then $B_1, \dots, B_n$ are disjoint and their union is the same as the union of $A_1, \dots, A_n$. Therefore by the finite additivity and monotonicity of λ ,

$$\lambda(A) = \sum_{i=1}^n \lambda(A \cap B_i) \leq \sum_{i=1}^n \lambda(B_i) \leq \sum_{i=1}^n \lambda(A_i),$$

where the last inequality holds because $B_i \subseteq A_i$ for each i . $\square$

Proposition A.1.14. *The functional λ defined above is a σ -finite measure on $\mathcal{A}$.*

Proof. Suppose that an element $A \in \mathcal{A}$ is a countable disjoint union of elements $A_1, A_2, \dots \in \mathcal{A}$. We have to show that

$$\lambda(A) = \sum_{i=1}^\infty \lambda(A_i). \tag{A.1.3}$$

It suffices to show this when $A = (a, b] \cap \mathbb{R}$ and $A_i = (a_i, b_i] \cap \mathbb{R}$ for each i , since each element of $\mathcal{A}$ is a finite disjoint union of such intervals. There is nothing to prove if $a = b$, so assume that $a < b$.

First suppose that $-\infty < a < b < \infty$. Take any $\delta > 0$ such that $a + \delta < b$, and take any $\epsilon > 0$. Then the closed interval $[a + \delta, b]$ is contained in the union of $(a_i, b_i + 2^{-i}\epsilon)$, $i \geq 1$. To see this, take any $x \in [a + \delta, b]$. Then $x \in (a, b]$, and hence $x \in (a_i, b_i]$ for some i . Thus, $x \in (a_i, b_i + 2^{-i}\epsilon)$.

Since $[a + \delta, b]$ is compact, it is therefore contained in the union of finitely many $(a_i, b_i + 2^{-i}\epsilon)$. Consequently, there exists k such that

$$(a + \delta, b] \subseteq \bigcup_{i=1}^k (a_i, b_i + 2^{-i}\epsilon].$$

Thus, by Lemma A.1.13,

$$b - a - \delta \leq \sum_{i=1}^k (b_i + 2^{-i}\epsilon - a_i) \leq \epsilon + \sum_{i=1}^{\infty} (b_i - a_i).$$

Since this holds for any ϵ and δ , we get

$$b - a \leq \sum_{i=1}^{\infty} (b_i - a_i). \quad (\text{A.1.4})$$

On other hand, for any k , finite additivity and monotonicity of λ implies that

$$b - a = \lambda(A) \geq \sum_{i=1}^k \lambda(A_i) = \sum_{i=1}^k (b_i - a_i).$$

Thus,

$$b - a \geq \sum_{i=1}^{\infty} (b_i - a_i), \quad (\text{A.1.5})$$

which proves (A.1.3) when a and b are finite. If either a or b is infinite, we find finite a', b' such that $(a', b'] \subseteq (a, b] \cap \mathbb{R}$. Repeating the above steps, we arrive at the inequality

$$b' - a' \leq \sum_{i=1}^{\infty} (b_i - a_i).$$

Since this hold for any finite $a' > a$ and $b' < b$, we recover (A.1.4), and (A.1.5) continues to hold as before. This completes the proof of countable additivity of λ . The σ -finiteness is trivial. $\square$

Corollary A.1.15. *The functional λ has a unique extension to a measure on $\mathcal{B}(\mathbb{R})$.*

Proof. It is easy to check that the algebra $\mathcal{A}$ generates the Borel σ -algebra. The existence

and uniqueness of the extension now follows by Proposition A.1.14 and Carathéodory's extension theorem. $\square$

The unique extension of λ given by Corollary A.1.15 is called the Lebesgue measure on the real line. The outer measure defined by the formula (A.1.2) in this case is called Lebesgue outer measure, and the σ -algebra induced by this outer measure is called the Lebesgue σ -algebra. Lebesgue measure is the measure that corresponds to our intuitive notion of the 'length' of a set, because $\lambda((a, b]) = b - a$ for any interval $(a, b]$, and the same holds for closed and open intervals.

A.1.8 Measurable functions

Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be two measurable spaces. A function $f : \Omega \rightarrow \Omega'$ is called measurable if $f^{-1}(A) \in \mathcal{F}$ for every $A \in \mathcal{F}'$. Here, as usual, $f^{-1}(A)$ denotes the set of all $x \in \Omega$ such that $f(x) \in A$. The main way to check that a function is measurable is the following lemma.

Lemma A.1.16. *Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be two measurable spaces and $f : \Omega \rightarrow \Omega'$ be a function. Suppose that there is a set $\mathcal{A} \subseteq \mathcal{F}'$ that generates $\mathcal{F}'$, and suppose that $f^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{A}$. Then f is measurable.*

Proof. It is easy to verify that the set of all $B \subseteq \Omega'$ such that $f^{-1}(B) \in \mathcal{F}$ is a σ -algebra. Since this set contains $\mathcal{A}$, it must also contain the σ -algebra generated by $\mathcal{A}$, which is $\mathcal{F}'$. Thus, f is measurable. $\square$

Essentially all functions that arise in practice are measurable. Let us now see why that is the case by identifying some large classes of measurable functions.

Proposition A.1.17. *Suppose that Ω and Ω' are topological spaces, and $\mathcal{F}$ and $\mathcal{F}'$ are their Borel σ -algebras. Then any continuous function from Ω into Ω' is measurable.*

Proof. Use Lemma A.1.16 with $\mathcal{A}$ = the set of all open subsets of Ω' . $\square$

It is easy to show that compositions of measurable functions are measurable. Combining this with Proposition A.1.17 shows, for example, that sums and products of real-valued measurable functions are measurable, since addition and multiplication are continuous maps from $\mathbb{R}^2$ to $\mathbb{R}$, and if $f, g : \Omega \rightarrow \mathbb{R}$ are measurable, then $(f, g) : \Omega \rightarrow \mathbb{R}^2$ is measurable with respect to $\mathcal{B}(\mathbb{R}^2)$ (easy to show).

Often, we will have occasion to consider measurable functions that take value in the set $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, \infty\}$, equipped with the σ -algebra generated by all intervals of the form $[a, b]$ where $-\infty \leq a \leq b \leq \infty$.

Proposition A.1.18. *Let $(\Omega, \mathcal{F})$ be a measurable space and let $\{f_n\}_{n \geq 1}$ be a sequence of measurable functions from Ω into $\mathbb{R}^*$. Let $g(\omega) := \inf_{n \geq 1} f_n(\omega)$ and $h(\omega) := \sup_{n \geq 1} f_n(\omega)$ for $\omega \in \Omega$. Then g and h are also measurable functions. As a consequence, the functions $\liminf_{n \rightarrow \infty} f_n$ and $\limsup_{n \rightarrow \infty} f_n$ are also measurable.*

Proof. For any $t \in \mathbb{R}^*$, $g(\omega) \geq t$ if and only if $f_n(\omega) \geq t$ for all n . Thus,

$$g^{-1}([t, \infty]) = \bigcap_{n=1}^{\infty} f_n^{-1}([t, \infty]),$$

which shows that $g^{-1}([t, \infty]) \in \mathcal{F}$. It is straightforward to verify that sets of the form $[t, \infty]$ generate the σ -algebra of $\mathbb{R}^*$. Thus, g is measurable. The proof for h is similar. For $\limsup$ and $\liminf$, simply recall that

$$\limsup_{n \rightarrow \infty} f_n(x) = \inf_{n \geq 1} \sup_{k \geq 1} f_k(x), \quad \liminf_{n \rightarrow \infty} f_n(x) = \sup_{n \geq 1} \inf_{k \geq 1} f_k(x),$$

and iterate the results for $\sup$ and $\inf$. $\square$

The above result shows that if $\{f_n\}_{n \geq 1}$ is a sequence of real-valued measurable functions converging pointwise to a limit f , then f is measurable. For functions that take values in more abstract spaces, the following result is often useful.

Proposition A.1.19. *Let $(\Omega, \mathcal{F})$ be a measurable space and let S be a separable metric space endowed with its Borel σ -algebra. If $\{f_n\}_{n \geq 1}$ is a sequence of measurable functions from Ω into S that converge pointwise to a limit function f , then f is also measurable.*

Proof. Let $B(x, r)$ denote the open ball with center x and radius r in S . Since S is separable, any open set is a countable union of open balls. Therefore by Lemma A.1.16 it suffices to check that $f^{-1}(B) \in \mathcal{F}$ for any open ball B . Take such a ball $B(x, r)$. For any $\omega \in \Omega$, if $f(\omega) \in B(x, r)$, then there is some large enough integer k such that $f(\omega) \in B(x, r - k^{-1})$. Since $f_n(\omega) \rightarrow f(\omega)$, this implies that $f_n(\omega) \in B(x, r - k^{-1})$ for all large enough n . On the other hand, if there exists $k \geq 1$ such that $f_n(\omega) \in B(x, r - k^{-1})$ for all large enough n , then $f(\omega) \in B(x, r)$. Thus, we have shown that $f(\omega) \in B(x, r)$ if and only if there is some integer $k \geq 1$ such that $f_n(\omega) \in B(x, r - k^{-1})$ for all large enough n . In set theoretic notation, this statement can be written as

$$f^{-1}(B(x, r)) = \bigcup_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} f_n^{-1}(B(x, r - k^{-1})).$$

Since each f_n is measurable, and σ -algebras are closed under countable unions and intersections, this shows that $f^{-1}(B(x, r)) \in \mathcal{F}$, completing the proof. $\square$

Measurable maps generate σ -algebras of their own, which are important for various purposes. Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be two measurable spaces and let $f : \Omega \rightarrow \Omega'$ be a measurable function. The σ -algebra generated by f is defined as

$$\sigma(f) := \{f^{-1}(A) : A \in \mathcal{F}'\}.$$

A.1.9 Lebesgue integration

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. This space will be fixed throughout this section. Let $f : \Omega \rightarrow \mathbb{R}^*$ be a measurable function, where $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, \infty\}$, as defined in the previous section. Our goal in this section is to define the Lebesgue integral

$$\int_{\Omega} f(\omega) d\mu(\omega).$$

The definition comes in several steps. For $A \in \mathcal{F}$, define the indicator function 1_A as

$$1_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Suppose that

$$f = \sum_{i=1}^n a_i 1_{A_i} \tag{A.1.6}$$

for some $a_1, \dots, a_n \in [0, \infty)$ and disjoint sets $A_1, \dots, A_n \in \mathcal{F}$. Such functions are called ‘nonnegative simple functions’. For a nonnegative simple function f as above, define

$$\int_{\Omega} f(\omega) d\mu(\omega) := \sum_{i=1}^n a_i \mu(A_i). \tag{A.1.7}$$

A subtle point here is that the same simple function can have many different representations like (A.1.6). It is not difficult to show that all of them yield the same answer in (A.1.7).

Next, take any measurable $f : \Omega \rightarrow [0, \infty]$. Let $\text{SF}^+(f)$ be the set of all nonnegative simple functions g such that $g(\omega) \leq f(\omega)$ for all ω . Define

$$\int_{\Omega} f(\omega) d\mu(\omega) := \sup_{g \in \text{SF}^+(f)} \int_{\Omega} g(\omega) d\mu(\omega).$$

It is not difficult to prove that if f is a nonnegative simple function, then this definition gives the same answer as (A.1.7), so there is no inconsistency.

Finally, take any measurable $f : \Omega \rightarrow \mathbb{R}^*$. Define $f^+(\omega) = \max\{f(\omega), 0\}$ and $f^-(\omega) = -\min\{f(\omega), 0\}$. Then f^+ and f^- are nonnegative measurable functions (easy to show), and $f = f^+ - f^-$. We say that the integral of f is defined when the integrals of at least one of f^+ and f^- is finite. In this situation, we define

$$\int_{\Omega} f(\omega) d\mu(\omega) := \int_{\Omega} f^+(\omega) d\mu(\omega) - \int_{\Omega} f^-(\omega) d\mu(\omega).$$

Often, we will simply write $\int f d\mu$ for the integral of f .

A measurable function $f : \Omega \rightarrow \mathbb{R}^*$ is called integrable if $\int f^+ d\mu$ and $\int f^- d\mu$ are both finite. Note that for the integral $\int f d\mu$ to be defined, the function f need not be integrable. As defined above, it suffices to have at least one of $\int f^+ d\mu$ and $\int f^- d\mu$ finite.

Sometimes, we will need to integrate a function f over a measurable subset $S \subseteq \Omega$ rather than the whole of Ω . This is defined simply by making f zero outside S and integrating the resulting function. That is, we define

$$\int_S f d\mu := \int_{\Omega} f 1_S d\mu,$$

provided that the right side is defined. Here and everywhere else, we use the convention $\infty \cdot 0 = 0$, which is a standard convention in measure theory and probability. The integral over S can also be defined in a different way, by considering S itself as a set endowed with a σ -algebra and a measure. To be precise, let $\mathcal{F}_S$ be the restriction of $\mathcal{F}$ to S , that is, the set of all subsets of S that belong to $\mathcal{F}$. Let μ_S be the restriction of μ to $\mathcal{F}_S$. Let f_S be the restriction of f to S . It is easy to see that $(S, \mathcal{F}_S, \mu_S)$ is a measure space and f_S is a measurable function from this space into $\mathbb{R}^*$. The integral of f over the set S can then be defined as the integral of f_S on the measure space $(S, \mathcal{F}_S, \mu_S)$, provided that the integral exists. When $S = \emptyset$, the integral can be defined to be zero.

A.1.10 The monotone convergence theorem

The following fundamental result of measure theory is known as the monotone convergence theorem.

Theorem A.1.20 (Monotone convergence theorem). *Suppose that $\{f_n\}_{n \geq 1}$ is a sequence of measurable functions from Ω into $[0, \infty]$, which are increasing pointwise to a limit function f . Then f is measurable, and $\int f d\mu = \lim \int f_n d\mu$.*

We need two lemmas. Throughout, $(\Omega, \mathcal{F}, \mu)$ denotes a measure space.

Lemma A.1.21. *If $f, g : \Omega \rightarrow [0, \infty]$ are two measurable functions such that $f \leq g$ everywhere, then $\int f d\mu \leq \int g d\mu$.*

Proof. If $s \in \text{SF}^+(f)$, then s is also in $\text{SF}^+(g)$. Thus, the definition of $\int g d\mu$ takes supremum over a larger set than $\int f d\mu$. This proves the inequality. $\square$

Lemma A.1.22. *Let $s : \Omega \rightarrow [0, \infty)$ be a measurable simple function. For each $S \in \mathcal{F}$, let $\nu(S) := \int_S s d\mu$. Then ν is a measure on $(\Omega, \mathcal{F})$.*

Proof. Suppose that

$$s = \sum_{i=1}^n a_i 1_{A_i}.$$

Since $\nu(\emptyset) = 0$ by definition, it suffices to show countable additivity of ν . Let $S_1, S_2, \dots$ be a sequence of disjoint sets in $\mathcal{F}$, and let S be their union. Then

$$\begin{aligned} \nu(S) &= \sum_{i=1}^n a_i \mu(A_i \cap S) \\ &= \sum_{i=1}^n a_i \left(\sum_{j=1}^{\infty} \mu(A_i \cap S_j) \right). \end{aligned}$$

Since an infinite series with nonnegative summands may be rearranged any way we like without altering the result, this gives

$$\nu(S) = \sum_{j=1}^{\infty} \sum_{i=1}^n a_i \mu(A_i \cap S_j) = \sum_{j=1}^{\infty} \nu(S_j).$$

This proves the countable additivity of ν . $\square$

We are now ready to prove Theorem A.1.20.

Proof of Theorem A.1.20. The measurability of f follows from Proposition A.1.18. Since $f \geq f_n$ for every n , Lemma A.1.21 gives $\int f d\mu \geq \lim \int f_n d\mu$, where the limit on the right exists because the integrals on the right are increasing with n . For the opposite inequality, take any $s \in \text{SF}^+(f)$. Define

$$\nu(S) := \int_S s d\mu,$$

so that by Lemma A.1.22, ν is a measure on Ω . Take any $\alpha \in (0, 1)$. Let

$$S_n := \{\omega : \alpha s(\omega) \leq f_n(\omega)\}.$$

It is easy to see that these sets are measurable and increasing with n . Moreover, note that any $\omega \in \Omega$ belongs to S_n for all sufficiently large n , because $f_n(\omega)$ increases to $f(\omega)$ and $\alpha s(\omega) < f(\omega)$ (unless $f(\omega) = 0$, in which case $\alpha s(\omega) = f_n(\omega) = 0$ for all n). Thus, Ω is the union of the increasing sequence $S_1, S_2, \dots$. Since ν is a measure, this shows that

$$\int s d\mu = \nu(\Omega) = \lim_{n \rightarrow \infty} \nu(S_n) = \lim_{n \rightarrow \infty} \int_{S_n} s d\mu.$$

But $\alpha s \leq f_n$ on S_n . Moreover, it is easy to see that

$$\int_{S_n} \alpha s d\mu = \alpha \int_{S_n} s d\mu$$

since s is a simple function. Thus, by Lemma A.1.21,

$$\alpha \int_{S_n} s d\mu = \int_{S_n} \alpha s d\mu \leq \int_{S_n} f_n d\mu \leq \int_{\Omega} f_n d\mu.$$

Combining the last two steps, we get $\alpha \int s d\mu \leq \lim \int f_n d\mu$. Since $\alpha \in (0, 1)$ is arbitrary, this shows that $\int s d\mu \leq \lim \int f_n d\mu$. Taking supremum over s , we get the desired result. $\square$

A.1.11 An approximation theorem for measurable functions

The following result is often useful in applications of the monotone convergence theorem.

Theorem A.1.23. *Given any measurable $f : \Omega \rightarrow [0, \infty]$, there is a sequence of nonnegative simple functions increasing pointwise to f .*

Proof. Take any n . If $k2^{-n} \leq f(\omega) < (k+1)2^{-n}$ for some integer $0 \leq k < n2^n$, let $f_n(\omega) = k2^{-n}$. If $f(\omega) \geq n$, let $f_n(\omega) = n$. It is easy to check that the sequence $\{f_n\}_{n \geq 1}$ is a sequence of nonnegative simple functions that increase pointwise to f . $\square$

An important corollary is the following fact about Lebesgue integration on the real line, which will be needed later.

Corollary A.1.24. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Lebesgue integrable function. Take any $a \in \mathbb{R}$ and let $g(x) := f(x + a)$. Then $\int g d\lambda = \int f d\lambda$, where λ is Lebesgue measure.*

Proof. For indicator functions, this follows from translation invariance of Lebesgue measure. This implies the claim for simple functions. We can then use Theorem A.1.23 and the monotone convergence theorem to pass to nonnegative measurable functions. Finally, for integrable functions, we break it into positive and negative parts, and apply the result for nonnegative functions. $\square$

A.1.12 Linearity of the Lebesgue integral

A basic result about Lebesgue integration, which is not quite obvious from the definition, is that the map $f \mapsto \int f d\mu$ is linear. This is a little surprising, because the Lebesgue integral is defined as a supremum, and functionals that are defined as suprema or infima are rarely linear. In the following, when defining the sum of two functions, we adopt the convention that $\infty - \infty = 0$. Typically such a convention can lead to inconsistencies, but if the functions are integrable then such occurrences will only take place on sets of measure zero and will not cause any problems.

Proposition A.1.25. *If f and g are two integrable functions from Ω into $\mathbb{R}^*$, then for any $\alpha, \beta \in \mathbb{R}$, the function $\alpha f + \beta g$ is integrable and $\int (\alpha f + \beta g) d\mu = \alpha \int f d\mu + \beta \int g d\mu$. Moreover, if f and g are measurable functions from Ω into $[0, \infty]$ (but not necessarily integrable), then $\int (f + g) d\mu = \int f d\mu + \int g d\mu$, and for any $\alpha \in \mathbb{R}$, $\int \alpha f d\mu = \alpha \int f d\mu$.*

Proof. It is easy to check that additivity of the integral holds for nonnegative simple functions. Take any measurable $f, g : \Omega \rightarrow [0, \infty]$. Then by Theorem A.1.23, there exist sequences of nonnegative simple functions $\{u_n\}_{n \geq 1}$ and $\{v_n\}_{n \geq 1}$ increasing to f and g pointwise. Then $u_n + v_n$ increases to $f + g$ pointwise. Thus, by the linearity of integral for nonnegative simple functions and the monotone convergence theorem,

$$\begin{aligned} \int (f + g) d\mu &= \lim_{n \rightarrow \infty} \int (u_n + v_n) d\mu \\ &= \lim_{n \rightarrow \infty} \left(\int u_n d\mu + \int v_n d\mu \right) = \int f d\mu + \int g d\mu. \end{aligned}$$

Next, take any $\alpha \geq 0$ and any measurable $f : \Omega \rightarrow [0, \infty]$. Any element of $\text{SF}^+(\alpha f)$ must be of the form αg for some $g \in \text{SF}^+(f)$. Thus

$$\int \alpha f d\mu = \sup_{g \in \text{SF}^+(f)} \int \alpha g d\mu = \alpha \sup_{g \in \text{SF}^+(f)} \int g d\mu = \alpha \int f d\mu.$$

If $\alpha \leq 0$, then $(\alpha f)^+ \equiv 0$ and $(\alpha f)^- = -\alpha f$. Thus,

$$\int \alpha f d\mu = - \int (-\alpha f) d\mu = \alpha \int f d\mu.$$

This completes the proofs of the assertions about nonnegative functions. Next, take any integrable f and g . It is easy to see that a consequence of integrability is that set where f or g take infinite values is a set of measure zero. The only problem in defining the function $f + g$ is that at some ω , it may happen that one of $f(\omega)$ and $g(\omega)$ is ∞ and the other is $-\infty$. At any such point, define $(f + g)(\omega) = 0$. Then $f + g$ is defined everywhere, is measurable, and $(f + g)^+ \leq f^+ + g^+$. Therefore, by the additivity of integration for nonnegative functions and the integrability of f and g , $\int (f + g)^+ d\mu$ is finite. Similarly, $\int (f + g)^- d\mu$ is finite. Thus, $f + g$ is integrable. Now,

$$(f + g)^+ - (f + g)^- = f + g = f^+ - f^- + g^+ - g^-,$$

which can be written as

$$(f + g)^+ + f^- + g^- = (f + g)^- + f^+ + g^+.$$

Note that the above identity holds even if one or more of the summands on either side are infinity. Thus, by additivity of integration for nonnegative functions,

$$\int (f + g)^+ d\mu + \int f^- d\mu + \int g^- d\mu = \int (f + g)^- d\mu + \int f^+ d\mu + \int g^+ d\mu,$$

which rearranges to give $\int (f + g) d\mu = \int f d\mu + \int g d\mu$. Finally, if f is integrable and $\alpha \geq 0$, then $(\alpha f)^+ = \alpha f^+$ and $(\alpha f)^- = \alpha f^-$, which shows that αf is integrable and $\int \alpha f d\mu = \alpha \int f d\mu$. Similarly, $\alpha \leq 0$, then $(\alpha f)^+ = -\alpha f^-$ and $(\alpha f)^- = -\alpha f^+$, and the proof can be completed as before. $\square$

An immediate consequence of Proposition A.1.25 is that if f and g are measurable real-valued functions such that $f \geq g$ everywhere, then $\int f d\mu \geq \int g d\mu$. This can be seen easily by writing $f = (f - g) + g$, and observing that $f - g$ is nonnegative. Another immediate consequence is that integrability of f is equivalent to the condition that $\int |f| d\mu$ is finite, since $|f| = f^+ + f^-$. Finally, another inequality that we will often use, which is an easy consequence of the triangle inequality, the definition of the Lebesgue integral, and the fact that $|f| = f^+ + f^-$, is that for any $f : \Omega \rightarrow \mathbb{R}^*$ such that $\int f d\mu$ is defined, $|\int f d\mu| \leq \int |f| d\mu$.

A.1.13 Fatou's lemma and dominated convergence

In this section we will establish two results about sequences of integrals that are widely used in probability theory. Throughout, $(\Omega, \mathcal{F}, \mu)$ is a measure space.

Theorem A.1.26 (Fatou's lemma). *Let $\{f_n\}_{n \geq 1}$ be a sequence of measurable functions from Ω into $[0, \infty]$. Then $\int (\liminf f_n) d\mu \leq \liminf \int f_n d\mu$.*

Proof. For each n , let $g_n := \inf_{m \geq n} f_m$. Then, as $n \rightarrow \infty$, g_n increases pointwise to $g := \liminf_{n \rightarrow \infty} f_n$. Moreover, $g_n \leq f_n$ everywhere and g_n and g are measurable by Proposition A.1.18. Therefore by the monotone convergence theorem,

$$\int g d\mu = \lim_{n \rightarrow \infty} \int g_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu,$$

which completes the proof. $\square$

Theorem A.1.27 (Dominated convergence theorem). *Let $\{f_n\}_{n \geq 1}$ be a sequence of measurable functions from Ω into $\mathbb{R}^*$ that converge pointwise to a limit function f . Suppose that there exists an integrable function h such that for each n , $|f_n| \leq h$ everywhere. Then $\lim \int f_n d\mu = \int f d\mu$. Moreover, we also have the stronger result $\lim \int |f_n - f| d\mu = 0$.*

Proof. We know that f is measurable since it is the limit of a sequence of measurable functions (Proposition A.1.18). Moreover, $|f| \leq h$ everywhere. Thus, $f_n + h$ and $f + h$ are nonnegative integrable functions and $f_n + h \rightarrow f + h$ pointwise. Therefore by Fatou's lemma, $\int (f + h) d\mu \leq \liminf \int (f_n + h) d\mu$. Since all integrals are finite (due to the integrability of h and the fact that $|f_n|$ and $|f|$ are bounded by h), this gives $\int f d\mu \leq \liminf \int f_n d\mu$.

Now replace f_n by $-f_n$ and f by $-f$. All the conditions still hold, and therefore the same deduction shows that $\int (-f) d\mu \leq \liminf \int (-f_n) d\mu$, which is the same as $\int f d\mu \geq \limsup \int f_n d\mu$. Thus, we get $\int f d\mu = \lim \int f_n d\mu$.

Finally, to show that $\lim \int |f_n - f| d\mu = 0$, observe that $|f_n - f| \rightarrow 0$ pointwise, and $|f_n - f|$ is bounded by the integrable function $2h$ everywhere. Then apply the first part. $\square$

A.1.14 The concept of almost everywhere

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. An event $A \in \mathcal{F}$ is said to ‘happen almost everywhere’ if $\mu(A^c) = 0$. Almost everywhere is usually abbreviated as a.e. Sometimes, in probability theory, we say ‘almost surely (a.s.)’ instead of a.e. When the measure μ may not be specified from the context, we say μ -a.e. If f and g are two functions such that the $\{\omega : f(\omega) = g(\omega)\}$ is an a.e. event, we say that $f = g$ a.e. Similarly, if $\{f_n\}_{n \geq 1}$ is a sequence of functions such that $\{\omega : \lim_{n \rightarrow \infty} f_n(\omega) = f(\omega)\}$ is an a.e. event, we say $f_n \rightarrow f$ a.e. The following lemma is a basic fact.

Proposition A.1.28. *Let $f : \Omega \rightarrow [0, \infty]$ be a measurable function. Then $\int f d\mu = 0$ if and only if $f = 0$ a.e.*

Proof. First suppose that $\mu(\{\omega : f(\omega) > 0\}) = 0$. Take any nonnegative measurable simple function g such that $g \leq f$ everywhere. Then $g = 0$ whenever $f = 0$. Thus, $\mu(\{\omega : g(\omega) > 0\}) = 0$. Since g is a simple function, this implies that $\int g d\mu = 0$. Taking

supremum over all such g gives $\int f d\mu = 0$. Conversely, suppose that $\mu(\{\omega : f(\omega) > 0\}) > 0$. Then

$$\begin{aligned}\mu(\{\omega : f(\omega) > 0\}) &= \mu\left(\bigcup_{n=1}^{\infty} \{\omega : f(\omega) > n^{-1}\}\right) \\ &= \lim_{n \rightarrow \infty} \mu(\{\omega : f(\omega) > n^{-1}\}) > 0,\end{aligned}$$

where the second equality follows from the observation that the sets in the union form an increasing sequence. Therefore for some n , $\mu(A_n) > 0$, where $A_n := \{\omega : f(\omega) > n^{-1}\}$. But then

$$\int f d\mu \geq \int f 1_{A_n} d\mu \geq \int n^{-1} 1_{A_n} d\mu = n^{-1} \mu(A_n) > 0.$$

This completes the proof of the proposition. $\square$

Broadly speaking, the idea is that ‘almost everywhere’ and ‘everywhere’ can be treated as basically the same thing. There are some exceptions to this rule of thumb, but in most situations it is valid. Often, when a function or a limit is defined almost everywhere, we will treat it as being defined everywhere by defining it arbitrarily (for example, equal to zero) on the set where it is undefined.

A.1.15 Finite dimensional product spaces

Let $(\Omega_1, \mathcal{F}_1), \dots, (\Omega_n, \mathcal{F}_n)$ be measurable spaces. Let $\Omega = \Omega_1 \times \dots \times \Omega_n$ be the Cartesian product of $\Omega_1, \dots, \Omega_n$. The product σ -algebra $\mathcal{F}$ on Ω is defined as the σ -algebra generated by sets of the form $A_1 \times \dots \times A_n$, where $A_i \in \mathcal{F}_i$ for $i = 1, \dots, n$. It is usually denoted by $\mathcal{F}_1 \times \dots \times \mathcal{F}_n$.

Proposition A.1.29. *Let Ω and $\mathcal{F}$ be as above. If each Ω_i is endowed with a σ -finite measure μ_i , then there is a unique measure μ on Ω which satisfies*

$$\mu(A_1 \times \dots \times A_n) = \prod_{i=1}^n \mu_i(A_i) \tag{A.1.8}$$

for each $A_1 \in \mathcal{F}_1, \dots, A_n \in \mathcal{F}_n$.

Proof. It is easy to check that the collection of finite disjoint unions of sets of the form $A_1 \times \dots \times A_n$ (sometimes called ‘rectangles’) form an algebra. Therefore by Carathéodory’s theorem, it suffices to show that μ defined through (A.1.8) is a countably additive measure on this algebra.

We will prove this by induction on n . It is true for $n = 1$ by the definition of a measure. Suppose that it holds for $n - 1$. Take any rectangular set $A_1 \times \dots \times A_n$. Suppose that this set is a disjoint union of $A_{i,1} \times \dots \times A_{i,n}$ for $i = 1, 2, \dots$, where $A_{i,j} \in \mathcal{F}_j$ for each i and j .

It suffices to show that

$$\mu(A_1 \times \cdots \times A_n) = \sum_{i=1}^{\infty} \mu(A_{i,1} \times \cdots \times A_{i,n}).$$

Take any $x \in A_1 \times \cdots \times A_{n-1}$. Let I be the collection of indices i such that $x \in A_{i,1} \times \cdots \times A_{i,n-1}$. Take any $y \in A_n$. Then $(x, y) \in A_1 \times \cdots \times A_n$, and hence $(x, y) \in A_{i,1} \times \cdots \times A_{i,n}$ for some i . In particular, $x \in A_{i,1} \times \cdots \times A_{i,n-1}$ and hence $i \in I$. Also, $y \in A_{i,n}$. Thus,

$$A_n \subseteq \bigcup_{i \in I} A_{i,n}.$$

On the other hand, if $y \in A_{i,n}$ for some $i \in I$, then $(x, y) \in A_{i,1} \times \cdots \times A_{i,n}$. Thus, $(x, y) \in A_1 \times \cdots \times A_n$, and therefore $y \in A_n$. This shows that A_n is the union of $A_{i,n}$ over $i \in I$. Now, if $y \in A_{i,n} \cap A_{j,n}$ for some distinct $i, j \in I$, then since $x \in (A_{i,1} \times \cdots \times A_{i,n-1}) \cap (A_{j,1} \times \cdots \times A_{j,n-1})$, we get that $(x, y) \in (A_{i,1} \times \cdots \times A_{i,n}) \cap (A_{j,1} \times \cdots \times A_{j,n})$, which is impossible. Thus, the sets $A_{i,n}$, as i ranges over I , are disjoint. This gives

$$\mu_n(A_n) = \sum_{i \in I} \mu_n(A_{i,n}).$$

Finally, if $x \notin A_1 \times \cdots \times A_{n-1}$ and $x \in A_{i,1} \times \cdots \times A_{i,n-1}$ for some i , then $A_{i,n}$ must be empty, because otherwise $(x, y) \in A_{i,1} \times \cdots \times A_{i,n}$ for any $y \in A_{i,n}$ and so $(x, y) \in A_1 \times \cdots \times A_n$, which implies that $x \in A_1 \times \cdots \times A_{n-1}$. Therefore we have proved that

$$1_{A_1 \times \cdots \times A_{n-1}}(x) \mu_n(A_n) = \sum_{i=1}^{\infty} 1_{A_{i,1} \times \cdots \times A_{i,n-1}}(x) \mu_n(A_{i,n}).$$

Let $\mu' := \mu_1 \times \cdots \times \mu_{n-1}$, which exists by the induction hypothesis. Integrating both sides with respect to the measure μ' on $\Omega_1 \times \cdots \times \Omega_{n-1}$, and applying the monotone convergence theorem to the right, we get

$$\mu'(A_1 \times \cdots \times A_{n-1}) \mu_n(A_n) = \sum_{i=1}^{\infty} \mu'(A_{i,1} \times \cdots \times A_{i,n-1}) \mu_n(A_{i,n}).$$

The desired result now follows by applying the induction hypothesis to μ' on both sides. $\square$

The measure μ of Proposition A.1.15 is called the product of $\mu_1, \dots, \mu_n$, and is usually denoted by $\mu_1 \times \cdots \times \mu_n$.

A.1.16 Fubini's theorem

We will now learn how to integrate with respect to product measures. The natural idea is to compute an integral with respect to a product measure as an iterated integral, integrating with respect to one coordinate at a time. This works under certain conditions. The conditions are given by Fubini's theorem, stated below.

Theorem A.1.30 (Fubini's theorem). *Let $(\Omega_1, \mathcal{F}_1, \mu_1)$ and $(\Omega_2, \mathcal{F}_2, \mu_2)$ be two σ -finite measure spaces. Let $\mu = \mu_1 \times \mu_2$, and let $f : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}^*$ be a measurable function. If f is either nonnegative or integrable, then the map $x \mapsto \int_{\Omega_2} f(x, y) d\mu_2(y)$ on Ω_1 and the map $y \mapsto \int_{\Omega_1} f(x, y) d\mu_1(x)$ on Ω_2 are well-defined and measurable (when set equal to zero if the integral is undefined). Moreover, we have*

$$\begin{aligned} \int_{\Omega_1 \times \Omega_2} f(x, y) d\mu(x, y) &= \int_{\Omega_1} \int_{\Omega_2} f(x, y) d\mu_2(y) d\mu_1(x) \\ &= \int_{\Omega_2} \int_{\Omega_1} f(x, y) d\mu_1(x) d\mu_2(y), \end{aligned}$$

Finally, if either of

$$\int_{\Omega_1} \int_{\Omega_2} |f(x, y)| d\mu_2(y) d\mu_1(x) \text{ or } \int_{\Omega_2} \int_{\Omega_1} |f(x, y)| d\mu_1(x) d\mu_2(y)$$

is finite, then f is integrable.

We need a preparatory lemma.

Lemma A.1.31. *Let $(\Omega_i, \mathcal{F}_i)$ be measurable spaces for $i = 1, 2, 3$. Let $f : \Omega_1 \times \Omega_2 \rightarrow \Omega_3$ be a measurable function. Then for each $x \in \Omega_1$, the map $y \mapsto f(x, y)$ is measurable on Ω_2 .*

Proof. Take any $A \in \mathcal{F}_3$ and $x \in \Omega_1$. Let $B := f^{-1}(A)$ and

$$B_x := \{y \in \Omega_2 : (x, y) \in B\} = \{y \in \Omega_2 : f(x, y) \in A\}.$$

We want to show that $B_x \in \mathcal{F}_2$. For the given x , let $\mathcal{G}$ be the set of all $E \in \mathcal{F}_1 \times \mathcal{F}_2$ such that $E_x \in \mathcal{F}_2$, where $E_x := \{y \in \Omega_2 : (x, y) \in E\}$. An easy verification shows that $\mathcal{G}$ is a σ -algebra. Moreover, it contains every rectangular set. Thus, $\mathcal{G} \supseteq \mathcal{F}_1 \times \mathcal{F}_2$, which shows in particular that $B_x \in \mathcal{F}_2$ for every $x \in \Omega_1$, since $B \in \mathcal{F}_1 \times \mathcal{F}_2$ due to the measurability of f . $\square$

We are now ready to prove Fubini's theorem.

Proof of Fubini's theorem. First, suppose that $f = 1_A$ for some $A \in \mathcal{F}_1 \times \mathcal{F}_2$. Then notice that for any $x \in \Omega_1$,

$$\int_{\Omega_2} f(x, y) d\mu_2(y) = \mu_2(A_x),$$

where $A_x := \{y \in \Omega_2 : (x, y) \in A\}$. The integral is well-defined by Lemma A.1.31. We will first prove that $x \mapsto \mu_2(A_x)$ is a measurable map, and its integral equals $\mu(A)$. This will prove Fubini's theorem for this f .

Let $\mathcal{L}$ be the set of all $E \in \mathcal{F}_1 \times \mathcal{F}_2$ such that the map $x \mapsto \mu_2(E_x)$ is measurable and integrates to $\mu(E)$. We will now show that $\mathcal{L}$ is a λ -system. We will first prove this under the assumption that μ_1 and μ_2 are finite measures. Clearly $\Omega_1 \times \Omega_2 \in \mathcal{L}$. If $E_1, E_2, \dots \in \mathcal{L}$

are disjoint, and E is their union, then for any x , E_x is the disjoint union of $(E_1)_x, (E_2)_x, \dots$, and hence

$$\mu_2(E_x) = \sum_{i=1}^{\infty} \mu_2((E_i)_x).$$

This shows that $x \mapsto \mu_2(E_x)$ is measurable. The monotone convergence theorem and the fact that $E_i \in \mathcal{L}$ for each i show that

$$\int_{\Omega_1} \mu_2(E_x) d\mu_1(x) = \sum_{i=1}^{\infty} \int_{\Omega_1} \mu_2((E_i)_x) d\mu_1(x) = \sum_{i=1}^{\infty} \mu(E_i) = \mu(E).$$

Thus, $E \in \mathcal{L}$, and therefore $\mathcal{L}$ is closed under countable disjoint unions.

Finally, take any $E \in \mathcal{L}$. Since μ_1 and μ_2 are finite measures, we have

$$\mu_2((E^c)_x) = \mu_2((E_x)^c) = \mu_2(\Omega_2) - \mu_2(E_x),$$

which proves that $x \mapsto \mu_2((E^c)_x)$ is measurable. It also proves that

$$\begin{aligned} \int_{\Omega_1} \mu_2((E^c)_x) d\mu_1(x) &= \mu_1(\Omega_1) \mu_2(\Omega_2) - \int_{\Omega_1} \mu_2(E_x) d\mu_1(x) \\ &= \mu(\Omega) - \mu(E) = \mu(E^c). \end{aligned}$$

Thus, $\mathcal{L}$ is a λ -system. Since it contains the π -system of all rectangles, which generates $\mathcal{F}_1 \times \mathcal{F}_2$, we have now established the claim that for any $E \in \mathcal{F}_1 \times \mathcal{F}_2$, the map $x \mapsto \mu_2(E_x)$ is measurable and integrates to $\mu(E)$, provided that μ_1 and μ_2 are finite measures.

Now let μ_1 and μ_2 be σ -finite measures. Let $\{E_{n,1}\}_{n \geq 1}$ and $\{E_{n,2}\}_{n \geq 2}$ be sequences of measurable sets of finite measure increasing to Ω_1 and Ω_2 , respectively. For each n , let $E_n := E_{n,1} \times E_{n,2}$, and define the functionals $\mu_{n,1}(A) := \mu_1(A \cap E_{n,1})$, $\mu_{n,2}(B) := \mu_2(B \cap E_{n,2})$ and $\mu_n(E) := \mu(E \cap E_n)$ for $A \in \mathcal{F}_1$, $B \in \mathcal{F}_2$ and $E \in \mathcal{F}_1 \times \mathcal{F}_2$. It is easy to see that these are finite measures, increasing to μ_1 , μ_2 and μ .

If $f : \Omega_1 \rightarrow [0, \infty]$ is a measurable function, it is easy to see that

$$\int_{\Omega_1} f(x) d\mu_{n,1}(x) = \int_{\Omega_1} f(x) 1_{E_{n,1}}(x) d\mu_1(x), \quad (\text{A.1.9})$$

where we use the convention $\infty \cdot 0 = 0$ on the right. To see this, first note that it holds for indicator functions by the definition of $\mu_{n,1}$. From this, pass to simple functions by linearity and then to nonnegative measurable functions by the monotone convergence theorem and Theorem A.1.23.

Next, note that for any $E \in \mathcal{F}_1 \times \mathcal{F}_2$ and any $x \in \Omega_1$, $\mu_2(E_x)$ is the increasing limit of $\mu_{n,2}(E_x) 1_{E_{n,1}}(x)$ as $n \rightarrow \infty$. Firstly, this shows that $x \mapsto \mu_2(E_x)$ is a measurable map. Moreover, for each n , $\mu_n = \mu_{n,1} \times \mu_{n,2}$, as can be seen by verifying on rectangles and using the uniqueness of product measures on σ -finite spaces. Therefore by the monotone

convergence theorem and equation (A.1.9),

$$\begin{aligned} \int_{\Omega_1} \mu_2(E_x) d\mu_1(x) &= \lim_{n \rightarrow \infty} \int_{\Omega_1} \mu_{n,2}(E_x) 1_{E_{n,1}}(x) d\mu_1(x) \\ &= \lim_{n \rightarrow \infty} \int_{\Omega_1} \mu_{n,2}(E_x) d\mu_{n,1}(x) \\ &= \lim_{n \rightarrow \infty} \mu_n(E) = \mu(E). \end{aligned}$$

This completes the proof of Fubini's theorem for all indicator functions. By linearity, this extends to all simple functions. Using the monotone convergence theorem and Theorem A.1.23, it is straightforward to extend the result to all nonnegative measurable functions.

Now take any integrable $f : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}^*$. Applying Fubini's theorem to f^+ and f^- , we get

$$\begin{aligned} \int f d\mu &= \int_{\Omega_1} \int_{\Omega_2} f^+(x, y) d\mu_2(y) d\mu_1(x) - \int_{\Omega_1} \int_{\Omega_2} f^-(x, y) d\mu_2(y) d\mu_1(x) \\ &= \int_{\Omega_1} g(x) d\mu_1(x) - \int_{\Omega_1} h(x) d\mu_1(x), \end{aligned}$$

where

$$g(x) := \int_{\Omega_2} f^+(x, y) d\mu_2(y), \quad h(x) := \int_{\Omega_2} f^-(x, y) d\mu_2(y).$$

By Fubini's theorem for nonnegative functions and the integrability of f , it follows that g and h are integrable functions. Thus,

$$\int f d\mu = \int_{\Omega_1} (g(x) - h(x)) d\mu_1(x),$$

where we have adopted the convention that $\infty - \infty = 0$, as in Proposition A.1.25.

Now, at any x where at least one of $g(x)$ and $h(x)$ is finite,

$$g(x) - h(x) = \int_{\Omega_2} (f^+(x, y) - f^-(x, y)) d\mu_2(y) = \int_{\Omega_2} f(x, y) d\mu_2(y). \quad (\text{A.1.10})$$

On the other hand, the set where $g(x)$ and $h(x)$ are both infinite is a set of measure zero, and therefore we may define

$$\int_{\Omega_2} f(x, y) d\mu_2(y) = 0$$

on this set, so that again (A.1.10) is valid under the convention $\infty - \infty = 0$. (The construction ensures, among other things, that $x \mapsto \int_{\Omega_2} f(x, y) d\mu_2(y)$ is a measurable function.) This concludes the proof of Fubini's theorem for integrable functions. The final assertion of the theorem follows easily from Fubini's theorem for nonnegative functions. $\square$

A.1.17 Infinite dimensional product spaces

Suppose now that $\{(\Omega_i, \mathcal{F}_i, \mu_i)\}_{i=1}^\infty$ is a countable sequence of σ -finite measure spaces. Let $\Omega = \Omega_1 \times \Omega_2 \times \cdots$. The product σ -algebra $\mathcal{F} = \mathcal{F}_1 \times \mathcal{F}_2 \times \cdots$ is defined to be the σ -algebra generated by all sets of the form $A_1 \times A_2 \times \cdots$, where $A_i = \Omega_i$ for all but finitely many i . Such sets generalize the notion of rectangles to infinite product spaces.

Although the definition of the product σ -algebra is just as before, the existence of a product measure is a slightly trickier question. In particular, we need that each μ_i is a probability measure to even define the infinite product measure on rectangles in a meaningful way. To see why this condition is necessary, suppose that the measure spaces are all the same. Then if $\mu_i(\Omega_i) > 1$, each rectangle must have measure ∞ , and if $\mu_i(\Omega_i) < 1$, each rectangle must have measure zero. To avoid these trivialities, we need to have $\mu_i(\Omega_i) = 1$.

Theorem A.1.32. *Let all notation be as above. Suppose that $\{\mu_i\}_{i=1}^\infty$ are probability measures. Then there exists a unique probability measure μ on $(\Omega, \mathcal{F})$ that satisfies*

$$\mu(A_1 \times A_2 \times \cdots) = \prod_{i=1}^{\infty} \mu_i(A_i)$$

for every $A_1 \times A_2 \times \cdots$ such that $A_i = \Omega_i$ for all but finitely many i .

Proof. First note that by the existence of finite dimensional product measures, the measure $\nu_n := \mu_1 \times \cdots \times \mu_n$ is defined for each n .

Next, define $\Omega^{(n)} := \Omega_{n+1} \times \Omega_{n+2} \times \cdots$. Let $A \in \mathcal{F}$ be called a *cylinder set* if it is of the form $B \times \Omega^{(n)}$ for some n and some $B \in \mathcal{F}_1 \times \cdots \times \mathcal{F}_n$. Define $\mu(A) := \nu_n(B)$. It is easy to see that this is a well-defined functional on $\mathcal{A}$, and that it satisfies the product property for rectangles.

Let $\mathcal{A}$ be the collection of all cylinder sets. It is not difficult to check that $\mathcal{A}$ is an algebra, and that μ is finitely additive and σ -finite on $\mathcal{A}$. Therefore by Carathéodory's theorem, we only need to check that μ is countably additive on $\mathcal{A}$. Suppose that $A \in \mathcal{A}$ is the union of a sequence of disjoint sets $A_1, A_2, \dots \in \mathcal{A}$. For each n , let $B_n := A \setminus (A_1 \cup \cdots \cup A_n)$. Since $\mathcal{A}$ is an algebra, each $B_n \in \mathcal{A}$. Since μ is finitely additive on $\mathcal{A}$, $\mu(A) = \mu(B_n) + \mu(A_1) + \cdots + \mu(A_n)$ for each n . Therefore we have to show that $\lim \mu(B_n) = 0$. Suppose that this is not true. Since $\{B_n\}_{n \geq 1}$ is a decreasing sequence of sets, this implies that there is some $\epsilon > 0$ such that $\mu(B_n) \geq \epsilon$ for all n . Using this, we will now get a contradiction to the fact that $\cap B_n = \emptyset$.

For each n , let $\mathcal{A}^{(n)}$ be the algebra of all cylinder sets in $\Omega^{(n)}$, and let $\mu^{(n)}$ be defined on $\mathcal{A}^{(n)}$ using $\mu_{n+1}, \mu_{n+2}, \dots$, the same way we defined μ on $\mathcal{A}$. For any n, m , and $(x_1, \dots, x_m) \in \Omega_1 \times \cdots \times \Omega_m$, define

$$B_n(x_1, \dots, x_m) := \{(x_{m+1}, x_{m+2}, \dots) \in \Omega^{(m)} : (x_1, x_2, \dots) \in B_n\}.$$

Since B_n is a cylinder set, it is of the form $C_n \times \Omega^{(m)}$ for some m and some $C_n \in \mathcal{F}_1 \times \cdots \times \mathcal{F}_m$. Therefore by Lemma A.1.31, $B_n(x_1) \in \mathcal{A}^{(1)}$ for any $x_1 \in \Omega_1$, and by Fubini's theorem, the

map $x_1 \mapsto \mu^{(1)}(B_n(x_1))$ is measurable. (Although we have not yet shown that $\mu^{(1)}$ is a measure on the product σ -algebra of $\Omega^{(1)}$, it is evidently a measure on the σ -algebra $\mathcal{G}$ of all sets of the form $D \times \Omega^{(m)} \subseteq \Omega^{(1)}$, where $D \in \mathcal{F}_2 \times \cdots \times \mathcal{F}_m$. Moreover, $B_n \in \mathcal{F}_1 \times \mathcal{G}$. This allows us to use Fubini's theorem to reach the above conclusion.) Thus, the set

$$F_n := \{x_1 \in \Omega_1 : \mu^{(1)}(B_n(x_1)) \geq \epsilon/2\}$$

is an element of $\mathcal{F}_1$. But again by Fubini's theorem,

$$\begin{aligned} \mu(B_n) &= \int \mu^{(1)}(B_n(x_1)) d\mu_1(x_1) \\ &= \int_{F_n} \mu^{(1)}(B_n(x_1)) d\mu_1(x_1) + \int_{F_n^c} \mu^{(1)}(B_n(x_1)) d\mu_1(x_1) \\ &\leq \mu_1(F_n) + \frac{\epsilon}{2}. \end{aligned}$$

Since $\mu(B_n) \geq \epsilon$, this shows that $\mu_1(F_n) \geq \epsilon/2$. Since $\{F_n\}_{n \geq 1}$ is a decreasing sequence of sets, this shows that $\cap F_n \neq \emptyset$. Choose a point $x_1^* \in \cap F_n$.

Note that $\{B_n(x_1^*)\}_{n \geq 1}$ is a decreasing sequence of sets in $\mathcal{F}^{(1)}$, such that $\mu^{(1)}(B_n(x_1^*)) \geq \epsilon/2$ for each n . Repeating the above argument for the product space $\Omega^{(1)}$ and the sequence $\{B_n(x_1^*)\}_{n \geq 1}$, we see that there exists $x_2^* \in \Omega_2$ such that $\mu^{(2)}(B_n(x_1^*, x_2^*)) \geq \epsilon/4$ for every n .

Proceeding like this, we obtain a point $x = (x_1^*, x_2^*, \dots) \in \Omega$ such that for any m and n ,

$$\mu^{(m)}(B_n(x_1^*, \dots, x_m^*)) \geq 2^{-m}\epsilon.$$

Take any n . Since B_n is a cylinder set, it is of the form $C_n \times \Omega^{(m)}$ for some m and some $C_n \in \mathcal{F}_1 \times \cdots \times \mathcal{F}_m$. Since $\mu^{(m)}(B_n(x_1^*, \dots, x_m^*)) > 0$, there is some $(x_{m+1}, x_{m+2}, \dots) \in \Omega^{(m)}$ such that $(x_1^*, \dots, x_m^*, x_{m+1}, x_{m+2}, \dots) \in B_n$. But by the form of B_n , this implies that $x \in B_n$. Thus, $x \in \cap B_n$. This completes the proof. $\square$

A.1.18 Cylinder σ -algebras

Having constructed products of countably many probability spaces, one may now wonder about uncountable products. Surprisingly, this is quite simple, given that we know how to handle the countable case. Suppose that $\{(\Omega_i, \mathcal{F}_i, \mu_i)\}_{i \in I}$ is an arbitrary collection of probability spaces. Let $\Omega := \times_{i \in I} \Omega_i$, and let $\mathcal{F} := \times_{i \in I} \mathcal{F}_i$ be defined using rectangles as in the countable case. Now take any countable set $J \subseteq I$, and let $\mathcal{F}_J := \times_{j \in J} \mathcal{F}_j$. Consider a set $A \in \mathcal{F}$ of the form $\{(\omega_i)_{i \in I} : (\omega_j)_{j \in J} \in B\}$, where B is some element of $\mathcal{F}_J$. Let $\mathcal{G}$ be the collection of all such A , as B varies in $\mathcal{F}_J$ and J varies over all countable subsets of I . It is not hard to check that $\mathcal{G}$ is in fact a σ -algebra. Moreover, it is contained in $\mathcal{F}$, and it contains all rectangles. Therefore $\mathcal{G} = \mathcal{F}$. Thus, we can define μ on this σ -algebra simply using the definition of the product measure on countable product spaces. The σ -algebra $\mathcal{G}$ is sometimes called the *cylinder σ -algebra* on Ω . We will see later that cylinder σ -algebras and probability measures on cylinder σ -algebras are important in probabilistic constructions

of quantum field theories.

A.2 L^p spaces

A.2.1 Markov's inequality and Jensen's inequality

The following simple but important inequality is called Markov's inequality in the probability literature.

Theorem A.2.1 (Markov's inequality). *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $f : \Omega \rightarrow [0, \infty]$ be a measurable function. Then for any $t > 0$,*

$$\mu(\{\omega : f(\omega) \geq t\}) \leq \frac{1}{t} \int f d\mu.$$

Proof. Take any $t > 0$. Define a function g as

$$g(\omega) := \begin{cases} 1 & \text{if } f(\omega) \geq t, \\ 0 & \text{if } f(\omega) < t. \end{cases}$$

Then $g \leq t^{-1}f$ everywhere. Thus,

$$\int g d\mu \leq \frac{1}{t} \int f d\mu.$$

The proof is completed by observing that $\int g d\mu = \mu(\{\omega : f(\omega) \geq t\})$ by the definition of Lebesgue integral for simple functions. $\square$

Jensen's inequality is another basic tool in probability theory. Unlike Markov's inequality, this inequality holds for probability measures only. First, recall that a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is called convex if for all $x, y \in \mathbb{R}$ and $t \in [0, 1]$,

$$\phi(tx + (1-t)y) \leq t\phi(x) + (1-t)\phi(y). \quad (\text{A.2.1})$$

A simple criterion that is often used to prove convexity is that ϕ'' is continuous and nonnegative everywhere. Indeed, if this holds, then a simple application of second order Taylor expansion at x immediately implies the above inequality.

The inequality (A.2.1) implies that for $y > x$ and $t \in (0, 1)$,

$$\frac{\phi(tx + (1-t)y) - \phi(x)}{(tx + (1-t)y) - x} \leq \frac{\phi(y) - \phi(x)}{y - x},$$

and for $y < x$, the inequality is reversed. Thus, the fraction $(\phi(y) - \phi(x))/(y - x)$ decreases as y decreases to x , and increases as y increases to x . Thus, the left and right derivatives exist at x :

$$\partial_L \phi(x) := \lim_{y \uparrow x} \frac{\phi(y) - \phi(x)}{y - x}, \quad \partial_R \phi(x) := \lim_{y \downarrow x} \frac{\phi(y) - \phi(x)}{y - x}.$$

(This proves, in particular, that ϕ is continuous.) Now, for $x < y$ and $t \in (0, 1)$, the inequality (A.2.1) can also be rewritten as

$$\frac{\phi(tx + (1-t)y) - \phi(x)}{(tx + (1-t)y) - x} \leq \frac{\phi(y) - \phi(tx + (1-t)y)}{y - (tx + (1-t)y)}.$$

This shows that for $x < y < z$, $(\phi(z) - \phi(y))/(z - y) \geq (\phi(y) - \phi(x))/(y - x)$. This shows, in particular, that $\partial_L \phi(x) \leq \partial_R \phi(x)$ for any x .

Now, take any x and a number $a \in [\partial_L \phi(x), \partial_R \phi(x)]$. Then for any $y > y' > x$,

$$\frac{\phi(y) - \phi(y')}{y - y'} \geq \frac{\phi(y') - \phi(x)}{y' - x},$$

as we deduced above. Taking $y' \downarrow x$, and recalling that ϕ is continuous, we get

$$\frac{\phi(y) - \phi(x)}{y - x} \geq \partial_R \phi(x) \geq a.$$

By a similar argument, for any $y < x$,

$$\frac{\phi(y) - \phi(x)}{y - x} \leq a.$$

Thus, $\phi(y) \geq \phi(x) + a(y - x)$ for all $y \in \mathbb{R}$. This is a fundamental fact about convex functions that we will use in the proof of Jensen's inequality below.

Theorem A.2.2 (Jensen's inequality). *Let X be an integrable random variable, and let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function such that $\phi(X)$ is also integrable. Then $\mathbb{E}(\phi(X)) \geq \phi(\mathbb{E}(X))$.*

Proof. Let $x := \mathbb{E}(X)$. Choose $a \in [\partial_L \phi(x), \partial_R \phi(x)]$. Then by the above deduction,

$$\phi(X) \geq \phi(x) + a(X - x).$$

Taking expectation on both sides completes the proof. $\square$

A function ϕ is called concave if $-\phi$ is convex. Clearly, the opposite of Jensen's inequality holds for concave functions. An important concave function is the logarithm, whose concavity can be verified by simply noting that its second derivative is negative everywhere on the positive real line. A consequence of the concavity of the logarithm is Young's inequality: If x and y are positive real numbers, and $p, q \in (1, \infty)$ are number such that $1/p + 1/q = 1$, then

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}.$$

To show this, simply take the logarithm of the right side and apply the definition of concavity. We will use Young's inequality later in the proof of Hölder's inequality.

A.2.2 The first Borel–Cantelli lemma

The first Borel–Cantelli lemma is an important tool for proving limit theorems in probability theory. We will use it in the next section to prove the completeness of L^p spaces.

Theorem A.2.3 (The first Borel–Cantelli lemma). *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, and let $A_1, A_2, \dots \in \mathcal{F}$ be events such that $\sum_{n=1}^{\infty} \mu(A_n) < \infty$. Then $\mu(\{\omega : \omega \in \text{infinitely many } A_n\text{'s}\}) = 0$.*

Proof. It is not difficult to see that in set theoretic notation,

$$\{\omega : \omega \in \text{infinitely many } A_n\text{'s}\} = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k.$$

Since the inner union on the right is decreasing in n ,

$$\begin{aligned} \mu(\{\omega : \omega \in \text{infinitely many } A_n\text{'s}\}) &\leq \inf_{n \geq 1} \mu\left(\bigcup_{k=n}^{\infty} A_k\right) \\ &\leq \inf_{n \geq 1} \sum_{k=n}^{\infty} \mu(A_k) = \lim_{n \rightarrow \infty} \sum_{k=n}^{\infty} \mu(A_k). \end{aligned}$$

The finiteness of $\sum \mu(A_n)$ shows that the limit on the right equals zero, completing the proof. $\square$

A.2.3 L^p spaces and inequalities

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, to be fixed throughout this section. Let $f : \Omega \rightarrow \mathbb{C}$ be a measurable function. Given any $p \in [1, \infty)$, the L^p norm of f is defined as

$$\|f\|_{L^p} := \left(\int |f|^p d\mu \right)^{1/p}.$$

The space $L^p(\Omega, \mathcal{F}, \mu)$ (or simply $L^p(\Omega)$ or $L^p(\mu)$) is the set of all measurable $f : \Omega \rightarrow \mathbb{C}$ such that $\|f\|_{L^p}$ is finite. In addition to the above, there is also the L^∞ norm, defined as

$$\|f\|_{L^\infty} := \inf\{K \in [0, \infty] : |f| \leq K \text{ a.e.}\}.$$

The right side is called the ‘essential supremum’ of the function $|f|$. It is not hard to see that $|f| \leq \|f\|_{L^\infty}$ a.e. As before, $L^\infty(\Omega, \mathcal{F}, \mu)$ is the set of all f with finite L^∞ norm.

It turns out that for any $1 \leq p \leq \infty$, $L^p(\Omega, \mathcal{F}, \mu)$ is actually a vector space and the L^p is actually a norm on this space, provided that we first quotient out the space by the equivalence relation of being equal almost everywhere. Moreover, these norms are complete (that is, Cauchy sequences converge). The only thing that is obvious is that $\|\alpha f\|_{L^p} = |\alpha| \|f\|_{L^p}$ for any $\alpha \in \mathbb{C}$ and any p and f . Proving the other claims, however, requires some work, which we do below.

Theorem A.2.4 (Hölder's inequality). *Take any measurable $f, g : \Omega \rightarrow \mathbb{C}$, and any $p \in [1, \infty]$. Let q be the solution of $1/p + 1/q = 1$. Then $\|fg\|_{L^1} \leq \|f\|_{L^p} \|g\|_{L^q}$.*

Proof. If $p = 1$, then $q = \infty$. The claimed inequality then follows simply as

$$\int |fg| d\mu \leq \|g\|_{L^\infty} \int |f| d\mu = \|f\|_{L^1} \|g\|_{L^\infty}.$$

If $p = \infty$, then $q = 1$ and the proof is exactly the same. So assume that $p \in (1, \infty)$, which implies that $q \in (1, \infty)$. Let $\alpha := 1/\|f\|_{L^p}$ and $\beta := 1/\|g\|_{L^q}$, and let $u := \alpha f$ and $v := \beta g$. Then $\|u\|_{L^p} = \alpha \|f\|_{L^p} = 1$ and $\|v\|_{L^q} = \beta \|g\|_{L^q} = 1$. Now, by Young's inequality,

$$|uv| \leq \frac{|u|^p}{p} + \frac{|v|^q}{q}$$

everywhere. Integrating both sides, we get

$$\|uv\|_{L^1} \leq \frac{\|u\|_{L^p}^p}{p} + \frac{\|v\|_{L^q}^q}{q} = \frac{1}{p} + \frac{1}{q} = 1.$$

It is now easy to see that this is precisely the inequality that we wanted to prove. $\square$

Using Hölder's inequality, it is now easy to prove that the L^p norms satisfy the triangle inequality.

Theorem A.2.5 (Minkowski's inequality). *For any $1 \leq p \leq \infty$ and any measurable $f, g : \Omega \rightarrow \mathbb{C}$, $\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$.*

Proof. If the right side is infinite, there is nothing to prove. So assume that the right side is finite. First, consider the case $p = \infty$. Since $|f| \leq \|f\|_{L^\infty}$ a.e. and $|g| \leq \|g\|_{L^\infty}$ a.e., it follows that $|f + g| \leq \|f\|_{L^\infty} + \|g\|_{L^\infty}$ a.e., which completes the proof by the definition of essential supremum. On the other hand, if $p = 1$, then the claimed inequality follows trivially from the triangle inequality and the additivity of integration. So, let us assume that $p \in (1, \infty)$. Then by Hölder's inequality,

$$\begin{aligned} \int |f + g|^p d\mu &= \int |f + g| |f + g|^{p-1} d\mu \\ &\leq \int |f| |f + g|^{p-1} d\mu + \int |g| |f + g|^{p-1} d\mu \\ &\leq \|f\|_{L^p} \| |f + g|^{p-1} \|_{L^q} + \|g\|_{L^p} \| |f + g|^{p-1} \|_{L^q}, \end{aligned}$$

where q solves $1/p + 1/q = 1$. But

$$\| |f + g|^{p-1} \|_{L^q} = \left(\int |f + g|^{(p-1)q} d\mu \right)^{1/q} = \left(\int |f + g|^p d\mu \right)^{1/q}.$$

Combining, and using the finiteness of $\|f + g\|_{L^p}$, we get

$$\|f + g\|_{L^p} = \left(\int |f + g|^p d\mu \right)^{1-1/q} \leq \|f\|_{L^p} + \|g\|_{L^p},$$

which is what we wanted to prove. $\square$

Minkowski's inequality shows, in particular, that $L^p(\Omega, \mathcal{F}, \mu)$ is a vector space, and the L^p norm satisfies two of the three required properties of a norm on this space. The only property that it does not satisfy is that f may be nonzero even if $\|f\|_{L^p} = 0$. But this is not a serious problem, since by Proposition A.1.28, the vanishing of the L^p norm implies that $f = 0$ a.e. More generally, $\|f - g\|_{L^p} = 0$ if and only if $f = g$ a.e. This shows that if we quotient out $L^p(\Omega, \mathcal{F}, \mu)$ by the equivalence relation of being equal a.e., the resulting quotient space is a vector space where the L^p norm is indeed a norm. Since we already think of two functions which are equal a.e. as effectively the same function, we will not worry about this technicality too much and continue to treat our definition of L^p space as a vector space with L^p norm.

The fact that is somewhat nontrivial, however, is that the L^p norm is complete. That is, any sequence of functions that is Cauchy in the L^p norm converges to a limit in L^p space. A first step towards the proof is the following lemma, which is important in its own right.

Lemma A.2.6. *If $\{f_n\}_{n \geq 1}$ is a Cauchy sequence in the L^p norm for some $1 \leq p \leq \infty$, then there is function $f \in L^p(\Omega, \mathcal{F}, \mu)$, and a subsequence $\{f_{n_k}\}_{k \geq 1}$, such that $f_{n_k} \rightarrow f$ a.e. as $k \rightarrow \infty$.*

Proof. First suppose that $p \in [1, \infty)$. It is not difficult to see that using the Cauchy criterion, we can extract a subsequence $\{f_{n_k}\}_{k \geq 1}$ such that $\|f_{n_k} - f_{n_{k+1}}\|_{L^p} \leq 2^{-k}$ for every k . Define the event

$$A_k := \{\omega : |f_{n_k}(\omega) - f_{n_{k+1}}(\omega)| \geq 2^{-k/2}\}.$$

Then by Markov's inequality,

$$\begin{aligned} \mu(A_k) &= \mu(\{\omega : |f_{n_k}(\omega) - f_{n_{k+1}}(\omega)|^p \geq 2^{-kp/2}\}) \\ &\leq 2^{kp/2} \int |f_{n_k} - f_{n_{k+1}}|^p d\mu \\ &= 2^{kp/2} \|f_{n_k} - f_{n_{k+1}}\|_{L^p}^p \leq 2^{-kp/2}. \end{aligned}$$

Thus, $\sum \mu(A_k) < \infty$. Therefore by the first Borel–Cantelli lemma, $\mu(B) = 0$, where $B := \{\omega : \omega \in \text{infinitely many } A_k\}$. If $\omega \in B^c$, then ω belongs to only finitely many of the A_k 's. This means that $|f_{n_k}(\omega) - f_{n_{k+1}}(\omega)| \leq 2^{-k/2}$ for all sufficiently large k . From this, it follows that $\{f_{n_k}(\omega)\}_{k \geq 1}$ is a Cauchy sequence of complex numbers. Define $f(\omega)$ to be the limit of this sequence. For $\omega \in B$, define $f(\omega) = 0$. Then f is measurable and $f_{n_k} \rightarrow f$ a.e. Moreover, by the a.e. version of Fatou's lemma,

$$\int |f|^p d\mu \leq \liminf_{k \rightarrow \infty} \int |f_{n_k}|^p d\mu = \liminf_{k \rightarrow \infty} \|f_{n_k}\|_{L^p}^p < \infty,$$

since a Cauchy sequence of real numbers must be bounded. Thus, f has finite L^p norm.

Next, suppose that $p = \infty$. Extract a subsequence $\{f_{n_k}\}_{k \geq 1}$ as before. Then for each k , $|f_{n_k} - f_{n_{k+1}}| \leq 2^{-k}$ a.e. Therefore, if we define

$$E := \{\omega : |f_{n_k}(\omega) - f_{n_{k+1}}(\omega)| \leq 2^{-k} \text{ for all } k\},$$

then $\mu(E^c) = 0$. For any $\omega \in E$, $\{f_{n_k}(\omega)\}_{k \geq 1}$ is a Cauchy sequence of complex numbers. Define $f(\omega)$ to be the limit of this sequence. For $\omega \in E^c$, define $f(\omega) = 0$. Then f is measurable and $f_{n_k} \rightarrow f$ a.e. Moreover, on E ,

$$|f| \leq |f_{n_1}| + \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}| \leq |f_{n_1}| + \sum_{k=1}^{\infty} 2^{-k},$$

which shows that f has finite L^∞ norm. $\square$

Theorem A.2.7 (Riesz–Fischer theorem). *For any $1 \leq p \leq \infty$, the L^p norm on $L^p(\Omega, \mathcal{F}, \mu)$ is complete.*

Proof. Take any sequence $\{f_n\}_{n \geq 1}$ that is Cauchy in L^p . Then note that by Lemma A.2.6, there is a subsequence $\{f_{n_k}\}_{k \geq 1}$ that converges a.e. to a function f which is also in L^p . First, suppose that $p \in [1, \infty)$. Take any $\epsilon > 0$, and find N such that for all $m, n \geq N$, $\|f_n - f_m\|_{L^p} < \epsilon$. Take any $n \geq N$. Then by the a.e. version of Fatou's lemma,

$$\begin{aligned} \int |f_n - f|^p d\mu &= \int \lim_{k \rightarrow \infty} |f_n - f_{n_k}|^p d\mu \\ &\leq \liminf_{k \rightarrow \infty} \int |f_n - f_{n_k}|^p d\mu \leq \epsilon^p. \end{aligned}$$

This shows that $f_n \rightarrow f$ in the L^p norm. Next, suppose that $p = \infty$. Take any $\epsilon > 0$ and find N as before. Let

$$E := \{\omega : |f_n(\omega) - f_m(\omega)| \leq \epsilon \text{ for all } m, n \geq N\},$$

so that $\mu(E^c) = 0$. Take any $n \geq N$. Then for any ω such that $\omega \in E$ and $f_{n_k}(\omega) \rightarrow f(\omega)$,

$$|f_n(\omega) - f(\omega)| = \lim_{k \rightarrow \infty} |f_n(\omega) - f_{n_k}(\omega)| \leq \epsilon.$$

This shows that $\|f_n - f\|_{L^\infty} \leq \epsilon$, completing the proof that $f_n \rightarrow f$ in the L^∞ norm. $\square$

The following fact about L^p spaces of probability measures is very important in probability theory. It does not hold for general measures.

Theorem A.2.8 (Monotonicity of L^p norms for probability measures). *Suppose that μ is a probability measure. Then for any measurable $f : \Omega \rightarrow \mathbb{C}$ and any $1 \leq p \leq q \leq \infty$, $\|f\|_{L^p} \leq \|f\|_{L^q}$.*

Proof. If $p = q = \infty$, there is nothing to prove. So let p be finite. If $q = \infty$, then

$$\int |f|^p d\mu \leq \|f\|_{L^\infty}^p \mu(\Omega) = \|f\|_{L^\infty}^p,$$

which proves the claim. So let q be also finite. First assume that $\|f\|_{L^p}$ and $\|f\|_{L^q}$ are both finite. Applying Jensen's inequality with the convex function $\phi(x) = |x|^{q/p}$, we get the desired inequality

$$\left(\int |f|^p d\mu \right)^{q/p} \leq \int |f|^q d\mu.$$

Finally, let us drop the assumption of finiteness of $\|f\|_{L^p}$ and $\|f\|_{L^q}$. Take a sequence of nonnegative simple functions $\{g_n\}_{n \geq 1}$ increasing pointwise to $|f|$ (which exists by Theorem A.1.23). Since μ is a probability measure, $\|g_n\|_{L^p}$ and $\|g_n\|_{L^q}$ are both finite. Therefore $\|g_n\|_{L^p} \leq \|g_n\|_{L^q}$ for each n . We can now complete the proof by applying the monotone convergence theorem to both sides. $\square$

The space L^2 holds a special status among all the L^p spaces, because it has a natural rendition as a Hilbert space with respect to the inner product

$$\langle f, g \rangle := \int f \bar{g} d\mu, \tag{A.2.2}$$

where $\bar{g}$ denotes the complex conjugate of g . It is easy to verify that this is indeed an inner product on $L^2(\Omega, \mathcal{F}, \mu)$, and the norm generated by this inner product is the L^2 norm (after quotienting out by the equivalence relation of a.e. equality, as usual). The completeness of the L^2 norm guarantees that this is indeed a Hilbert space. The Cauchy–Schwarz inequality on this Hilbert space is a special case of Hölder's inequality with $p = q = 2$.

A.2.4 Approximation theorems for L^p functions

For $p, q \geq 1$ and a measure μ , define the $L^{p,q}(\mu)$ norm as

$$\|f\|_{L^{p,q}(\mu)} := \|f\|_{L^p(\mu)} + \|f\|_{L^q(\mu)}.$$

Note that convergence happens in $L^{p,q}$ if and only if it happens in both L^p and L^q . Recall the approximation theorem that we proved for measurable sets in Section A.1.4 (Theorem A.1.5). The following is an analogous result for functions in L^p and in $L^{p,q}$.

Theorem A.2.9. *Let $(\Omega, \mathcal{F}, \mu)$ be a probability space and f be a function in $L^p(\Omega, \mathcal{F}, \mu)$ for some $p \in [1, \infty)$. Let $\mathcal{A}$ be an algebra that generates $\mathcal{F}$. Then there is a sequence of $\mathcal{A}$ -measurable complex-valued simple functions $\{f_n\}_{n \geq 1}$ converging to f in L^p . The same holds for $L^{p,q}$, for any $p, q \in [1, \infty)$.*

Proof. Breaking up into real and imaginary parts, it is clear that it suffices to prove the theorem for real-valued f . Let f^+ and f^- denote the positive and negative parts of f . By Theorem A.1.23, we know that there is a sequence of nonnegative simple functions

$\{g_n\}_{n \geq 1}$ that are increasing to f^+ . Since f^+ is in L^p and $|f^+(x) - g_n(x)| \leq f^+(x)$ for each n and x , the dominated convergence theorem implies that $g_n \rightarrow f^+$ in L^p . Similarly, there is a sequence of nonnegative simple functions converging to f^- in L^p . Thus, it suffices to prove the theorem under the assumption that f is a nonnegative simple function. Clearly, we can further reduce this to the case that f is an indicator function. But for indicator functions, this is a simple consequence of Theorem A.1.5. Exactly the same proof goes through for $L^{p,q}$. $\square$

The following is an important corollary of Theorem A.2.9.

Corollary A.2.10. *Take any $p \geq 1$. Let $\{(\Omega_i, \mathcal{F}_i)\}_{i \in I}$ be an arbitrary collection of measure spaces, and let Ω be the product space with cylinder σ -algebra $\mathcal{F}$. Let μ be a probability measure on $(\Omega, \mathcal{F})$. Then for any $f \in L^p(\Omega, \mathcal{F}, \mu)$, there is a sequence of function $\{f_n\}_{n \geq 1}$ converging to f in L^p , such that for each n , f_n is measurable with respect to the sub- σ -algebra generated by a finite collection of coordinate maps.*

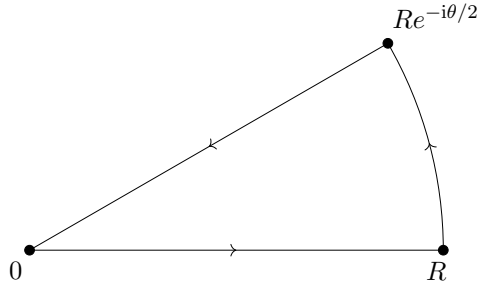
Proof. For any finite set $J \subseteq I$, let $\mathcal{F}_J$ be the σ -algebra generated by the coordinate maps $(x_i)_{i \in I} \mapsto x_j, j \in J$. Let $\mathcal{A}$ be the union of $\mathcal{F}_J$ over all finite subsets $J \subseteq I$. It is easy to see that $\mathcal{A}$ is algebra, and that it generates $\mathcal{F}$. Thus, we get the claimed result by Theorem A.2.9, because any $\mathcal{A}$ -measurable simple function is necessarily a function of finitely many coordinates. The same holds for $L^{p,q}$ for any $p, q \in [1, \infty)$. $\square$

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be *rapidly decaying* if for any $\alpha > 0$, there is some C such that $|f(x)| \leq C(1 + \|x\|)^{-\alpha}$ for all $x \in \mathbb{R}^n$. Let $\mathcal{S}(\mathbb{R}^n, \mathbb{R})$ denote the space of real-valued *Schwartz functions* on $\mathbb{R}^n$, that is, infinitely differentiable functions f such that f and all its derivatives of all orders are rapidly decaying functions. Let $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$ denote the space of all complex-valued functions on $\mathbb{R}^n$, that is, functions whose real and imaginary parts are Schwartz functions. We use different notations for real- and complex-valued Schwartz functions because Schwartz functions are typically assume to be real-valued; but we will need complex-valued Schwartz functions for certain purposes, such as the following approximation theorem.

Theorem A.2.11. *Let $L^p(\mathbb{R}^n)$ denote the L^p space of Lebesgue measure on $\mathbb{R}^n$. Then $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$ is a dense subspace of $L^p(\mathbb{R}^n)$ for each $p \in [1, \infty)$. The same holds for $L^{p,q}(\mathbb{R}^n)$, for any $p, q \in [1, \infty)$.*

Proof. As already observed in the proof of Theorem A.2.9, it is a consequence of Theorem A.1.23 that simple functions are dense in $L^p(\mathbb{R}^n)$. Thus, it suffices to prove that for any $A \in \mathcal{B}(\mathbb{R}^n)$ with finite Lebesgue measure, there is a sequence $\{f_n\}_{n \geq 1}$ in $\mathcal{S}(\mathbb{R}^n, \mathbb{R})$ converging to 1_A in L^p . But if A has finite Lebesgue measure, it is easy to see by the dominated convergence theorem that we can construct a sequence of bounded Borel sets $\{A_n\}_{n \geq 1}$ such that $1_{A_n} \rightarrow 1_A$ in L^p . Thus, it suffices to prove the above claim when A is a bounded Borel set.

Let A be a bounded Borel subset of $\mathbb{R}^n$. Let $\mathcal{A}$ be the collection of subsets of $\mathbb{R}^n$ that are disjoint unions of sets of the form $[a_1, b_1) \times \cdots \times [a_n, b_n)$ for some finite $a_1, \dots, a_n$ and


 Figure A.1: The contour C from the proof of Lemma A.3.2.

$b_1, \dots, b_n$. As we observed earlier in our construction of Lebesgue measure, $\mathcal{A}$ is an algebra of sets. Therefore by Theorem A.1.5 applied to the restriction of Lebesgue measure to a large enough cube that contains A (and recalling Lemma A.1.1), we can find a sequence of sets $\{A_n\}_{n \geq 1}$ in $\mathcal{A}$ such that $1_{A_n} \rightarrow 1_A$ in L^p . This allows us to further reduce to the case where A is of the form $[a_1, b_1) \times \dots \times [a_n, b_n)$. But for sets of this form, it is easy to construct a sequence in $\mathcal{S}(\mathbb{R}^n, \mathbb{R})$ that converges to 1_A in L^p . The proof for $L^{p,q}(\mathbb{R}^n)$ is exactly the same. $\square$

A.3 Fourier transforms

A.3.1 The inversion formula

Let $C(\mathbb{R}^n)$ denote the space of continuous complex-valued functions on $\mathbb{R}^n$. For $f \in C(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$, define

$$\widehat{f}(t) := \int_{\mathbb{R}^n} e^{it \cdot x} f(x) dx$$

to be the *Fourier transform* of f , where the integral on the right is the sum of the integrals of the real and imaginary parts of the integrand. The following results is known as the Fourier inversion theorem.

Theorem A.3.1. *For any $f \in C(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$, $\widehat{f}(t)$ is well-defined, continuous and integrable on $\mathbb{R}^n$, and satisfies the Fourier inversion formula*

$$f(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(t) e^{-it \cdot x - \epsilon \|x\|^2} dt.$$

Moreover, if $f \in \mathcal{S}(\mathbb{R}^n, \mathbb{C})$, then $\widehat{f} \in \mathcal{S}(\mathbb{R}^n, \mathbb{C})$, and the inversion formula reduces to

$$f(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(t) e^{-it \cdot x} dt.$$

We need the following lemma, which will also have other uses later.

Lemma A.3.2. *Let a and b be two complex numbers such that $a = re^{i\theta}$ for some $r > 0$ and $\theta \in (-\pi/2, \pi/2)$. Then*

$$\int_{-\infty}^{\infty} e^{-ax^2-bx} dx = e^{b^2/4a} e^{-i\theta/2} \sqrt{\frac{\pi}{r}}.$$

Proof. First of all, note that the integral is absolutely convergent since $\operatorname{Re}(a) > 0$. Next, let L be the line in the complex plane that is parallel to the real line and passes through $-b/2a$. Since $\operatorname{Re}(a) > 0$, it is easy to justify that the contour of integration can be shifted from the real line to the line L without changing the value of the integral. The value, therefore, is equal to

$$\int_{-\infty}^{\infty} e^{-a(x-b/2a)^2-b(x-b/2a)} dx = e^{b^2/4a} \int_{-\infty}^{\infty} e^{-ax^2} dx.$$

Again, the integral on the right is absolutely convergent because $\operatorname{Re}(a) > 0$. Moreover, by symmetry, it is equal to

$$2 \int_0^{\infty} e^{-ax^2} dx.$$

Recall that $a = re^{i\theta}$ for some $r > 0$ and $\theta \in (-\pi/2, \pi/2)$. Take any $R > 0$. Let C denote the closed contour which goes from 0 to R , then along a circular arc from R to $Re^{-i\theta/2}$, and then along a straight line from $Re^{-i\theta/2}$ back to 0 (see Figure A.1). Since the function e^{-az^2} is analytic on $\mathbb{C}$, we have that

$$\oint_C e^{-az^2} dz = 0.$$

Let C_1 , C_2 and C_3 denote the three parts of C described above. Then

$$\int_{C_1} e^{-az^2} dz = \int_0^R e^{-ax^2} dx,$$

and

$$\int_{C_3} e^{-az^2} dz = - \int_0^R e^{-a(xe^{-i\theta/2})^2} e^{-i\theta/2} dx = -e^{-i\theta/2} \int_0^R e^{-rx^2} dx.$$

Next, note that

$$\int_{C_2} e^{-az^2} dz = \int_0^{\theta/2} e^{-a(Re^{-is})^2} (-iRe^{-is}) ds.$$

Thus, if $\theta \in [0, \pi/2)$, then

$$\begin{aligned} \left| \int_{C_2} e^{-az^2} dz \right| &\leq \int_0^{\theta/2} |e^{-a(Re^{-is})^2} (-iRe^{-is})| ds \\ &= R \int_0^{\theta/2} e^{-\operatorname{Re}(aR^2 e^{-2is})} ds \\ &= R \int_0^{\theta/2} e^{-R^2 r \cos(\theta-2s)} ds. \end{aligned}$$

Now, since $\theta \in [0, \pi/2)$, it follows that for all $s \in [0, \theta/2]$, $\cos(\theta - 2s) \geq \cos \theta > 0$. Using this in the above expression, we get

$$\left| \int_{C_2} e^{-az^2} dz \right| \leq \frac{1}{2} R \theta e^{-R^2 r \cos \theta}.$$

If $\theta \in (-\pi/2, 0]$, a similar argument shows that

$$\left| \int_{C_2} e^{-az^2} dz \right| \leq \frac{1}{2} R |\theta| e^{-R^2 r \cos \theta}.$$

Thus, in either case, the integral over C_2 tends to zero as $R \rightarrow \infty$. Combining this with the expressions for the integrals over C_1 and C_3 obtained above, we get that

$$\int_0^\infty e^{-ax^2} dx = e^{-i\theta/2} \int_0^\infty e^{-rx^2} dx = \frac{e^{-i\theta/2}}{2} \sqrt{\frac{\pi}{r}},$$

where the last identity follows by a standard formula for Gaussian integrals. Together with the computation at the beginning of the proof, this completes the argument. $\square$

We are now ready to prove Theorem A.3.1.

Proof of Theorem A.3.1. First, note that since $f \in L^1(\mathbb{R}^n)$, $|\widehat{f}|$ is uniformly bounded. Thus, for any $\epsilon > 0$, it makes sense to write

$$\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(t) e^{-it \cdot x - \epsilon \|t\|^2} dt = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y) e^{it \cdot y} dy \right) e^{-it \cdot x - \epsilon \|t\|^2} dt. \quad (\text{A.3.1})$$

Now, since $f \in L^1(\mathbb{R}^n)$ and $\epsilon > 0$,

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x)| e^{-\epsilon \|t\|^2} dx dt < \infty.$$

Thus, we can apply Fubini's theorem to the right side in (A.3.1) and get

$$\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(t) e^{-it \cdot x - \epsilon \|t\|^2} dt = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(y) e^{it \cdot (y-x) - \epsilon \|t\|^2} dt dy.$$

By Lemma A.3.2,

$$\int_{\mathbb{R}^n} e^{it \cdot (y-x) - \epsilon \|t\|^2} dt = (2\pi)^{n/2} (2\epsilon)^{-n/2} e^{-\frac{1}{4\epsilon} \|y-x\|^2}.$$

Thus,

$$\begin{aligned} \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(x) e^{it \cdot (y-x) - \epsilon \|t\|^2} dt dx &= \frac{1}{(2\pi)^{n/2} (2\epsilon)^{n/2}} \int_{\mathbb{R}^n} f(y) e^{-\frac{1}{4\epsilon} \|y-x\|^2} dy \\ &= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(x + \sqrt{2\epsilon} z) e^{-\frac{1}{2} \|z\|^2} dz. \end{aligned}$$

Since f is continuous and in L^1 , the above identity and the dominated convergence theorem imply that

$$\lim_{\epsilon \rightarrow 0} \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(t) e^{-it \cdot x - \epsilon \|t\|^2} dt = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(x) e^{-\frac{1}{2} \|z\|^2} dz = f(x).$$

This completes the proof of the first claim. For the second claim, note that if $f \in \mathcal{S}(\mathbb{R}^n, \mathbb{C})$, then we can apply integration by parts to deduce that for any $1 \leq i \leq n$ and for any t such that $t_i \neq 0$,

$$\widehat{f}(t) = \frac{1}{-it_i} \int_{\mathbb{R}^n} e^{-it \cdot x} \partial_i f(x) dx,$$

where ∂_i denotes the partial derivative in coordinate i . Iterating this, it is easy to show that $\widehat{f}$ decays faster than any polynomial in $1/\|t\|$ as $\|t\| \rightarrow \infty$. Iterating this argument, one can show that $\widehat{f}$ is a Schwartz function. The second claim follows from this observation and the dominated convergence theorem. $\square$

A.3.2 The Plancherel identity

The inversion formula yields, among other things, the Plancherel identity.

Theorem A.3.3. *For $f, g \in \mathcal{S}(\mathbb{R}^n, \mathbb{C})$,*

$$\int_{\mathbb{R}^n} f(x) \overline{g(x)} dx = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(t) \overline{\widehat{g}(t)} dt.$$

Proof. For $\epsilon > 0$, define

$$\begin{aligned} f_\epsilon(x) &:= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(t) e^{-it \cdot x - \epsilon \|x\|^2} dt, \\ g_\epsilon(x) &:= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{g}(t) e^{-it \cdot x - \epsilon \|x\|^2} dt. \end{aligned}$$

By the decay rates of $\widehat{f}$ and $\widehat{g}$ from Theorem A.3.1, we can apply Fubini's theorem to deduce that

$$\int_{\mathbb{R}^n} f_\epsilon(x) \overline{g_\epsilon(x)} dx = \frac{1}{(2\pi)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \widehat{f}(t) \overline{\widehat{g}(s)} \left(\int_{\mathbb{R}^n} e^{-i(t-s) \cdot x - 2\epsilon \|x\|^2} dx \right) ds dt.$$

By Lemma A.3.2,

$$\int_{\mathbb{R}^n} e^{-i(t-s) \cdot x - 2\epsilon \|x\|^2} dx = e^{-(t-s)^2/8\epsilon} \left(\frac{\pi}{2\epsilon} \right)^{n/2}.$$

Combining the above displays and applying a change of variable, we get

$$\begin{aligned} \int_{\mathbb{R}^n} f_\epsilon(x) \overline{g_\epsilon(x)} dx &= \frac{1}{(2\pi)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \widehat{f}(t) \overline{\widehat{g}(s)} e^{-(t-s)^2/8\epsilon} \left(\frac{\pi}{2\epsilon} \right)^{n/2} ds dt \\ &= \frac{1}{(2\pi)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \widehat{f}(t) \overline{\widehat{g}(t + \sqrt{8\epsilon}u)} e^{-u^2} (4\pi)^{n/2} du dt. \end{aligned}$$

By the decay properties of $\widehat{f}$ and $\widehat{g}$, the right side tends to

$$\frac{1}{(2\pi)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \widehat{f}(t) \overline{\widehat{g}(t)} e^{-u^2} (4\pi)^{n/2} du dt = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \widehat{f}(t) \overline{\widehat{g}(t)} dt$$

as $\epsilon \rightarrow 0$. On the other hand, by Theorem A.3.1 and the dominated convergence theorem, it is easy to see that

$$\lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^n} f_\epsilon(x) \overline{g_\epsilon(x)} dx = \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^n} \lim_{\epsilon \rightarrow 0} f_\epsilon(x) \overline{g_\epsilon(x)} dx = \int_{\mathbb{R}^n} f(x) \overline{g(x)} dx.$$

This completes the proof. $\square$

A consequence of Theorem A.3.3 is that for all $f, g \in \mathcal{S}(\mathbb{R}^n, \mathbb{C})$,

$$\|f - g\|_{L^2}^2 = \frac{1}{(2\pi)^n} \|\widehat{f} - \widehat{g}\|_{L^2}^2.$$

Since $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$ is dense in $L^2(\mathbb{R}^n)$ by Theorem A.2.11, this allows us to extend the definition of the Fourier transform to all of $L^2(\mathbb{R}^n)$, as follows. For any $f \in L^2(\mathbb{R}^n)$, take a sequence $\{f_k\}_{k \geq 1}$ of Schwartz functions converging to f in L^2 . By the above identity, the sequence $\{\widehat{f}_k\}_{k \geq 1}$ is Cauchy in L^2 . Thus, there is some function $\widehat{f} \in L^2(\mathbb{R}^n)$ such that $\widehat{f}_k \rightarrow \widehat{f}$ in L^2 . We declare $\widehat{f}$ to be the Fourier transform of f . It is easy to see by the above identity that $\widehat{f}$ does not depend on the choice of the approximating sequence. Moreover, Theorem A.3.3 strengthens to the following, whose proof is obvious from the above construction.

Theorem A.3.4. For $f, g \in L^2(\mathbb{R}^n)$,

$$\int_{\mathbb{R}^n} f(x) \overline{g(x)} dx = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(t) \overline{\widehat{g}(t)} dt.$$

Although the Fourier transform does not always have an explicit form for functions in L^2 , it does have the usual form for functions that are both in L^1 and L^2 .

Theorem A.3.5. For $f \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$ and almost every $t \in \mathbb{R}^n$, we have

$$\widehat{f}(t) = \int_{\mathbb{R}^n} f(x) e^{it \cdot x} dx.$$

Proof. By Theorem A.2.11, we can choose a sequence $\{f_k\}_{k \geq 1}$ of Schwartz functions converging to f in both L^1 and L^2 . From L^1 convergence, it is clear that for each t ,

$$\lim_{k \rightarrow \infty} \widehat{f_k}(t) = \int_{\mathbb{R}^n} f(x) e^{it \cdot x} dx.$$

But $\widehat{f_k} \rightarrow \widehat{f}$ in L^2 , since $f_k \rightarrow f$ in L^2 . Thus, by Lemma A.2.6, there is a subsequence that converges pointwise a.e. to $\widehat{f}$. This completes the proof. $\square$

A.3.3 Fourier transforms in Minkowski space

For various reasons, we will need to work with the notion of Fourier transforms under the Minkowski inner product. The Minkowski inner product on $\mathbb{R}^4$ is defined as

$$(x, y) := x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3.$$

The norm of a vector x under this inner product is defined as

$$x^2 := (x, x) = x_0^2 - x_1^2 - x_2^2 - x_3^2.$$

Note that this norm is not positive definite. For a function $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$, one defines its Fourier transform under the Minkowski inner product as

$$\widetilde{f}(p) := \int_{\mathbb{R}^4} e^{i(x,p)} f(x) dx.$$

Clearly, this is well-defined since f is a Schwartz function. Moreover, it has its own version of the inversion formula, as follows.

Theorem A.3.6. *Let $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$ and $\widetilde{f}$ be its Fourier transform under the Minkowski inner product. Then $\widetilde{f} \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$, and*

$$f(x) = \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} e^{-i(x,p)} \widetilde{f}(p) dp.$$

Proof. Let $g(x_0, x_1, x_2, x_3) := f(x_0, -x_1, -x_2, -x_3)$. Then g is a Schwartz function, and

$$\begin{aligned} \widehat{g}(p) &= \iiint e^{i(x_0 p_0 + x_1 p_1 + x_2 p_2 + x_3 p_3)} f(x_0, -x_1, -x_2, -x_3) dx_0 dx_1 dx_2 dx_3 \\ &= \iiint e^{i(y_0 p_0 - y_1 p_1 - y_2 p_2 - y_3 p_3)} f(y_0, y_1, y_2, y_3) dy_0 dy_1 dy_2 dy_3 \\ &= \widetilde{f}(p). \end{aligned}$$

This shows that $\widetilde{f} \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$. Moreover, by the inversion formula for the ordinary Fourier

transform, this gives

$$\begin{aligned} f(x_0, x_1, x_2, x_3) &= g(x_0, -x_1, -x_2, -x_3) \\ &= \frac{1}{(2\pi)^4} \int e^{-i(x,p)} \widehat{g}(p) dp = \frac{1}{(2\pi)^4} \int e^{-i(x,p)} \widetilde{f}(p) dp. \end{aligned}$$

This completes the proof. $\square$

The Minkowski space Fourier transform has its own version of the Plancherel identity:

Theorem A.3.7. For $f, g \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$,

$$\int_{\mathbb{R}^4} f(x) \overline{g(x)} dx = \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} \widetilde{f}(p) \overline{\widetilde{g}(p)} dp.$$

Proof. As in the proof of Theorem A.3.6, if we take $u(x_0, x_1, x_2, x_3) := f(x_0, -x_1, -x_2, -x_3)$ and $v(x_0, x_1, x_2, x_3) := g(x_0, -x_1, -x_2, -x_3)$, then $\widetilde{f} = \widehat{u}$ and $\widetilde{g} = \widehat{v}$. Thus, by the Plancherel identity for the ordinary Fourier transform (Theorem A.3.3),

$$\begin{aligned} \int_{\mathbb{R}^4} f(x) \overline{g(x)} dx &= \int_{\mathbb{R}^4} u(x) \overline{v(x)} dx \\ &= \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} \widehat{u}(t) \overline{\widehat{v}(t)} dt \\ &= \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} \widetilde{f}(p) \overline{\widetilde{g}(p)} dp, \end{aligned}$$

which completes the proof. $\square$

Theorem A.3.7 yields the following corollary.

Corollary A.3.8. For $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{R})$,

$$\int_{\mathbb{R}^4} (\nabla f(x))^2 dx = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^4} p^2 |\widetilde{f}(p)|^2 dp.$$

Proof. Let ∂_j denote partial derivative in coordinate j . Then by integration by parts,

$$\widetilde{\partial_j f}(p) = -i\eta_j p_j \widetilde{f}(p),$$

where $\eta_0 = 1, \eta_1 = \eta_2 = \eta_3 = -1$. Thus, by Theorem A.3.7,

$$\int_{\mathbb{R}^4} (\partial_j f(x))^2 dx = \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} p_j^2 |\widetilde{f}(p)|^2 dp.$$

This gives

$$\begin{aligned}
 \int_{\mathbb{R}^4} (\nabla f(x))^2 dx &= \sum_{j=0}^3 \eta_j \int_{\mathbb{R}^4} (\partial_j f(x))^2 dx \\
 &= \frac{1}{(2\pi)^4} \sum_{j=0}^3 \eta_j \int_{\mathbb{R}^4} p_j^2 |\tilde{f}(p)|^2 dp \\
 &= \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} p^2 |\tilde{f}(p)|^2 dp,
 \end{aligned}$$

which completes the proof. $\square$

A.3.4 Fourier transforms on the unit circle

Let $S^1 := \{x \in \mathbb{R}^2 : \|x\| = 1\}$ denote the unit circle in $\mathbb{R}^2$, equipped with its natural Riemannian metric and the corresponding volume form. It is not hard to see that if we identify S^1 as $[-\pi, \pi)$ through the correspondence $\theta \mapsto (\cos \theta, \sin \theta)$, then integration with respect to the volume form on S^1 corresponds to integration with respect to Lebesgue measure on $[-\pi, \pi)$. A function $f \in C^\infty(S^1, \mathbb{C})$ corresponds to a map $f : [-\pi, \pi) \rightarrow \mathbb{C}$ which is infinitely differentiable in $(-\pi, \pi)$, and for each $k \geq 0$,

$$\lim_{x \downarrow -\pi} f^{(k)}(x) = \lim_{x \uparrow \pi} f^{(k)}(x) \in \mathbb{C},$$

where $f^{(k)}$ denotes the k^{th} derivative of f (with $f^{(0)} := f$).

For a function $f : [-\pi, \pi) \rightarrow \mathbb{C}$ that is integrable with respect to Lebesgue measure, the n^{th} Fourier coefficient of f , for $n \in \mathbb{Z}$, is defined as

$$\widehat{f}(n) := \int_{-\pi}^{\pi} f(x) e^{inx} dx.$$

The following lemma shows that the coefficients are rapidly decaying if f is smooth.

Lemma A.3.9. *If $f \in C^\infty(S^1, \mathbb{C})$, then for any $\alpha > 0$, $|\widehat{f}(n)| = o(|n|^{-\alpha})$ as $|n| \rightarrow \infty$.*

Proof. Since $f(x)$ has the same limit as $x \rightarrow \pm\pi$, integration by parts gives

$$\begin{aligned}
 \widehat{f}(n) &= \left[\frac{f(x) e^{inx}}{in} \right]_{-\pi}^{\pi} - \frac{1}{in} \int_{-\pi}^{\pi} f'(x) e^{inx} dx \\
 &= -\frac{1}{in} \int_{-\pi}^{\pi} f'(x) e^{inx} dx.
 \end{aligned}$$

Thus, $|\widehat{f}(n)| = O(|n|^{-1})$ as $|n| \rightarrow \infty$. Since for each k , $\widehat{f^{(k)}}(x)$ has the same limit as $x \rightarrow \pm\pi$, we can iterate the above argument to show that $|\widehat{f}(n)| = O(|n|^{-k})$ for any k . This completes the proof. $\square$

For $f \in C^\infty(S^1, \mathbb{C})$, the above lemma allows us to define

$$g(x) := \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \widehat{f}(n) e^{-inx},$$

and deduce that g is also in $C^\infty(S^1, \mathbb{C})$. It turns out that g is equal to f . This is the Fourier inversion lemma for the unit circle:

Lemma A.3.10. *Let f and g be as above. Then $f = g$.*

Proof. For $N \geq 0$, define

$$g_N(x) := \frac{1}{2\pi} \sum_{n=-N}^N \widehat{f}(n) e^{-inx},$$

and then define

$$h_N(x) := \frac{1}{N+1} \sum_{k=0}^N g_k(x).$$

We know that $g_N(x) \rightarrow g(x)$ as $N \rightarrow \infty$. This implies that $h_N(x)$ also converges to $g(x)$ as $N \rightarrow \infty$. Thus, it suffices to prove that $h_N(x) \rightarrow f(x)$. Moreover, by the continuity of f and g , it suffices to prove this for $|x| < \pi$. Fix some $x \in (-\pi, \pi)$, and note that

$$\begin{aligned} h_N(x) &= \frac{1}{2\pi(N+1)} \sum_{k=0}^N \sum_{n=-k}^k \widehat{f}(n) e^{-inx} \\ &= \frac{1}{2\pi(N+1)} \sum_{n=-N}^N \sum_{k=|n|}^N \widehat{f}(n) e^{-inx} \\ &= \frac{1}{2\pi} \sum_{n=-N}^N \widehat{f}(n) \frac{N - |n| + 1}{N+1} e^{-inx}. \end{aligned}$$

Plugging in the expression for $\widehat{f}(n)$, we get

$$h_N(x) = \int_{-\pi}^{\pi} f(y) F_N(x-y) dy, \tag{A.3.2}$$

where

$$F_N(u) := \frac{1}{2\pi(N+1)} \sum_{n=-N}^N (N - |n| + 1) e^{-inu}.$$

Now, by the formula for the sum of a geometric series, we have, for $u \not\equiv 0 \pmod{2\pi}$,

$$\sum_{n=0}^N e^{-inu} = \frac{1 - e^{-i(N+1)u}}{1 - e^{-iu}} = e^{-\frac{1}{2}iNu} \frac{\sin(\frac{1}{2}(N+1)u)}{\sin(\frac{1}{2}u)}.$$

This gives

$$\begin{aligned} \frac{\sin^2(\frac{1}{2}(N+1)u)}{\sin^2(\frac{1}{2}u)} &= \left| \sum_{n=0}^N e^{-inu} \right|^2 \\ &= \sum_{m,n=0}^N e^{-i(m-n)u} = \sum_{n=-N}^N (N - |n| + 1) e^{-inu}. \end{aligned}$$

Thus, we get

$$F_N(u) = \frac{\sin^2(\frac{1}{2}(N+1)u)}{2\pi(N+1)\sin^2(\frac{1}{2}u)}.$$

Plugging this into equation (A.3.2) gives

$$h_N(x) = \int_{-\pi}^{\pi} f(y) \frac{\sin^2(\frac{1}{2}(N+1)(x-y))}{2\pi(N+1)\sin^2(\frac{1}{2}(x-y))} dy.$$

Making the change of variable $z := \frac{1}{2}(N+1)(x-y)$, we get

$$h_N(x) = \int_{\frac{1}{2}(N+1)(x-\pi)}^{\frac{1}{2}(N+1)(x+\pi)} f\left(x - \frac{2z}{N+1}\right) \frac{\sin^2 z}{\pi(N+1)^2 \sin^2(\frac{z}{N+1})} dz.$$

As $N \rightarrow \infty$, the integrand converges to

$$f(x) \frac{\sin^2 z}{\pi z^2}.$$

Moreover, within the range of integration we have

$$\frac{|z|}{N+1} = \frac{1}{2}|x-y| \leq \frac{1}{2}(\pi + |x|) < \pi,$$

by our assumption that $|x| < \pi$. Thus, there is a positive constant C depending on x such that throughout the range of integration, we have

$$\left| \sin\left(\frac{z}{N+1}\right) \right| \geq \frac{C|z|}{N+1}.$$

Thus, the absolute value of the integrand is bounded above by

$$\frac{C' \sin^2 z}{z^2},$$

where C' is a finite constant depending on f and x . This allows us to apply the dominated convergence theorem and conclude that

$$\lim_{N \rightarrow \infty} h_N(x) = \int_{-\infty}^{\infty} f(x) \frac{\sin^2 z}{\pi z^2} dz = f(x),$$

where the last identity follows by considering the case $f \equiv 1$, where $h_N \equiv 1$ for each N . $\square$

We will now extend this result to smooth functions on $(S^1)^d$ for arbitrary d . Let $f : [-\pi, \pi]^d \rightarrow \mathbb{R}$

A.3.5 The Poisson summation formula

The following important result is known as the Poisson summation formula.

Theorem A.3.11. *For $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{C})$, we have*

$$\sum_{n \in \mathbb{Z}^d} f(n) = \sum_{n \in \mathbb{Z}^d} \widehat{f}(2\pi n).$$

Proof. Define a function $g : [-\pi, \pi]^d \rightarrow \mathbb{C}$ as

$$g(x) := \sum_{n \in \mathbb{Z}^d} f((2\pi)^{-1}x + n).$$

This is well-defined since f is a Schwartz function. Moreover, the Schwartzian nature of f allows us to differentiate g as many times as we want, by taking the derivatives inside the sum. If $g^{(k)}$ and $f^{(k)}$ denote the k^{th} derivatives of g and f , then we get

$$g^{(k)}(x) := (2\pi)^{-k} \sum_{n \in \mathbb{Z}^d} f^{(k)}((2\pi)^{-1}x + n).$$

This shows, in particular, that

$$\lim_{x \uparrow \pi} g^{(k)}(x) = \lim_{x \downarrow -\pi} g^{(k)}(x)$$

for each k . Thus, by Lemma A.3.10,

$$\sum_{n \in \mathbb{Z}} f(n) = g(0) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \widehat{g}(n). \quad (\text{A.3.3})$$

Now note that by the dominated convergence theorem,

$$\begin{aligned} \widehat{g}(n) &= \int_{-\pi}^{\pi} \left(\sum_{m \in \mathbb{Z}} f((2\pi)^{-1}x + m) \right) e^{inx} dx \\ &= \sum_{m \in \mathbb{Z}} \int_{-\pi}^{\pi} f((2\pi)^{-1}x + m) e^{inx} dx \\ &= 2\pi \sum_{m \in \mathbb{Z}} \int_{-\frac{1}{2}}^{\frac{1}{2}} f(y + m) e^{2\pi i n y} dy. \end{aligned}$$

But, we have

$$\begin{aligned}\int_{-\frac{1}{2}}^{\frac{1}{2}} f(y+m)e^{2\pi i n y} dy &= \int_{-\frac{1}{2}}^{\frac{1}{2}} f(y+m)e^{2\pi i n (y+m)} dy \\ &= \int_{m-\frac{1}{2}}^{m+\frac{1}{2}} f(y)e^{2\pi i n y} dy.\end{aligned}$$

Plugging this into the previous display and again applying the dominated convergence theorem, we get

$$\widehat{g}(n) = 2\pi \sum_{m \in \mathbb{Z}} \int_{m-\frac{1}{2}}^{m+\frac{1}{2}} f(y)e^{2\pi i n y} dy = 2\pi \int_{-\infty}^{\infty} f(y)e^{2\pi i n y} dy = 2\pi \widehat{f}(2\pi n).$$

By equation (A.3.3), this completes the proof. □

Appendix B

Probability Theory

B.1 Basics of probability theory

B.1.1 Random variables

If $(\Omega, \mathcal{F})$ is a measurable space, a measurable function X from Ω into $\mathbb{R}$ or $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, \infty\}$ is called a random variable. More generally, if $(\Omega', \mathcal{F}')$ is another measurable space, then a measurable function X from Ω into Ω' is called a Ω' -valued random variable defined on Ω . In particular, if $\Omega' = \mathbb{R}^n$, we say that X is a random vector.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let X be a random variable defined on Ω . It is the common convention in probability theory to write $\mathbb{P}(X \in A)$ instead of $\mathbb{P}(\{\omega \in \Omega : X(\omega) \in A\})$. Similarly, we write $\{X \in A\}$ to denote the set $\{\omega \in \Omega : X(\omega) \in A\}$. Similarly, if X and Y are two random variables, the event $\{X \in A, Y \in B\}$ is the set $\{\omega : X(\omega) \in A, Y(\omega) \in B\}$.

Another commonly used convention is that if $f : \Omega' \rightarrow \mathbb{R}$ is a measurable function and X is an Ω' -valued random variable, $f(X)$ denotes the real-valued random variable $f \circ X$. The σ -algebra generated by a random variable X is denoted by $\sigma(X)$, and if $\{X_i\}_{i \in I}$ is a collection of random variables defined on the same probability space, then the σ -algebra $\sigma(\{X_i\}_{i \in I})$ generated by the collection $\{X_i\}_{i \in I}$ is defined to be the σ -algebra generated by the union of the sets $\sigma(X_i)$, $i \in I$. We will have use for the following result.

Proposition B.1.1. *Let $\{X_i\}_{i \in I}$ be a collection of $(\mathbb{R}^*)^I$ -valued random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Define a map $X : \Omega \rightarrow (\mathbb{R}^*)^I$ as $X(\omega) := (X_i(\omega))_{i \in I}$. Suppose we endow $(\mathbb{R}^*)^I$ with its cylinder σ -algebra. Then X is a measurable map, and $\sigma(X) = \sigma(\{X_i : i \in I\})$. Moreover, two $(\mathbb{R}^*)^I$ -valued random variables X and Y have the same law if and only if for every finite set $J \subseteq I$, the random vectors $(X_j)_{j \in J}$ and $(Y_j)_{j \in J}$ have the same distribution*

Proof. To prove that X is measurable, it suffices to show that $X^{-1}(A) \in \mathcal{F}$ for every rectangular set A . This follows easily from the fact that each X_i is measurable. This also implies that $\sigma(X) \subseteq \sigma(\{X_i : i \in I\})$. The reverse inclusion follows from the measurability (with respect to the cylinder σ -algebra) of coordinate maps on $(\mathbb{R}^*)^I$. The second assertion

follows by Theorem A.1.4, the measurability of the coordinate maps, and the fact that the coordinate maps generate the cylinder σ -algebra. $\square$

We will have use for the following result about real-valued random variables that are measurable with respect to the σ -algebra generated by a random variable taking value in some measurable space.

Proposition B.1.2. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let X be an Ω' -valued random variable, where $(\Omega', \mathcal{F}')$ is a measurable space. Let Y be an $\mathbb{R}^*$ -valued random variable defined on Ω that is measurable with respect to $\sigma(X)$. Then there exists a measurable map $F : \Omega' \rightarrow \mathbb{R}^*$ such that $Y = F(X)$. Moreover, if F and G are two such maps, then $F = G$ a.e. with respect to the law of X .*

Proof. First, suppose that $Y = 1_A$ for some $A \in \sigma(X)$. Then $A = X^{-1}(B)$ for some $B \in \mathcal{F}'$. Define $F := 1_B$. Then F is a measurable map, and $Y(\omega) = 1_A(\omega) = 1_B(X(\omega)) = F(X(\omega))$. Thus, $Y = F(X)$. This proves the result when Y is an indicator function. Clearly, this extends to nonnegative simple functions. Next, let Y be any $[0, \infty]$ -valued random variable defined on Ω that is $\sigma(X)$ -measurable. By Theorem A.1.23, we can find an increasing sequence of nonnegative simple $\sigma(X)$ -measurable random variables $\{Y_n\}_{n \geq 1}$ increasing to Y . For each n , let $F_n : \Omega' \rightarrow \mathbb{R}^*$ be a measurable map such that $Y_n = F_n(X)$. Define $F := \limsup_{n \rightarrow \infty} F_n$ pointwise. Then F is measurable. Note that for all $\omega \in \Omega$,

$$\begin{aligned} F(X(\omega)) &= \limsup_{n \rightarrow \infty} F_n(X(\omega)) \\ &= \limsup_{n \rightarrow \infty} Y_n(\omega) = \lim_{n \rightarrow \infty} Y_n(\omega) = Y(\omega). \end{aligned}$$

This completes the proof when Y is nonnegative. Lastly, for a general $\mathbb{R}^*$ -valued $\sigma(X)$ -measurable random variable Y , let Y^+ and Y^- denote the positive and negative parts of Y . Let F_1 and F_2 denote measurable functions on Ω' such that $Y^+ = F_1(X)$ and $Y^- = F_2(X)$, constructed as above. The construction guarantees that F_1 and F_2 cannot be simultaneously ∞ at any point in Ω' . It is easy to argue from this that $F := F_1 - F_2$ is a well-defined measurable map. Clearly, we have $Y = F(X)$. For the last assertion, note that if $Y = F(X)$ and $Y = G(X)$, then $F(X) = G(X)$ and hence

$$\mathbb{P}(X^{-1}(\{\omega' \in \Omega' : F(\omega') = G(\omega')\})) = \mathbb{P}(\{\omega \in \Omega : F(X(\omega)) = G(X(\omega))\}) = 1.$$

Thus, $F = G$ a.e. with respect to the law of X . $\square$

Let X be a random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and taking values in some measurable space $(\Omega', \mathcal{F}')$. Then X induces a probability measure μ_X on this space, defined as

$$\mu_X(A) := \mathbb{P}(X \in A).$$

The probability measure μ_X is called the law of X or the distribution of X .

The expected value (or expectation, or mean) of an $\mathbb{R}^*$ -valued random variable X , denoted by $\mathbb{E}(X)$, is simply the integral $\int X d\mathbb{P}$, provided that the integral exists. A random variable X is called integrable if it is integrable as a measurable function, that is, if $\mathbb{E}|X| < \infty$. A notation that is often used is that if X is a random variable and A is an event (defined on the same space), then

$$\mathbb{E}(X; A) := \mathbb{E}(X1_A).$$

The variance of X is defined as

$$\text{Var}(X) := \mathbb{E}(X^2) - (\mathbb{E}(X))^2,$$

provided that $\mathbb{E}(X^2)$ is finite (such random variables are called ‘square integrable’). Note that by the monotonicity of Hölder norms for probability measures, the finiteness of $\mathbb{E}(X^2)$ automatically implies the finiteness of $\mathbb{E}|X|$ and in particular the existence of $\mathbb{E}(X)$. The covariance of two random variables X and Y defined on the same probability space is defined as

$$\text{Cov}(X, Y) := \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y),$$

provided that both X , Y and XY are integrable. Notice that $\text{Var}(X) = \text{Cov}(X, X)$. It is easy to see that covariance is a bilinear form on square integrable random variables.

B.1.2 Independence

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and suppose that $\mathcal{G}_1, \dots, \mathcal{G}_n$ are sub- σ -algebras of $\mathcal{F}$. We say that $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent if for any $A_1 \in \mathcal{G}_1, A_2 \in \mathcal{G}_2, \dots, A_n \in \mathcal{G}_n$,

$$\mathbb{P}(A_1 \cap \dots \cap A_n) = \prod_{i=1}^n \mathbb{P}(A_i).$$

More generally, an arbitrary collection $\{\mathcal{G}_i\}_{i \in I}$ of sub- σ -algebras are called independent if any finitely many of them are independent.

The independence of σ -algebras is used to define the independence of random variables and events. A collection of events $\{A_i\}_{i \in I}$ in $\mathcal{F}$ are said to be independent if the σ -algebras $\mathcal{G}_i := \{\emptyset, A_i, A_i^c, \Omega\}$ generated by the A_i ’s are independent. Moreover, an event A is said to be independent of a σ -algebra $\mathcal{G}$ if the σ -algebras $\{\emptyset, A, A^c, \Omega\}$ and $\mathcal{G}$ are independent.

A collection of random variables $\{X_i\}_{i \in I}$ defined on Ω are said to be independent if $\{\sigma(X_i)\}_{i \in I}$ are independent sub- σ -algebras of $\mathcal{F}$. A random variable X is said to be independent of a σ -algebra $\mathcal{G}$ if the σ -algebras $\sigma(X)$ and $\mathcal{G}$ are independent.

A particularly important definition in probability theory is the notion of an independent and identically distributed (i.i.d.) sequence of random variables. A sequence $\{X_i\}_{i=1}^\infty$ is said to be i.i.d. if the X_i ’s are independent and all have the same distribution.

An important property of independent random variables is the following.

Proposition B.1.3. *If X and Y are independent random variables such that X and Y are integrable, then the product XY is also integrable and $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$. The identity also holds if X and Y are nonnegative but not necessarily integrable.*

Proof. First, suppose that $X = \sum_{i=1}^k a_i 1_{A_i}$ and $Y = \sum_{j=1}^m b_j 1_{B_j}$ for some nonnegative a_i 's and b_j 's, and measurable A_i 's and B_j 's. Without loss of generality, assume that all the a_i 's are distinct and all the b_j 's are distinct. Then each $A_i \in \sigma(X)$ and each $B_j \in \sigma(Y)$, because $A_i = X^{-1}(\{a_i\})$ and $B_j = Y^{-1}(\{b_j\})$. Thus, for each i and j , A_i and B_j are independent. This gives

$$\begin{aligned} \mathbb{E}(XY) &= \sum_{i=1}^k \sum_{j=1}^m a_i b_j \mathbb{E}(1_{A_i} 1_{B_j}) = \sum_{i=1}^k \sum_{j=1}^m a_i b_j \mathbb{P}(A_i \cap B_j) \\ &= \sum_{i=1}^k \sum_{j=1}^m a_i b_j \mathbb{P}(A_i) \mathbb{P}(B_j) = \mathbb{E}(X) \mathbb{E}(Y). \end{aligned}$$

Next take any nonnegative X and Y that are independent. Construct nonnegative simple random variables X_n and Y_n increasing to X and Y , using the method from the proof of Theorem A.1.23. From the construction, it is easy to see that each X_n is $\sigma(X)$ -measurable and each Y_n is $\sigma(Y)$ -measurable. Therefore X_n and Y_n are independent, and hence $\mathbb{E}(X_n Y_n) = \mathbb{E}(X_n) \mathbb{E}(Y_n)$, since we have already proved this identity for nonnegative simple random variables. Now note that since $X_n \uparrow X$ and $Y_n \uparrow Y$, we have $X_n Y_n \uparrow XY$. Therefore by the monotone convergence theorem,

$$\mathbb{E}(XY) = \lim_{n \rightarrow \infty} \mathbb{E}(X_n Y_n) = \lim_{n \rightarrow \infty} \mathbb{E}(X_n) \mathbb{E}(Y_n) = \mathbb{E}(X) \mathbb{E}(Y)$$

Finally, take any independent X and Y . It is easy to see that X^+ and X^- are $\sigma(X)$ -measurable, and Y^+ and Y^- are $\sigma(Y)$ -measurable. Therefore

$$\begin{aligned} \mathbb{E}|XY| &= \mathbb{E}((X^+ + X^-)(Y^+ + Y^-)) \\ &= \mathbb{E}(X^+ Y^+) + \mathbb{E}(X^+ Y^-) + \mathbb{E}(X^- Y^+) + \mathbb{E}(X^- Y^-) \\ &= \mathbb{E}(X^+) \mathbb{E}(Y^+) + \mathbb{E}(X^+) \mathbb{E}(Y^-) + \mathbb{E}(X^-) \mathbb{E}(Y^+) + \mathbb{E}(X^-) \mathbb{E}(Y^-) \\ &= \mathbb{E}|X| \mathbb{E}|Y|. \end{aligned}$$

Since $\mathbb{E}|X|$ and $\mathbb{E}|Y|$ are both finite, this shows that XY is integrable. Repeating the steps in the above display starting with $\mathbb{E}(XY)$ instead of $\mathbb{E}|XY|$, we get $\mathbb{E}(XY) = \mathbb{E}(X) \mathbb{E}(Y)$. $\square$

The next result is often useful for proving independence of σ -algebras.

Theorem B.1.4. *Let $\{X_i\}_{i \in I}$ and $\{Y_j\}_{j \in J}$ be two collections of random variables defined on the same probability space. Suppose that for any finite $F \subseteq I$ and $G \subseteq J$, the collections $\{X_i\}_{i \in F}$ and $\{Y_j\}_{j \in G}$ are independent. Then $\sigma(\{X_i\}_{i \in I})$ and $\sigma(\{Y_j\}_{j \in J})$ are independent.*

To prove the theorem, we need the following lemma.

Lemma B.1.5. *If $\mathcal{A}$ is a π -system of sets and B is an event such that B and A are independent for every $A \in \mathcal{A}$, show that B and $\sigma(\mathcal{A})$ are independent.*

Proof. Let $\mathcal{L}$ be the set of all $A \in \sigma(\mathcal{A})$ such that B and A are independent. By assumption, $\mathcal{A} \subseteq \mathcal{L}$. Since $\mathcal{A}$ is a π -system, Dynkin's theorem tells us that it suffices to prove that $\mathcal{L}$ is a λ -system. Since $\Omega \in \sigma(\mathcal{A})$ and $\mathbb{P}(B \cap \Omega) = \mathbb{P}(B) = \mathbb{P}(B) \cap \mathbb{P}(\Omega)$, we conclude that $\Omega \in \mathcal{L}$. Next, take any $A \in \mathcal{L}$. Note that

$$\begin{aligned} \mathbb{P}(B \cap A^c) &= \mathbb{P}(B) - \mathbb{P}(B \cap A) \\ &= \mathbb{P}(B) - \mathbb{P}(B)\mathbb{P}(A) = \mathbb{P}(B)(1 - \mathbb{P}(A)) = \mathbb{P}(B)\mathbb{P}(A^c). \end{aligned}$$

Thus, $\mathcal{L}$ is closed under taking complements. A similar argument shows that $\mathcal{L}$ is closed under taking countable disjoint unions. $\square$

We are now ready to prove Theorem B.1.4.

Proof of Theorem B.1.4. Let $\mathcal{A}$ be the union of $\sigma(\{X_i\}_{i \in F})$ over all finite sets $F \subseteq I$, and let $\mathcal{B}$ be the union of $\sigma(\{Y_i\}_{i \in G})$ over all finite sets $G \subseteq J$. It is easy to see that $\mathcal{A}$ and $\mathcal{B}$ are algebras; in particular, they are π -systems. Take any $B \in \mathcal{B}$. Then B and A are independent for every $A \in \mathcal{A}$. Thus, by Lemma B.1.5, B and A are independent for every $A \in \sigma(\mathcal{A}) = \sigma(\{X_i\}_{i \in I})$. Next, take any $A \in \sigma(\{X_i\}_{i \in I})$. We have already established that A and B are independent for any $B \in \mathcal{B}$. By another application of Dynkin's theorem, we conclude that A and B are independent for all $B \in \sigma(\mathcal{B})$. $\square$

B.1.3 Probability densities

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let ν be another measure on $(\Omega, \mathcal{F})$. We say that ν has a density ρ with respect to μ if $\rho : \Omega \rightarrow [0, \infty)$ is a measurable function such that for all $A \in \mathcal{F}$,

$$\nu(A) = \int_A \rho(x) d\mu(x).$$

We sometimes denote this relation by

$$d\nu(x) = \rho(x) d\mu(x), \quad \text{or} \quad \frac{d\nu}{d\mu} = \rho.$$

If ν is a probability measure, we say that ρ is its probability density function (p.d.f.). Conversely, if $\rho : \Omega \rightarrow [0, \infty)$ is any measurable function such that

$$\int_{\Omega} \rho(x) d\mu(x) = 1,$$

then it is easy to check that the functional

$$\nu(A) := \int_A \rho(x) d\mu(x)$$

defines a probability measure on $(\Omega, \mathcal{F})$.

Lemma B.1.6. *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let ν be a probability measure $(\Omega, \mathcal{F})$ that has density ρ with respect to μ . Then for any measurable $g : \Omega \rightarrow \mathbb{R}$ that is integrable with respect to ν , we have*

$$\int_{\Omega} g(x) d\nu(x) = \int_{\Omega} g(x) \rho(x) d\mu(x).$$

Proof. When g is an indicator function, the result follows from the definition of p.d.f. Therefore, the result holds when g is a simple function, by the linearity of Lebesgue integrals. We can now pass to nonnegative measurable g using Theorem A.1.5 and the monotone convergence theorem, and finally to integrable g by decomposing into positive and negative parts. $\square$

B.1.4 Total variation distance

Let $(\Omega, \mathcal{F})$ be a measurable space and ν_1, ν_2 be two probability measures on this space. The total variation distance between ν_1, ν_2 is defined as

$$d_{\text{TV}}(\nu_1, \nu_2) = \inf \mathbb{P}(X \neq Y),$$

where the infimum is take over all pairs of Ω -valued random variables (X, Y) defined on the same probability space, such that $X \sim \nu_1$ and $Y \sim \nu_2$. Another way to say this is that $d_{\text{TV}}(\nu_1, \nu_2)$ is the infimum of $\gamma(\{(x, y) \in \Omega^2 : x \neq y\})$ over all probability measures γ on $(\Omega^2, \mathcal{F}^2)$ that has marginals ν_1 and ν_2 (meaning that $\nu_1(A) = \gamma(A \times \Omega)$ and $\nu_2(A) = \gamma(\Omega \times A)$ for all $A \in \mathcal{F}$). The following lemma gives a useful alternative formula for the total variation distance.

Lemma B.1.7. *Let ν_1, ν_2 be as above, and let μ be a measure on $(\Omega, \mathcal{F})$ such that ν_1 has density ρ_1 with respect to μ and ν_2 has density ρ_2 with respect to μ . Then*

$$d_{\text{TV}}(\nu_1, \nu_2) = \frac{1}{2} \int_{\Omega} |\rho_1(x) - \rho_2(x)| d\mu(x).$$

Moreover, the infimum in the definition of total variation distance is attained at some probability measure on Ω^2 .

Proof. Let $\alpha(x) := \min\{\rho_1(x), \rho_2(x)\}$, and let $p := \int_{\Omega} \alpha(x) d\mu(x)$. Construct a pair of Ω -valued random variables (X, Y) as follows. If $p = 0$, generate $X \sim \nu_1$ and $Y \sim \nu_2$ independently. If $p = 1$, generate $X \sim \nu_1$ and let $Y = X$. If $0 < p < 1$, let β be the probability density function α/p , and let θ_1 and θ_2 be the probability density functions $(\rho_1 - \alpha)/(1 - p)$ and $(\rho_2 - \alpha)/(1 - p)$. With probability p , generate X from β and let $Y = X$. With probability $1 - p$, generate X and Y independently from θ_1 and θ_2 . Note that θ_1 and θ_2 cannot be simultaneously nonzero anywhere. This implies that

$$\mathbb{P}(X \neq Y) = 1 - p. \tag{B.1.1}$$

We claim that $X \sim \nu_1$ and $Y \sim \nu_2$. To see this, note that if $0 < p < 1$, then for any $A \in \mathcal{F}$,

$$\begin{aligned}\mathbb{P}(X \in A) &= p \int_A \frac{\alpha(x)}{p} d\mu(x) + (1-p) \int_A \frac{\rho_1(x) - \alpha(x)}{1-p} d\mu(x) \\ &= \int_A \rho_1(x) d\mu(x) = \nu_1(A),\end{aligned}$$

and by a similar argument $\mathbb{P}(Y \in A) = \nu_2(A)$. If $p = 0$, then $\alpha = 0$ a.e., which again implies the above identity. Finally, if $p = 1$, then $\int (\rho_j - \alpha) d\mu = 0$ for $j = 1, 2$. Since $\rho_j \geq \alpha$, this implies that $\alpha = \rho_1 = \rho_2$, which ensures that $X \sim \nu_1$ and $Y \sim \nu_2$.

Next, note that

$$\begin{aligned}& \frac{1}{2} \int_{\Omega} |\rho_1(x) - \rho_2(x)| d\mu(x) \\ &= \frac{1}{2} \int_{\Omega} (\rho_1(x) + \rho_2(x) - 2 \min\{\rho_1(x), \rho_2(x)\}) d\mu(x) \\ &= 1 - \int_{\Omega} \alpha(x) d\mu(x) = 1 - p.\end{aligned}$$

Combining this with equation (B.1.1) and the observation from the previous paragraph, we get that

$$d_{\text{TV}}(\nu_1, \nu_2) \leq \mathbb{P}(X \neq Y) = 1 - p = \frac{1}{2} \int_{\Omega} |\rho_1(x) - \rho_2(x)| d\mu(x).$$

For the reverse inequality, let $A := \{x \in \Omega : \rho_1(x) \geq \rho_2(x)\}$. Take any (X, Y) such that $X \sim \nu_1$ and $Y \sim \nu_2$. Since

$$\begin{aligned}0 &= \int_{\Omega} (\rho_1(x) - \rho_2(x)) d\mu(x) \\ &= \int_{\Omega} (\rho_1(x) - \rho_2(x))^+ d\mu(x) - \int_{\Omega} (\rho_1(x) - \rho_2(x))^- d\mu(x),\end{aligned}$$

we get

$$\begin{aligned}\int_{\Omega} |\rho_1(x) - \rho_2(x)| d\mu(x) &= \int_{\Omega} (\rho_1(x) - \rho_2(x))^+ d\mu(x) + \int_{\Omega} (\rho_1(x) - \rho_2(x))^- d\mu(x) \\ &= 2 \int_{\Omega} (\rho_1(x) - \rho_2(x))^+ d\mu(x) \\ &= 2 \int_A (\rho_1(x) - \rho_2(x)) d\mu(x) = 2(\mathbb{P}(X \in A) - \mathbb{P}(Y \in A)).\end{aligned}$$

But, we have

$$\begin{aligned}\mathbb{P}(X \in A) - \mathbb{P}(Y \in A) &\leq \mathbb{P}(X \in A) - \mathbb{P}(X \in A, Y \in A) \\ &= \mathbb{P}(X \in A, Y \notin A) \leq \mathbb{P}(X \neq Y).\end{aligned}$$

Thus, we get

$$\int_{\Omega} |\rho_1(x) - \rho_2(x)| d\mu(x) \leq \mathbb{P}(X \neq Y).$$

Taking infimum over (X, Y) proves the reverse inequality. $\square$

B.2 Gaussian random variables

B.2.1 Real-valued Gaussian random variables

It is common to say that a real-valued random variable has a p.d.f. if its law has a density with respect to Lebesgue measure. An important example of such a p.d.f. is the density function for the normal (or Gaussian) distribution with mean parameter $\mu \in \mathbb{R}$ and standard deviation parameter $\sigma > 0$, given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

If a random variable X has this distribution, we write $X \sim N(\mu, \sigma^2)$. The special case of $\mu = 0$ and $\sigma = 1$ is known as the standard normal (or standard Gaussian) distribution. It is easy to check that the above function is indeed a probability density function, and that its expected value is μ and variance is σ^2 .

B.2.2 Gaussian random vectors

Recall that a random vector X is simply an $\mathbb{R}^n$ -valued random variable for some positive integer n . The probability density function of X , if it exists, is a measurable function $f : \mathbb{R}^n \rightarrow [0, \infty)$ such that for any $A \in \mathcal{B}(\mathbb{R}^n)$,

$$\mathbb{P}(X \in A) = \int_A f(x_1, \dots, x_n) dx_1 \cdots dx_n,$$

where $dx_1 \cdots dx_n$ denotes integration with respect to Lebesgue measure on $\mathbb{R}^n$. The law μ_X of X the probability measure on $\mathbb{R}^n$ induced by X , that is, $\mu_X(A) := \mathbb{P}(X \in A)$. The mean vector of a random vector $X = (X_1, \dots, X_n)$ is the vector $\mu = (\mu_1, \dots, \mu_n)$, where $\mu_i = \mathbb{E}(X_i)$, assuming that the means exist. The covariance matrix of a random vector $X = (X_1, \dots, X_n)$ is the $n \times n$ matrix $\Sigma = (\sigma_{ij})_{i,j=1}^n$, where $\sigma_{ij} = \text{Cov}(X_i, X_j)$, provided that these covariances exist. The covariance matrix is sometimes denoted by $\text{Cov}(X)$.

The higher dimensional analogues of normal random variables are multivariate normal (or multivariate Gaussian) random vectors. Given $\mu \in \mathbb{R}^n$ and a strictly positive definite matrix Σ of order n , the multivariate normal distribution with mean μ and covariance matrix Σ is the probability measure on $\mathbb{R}^n$ with probability density function

$$\frac{1}{(2\pi)^{n/2}(\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right).$$

If X has this law, we write $X \sim N_n(\mu, \Sigma)$. An easy computation shows that if $X \sim N_n(\mu, \Sigma)$, then $\mathbb{E}(X) = \mu$ and $\text{Cov}(X) = \Sigma$. The $N_n(0, I)$ distribution is called the n -dimensional standard normal distribution (where I denotes the identity matrix). Note that if $X \sim N_n(0, I)$, the probability density function of X factorizes into the product of $N(0, 1)$ densities, one for each coordinate. From this, it follows that the coordinates of X are i.i.d. $N(0, 1)$ random variables.

B.2.3 Gaussian fields

A Gaussian field or a Gaussian process is any collection of Gaussian random variables $X = (X_i)_{i \in I}$ such that for any finite set $J \subseteq I$, the random vector $(X_j)_{j \in J}$ has a multivariate Gaussian distribution. Recall that X can be viewed as an $\mathbb{R}^I$ -valued random variable, and by Proposition B.1.1, that its distribution is completely determined by its finite-dimensional distributions (i.e., the distributions of $(X_j)_{j \in J}$ for finite subsets $J \subseteq I$). We will make frequent use of the following result.

Proposition B.2.1. *Let $X = (X_i)_{i \in I}$ be a Gaussian field. Suppose that J and K are disjoint subsets of I such that $\text{Cov}(X_j, X_k) = 0$ for every $j \in J$ and $k \in K$. Then the Gaussian fields $(X_j)_{j \in J}$ and $(X_k)_{k \in K}$ are independent.*

Proof. If $J \cup K$ is finite, then this is easy to see that this is true simply by writing down the probability density of $(X_l)_{l \in J \cup K}$ and observing that the density factorizes into a product of two densities. For the general case, the same argument establishes that $(X_j)_{j \in J'}$ and $(X_k)_{k \in K'}$ are independent for any finite $J' \subseteq J$ and any finite $K' \subseteq K$. By Theorem B.1.4, the result follows. $\square$

B.2.4 Characteristic functions of Gaussian random variables

An important function associated with an n -dimensional random vector X is its characteristic function $\phi_X : \mathbb{R}^n \rightarrow \mathbb{C}$, defined as

$$\phi_X(t) := \mathbb{E}(e^{it \cdot X}),$$

where $i = \sqrt{-1}$ and $t \cdot X$ is the inner product of t and X . The right side of the above expression is defined as

$$\mathbb{E}(e^{it \cdot X}) := \mathbb{E}(\cos t \cdot X) + i\mathbb{E}(\sin t \cdot X),$$

and the two expectations on the right always exist because $\cos$ and $\sin$ are bounded functions. In fact, the expected value of any complex random variable can be defined in the same manner, provided that the expected values of the real and imaginary parts exist and are finite.

Proposition B.2.2. *If X and Y are integrable random variables and $Z = X + iY$, and $\mathbb{E}(Z)$ is defined as $\mathbb{E}(X) + i\mathbb{E}(Y)$, then $|\mathbb{E}(Z)| \leq \mathbb{E}|Z|$.*

Proof. Let $\alpha := \mathbb{E}(Z)$ and $\bar{\alpha}$ denote the complex conjugate of α . It is easy to check the linearity of expectation holds for complex random variables. Thus,

$$\begin{aligned} |\mathbb{E}(Z)|^2 &= \bar{\alpha}\mathbb{E}(Z) = \mathbb{E}(\bar{\alpha}Z) \\ &= \mathbb{E}(\operatorname{Re}(\bar{\alpha}Z)) + i\mathbb{E}(\operatorname{Im}(\bar{\alpha}Z)), \end{aligned}$$

where $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ denote the real and imaginary parts of a complex number z . Since $|\mathbb{E}(Z)|^2$ is real, the above identity show that $|\mathbb{E}(Z)|^2 = \mathbb{E}(\operatorname{Re}(\bar{\alpha}Z))$. But $\operatorname{Re}(\bar{\alpha}Z) \leq |\bar{\alpha}Z| = |\alpha||Z|$. Thus, $|\mathbb{E}(Z)|^2 \leq |\alpha|\mathbb{E}|Z|$, which completes the proof. $\square$

The above result shows, in particular, that $|\phi_X(t)| \leq 1$ for any random vector X and any t . The next result shows that the characteristic function characterizes the law of X .

Theorem B.2.3. *If two n -dimensional random vectors X and Y have the same characteristic function, then they have the same law.*

Proof. Take any Schwartz function f on $\mathbb{R}^n$. By Theorem A.3.1 and Fubini's theorem,

$$\begin{aligned} \mathbb{E}(f(X)) &= \mathbb{E}\left(\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(t) e^{-it \cdot X} dt\right) \\ &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(t) \phi_X(t) dt \\ &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(t) \phi_Y(t) dt = \mathbb{E}(f(Y)). \end{aligned}$$

Now, it is easy to see that the indicator function of any set of the form $[a_1, b_1) \times \cdots \times [a_n, b_n)$ can be approach pointwise almost everywhere by a sequence of uniformly bounded Schwartz functions. Thus, the above identity shows that $\mathbb{P}(X \in A) = \mathbb{P}(Y \in A)$ for any such set A . By Theorem A.1.4, this proves that X and Y have the same law. $\square$

We now compute the characteristic function of the $N_n(\mu, \Sigma)$ distribution. Let X be a random vector with this distribution. Let $\Sigma^{1/2}$ denote the positive definite square-root of Σ and $\Sigma^{-1/2}$ denote its inverse. Then applying the change of variable $y = \Sigma^{-1/2}(x - \mu)$, we get

$$\begin{aligned} \phi_X(t) &= \int_{\mathbb{R}^n} \frac{1}{(2\pi)^{n/2} \sqrt{\det \Sigma}} \exp\left(it \cdot x - \frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right) dx \\ &= \int_{\mathbb{R}^n} \frac{1}{(2\pi)^{n/2}} \exp\left(it \cdot (\mu + \Sigma^{1/2}y) - \frac{\|y\|^2}{2}\right) dy \\ &= e^{it \cdot \mu} \int_{\mathbb{R}^n} \frac{1}{(2\pi)^{n/2}} \exp\left(ia \cdot y - \frac{\|y\|^2}{2}\right) dy, \end{aligned}$$

where $a := \Sigma^{1/2}t$. Thus, by Lemma A.3.2,

$$\begin{aligned}\phi_X(t) &= e^{it \cdot \mu} \prod_{i=1}^n \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} \exp\left(ia_i y - \frac{y^2}{2}\right) dy \\ &= e^{it \cdot \mu - \frac{1}{2}\|a\|^2} = e^{it \cdot \mu - \frac{1}{2}t^T \Sigma t}.\end{aligned}$$

An important consequence of the above calculation is the following fact. Suppose that $X \sim N_n(\mu, \Sigma)$, and A is an $m \times n$ real matrix for some positive integer m . Let $b \in \mathbb{R}^m$, and let $Y := AX + b$. Then

$$\begin{aligned}\phi_Y(t) &= \mathbb{E}(e^{it \cdot Y}) \\ &= \mathbb{E}(e^{it \cdot (AX+b)}) \\ &= e^{it \cdot b} \mathbb{E}(e^{i(A^T t) \cdot X}) \\ &= e^{it \cdot (A\mu+b) - \frac{1}{2}A\Sigma A^T}.\end{aligned}$$

Thus, by Theorem B.2.3, we deduce that if A has rank m , then $Y \sim N_m(A\mu + b, A\Sigma A^T)$. But even if A does not have full rank, the above formula for the characteristic function of Y is still valid. This allows us to define the $N_n(\mu, \Sigma)$ distribution even if Σ is only positive semidefinite and not positive definite. Namely, for any $\mu \in \mathbb{R}^n$ and any $n \times n$ positive semidefinite matrix Σ , the $N_n(\mu, \Sigma)$ is the unique probability distribution on $\mathbb{R}^n$ that has characteristic function $e^{i\mu \cdot t - \frac{1}{2}t^T \Sigma t}$.

The most important consequence of the above discussion is that if we have a Gaussian random vector, and take any linear transformation of it, possibly mapping to a different dimension, the result is still a Gaussian random vector. Its distribution is determined simply by calculating its mean vector and covariance matrix.

A result about Gaussian random vectors that we will use later is the following.

Theorem B.2.4. *Let $X \sim N_n(\mu, \Sigma)$, where $\mu \in \mathbb{R}^n$ and Σ is an $n \times n$ positive semidefinite matrix. Let ν be the law of X . Then the span of the functions $x \mapsto e^{it \cdot x}$, as t ranges over $\mathbb{R}^n$, is a dense subspace of $L^2(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \nu)$.*

Proof. Let m be the rank of Σ . Then there is an $n \times m$ matrix A of rank m such that $\Sigma = AA^T$. Thus, if $Y \sim N_m(0, I)$, then $\mu + AY \sim N_n(\mu, \Sigma)$. So let us assume that $X = \mu + AY$. Take any $f \in L^2(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \nu, \mathbb{C})$, so that $\mathbb{E}|f(X)|^2 < \infty$. Define $g : \mathbb{R}^m \rightarrow \mathbb{R}$ as

$$g(x) := f(\mu + Ax).$$

Suppose we can show that for any $\epsilon > 0$, there exist some $k \geq 1$, some $s_1, \dots, s_k \in \mathbb{R}^m$, and some $a_1, \dots, a_k \in \mathbb{C}$ such that

$$\mathbb{E} \left| g(Y) - \sum_{j=1}^k a_j e^{is_j \cdot Y} \right|^2 < \epsilon.$$

Since $\text{rank}(A) = m$ and $m \leq n$, there exist $t_1, \dots, t_k \in \mathbb{R}^n$ such that $A^T t_j = s_j$ for $j = 1, \dots, k$. Thus,

$$\mathbb{E} \left| f(X) - \sum_{j=1}^k a_j e^{-it_j \cdot \mu} e^{it_j \cdot X} \right|^2 = \mathbb{E} \left| g(Y) - \sum_{j=1}^k a_j e^{is_j \cdot Y} \right|^2 < \epsilon.$$

This shows that it suffices to prove the following: The span of the functions $x \mapsto e^{is \cdot x}$, as t ranges over $\mathbb{R}^m$, is a dense subspace of $L^2(\mathbb{R}^m, \mathcal{B}(\mathbb{R}^m), \gamma, \mathbb{C})$, where γ is the standard Gaussian measure on $\mathbb{R}^m$. Let us denote this L^2 space by $L^2(\gamma)$, and let $L^2(\mathbb{R}^m)$ denote the L^2 space of complex-valued functions on $\mathbb{R}^m$ under Lebesgue measure. Let S denote the closure of the span of the functions $x \mapsto e^{is \cdot x}$ in $L^2(\gamma)$. Take any $g \in L^2(\gamma)$. Let $g_N(x) := g(x)1_{|x| \leq N}$. Then by the dominated convergence theorem, $g_N \rightarrow g$ in $L^2(\gamma)$ as $N \rightarrow \infty$. Thus, it suffices to prove that any $g \in L^2(\gamma)$ that vanishes outside a bounded region is in S .

Now, any $g \in L^2(\gamma)$ that vanishes outside a bounded region is in $L^2(\mathbb{R}^m)$, and convergence in $L^2(\mathbb{R}^m)$ implies convergence in $L^2(\gamma)$. By Theorem A.2.11, Schwartz functions are dense in $L^2(\mathbb{R}^m)$. Combining these observations, we see that it suffices to prove that any Schwartz function is in S .

Take any Schwartz function h . Recall that by Theorem A.3.1, its Fourier transform $\hat{h}$ is also a Schwartz function, and the Fourier inversion formula holds. Using smoothness and decay properties of Schwartz functions and their derivatives, it is now easy to approximate the inversion formula by a finite linear combination of functions of the form $e^{is \cdot x}$ up to any desired degree of accuracy in the $L^2(\gamma)$ norm. $\square$

We will also use the following corollary of Theorem B.2.4.

Corollary B.2.5. *Let $X \sim N_n(\mu, \Sigma)$, where $\mu \in \mathbb{R}^n$ and Σ is an $n \times n$ positive semidefinite matrix. Let ν be the law of X . Then the span of polynomial functions of X (with complex coefficients) is a dense subspace of $L^2(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \nu)$.*

Proof. A straightforward calculation using the classical gamma integral shows that if $Z \sim N(0, 1)$, then $\mathbb{E}|Z|^p$ grows like $p^{p/2+o(1)}$ as $p \rightarrow \infty$. Using this and various applications of Hölder's inequality, it follows that any function like $e^{it \cdot X}$ can be arbitrarily well-approximated in L^2 by a polynomial, simply by truncating the power series for $e^{it \cdot X}$. By Theorem B.2.4, this completes the proof. $\square$

B.2.5 The Wick product formula

The Wick product formula is an important fact about Gaussian random variables that is widely used in quantum field theory. In the following theorem and elsewhere, we will say that Gaussian random variable is ‘centered’ if it has expected value zero. Recall that a perfect matching of a vertex set V is a set of undirected edges E on V such that every vertex has degree exactly equal to 1. A perfect matching exists if and only if the size of V is even.

Theorem B.2.6 (Wick product formula). *Suppose that $X_1, \dots, X_n$ are jointly centered Gaussian random variables, where n is even. Then*

$$\mathbb{E}(X_1 X_2 \cdots X_n) = \sum_{\substack{\text{all perfect matchings} \\ E \text{ of } 1, \dots, n}} \prod_{\{i,j\} \in E} \mathbb{E}(X_i X_j).$$

If n is odd, the above expectation is equal to zero.

Proof. We will prove the Wick product formula by induction on n . For $n = 1$ and $n = 2$, the result is trivial. Let us take $n \geq 3$ and assume that the claim is true for smaller values of n . Note that for $Z \sim N(0, 1)$ and any integer $k \geq 2$, integration by parts gives

$$\begin{aligned} \mathbb{E}(Z^k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^k e^{-\frac{1}{2}x^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (k-1)x^{k-2} e^{-\frac{1}{2}x^2} dx = \mathbb{E}((k-1)Z^{k-2}). \end{aligned}$$

In other words, $\mathbb{E}(Zf(Z)) = \mathbb{E}(f'(Z))$ for any monomial f . Taking linear combinations, we see that this holds for any polynomial f . By Fubini's theorem, this implies that if $Z_1, \dots, Z_n$ are independent $N(0, 1)$ random variables, then for any polynomial f in n variables and any j ,

$$\mathbb{E}(Z_j f(Z_1, \dots, Z_n)) = \mathbb{E}(\partial_j f(Z_1, \dots, Z_n)), \quad (\text{B.2.1})$$

where $\partial_j f$ is the partial derivative of f in coordinate j .

Let $\Sigma = (\sigma_{ij})_{1 \leq i, j \leq n}$ denote the covariance matrix of $X = (X_1, \dots, X_n)$, so that $\sigma_{ij} = \mathbb{E}(X_i X_j)$. Let $A = (a_{ij})_{1 \leq i, j \leq n}$ be an $n \times n$ matrix such that $\Sigma = AA^T$. (Such a matrix exists because Σ is positive semidefinite.) Then we can express X as AZ , where $Z \sim N_n(0, I)$. Take any polynomial f in n variables. Then $f(X) = f(AZ)$ is also a polynomial in Z , and by an abuse of notation,

$$\frac{\partial}{\partial Z_j} f(X) = \sum_{i=1}^n \partial_i f(X) \frac{\partial X_i}{\partial Z_j} = \sum_{i=1}^n a_{ij} \partial_i f(X).$$

Thus, by equation (B.2.1),

$$\begin{aligned} \mathbb{E}(X_i f(X)) &= \sum_{j=1}^n a_{ij} \mathbb{E}(Z_j f(X)) \\ &= \sum_{j=1}^n a_{ij} \mathbb{E}\left(\frac{\partial}{\partial Z_j} f(X)\right) \\ &= \sum_{j,k=1}^n a_{ij} a_{kj} \mathbb{E}(\partial_k f(X)) = \sum_{k=1}^n \sigma_{ik} \mathbb{E}(\partial_k f(X)). \end{aligned} \quad (\text{B.2.2})$$

In particular, taking $i = 1$ and $f(X) = X_2 X_3 \cdots X_n$, we get

$$\mathbb{E}(X_1 X_2 \cdots X_n) = \sum_{k=1}^n \sigma_{1k} \mathbb{E} \left(\prod_{\substack{1 \leq j \leq n, \\ j \neq 1, k}} X_j \right).$$

It is now easy to see how the claim follows from the induction hypothesis. $\square$

The Wick product formula stated in the above theorem is for products of real Gaussian random variables. We will need a generalization of that to complex Gaussian random variables. A complex Gaussian random variable is simply a complex random variable whose real and imaginary parts are jointly Gaussian. The Wick product formula has exactly the same form in the complex case as in the real case.

Theorem B.2.7 (Wick product formula, complex case). *Suppose that $X_1, \dots, X_n$ are jointly centered complex Gaussian random variables, where n is even. Then*

$$\mathbb{E}(X_1 X_2 \cdots X_n) = \sum_{\substack{\text{all perfect matchings } E \\ \text{of } 1, \dots, n}} \prod_{\{i, j\} \in E} \mathbb{E}(X_i X_j).$$

If n is odd, the above expectation is equal to zero.

Proof. Let $\mathcal{P}$ denote the set of perfect matchings of $1, \dots, n$. Let X_i^0 and X_i^1 denote the real and imaginary parts of X_i . Then

$$\begin{aligned} \mathbb{E}(X_1 X_2 \cdots X_n) &= \mathbb{E}((X_1^0 + iX_1^1)(X_2^0 + iX_2^1) \cdots (X_n^0 + iX_n^1)) \\ &= \sum_{s \in \{0,1\}^n} i^{s_1 + \cdots + s_n} \mathbb{E}(X_1^{s_1} X_2^{s_2} \cdots X_n^{s_n}). \end{aligned}$$

If n is odd, Theorem B.2.6 shows that each term on the right is zero. If n is even, then again by Theorem B.2.6, this gives

$$\mathbb{E}(X_1 X_2 \cdots X_n) = \sum_{s \in \{0,1\}^n} i^{s_1 + \cdots + s_n} \left(\sum_{E \in \mathcal{P}} \prod_{\{i, j\} \in E} \mathbb{E}(X_i^{s_i} X_j^{s_j}) \right).$$

Exchanging the order of summation, we get

$$\begin{aligned} \mathbb{E}(X_1 X_2 \cdots X_n) &= \sum_{E \in \mathcal{P}} \sum_{s \in \{0,1\}^n} i^{s_1 + \cdots + s_n} \prod_{\{i, j\} \in E} \mathbb{E}(X_i^{s_i} X_j^{s_j}) \\ &= \sum_{E \in \mathcal{P}} \prod_{\{i, j\} \in E} \mathbb{E}((X_i^0 + iX_i^1)(X_j^0 + iX_j^1)) \\ &= \sum_{E \in \mathcal{P}} \prod_{\{i, j\} \in E} \mathbb{E}(X_i X_j). \end{aligned}$$

This completes the proof. $\square$

The following corollary is sometimes useful.

Corollary B.2.8. *Suppose that $X_1, \dots, X_{m+n}$ are jointly centered complex Gaussian random variables, where $m+n$ is even and $m \leq n$. Let $\mathcal{P}'$ be the set of all perfect matchings of $1, \dots, m+n$ where at least one $i \leq m$ is matched with some $j > m$. Then*

$$\text{Cov}(X_1 \cdots X_m, X_{m+1} \cdots X_{m+n}) = \sum_{E \in \mathcal{P}'} \prod_{\{i,j\} \in E} \mathbb{E}(X_i X_j).$$

If $m+n$ is odd, the covariance is zero.

Proof. Writing $\mathbb{E}(X_1 \cdots X_{m+n})$ using the Wick product formula, and subtracting off $\mathbb{E}(X_1 \cdots X_m)\mathbb{E}(X_{m+1} \cdots X_{m+n})$, where both expectations are written using the Wick product formula, we see that terms corresponding to matchings outside $\mathcal{P}'$ cancel out. $\square$

B.2.6 Wick ordered monomials of Gaussian random variables

Let $\{X_i\}_{i \in I}$ be a centered complex Gaussian field — that is, a collection of jointly centered complex Gaussian random variables. Given any $i_1, \dots, i_n \in I$ (not necessarily distinct), we say that the product $X_{i_1} X_{i_2} \cdots X_{i_n}$ is a monomial of degree n . The *Wick ordered* version of this monomial is defined as follows. For any finite set V , let $\mathcal{P}(V)$ denote the set of perfect matchings of V . Denoting $V := \{1, \dots, n\}$, we define the Wick ordered version of the monomial $X_{i_1} X_{i_2} \cdots X_{i_n}$ as

$$:X_{i_1} X_{i_2} \cdots X_{i_n}: = \sum_{\substack{U \subseteq V, \\ |U| \text{ even}}} (-1)^{|U|/2} \left(\sum_{E \in \mathcal{P}(U)} \prod_{\{a,b\} \in E} \mathbb{E}(X_{i_a} X_{i_b}) \right) \left(\prod_{a \in V \setminus U} X_{i_a} \right).$$

To understand this definition, let us look at the case $n = 4$. Then $V = \{1, 2, 3, 4\}$ has 8 subsets of even size. Therefore, we have 8 choices of U . The term corresponding to $U = \emptyset$ is equal to $X_{i_1} X_{i_2} X_{i_3} X_{i_4}$. Next, let us look at the case $U = \{1, 2\}$. This U has exactly one perfect matching, which implies that the contribution from this term is $-\mathbb{E}(X_{i_1} X_{i_2}) X_{i_3} X_{i_4}$. Similar contributions come from the other subsets of size 2. Finally, the contribution from $U = \{1, 2, 3, 4\}$ is

$$\sum_{E \in \mathcal{P}(V)} \prod_{\{a,b\} \in E} \mathbb{E}(X_{i_a} X_{i_b}).$$

Putting it all together, we have

$$\begin{aligned} & :X_{i_1} X_{i_2} X_{i_3} X_{i_4}: \\ &= X_{i_1} X_{i_2} X_{i_3} X_{i_4} - \mathbb{E}(X_{i_1} X_{i_2}) X_{i_3} X_{i_4} - \mathbb{E}(X_{i_1} X_{i_3}) X_{i_2} X_{i_4} - \mathbb{E}(X_{i_1} X_{i_4}) X_{i_2} X_{i_3} \\ &\quad - \mathbb{E}(X_{i_2} X_{i_3}) X_{i_1} X_{i_4} - \mathbb{E}(X_{i_2} X_{i_4}) X_{i_1} X_{i_3} - \mathbb{E}(X_{i_3} X_{i_4}) X_{i_1} X_{i_2} \\ &\quad + \mathbb{E}(X_{i_1} X_{i_2}) \mathbb{E}(X_{i_3} X_{i_4}) + \mathbb{E}(X_{i_1} X_{i_3}) \mathbb{E}(X_{i_2} X_{i_4}) + \mathbb{E}(X_{i_1} X_{i_4}) \mathbb{E}(X_{i_2} X_{i_3}) \\ &\quad + \mathbb{E}(X_{i_2} X_{i_3}) \mathbb{E}(X_{i_1} X_{i_4}) + \mathbb{E}(X_{i_2} X_{i_4}) \mathbb{E}(X_{i_1} X_{i_3}) + \mathbb{E}(X_{i_3} X_{i_4}) \mathbb{E}(X_{i_1} X_{i_2}). \end{aligned}$$

So, for example, for a single centered Gaussian random variable X ,

$$:X^4: = X^4 - 6\mathbb{E}(X^2)X^2 + 6(\mathbb{E}(X^2))^2.$$

The key property of Wick ordered monomials is captured by the following result.

Theorem B.2.9. *Let $\{X_i\}_{i \in I}$ be a centered complex Gaussian field. Any Wick ordered monomial associated with this field has mean zero, and the expected value of the product of two Wick ordered monomials of different degrees is equal to zero. Lastly, for any n and any $i_1, \dots, i_n \in I$ and $j_1, \dots, j_n \in I$,*

$$\mathbb{E}(:X_{i_1} \cdots X_{i_n} :: X_{j_1} \cdots X_{j_n} :) = \sum_{\sigma \in S_n} \prod_{a=1}^n \mathbb{E}(X_{i_a} X_{j_{\sigma(a)}}).$$

Proof. Take any $n \geq 1$, $m \geq 0$, and $i_1, \dots, i_{m+n} \in I$, not necessarily distinct. Let $V := \{1, \dots, n\}$ and $W := \{1, \dots, m+n\}$. Then note that

$$:X_{i_1} \cdots X_{i_n} : X_{i_{n+1}} \cdots X_{i_{m+n}} = \sum_{\substack{U \subseteq V, \\ |U| \text{ even}}} (-1)^{|U|/2} \left(\sum_{E \in \mathcal{P}(U)} \prod_{\{a,b\} \in E} \mathbb{E}(X_{i_a} X_{i_b}) \right) \left(\prod_{a \in W \setminus U} X_{i_a} \right).$$

By the Wick product formula for complex Gaussian random variables,

$$\mathbb{E} \left(\prod_{a \in W \setminus U} X_{i_a} \right) = \sum_{F \in \mathcal{P}(W \setminus U)} \prod_{\{c,d\} \in F} \mathbb{E}(X_{i_c} X_{i_d}).$$

Combining the last two displays, we get

$$\mathbb{E}(:X_{i_1} \cdots X_{i_n} : X_{i_{n+1}} \cdots X_{i_{m+n}}) = \sum_{\substack{U \subseteq V, \\ |U| \text{ even}}} (-1)^{|U|/2} \sum_{E \in \mathcal{P}(U)} \prod_{\{a,b\} \in E \cup F} \mathbb{E}(X_{i_a} X_{i_b}).$$

The right side can be rewritten as

$$\sum_{G \in \mathcal{P}(W)} c(G) \prod_{\{a,b\} \in G} \mathbb{E}(X_{i_a} X_{i_b}),$$

where the coefficient $c(G)$ is given by

$$c(G) = \sum_{\substack{U \subseteq V, \\ |U| \text{ even}}} (-1)^{|U|/2} c(G, U),$$

where $c(G, U) = 1$ if G can be decomposed as $E \cup F$ for some $E \in \mathcal{P}(U)$ and $F \in \mathcal{P}(W \setminus U)$, and 0 otherwise.

Take any $G \in \mathcal{P}(W)$. Let G' be the subset of edges in G whose endpoints are both in V . If G' is empty, then no nonempty $U \subseteq V$ can have $c(G, U) = 1$. Also, trivially, $c(G, \emptyset) = 1$. Therefore in this case, $c(G) = 1$. Next, suppose that $|G'| = k > 0$. Then $c(G, U) = 1$ if and

only if U consists of the endpoints of some subset of edges in G' . This shows that

$$c(G) = \sum_{j=0}^k \binom{k}{j} (-1)^j = 0.$$

Thus, we conclude that

$$\mathbb{E}(:X_{i_1} \cdots X_{i_n} : X_{i_{n+1}} \cdots X_{i_{n+m}}) = \sum_{G \in \mathcal{P}'(W)} \prod_{\{a,b\} \in G} \mathbb{E}(X_{i_a} X_{i_b}), \quad (\text{B.2.3})$$

where $\mathcal{P}'(W)$ consists of all $G \in \mathcal{P}(W)$ such that every edge in G connects a vertex in V to a vertex in $W \setminus V$. Note that $\mathcal{P}'(W)$ is empty if $m < n$. In particular, it is empty if $m = 0$. This proves that $\mathbb{E}(:X_{i_1} \cdots X_{i_n} :) = 0$. Moreover, it shows that

$$\mathbb{E}(:X_{i_1} \cdots X_{i_n} : X_{i_{n+1}} \cdots X_{i_{n+m}}) = 0$$

whenever $m < n$, which implies that

$$\mathbb{E}(:X_{i_1} \cdots X_{i_n} :: X_{i_{n+1}} \cdots X_{i_{n+m}} :) = 0$$

when $m < n$, and that

$$\mathbb{E}(:X_{i_1} \cdots X_{i_n} :: X_{i_{n+1}} \cdots X_{i_{2n}} :) = \mathbb{E}(:X_{i_1} \cdots X_{i_n} : X_{i_{n+1}} \cdots X_{i_{2n}}).$$

But by equation (B.2.3), it is easy to see that the right side is equal to the claimed value. $\square$

B.3 Some other distributions

B.3.1 Convolutions

Given two probability measures μ_1 and μ_2 on $\mathbb{R}$, their convolution $\mu_1 * \mu_2$ is defined to be the push-forward of $\mu_1 \times \mu_2$ under the addition map from $\mathbb{R}^2$ to $\mathbb{R}$. That is, if $\phi(x, y) := x + y$, then for any $A \in \mathcal{B}(\mathbb{R})$,

$$\mu_1 * \mu_2(A) := \mu_1 \times \mu_2(\phi^{-1}(A)).$$

It is not hard to show that in the language of random variables, the convolution of two probability measures has the following description: If X and Y are independent random variables with laws μ_1 and μ_2 , then $\mu_1 * \mu_2$ is the law of $X + Y$.

Proposition B.3.1. *Let X and Y be independent random variables. Suppose that Y has probability density function g . Then the sum $Z := X + Y$ has probability density function $h(z) = \mathbb{E}g(z - X)$. Consequently, if X has probability density function f , then*

$$h(z) = \int_{-\infty}^{\infty} f(x)g(z - x)dx.$$

Proof. Let μ_1 be the law of X and μ_2 be the law of Y . Let $\mu := \mu_1 \times \mu_2$. Then the discussion preceding the statement of the proposition shows that for any $A \in \mathcal{B}(\mathbb{R})$,

$$\begin{aligned}\mathbb{P}(Z \in A) &= \mu_1 * \mu_2(A) = \mu(\phi^{-1}(A)) \\ &= \int_{\mathbb{R}^2} 1_{\phi^{-1}(A)}(x, y) d\mu(x, y).\end{aligned}$$

By Fubini's theorem, this integral equals

$$\int_{\mathbb{R}} \int_{\mathbb{R}} 1_{\phi^{-1}(A)}(x, y) d\mu_2(y) d\mu_1(x).$$

But for any x ,

$$\begin{aligned}\int_{\mathbb{R}} 1_{\phi^{-1}(A)}(x, y) d\mu_2(y) &= \int_{\mathbb{R}} 1_A(x + y) d\mu_2(y) \\ &= \int_{\mathbb{R}} 1_A(x + y) g(y) dy \\ &= \int_{\mathbb{R}} 1_A(z) g(z - x) dz = \int_A g(z - x) dz,\end{aligned}$$

where the last step follows by the translation invariance of Lebesgue measure (Corollary A.1.24). Thus again by Fubini's theorem,

$$\begin{aligned}\mathbb{P}(Z \in A) &= \int_{\mathbb{R}} \int_A g(z - x) dz d\mu_1(x) \\ &= \int_A \int_{\mathbb{R}} g(z - x) d\mu_1(x) dz \\ &= \int_A \mathbb{E}g(z - X) dz.\end{aligned}$$

This proves the first claim. The second claim follows by combining the first claim and Lemma B.1.6. $\square$

B.3.2 The Cauchy distribution

The standard Cauchy distribution is the probability measure on the real line that has probability density function

$$\frac{1}{\pi(1 + x^2)}, \quad x \in \mathbb{R}.$$

Note that this is a probability density since it is the derivative of $\frac{1}{\pi} \arctan x$, which goes from $-\frac{1}{2}$ to $\frac{1}{2}$ as x goes from $-\infty$ to ∞ . We will now compute the characteristic function of the standard Cauchy distribution.

Take any $t \in \mathbb{R}$. First, suppose that $t > 0$. Take any $R > 1$. Consider the semicircular contour C on the upper half-plane with vertices at $-R$ and R . Let

$$f(z) := \frac{e^{itz}}{\pi(1 + z^2)}, \quad z \in \mathbb{C} \setminus \{-i, i\}.$$

The contour C encloses the pole i of f . Therefore, by the residue theorem,

$$\oint_C f(z)dz = 2\pi i \lim_{z \rightarrow i} (z - i)f(z) = 2\pi i \lim_{z \rightarrow i} \frac{e^{itz}}{\pi(z + i)} = e^{-t}.$$

Now, it is easy to see that the contribution to integral from the circular part is of order $1/R$. Thus,

$$\oint_C f(z)dz = \int_{-R}^R f(x)dx + O(1/R).$$

Taking $R \rightarrow \infty$, we see that for $t > 0$,

$$\int_{-\infty}^{\infty} \frac{e^{itx}}{\pi(1+x^2)}dx = e^{-t}.$$

A similar argument using the semicircular contour in the lower half-plane with vertices at $-R$ and R shows that for $t < 0$,

$$\int_{-\infty}^{\infty} \frac{e^{itx}}{\pi(1+x^2)}dx = e^t.$$

Combining, we see that the characteristic function of the standard Cauchy distribution is given by $\phi(t) = e^{-|t|}$. We will need this fact at a later point.

B.3.3 The Lévy distribution

The standard Lévy distribution is the probability measure on $(0, \infty)$ with probability density function

$$\frac{e^{-\frac{1}{2x}}}{\sqrt{2\pi x^3/2}}, \quad x > 0.$$

To see that this is indeed a p.d.f., note that by the change of variable $y = x^{-1/2}$, we get

$$\int_0^{\infty} \frac{e^{-\frac{1}{2x}}}{\sqrt{2\pi x^3/2}}dx = \int_0^{\infty} \frac{2e^{-\frac{1}{2}y^2}}{\sqrt{2\pi}}dy = 1.$$

Let us now calculate the characteristic function of this distribution. The characteristic function is given by

$$\phi(t) = \int_0^{\infty} \frac{e^{itx - \frac{1}{2x}}}{\sqrt{2\pi x^3/2}}dx = \int_0^{\infty} \frac{2}{\sqrt{2\pi}} \exp\left(\frac{it}{y^2} - \frac{y^2}{2}\right)dy, \quad (\text{B.3.1})$$

where again, we used the change of variable $y = x^{-1/2}$.

When $t = 0$, we already know that $\phi(t) = 0$. So let us take nonzero t . The imaginary number $-2it$ has two square-roots. One square-root has positive real part, and the other has negative real part. Let $\sqrt{-2it}$ denote the square-root that has positive real part, and let $(-2it)^{1/4}$ denote the square-root of this $\sqrt{-2it}$ that has positive real part. A simple argument shows that in the rightmost integral in equation (B.3.1), the integration can be

shifted to any ray from the origin that goes rightward. In particular, we can choose the ray that passes through $(-2it)^{1/4}$. This gives

$$\begin{aligned}\phi(t) &= \sqrt{\frac{2}{\pi}}(-2it)^{1/4} \int_0^\infty \exp\left(\frac{it}{\sqrt{-2it}y^2} - \frac{\sqrt{-2it}y^2}{2}\right) dy \\ &= \sqrt{\frac{2}{\pi}}(-2it)^{1/4} \int_0^\infty \exp\left(-\frac{\sqrt{-2it}}{2}\left(y^2 + \frac{1}{y^2}\right)\right) dy \\ &= \sqrt{\frac{2}{\pi}}(-2it)^{1/4} e^{-\sqrt{-2it}} \int_0^\infty \exp\left(-\frac{\sqrt{-2it}}{2}\left(y - \frac{1}{y}\right)^2\right) dy.\end{aligned}$$

Let I denote the integral in the last line. Then the change of variable $z = 1/y$ shows that

$$I = \int_0^\infty \frac{1}{z^2} \exp\left(-\frac{\sqrt{-2it}}{2}\left(z - \frac{1}{z}\right)^2\right) dz.$$

Averaging this new expression for I with the original expression, we get

$$I = \frac{1}{2} \int_0^\infty \left(1 + \frac{1}{y^2}\right) \exp\left(-\frac{\sqrt{-2it}}{2}\left(y - \frac{1}{y}\right)^2\right) dy.$$

Doing the change of variable $z = y - \frac{1}{y}$ in the above integral, we get

$$I = \frac{1}{2} \int_{-\infty}^\infty \exp\left(-\frac{\sqrt{-2it}}{2}z^2\right) dz$$

By Lemma A.3.2, the above integral is equal to $\sqrt{2\pi}/(-2it)^{1/4}$. Substituting this value, we get

$$\phi(t) = e^{-\sqrt{-2it}}.$$

The above form of the characteristic function implies that if X and Y are independent random variables from the Lévy distribution, then for any $a, b > 0$, $a^2X + b^2Y$ has the same law as $(a + b)^2X$. This is because

$$\begin{aligned}\phi_{a^2X+b^2Y}(t) &= \mathbb{E}(e^{it(a^2X+b^2Y)}) \\ &= \mathbb{E}(e^{ita^2X})\mathbb{E}(e^{itb^2Y}) \\ &= \phi(ta^2)\phi(tb^2) = e^{-a\sqrt{-2it}}e^{-b\sqrt{-2it}} = e^{-(a+b)\sqrt{-2it}} = \phi_{(a+b)^2X}(t).\end{aligned}$$

It is easy to see that the p.d.f. of a^2X is the function

$$p_a(x) = \frac{a}{\sqrt{2\pi}x^{3/2}}e^{-\frac{a^2}{2x}}$$

on the nonnegative axis. Thus, the convolution of p_a and p_b is p_{a+b} . We will need this fact later.

B.4 Conditional probability

B.4.1 Conditional expectation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let X be a real-valued random variable defined on this space. Suppose that X is integrable (that is, $\mathbb{E}|X|$ is finite). Let $\mathcal{G}$ be a sub- σ -algebra of $\mathcal{F}$. The conditional expectation of X given $\mathcal{G}$, denoted by $\mathbb{E}(X|\mathcal{G})$, is a $\mathcal{G}$ -measurable integrable random variable Y such that for any $B \in \mathcal{G}$, $\mathbb{E}(X; B) = \mathbb{E}(Y; B)$.

It is not clear from the above definition whether such a random variable exists, but if it does, then almost sure uniqueness is not difficult to prove.

Lemma B.4.1. *If Y and Z are two random variables that qualify as $\mathbb{E}(X|\mathcal{G})$, then $Y = Z$ a.s.*

Proof. Let $B := \{\omega : Y(\omega) > Z(\omega)\}$. Note that $B \in \mathcal{G}$. Since $\mathbb{E}(X; B) = \mathbb{E}(Y; B) = \mathbb{E}(Z; B)$, we get $\mathbb{E}((Y - Z); B) = 0$. But $(Y - Z)1_B$ is a nonnegative random variable. Thus, $(Y - Z)1_B = 0$ a.s., which implies that $Y \leq Z$ a.s. Similarly, $Y \geq Z$ a.s. $\square$

We will now show that conditional expectation always exists. To do this, we will follow the usual route — that is, first show it for simple random variables, then use monotonicity to show it for all nonnegative random variables, and finally use positive and negative parts to cover all integrable random variables.

Lemma B.4.2. *If X is a simple random variable, then $\mathbb{E}(X|\mathcal{G})$ exists.*

Proof. Since X is simple, it is square-integrable. Let $\mathcal{H}$ denote the Hilbert space $L^2(\Omega, \mathcal{G}, \mathbb{P})$. Let $\{Y_n\}_{n \geq 1}$ be a sequence of elements of $\mathcal{H}$ such that

$$\lim_{n \rightarrow \infty} \mathbb{E}(X - Y_n)^2 = s := \inf_{Y \in \mathcal{H}} \mathbb{E}(X - Y)^2. \quad (\text{B.4.1})$$

Take any m and n . Let $Z := (Y_m + Y_n)/2$. Recall the parallelogram identity $(a - b)^2 + (a + b)^2 = 2a^2 + 2b^2$. Taking $a = (X - Y_m)/2$ and $b = (X - Y_n)/2$, we get

$$\mathbb{E}(X - Z)^2 + \frac{1}{4}\mathbb{E}(Y_m - Y_n)^2 = \frac{1}{2}\mathbb{E}(X - Y_m)^2 + \frac{1}{2}\mathbb{E}(X - Y_n)^2.$$

Since $Z \in \mathcal{H}$, $\mathbb{E}(X - Z)^2 \geq s$. Thus,

$$\frac{1}{4}\mathbb{E}(Y_m - Y_n)^2 \leq \frac{1}{2}\mathbb{E}(X - Y_m)^2 + \frac{1}{2}\mathbb{E}(X - Y_n)^2 - s.$$

By (B.4.1), this proves that $\{Y_n\}_{n \geq 1}$ is a Cauchy sequence in $\mathcal{H}$. Since $\mathcal{H}$ is a Hilbert space, this sequence has a $\mathcal{G}$ -measurable L^2 limit Y . Clearly,

$$\mathbb{E}(X - Y)^2 = \lim_{n \rightarrow \infty} \mathbb{E}(X - Y_n)^2 = s.$$

Now take any $Z \in \mathcal{H}$. Then for any $\lambda \in \mathbb{R}$, $Y + \lambda Z \in \mathcal{H}$, and so by the above property of Y , $\mathbb{E}(X - Y - \lambda Z)^2$ is minimized when $\lambda = 0$. But this is just a quadratic polynomial in

λ , and so the condition that it is minimized at 0 implies that the coefficient of λ , namely $\mathbb{E}((X - Y)Z)$, is zero. Taking $Z = 1_B$ for any $B \in \mathcal{G}$ proves that $Y = \mathbb{E}(X|\mathcal{G})$. $\square$

Lemma B.4.3. *If X_1 and X_2 are simple random variables and $a, b \in \mathbb{R}$, then $\mathbb{E}(aX_1 + bX_2|\mathcal{G}) = a\mathbb{E}(X_1|\mathcal{G}) + b\mathbb{E}(X_2|\mathcal{G})$ a.s.*

Proof. It is easy to see from the definition that $a\mathbb{E}(X_1|\mathcal{G}) + b\mathbb{E}(X_2|\mathcal{G})$ is a valid candidate for $\mathbb{E}(aX_1 + bX_2|\mathcal{G})$, and therefore the result follows by the uniqueness of conditional expectation. $\square$

Lemma B.4.4. *If X is a nonnegative simple random variable, then $\mathbb{E}(X|\mathcal{G})$ is also a nonnegative random variable.*

Proof. Let $Y = \mathbb{E}(X|\mathcal{G})$. Let $B := \{\omega : Y(\omega) < 0\}$. Then $B \in \mathcal{G}$, and hence $\mathbb{E}(Y; B) = \mathbb{E}(X; B)$. Since X is nonnegative, $\mathbb{E}(X; B) \geq 0$. Thus, $\mathbb{E}(Y; B) \geq 0$. But this implies that $Y \geq 0$ a.s. $\square$

Lemma B.4.5. *If X is $[0, \infty]$ -valued, $\mathbb{E}(X|\mathcal{G})$ exists and is $[0, \infty]$ -valued. If X is integrable, then so is $\mathbb{E}(X|\mathcal{G})$, and these random variables have the same expected value. Lastly, if X_1 and X_2 are $[0, \infty]$ -valued random variables and $a, b \geq 0$, then $\mathbb{E}(aX_1 + bX_2|\mathcal{G}) = a\mathbb{E}(X_1|\mathcal{G}) + b\mathbb{E}(X_2|\mathcal{G})$ a.s.*

Proof. By Theorem A.1.23, we can find a sequence of nonnegative simple functions $\{X_n\}_{n \geq 1}$ increasing to X . By Lemma B.4.2, $Y_n := \mathbb{E}(X_n|\mathcal{G})$ exists for every n . By Lemma B.4.3, $\mathbb{E}(X_n - X_{n-1}|\mathcal{G}) = Y_n - Y_{n-1}$ for every n . Therefore by Lemma B.4.4, $\{Y_n\}_{n \geq 1}$ is also an increasing sequence of nonnegative random variables. Let Y be the pointwise limit of this sequence. Then for any $B \in \mathcal{G}$, the monotone convergence theorem gives us

$$\begin{aligned} \mathbb{E}(X; B) &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n; B) \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(Y_n; B) = \mathbb{E}(Y; B). \end{aligned}$$

Thus, $Y = \mathbb{E}(X|\mathcal{G})$. Clearly, Y is $[0, \infty]$ -valued. If $\mathbb{E}(X) < \infty$, then $\mathbb{E}(Y) = \mathbb{E}(Y; \Omega) = \mathbb{E}(X; \Omega) = \mathbb{E}(X)$ is also finite. Linearity is proved exactly as in Lemma B.4.3. $\square$

Finally, we arrive at the main result of this section.

Theorem B.4.6. *For any integrable random variable X , $\mathbb{E}(X|\mathcal{G})$ exists and is unique almost everywhere. Moreover, $|\mathbb{E}(X|\mathcal{G})| \leq \mathbb{E}(|X||\mathcal{G})$ a.s.*

Proof. We have already established uniqueness in Lemma B.4.1. To prove existence, let X^+ and X^- denote the positive and negative parts of X . Then by Lemma B.4.5, $Y_1 := \mathbb{E}(X^+|\mathcal{G})$ and $Y_2 := \mathbb{E}(X^-|\mathcal{G})$ exist. Let $Y := Y_1 - Y_2$. Since X is integrable, so are X^+ and X^- . So by the last assertion of Lemma B.4.5, Y_1 and Y_2 are integrable, and thus, so is Y . Therefore for any $B \in \mathcal{G}$,

$$\begin{aligned} \mathbb{E}(X; B) &= \mathbb{E}(X^+; B) - \mathbb{E}(X^-; B) \\ &= \mathbb{E}(Y_1; B) - \mathbb{E}(Y_2; B) = \mathbb{E}((Y_1 - Y_2); B) = \mathbb{E}(Y; B). \end{aligned}$$

Thus, Y qualifies as $\mathbb{E}(X|\mathcal{G})$. Finally, by the linearity of conditional expectation for nonnegative random variables, $|Y| \leq Y_1 + Y_2 = \mathbb{E}(X^+|\mathcal{G}) + \mathbb{E}(X^-|\mathcal{G}) = \mathbb{E}(|X||\mathcal{G})$ a.s. $\square$

Thus, we have established the existence and uniqueness of the conditional expectation of an integrable random variable given a σ -algebra. One can similarly define the conditional probability of an event, as $\mathbb{P}(A|\mathcal{G}) := \mathbb{E}(1_A|\mathcal{G})$. Another common notation is that if X and Y are two random variables, then $\mathbb{E}(Y|\sigma(X))$ is written as $\mathbb{E}(Y|X)$. Recall that by Proposition B.1.2, any $\sigma(X)$ -measurable random variable can be expressed as $F(X)$ for some function F that is unique up to a.e. equivalences with respect to the law of X . We can use this function F to write

$$\mathbb{E}(Y|X = x) = F(x), \quad (\text{B.4.2})$$

with the understanding that both sides are equivalence classes of functions (equivalent up to a.e. equality under the law of X) rather than actual functions. A case where F can be easily computed is the following.

Lemma B.4.7. *Let X_1 and X_2 be independent random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and taking values in some measurable spaces $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$, respectively. Let $F : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}^*$ be a measurable map such that the random variable $Y := F(X_1, X_2)$ is integrable. Then*

$$\mathbb{E}(Y|X_1 = x) = \int_{\Omega_2} F(x, y) d\mu_{X_2}(y).$$

Proof. Let $g(x)$ denote the integral displayed in the statement of the lemma. The integrability of Y means that F is an integrable map. Therefore g is well-defined and integrable with respect to the law of X_1 , and by Fubini's theorem, we have that for any $B = X_1^{-1}(A) \in \sigma(X_1)$,

$$\begin{aligned} \mathbb{E}(g(X_1); B) &= \int_A g(x) d\mu_{X_1}(x) \\ &= \int_A \int_{\Omega_2} F(x, y) d\mu_{X_2}(y) d\mu_{X_1}(x) = \mathbb{E}(Y; X_1 \in B). \end{aligned}$$

This shows that $g(X_1) = \mathbb{E}(Y|X_1)$. $\square$

It is clear from the definition that if conditional expectation exists, it must be linear. That is,

$$\mathbb{E}(aX_1 + bX_2|\mathcal{G}) = a\mathbb{E}(X_1|\mathcal{G}) + b\mathbb{E}(X_2|\mathcal{G}) \text{ a.s.}$$

This holds because the right side is easily seen to be a valid candidate for the conditional expectation of $aX_1 + bX_2$ given $\mathcal{G}$, and we know that conditional expectation is unique. By Lemma B.4.5, the conditional expectation of a nonnegative random variable is a nonnegative random variable. Together with linearity, this implies that conditional expectation is

monotone. That is, if $X \leq Y$ a.s., then $\mathbb{E}(X|\mathcal{G}) \leq \mathbb{E}(Y|\mathcal{G})$. This implies Jensen's inequality for conditional expectation:

Theorem B.4.8 (Jensen's inequality for conditional expectation). *Let X be an integrable random variable defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function such that $\phi(X)$ is also integrable. Then for any σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}(\phi(X)|\mathcal{G}) \geq \phi(\mathbb{E}(X|\mathcal{G}))$ a.s.*

Proof. Let $r_1, r_2, \dots$ be an enumeration of the rationals on the real line. For each n , choose $a_n \in [\partial_L \phi(r_n), \partial_R \phi(r_n)]$. Then, as observed earlier,

$$\phi(X) \geq \phi(r_n) + a_n(X - r_n).$$

Thus, by the monotonicity of conditional expectation,

$$\mathbb{E}(\phi(X)|\mathcal{G}) \geq \phi(r_n) + a_n(\mathbb{E}(X|\mathcal{G}) - r_n) \quad \text{a.s.}$$

Since the intersection of countably many event of probability one has probability one, we get that

$$\mathbb{E}(\phi(X)|\mathcal{G}) \geq \sup_{n \geq 1} (\phi(r_n) + a_n(\mathbb{E}(X|\mathcal{G}) - r_n)) \quad \text{a.s.}$$

Since there are rational numbers arbitrarily close to $\mathbb{E}(X|\mathcal{G})$, and ϕ is continuous, the above inequality implies that $\mathbb{E}(\phi(X)|\mathcal{G}) \geq \phi(\mathbb{E}(X|\mathcal{G}))$ a.s. $\square$

We end this section with two further properties of conditional expectation.

Proposition B.4.9 (Tower property of conditional expectation). *Let X be a random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G}' \subset \mathcal{G}$ be two sub- σ -algebras of $\mathcal{F}$. Then*

$$\mathbb{E}(X|\mathcal{G}') = \mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{G}') \quad \text{a.s.}$$

Proof. Take any $B \in \mathcal{G}'$. Then $B \in \mathcal{G}$, and hence

$$\mathbb{E}(\mathbb{E}(X|\mathcal{G}); B) = \mathbb{E}(X; B) = \mathbb{E}(\mathbb{E}(X|\mathcal{G}'); B).$$

Thus, $\mathbb{E}(X|\mathcal{G}')$ satisfies the defining property of the conditional expectation of $\mathbb{E}(X|\mathcal{G})$ given $\mathcal{G}'$, which proves the claim. $\square$

Proposition B.4.10. *If Y is $\mathcal{G}$ -measurable and XY is integrable (in addition to X being integrable), then*

$$\mathbb{E}(XY|\mathcal{G}) = Y\mathbb{E}(X|\mathcal{G}) \quad \text{a.s.}$$

Proof. Let us first prove the weaker statement

$$\mathbb{E}(XY) = \mathbb{E}(Y\mathbb{E}(X|\mathcal{G})). \tag{B.4.3}$$

Note that this follows from the definition of conditional expectation if $Y = 1_B$ for some $B \in \mathcal{G}$. Therefore it holds for any simple $\mathcal{G}$ -measurable Y . Next, if X and Y are both nonnegative, we can take a sequence of nonnegative, simple, $\mathcal{G}$ -measurable random variables increasing to Y and apply the monotone convergence theorem on both sides to get (B.4.3). In particular, this shows that if X and Y are nonnegative and XY is integrable, then so is $Y\mathbb{E}(X|\mathcal{G})$.

Finally, in the general case, note that the integrability of XY implies that X^+Y^+ , X^+Y^- , X^-Y^+ and X^-Y^- are all integrable. By linearity, this gives us (B.4.3) for any X and Y such that X and XY are integrable, and Y is $\mathcal{G}$ -measurable.

Next, replace Y by $Y1_B$ in (B.4.3), where $B \in \mathcal{G}$. Since the integrability of XY implies that of $XY1_B$, we get

$$\mathbb{E}(XY; B) = \mathbb{E}(Y\mathbb{E}(X|\mathcal{G}); B).$$

Since this holds for any $B \in \mathcal{G}$, this completes the proof. $\square$

B.4.2 Conditional independence

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We say that two sub- σ -algebras $\mathcal{G}_1$ and $\mathcal{G}_2$ of $\mathcal{F}$ are conditionally independent given a third sub- σ -algebra $\mathcal{G}_0$ if for every $A \in \mathcal{G}_1$ and $B \in \mathcal{G}_2$, we have

$$\mathbb{P}(A \cap B | \mathcal{G}_0) = \mathbb{P}(A | \mathcal{G}_0) \mathbb{P}(B | \mathcal{G}_0) \quad \text{a.s.}$$

In the same spirit, we say that two random variables X and Y are conditionally independent given a third random variable Z if $\sigma(X)$ and $\sigma(Y)$ are conditionally independent given $\sigma(Z)$. The following lemma gives a useful criterion for proving conditional independence.

Lemma B.4.11. *Let X_0 , X_1 and X_2 be three random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and taking values in some measurable spaces $(\Omega_0, \mathcal{F}_0)$, $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$, respectively. Suppose that X_2 can be expressed as $F(X_0, Z)$, where Z is an $(\Omega_3, \mathcal{F}_3)$ -valued random variable (which is another measurable space), Z is independent of (X_0, X_1) , and $F : \Omega_0 \times \Omega_3 \rightarrow \Omega_2$ is a measurable map. Then X_2 is conditionally independent of X_1 given X_0 .*

Proof. Take any $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$. Then for any $C \in \mathcal{F}_0$, we have

$$\begin{aligned} \mathbb{E}(\mathbb{P}(X_1 \in A, X_2 \in B | X_0); X_0 \in C) &= \mathbb{P}(X_1 \in A, X_2 \in B, X_0 \in C) \\ &= \mathbb{P}(X_1 \in A, F(X_0, Z) \in B, X_0 \in C) \\ &= \int_{A \times C} \int_{\{z: F(x_0, z) \in B\}} d\mu_Z(z) d\mu_{(X_0, X_1)}(x_0, x_1). \end{aligned}$$

But by Lemma B.4.7, the function

$$G(x_0) := \int_{\{z: F(x_0, z) \in B\}} d\mu_Z(z)$$

is a representation of $\mathbb{P}(X_2 \in B|X_0 = x_0)$. Thus, we get

$$\begin{aligned}\mathbb{E}(\mathbb{P}(X_1 \in A, X_2 \in B|X_0); X_0 \in C) &= \mathbb{P}(\mathbb{P}(X_2 \in B|X_0); X_1 \in A, X_0 \in C) \\ &= \mathbb{P}(\mathbb{P}(X_2 \in B|X_0)\mathbb{P}(X_1 \in A|X_0); X_0 \in C)\end{aligned}$$

This proves the claim. $\square$

B.4.3 A criterion for independence

Conditional distributions can be used to give the following useful criterion for two random variables to be independent.

Lemma B.4.12. *Let X and Y be two random variables defined on the same probability space, but possibly taking values in different spaces. Suppose that the conditional law of Y given X does not depend on the value of X , in the sense that for any f such that $f(Y)$ is integrable, we have $\mathbb{E}(f(Y)|X)$ is a constant almost surely. Then X and Y are independent.*

Proof. Since $\mathbb{E}(\mathbb{E}(f(Y)|X)) = \mathbb{E}(f(Y))$, the given condition implies that $\mathbb{E}(f(Y)|X) = \mathbb{E}(f(Y))$ almost surely. Thus, for any A and B ,

$$\begin{aligned}\mathbb{P}(X \in A, Y \in B) &= \mathbb{E}(1_A(X)1_B(Y)) \\ &= \mathbb{E}(1_A(X)\mathbb{E}(1_B(Y)|X)) = \mathbb{E}(1_A(X)\mathbb{E}(1_B(Y))) \\ &= \mathbb{E}(1_A(X))\mathbb{E}(1_B(Y)) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).\end{aligned}$$

Therefore, X and Y are independent. $\square$

B.4.4 Conditional probability densities

Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2$, be two measurable spaces. Let μ_1, μ_2 be measures on Ω_1, Ω_2 . Suppose that we have a probability measure ν on $\Omega_1 \times \Omega_2$ that has density ρ with respect to $\mu_1 \times \mu_2$. Let (X_1, X_2) be a $\Omega_1 \times \Omega_2$ -valued random variable, so that ρ is the p.d.f. of (X_1, X_2) with respect to $\mu_1 \times \mu_2$. Then it is an easy consequence of Fubini's theorem that the p.d.f. of X_1 with respect to μ_1 is given by

$$\rho_1(x) = \int_{\Omega_2} \rho(x_1, x_2) d\mu_2(x_2).$$

Similarly, the p.d.f. of X_2 with respect to μ_2 is

$$\rho_2(x) = \int_{\Omega_1} \rho(x_1, x_2) d\mu_1(x_1).$$

The densities ρ_1 and ρ_2 are called the marginal densities of ρ . The *conditional probability density* of X_1 given $X_2 = x_2$ is defined as

$$\rho_{1|2}(x_1|x_2) := \frac{\rho(x_1, x_2)}{\rho_2(x_2)}$$

if $\rho(x_2) > 0$, and arbitrarily if $\rho(x_2) = 0$. The reason why we call this the conditional density is because of the following fact (recalling the definition of $\mathbb{E}(Y|X = x)$ from Subsection B.4.1).

Lemma B.4.13. *Take any bounded measurable function $f : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$. For $x_2 \in \Omega_2$, define*

$$g(x_2) := \int_{\Omega_1} f(x_1, x_2) \rho_{1|2}(x_1|x_2) d\mu_1(x_1).$$

Then g is well-defined, and $g(X_2) = \mathbb{P}(f(X_1, X_2)|X_2)$ a.s.

Proof. Take any $B \in \mathcal{F}_2$. Then by Fubini's theorem,

$$\begin{aligned} \mathbb{E}(1_B(X_2)g(X_2)) &= \int_{\Omega_2} 1_B(x_2)g(x_2)\rho_2(x_2)d\mu_2(x_2) \\ &= \int_{\Omega_2} 1_B(x_2) \left(\int_{\Omega_1} f(x_1, x_2)\rho(x_1, x_2)d\mu_1(x_1) \right) d\mu_2(x_2) \\ &= \int_{\Omega_1 \times \Omega_2} 1_B(x_2)f(x_1, x_2)d\nu(x_1, x_2). \end{aligned}$$

This proves that $g(X_2) = \mathbb{P}(f(X_1, X_2)|X_2)$ a.s. □

B.5 Markov chains and Markov processes

B.5.1 Markov chains

A Markov chain is defined by its transition kernel. A transition kernel, or transition probability, is defined as follows. Let $(S, \mathcal{S})$ be a measurable space. A function $K : S \times \mathcal{S} \rightarrow \mathbb{R}$ is called a ‘Markov transition probability’ or ‘Markov transition kernel’ if

1. for each $x \in S$, $A \mapsto K(x, A)$ is probability measure on $(S, \mathcal{S})$, and
2. for each $A \in \mathcal{S}$, $x \mapsto K(x, A)$ is a measurable function.

Let $(S, \mathcal{S})$ be a measurable space. Let $\{X_n\}_{n \geq 0}$ be a sequence of S -valued random variables adapted to a filtration of σ -algebras $\{\mathcal{F}_n\}_{n \geq 0}$. Let K be a Markov transition kernel on S and σ be a probability measure on σ . We say that $\{X_n\}_{n \geq 0}$ is a time-homogeneous Markov chain with kernel K and initial distribution σ if $X_0 \sim \sigma$, and for all $n \geq 0$ and $B \in \mathcal{S}$, $\mathbb{P}(X_{n+1} \in B | \mathcal{F}_n) = K(X_n, B)$ a.s.

The n -step transition kernel of the above Markov chain is a kernel K^n such that

$$K^n(X_0, B) = \mathbb{P}(X_n \in B | \mathcal{F}_0).$$

An easy inductive argument via the tower property of conditional expectation shows that

$$K^n(x_0, B) = \int_S \int_S \cdots \int_B K(x_{n-1}, dx_n) \cdots K(x_1, dx_2) K(x_0, dx_1),$$

where $\int \cdots K(x, dy)$ denotes integration with respect to the measure $K(x, \cdot)$.

Suppose that we have a measure μ on $(S, \mathcal{S})$ such that for each $x \in S$, the probability measure $K(x, \cdot)$ has a probability density $p(x, \cdot)$ with respect to μ . That is, for all $A \in \mathcal{S}$,

$$K(x, A) = \int_A p(x, y) d\mu(y).$$

In this case, we say that p is the transition probability density function of our Markov chain. One can show that in this case, $K^n(x, \cdot)$ also has a density with respect to μ , given by

$$p^n(x, y) = \int_S \cdots \int_S p(x, x_1) p(x_1, x_2) \cdots p(x_{n-1}, y) d\mu(x_1) \cdots d\mu(x_{n-1}).$$

A probability measure ν on $(S, \mathcal{S})$ is said to be an invariant measure for the Markov chain if for any $A \in \mathcal{S}$,

$$\int_S K(x, A) d\nu(x) = \nu(A).$$

In case there is a measure μ as above, and ν has a density f with respect to μ , we say that f is an invariant probability density for the Markov chain. We will need the following special case of a theorem of Doeblin.

Theorem B.5.1 (Special case of Doeblin's theorem). *Consider a Markov chain on a state space $(S, \mathcal{S})$ that has transition probability density p with respect to some measure μ . Suppose also that it has an invariant probability measure ν that has density f with respect to μ . Suppose that there is some $\epsilon > 0$ such that for all $x, y \in S$,*

$$p(x, y) \geq \epsilon f(y). \tag{B.5.1}$$

Then for any $n \geq 1$, $x \in S$,

$$\int_S |p^n(x, y) - f(y)| \leq 2(1 - \epsilon)^n.$$

Proof. Define

$$q(x, y) := \frac{p(x, y) - \epsilon f(y)}{1 - \epsilon}.$$

Then note that $q(x, y) \geq 0$, and

$$\int_S q(x, y) d\mu(y) = \frac{1}{1 - \epsilon} \left(\int_S p(x, y) d\mu(y) - \epsilon \int_S f(y) d\mu(y) \right) = 1. \tag{B.5.2}$$

Next, define

$$q^n(x, y) := \int_S \cdots \int_S q(x, x_1) q(x_1, x_2) \cdots q(x_{n-1}, y) d\mu(x_1) \cdots d\mu(x_{n-1}).$$

Iterating (B.5.2) gives

$$\int_S q^n(x, y) d\mu(y) = 1. \quad (\text{B.5.3})$$

Now, note that since f is an invariant density, we have

$$\begin{aligned} \int_S f(x) q(x, y) d\mu(x) &= \frac{1}{1-\epsilon} \left(\int_S f(x) p(x, y) d\mu(x) - \epsilon \int_S f(x) f(y) d\mu(x) \right) \\ &= \frac{1}{1-\epsilon} (f(y) - \epsilon f(y)) = f(y). \end{aligned} \quad (\text{B.5.4})$$

Next, we claim that for each n ,

$$p^n(x, y) = (1-\epsilon)^n q^n(x, y) + (1 - (1-\epsilon)^n) f(y). \quad (\text{B.5.5})$$

We will prove this by induction on n . Since

$$p(x, y) = (1-\epsilon) q(x, y) + \epsilon f(y),$$

the claim is true for $n = 1$. Suppose that it holds for $n - 1$. Then

$$\begin{aligned} p^n(x, y) &= \int_S p^{n-1}(x, z) p(z, y) d\mu(z) \\ &= \int_S ((1-\epsilon)^{n-1} q^{n-1}(x, z) + (1 - (1-\epsilon)^{n-1}) f(z)) ((1-\epsilon) q(z, y) + \epsilon f(y)) d\mu(z). \end{aligned}$$

Let us expand the brackets and evaluate the terms one by one. First, note that

$$\int_S q^{n-1}(x, z) q(z, y) d\mu(z) = q^n(x, y).$$

Next, by equation (B.5.4),

$$\int_S f(z) q(z, y) d\mu(z) = f(y).$$

Lastly, by equation (B.5.3) and the fact that f is a probability density,

$$\int_S q^{n-1}(x, z) f(y) d\mu(z) = f(y), \quad \int_S f(z) f(y) d\mu(z) = f(y).$$

By the above calculations, we get

$$\begin{aligned} p^n(x, y) &= (1-\epsilon)^n q^n(x, y) + (1 - (1-\epsilon)^{n-1}) (1-\epsilon) f(y) \\ &\quad + (1-\epsilon)^{n-1} \epsilon f(y) + (1 - (1-\epsilon)^{n-1}) \epsilon f(y) \\ &= (1-\epsilon)^n q^n(x, y) + (1 - (1-\epsilon)^n) f(y). \end{aligned}$$

This completes the induction step, proving the claim (B.5.5). Now, equation (B.5.5) gives

$$\begin{aligned} \int_S |p^n(x, y) - f(y)| d\mu(y) &= (1 - \epsilon)^n \int_S |q^n(x, y) - f(y)| d\mu(y) \\ &\leq (1 - \epsilon)^n \int_S (|q^n(x, y)| + |f(y)|) d\mu(y). \end{aligned}$$

But q^n and f are both nonnegative functions and integrate to 1. This completes the proof. $\square$

A corollary of Doeblin's theorem is that Markov chains that satisfy the Doeblin condition (B.5.1) have *exponentially decaying correlations*, as given by the following result.

Corollary B.5.2. *Let $\{X_n\}_{n \geq 0}$ be a Markov chain satisfying the conditions of Theorem B.5.1, and suppose that X_0 has a probability density with respect to μ . Let g_1 be the joint probability density of $X_0, \dots, X_i$, and g_2 be the joint probability density function of $X_j, \dots, X_k$, where $i < j < k$. Let g be the joint probability density function of $X_0, \dots, X_i, X_j, \dots, X_k$. Then*

$$\begin{aligned} \int |g(x_0, \dots, x_i, x_j, \dots, x_k) - g_1(x_0, \dots, x_i)g_2(x_j, \dots, x_k)| \\ \cdot d\mu(x_0) \cdots d\mu(x_i) d\mu(x_j) \cdots d\mu(x_k) \leq 4(1 - \epsilon)^{j-i}. \end{aligned}$$

Consequently, if F is a measurable $[-1, 1]$ -valued function of $X_0, \dots, X_i$ and G is a measurable $[-1, 1]$ -valued function of $X_j, \dots, X_k$, then

$$|\mathbb{E}(FG) - \mathbb{E}(F)\mathbb{E}(G)| \leq 4(1 - \epsilon)^{j-i}.$$

Proof. Let h be the probability density function of X_0 . Then note that

$$g_1(x_0, \dots, x_i) = h(x_0)p(x_0, x_1) \cdots p(x_{i-1}, x_i).$$

Similarly,

$$\begin{aligned} g(x_0, \dots, x_i, x_j, \dots, x_k) \\ = h(x_0)p(x_0, x_1) \cdots p(x_{i-1}, x_i)p^{j-i}(x_i, x_j)p(x_j, x_{j+1}) \cdots p(x_{k-1}, x_k), \end{aligned}$$

and

$$g_2(x_j, \dots, x_k) = w(x_j)p(x_j, x_{j+1}) \cdots p(x_{k-1}, x_k),$$

where w is the probability density of X_j , given by

$$w(x) = \int h(x_0)p^j(x_0, x)d\mu(x_0).$$

These expressions show that

$$\begin{aligned}
 & \int |g(x_0, \dots, x_i, x_j, \dots, x_k) - g_1(x_0, \dots, x_i)g_2(x_j, \dots, x_k)| \\
 & \quad \cdot d\mu(x_0) \cdots d\mu(x_i)d\mu(x_j) \cdots d\mu(x_k) \\
 & \leq \int h(x_0)p(x_0, x_1) \cdots p(x_{i-1}, x_i) |p^{j-i}(x_i, x_j) - w(x_j)| p(x_j, x_{j+1}) \cdots p(x_{k-1}, x_k) \\
 & \quad \cdot d\mu(x_0) \cdots d\mu(x_i)d\mu(x_j) \cdots d\mu(x_k).
 \end{aligned}$$

Now, we can successively integrate out $x_k, x_{k-1}, \dots, x_{j+1}$, and reduce the above expression to

$$\int h(x_0)p(x_0, x_1) \cdots p(x_{i-1}, x_i) |p^{j-i}(x_i, x_j) - w(x_j)| d\mu(x_0) \cdots d\mu(x_i)d\mu(x_j). \quad (\text{B.5.6})$$

For a fixed x_i , Theorem B.5.1 gives

$$\int |p^{j-i}(x_i, x_j) - f(x_j)| d\mu(x_j) \leq 2(1 - \epsilon)^{j-i},$$

and

$$\begin{aligned}
 \int |w(x_j) - f(x_j)| d\mu(x_j) &= \int \left| \int h(x_0)(p^j(x_0, x_j) - f(x_j)) d\mu(x_0) \right| d\mu(x_j) \\
 &\leq \int \int h(x_0) |p^j(x_0, x_j) - f(x_j)| d\mu(x_0) d\mu(x_j) \\
 &\leq 2(1 - \epsilon)^j \int h(x_0) d\mu(x_0) = 2(1 - \epsilon)^j.
 \end{aligned}$$

Combining the last two displays and applying the triangle inequality, we get

$$\int |p^{j-i}(x_i, x_j) - w(x_j)| d\mu(x_j) \leq 4(1 - \epsilon)^{j-i}.$$

Using this in equation (B.5.6), we get

$$\begin{aligned}
 & \int |g(x_0, \dots, x_i, x_j, \dots, x_k) - g_1(x_0, \dots, x_i)g_2(x_j, \dots, x_k)| \\
 & \quad \cdot d\mu(x_0) \cdots d\mu(x_i)d\mu(x_j) \cdots d\mu(x_k) \\
 & \leq 4(1 - \epsilon)^{j-i} \int h(x_0)p(x_0, x_1) \cdots p(x_{i-1}, x_i) d\mu(x_0) \cdots d\mu(x_i) \\
 & = 4(1 - \epsilon)^{j-i}.
 \end{aligned}$$

This completes the proof. □

B.5.2 The Gibbs sampling algorithm

Let $(\Omega, \mathcal{F})$ be a measurable space and let ν be a probability measure on $(\Omega^n, \mathcal{F}^n)$ for some $n \geq 1$. The Gibbs sampler is an algorithm for generating a random vector $X = (X_1, \dots, X_n)$

with law ν if we know how to generate, for each i , from the conditional distribution of X_i given $(X_j)_{j \neq i}$. The algorithm produces a Markov chain $X^0, X^1, \dots$ on Ω^n , as follows. For simplicity, let us work under the assumption that ν has a density ρ with respect to some product measure μ^n on Ω^n . Then the marginal density of $(X_j)_{j \neq i}$ is

$$\rho_{-i}((x_j)_{j \neq i}) = \int \rho(x_1, \dots, x_n) d\mu(x_i), \quad (\text{B.5.7})$$

and the conditional density of X_i given $(X_j)_{j \neq i}$ is

$$\rho_i(x_i | (x_j)_{j \neq i}) = \frac{\rho(x_1, \dots, x_n)}{\rho_{-i}((x_j)_{j \neq i})} \quad (\text{B.5.8})$$

if $\rho_{-i}((x_j)_{j \neq i}) \neq 0$, and defined arbitrarily (for example, zero) if the denominator is zero. (See Subsection B.4.4 for a discussion on conditional probability densities.) The transition kernel of the Gibbs sampler is given by

$$K(x, A) = \frac{1}{n} \sum_{i=1}^n \int_{\Omega} \rho_i(y | (x_j)_{j \neq i}) 1_A(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) d\mu(y)$$

for $x = (x_1, \dots, x_n) \in \Omega^n$ and $A \in \mathcal{F}^n$. In words, this means that to transition from X^k to X^{k+1} , we choose a coordinate $i \in \{1, \dots, n\}$ uniformly at random, draw X_i^{k+1} from the conditional distribution of X_i^k given $(X_j^k)_{j \neq i}$, and let $X_j^{k+1} = X_j^k$ for all $j \neq i$.

Proposition B.5.3. *The probability measure ν is an invariant measure for the Gibbs sampler defined above.*

Proof. Note that for any $A \in \mathcal{F}^n$,

$$\begin{aligned} & \int_{\Omega^n} \rho(x) K(x, A) d\mu^n(x) \\ &= \frac{1}{n} \sum_{i=1}^n \int_{\Omega^n} \int_{\Omega} \rho(x) \rho_i(y | (x_j)_{j \neq i}) 1_A(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) d\mu(y) d\mu^n(x). \end{aligned}$$

Since the integrand on the right depends on x_i only through $\rho(x)$, integrating out x_i leaves us with

$$\begin{aligned} & \int_{\Omega^n} \rho(x) K(x, A) d\mu^n(x) \\ &= \frac{1}{n} \sum_{i=1}^n \int_{\Omega^{n-1}} \int_{\Omega} \rho_{-i}((x_j)_{j \neq i}) \rho_i(y | (x_j)_{j \neq i}) \\ & \quad \cdot 1_A(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) d\mu(y) d\mu^{n-1}((x_j)_{j \neq i}). \end{aligned} \quad (\text{B.5.9})$$

Now, note that

$$\rho_{-i}((x_j)_{j \neq i}) \rho_i(y | (x_j)_{j \neq i}) = \rho(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) \quad (\text{B.5.10})$$

for μ -a.e. y . This follows from the definition (B.5.8) of $\rho_i(y|(x_j)_{j \neq i})$ if $\rho_{-i}((x_j)_{j \neq i}) \neq 0$. On the other hand, if $\rho_{-i}((x_j)_{j \neq i}) = 0$, then the definition (B.5.7) of $\rho_{-i}((x_j)_{j \neq i})$ shows that $\rho(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) = 0$ for μ -a.e. y . Thus, equation (B.5.10) holds for μ -a.e. y . Using this in equation (B.5.9), we get

$$\begin{aligned} & \int_{\Omega^n} \rho(x) K(x, A) d\mu^n(x) \\ &= \frac{1}{n} \sum_{i=1}^n \int_{\Omega^{n-1}} \int_{\Omega} \rho(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) \\ & \quad \cdot 1_A(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) d\mu(y) d\mu^{n-1}((x_j)_{j \neq i}). \end{aligned}$$

But each term on the right is equal to $\nu(A)$. This completes the proof. $\square$

B.5.3 Markov processes

Let $\{X_t\}_{t \in \mathbb{R}}$ be random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and taking values in some measurable space $(\Omega', \mathcal{F}')$. For each t , let $\mathcal{F}_t := \sigma(X_t)$, $\mathcal{F}_{\leq t} := \sigma(\{X_s\}_{s \leq t})$, and $\mathcal{F}_{\geq t} := \sigma(\{X_s\}_{s \geq t})$. We say that $\{X_t\}_{t \in \mathbb{R}}$ is a Markov process if for every t , the σ -algebras $\mathcal{F}_{\geq t}$ and $\mathcal{F}_{\leq t}$ are conditionally independent given $\mathcal{F}_t$. The following lemma is often useful.

Lemma B.5.4. *Let $\{X_t\}_{t \in \mathbb{R}}$ be a Markov process as above. Then for any $t \in \mathbb{R}$, and any integrable random variable Z that is $\mathcal{F}_{\geq t}$ -measurable, we have*

$$\mathbb{E}(Z|\mathcal{F}_{\leq t}) = \mathbb{E}(Z|\mathcal{F}_t) \quad a.s.$$

Proof. By the Markov property, we have that for any $A \in \mathcal{F}_{\leq s}$,

$$\mathbb{E}(Z 1_A | \mathcal{F}_t) = \mathbb{E}(Z | \mathcal{F}_t) \mathbb{P}(A | \mathcal{F}_t).$$

By Proposition B.4.10, this shows that

$$\mathbb{E}(Z 1_A) = \mathbb{E}[\mathbb{E}(Z | \mathcal{F}_t) \mathbb{P}(A | \mathcal{F}_t)] = \mathbb{E}[\mathbb{E}(Z | \mathcal{F}_s) 1_A].$$

This proves the claim. $\square$

A Markov process as above is called *stationary* if for any $s \in \mathbb{R}$, the process $\{X_{t+s}\}_{t \in \mathbb{R}}$, as an $(\Omega')^{\mathbb{R}}$ -valued random variable, has the same law as the original process $\{X_t\}_{t \in \mathbb{R}}$. Note that by Proposition B.1.1, this simply means that for any n and any $t_1, t_2, \dots, t_n \in \mathbb{R}$, the random vector $(X_{t_1+s}, \dots, X_{t_n+s})$ has the same law as the random vector $(X_{t_1}, \dots, X_{t_n})$. Note that, in particular, the law of X_t is the same for all t if $\{X_t\}_{t \in \mathbb{R}}$ is a stationary process.

The Markov process $\{X_t\}_{t \in \mathbb{R}}$ is called *reversible* if it has the same law as the time-reversed process $\{X_{-t}\}_{t \in \mathbb{R}}$.

Let $\{X_t\}_{t \in \mathbb{R}}$ be a stationary, reversible Markov process as above and μ be the law of X_0 . Let $\mathcal{H}$ denote the Hilbert space $L^2(\Omega', \mathcal{F}', \mu, \mathbb{C})$ (where the $\mathbb{C}$ denotes that we are

considering complex-valued L^2 functions), with inner product

$$\langle F, G \rangle := \int_{\Omega'} F(x) \overline{G(x)} d\mu(x).$$

For $F \in \mathcal{H}$ and $t \geq 0$, define

$$P(t)F(x) := \mathbb{E}(F(X_t) | X_0 = x).$$

By the definition of conditional expectation and Theorem B.4.6, we know that $P(t)F$ is an integrable function on $(\Omega', \mathcal{F}', \mu)$. The following theorem establishes some properties that will be useful later.

Theorem B.5.5. *For each $t \geq 0$, $P(t)$ is a linear map from $\mathcal{H}$ into $\mathcal{H}$, with norm ≤ 1 , which is symmetric, meaning that $\langle P(t)F, G \rangle = \langle F, P(t)G \rangle$ for all $F, G \in \mathcal{H}$. Moreover, for all $s, t \geq 0$, $P(s+t) = P(s)P(t)$.*

Proof. The fact that $P(t)F \in \mathcal{H}$ follows from Jensen's inequality for conditional expectation, which also shows that $\|P(t)\| \leq 1$, because for any $F \in \mathcal{H}$,

$$\|P(t)F\|_{L^2}^2 = \mathbb{E}[(\mathbb{E}(F(X_t) | X_0))^2] \leq \mathbb{E}[\mathbb{E}(F(X_t)^2 | X_0)] = \mathbb{E}(F(X_t)^2) = \|F\|_{L^2}^2.$$

Linearity of $P(t)$ follows from linearity of conditional expectation. To see that $P(t)$ is a symmetric operator, notice that by reversibility, (X_0, X_t) has the same law as (X_0, X_{-t}) ; and by stationarity, (X_0, X_{-t}) has the same law as (X_t, X_0) . Thus, (X_0, X_t) has the same law as (X_t, X_0) . By Proposition B.4.10, this shows that

$$\begin{aligned} \langle P(t)F, G \rangle &= \mathbb{E}(P(t)F(X_0)G(X_0)) \\ &= \mathbb{E}(F(X_t)G(X_0)) = \mathbb{E}(F(X_0)G(X_t)) = \mathbb{E}(F(X_0)P(t)G(X_0)) = \langle F, P(t)G \rangle. \end{aligned}$$

It remains to prove the last assertion. Take any $F \in \mathcal{H}$ and any $s, t \geq 0$. Let $G := P(t)F$. We claim that

$$G(X_s) = \mathbb{E}(F(X_{s+t}) | \mathcal{F}_s). \tag{B.5.11}$$

To see this, note that for any $H \in \mathcal{H}$, two applications of stationarity give

$$\begin{aligned} \mathbb{E}(F(X_{s+t})H(X_s)) &= \mathbb{E}(F(X_t)H(X_0)) \\ &= \mathbb{E}(G(X_0)H(X_0)) \\ &= \mathbb{E}(G(X_s)H(X_s)). \end{aligned}$$

This proves equation (B.5.11). Therefore, by the tower property of conditional expectation

and Lemma B.5.4,

$$\begin{aligned}
 P(s+t)F(X_0) &= \mathbb{E}(F(X_{s+t})|\mathcal{F}_0) \\
 &= \mathbb{E}(F(X_{s+t})|\mathcal{F}_{\leq 0}) \\
 &= \mathbb{E}[\mathbb{E}(F(X_{s+t})|\mathcal{F}_{\leq s})|\mathcal{F}_{\leq 0}] \\
 &= \mathbb{E}[G(X_s)|\mathcal{F}_{\leq 0}] = \mathbb{E}[G(X_s)|\mathcal{F}_0] = P(s)G(X_0).
 \end{aligned}$$

This proves that $P(s+t)F = P(s)P(t)F$. $\square$

B.6 Brownian motion

We define a *random curve* on the time interval $[0, t]$ to be a measurable map $B : (\Omega, \mathcal{F}) \rightarrow (C[0, t], \mathcal{B}(C[0, t]))$, where $C[0, t]$ is the set of continuous functions from $[0, t]$ into $\mathbb{R}$, and $\mathcal{B}(C[0, t])$ is the Borel σ -algebra of $C[0, t]$, defined as follows. First, we define a norm on $C[0, t]$ as

$$\|f\| := \sup_{0 \leq s \leq t} |f(s)|.$$

This is sometimes called the supremum norm or the sup-norm. This norm induces a metric on $C[0, t]$ in the usual way, that is, the distance between f and g is $\|f - g\|$. The metric defines a topology on $C[0, t]$, that is, the topology generated by the open balls of this metric. Finally, the Borel σ -algebra of $C[0, t]$ is the σ -algebra generated by the open sets in this topology.

A $C[0, t]$ -valued random curve B is called standard Brownian motion on the time interval $[0, t]$ if for any n and any $0 \leq s_1, \dots, s_n \leq t$, the random vector $(B(s_1), \dots, B(s_n))$ is multivariate normal with $\mathbb{E}(B(s_i)) = 0$ for each i and $\mathbb{E}(B(s_i)B(s_j)) = \min\{s_i, s_j\}$ for each $1 \leq i, j \leq n$. It is a simple consequence of Theorem A.1.4 that these finite-dimensional distributions uniquely determine the law of Brownian motion.

It is not immediately clear that one can indeed construct Brownian motion as described above. We now show that this is possible, by an explicit construction due to Paul Lévy. Let ξ and $\{\xi_{k,j} : k \geq 1, j = 1, 3, 5, \dots, 2^k - 1\}$ be independent $N(0, 1)$ random variables, all defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Standard facts from probability theory show that such a construction is possible. We will now define a sequence of random curves $\{B_k\}_{k \geq 0}$ on the time interval $[0, t]$ using these random variables, and then show that this sequence converges to Brownian motion as $k \rightarrow \infty$.

Let $B_0(0) = 0$ and $B_0(t) = \sqrt{t}\xi$, and define $B_0(s)$ for all $s \in (0, t)$ by linear interpolation. That is, $B_0(s) = (s/t)B_0(t)$ for all $s \in [0, t]$. Then define B_k inductively as follows, such that for each k , the random curve B_k is continuous on $[0, t]$ and piecewise linear in each interval of the form $[t(j-1)2^{-k}, tj2^{-k}]$ for $j = 1, \dots, 2^k$. Note that this is true for $k = 0$. Suppose we have constructed up to B_{k-1} . Take any $0 \leq j \leq 2^k$. Let $B_k(tj2^{-k}) = B_{k-1}(tj2^{-k})$ if j

is even. If j is odd, let

$$B_k(tj2^{-k}) := B_{k-1}(tj2^{-k}) + \sqrt{t}2^{-(k+1)/2}\xi_{k,j}. \quad (\text{B.6.1})$$

This defines $B_k(s)$ at every s of the form $tj2^{-k}$ for $j = 0, 1, \dots, 2^k$. Within each interval of the form $(t(j-1)2^{-k}, tj2^{-k})$, define B_k by linearly interpolating the values at the endpoints of the interval. That is, if $s = \alpha t(j-1)2^{-k} + (1-\alpha)tj2^{-k}$ for some $\alpha \in (0, 1)$, we define

$$B_k(s) := \alpha B_k(t(j-1)2^{-k}) + (1-\alpha)B_k(tj2^{-k}).$$

The following lemma shows that B_k converges to a limiting random curve B as k goes to infinity.

Lemma B.6.1. *With probability one, the sequence $\{B_k\}_{k \geq 1}$ is Cauchy in the supremum norm on $C[0, t]$. Consequently, there is a $C[0, t]$ -valued random curve B defined on $(\Omega, \mathcal{F}, \mathbb{P})$ such that with probability one, $\|B_k - B\| \rightarrow 0$ as $k \rightarrow \infty$.*

Proof. Take any even number j between 0 and 2^k . Then note that by construction, B_{k-1} is linear in the interval $[tj2^{-k}, t(j+2)2^{-k}]$. Also by construction, B_k is equal to B_{k-1} at the endpoints of this interval, and linear in each half of the interval. This shows that the maximum value of $|B_k(s) - B_{k-1}(s)|$ for s in this interval is attained at the midpoint of the interval, that is, at $s = t(j+1)2^{-k}$. Now, by equation (B.6.1),

$$B_k(t(j+1)2^{-k}) - B_{k-1}(t(j+1)2^{-k}) = \sqrt{t}2^{-(k+1)/2}\xi_{k,j+1}.$$

Thus, we conclude that

$$\sup_{s \in [tj2^{-k}, t(j+2)2^{-k}]} |B_k(s) - B_{k-1}(s)| = \sqrt{t}2^{-(k+1)/2}|\xi_{k,j+1}|.$$

Taking supremum over all even j between 0 and 2^k , we get

$$\|B_k - B_{k-1}\| = \sqrt{t}2^{-(k+1)/2} \max_{j=1,3,5,\dots,2^k-1} |\xi_{k,j}|. \quad (\text{B.6.2})$$

Now, if $\xi \sim N(0, 1)$, then for any $x \geq 0$,

$$\mathbb{P}(|\xi| \geq x) = \mathbb{P}(e^{\xi^2/4} \geq e^{x^2/4}) \leq \frac{\mathbb{E}(e^{\xi^2/4})}{e^{x^2/4}} = Ce^{-x^2/4}, \quad (\text{B.6.3})$$

where C is a finite constant that does not depend on x . Thus,

$$\mathbb{P}\left(\max_{j=1,3,5,\dots,2^k-1} |\xi_{k,j}| > k\right) \leq \sum_{j=1,3,5,\dots,2^k-1} \mathbb{P}(|\xi_{k,j}| > k) \leq C2^k e^{-k^2/4}.$$

Thus, we conclude that

$$\mathbb{P}(\|B_k - B_{k-1}\| > \sqrt{t}2^{-(k+1)/2}k) \leq C2^k e^{-k^2/4}.$$

Note that the bound on the right is summable in k . Thus, by the first Borel–Cantelli lemma from basic probability theory,

$$\mathbb{P}(\|B_k - B_{k-1}\| > \sqrt{t}2^{-(k+1)/2}k \text{ for only finitely many } k) = 1.$$

But if $\|B_k - B_{k-1}\| > \sqrt{t}2^{-(k+1)/2}k$ for only finitely many k , then the sequence $\{B_k\}_{k \geq 1}$ is Cauchy in the supremum norm. Let B denote the limit of B_k on the set where this sequence is Cauchy, and let $B \equiv 0$ on the set where it is not. Note that B is a $C[0, t]$ -valued random curve, since the uniform limit of a limit of continuous functions on $[0, t]$ is continuous. It is a standard exercise to show that the random curve B so constructed is a measurable map from $(\Omega, \mathcal{F})$ into $(C[0, t], \mathcal{B}(C[0, t]))$. This completes the proof. $\square$

The following lemma shows that the random continuous function B constructed in Lemma B.6.1 is standard Brownian motion on the time interval $[0, t]$.

Lemma B.6.2. *For any $0 \leq s_1, \dots, s_n \leq t$, the random vector $(B(s_1), \dots, B(s_n))$ is multivariate normal with $\mathbb{E}(B(s_i)) = 0$ for each i and $\mathbb{E}(B(s_i)B(s_j)) = \min\{s_i, s_j\}$ for each $1 \leq i, j \leq n$.*

Proof. First, let us assume that $s_1, \dots, s_n$ are of the form $tj_12^{-k}, \dots, tj_n2^{-k}$ for some k and some $0 \leq j_1, \dots, j_n \leq 2^k$. Then note that by construction, for each i , the value of $B_l(s_i)$ remains the same for all $l \geq k$. Consequently, $B(s_i) = B_k(s_i)$ for each i .

Again, from construction and the fact that affine transformations preserve normality, it is clear that $(B_k(s_1), \dots, B_k(s_n))$ is a normal random vector and that $\mathbb{E}(B_k(s_i)) = 0$ for each i . We will now show by induction on k that each B_k has the covariance structure of Brownian motion in the sense that for each k and each $0 \leq i, j \leq 2^k$,

$$\mathbb{E}(B_k(ti2^{-k})B_k(tj2^{-k})) = t2^{-k} \min\{i, j\}. \quad (\text{B.6.4})$$

This true for $k = 0$ because $B_0(0) = 0$ and $B_0(t) \sim N(0, t)$. Suppose that this holds up to $k - 1$. Take any $0 \leq i, j \leq 2^k$. If i and j are both even, then

$$\begin{aligned} B_k(ti2^{-k}) &= B_{k-1}(ti2^{-k}) = B_{k-1}(t(i/2)2^{-(k-1)}), \\ B_k(tj2^{-k}) &= B_{k-1}(tj2^{-k}) = B_{k-1}(t(j/2)2^{-(k-1)}), \end{aligned}$$

and thus, by the induction hypothesis,

$$\begin{aligned} \mathbb{E}(B_k(ti2^{-k})B_k(tj2^{-k})) &= \mathbb{E}(B_{k-1}(t(i/2)2^{-(k-1)})B_{k-1}(t(j/2)2^{-(k-1)})) \\ &= t2^{-(k-1)} \min\{i/2, j/2\} = t2^{-k} \min\{i, j\}. \end{aligned}$$

Next, suppose that i is even and j is odd. Then by equation (B.6.1),

$$\begin{aligned} B_k(tj2^{-k}) &= B_{k-1}(tj2^{-k}) + \sqrt{t}2^{-k/2}\xi_{k,j} \\ &= \frac{1}{2}(B_{k-1}(t(j-1)2^{-k}) + B_{k-1}(t(j+1)2^{-k})) + \sqrt{t}2^{-(k+1)/2}\xi_{k,j} \\ &= \frac{1}{2}(B_{k-1}(t((j-1)/2)2^{-(k-1)}) + B_{k-1}(t((j+1)/2)2^{-(k-1)})) + \sqrt{t}2^{-(k+1)/2}\xi_{k,j}. \end{aligned} \quad (\text{B.6.5})$$

The formula for $B_k(ti2^{-k})$ remains the same as before. Thus,

$$\mathbb{E}(B_k(ti2^{-k})B_k(tj2^{-k})) = \frac{1}{2}t2^{-(k-1)}(\min\{i/2, (j-1)/2\} + \min\{i/2, (j+1)/2\}).$$

If $j > i$, then the right side is $t2^{-k}i$, whereas if $j < i$, then the right side is $t2^{-k}j$. This takes care of this case. Next, suppose that i and j are both odd, and not equal. Then by the above calculations.

$$\begin{aligned} \mathbb{E}(B_k(ti2^{-k})B_k(tj2^{-k})) &= \frac{1}{4}t2^{-(k-1)}(\min\{(i-1)/2, (j-1)/2\} + \min\{(i+1)/2, (j-1)/2\} \\ &\quad + \min\{(i-1)/2, (j+1)/2\}, \min\{(i+1)/2, (j+1)/2\}). \end{aligned}$$

Again, separately considering the cases $i < j$ and $i > j$, we see that equation (B.6.4) holds for both. The final possibility is that i is odd and $j = i$. Then by the formula (B.6.5) and the induction hypothesis, we get that

$$\begin{aligned} \mathbb{E}(B_k(tj2^{-k})^2) &= \frac{1}{4}(3t((j-1)/2)2^{-(k-1)} + t((j+1)/2)2^{-(k-1)} + t2^{-(k+1)}) \\ &= \frac{1}{4}t(4j-2)2^{-k} + t2^{-(k+1)} = tj2^{-k}. \end{aligned}$$

This completes the induction step and validates the claim of the theorem when $s_1, \dots, s_n$ are of the form $tj_12^{-k}, \dots, tj_n2^{-k}$ for some k and some $0 \leq j_1, \dots, j_n \leq 2^k$. Now, given $s_1, \dots, s_n \in [0, t]$, one can find sequence $(s_1^m, \dots, s_n^m)$, for $m = 1, 2, \dots$, such that each $(s_1^m, \dots, s_n^m)$ is of the above form and $s_i^m \rightarrow s_i$ as $m \rightarrow \infty$ for each i . Since B is a continuous function, this means that the random vector $(B(s_1^m), \dots, B(s_n^m))$ converges to the random vector $(B(s_1), \dots, B(s_n))$ as $m \rightarrow \infty$. Computing the characteristic function of $(B(s_1), \dots, B(s_n))$ immediately shows that it has the claimed distribution. $\square$

We will need the following property of standard Brownian motion later.

Lemma B.6.3. *Let B be standard Brownian motion on $[0, t]$. Let $M(t) := \max_{0 \leq s \leq t} |B(s)|$. There are positive universal constants C_1 and C_2 such that for any $x \geq 0$,*

$$\mathbb{P}(M(t) > x) \leq C_1 e^{-C_2 x^2/t}.$$

Proof. Since $\mathbb{P}(M \geq x) \leq 1$, it suffices to prove the claim for $x \geq \sqrt{t}$. So let us assume

henceforth that $x \geq \sqrt{t}$. Since $B_k \rightarrow B$ in the supremum norm, it is easy to see that

$$M(t) \leq \|B_0\| + \sum_{k=1}^{\infty} \|B_k - B_{k-1}\|.$$

Therefore,

$$\mathbb{P}(M(t) > x) \leq \mathbb{P}(\|B_0\| > C_0 x) + \sum_{k=1}^{\infty} \mathbb{P}(\|B_k - B_{k-1}\| > 2^{-(k+1)/4} x),$$

where

$$C_0 := 1 - \sum_{k=1}^{\infty} 2^{-(k+1)/4}.$$

Thus, by the inequalities (B.6.2) and (B.6.3), and the fact that $\|B_0\| = \sqrt{t}|\xi|$, we get

$$\begin{aligned} \mathbb{P}(M(t) > x) &\leq \mathbb{P}(|\xi| > t^{-1/2} C_0 x) + \sum_{k=1}^{\infty} \mathbb{P}\left(\max_{j=1,3,5,\dots,2^k-1} |\xi_{k,j}| > t^{-1/2} 2^{(k+1)/4} x\right) \\ &\leq \mathbb{P}(|\xi| > t^{-1/2} C_0 x) + \sum_{k=1}^{\infty} 2^k \mathbb{P}(|\xi| > t^{-1/2} 2^{(k+1)/4} x) \\ &\leq C e^{-t^{-1} C_0^2 x^2 / 4} + \sum_{k=1}^{\infty} 2^k e^{-t^{-1} 2^{(k+1)/2} x^2 / 4}. \end{aligned} \quad (\text{B.6.6})$$

Since $x \geq \sqrt{t}$, we have

$$\frac{x^2}{t} \geq \frac{1}{2} \left(1 + \frac{x^2}{t}\right).$$

Using this, we get

$$\begin{aligned} \sum_{k=1}^{\infty} 2^k e^{-t^{-1} 2^{(k+1)/2} x^2 / 4} &\leq \sum_{k=1}^{\infty} 2^k e^{-2^{(k+1)/2} / 8 - t^{-1} 2^{(k+1)/2} x^2 / 8} \\ &\leq e^{-x^2 / 8t} \sum_{k=1}^{\infty} 2^k e^{-2^{(k+1)/2} / 8}. \end{aligned}$$

Plugging this into equation (B.6.6) completes the proof. $\square$

We will now construct Brownian motion on the time interval $[0, \infty)$. Let $C[0, \infty)$ be the space of all continuous functions from $[0, \infty)$ into $\mathbb{R}$, with the metric

$$\rho(f, g) := \sum_{j=0}^{\infty} 2^{-j} \frac{\|f - g\|_{[j, j+1]}}{1 + \|f - g\|_{[j, j+1]}},$$

where

$$\|f - g\|_{[j, j+1]} := \sup_{x \in [j, j+1]} |f(x) - g(x)|.$$

It is easy to show that this is a Polish space, and moreover, that $f_n \rightarrow f$ on this space if

and only if $f_n \rightarrow f$ uniformly on compact sets.

Let $B^1, B^2, \dots$ be a sequence of i.i.d. Brownian motions on the time interval $[0, 1]$. Define a $C[0, \infty)$ -valued random variable B as follows. If $k \leq t < k + 1$ for some integer $k \geq 0$, define

$$B(t) := \sum_{j=1}^{k-1} B^j(1) + B^k(t - k).$$

The random function B is called standard Brownian motion on the time interval $[0, \infty)$. It is easy to check that the process B defined above is indeed a $C[0, \infty)$ -valued random variable, and that for any n and any $0 \leq t_1 \leq \dots \leq t_n < \infty$, the vector $(B(t_1), \dots, B(t_n))$ is a multivariate Gaussian random vector with mean vector zero and covariance structure given by $\text{Cov}(B(t_i), B(t_j)) = \min\{t_i, t_j\}$.

Appendix C

Differential Geometry

C.1 Review of topics from linear algebra

C.1.1 Tensor products of vector spaces

Let $V_1, \dots, V_k$ be vector spaces over a field $\mathbb{F}$. The tensor product $V_1 \otimes V_2 \otimes \dots \otimes V_k$ is a vector space over $\mathbb{F}$ defined as follows. Consider the Cartesian product $V_1 \times \dots \times V_k$. Let L be the vector space of $\mathbb{F}$ with $V_1 \times \dots \times V_k$ as its basis. That is, the elements of L are finite linear combinations of the form

$$\sum_{i=1}^n a_i(v_{i,1}, \dots, v_{i,k}),$$

where n is arbitrary, $a_1, \dots, a_n \in \mathbb{F}$, and $(v_{i,1}, \dots, v_{i,k}) \in V_1 \times \dots \times V_k$ for each i . Let R be the subspace of L spanned by the following elements:

1. Elements of the form

$$(u_1, \dots, u_{i-1}, u_i + v_i, u_{i+1}, \dots, u_k) - (u_1, \dots, u_k) - (u_1, \dots, u_{i-1}, v_i, u_{i+1}, \dots, u_k),$$

for each $1 \leq i \leq k$ and $u_1, \dots, u_k, v_i \in V$.

2. Elements of the form

$$(u_1, \dots, u_{i-1}, au_i, u_{i+1}, \dots, u_k) - a(u_1, \dots, u_k),$$

for each $1 \leq i \leq k$, $a \in \mathbb{F}$, and $u_1, \dots, u_k \in V$.

We define $V_1 \otimes \dots \otimes V_k$ to be the quotient space L/R . We denote the image of $(v_1, \dots, v_k)$ in this quotient space by

$$v_1 \otimes \dots \otimes v_k.$$

The construction implies that

$$\begin{aligned} u_1 \otimes \cdots \otimes u_{i-1} \otimes (u_i + v_i) \otimes u_{i+1} \cdots \otimes u_k \\ = u_1 \otimes \cdots \otimes u_k + u_1 \otimes \cdots \otimes u_{i-1} \otimes v_i \otimes u_{i+1} \otimes \cdots \otimes u_k, \end{aligned}$$

and

$$u_1 \otimes \cdots \otimes u_{i-1} \otimes (au_i) \otimes u_{i+1} \cdots \otimes u_k = a(u_1 \otimes \cdots \otimes u_k).$$

As an example, consider $\mathbb{R}^n$ as a vector space over $\mathbb{R}$. We claim that for any $k \geq 1$, the tensor power $(\mathbb{R}^n)^{\otimes k}$ (that is, the k -fold tensor product of $\mathbb{R}^n$ with itself) can be identified with the space of k -tensors of order n , that is, arrays of the form

$$(a_{i_1, \dots, i_k})_{1 \leq i_1, \dots, i_k \leq n}.$$

To see this, let $e_1, \dots, e_n$ denote the standard basis vectors of $\mathbb{R}^n$. It is easy to check that the vectors

$$e_{i_1} \otimes \cdots \otimes e_{i_k}, \quad 1 \leq i_1, \dots, i_k \leq n$$

form a basis of $(\mathbb{R}^n)^{\otimes k}$. Thus, any element of $(\mathbb{R}^n)^{\otimes k}$ can be uniquely represented as

$$\sum_{1 \leq i_1, \dots, i_k \leq n} a_{i_1, \dots, i_k} e_{i_1} \otimes \cdots \otimes e_{i_k}$$

for some coefficients $a_{i_1, \dots, i_k} \in \mathbb{R}$. This gives the required isomorphism. Note that this shows, in particular, that $\dim((\mathbb{R}^n)^{\otimes k}) = n^k$.

A useful way to think about the tensor product of two finite dimensional vector spaces U and V is the following. Let $u_1, \dots, u_m$ be a basis of U and $v_1, \dots, v_n$ be a basis of V . Then the vectors $u_i \otimes v_j$ form a basis of $U \otimes V$. Thus, any element of $U \otimes V$ can be uniquely written as a linear combination

$$\sum_{i=1}^m \sum_{j=1}^n a_{ij} u_i \otimes v_j,$$

which can be rewritten as

$$\sum_{j=1}^n x_j \otimes v_j,$$

where

$$x_j = \sum_{i=1}^m a_{ij} u_i \in U.$$

Seen this way, each element of $U \otimes V$ is sort of a linear combination of the basis elements of V , but with coefficients in U instead of in $\mathbb{F}$.

We use $V^{\otimes k}$ to denote the tensor product of V with itself k times. It is the convention to let $V^{\otimes}$ denote the vector space $\mathbb{F}$ over itself.

C.1.2 Exterior powers of a vector space

Let V be a vector space over a field $\mathbb{F}$. For an integer $k \geq 1$, the k^{th} exterior power of V , denoted $\Lambda^k(V)$, is a vector space over $\mathbb{F}$ defined as follows. Let $V^{\otimes k}$ be the k^{th} tensor power of V . Let R be the subspace of $V^{\otimes k}$ spanned by elements of the form $v_1 \otimes \cdots \otimes v_k$ with $v_i = v_{i+1}$ for some $1 \leq i \leq k-1$. We define $\Lambda^k(V)$ to be the quotient space $V^{\otimes k}/R$. The image of $v_1 \otimes \cdots \otimes v_k$ in this quotient space is denoted by

$$v_1 \wedge v_2 \wedge \cdots \wedge v_k.$$

The quotienting by R has the following consequence. With $k = 2$, note that by the definition of R , we have

$$\begin{aligned} 0 &= (v_1 + v_2) \wedge (v_1 + v_2) \\ &= v_1 \wedge v_1 + v_1 \wedge v_2 + v_2 \wedge v_1 + v_2 \wedge v_2 \\ &= v_1 \wedge v_2 + v_2 \wedge v_1. \end{aligned}$$

Thus, for any $v_1, v_2 \in V$,

$$v_1 \wedge v_2 = -v_2 \wedge v_1.$$

This extends more generally to

$$v_{\pi(1)} \wedge \cdots \wedge v_{\pi(k)} = \text{sign}(\pi) v_1 \wedge \cdots \wedge v_k \quad (\text{C.1.1})$$

for any k and any permutation $\pi \in S_k$, where S_k is the group of all permutations of $1, \dots, k$, and $\text{sign}(\pi)$ is the sign of π . Lastly, by convention, $\Lambda^0(V)$ denotes the vector space $\mathbb{F}$ over itself.

As an example, consider $\mathbb{R}^n$ as a vector space over $\mathbb{R}$. We claim that for $1 \leq k \leq n$, the exterior power $\Lambda^k(\mathbb{R}^n)$ can be identified with the space of antisymmetric k -tensors of order n , that is, arrays of the form

$$(a_{i_1, \dots, i_k})_{1 \leq i_1, \dots, i_k \leq n}$$

satisfying

$$a_{i_{\pi(1)}, \dots, i_{\pi(k)}} = \text{sign}(\pi) a_{i_1, \dots, i_k} \quad (\text{C.1.2})$$

for all $\pi \in S_n$. To see this, let $e_1, \dots, e_n$ denote the standard basis vectors of $\mathbb{R}^n$. It is easy to check that the vectors

$$e_{i_1} \wedge \cdots \wedge e_{i_k}, \quad 1 \leq i_1 < \cdots < i_k \leq n$$

form a basis of $\Lambda^k(\mathbb{R}^n)$. Thus, any element of $\Lambda^k(\mathbb{R}^n)$ can be uniquely represented as

$$\sum_{1 \leq i_1 < \cdots < i_k \leq n} a_{i_1, \dots, i_k} e_{i_1} \wedge \cdots \wedge e_{i_k}$$

for some coefficients $a_{i_1, \dots, i_k} \in \mathbb{R}$. The definition of $a_{i_1, \dots, i_k}$ can then be extended to all $1 \leq i_1, \dots, i_k \leq n$ using the relation (C.1.2). This gives the required isomorphism. Note that this shows, in particular, that $\dim(\Lambda^k(\mathbb{R}^n)) = \binom{n}{k}$ (which coheres the convention that $\Lambda^0(V) = \mathbb{F}$). For $k > n$, $\Lambda^k(\mathbb{R}^n) = \{0\}$.

C.1.3 Exterior algebra of a vector space

Let V be a vector space over a field $\mathbb{F}$. The exterior algebra $\Lambda(V)$ of V , also known as the Grassmann algebra, is an algebra obtained by ‘putting together’ all the exterior powers in a certain way. First, we define the tensor algebra of V as the direct sum

$$T(V) := \bigoplus_{k=0}^{\infty} V^{\otimes k},$$

where $V^{\otimes 0} := \mathbb{F}$ (meaning that the elements of $T(V)$ are formal linear combinations of elements from $V^{\otimes k}$, $k = 0, 1, \dots$, with coefficients from $\mathbb{F}$, only finitely many of which are nonzero), and the product that makes this an algebra, denoted by $\otimes$, is defined via the prescription

$$(v_1 \otimes \dots \otimes v_k) \otimes (w_1 \otimes \dots \otimes w_l) := v_1 \otimes \dots \otimes v_k \otimes w_1 \otimes \dots \otimes w_l. \quad (\text{C.1.3})$$

It is not hard to show that this gives a well-defined product on $T(V)$ that makes $T(V)$ an algebra. A consequence of this definition is that if $v \in V^{\otimes k}$ and $w \in V^{\otimes l}$, then $v \otimes w \in V^{\otimes(k+l)}$. Now, let I be the two-sided ideal of this algebra generated by elements of the form $v \otimes v$, $v \in V$. Explicitly,

$$I = \left\{ \sum_{i=1}^m s_i \otimes v_i \otimes v_i \otimes t_i : m \geq 0, s_i, t_i \in T(V), v_i \in V \text{ for } i = 1, \dots, m \right\}.$$

Definition C.1.1. Let V be a vector space over a field $\mathbb{F}$, and let $T(V)$ and I be defined as above. We define the exterior algebra of V , denoted $\Lambda(V)$, to be the quotient $T(V)/I$.

The product of two elements $v, w \in \Lambda(V)$, induced from the product (C.1.3) on $T(V)$, is denoted by $v \wedge w$. This is called the exterior product or the wedge product. The exterior algebra decomposes — as a vector space — as the direct sum

$$\Lambda(V) = \bigoplus_{k=0}^{\infty} \Lambda^k(V), \quad (\text{C.1.4})$$

where $\Lambda^0(V) = \mathbb{F}$. It is not hard to check that if $v \in \Lambda^k(V)$ and $w \in \Lambda^l(V)$, then $v \wedge w \in \Lambda^{k+l}(V)$. An important observation is that $w \wedge v$ is also in $\Lambda^{k+l}(V)$, and it is related to $v \wedge w$ by the relation

$$w \wedge v = (-1)^{kl} v \wedge w. \quad (\text{C.1.5})$$

This can be easily verified by decomposing v and w as linear combinations of basis elements.

C.2 Review of basic differential geometry

C.2.1 Smooth manifolds

Let us recall some basic definitions from differential geometry. We assume that the reader is familiar with basic topological terminology. A smooth n -dimensional manifold is a set which locally looks like an open subset of $\mathbb{R}^n$. More precisely:

Definition C.2.1. *An n -dimensional smooth manifold M with an atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{A}}$ is a second countable Hausdorff topological space, such that*

1. *Each U_α is an open subset of M .*
2. *Each φ_α is a homeomorphism between U_α and an open set $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.*
3. *The union of U_α over all $\alpha \in \mathcal{A}$ is equal to M .*
4. *If $U_\alpha \cap U_\beta \neq \emptyset$ for some $\alpha, \beta \in \mathcal{A}$, then the map $\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta)$ is smooth (i.e., infinitely differentiable).*

For example, $\mathbb{R}^n$ is an n -dimensional smooth manifold, with one possible atlas being just the pair consisting of $\mathbb{R}^n$ and the identity map on $\mathbb{R}^n$. The unit sphere $S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$ is an $(n-1)$ -dimensional smooth manifold.

The elements of an atlas are called charts. A chart (U, φ) is said to be ‘around’ a point $x \in M$ if $x \in U$.

The atlas of a manifold is not unique. Two atlases are said to be compatible if their union is also an atlas. The union of all atlases that are compatible with a given atlas is called a maximal atlas. The maximal atlas is unique and determines the structure of the manifold.

Recall that a map f from a smooth manifold M into a smooth manifold N is said to be smooth if it is smooth in local neighborhoods of points of M , viewing them as subsets of $\mathbb{R}^n$. More precisely:

Definition C.2.2. *Let M and N be two smooth manifolds. A map $f : M \rightarrow N$ is said to be smooth if for any $x \in M$, there is a chart (U, φ) around x and a chart (V, ψ) around $f(x)$ such that $\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \rightarrow \psi(V)$ is smooth. A smooth map $f : M \rightarrow N$ is called a diffeomorphism if f is a bijection and f^{-1} is also smooth. The set of all smooth maps from M into $\mathbb{R}$ is denoted by $C^\infty(M)$. The set of all smooth maps from M into N is denoted by $C^\infty(M, N)$.*

Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{A}}$ be an atlas for a manifold M . For any $\alpha, \beta \in \mathcal{A}$, let $\varphi_\alpha \times \varphi_\beta : U_\alpha \times U_\beta \rightarrow \mathbb{R}^{2n}$ denote the map

$$(\varphi_\alpha \times \varphi_\beta)(x, y) := (\varphi_\alpha(x), \varphi_\beta(y)).$$

Then, endowing $M \times M$ with the atlas $\{(U_\alpha \times U_\beta, \varphi_\alpha \times \varphi_\beta)\}_{\alpha, \beta \in \mathcal{A}}$ makes it a smooth $2n$ -dimensional manifold. We can similarly endow manifold structures to the Cartesian product of any finite number of manifolds.

We will need the concept of an *orientable manifold* later. The definition is as follows. Recall that the Jacobian of a smooth map $f = (f_1, \dots, f_n) : U \rightarrow \mathbb{R}^n$, where U is an open subset of $\mathbb{R}^n$, is defined as

$$J(x) := \det \left(\left(\frac{\partial f_i}{\partial x_j} \right)_{1 \leq i, j \leq n} \right).$$

Definition C.2.3. Let M be a smooth n -dimensional manifold. We say that M is *orientable* if there is an atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{A}}$ of M such that for any $\alpha, \beta \in \mathcal{A}$ such that $U_\alpha \cap U_\beta \neq \emptyset$, the map $\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta)$ has a positive Jacobian everywhere on $\varphi_\beta(U_\alpha \cap U_\beta)$.

An orientable manifold has the notion of an ‘outward direction’ everywhere. The Möbius strip, for example, is not orientable.

C.2.2 Fiber bundles and vector bundles

Let M and F be two smooth manifolds. A fiber bundle is essentially a smooth varying collection of manifolds (called ‘fibers’), each having the same structure as F , and indexed by elements of M . The formal definition is as follows.

Definition C.2.4. A smooth fiber bundle is a 4-tuple (E, B, π, F) , where

1. the ‘total space’ E , the ‘base space’ B , and the ‘typical fiber’ F are smooth manifolds,
2. the ‘projection map’ π is a smooth surjective map from E onto B , and
3. for each $x \in B$, there is an open neighborhood U containing x and a diffeomorphism $\varphi : \pi^{-1}(U) \rightarrow U \times F$ such that $\pi(\varphi^{-1}(z, y)) = z$ for all $z \in U$ and $y \in F$.

The pair (U, φ) in the above definition is called a ‘local trivialization’ of the fiber bundle. Note that the condition $\pi(\varphi^{-1}(z, y)) = z$ implies that if $u \in \pi^{-1}(\{z\})$ for some $z \in U$, then $\varphi(u)$ must be of the form (z, v) for some $v \in F$. We will sometimes denote this v by $\tilde{\varphi}(u)$.

The inverse images $\pi^{-1}(\{x\})$, for $x \in B$, are called the ‘fibers’ of the fiber bundle. The simplest kind of fiber bundle is just the Cartesian product $E = B \times F$ (endowed with the product manifold structure), with $\pi(x, y) = x$, $U = B$, and $\varphi(x, y) = (x, y)$. These are called ‘trivial’ bundles.

Definition C.2.5. Given a fiber bundle (E, B, π, F) , a smooth section of the bundle is a smooth map $s : B \rightarrow E$ such that $\pi(s(x)) = x$ for all $x \in B$.

In other words, a smooth section is simply a way of choosing one element from each fiber in a ‘smooth’ way. Often, we simply refer to the total space E , instead of the 4-tuple (E, B, π, F) , as a fiber bundle.

Vector bundles are special kinds of fiber bundles where the typical fibers are finite dimensional vector spaces over $\mathbb{R}$. The precise definition is as follows.

Definition C.2.6. *A fiber bundle is called a vector bundle if*

1. *there is some n such that each fiber $\pi^{-1}(\{x\})$ is a vector space of dimension n over $\mathbb{R}$,*
2. *the typical fiber is $\mathbb{R}^n$, and*
3. *each local trivialization $\varphi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$ has the property that $\varphi^{-1}(x, \cdot)$ is a linear isomorphism between the vector spaces $\mathbb{R}^n$ and $\pi^{-1}(\{x\})$, for each $x \in U$.*

Let M be a smooth n -dimensional manifold and V be an m -dimensional vector space over $\mathbb{R}$. The trivial bundle $M \times V$ is defined as follows. Take any linear isomorphism $\phi : V \rightarrow \mathbb{R}^m$. Then ϕ can be used to define a topology and a manifold structure on V . It is easy to see that this structure does not depend on the choice of the isomorphism. Once that is defined, one can define the bundle $M \times V$ as we defined trivial fiber bundles above.

Besides trivial vector bundles, two other important examples of vector bundles are the tangent and cotangent bundles of a manifold. We will encounter them in the next section.

We will sometimes need to work with tensor products of vector bundles. The tensor product of two vector bundles is defined as follows.

Definition C.2.7. *Let $(E_1, B, \pi_1, \mathbb{R}^m)$ and $(E_2, B, \pi_2, \mathbb{R}^n)$ be two vector bundles over the same base space B . We define the tensor product $(E, B, \pi, \mathbb{R}^m \otimes \mathbb{R}^n)$ as:*

1. *The total space E consists of all pairs (x, w) , where $x \in B$, $w \in \pi_1^{-1}(\{x\}) \otimes \pi_2^{-1}(\{x\})$.*
2. *The projection map π is given by $\pi(x, w) = x$.*
3. *Take any $x \in B$, an open neighborhood U of x , and local trivializations φ_1 and φ_2 of E_1 and E_2 on U . A local trivialization $\varphi : \pi^{-1}(U) \rightarrow U \times (\mathbb{R}^m \otimes \mathbb{R}^n)$ is defined as $\varphi(y, u \otimes v) := (y, \tilde{\varphi}_1(u) \otimes \tilde{\varphi}_2(v))$, where $y \in U$, $u \in \pi_1^{-1}(\{y\})$, $v \in \pi_2^{-1}(\{y\})$, and for $i = 1, 2$, $\tilde{\varphi}_i$ is the map defined through $\varphi_i(u) = (\pi_i(u), \tilde{\varphi}_i(u))$.*

We denote the total space E by $E_1 \otimes E_2$.

C.2.3 Tangent and cotangent bundles

Let M be an n -dimensional smooth manifold. A tangent vector at a point $x \in M$ is defined as follows. Let $\gamma : (-\epsilon, \epsilon) \rightarrow M$ be a smooth map (considering $(-\epsilon, \epsilon)$ as a 1-dimensional smooth manifold). We will refer to γ as a ‘curve’ in M . Suppose that $\gamma(0) = x$. Take any $f \in C^\infty(M)$. The derivative of f at x in the direction γ is the real number

$$\left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0}.$$

(It is not hard to verify from the given conditions that this derivative exists.) Let us say that two curves γ_1 and γ_2 are equivalent if the derivatives of any f at x in the directions γ_1 and γ_2 are the same. The set of equivalence classes is known as the tangent space of M at x , denoted by $T_x M$. For any $v \in T_x M$ and any $f \in C^\infty(M)$, we define

$$v(f) := \left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0}, \quad (\text{C.2.1})$$

where γ is any element of the equivalence class v .

Recall that for any $x \in M$, $T_x M$ is an n -dimensional vector space, isomorphic to $\mathbb{R}^n$. Indeed, given a chart (U, φ) around x , $T_x M$ has the following explicit description. For $1 \leq i \leq n$, let $\gamma_i : \mathbb{R} \rightarrow M$ be the curve $\gamma_i(t) = \varphi^{-1}(\varphi(x) + te_i)$, where $e_1, \dots, e_n$ denote the standard basis vectors of $\mathbb{R}^n$. Note that $\gamma_i(0) = x$ for each i . The element of $T_x M$ that contains γ_i is denoted by

$$\frac{\partial}{\partial x_i},$$

and we denote derivative of $t \mapsto f(\gamma_i(t))$ at $t = 0$ by

$$\frac{\partial f}{\partial x_i}(x).$$

It is not hard to verify using the above definition that

$$\frac{\partial f}{\partial x_i}(x) = \left. \frac{\partial}{\partial y_i} (f \circ \varphi^{-1})(y) \right|_{y=\varphi(x)}. \quad (\text{C.2.2})$$

where $f \circ \varphi^{-1}$ is from $\varphi(U) \subseteq \mathbb{R}^n$ into $\mathbb{R}$.

Then $T_x M$ is a vector space with basis $\frac{\partial}{\partial x_i}$, $i = 1, \dots, n$. That is, for any $v \in T_x M$ represented by a curve γ , there exist unique $a_1, \dots, a_n \in \mathbb{R}$ such that for any f ,

$$\left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0} = \sum_{i=1}^n a_i \frac{\partial f}{\partial x_i}(x),$$

or in other words,

$$v = \sum_{i=1}^n a_i \frac{\partial}{\partial x_i}.$$

Conversely, given any $a_1, \dots, a_n \in \mathbb{R}$, the curve $\gamma(t) = \varphi^{-1}(ta_1, \dots, ta_n)$ satisfies the above identity. Note that the basis $\frac{\partial}{\partial x_i}$, $i = 1, \dots, n$ is dependent on the choice of chart (U, φ) , but the definition of $T_x M$ is not dependent on the choice of the chart.

The cotangent space at a point x , denoted $T_x^* M$, is the dual of the vector space $T_x M$ (that is the space of all linear maps from $T_x M$ into $\mathbb{R}$). Given a chart (U, φ) around x , a canonical basis of $T_x^* M$ is given by $dx_1, \dots, dx_n$, where $dx_i : T_x M \rightarrow \mathbb{R}$ is the linear map defined by

$$dx_i \left(\frac{\partial}{\partial x_j} \right) = \delta_{ij},$$

where $\delta_{ij} = 1$ if $i = j$ and 0 if $i \neq j$.

Definition C.2.8. *The tangent bundle of M , denoted TM , is the set of all pairs (x, v) where $x \in M$ and $v \in T_x M$. Similarly, the cotangent bundle of M , denoted T^*M , is the set of all pairs (x, w) where $x \in M$ and $w \in T_x^* M$.*

The sets TM and T^*M are the total spaces of two vector bundles, which we now define. First, we give manifold structures to TM and T^*M . Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{A}}$ be an atlas of M . We use this atlas to define atlases for TM and T^*M . For TM , the set of neighborhoods are $\{V_\alpha\}_{\alpha \in \mathcal{A}}$, where

$$V_\alpha := \{(x, v) : x \in U_\alpha, v \in T_x M\}.$$

Take any $x \in U_\alpha$ and $v \in T_x M$. Then there is a unique set of coefficients $a_1, \dots, a_n \in \mathbb{R}$ such that

$$v = \sum_{i=1}^n a_i \frac{\partial}{\partial x_i}.$$

Define $(\varphi_\alpha)_*(v) := (a_1, \dots, a_n) \in \mathbb{R}^n$, and let

$$\psi_\alpha(x, v) := (\varphi_\alpha(x), (\varphi_\alpha)_*(v)) \in \mathbb{R}^{2n}.$$

The collection $\{(V_\alpha, \psi_\alpha)\}_{\alpha \in \mathcal{A}}$ gives the required atlas for TM . Similarly, for T^*M , let

$$W_\alpha := \{(x, w) : x \in U_\alpha, w \in T_x^* M\}.$$

For any $x \in U_\alpha$ and $w \in T_x^* M$, there is a unique set of coefficients $b_1, \dots, b_n$ such that

$$w = \sum_{i=1}^n b_i dx_i.$$

Define $\varphi_\alpha^*(w) := (b_1, \dots, b_n)$, and let

$$\xi_\alpha(x, w) := (\varphi_\alpha(x), \varphi_\alpha^*(w)) \in \mathbb{R}^{2n}.$$

The collection $\{(W_\alpha, \xi_\alpha)\}_{\alpha \in \mathcal{A}}$ gives the required atlas for T^*M .

The projection maps, fibers and local trivializations should now be obvious. For the tangent bundle, the projection map is simply the map that takes $(x, v) \in TM$ to x . Similarly, the projection map for the cotangent bundle takes $(x, w) \in T^*M$ to x . The typical fibers in both cases is just $\mathbb{R}^n$, and the fibers are tangent and cotangent spaces at points of M , which are n -dimensional vector spaces over $\mathbb{R}$. The pairs $(U_\alpha, (\varphi_\alpha)_*)$ and $(U_\alpha, \varphi_\alpha^*)$ are local trivializations of TM and T^*M .

For $1 \leq k \leq n$, the k^{th} exterior power of the cotangent bundle, denoted $\Lambda^k T^*M$, is defined to be the set of all pairs of the form (x, w) , where $x \in M$ and $w \in \Lambda^k T_x^* M$. We make this a smooth manifold in the same way that did for T^*M above. Namely, for each α , define

$$Z_\alpha := \{(x, w) : x \in U_\alpha, w \in \Lambda^k T_x^* M\}.$$

Take any $x \in M$ and $w \in \Lambda^k T_x^* M$. Then by the discussion at the end of Section C.1.2, it is easy to see that w can be represented as

$$w = \sum_{1 \leq i_1 < \dots < i_k \leq n} a_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}$$

for a unique set of coefficients $a_{i_1, \dots, i_k}$. This gives us a natural way to define a homeomorphism η_α from Z_α into an open subset of $\mathbb{R}^{n + \binom{n}{k}}$. The collection $\{(Z_\alpha, \eta_\alpha)\}_{\alpha \in \mathcal{A}}$ gives an atlas for $\Lambda^k T^* M$.

As before, we can construct a fiber bundle with $\Lambda^k T^* M$ as the total space and M as the base space. For example, the projection map $\pi : \Lambda^k T^* M \rightarrow M$ is defined as $\pi(x, w) = x$. It is not hard to show that π is a smooth map between the two manifolds.

C.2.4 Vector fields on a manifold

A vector field X on a smooth manifold is a map that assigns an element of $T_x M$ to each $x \in M$, in such a way that $X(x)$ varies smoothly with x . The precise definition is as follows.

Definition C.2.9. *Let M be a smooth manifold. A smooth vector field X on M is a smooth section of the tangent bundle TM . The space of all vector fields on M is denoted by $\mathcal{X}(M)$.*

A vector field on M acts on $C^\infty(M)$ in a natural way. Given $f \in C^\infty(M)$ and a smooth vector field X on M , we define $X(f) \in C^\infty(M)$ as

$$X(f)(x) := X(x)(f),$$

where the action of the tangent vector $X(x)$ on the function f is defined as in equation (C.2.1). It is easy to verify that $X(f)$ is indeed an element of $C^\infty(M)$.

Let n be the dimension of M . Let (U, φ) be a chart on M . With respect to the coordinates given by this chart, we can write X at a point $x \in U$ as a vector $(X_1(x), \dots, X_n(x)) \in \mathbb{R}^n$. Then

$$X(f) = \sum_{i=1}^n X_i \frac{\partial f}{\partial x_i}, \tag{C.2.3}$$

where we are suppressing the argument x on both sides. Thus, locally, a vector field X can be viewed as a smooth map from U into $\mathbb{R}^n$. But the abstract definition is necessary for defining vector fields in a coordinate-free manner.

Given two smooth vectors fields X and Y on a smooth manifold M , we define the Lie bracket of X and Y as the map $[X, Y] : C^\infty(M) \rightarrow C^\infty(M)$ given by

$$[X, Y](f) := X(Y(f)) - Y(X(f)).$$

Recall the following fundamental lemma. Its proof gives the explicit formula for $[X, Y]$ in coordinates.

Lemma C.2.10. *For any two smooth vector field X, Y on a smooth manifold M , there is a smooth vector field Z on M such that $Z = [X, Y]$ as a map on $C^\infty(M)$.*

Proof. Let (U, φ) be a chart on M . In the coordinates defined by this chart, the formula (C.2.3) shows that we can write

$$\begin{aligned} [X, Y](f) &= X(Y(f)) - Y(X(f)) \\ &= \sum_{i=1}^n X_i \frac{\partial}{\partial x_i} (Y(f)) - \sum_{i=1}^n Y_i \frac{\partial}{\partial x_i} (X(f)) \\ &= \sum_{i=1}^n X_i \frac{\partial}{\partial x_i} \left(\sum_{j=1}^n Y_j \frac{\partial f}{\partial x_j} \right) - \sum_{i=1}^n Y_i \frac{\partial}{\partial x_i} \left(\sum_{j=1}^n X_j \frac{\partial f}{\partial x_j} \right) \\ &= \sum_{j=1}^n \left[\sum_{i=1}^n \left(X_i \frac{\partial Y_j}{\partial x_i} - Y_i \frac{\partial X_j}{\partial x_i} \right) \right] \frac{\partial f}{\partial x_j}, \end{aligned}$$

where the last identity holds because

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i},$$

which causes the terms involving second derivatives of f to cancel out. Thus, if we define the field

$$Z := \sum_{i=1}^n \left(X_i \frac{\partial Y_j}{\partial x_i} - Y_i \frac{\partial X_j}{\partial x_i} \right)$$

on the neighborhood U , then $Z(f) = [X, Y](f)$ on U . It is not hard to verify that the above definition of Z remains invariant under any local change of coordinates, and if we define Z like this on every chart, then Z is a smooth vector field on M and $Z = [X, Y]$ as a map on $C^\infty(M)$. $\square$

C.3 Lie groups and Lie algebras

C.3.1 Lie groups

Recall the following definition of Lie group.

Definition C.3.1. *Let G be a group that is also a smooth n -dimensional manifold. We say that G is a Lie group if the following hold:*

1. *The map $(g, h) \mapsto gh$ from $G \times G$ into G is smooth.*
2. *The map $g \mapsto g^{-1}$ from G into G is smooth.*

Lie groups are central to gauge theory. Some Lie groups that will be important for us are the following:

1. The group $\mathrm{GL}(n, \mathbb{C})$ of all $n \times n$ invertible complex matrices, with the group operation of matrix multiplication.

2. The group $U(n)$ of all $n \times n$ unitary matrices.
3. The group $SU(n)$ of all $n \times n$ unitary matrices with determinant 1.

It is not entirely trivial to verify that the above groups are indeed Lie groups. Proofs can be found in any standard textbook on Lie theory. It is important to note that although $GL(n, \mathbb{C})$, $U(n)$ and $SU(n)$ are groups of matrices with complex entries, as manifolds they are still viewed as locally like $\mathbb{R}^m$, where m is the dimension of the manifold ($m = 2n^2$ for $GL(n, \mathbb{C})$, $m = n^2$ for $U(n)$ and $m = n^2 - 1$ for $SU(n)$).

For example, to see that $GL(n, \mathbb{C})$ has dimension $2n^2$, observe that by considering the real and complex parts of the entries of a matrix in $GL(n, \mathbb{C})$, the matrix can be identified with a collection of $2n^2$ real numbers. Moreover, since any matrix that is sufficiently close to an invertible matrix is also invertible, it is easy to show that the single chart $(GL(n, \mathbb{C}), \varphi)$ gives an atlas for $GL(n, \mathbb{C})$, where φ takes a matrix to a point in $\mathbb{R}^{2n^2}$ as above.

C.3.2 Lie algebras

In general, one can compute the dimension of a Lie group by identifying its Lie algebra, which is defined as follows.

Definition C.3.2. *Let G be a Lie group. The Lie algebra $\mathfrak{g}$ of G is the tangent space of G at the identity element.*

We will mostly be concerned with Lie groups that are subgroups of $GL(n, \mathbb{C})$ and inherit the manifold structure from $GL(n, \mathbb{C})$. These groups are sometimes called ‘matrix Lie groups’. Let G be such a group, of dimension $m \leq 2n^2$. For $g \in G$, the tangent space $T_g G$ can be represented as a subspace of the vector space $M(n, \mathbb{C})$ of all $n \times n$ complex matrices, as follows. Let $\gamma : (-\epsilon, \epsilon) \rightarrow G$ be a smooth curve such that $\gamma(0) = g$. One can show that the element $v \in T_g G$ corresponding to γ is determined by the matrix $X = (\gamma'_{ij}(0))_{1 \leq i, j \leq n} \in M(n, \mathbb{C})$. Moreover, the mapping $v \mapsto X$ is injective and linear. Thus, one can use the matrix X to represent the tangent space element v . In particular, $\mathfrak{g}$ can be represented as an m -dimensional subspace of $M(n, \mathbb{C})$. For example, it turns out that the Lie algebra $\mathfrak{u}(n)$ of $U(n)$ is the space of all $n \times n$ complex skew-Hermitian matrices, and the Lie algebra $\mathfrak{su}(n)$ of $SU(n)$ is the space of all $n \times n$ complex skew-Hermitian matrices with trace zero.

For example, to see that $\mathfrak{u}(n)$ is the space of all $n \times n$ skew-Hermitian matrices, observe that for any curve γ in $U(n)$ with $\gamma(0) = I$ (where I is the identity matrix), we must have $\gamma(t)\gamma(t)^* = I$ for all t , where $\gamma(t)^*$ denotes the adjoint of $\gamma(t)$. Differentiating on both sides, we get

$$\gamma'(t)\gamma(t)^* + \gamma(t)\gamma'(t)^* = 0$$

for all t . Since $\gamma(0) = I$, this implies that

$$\gamma'(0) + \gamma'(0)^* = 0.$$

That is, $\gamma'(0)$ is skew-Hermitian. Therefore, every element of $\mathfrak{u}(n)$ is skew-Hermitian. To show that every skew-Hermitian matrix is in $\mathfrak{u}(n)$, the simplest way is to consider the curve $\gamma(t) := \exp(tX)$ (see below for the definition of $\exp$), where X is a skew-Hermitian matrix, and deduce that γ is a curve in $U(n)$ satisfying $\gamma(0) = I$ and $\gamma'(0) = X$.

Next, let G be any Lie group and $\mathfrak{g}$ denote its Lie algebra. A Lie algebra element $X \in \mathfrak{g}$ defines a smooth vector field on G , which we also denote by X , as follows. Let e be the identity element of G , and take any $g \in G$. Take any smooth curve $\gamma : (-\epsilon, \epsilon) \rightarrow G$ with $\gamma(0) = e$, representing the vector X . We ‘push forward’ γ to a curve η passing through g by defining

$$\eta(t) := g\gamma(t).$$

The tangent vector X is then pushed forward to a tangent vector $X(g)$ at g corresponding to the equivalence class of the curve η . It is not hard to show that $X(g)$ does not depend on the choice of the curve γ representing X . If G is a matrix Lie group, then $X(g)$ is explicitly given by $g\gamma'(0) = gX$.

Given $X, Y \in \mathfrak{g}$, and viewing them as vector fields on M in the above manner, we can define the Lie bracket $[X, Y]$ as in Section C.2.4. The value of this field at e is an element of $\mathfrak{g}$, which is also denoted by $[X, Y]$. The mapping $(X, Y) \mapsto [X, Y]$ is bilinear and satisfies the Jacobi identity

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

This defines an algebraic structure on $\mathfrak{g}$, which is why $\mathfrak{g}$ is called an algebra instead of just a vector space. When G is a matrix Lie group, the element $[X, Y] \in \mathfrak{g}$ turns out to be simply the commutator of the matrices X and Y , that is, $[X, Y] = XY - YX$. The following definition generalizes this observation.

Definition C.3.3. *Let V be a vector space over a field $\mathbb{F}$. The Lie algebra $\mathfrak{gl}(V)$ is the vector space $\text{GL}(V)$ endowed with the Lie product $[X, Y] := XY - YX$ for $X, Y \in \text{GL}(V)$.*

Take a Lie group G with Lie algebra $\mathfrak{g}$ and identity element e . Given $X \in \mathfrak{g}$, there is a natural curve through e that represents X , given by the *exponential map* $\gamma(t) = \exp(tX)$. This is the unique smooth curve from $\mathbb{R}$ into G such that $\gamma(0) = e$, $\gamma(t)\gamma(s) = \gamma(t+s)$ for all $t, s \in \mathbb{R}$, and γ represents X . It can be shown that for any $X \in \mathfrak{g}$, $\exp(tX)$ exists and is unique. When G is a matrix Lie group, the exponential map is explicitly given by

$$\exp(tX) = \sum_{k=0}^{\infty} \frac{t^k}{k!} X^k. \quad (\text{C.3.1})$$

Note that in this case, the derivative of $\exp(tX)$ with respect to t is equal to X at $t = 0$.

C.3.3 Representations of Lie groups and Lie algebras

Representations of Lie groups and Lie algebras are central to the study of Yang–Mills theories. Recall the following definition of representation of a Lie group.

Definition C.3.4. Let G be a Lie group and V be a vector space over $\mathbb{R}$ or $\mathbb{C}$. Let $\mathrm{GL}(V)$ denote the group of invertible linear maps from V into V . A representation of G in V is a group homomorphism $\rho : G \rightarrow \mathrm{GL}(V)$, that is, a map satisfying

1. $\rho(e) = I$, where e is the identity of G and I is the identity of $\mathrm{GL}(V)$, and
2. $\rho(gh) = \rho(g)\rho(h)$ for all $g, h \in G$.

A representation is called finite-dimensional if the vector space is finite-dimensional. Henceforth, we will assume that all representations are finite-dimensional unless otherwise stated. Two kinds of representations will occur frequently:

1. Let G be a Lie group that is a subgroup of $\mathrm{GL}(n, \mathbb{C})$ for some n . The fundamental representation (or standard representation) of G is the map $\rho : G \rightarrow \mathrm{GL}(n, \mathbb{C})$ that takes $g \in G$ to the linear map $x \mapsto gx$ on $\mathbb{C}^n$, where gx is the product of the matrix g and the column vector x .
2. Let G be a Lie group with identity element e , and let $\mathfrak{g}$ be its Lie algebra. The adjoint representation of G is the map $\mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g})$ defined as follows. Take any $g \in G$ and $X \in \mathfrak{g}$. Let $\gamma : (-\epsilon, \epsilon) \rightarrow G$ be a smooth curve representing X in $\mathfrak{g} = T_e G$. Let $\eta(t) := g\gamma(t)g^{-1}$, so that η is also a smooth curve with $\eta(0) = e$. We define $\mathrm{Ad}(g)(X)$ to be the element of $T_e G$ corresponding to the curve η . When G is a matrix Lie group, we simply have $\mathrm{Ad}(g)(X) = gXg^{-1}$. Usually, we write Ad_g instead of $\mathrm{Ad}(g)$.

We will also need to deal with representations of Lie algebras. Let V be a finite-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$. Recall that the Lie algebra $\mathfrak{gl}(V)$ of the Lie group $\mathrm{GL}(V)$ is simply the vector space (over $\mathbb{R}$) of all linear maps from V into V . Lie algebra representations are defined as follows.

Definition C.3.5. Let G be a Lie group with Lie algebra $\mathfrak{g}$. A representation of $\mathfrak{g}$ in a vector space V is an algebra homomorphism from $\mathfrak{g}$ into $\mathfrak{gl}(V)$, meaning that it is linear, and for any $X, Y \in \mathfrak{g}$, $\rho([X, Y]) = [\rho(X), \rho(Y)]$.

An important example is the adjoint representation of a Lie algebra. This is defined as follows. Let G be a Lie group with Lie algebra $\mathfrak{g}$. The adjoint representation of $\mathfrak{g}$, denoted ad , is the map from $\mathfrak{g}$ into $\mathfrak{gl}(\mathfrak{g})$ given by

$$\mathrm{ad}(X)(Y) := [X, Y].$$

(Note that $\mathfrak{gl}(\mathfrak{g})$ is the real vector space of all linear maps from $\mathfrak{g}$ into $\mathfrak{g}$, and $\mathrm{ad}(X)$, as defined above, is an element of this space.) Usually, we write ad_X instead of $\mathrm{ad}(X)$. Note that this is a representation of $\mathfrak{g}$ because it is obviously linear, and by the Jacobi identity

for the Lie product,

$$\begin{aligned}
 \operatorname{ad}_{[X,Y]}(Z) &= [[X, Y], Z] \\
 &= [X, [Y, Z]] - [Y, [X, Z]] \\
 &= \operatorname{ad}_X([Y, Z]) - \operatorname{ad}_Y([Z, X]) \\
 &= \operatorname{ad}_X(\operatorname{ad}_Y(Z)) - \operatorname{ad}_Y(\operatorname{ad}_X(Z)),
 \end{aligned}$$

which shows that $\operatorname{ad}_{[X,Y]} = [\operatorname{ad}_X, \operatorname{ad}_Y]$.

The adjoint representation is used to define a certain symmetric bilinear form on $\mathfrak{g}$, known as the *Killing form*. It is defined as

$$B(X, Y) := \operatorname{Tr}(\operatorname{ad}_X \operatorname{ad}_Y).$$

Let $\mathfrak{g}^*$ denote the dual space $\mathfrak{g}$, that is, the space of all linear maps from $\mathfrak{g}$ into $\mathbb{R}$. The Killing form B can be alternatively viewed as a linear map $B : \mathfrak{g} \rightarrow \mathfrak{g}^*$, defined as

$$B(X)(Y) := B(X, Y).$$

For any $X \in \mathfrak{g}$, we define a linear map $\operatorname{ad}_X^* : \mathfrak{g}^* \rightarrow \mathfrak{g}^*$ as

$$\operatorname{ad}_X^*(\phi)(Y) := \phi(\operatorname{ad}_X(Y)) = \phi([X, Y]),$$

where $\phi \in \mathfrak{g}^*$. The following lemma establishes an important ‘intertwining’ property of the Killing form.

Lemma C.3.6. *In the above setting, we have that for any $X \in \mathfrak{g}$, the Killing form B satisfies $B\operatorname{ad}_X = -\operatorname{ad}_X^*B$.*

Proof. Note that for any $X, Y, Z \in \mathfrak{g}$,

$$\begin{aligned}
 B([X, Y], Z) &= \operatorname{Tr}(\operatorname{ad}_{[X,Y]} \operatorname{ad}_Z) \\
 &= \operatorname{Tr}([\operatorname{ad}_X, \operatorname{ad}_Y] \operatorname{ad}_Z) \\
 &= \operatorname{Tr}(\operatorname{ad}_X \operatorname{ad}_Y \operatorname{ad}_Z) - \operatorname{Tr}(\operatorname{ad}_Y \operatorname{ad}_X \operatorname{ad}_Z).
 \end{aligned}$$

But $\operatorname{Tr}(\operatorname{ad}_X \operatorname{ad}_Y \operatorname{ad}_Z) = \operatorname{Tr}(\operatorname{ad}_Y \operatorname{ad}_Z \operatorname{ad}_X)$. Thus, we get

$$\begin{aligned}
 B([X, Y], Z) &= \operatorname{Tr}(\operatorname{ad}_Y \operatorname{ad}_Z \operatorname{ad}_X) - \operatorname{Tr}(\operatorname{ad}_Y \operatorname{ad}_X \operatorname{ad}_Z) \\
 &= -\operatorname{Tr}(\operatorname{ad}_Y [\operatorname{ad}_X, \operatorname{ad}_Z]) = -\operatorname{Tr}(\operatorname{ad}_Y \operatorname{ad}_{[X,Z]}) \\
 &= -\operatorname{Tr}(\operatorname{ad}_{[X,Z]} \operatorname{ad}_Y) = -B([X, Z], Y).
 \end{aligned}$$

Now, for any $X, Y, Z \in \mathfrak{g}$,

$$(B\operatorname{ad}_X(Y))(Z) = B([X, Y])(Z) = B([X, Y], Z),$$

and

$$(\operatorname{ad}_X^* B(Y))(Z) = B(Y)([X, Z]) = B([X, Z], Y).$$

Combining the last three displays completes the proof. $\square$

C.3.4 Structure constants of a Lie algebra

Let r be the dimension of a Lie algebra $\mathfrak{g}$. A set of *generators* T_a , $a = 1, 2, \dots, r$ of $\mathfrak{g}$ is simply any basis of the vector space $\mathfrak{g}$. The *structure constants* of a given set of generators are real numbers f_{ab}^c such that for any a, b, c ,

$$[T_a, T_b] = \sum_{c=1}^r f_{ab}^c T_c.$$

For example, consider the Lie algebra $\mathfrak{su}(2)$. This is the space of all 2×2 skew-Hermitian matrices with trace zero. A well-known basis of $\mathfrak{su}(2)$ is the set $i\sigma_1, i\sigma_2, i\sigma_3$, where $\sigma_1, \sigma_2, \sigma_3$ are the three Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

An easy verification shows that $[\sigma_a, \sigma_a] = 0$ for each a , and

$$[\sigma_1, \sigma_2] = 2i\sigma_3, \quad [\sigma_2, \sigma_3] = 2i\sigma_1, \quad [\sigma_3, \sigma_1] = 2i\sigma_2.$$

Thus, in this example, $f_{ab}^c = -2 \operatorname{sign}(\pi)$, where $\pi \in S_3$ is the permutation given by $\pi(1) = a$, $\pi(2) = b$, $\pi(3) = c$.

It is clear that f_{ab}^c is antisymmetric in a, b ; that is, $f_{ba}^c = -f_{ab}^c$. There is a special situation in which f_{ab}^c also changes sign upon exchanging c and a , or c and b . If this holds, then we say that the structure constants are totally antisymmetric. A sufficient condition for total antisymmetry of the structure constants is given by the following lemma.

Lemma C.3.7. *Suppose that G is a compact Lie subgroup of $U(n)$ for some n . Let $\mathfrak{g}$ be its Lie algebra, so that the elements of $\mathfrak{g}$ are skew-Hermitian matrices of order n . Define an inner product on $\mathfrak{g}$ as*

$$\langle X, Y \rangle := \operatorname{Tr}(XY^*).$$

Let $T_1, \dots, T_r$ be an orthonormal basis of $\mathfrak{g}$ with respect to this inner product. Then the structure constants for this basis are totally antisymmetric.

Proof. It is easy to see that the inner product defined above is indeed a positive definite inner product on $\mathfrak{g}$. Since $T_1, \dots, T_r$ is an orthonormal basis and the elements of $\mathfrak{g}$ are skew-Hermitian, we have

$$f_{ab}^c = \langle [T_a, T_b], T_c \rangle = \operatorname{Tr}([T_a, T_b]T_c^*) = -\operatorname{Tr}([T_a, T_b]T_c).$$

Thus,

$$\begin{aligned}
 f_{ab}^c &= -\operatorname{Tr}(T_a T_b T_c) + \operatorname{Tr}(T_b T_a T_c) \\
 &= -\operatorname{Tr}(T_b T_c T_a) + \operatorname{Tr}(T_c T_b T_a) \\
 &= -\operatorname{Tr}([T_b, T_c] T_a) = \langle [T_b, T_c] T_a \rangle = f_{bc}^a = -f_{cb}^a.
 \end{aligned}$$

Thus, f_{ab}^c changes sign upon exchanging c and a . Consequently, we also have

$$f_{ac}^b = -f_{ca}^b = f_{ba}^c = -f_{ab}^c.$$

We conclude that the structure constants are totally antisymmetric. $\square$

C.3.5 Structure constants of simple Lie algebras

Let $\mathfrak{g}$ be the Lie algebra of a Lie group G . An ideal of $\mathfrak{g}$ is a subspace $\mathfrak{I}$ such that for any $x \in \mathfrak{I}$ and $y \in \mathfrak{g}$, $[x, y]$ is an element of $\mathfrak{I}$. A Lie algebra is called ‘simple’ if it is non-Abelian and the only ideals of $\mathfrak{g}$ are $\{0\}$ and $\mathfrak{g}$. For example, it is well known that $\mathfrak{su}(n)$ is simple for every $n \geq 2$. The following lemma is a basic property of simple Lie algebras, which says that the adjoint representation of a simple Lie algebra is irreducible.

Lemma C.3.8. *Let $\mathfrak{g}$ be a simple Lie algebra. If a subspace $\mathfrak{h}$ of $\mathfrak{g}$ is mapped into itself by ad_X for every $X \in \mathfrak{g}$, then $\mathfrak{h}$ is either $\{0\}$ or $\mathfrak{g}$.*

Proof. Take any $Y \in \mathfrak{h}$ and $X \in \mathfrak{g}$. Then $[X, Y] = \operatorname{ad}_X(Y) \in \mathfrak{h}$. Thus, $\mathfrak{h}$ is an ideal. This proves the claim. $\square$

Let us now specialize to the following setting. Let $\mathfrak{g}$ be a simple matrix Lie algebra contained in $U(n)$ for some n . Let f_{bc}^a be its structure constants in a basis $T_1, \dots, T_m$ that is orthonormal with respect to the inner product $\langle X, Y \rangle = \operatorname{Tr}(XY^*)$. Then note that the dual space $\mathfrak{g}^*$ can be identified with $\mathfrak{g}$ via the map $X \mapsto \langle X, \cdot \rangle$, and each element of $\mathfrak{g}$ can be viewed as an element of $\mathbb{R}^m$ by identifying it with its coefficients in the above basis. Under this correspondence, the Killing for B is a symmetric $m \times m$ real matrix, and for each $X \in \mathfrak{g}$, ad_X is an $m \times m$ real matrix. We claim that ad_X^* is the transpose of ad_X . To see this, take any $\phi \in \mathfrak{g}^*$ and $X, Y \in \mathfrak{g}$. Let $u, v, w \in \mathbb{R}^m$ be the representatives of ϕ, X, Y , respectively, and let M be the matrix representing ad_X . Then

$$\operatorname{ad}_X^*(\phi)(Y) = \phi(\operatorname{ad}_X(Y)) = u^T M w,$$

where u^T denotes the transpose of the column vector u . This shows that $\operatorname{ad}_X^*(\phi)$ is represented by the vector $M^T u$. Thus, ad_X^* is represented by the matrix M^T . This observation lets us prove the following lemma.

Lemma C.3.9. *In the above setting, the matrix representing the Killing form is a constant multiple of the identity matrix.*

Proof. For simplicity, let B denote the $m \times m$ matrix that represents the Killing form, and for any X , let ad_X denote the $m \times m$ matrix that represents ad_X . Since B is a real symmetric matrix, it has a spectral decomposition $B = UDU^T$, where U is an orthogonal matrix and D is a diagonal matrix. Lemma C.3.6 implies that $B\text{ad}_X = -(\text{ad}_X)^T B$, which can be rewritten as

$$UDU^T \text{ad}_X = -(\text{ad}_X)^T UDU^T.$$

Multiplying both sides by U^T on the left and U on the right, we get

$$DV_X = -V_X^T D,$$

where $V_X := U^T \text{ad}_X U$. Let $V_X(i, j)$ be the $(i, j)^{\text{th}}$ entry of V_X , and D_i be the i^{th} diagonal entry of D . The above identity shows that for all i, j ,

$$D_i V_X(i, j) = D_j V_X(i, j).$$

Define an undirected graph on $\{1, \dots, m\}$ by putting an edge between two distinct vertices i and j whenever there exists some $X \in \mathfrak{g}$ such that $V_X(i, j) \neq 0$. The above identity implies that whenever i and j are connected by an edge, D_i must be equal to D_j . Thus, if this graph is connected, then we can conclude that the D_i 's are all equal, which implies that B is a constant multiple of the identity matrix. We claim that the graph is indeed connected. To see this, suppose that the graph is not connected. Then we can partition $\{1, \dots, m\}$ into a pair of nonempty disjoint subsets A and B , such that there is no edge from A to B . In other words, for any $i \in A$ and $j \in B$, $V_X(i, j) = 0$ for all $X \in \mathfrak{g}$. Since $V_X = U^T \text{ad}_X U$, this means that $u_i^T \text{ad}_X u_j = 0$ for each $i \in A$, $j \in B$, and $X \in \mathfrak{g}$, where $u_1, \dots, u_m$ are the columns of U . Let $\mathfrak{h}$ be the span of u_j , $j \in B$. Then the orthogonal complement $\mathfrak{h}^\perp$ is the span of u_i , $i \in A$. But by the above observation, ad_X maps $\mathfrak{h}$ into $(\mathfrak{h}^\perp)^\perp = \mathfrak{h}$. Thus, by Lemma C.3.8, $\mathfrak{h}$ is either $\{0\}$ or $\mathfrak{g}$. But this contradicts the hypothesis that A and B are both nonempty. This completes the proof. $\square$

The above lemma yields the following corollary, which is important in the study of Yang–Mills theories.

Corollary C.3.10. *In the above setting, we have*

$$\sum_{b,c=1}^m f_{bc}^a f_{bc}^{a'} = C \delta_{aa'}$$

for some constant C that depends only on $\mathfrak{g}$.

Proof. Note that

$$\begin{aligned} \operatorname{ad}_{T_a} \left(\sum_{b=1}^m x_b T_b \right) &= \sum_{b=1}^m x_b [T_a, T_b] \\ &= \sum_{b=1}^m x_b \left(\sum_{c=1}^m f_{ab}^c T_c \right) \\ &= \sum_{c=1}^m \left(\sum_{b=1}^m f_{ab}^c x_b \right) T_c. \end{aligned}$$

Thus, the matrix representation of ad_{T_a} is given by $(f_{ab}^c)_{c,b=1,\dots,m}$. Since the $(a, a')^{\text{th}}$ entry of B is given by $\operatorname{Tr}(\operatorname{ad}_{T_a} \operatorname{ad}_{T_{a'}})$, this shows that this entry is equal to

$$\sum_{c,b=1}^m f_{ab}^c f_{a'b}^c.$$

But by Lemma C.3.7, this is equal to

$$- \sum_{c,b=1}^m f_{bc}^a f_{bc}^{a'}.$$

By Lemma C.3.9, this completes the proof. $\square$

C.4 Differential forms

C.4.1 Differential forms on a smooth manifold

Recall the definition of smooth section of a vector bundle from Section C.2.2.

Definition C.4.1. *Let M be an n -dimensional smooth manifold. A differential 0-form on M is a smooth map from M into $\mathbb{R}$. For $1 \leq k \leq n$, a differential k -form ω is a smooth section of $\Lambda^k T^*M$. The set of all k -forms is denoted by $\Omega^k(M)$.*

Let ω be a k -form for some $1 \leq k \leq n$. Given a chart (U, φ) , we can represent ω on U as

$$\omega(x) = \sum_{1 \leq i_1 < \dots < i_k \leq n} \omega_{i_1, \dots, i_k}(x) dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

by a slight abuse of notation, because the right side is the second coordinate of $\omega(x)$, and we are ignoring the first coordinate because it is simply x . The maps $\omega_{i_1, \dots, i_k}$ are smooth functions from U into $\mathbb{R}$. (Note that $dx_1, \dots, dx_n$ on the right are also dependent on x , but we suppress that for simplicity of notation.) Thus, within a chart (U, φ) , a differential k -form can be viewed simply as a collection of smooth functions $(\omega_{i_1, \dots, i_k})_{1 \leq i_1 < \dots < i_k \leq n}$. But the abstract definition k -forms is necessary for defining them in a way that does not depend on the choice of charts.

We will also need to work with vector space-valued k forms, where $\omega_{i_1, \dots, i_k}(x)$ is an element of a vector space instead of being a real number. To make sense of this, let us recall the discussion in Section C.1.1, where we observed that an element of a tensor product space $U \otimes V$ can be viewed as a linear combination of basis elements of V with coefficients in U . In the same way, we define:

Definition C.4.2. *Let M be an n -dimensional smooth manifold, and V be a finite-dimensional vector space over $\mathbb{R}$. For $k \geq 0$, a V -valued k -form is a smooth section of the vector bundle $(M \times V) \otimes \Lambda^k T^*M$, where $M \times V$ is the trivial vector bundle with base space M and typical fiber V . The set of all V -valued k -forms is denoted by $\Omega^k(M, V)$.*

On a chart (U, φ) , an element of $\Omega^k(M, V)$ takes the form

$$A(x) = \sum_{1 \leq i_1 < \dots < i_k \leq n} A_{i_1, \dots, i_k}(x) \otimes (dx_{i_1} \wedge \dots \wedge dx_{i_k}),$$

for some smooth functions $A_{i_1, \dots, i_k} : U \rightarrow V$. For simplicity of notation, we usually omit the $\otimes$ symbol, and write

$$A = \sum_{1 \leq i_1 < \dots < i_k \leq n} A_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

Of particular importance is the case $V = \mathfrak{g}$, where $\mathfrak{g}$ is a Lie algebra. We will return to Lie algebra-valued differential forms later.

C.4.2 Wedge product and exterior derivative

As noted in Section C.4, a differential k -form ω on a smooth manifold M can be viewed as a map that sends each $x \in M$ to a vector $\omega(x) \in \Lambda_k T_x^*M$. In particular, $\omega(x)$ is an element of the exterior algebra $\Lambda(T_x^*M)$. As such, we can define the wedge product of a k -form ω_1 and an l -form ω_2 pointwise as

$$(\omega_1 \wedge \omega_2)(x) := \omega_1(x) \wedge \omega_2(x) \in \Lambda^{k+l} T_x^*M.$$

It turns out that the object $\omega_1 \wedge \omega_2$ defined above is indeed a smooth differential $(k+l)$ -form on M . Thus, the wedge product defined above is a valid operation on differential forms. An important observation is that due to the relation (C.1.5), the wedge products $\omega_1 \wedge \omega_2$ and $\omega_2 \wedge \omega_1$ are related by the identity

$$\omega_2 \wedge \omega_1 = (-1)^{kl} \omega_1 \wedge \omega_2. \quad (\text{C.4.1})$$

The exterior derivative df of a k -form f is a $(k+1)$ -form, defined as follows. Let n be the dimension of M . Axiomatically, the mapping $f \mapsto df$ should have the following properties:

1. (Definition for 0-forms.) If f is a 0-form, df should be the differential of f , defined in

local coordinates as

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i,$$

where $\frac{\partial f}{\partial x_i}$ is defined as in equation (C.2.2).

2. (Linearity.) If f and g are two k -forms, then $d(af + bg) = adf + bdf$ for any $a, b \in \mathbb{R}$.
3. (Leibniz rule.) If f is a k -form and g is an l -form, then $d(f \wedge g) = df \wedge g + (-1)^k f \wedge dg$.
4. (Poincaré lemma.) For any differential form f , $ddf = 0$.

It turns out that there is a unique definition of d that satisfies the above properties. Indeed, if f is a k -form for some $1 \leq k \leq n-1$, let us write f as

$$f = \sum_{1 \leq i_1 < \dots < i_k \leq n} f_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

Let (U, φ) be a chart of M . We can consider U itself to be a manifold, with atlas $\{(U, \varphi)\}$. Then the summand on the right side above is actually the wedge product of the 0-form $f_{i_1, \dots, i_k}$ and the 1-forms $dx_{i_1}, \dots, dx_{i_k}$ on this manifold. Thus, by the properties of d listed above, we get

$$\begin{aligned} df &= \sum_{1 \leq i_1 < \dots < i_k \leq n} d(f_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}) \\ &= \sum_{1 \leq i_1 < \dots < i_k \leq n} (df_{i_1, \dots, i_k}) \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k} \\ &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \left(\sum_{i=1}^n \frac{\partial f_{i_1, \dots, i_k}}{\partial x_i} dx_i \right) \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}. \end{aligned}$$

Take any $1 \leq i_1 < \dots < i_k \leq n$ and any $1 \leq i \leq n$. Note that if $i \in \{i_1, \dots, i_k\}$, then

$$dx_i \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k} = 0.$$

On the other hand, if $i_j < i < i_{j+1}$ for some j , then

$$dx_i \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k} = (-1)^j dx_{i_1} \wedge \dots \wedge dx_{i_j} \wedge dx_i \wedge dx_{i_{j+1}} \wedge \dots \wedge dx_{i_k}.$$

Combining these observations, we get that

$$df = \sum_{1 \leq i_1 < \dots < i_{k+1} \leq n} \left(\sum_{j=1}^{k+1} (-1)^{j-1} \frac{\partial f_{i_1, \dots, i_{j-1}, i_{j+1}, \dots, i_{k+1}}}{\partial x_{i_j}} \right) dx_{i_1} \wedge \dots \wedge dx_{i_{k+1}}. \quad (\text{C.4.2})$$

Equation (C.4.2) gives the formula for df in local coordinates. One can show that this is an intrinsic definition, that is, independent of the choice of coordinates.

As an example, let us explicitly write down the formula for df when f is the 1-form

$$f = \sum_{i=1}^n f_i dx_i.$$

in local coordinates. Then by the formula (C.4.2), we get

$$df = \sum_{1 \leq i < j \leq n} \left(\frac{\partial f_j}{\partial x_i} - \frac{\partial f_i}{\partial x_j} \right) dx_i \wedge dx_j.$$

The exterior derivative can be defined also for vector space-valued differential forms, as follows. Let V be a vector space over $\mathbb{R}$. Let $v_1, \dots, v_m$ be a basis of V . Then any V -valued k -form ω can be written as

$$\omega = \sum_{i=1}^m v_i \omega_i,$$

where $\omega_1, \dots, \omega_m$ are real-valued k -forms. We define

$$d\omega := \sum_{i=1}^m v_i d\omega_i.$$

To define wedge products of vector space-valued differential forms, we need a product on the vector space. That is, we need V to be an algebra over $\mathbb{R}$. Recall that the *rank* of such an algebra is its dimension as a vector space. Let V be an algebra over $\mathbb{R}$ of rank m . Let $v_1, \dots, v_m$ be a linear basis of V . Let ω be a V -valued k -form and η be a V -valued l -form. Let us write

$$\omega = \sum_{i=1}^m v_i \omega_i, \quad \eta = \sum_{i=1}^m v_i \eta_i,$$

where ω_i 's and η_i 's are real-valued differential forms. Then we define

$$\omega \wedge \eta := \sum_{i,j=1}^m v_i v_j \omega_i \wedge \eta_j.$$

An important example is the algebra $M(m, \mathbb{C})$ of all $m \times m$ complex matrices under ordinary matrix multiplication, which is an algebra of rank $2m^2$ over $\mathbb{R}$.

C.4.3 Wedge products and exterior derivatives of Lie algebra-valued differential forms

Recall the wedge product on matrix-valued differential forms defined in the previous section, where the product on $M(m, \mathbb{C})$ was taken to be the ordinary matrix product. One can alternatively take the Lie product on this vector space, which also makes it an algebra over $\mathbb{R}$. If G is a matrix Lie group contained in $GL(m, \mathbb{C})$, its Lie algebra $\mathfrak{g}$ is a Lie sub-algebra of $M(m, \mathbb{C})$. The Lie product gives a definition of the wedge product of $\mathfrak{g}$ -valued differential forms that is different than the one given by ordinary matrix multiplication. To distinguish

the two, we use the notation $[\omega \wedge \eta]$ or $[\omega, \eta]$ to denote the wedge product of two $\mathfrak{g}$ -valued differential forms when using the Lie product structure of $\mathfrak{g}$, and use the notation $\omega \wedge \eta$ when using ordinary matrix multiplication.

Lemma C.4.3. *Let $\mathfrak{g}$ be a matrix Lie algebra. If ω is a $\mathfrak{g}$ -valued k -form and η is a $\mathfrak{g}$ -valued l -form, then*

$$[\omega \wedge \eta] = [\omega, \eta] = \omega \wedge \eta - (-1)^{kl} \eta \wedge \omega.$$

Proof. Let us write in local coordinates

$$\begin{aligned} \omega &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \omega_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k} \\ \eta &= \sum_{1 \leq i_1 < \dots < i_l \leq n} \eta_{i_1, \dots, i_l} dx_{i_1} \wedge \dots \wedge dx_{i_l}, \end{aligned}$$

where $\omega_{i_1, \dots, i_k}$ and $\eta_{i_1, \dots, i_l}$ are smooth matrix-valued functions defined locally. Then

$$\begin{aligned} [\omega, \eta] &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{1 \leq j_1 < \dots < j_l \leq n} [\omega_{i_1, \dots, i_k}, \eta_{j_1, \dots, j_l}] dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_l} \\ &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{1 \leq j_1 < \dots < j_l \leq n} \omega_{i_1, \dots, i_k} \eta_{j_1, \dots, j_l} dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_l} \\ &\quad - \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{1 \leq j_1 < \dots < j_l \leq n} \eta_{j_1, \dots, j_l} \omega_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_l}. \end{aligned}$$

Now note that for any $1 \leq i_1 < \dots < i_k \leq n$ and $1 \leq j_1 < \dots < j_l \leq n$,

$$dx_{j_1} \wedge \dots \wedge dx_{j_l} \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k} = (-1)^{kl} dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_l}.$$

Plugging this into the previous expression gives the desired result. $\square$

When $\mathfrak{g}$ is a matrix Lie algebra, the fact that $[\omega \wedge \eta]$ and $\omega \wedge \eta$ are two different objects is admittedly quite confusing; but it is standard. An important point to observe is that while $[\omega \wedge \eta]$ is defined for an arbitrary Lie algebra, $\omega \wedge \eta$ is defined only for matrix Lie algebras. Moreover, $[\omega \wedge \eta]$ is a $\mathfrak{g}$ -valued differential form, but $\omega \wedge \eta$ may not be $\mathfrak{g}$ -valued. Lastly, note that when ω is a $\mathfrak{g}$ -valued 1-forms and $\mathfrak{g}$ is a matrix Lie algebra, $\omega \wedge \omega = \frac{1}{2}[\omega \wedge \omega]$. In particular, $\omega \wedge \omega$ is a $\mathfrak{g}$ -valued 2-form. This will be important later.

Let $\mathfrak{g}$ be a Lie algebra with generators T_a , $a = 1, \dots, r$. If ω is a $\mathfrak{g}$ -valued k -form on a manifold M , we can write ω as

$$\omega = \sum_{a=1}^r T_a \omega_a$$

for some real-valued k -forms ω_a , $a = 1, \dots, r$. (Recall that the summand is actually $T_a \otimes \omega_a$, but we omit the $\otimes$ symbol.) The exterior derivative of ω is defined as

$$d\omega := \sum_{a=1}^r T_a d\omega_a.$$

When $\mathfrak{g}$ is a matrix Lie algebra, it is not hard to verify that this is the same as the definition of the exterior derivative of a matrix-valued k -form given in Section C.4.2. It can be shown that the exterior derivative of Lie algebra-valued differential forms satisfies the Leibniz rule and the Poincaré lemma, just like real-valued differential forms.

Let ω be a $\mathfrak{g}$ -valued k -form and η be a $\mathfrak{g}$ -valued l -form. From the definition of $[\omega \wedge \eta]$ and the identity (C.4.1), it follows that

$$\begin{aligned} [\eta \wedge \omega] &= \sum_{a,b=1}^r [T_b, T_a](\eta_b \wedge \omega_a) \\ &= \sum_{a,b=1}^r (-[T_a, T_b])(-1)^{kl} \omega_a \wedge \eta_b \\ &= -(-1)^{kl} [\omega \wedge \eta]. \end{aligned} \tag{C.4.3}$$

Another identity that will be useful for us is the following.

Lemma C.4.4. *Let $\mathfrak{g}$ be a Lie algebra and M be a smooth manifold. Let ω be a $\mathfrak{g}$ -valued k -form on M , for some $1 \leq k \leq n$. Then $[\omega \wedge [\omega \wedge \omega]] = 0$.*

Proof. Let $\omega = \sum_{a=1}^r T_a \omega_a$ be the decomposition of ω in terms of a set of generators $T_1, \dots, T_m$ of $\mathfrak{g}$. Then by the associativity of the wedge product on exterior algebras, the relation (C.4.1), and the Jacobi identity for the Lie product, we get

$$\begin{aligned} [\omega \wedge [\omega \wedge \omega]] &= \sum_{a,b,c=1}^r [T_a, [T_b, T_c]](\omega_a \wedge (\omega_b \wedge \omega_c)) \\ &= - \sum_{a,b,c=1}^r [T_b, [T_c, T_a]](\omega_a \wedge (\omega_b \wedge \omega_c)) - \sum_{a,b,c=1}^r [T_c, [T_a, T_b]](\omega_a \wedge (\omega_b \wedge \omega_c)) \\ &= - \sum_{a,b,c=1}^r [T_b, [T_c, T_a]](\omega_b \wedge (\omega_c \wedge \omega_a)) - \sum_{a,b,c=1}^r [T_c, [T_a, T_b]](\omega_c \wedge (\omega_a \wedge \omega_b)) \\ &= -2[\omega \wedge [\omega \wedge \omega]]. \end{aligned}$$

This proves the claim. $\square$

C.4.4 Integrals of differential forms

Let M be an orientable, smooth n -dimensional manifold. Let (U, φ) be a chart of M . Let ω be an n -form on M which vanishes outside U . The *integral* of ω on M is defined as follows. Let us express ω in the coordinates defined by (U, φ) as

$$\omega(x) = a(x) dx_1 \wedge \cdots \wedge dx_n.$$

Then we define

$$\int_M \omega := \int_{\varphi(U)} a(\varphi^{-1}(x)) dx,$$

where the integral on the right is just an ordinary Riemann integral on $\mathbb{R}^n$. It is instructive to verify that this definition does not depend on the choice of the chart. Suppose that (V, ψ) is another chart such that ω vanishes outside V and the Jacobian J of $\varphi \circ \psi^{-1}$ is positive everywhere. A simple calculation shows that the basis vectors $dx_1, \dots, dx_n$ of T_x^*M in the coordinates specified by (U, φ) are related to the basis vector $dx'_1, \dots, dx'_n$ of T_x^*M in the coordinates specified by (V, ψ) through the relation

$$dx_i = \sum_{j=1}^n \frac{\partial(\varphi \circ \psi^{-1})_i}{\partial x_j} dx'_j. \quad (\text{C.4.4})$$

Thus,

$$\begin{aligned} \omega(x) &= a(x) dx_1 \wedge \cdots \wedge dx_n \\ &= a(x) \left(\sum_{j=1}^n \frac{\partial(\varphi \circ \psi^{-1})_1}{\partial x_j} dx'_j \right) \wedge \cdots \wedge \left(\sum_{j=1}^n \frac{\partial(\varphi \circ \psi^{-1})_n}{\partial x_j} dx'_j \right) \\ &= a(x) \sum_{1 \leq j_1, \dots, j_n \leq n} \left(\prod_{i=1}^n \frac{\partial(\varphi \circ \psi^{-1})_i}{\partial x_{j_i}} \right) dx'_{j_1} \wedge \cdots \wedge dx'_{j_n}. \end{aligned}$$

Now, note that $dx'_{j_1} \wedge \cdots \wedge dx'_{j_n} = \text{sign}(j) dx'_1 \wedge \cdots \wedge dx'_n$ if j is a permutation of $1, \dots, n$, and 0 otherwise. Thus,

$$\omega(x) = a(x) J(x) dx'_1 \wedge \cdots \wedge dx'_n. \quad (\text{C.4.5})$$

This shows that in the coordinates specified by (V, ψ) , ω is expressed as $b(x) dx'_1 \wedge \cdots \wedge dx'_n$, where

$$b(x) = a(x) J(x).$$

But by the change of variable formula for Riemann integrals and the positivity of J ,

$$\begin{aligned} \int_{\varphi(U \cap V)} a(\varphi^{-1}(x)) dx &= \int_{\psi(U \cap V)} a(\varphi^{-1}(\varphi \circ \psi^{-1}(x))) J(x) dx \\ &= \int_{\psi(U \cap V)} a(\psi^{-1}(x)) J(x) dx = \int_{\psi(U \cap V)} b(x) dx. \end{aligned}$$

This shows that $\int_M \omega$ does not depend on the choice of the chart, as long as ω vanishes outside the chart.

For a general n -form ω on an orientable manifold, $\int_M \omega$ is defined as follows. Suppose that it is possible to find a finite collection of charts $\{(U_i, \varphi_i)\}_{i=1, \dots, N}$ such that ω vanishes outside $U_1 \cup \cdots \cup U_N$. This is always possible if M is compact, but may be possible even if M is not compact — for example, for $M = \mathbb{R}^n$. Suppose further that one can find a finite collection of smooth maps $\rho_i : M \rightarrow [0, 1]$, $i = 1, \dots, N$, such that for each i , ρ_i is zero outside U_i , and for each x , $\sum_{i=1}^N \rho_i(x) = 1$. Such a collection is known as a partition of unity subordinate to the collection $\{(U_i, \varphi_i)\}_{i=1, \dots, N}$. Again, a partition of unity always

exists when M is compact, but may exist even for non-compact manifolds, such as $\mathbb{R}^n$. Given a partition of unity, we define

$$\int_M \omega := \sum_{i=1}^N \int_M \rho_i \omega,$$

where, for each x , $\rho_i \omega(x)$ is the scalar multiple of $\omega(x)$ by $\rho_i(x)$. It can be shown that the choice of the partition of unity or the finite atlas does not affect the outcome.

An important fact about integrals of differential forms is Stokes' theorem.

Theorem C.4.5 (Stokes' theorem for manifolds without boundary). *Let M be a smooth n dimensional manifold, and let ω be an $(n-1)$ -form on M . Suppose that ω vanishes outside $U_1 \cup \dots \cup U_N$, where $\{(U_i, \varphi_i)\}_{i=1, \dots, N}$ is a finite collection of charts that admits a partition of unity $\{\rho_i\}_{i=1, \dots, N}$. Suppose that for each i , $(\rho_i \omega)(x_m) \rightarrow 0$ for any sequence $\{x_m\}_{m \geq 1}$ that eventually exits any compact set contained in U_i . Further, assume that if*

$$\rho_i \omega = \sum_{j=1}^n f_j dx_1 \wedge \dots \wedge dx_{j-1} \wedge dx_{j+1} \wedge \dots \wedge dx_n \quad (\text{C.4.6})$$

in local coordinates on the chart (U_i, φ_i) , then for each j ,

$$\int_{\varphi_i(U_i)} \left| \frac{\partial}{\partial x_j} (f_j \circ \varphi_i^{-1})(x) \right| dx < \infty. \quad (\text{C.4.7})$$

Then we have that

$$\int_M d\omega = 0.$$

Proof. Let us prove the theorem when $N = 1$ and $\rho_1 \omega = \omega$. The general case is similar. Note that by the definition of exterior derivative,

$$d\omega = \left(\sum_{j=1}^n (-1)^{j-1} \frac{\partial f_j}{\partial x_j} \right) dx_1 \wedge \dots \wedge dx_n.$$

Thus, by the identity (C.2.2), we get

$$\begin{aligned} \int_M d\omega &= \int_{U_1} d\omega \\ &= \int_{\varphi_1(U_1)} \sum_{j=1}^n (-1)^{j-1} \frac{\partial}{\partial x_j} (f_j \circ \varphi_1^{-1})(x) dx \\ &= \sum_{j=1}^n (-1)^{j-1} \int_{\varphi_1(U_1)} \frac{\partial}{\partial x_j} (f_j \circ \varphi_1^{-1})(x) dx. \end{aligned}$$

Now take any j , and any $x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n \in \mathbb{R}$. Let

$$V := \{x \in \mathbb{R} : (x_1, \dots, x_{j-1}, x, x_{j+1}, \dots, x_n) \in \varphi_1(U_1)\}.$$

Then V is an open subset of $\mathbb{R}$, and hence, a finite or countable disjoint union of open intervals which are the connected components of V . By the fundamental theorem of calculus and the fact that $f_j \circ \varphi_1^{-1}$ vanishes at the boundaries of these intervals (which is an easy consequence of the assumption that $\varphi_1(x_m) \rightarrow 0$ for any sequence $\{x_m\}_{m \geq 1}$ that eventually exits any given compact set contained in U_1), we get that

$$\int_V \frac{\partial}{\partial x_j} (f_j \circ \varphi_1^{-1})(x_1, \dots, x_{j-1}, x, x_{j+1}, \dots, x_n) dx = 0.$$

Since this holds for any $x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n$, and by the assumption (C.4.7) we can apply Fubini's theorem to integrate coordinate by coordinate, we get that

$$\int_{\varphi_1(U_1)} \frac{\partial}{\partial x_j} (f_j \circ \varphi_1^{-1})(x) dx = 0.$$

This completes the proof. $\square$

A simple situation where the assumptions of Theorem C.4.5 are satisfied for any $(n-1)$ -form is when M is compact. In that case, one can always find a finite atlas $\{(U_i, \varphi_i)\}_{i=1, \dots, N}$, and a partition of unity $\rho_1, \dots, \rho_N$ such that each ρ_i is zero outside a compact set contained in U_i .

Another simple case is when $M = \mathbb{R}^n$ with the usual manifold structure. In this case, the following condition on ω ensures that it satisfies the assumptions of Theorem C.4.5. Suppose that ω has the form displayed on the right side of equation (C.4.6), in the usual local coordinates on $\mathbb{R}^n$. Then we need that the condition (C.4.7) is satisfied (with φ_i being the identity map), and that each f_j vanishes at infinity.

C.5 Principal bundles

C.5.1 Definition of a principal bundle

Principal bundles are special cases of fiber bundles where the typical fiber is a Lie group. The precise definition is as follows.

Definition C.5.1. *A smooth principal bundle (P, M, π, G) is a smooth fiber bundle where G is a Lie group, which has a smooth right action on P (that is, a smooth map from $P \times G$ into P which has the properties of a group action) that satisfies the following conditions:*

1. *The action preserves each fiber $\pi^{-1}(x)$, $x \in M$, meaning that $\pi(pg) = \pi(p)$ for all $p \in P$.*
2. *The action is free, meaning that if g is not the identity element, then $pg \neq p$ for all $p \in P$.*
3. *The action is transitive on each fiber, meaning that for any $p, q \in P$ such that $\pi(p) = \pi(q)$, there exists $g \in G$ such that $pg = q$.*

The simplest kinds of principal bundles are trivial bundles, where $P = M \times G$, and $\pi(x, g) = x$ for all $x \in M$ and $g \in G$. For trivial bundles, local trivializations are obtained by simply taking $U = M$ and φ to be the identity map on $M \times G$. The group action for a trivial bundle is also very simple: $(x, g)h = (x, gh)$ for any $x \in M$ and $g, h \in G$. Note that for the trivial bundle, the tangent space $T_p P$ at a point $p = (x, g) \in P$ is simply the product vector space $T_x M \times T_g G$. We will exclusively deal with trivial principal bundles in these lectures.

C.5.2 Connections on a trivial principal bundle

The basic objects of study in quantum gauge theories are principal connections on principal bundles. While the definition of a principal connection on a general principal bundle is somewhat complicated, it reduces to something quite simple for trivial principal bundles.

Definition C.5.2. *Let $P = M \times G$ be a trivial principal bundle, and let $\mathfrak{g}$ be the Lie algebra of G . A principal G -connection on P is a $\mathfrak{g}$ -valued differential 1-form on M .*

Let $P = M \times G$ be a trivial principal bundle, and let $\mathfrak{g}$ be the Lie algebra of G . Let $A \in \Omega^1(M, \mathfrak{g})$ be a principal G -connection on P . The curvature form of A is the $\mathfrak{g}$ -valued 2-form defined as

$$F := dA + \frac{1}{2}[A \wedge A],$$

where dA is the exterior derivative of A and $[A \wedge A]$ denotes the wedge product of A with itself, as defined in Section C.4.3. Note that an alternative notation for $[A \wedge A]$ is $[A, A]$. Note also that when $\mathfrak{g}$ is a matrix Lie algebra, then $\frac{1}{2}[A \wedge A] = A \wedge A$, where $A \wedge A$ is the wedge product of A with itself when we view A as a matrix-valued differential form. Thus, we may rewrite F as

$$F = dA + A \wedge A$$

when $\mathfrak{g}$ is a matrix Lie algebra.

As an illustration, let us now work out the explicit formula for the curvature form when G is a matrix Lie group. Suppose that G is a Lie subgroup of $\mathrm{GL}(m, \mathbb{C})$, so that $\mathfrak{g}$ is a subspace of $M(m, \mathbb{C})$. Let n be the dimension of M . In local coordinates, let us write

$$A = \sum_{i=1}^n A_i dx_i,$$

where $A_1, \dots, A_n$ are smooth $\mathfrak{g}$ -valued maps. Then recall from the discussion at the end of Section C.4.2 that

$$dA = \sum_{1 \leq i < j \leq n} \left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} \right) dx_i \wedge dx_j.$$

Next, note that

$$\begin{aligned}
 A \wedge A &= \left(\sum_{i=1}^n A_i dx_i \right) \wedge \left(\sum_{j=1}^n A_j dx_j \right) \\
 &= \sum_{i,j=1}^n A_i A_j dx_i \wedge dx_j \\
 &= \sum_{1 \leq i < j \leq n} [A_i, A_j] dx_i \wedge dx_j,
 \end{aligned}$$

where $[A_i, A_j] := A_i A_j - A_j A_i$ is the commutator of A_i and A_j . Combining, we get that

$$F = \sum_{1 \leq i < j \leq n} \left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} + [A_i, A_j] \right) dx_i \wedge dx_j. \quad (\text{C.5.1})$$

The curvature form is crucial to the definition of Yang–Mills theories.

C.6 Riemannian manifolds

C.6.1 Definition of a Riemannian manifold

Let M be a smooth manifold. Let S be the set of all triples (x, u, v) where $p \in M$ and $u, v \in T_x M$. We can give a manifold structure to S in the same way that we did for $\Lambda^k T^*M$ in Section C.2.3. Having done that, we define a Riemannian metric g on M to be a smooth map from S into $\mathbb{R}$ satisfying the property that for each $x \in M$, the map $g(x, \cdot, \cdot) : T_x M \times T_x M \rightarrow \mathbb{R}$ is a positive definite inner product on the vector space $T_x M$. A smooth manifold endowed with a Riemannian metric is called a Riemannian manifold.

Given a coordinate chart on M , a Riemannian metric g can be described by an $n \times n$ matrix $g(x) = (g_{ij}(x))_{1 \leq i, j \leq n}$ (called the *metric tensor*), where

$$g_{ij}(x) = g\left(x, \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}\right).$$

Note that although the inner product $g(x, \cdot, \cdot)$ is an intrinsically defined object, the matrix $g(x)$ depends on the choice of a coordinate chart.

Recall the definition of orientability from Section C.2. Let M be an orientable compact Riemannian manifold with a smooth Riemannian metric g and dimension n . The *volume form* on M is the differential n -form given in local coordinates by

$$\omega(x) := \sqrt{\det(g(x))} dx_1 \wedge \cdots \wedge dx_n. \quad (\text{C.6.1})$$

Using the calculation in (C.4.5), one can check that this is indeed a valid definition — that is, it remains unaffected by a change of coordinates.

Definition C.6.1. Let M be a compact, orientable Riemannian manifold with metric g . For $f \in C^\infty(M)$, the integral of f over M , denoted $\int_M f$, is defined to be the integral

$\int_M f\omega$, where ω is the volume form of M and $f\omega$ is the differential form defined as $f\omega(x) := f(x)\omega(x)$.

If $M = \mathbb{R}^n$, for instance, the tangent space at each point is a copy of $\mathbb{R}^n$. The standard inner product on this tangent space defines the standard Riemannian metric on $\mathbb{R}^n$. Integration with respect to the volume form of this metric is ordinary Riemann integration on $\mathbb{R}^n$.

C.6.2 Hodge duals of differential forms

Let (M, g) be an n -dimensional Riemannian manifold. Let η be a k -form on M , for some $0 \leq k \leq n$. The Hodge dual of η , denoted $*\eta$, is an $(n - k)$ -form defined as follows. For any $i_1, \dots, i_n \in \{1, \dots, n\}$, let $\epsilon_{i_1, \dots, i_n}$ denote the Levi-Civita symbol, defined as

$$\epsilon_{i_1, \dots, i_n} = \begin{cases} \text{sign}(\pi) & \text{if } i_1, \dots, i_n \text{ are distinct, and } \pi \text{ denote the permutation } \pi(j) := i_j, \\ 0 & \text{if } i_1, \dots, i_n \text{ are not distinct.} \end{cases}$$

Take any coordinate chart (U, φ) . Let $g(x) = (g_{ij}(x))_{1 \leq i, j \leq n}$ denote the metric tensor at a point $x \in U$, with respect to this coordinate chart. Let $(g^{ij}(x))_{1 \leq i, j \leq n}$ denote the matrix $g(x)^{-1}$, following standard convention. Suppose that η has the local representation

$$\begin{aligned} \eta &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \eta_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k} \\ &= \frac{1}{k!} \sum_{1 \leq i_1, \dots, i_k \leq n} \eta_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}, \end{aligned}$$

where in the second line, we extended the definition of $\eta_{i_1, \dots, i_k}$ by defining

$$\eta_{i_{\pi(1)}, \dots, i_{\pi(k)}} := \text{sign}(\pi) \eta_{i_1, \dots, i_k}$$

when $i_1 < \dots < i_k$ and $\pi \in S_k$, and defining $\eta_{i_1, \dots, i_k} := 0$ when $i_1, \dots, i_k$ are not distinct. Then $*\eta$ is locally given by

$$*\eta = \frac{1}{(n - k)!} \sum_{1 \leq j_{k+1}, \dots, j_n \leq n} (*\eta)_{j_{k+1}, \dots, j_n} dx_{j_{k+1}} \wedge \dots \wedge dx_{j_n},$$

where

$$(*\eta)_{j_{k+1}, \dots, j_n} := \frac{\sqrt{\det(g)}}{k!} \sum_{1 \leq i_1, \dots, i_k \leq n} \sum_{1 \leq j_1, \dots, j_k \leq n} \epsilon_{j_1, \dots, j_n} g^{i_1 j_1} \dots g^{i_k j_k} \eta_{i_1, \dots, i_k}. \quad (\text{C.6.2})$$

It can be shown that this definition of $*\eta$ is intrinsic, that is, it does not depend on the choice of the coordinate chart. When $k = 0$, the above notation simply means that $*\eta$ is the n -form $\eta\omega$, where ω is the volume form defined in equation (C.6.1). The mapping $\eta \mapsto *\eta$ is called the action of the Hodge star operator on η .

The Hodge dual of a matrix-valued differential form is defined in the same manner, through the formula (C.6.2).

We now state and prove two lemmas about the Hodge dual that will be useful for us.

Lemma C.6.2. *Let η be a k -form or a matrix-valued k -form on an n -dimensional Riemannian manifold (M, g) . Then $**\eta = (-1)^{k(n-k)}\eta$.*

Proof. Let $\gamma := *\eta$, so that γ is represented in local coordinates by the formula displayed in equation (C.6.2). Let $\theta := *\gamma$, and let its coordinate representation be

$$\theta = \sum_{1 \leq l_1 < \dots < l_k \leq n} \theta_{l_1, \dots, l_k} dx_{l_1} \wedge \dots \wedge dx_{l_k}.$$

Then again by equation (C.6.2), we see that for any $1 \leq l_1 < \dots < l_k \leq n$,

$$\begin{aligned} \theta_{l_1, \dots, l_k} &= \frac{\sqrt{\det(g)}}{(n-k)!} \sum_{1 \leq j_{k+1}, \dots, j_n \leq n} \sum_{1 \leq l_{k+1}, \dots, l_n \leq n} \epsilon_{l_{k+1}, \dots, l_n, l_1, \dots, l_k} g^{j_{k+1} l_{k+1}} \dots g^{j_n l_n} \gamma_{j_{k+1}, \dots, j_n} \\ &= \frac{(-1)^{k(n-k)} \det(g)}{(n-k)! k!} \sum_{1 \leq j_{k+1}, \dots, j_n \leq n} \sum_{1 \leq l_{k+1}, \dots, l_n \leq n} \sum_{1 \leq i_1, \dots, i_k \leq n} \sum_{1 \leq j_1, \dots, j_k \leq n} \epsilon_{l_1, \dots, l_n} \epsilon_{j_1, \dots, j_n} \\ &\quad \cdot g^{j_{k+1} l_{k+1}} \dots g^{j_n l_n} g^{i_1 j_1} \dots g^{i_k j_k} \eta_{i_1, \dots, i_k}. \end{aligned}$$

Now, suppose that $l_1, \dots, l_n$ are distinct, and $i_1, \dots, i_k$ are distinct. It suffices to work under these conditions, because otherwise the summand in the above sum is zero. Note that by the symmetric nature of g^{-1} ,

$$\begin{aligned} &\sum_{1 \leq j_1, \dots, j_n \leq n} \epsilon_{j_1, \dots, j_n} g^{j_{k+1} l_{k+1}} \dots g^{j_n l_n} g^{i_1 j_1} \dots g^{i_k j_k} \\ &= \sum_{1 \leq j_1, \dots, j_n \leq n} \epsilon_{j_1, \dots, j_n} g^{l_{k+1} j_{k+1}} \dots g^{l_n j_n} g^{i_1 j_1} \dots g^{i_k j_k}. \end{aligned}$$

Now, if $\{i_1, \dots, i_k\} \cap \{l_{k+1}, \dots, l_n\} \neq \emptyset$, then the above sum is the determinant of a matrix with repeated rows, and hence zero. If this is not the case, then $i_1, \dots, i_k$ is a permutation of $l_1, \dots, l_k$. Let us call this permutation π . Then the above sum equals $\text{sign}(\pi) \epsilon_{l_1, \dots, l_n} \det(g^{-1})$. But

$$\text{sign}(\pi) \eta_{i_1, \dots, i_k} = \eta_{l_1, \dots, l_k}.$$

Putting it all together, we get $\theta_{l_1, \dots, l_k} = (-1)^{k(n-k)} \eta_{l_1, \dots, l_k}$, which completes the proof. $\square$

Lemma C.6.3. *Let η and γ be two k -forms or matrix-valued k -forms on an n -dimensional Riemannian manifold (M, g) with volume form ω . Then*

$$\eta \wedge *\gamma = \langle \eta, \gamma \rangle \omega,$$

where

$$\langle \eta, \gamma \rangle := \sum_{1 \leq i_1 < \dots < i_k \leq n} \eta_{i_1, \dots, i_k} \gamma^{i_1, \dots, i_k}, \quad (\text{C.6.3})$$

with

$$\gamma^{i_1, \dots, i_k} := \sum_{1 \leq j_1, \dots, j_k \leq n} g^{i_1 j_1} \dots g^{i_k j_k} \gamma_{j_1, \dots, j_k}.$$

Proof. First, note that in local coordinates

$$\begin{aligned} \eta \wedge * \gamma &= \frac{1}{k!(n-k)!} \sum_{1 \leq i_1, \dots, i_k \leq n} \sum_{1 \leq j_{k+1}, \dots, j_n \leq n} \eta_{i_1, \dots, i_k} (*\gamma)_{j_{k+1}, \dots, j_n} \\ &\quad dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_{k+1}} \wedge \dots \wedge dx_{j_n} \\ &= \frac{\sqrt{\det(g)}}{(k!)^2(n-k)!} \sum_{1 \leq i_1, \dots, i_k \leq n} \sum_{1 \leq i'_1, \dots, i'_k \leq n} \sum_{1 \leq j_1, \dots, j_n \leq n} g^{i'_1 j_1} \dots g^{i'_k j_k} \eta_{i_1, \dots, i_k} \gamma_{i'_1, \dots, i'_k} \epsilon_{j_1, \dots, j_n} \\ &\quad dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_{k+1}} \wedge \dots \wedge dx_{j_n}. \end{aligned}$$

Now, observe that

$$dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_{k+1}} \wedge \dots \wedge dx_{j_n} = \epsilon_{i_1, \dots, i_k, j_{k+1}, \dots, j_n} dx_1 \wedge \dots \wedge dx_n.$$

Next, observe that if $\epsilon_{j_1, \dots, j_n}$ and $\epsilon_{i_1, \dots, i_k, j_{k+1}, \dots, j_n}$ are both nonzero, then $j_1, \dots, j_k$ must be a permutation of $i_1, \dots, i_k$. Moreover, in this case, the product of $\epsilon_{j_1, \dots, j_n}$ and $\epsilon_{i_1, \dots, i_k, j_{k+1}, \dots, j_n}$ is equal to the sign of this permutation. Combining these observations, and further observing that $g^{ij} = g^{ji}$, we get

$$\begin{aligned} \eta \wedge * \gamma &= \left(\sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{1 \leq i'_1 < \dots < i'_k \leq n} \sum_{\pi \in S_k} g^{i'_1 i_{\pi(1)}} \dots g^{i'_k i_{\pi(k)}} \text{sign}(\pi) \eta_{i_1, \dots, i_k} \gamma_{i'_1, \dots, i'_k} \right) \omega \\ &= \left(\frac{1}{k!} \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{\pi \in S_k} \text{sign}(\pi) \eta_{i_1, \dots, i_k} \gamma^{i_{\pi(1)}, \dots, i_{\pi(k)}} \right) \omega. \end{aligned}$$

But for any $\pi \in S_k$,

$$\begin{aligned} \gamma^{i_{\pi(1)}, \dots, i_{\pi(k)}} &= \sum_{1 \leq j_1, \dots, j_k \leq n} g^{i_{\pi(1)} j_1} \dots g^{i_{\pi(k)} j_k} \gamma_{j_1, \dots, j_k} \\ &= \sum_{1 \leq j_1, \dots, j_k \leq n} g^{i_1 j_{\pi^{-1}(1)}} \dots g^{i_k j_{\pi^{-1}(k)}} \gamma_{j_{\pi^{-1}(1)}, \dots, j_{\pi^{-1}(k)}} \text{sign}(\pi^{-1}) \\ &= \text{sign}(\pi^{-1}) \gamma^{i_1, \dots, i_k}. \end{aligned}$$

Plugging this into the previous display completes the proof. $\square$

C.7 Haar measure on matrix Lie groups

Let G be a compact matrix Lie group, contained in $U(n)$ for some n . The space $M(n, \mathbb{C})$ of all $n \times n$ complex matrices has a natural inner product, given by

$$\langle A, B \rangle := \text{Tr}(AB^*), \quad (\text{C.7.1})$$

where B^* is the Hermitian conjugate of B . This inner product induces a Riemannian metric ρ on G , as follows. Recall from the discussion in Section C.3 that a tangent vector at $g \in G$ can be uniquely represented by a matrix $X \in M(n, \mathbb{C})$, where $X = \gamma'(0)$ for a curve $\gamma : (-\epsilon, \epsilon) \rightarrow G$ representing the tangent vector. For two tangent vectors at g represented by matrices X and Y , we define $\rho(X, Y) := \langle X, Y \rangle$.

Integration with respect to the volume form induced by the metric g , as defined in Definition C.6.1, induces a measure on G that is called the (unnormalized) ‘Haar measure’. It is not difficult to show that this measure has finite total mass. Upon dividing by the total mass, we get a probability measure on G , called the ‘normalized Haar measure’. Let μ denote the unnormalized Haar measure.

The key property of the Haar measure is that it is invariant under left- and right-actions of G on itself, as shown by the following lemma.

Lemma C.7.1. *For any smooth $f : G \rightarrow \mathbb{C}$ and any $h \in G$,*

$$\int f(hg) d\mu(g) = \int f(gh) d\mu(g) = \int f(g) d\mu(g).$$

Proof. To see this, suppose that f is supported on a coordinate chart (U, φ) , where φ maps U homeomorphically onto an open subset of $\mathbb{R}^m$, with m being the dimension of G . At a point $g \in U$, the tangent vector $\frac{\partial}{\partial x_i}$ is given by

$$\frac{\partial}{\partial x_i} = \partial_i \varphi^{-1}(\varphi(g)),$$

where $\partial_i \varphi^{-1}$ is the partial derivative of the map $\varphi^{-1} : \varphi(U) \rightarrow G$ in coordinate i . Note that the right side is an $n \times n$ matrix. Thus,

$$\begin{aligned} \rho\left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}\right) &= \langle \partial_i \varphi^{-1}(\varphi(g)), \partial_j \varphi^{-1}(\varphi(g)) \rangle \\ &= \text{Tr}(\partial_i \varphi^{-1}(\varphi(g)) \partial_j \varphi^{-1}(\varphi(g))^*). \end{aligned}$$

For $x \in \varphi(U)$, let $\rho(x)$ denote the $m \times m$ matrix whose $(i, j)^{\text{th}}$ element is

$$\rho_{ij}(x) := \text{Tr}(\partial_i \varphi^{-1}(x) \partial_j \varphi^{-1}(x)^*).$$

Then by the above calculation, the volume element of our Riemannian metric, in the local

coordinate system of the chart (U, φ) , is given by

$$\sqrt{\det \rho(x)} dx_1 \wedge \cdots \wedge dx_m.$$

Thus,

$$\int f(g) d\mu(g) = \int_{\varphi(U)} f(\varphi^{-1}(x)) \sqrt{\det \rho(x)} dx. \quad (\text{C.7.2})$$

Let $V := h^{-1}U = \{h^{-1}g : g \in U\}$, and define $\psi : V \rightarrow \mathbb{R}^m$ and $\psi(g) := \varphi(hg)$. Then (V, ψ) is a local coordinate chart of G , and the smooth function $f_1 : G \rightarrow \mathbb{R}$ given by $f_1(g) := f(hg)$ is supported on V . The analogue of ρ on this chart is given by τ , where

$$\tau_{ij}(x) := \text{Tr}(\partial_i \psi^{-1}(x) \partial_j \psi^{-1}(x)^*).$$

Then, as before,

$$\begin{aligned} \int f_1(g) d\mu(g) &= \int_{\psi(V)} f_1(\psi^{-1}(x)) \sqrt{\det \tau(x)} dx \\ &= \int_{\psi(V)} f(h\psi^{-1}(x)) \sqrt{\det \tau(x)} dx. \end{aligned}$$

But note that

$$\psi(V) = \varphi(hV) = \varphi(hh^{-1}U) = \varphi(U),$$

and for $x \in \varphi(U)$,

$$\psi^{-1}(x) = h^{-1}\varphi^{-1}(x).$$

Consequently, $\partial_i \psi^{-1} = h^{-1} \partial_i \varphi^{-1}$, and hence,

$$\begin{aligned} \tau_{ij}(x) &= \text{Tr}(h^{-1} \partial_i \psi^{-1}(x) \partial_j \psi^{-1}(x)^* (h^{-1})^*) \\ &= \text{Tr}(\partial_i \psi^{-1}(x) \partial_j \psi^{-1}(x)^*) = \rho_{ij}(x), \end{aligned}$$

where the second equality holds because $G \subseteq \text{U}(n)$. Combining the above displays, we get

$$\begin{aligned} \int f(hg) d\mu(g) &= \int_V f_1(g) d\mu(g) \\ &= \int_{\psi(V)} f_1(\psi^{-1}(x)) \sqrt{\det \tau(x)} dx \\ &= \int_{\varphi(U)} f(\varphi^{-1}(x)) \sqrt{\det \rho(x)} dx = \int f(g) d\mu(g). \end{aligned}$$

This proves the left-invariance of μ for functions supported on a single coordinate chart. It is easy to extend this to all smooth functions. Right-invariance follows similarly. $\square$

We will now discuss a specific choice of chart. Take any $g \in G$. Recall from the discussion in Section C.3 that since G is a matrix Lie group, the map $X \mapsto e^X$, where e^X

is simply the matrix exponential defined via series expansion, maps from $\mathfrak{g}$ into G . Let $\|X\|$ denote the Euclidean norm of X . The following lemma shows that $X \mapsto e^X$ is a homeomorphism when restricted to the set of all $X \in \mathfrak{g}$ with $\|X\|$ small enough.

Lemma C.7.2. *For any $X, Y \in \mathfrak{g}$ such that $\|X\|, \|Y\| \leq \frac{1}{2}$, we have*

$$(2 - e^{1/2})\|X - Y\| \leq \|e^X - e^Y\| \leq e^{1/2}\|X - Y\|.$$

Proof. The Cauchy-Schwarz inequality shows that the Euclidean norm on $n \times n$ matrices is a submultiplicative, that is, $\|XY\| \leq \|X\|\|Y\|$. Thus, if $\|X\|, \|Y\| \leq \frac{1}{2}$, then

$$\begin{aligned} \|e^X - e^Y\| &\leq \sum_{k=0}^{\infty} \frac{1}{k!} \|X^k - Y^k\| \\ &\leq \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{j=0}^{k-1} \|X^j(X - Y)Y^{k-j-1}\| \\ &\leq \sum_{k=0}^{\infty} \frac{1}{k!} k 2^{-(k-1)} \|X - Y\| = e^{1/2} \|X - Y\|. \end{aligned}$$

This proves the upper bound. For the lower bound, observe that

$$\begin{aligned} \|e^X - e^Y\| &= \left\| X - Y + \sum_{k=2}^{\infty} \frac{1}{k!} (X^k - Y^k) \right\| \\ &\geq \|X - Y\| - \left\| \sum_{k=2}^{\infty} \frac{1}{k!} (X^k - Y^k) \right\| \\ &\geq \|X - Y\| - \sum_{k=2}^{\infty} \frac{1}{k!} \|X^k - Y^k\| \\ &\geq \|X - Y\| - \sum_{k=2}^{\infty} \frac{1}{k!} k 2^{-(k-1)} \|X - Y\| = (2 - e^{1/2}) \|X - Y\|. \end{aligned}$$

This completes the proof. $\square$

Let $T_1, \dots, T_m$ be a set of generators of $\mathfrak{g}$ that are orthonormal with respect to the inner product displayed in equation (C.7.1). Define

$$\xi(x_1, \dots, x_m) := \exp\left(\sum_{a=1}^m x_a T_a\right)$$

for $(x_1, \dots, x_m) \in \mathbb{R}^m$, so that ξ is an isomorphism of the inner product spaces $\mathbb{R}^m$ and $\mathfrak{g}$.

Lemma C.7.3. *There is local coordinate chart (V, ψ) of G containing I , such that $\psi^{-1} = \xi$ on $\psi(V)$ and $\psi(V)$ is a bounded open neighborhood of zero in $\mathfrak{g}$.*

Proof. Take any chart (U, φ) of G that contains the identity. Since U is open and $I \in U$, U contains an open ball (with respect to the Euclidean metric on G) of some radius $r > 0$

centered at I . Without loss, let us assume that $r < \frac{1}{2}$. If B denotes the open ball of radius $e^{-1/2}r$ centered at the origin in $\mathbb{R}^m$, then by Lemma C.7.2, the restriction of ξ to B is an injective continuous map from B into U . Consequently, $\varphi \circ \xi$ is an injective continuous map from B into $\varphi(U) \subseteq \mathbb{R}^m$. By Brouwer's invariance of domain theorem, this implies that $\varphi \circ \xi$ is an open map on B . Consequently,

$$V := \xi(B) = \varphi^{-1}(\varphi \circ \xi(B))$$

is an open neighborhood of I . Again by Lemma C.7.2, $\psi := \xi^{-1}$ is a homeomorphism between V and B . Lastly, note that $V \subseteq U$ and $\varphi \circ \psi^{-1} = \varphi \circ \xi$ is a smooth map. Thus, (V, ψ) is a local coordinate chart. $\square$

Lemma C.7.3 leads to the following result.

Theorem C.7.4. *Let G be a compact matrix Lie group, contained in $U(n)$ for some n . Let $\mathfrak{g}$ denote the Lie algebra of G . Then there is a smooth function $J : \mathfrak{g} \rightarrow [0, \infty)$ such that $J(0) = 1$, and for any open neighborhood V of the identity in G with sufficiently small diameter, there is a bounded open neighborhood U of zero in $\mathfrak{g}$ such that*

$$\int_V f(g) d\mu(g) = \int_U f(e^X) J(X) dX$$

for any smooth function $f : G \rightarrow \mathbb{C}$, where μ is the unnormalized Haar measure on G and dX denotes integration with respect to Lebesgue measure on $\mathfrak{g}$ (upon identifying $\mathfrak{g}$ with $\mathbb{R}^m$ using an orthonormal basis, where m is the dimension of G). Moreover, the diameter of U goes to zero as the diameter of V goes to zero.

Proof. Let (V, ψ) be the chart from Lemma C.7.3. Recall the function ρ from the proof of Lemma C.7.1, with (V, ψ) in place of the chart (U, φ) . Then

$$\rho_{ij}(x) = \text{Tr}(\partial_i \xi(x) \partial_j \xi(x)^*).$$

From the explicit form of ξ , it is clear that ρ is a smooth function, and by the identity (C.7.2), we have that

$$\int f(g) d\mu(g) = \int_{\psi(V)} f(\xi(x)) \sqrt{\det \rho(x)} dx$$

for any f that vanishes outside V . Lastly, a simple calculation shows that

$$\partial_i \xi(0) = T_i$$

for $i = 1, \dots, m$. By our choice of $T_1, \dots, T_m$, this implies that $\rho(0)$ is the identity matrix. The claim follows by combining all of the above observations. $\square$

C.8 Discrete differential geometry

In this section, we develop the tools of discrete differential geometry that are needed for analyzing Abelian lattice gauge theories via duality transformations.

C.8.1 The cell complex of $\mathbb{Z}^n$

Take any $n \geq 1$ and any $x \in \mathbb{Z}^n$. There are n edges coming out of x in the positive direction. Let us denote these edges by $dx_1, \dots, dx_n$. For each $1 \leq k \leq n$ and $1 \leq i_1 < i_2 < \dots < i_k \leq n$, the edges $dx_{i_1}, \dots, dx_{i_k}$ define a positively oriented k -cell of $\mathbb{Z}^n$. (For example, a plaquette is a 2-cell.) We will denote this k -cell by the wedge product

$$dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k}.$$

A 0-cell is simply a vertex. We will use $-dx_{i_1} \wedge \dots \wedge dx_{i_k}$ to denote the negatively oriented version of $dx_{i_1} \wedge \dots \wedge dx_{i_k}$. The collection of all oriented k -cells, as k ranges from 0 to n , is called the cell complex of $\mathbb{Z}^n$.

Now take any arbitrary $1 \leq i_1, \dots, i_k \leq n$. Following the usual conventions for wedge products, $dx_{i_1} \wedge \dots \wedge dx_{i_k}$ is defined to be equal to zero if $i_1, \dots, i_k$ are not all distinct, and to be equal to $(-1)^m dx_{j_1} \wedge \dots \wedge dx_{j_k}$ otherwise, where $j_1, \dots, j_k$ is the increasing rearrangement of $i_1, \dots, i_k$ and m is the sign of the permutation that takes $i_1, \dots, i_k$ to $j_1, \dots, j_k$. For example, $dx_1 \wedge dx_1 = 0$ and $dx_2 \wedge dx_1 = -dx_1 \wedge dx_2$.

C.8.2 Differential forms on $\mathbb{Z}^n$

Let R be a ring. An R -valued k -form on $\mathbb{Z}^n$ is a R -valued function on the set of positively oriented k -cells, which is zero everywhere except at a finite number of k -cells. An R -valued k -form f will be denoted by the formal expression

$$f(x) = \sum_{1 \leq i_1 < \dots < i_k \leq n} f_{i_1 \dots i_k}(x) dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

where $f_{i_1 \dots i_k}(x)$ is the value assigned to the positively oriented k -cell $dx_{i_1} \wedge \dots \wedge dx_{i_k}$. We will use the convention that $f_{j_1 \dots j_k} = 0$ if $j_1, \dots, j_k$ are not distinct, and for distinct $j_1, \dots, j_k$, $f_{j_1 \dots j_k}$ denotes $s f_{i_1 \dots i_k}$, where $i_1, \dots, i_k$ is the increasing rearrangement of $j_1, \dots, j_k$ and s is the sign of the permutation that takes $(j_1, \dots, j_k)$ to $(i_1, \dots, i_k)$.

Note that a 0-form is simply a R -valued function on $\mathbb{Z}^n$. For $k > n$ or $k < 0$, there is only one k -form, which is denoted by 0. The *support* of a k -form f , denoted by $\text{supp } f$, is the set of $x \in \mathbb{Z}^n$ such that $f(x) \neq 0$.

If f is an R -valued k -form and c is the k -cell $dx_{i_1} \wedge \dots \wedge dx_{i_k}$, we will sometimes use the notation $f(c)$ to denote $f_{i_1 \dots i_k}(x)$.

C.8.3 Discrete exterior derivative

For any function $h : \mathbb{Z}^n \rightarrow R$, any $x \in \mathbb{Z}^n$ and any $1 \leq i \leq n$, define the discrete derivative of h in direction i at the point x as

$$\partial_i h(x) := h(x + e_i) - h(x),$$

where e_i is the vector that has 1 in coordinate i and 0 in all other coordinates. For $0 \leq k \leq n-1$, the exterior derivative of a k -form f is the $(k+1)$ -form

$$df(x) := \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{1 \leq i \leq n} \partial_i f_{i_1 \dots i_k}(x) dx_i \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}. \quad (\text{C.8.1})$$

In other words, if $g = df$, then for any $1 \leq i_1 < \dots < i_{k+1} \leq n$,

$$g_{i_1 \dots i_{k+1}}(x) = \sum_{1 \leq j \leq k+1} (-1)^{j-1} \partial_{i_j} f_{i_1 \dots \widehat{i_j} \dots i_{k+1}}(x), \quad (\text{C.8.2})$$

where $i_1 \dots \widehat{i_j} \dots i_{k+1}$ is the list obtained by omitting i_j from $i_1 i_2 \dots i_{k+1}$. For example, if $n = 4$ and

$$f(x) = f_{12}(x) dx_1 \wedge dx_2 + f_{13}(x) dx_1 \wedge dx_3, \quad (\text{C.8.3})$$

then

$$\begin{aligned} df(x) &= (\partial_3 f_{12}(x) - \partial_2 f_{13}(x)) dx_1 \wedge dx_2 \wedge dx_3 \\ &\quad + \partial_4 f_{12}(x) dx_1 \wedge dx_2 \wedge dx_4 + \partial_4 f_{13}(x) dx_1 \wedge dx_3 \wedge dx_4. \end{aligned}$$

The exterior derivative of any n -form is zero, consistent with our convention that 0 is the only k -form when $k > n$. A differential form f is called *closed* if $df = 0$, and *exact* if $f = dg$ for some g .

Lemma C.8.1. *For any R -valued differential form f on $\mathbb{Z}^n$, $ddf = 0$.*

Proof. If f is a k -form for $k \geq n-1$, then $ddf = 0$ automatically, so there is nothing to prove. Assume that $k < n-1$. Since the exterior derivative is a linear operator, the formula (C.8.1) implies that

$$ddf(x) := \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{1 \leq i, j \leq n} \partial_j \partial_i f_{i_1 \dots i_k}(x) dx_j \wedge dx_i \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

Inside the sum, if $i = j$, the term is zero since $dx_i \wedge dx_i = 0$. If $i \neq j$, the (i, j) -term cancels

the (j, i) -term, since $dx_j \wedge dx_i = -dx_i \wedge dx_j$ and for any h ,

$$\begin{aligned} \partial_j \partial_i h(x) &= \partial_i h(x + e_j) - \partial_i h(x) \\ &= (h(x + e_j + e_i) - h(x + e_j)) - (h(x + e_i) - h(x)) \\ &= (h(x + e_j + e_i) - h(x + e_i)) - (h(x + e_j) - h(x)) \\ &= \partial_i \partial_j h(x). \end{aligned}$$

Thus, all terms cancel, showing that $ddf = 0$. $\square$

C.8.4 Discrete Poincaré lemma

An important result about the discrete exterior derivative is the discrete Poincaré lemma, which says that every closed form is exact. In the following, a hypercube $B \subseteq \mathbb{Z}^n$ is a set of the form

$$([a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]) \cap \mathbb{Z}^n,$$

where the a_i 's and b_i 's are integers.

Lemma C.8.2 (Discrete Poincaré lemma). *Take any $1 \leq k \leq n - 1$. Let R be a ring. Then for any R -valued closed k -form f on $\mathbb{Z}^n$, there is an R -valued $(k - 1)$ -form g such that $dg = f$. Moreover, $\text{supp } g$ is contained in the smallest hypercube that contains $\text{supp } f$. The result also holds for $k = 0$, in the sense that $f = 0$ is the only 0-form that satisfies $df = 0$.*

Proof. First, let us do the case $k = 0$, since it is easy. Let f be a 0-form satisfying $df = 0$. This means that $f(x) = f(y)$ for any x, y that are neighbors on the lattice. Since f has finite support, it follows that $f = 0$ everywhere.

Next, let us do the case $1 \leq k \leq n - 1$. Let $a_1, \dots, a_n, b_1, \dots, b_n$ be integers such that $f(x) = 0$ whenever $x \notin [a_1, b_1] \times \cdots \times [a_n, b_n]$. For each $r \in \mathbb{Z}$, let

$$S_r := \mathbb{Z}^{n-1} \times \{r\}.$$

We define $g(x) = 0$ if $x \in S_r$ for some $r < a_n$. Then, starting from $r = a_n$, we define g inductively on S_r , given that it has been defined on S_{r-1} , as follows. For $1 \leq i_1 < \cdots < i_{k-1} \leq n$ and $x \in S_r$, let

$$g_{i_1 \dots i_{k-1}}(x) := g_{i_1 \dots i_{k-1}}(x - e_n) + (-1)^{k-1} f_{i_1 \dots i_{k-1} n}(x - e_n). \quad (\text{C.8.4})$$

Note that with this construction, $g = 0$ on S_r for all $r \leq a_n$, since $f = 0$ on S_r for all $r < a_n$. The construction also ensures that $g_{i_1 \dots i_{k-1}}(x) = 0$ for all x when $i_{k-1} = n$. Note that g may not have finite support, but we will not worry about that yet.

Let $h = dg$. We claim that $h = f$. Take any $x \in \mathbb{Z}$. Suppose that $x \in S_r$ for some $r < a_n$. Then $f(x) = 0$. On the other hand, $h(x)$ depends only on the values of g on S_r and S_{r+1} , which are all zero since $r < a_n$ and $g = 0$ on S_r for all $r \leq a_n$, as noted in the previous paragraph.

Next, take any $1 \leq i_1 < \dots < i_k \leq n$ and $x \in S_r$ for some $r \geq a_n$. We will prove by induction on r that

$$h_{i_1 \dots i_k}(x) = f_{i_1 \dots i_k}(x). \quad (\text{C.8.5})$$

First, suppose that $i_k = n$. Recall that

$$h_{i_1 \dots i_k}(x) = \sum_{1 \leq j \leq k} (-1)^{j-1} \partial_{i_j} g_{i_1 \dots \widehat{i_j} \dots i_k}(x).$$

Since $g_{i_1 \dots i_{k-1}}(x) = 0$ when $i_{k-1} = n$, we see that $\partial_{i_j} g_{i_1 \dots \widehat{i_j} \dots i_k}(x) = 0$ for $j < k$. Also, by equation (C.8.4),

$$\partial_n g_{i_1 \dots i_{k-1}}(x) = (-1)^{k-1} f_{i_1 \dots i_{k-1} n}(x).$$

Combining, we get

$$h_{i_1 \dots i_k}(x) = (-1)^{k-1} \partial_n g_{i_1 \dots i_{k-1}}(x) = f_{i_1 \dots i_{k-1} n}(x).$$

This completes the induction step when $i_k = n$.

Next, suppose that $i_k < n$. Let $y = x - e_n$, so that $y \in S_{r-1}$. Let $u = dh$ and $v = df$. Then by Lemma C.8.1, $u = 0$, and by assumption, $v = 0$. Thus,

$$0 = u_{i_1 \dots i_k n}(y) = (-1)^k \partial_n h_{i_1 \dots i_k}(y) + \sum_{1 \leq j \leq k} (-1)^{j-1} \partial_{i_j} h_{i_1 \dots \widehat{i_j} \dots i_k n}(y), \quad (\text{C.8.6})$$

and

$$0 = v_{i_1 \dots i_k n}(y) = (-1)^k \partial_n f_{i_1 \dots i_k}(y) + \sum_{1 \leq j \leq k} (-1)^{j-1} \partial_{i_j} f_{i_1 \dots \widehat{i_j} \dots i_k n}(y). \quad (\text{C.8.7})$$

Take any $1 \leq j \leq k$. Then

$$\begin{aligned} & \partial_{i_j} h_{i_1 \dots \widehat{i_j} \dots i_k n}(y) - \partial_{i_j} f_{i_1 \dots \widehat{i_j} \dots i_k n}(y) \\ &= (h_{i_1 \dots \widehat{i_j} \dots i_k n}(y + e_{i_j}) - h_{i_1 \dots \widehat{i_j} \dots i_k n}(y)) - (f_{i_1 \dots \widehat{i_j} \dots i_k n}(y + e_{i_j}) - f_{i_1 \dots \widehat{i_j} \dots i_k n}(y)). \end{aligned}$$

Since $i_k < n$, the point $z := y + e_{i_j}$ belongs to S_{r-1} , just like y . Therefore by the identity (C.8.5) for points in S_{r-1} , the above expression vanishes. Combining this with equations (C.8.6) and (C.8.7), we get

$$\partial_n f_{i_1 \dots i_k}(y) = \partial_n h_{i_1 \dots i_k}(y).$$

But by the induction hypothesis, $f_{i_1 \dots i_k} = h_{i_1 \dots i_k}$ on S_{r-1} . Since $y + e_n = x$, the above identity therefore reduces to equation (C.8.5) for points in S_r . This completes the proof of the claim that $h = f$.

It remains to show that g vanishes outside the hypercube $B := ([a_1, b_1] \times \dots \times [a_n, b_n]) \cap \mathbb{Z}^n$. Take any $x = (x_1, \dots, x_n) \in \mathbb{Z}^n$. Then the inductive construction (C.8.4) ensures that if $x_i \notin [a_i, b_i]$ for some $i < n$, then $g(x) = 0$. Also, we have that $g(x) = 0$ if $x_n \leq a_n$. Lastly,

if $g(x) = 0$ for all $x \in S_{b_n+1}$, then $g(x) = 0$ when $x_n > b_n + 1$, by equation (C.8.4) and the fact that f vanishes outside B . Thus, it suffices to show that $g(x) = 0$ if $x_n = b_n + 1$.

It turns out that this is true only if $k = 1$. For $k \geq 2$, we will have to replace g by a different $(k-1)$ -form g' , also satisfying $dg' = f$, which vanishes outside B . But first, let us complete the proof for $k = 1$. Take any point $x \in S_{b_n+1}$. Then $dg(x) = f(x) = 0$, which implies that

$$0 = \partial_i g(x) = g(x + e_i) - g(x)$$

for all $1 \leq i \leq n-1$. This shows that g is constant on S_{b_n+1} . But, as observed above, $g(x) = 0$ if $x_i \notin [a_i, b_i]$ for some $1 \leq i \leq n-1$. Thus, $g = 0$ on S_{b_n+1} . This completes the proof of the lemma when $k = 1$, for any $n \geq 2$. This also completes the proof of the lemma when $n = 2$, because k must be 1 if $n = 2$.

Next, take any $n \geq 3$ and suppose that the lemma has been proved for dimensions $\leq n-1$. Let g and f be as above, with $2 \leq k < n$. Define a $(k-1)$ -form q on $\mathbb{Z}^{n-1}$ as

$$q(y) := (-1)^k g(y, b_n + 1), \quad y \in \mathbb{Z}^{n-1}.$$

Since $dg = f = 0$ on S_{b_n+1} , it follows that $dq = 0$. Moreover, q vanishes outside $[a_1, b_1] \times \cdots \times [a_{n-1}, b_{n-1}]$. Thus, by the induction hypothesis, there is a $(k-2)$ -form p on $\mathbb{Z}^{n-1}$ such that $dp = q$, and p vanishes outside $[a_1, b_1] \times \cdots \times [a_{n-1}, b_{n-1}]$.

Define a $(k-1)$ -form w on $\mathbb{Z}^n$ as follows. Let $w(x) = g(x)$ if $x \in S_r$ for some $r \geq b_n + 1$, and $w(x) = 0$ if $x \in S_r$ for some $r < b_n$. If $x = (x_1, \dots, x_{n-1}, b_n) \in S_{b_n}$, we define

$$w_{i_1 \dots i_{k-2} n}(x) := p_{i_1 \dots i_{k-2}}(x_1, \dots, x_{n-1})$$

for $1 \leq i_1 < \cdots < i_{k-2} < n$, and $w_{i_1 \dots i_{k-1}}(x) = 0$ if $i_{k-1} < n$. We claim that $dw = 0$. To see this, let $s := dw$. First, note that if $x \in S_r$ for some $r \geq b_n + 1$, then $s(x) = dg(x) = 0$. Next, suppose that $x \in S_r$ for some $r < b_n$. Recall that for any $1 \leq i_1 < \cdots < i_k \leq n$,

$$s_{i_1 \dots i_k}(x) = \sum_{j=1}^n (-1)^{j-1} \partial_{i_j} w_{i_1 \dots \widehat{i_j} \dots i_k}(x)$$

If $i_k < n$, this is zero, since $x + e_{i_j} \in S_r$ for all j in this case, and $w = 0$ on S_r . For the same reason, if $i_k = n$, then

$$\begin{aligned} s_{i_1 \dots i_{k-1} n}(x) &= (-1)^{n-1} \partial_n w_{i_1 \dots i_{k-1}}(x) \\ &= (-1)^{n-1} (w_{i_1 \dots i_{k-1}}(x + e_n) - w_{i_1 \dots i_{k-1}}(x)). \end{aligned}$$

But $w_{i_1 \dots i_{k-1}}(x) = 0$ since $x \in S_r$, and $w_{i_1 \dots i_{k-1}}(x + e_n) = 0$ since $x + e_n \in S_{r+1}$, $r+1 \leq b_n$, and $i_{k-1} < n$. This shows that $s(x) = 0$ if $x \in S_r$ for some $r < b_n$. Lastly, take $x = (x_1, \dots, x_{n-1}, b_n) \in S_{b_n}$ and $1 \leq i_1 < \cdots < i_k \leq n$. If $i_k < n$, then $\partial_{i_j} w_{i_1 \dots \widehat{i_j} \dots i_k}(x) = 0$ for any j , which implies that $s_{i_1 \dots i_k}(x) = 0$. So, suppose that $i_k = n$. Let $x' := (x_1, \dots, x_{n-1})$.

Then for $1 \leq j \leq k-1$,

$$\begin{aligned}\partial_{i_j} w_{i_1 \dots \widehat{i_j} \dots i_{k-1} n}(x) &= w_{i_1 \dots \widehat{i_j} \dots i_{k-1} n}(x + e_{i_j}) - w_{i_1 \dots \widehat{i_j} \dots i_{k-1} n}(x) \\ &= p_{i_1 \dots \widehat{i_j} \dots i_{k-1}}(x' + e_{i_j}) - p_{i_1 \dots \widehat{i_j} \dots i_{k-1}}(x') \\ &= \partial_{i_j} p_{i_1 \dots \widehat{i_j} \dots i_{k-1}}(x'),\end{aligned}$$

and

$$\begin{aligned}\partial_n w_{i_1 \dots i_{k-1}}(x) &= w_{i_1 \dots i_{k-1}}(x + e_n) - w_{i_1 \dots i_{k-1}}(x) \\ &= g_{i_1 \dots i_{k-1}}(x + e_n).\end{aligned}$$

Thus, we get

$$\begin{aligned}s_{i_1 \dots i_{k-1} n}(x) &= \sum_{j=1}^{n-1} (-1)^{j-1} \partial_{i_j} p_{i_1 \dots \widehat{i_j} \dots i_{k-1}}(x') + (-1)^{k-1} g_{i_1 \dots i_{k-1}}(x + e_n) \\ &= q_{i_1 \dots i_{k-1}}(x') + (-1)^{k-1} g_{i_1 \dots i_{k-1}}(x + e_n) \\ &= (-1)^k g_{i_1 \dots i_{k-1}}(x + e_n) + (-1)^{k-1} g_{i_1 \dots i_{k-1}}(x + e_n) = 0.\end{aligned}$$

This completes the proof of the claim that $dw = 0$. Now define $g' := g - w$. Then $dg' = dg - dw = dg = f$. Also, note that g' vanishes outside the hypercube B , since $g(x) = w(x)$ for when $x \in S_r$ for some $r \geq b_n + 1$, $w(x) = 0$ when $x \in S_r$ for some $r < b_n$, and $w(x) = 0$ if $x \in S_{b_n}$ but $x_i \notin [a_i, b_i]$ for some $1 \leq i \leq n-1$. $\square$

C.8.5 Discrete coderivative

The exterior derivative operator d has an adjoint, denoted by δ and sometimes called the ‘codifferential operator’ or simply ‘coderivative’. Letting

$$\bar{\partial}_i h(x) := h(x) - h(x - e_i),$$

the operator δ is defined as

$$\delta f(x) := \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{1 \leq l \leq k} (-1)^l \bar{\partial}_{i_l} f_{i_1 \dots \widehat{i_l} \dots i_k}(x) dx_{i_1} \wedge \dots \wedge \widehat{dx_{i_l}} \wedge \dots \wedge dx_{i_k}. \quad (\text{C.8.8})$$

The operator δ takes k -forms to $(k-1)$ -forms. If $k = 0$, then $\delta f = 0$ by definition. As an example, suppose that $n = 4$ and f is the 2-form defined in equation (C.8.3). Then

$$\delta f(x) = (\bar{\partial}_2 f_{12}(x) + \bar{\partial}_3 f_{13}(x)) dx_1 - \bar{\partial}_1 f_{12}(x) dx_2 - \bar{\partial}_1 f_{13}(x) dx_3.$$

Note that on manifolds, ∂_i and $\bar{\partial}_i$ are the same. But in the discrete setting, the two operators are not the same, and we need to define δ using $\bar{\partial}_i$ as above.

The coderivative may be alternatively expressed as follows. Let $g := \delta f$. Then

equation (C.8.8) shows that for any $1 \leq i_1 < \cdots < i_{k-1} \leq n$,

$$g_{i_1 \cdots i_{k-1}}(x) = - \sum_{i=1}^n \bar{\partial}_i f_{ii_1 \cdots i_{k-1}}(x). \quad (\text{C.8.9})$$

C.8.6 Discrete Hodge dual

An important object for us is the dual of the lattice $\mathbb{Z}^n$, which we will denote by $*\mathbb{Z}^n$. This is just another copy of $\mathbb{Z}^n$, whose vertices are the centers of the n -cells of the primal lattice. This gives a correspondence between the n -cells of the primal lattice and the 0-cells of the dual lattice. This correspondence extends to a correspondence between the cell complex of the primal lattice and the cell complex of the dual lattice, where an oriented k -cell c of the primal lattice is paired with an oriented $(n-k)$ -cell $*c$ of the dual lattice. The cell $*c$ is called the Hodge dual of the cell c . The duality is defined as follows.

Take any x in the primal lattice. Let y denote the center of the n -cell $dx_1 \wedge \cdots \wedge dx_n$, so that $y = *(dx_1 \wedge \cdots \wedge dx_n)$ according to our convention. The vertex y is sometimes called the dual vertex of x . For any $\Lambda \subseteq \mathbb{Z}^n$, $*\Lambda$ denotes the set of duals of the vertices in Λ . Explicitly, if

$$\Lambda = \times_{i=1}^n \{a_i, a_i + 1, \dots, b_i - 1, b_i\},$$

then

$$*\Lambda = \times_{i=1}^n \left\{ a_i + \frac{1}{2}, a_i + \frac{3}{2}, \dots, b_i - \frac{1}{2}, b_i + \frac{1}{2} \right\}. \quad (\text{C.8.10})$$

Let $dy_1, \dots, dy_n$ be the edges coming out of y in the *negative* direction. Take any $1 \leq i_1 < \cdots < i_k \leq n$. Let $j_1, \dots, j_{n-k}$ be an enumeration of $\{1, \dots, n\} \setminus \{i_1, \dots, i_k\}$, and define

$$*(dx_{i_1} \wedge \cdots \wedge dx_{i_k}) := s dy_{j_1} \wedge \cdots \wedge dy_{j_{n-k}},$$

where s is the sign of the permutation $(i_1, \dots, i_k, j_1, \dots, j_{n-k})$. It is easy to see that the right side does not depend on our choice of $j_1, \dots, j_{n-k}$. By the same principle, define

$$*(dy_{j_1} \wedge \cdots \wedge dy_{j_{n-k}}) := (-1)^{k(n-k)} s dx_{i_1} \wedge \cdots \wedge dx_{i_k},$$

where $(-1)^{k(n-k)}$ is present because the sign of $(j_1, \dots, j_{n-k}, i_1, \dots, i_k)$ is $(-1)^{k(n-k)} s$. With x, y, s and $j_1, \dots, j_{n-k}$ as above, the Hodge dual of an R -valued k -form f is defined as the following $(n-k)$ -form on the dual lattice:

$$*f(y) := \sum_{1 \leq i_1 < \cdots < i_k \leq n} f_{i_1 \cdots i_k}(x) s dy_{j_1} \wedge \cdots \wedge dy_{j_{n-k}}.$$

(Note that s and $j_1, \dots, j_{n-k}$ depend on $i_1, \dots, i_k$ in the above sum, but this dependence has been suppressed for notational clarity.) For example, if $n = 4$ and f is the 2-form

defined in equation (C.8.3), then

$$*f(y) = f_{12}(x)dy_3 \wedge dy_4 - f_{13}(x)dy_2 \wedge dy_4.$$

One can similarly define the dual of a k -form on the dual lattice, which is an $(n - k)$ -form on the primal lattice. The prescription is exactly the same as above, except that now $x \in *\mathbb{Z}^n$ and y is the dual of x in $\mathbb{Z}^n$, defined to be the center of the n -cell $dx_1 \wedge \cdots \wedge dx_n$. It is easy to see that for a k -form f on the primal lattice,

$$**f = (-1)^{k(n-k)} f. \quad (\text{C.8.11})$$

The exterior derivative operator on the cell complex of the dual lattice is defined just the same way as on the primal lattice, but with ∂_i replaced by $\bar{\partial}_i$, since the edges in the dual lattice are oriented in the opposite direction. The exterior derivative on the dual lattice is also denoted by d . With this definition, the following lemma connects the Hodge star operator with the adjoint of the exterior derivative on the dual lattice. This is a discrete version of a well-known lemma about differential forms on manifolds.

Lemma C.8.3. *For any R -valued k -form f on $\mathbb{Z}^n$,*

$$\delta f(x) = (-1)^{n(k+1)+1} *d*f(y),$$

where y is center of the n -cell $dx_1 \wedge \cdots \wedge dx_n$.

Proof. Note that if f is a R -valued k -form on the primal lattice, and x, y, s and $j_1, \dots, j_{n-k}$ are as in the paragraphs preceding the statement of the lemma, then

$$d*f(y) = \sum_{1 \leq i_1 < \cdots < i_k \leq n} \sum_{1 \leq i \leq n} \bar{\partial}_i f_{i_1 \dots i_k}(x) s dy_i \wedge dy_{j_1} \wedge \cdots \wedge dy_{j_{n-k}}.$$

The summand on the right is zero unless i is one of $i_1, \dots, i_k$. If $i = i_l$ for some l , then

$$\begin{aligned} & *(dy_i \wedge dy_{j_1} \wedge \cdots \wedge dy_{j_{n-k}}) \\ &= (-1)^{k(n-k)+n-k+l-1} s dx_{i_1} \wedge \cdots \wedge \widehat{dx_{i_l}} \wedge \cdots \wedge dx_{i_k}, \end{aligned}$$

since the sign of $(i_l, j_1, \dots, j_{n-k}, i_1, \dots, \widehat{i_l}, \dots, i_k)$ equals

$$(-1)^{k(n-k)+n-k+l-1} s.$$

Thus, putting $r := k(n - k) + n - k - 1$, we get

$$\begin{aligned} & *d*f(y) \\ &= (-1)^r \sum_{1 \leq i_1 < \cdots < i_k \leq n} \sum_{1 \leq l \leq k} (-1)^l \bar{\partial}_{i_l} f_{i_1 \dots i_k}(x) dx_{i_1} \wedge \cdots \wedge \widehat{dx_{i_l}} \wedge \cdots \wedge dx_{i_k} \\ &= (-1)^r \delta f(x). \end{aligned}$$

Since $k(k+1)$ is even, $(-1)^r = (-1)^{n(k+1)+1}$. $\square$

C.8.7 Poincaré lemma for the coderivative

The following is a dual version of the Poincaré lemma. The proof is a combination of the Poincaré lemma and Lemma C.8.3.

Lemma C.8.4 (Poincaré lemma for the coderivative). *Take any $1 \leq k \leq n-1$. Let f be an R -valued k -form on $\mathbb{Z}^n$ with $\delta f = 0$. Then there is a $(k+1)$ -form h such that $f = \delta h$. Moreover, $\text{supp } h$ is contained in the smallest hypercube containing $\text{supp } f$. The result also holds for $k = n$, in the sense that $f = 0$ is the only n -form that satisfies $\delta f = 0$.*

Proof. Since $\delta f = 0$, Lemma C.8.3 implies that $d*f = 0$. Let B be the smallest hypercube that contains $\text{supp } f$. Let $*B$ be the hypercube in $*\mathbb{Z}^n$ consisting of the centers of all the k -cells in B . Then it is easy to check that $*B$ contains the support of $*f$. Thus, by Lemma C.8.2, there is an $(n-k-1)$ -form g on $*\mathbb{Z}^n$ such that $dg = *f$ and $\text{supp } g \subseteq *B$. Let

$$h := (-1)^{-(k-1)(n-k+1)-k(n-k)-nk-1} *g,$$

so that

$$*h = (-1)^{-k(n-k)-nk-1} g.$$

Thus, by Lemma C.8.3,

$$\delta h = (-1)^{nk+1} *d*h = (-1)^{-k(n-k)} *dg = f.$$

Lastly, note that by the definition of h , it follows that h vanishes outside B . $\square$

C.8.8 Inner product of differential forms

Let f and g be two R -valued k -forms on $\mathbb{Z}^n$, where R is a ring and $0 \leq k \leq n$. The inner product of f and g is defined as

$$(f, g) := \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq i_1 < \dots < i_k \leq n} f_{i_1 \dots i_k}(x) g_{i_1 \dots i_k}(x).$$

When $R = \mathbb{R}$, this inner product induces the norm

$$\|f\| := (f, f)^{1/2}.$$

The above inner product satisfies the following ‘summation by parts’ identity.

Lemma C.8.5. *Let R be a ring. For any R -valued $(k+1)$ -form f and k -form g on $\mathbb{Z}^n$, where $0 \leq k \leq n-1$, we have*

$$(f, dg) = (\delta f, g).$$

Proof. Let $h := \delta f$. Recalling the formula for the coderivative from equation (C.8.9), we see that

$$(h, g) = - \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{i=1}^n \bar{\partial}_i f_{ii_1 \dots i_k}(x) g_{i_1 \dots i_k}(x).$$

This gives

$$\begin{aligned} (h, g) &= - \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{i=1}^n (f_{ii_1 \dots i_k}(x) - f_{ii_1 \dots i_k}(x - e_i)) g_{i_1 \dots i_k}(x) \\ &= - \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{i=1}^n f_{ii_1 \dots i_k}(x) g_{i_1 \dots i_k}(x) \\ &\quad + \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{i=1}^n f_{ii_1 \dots i_k}(x - e_i) g_{i_1 \dots i_k}(x). \end{aligned}$$

But note that

$$\begin{aligned} &\sum_{x \in \mathbb{Z}^n} \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{i=1}^n f_{ii_1 \dots i_k}(x - e_i) g_{i_1 \dots i_k}(x) \\ &= \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{i=1}^n f_{ii_1 \dots i_k}(x) g_{i_1 \dots i_k}(x + e_i). \end{aligned}$$

Plugging this back into the previous display, we get

$$(h, g) = \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{i=1}^n f_{ii_1 \dots i_k}(x) \partial_i g_{i_1 \dots i_k}(x).$$

Take any $1 \leq j_1 < \dots < j_{k+1} \leq n$. Then $f_{j_1 \dots j_k}(x)$ occurs in the above sum as $(-1)^{l-1} f_{j_1 \dots \widehat{j_l} \dots j_{k+1}}(x)$ for $l = 1, \dots, k+1$. For each l , this comes multiplied by the factor $\partial_{j_l} g_{j_1 \dots \widehat{j_l} \dots j_{k+1}}(x)$. Thus, we get

$$\begin{aligned} (h, g) &= \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq j_1 < \dots < j_{k+1} \leq n} f_{j_1 \dots j_{k+1}}(x) \left(\sum_{l=1}^{k+1} (-1)^{l-1} \partial_{j_l} g_{j_1 \dots \widehat{j_l} \dots j_{k+1}}(x) \right) \\ &= \sum_{x \in \mathbb{Z}^n} \sum_{1 \leq j_1 < \dots < j_{k+1} \leq n} f_{j_1 \dots j_{k+1}}(x) u_{j_1 \dots j_{k+1}}(x), \end{aligned}$$

where $u := dg$. This completes the proof. $\square$

C.8.9 Poincaré lemma for 0-forms

Recall that $\delta f = 0$ for any 0-form f , by definition. But it is not hard to see that not all 0-forms can be expressed as δg for some 1-form g . Can we still have some version of the coderivative Poincaré lemma for 0-forms? The following result gives the necessary and sufficient condition.

Lemma C.8.6. *Let R be a ring. An R -valued 0-form f on $\mathbb{Z}^n$ satisfies $f = \delta g$ for some 1-form g if and only if $\sum_{x \in \mathbb{Z}^n} f(x) = 0$. Moreover, if this criterion is satisfied and $\text{supp } f$ is connected in the usual graph structure on $\mathbb{Z}^n$, then it can be arranged that $g(e) = 0$ unless both endpoints of e are in $\text{supp } f$.*

Proof. First, suppose that $f = \delta g$ for some 1-form g . Let h be the 0-form that is equal to 1 at any vertex that is an endpoint of some edge e with $g(e) \neq 0$, and 0 elsewhere. Note that the support of h contains the support of f . Thus, by Lemma C.8.5,

$$\sum_{x \in \mathbb{Z}^n} f(x) = (f, h) = (\delta g, h) = (g, dh).$$

But by the definition of h , $dh(e) = 0$ for any edge e such that $g(e) \neq 0$. Thus, the right side is zero. This proves one direction of the lemma.

For the other direction, suppose that $\sum_{x \in \mathbb{Z}^n} f(x) = 0$. For $x \in \mathbb{Z}^n$, let ϵ_x denote the 0-form that is 1 at x and 0 everywhere else. For any distinct $x, y \in \mathbb{Z}^n$, let $\epsilon_{x,y} := \epsilon_x - \epsilon_y$. For any such pair (x, y) , choose a path of distinct vertices $x_0, \dots, x_k$ from x to y , with $x_0 = x$, $x_k = y$, and x_{j+1} a neighbor of x_j for each j . Define an R -valued 1-form $\tau_{x,y}$ as

$$\tau_{x,y}(e) := \begin{cases} 1 & \text{if } e = (x_j, x_{j+1}) \text{ for some } 0 \leq j \leq k-1, \\ 0 & \text{otherwise.} \end{cases}$$

A simple verification shows that

$$\delta \tau_{x,y} = \epsilon_{x,y}. \quad (\text{C.8.12})$$

Next, let S denote the support of f . If $S = \emptyset$, we can simply take $g = 0$. Otherwise, let x_0 be the smallest element of S in the lexicographic ordering. We claim that

$$f = \sum_{x \in S \setminus \{x_0\}} f(x) \epsilon_{x,x_0}. \quad (\text{C.8.13})$$

To see this, note that by the condition $\sum_{x \in \mathbb{Z}^n} f(x) = 0$, we have

$$\begin{aligned} \sum_{x \in S \setminus \{x_0\}} f(x) \epsilon_{x,x_0} &= \sum_{x \in S \setminus \{x_0\}} f(x) (\epsilon_x - \epsilon_{x_0}) \\ &= \sum_{x \in S \setminus \{x_0\}} f(x) \epsilon_x - \left(\sum_{x \in S \setminus \{x_0\}} f(x) \right) \epsilon_{x_0} \\ &= \sum_{x \in S \setminus \{x_0\}} f(x) \epsilon_x + f(x_0) \epsilon_{x_0} = f. \end{aligned}$$

This proves equation (C.8.13). Define

$$g := \sum_{x \in S \setminus \{x_0\}} f(x) \tau_{x,x_0}.$$

Then by equation (C.8.12) and equation (C.8.13), we get

$$\delta g = \sum_{x \in S \setminus \{x_0\}} f(x) \delta \tau_{x, x_0} = \sum_{x \in S \setminus \{x_0\}} f(x) \epsilon_{x, x_0} = f.$$

This completes the proof of the converse direction. Finally, if S is connected, then the path between x and y for any $x, y \in S$ can be chosen to lie entirely in S . Thus, $\tau_{x, y}(e) = 0$ for any e whose endpoints are not both in S . Consequently, g inherits this property. $\square$

C.8.10 The discrete Hodge Laplacian

Let R be a ring, and consider R -valued differential forms on $\mathbb{Z}^n$. The discrete Hodge Laplacian is the operator

$$\Delta := d\delta + \delta d$$

that takes k -forms to k -forms. Let us now see what this turns out to be more explicitly. First, consider 0-forms. If f is a 0-form, then $\delta f = 0$ by definition. Let $g := df$ and $h := \delta g$, so that $h = \Delta f$. Then $g_i(x) = f(x + e_i) - f(x)$ for $x \in \mathbb{Z}^n$ and $1 \leq i \leq n$, and by the formula (C.8.9),

$$\begin{aligned} h(x) &= - \sum_{i=1}^n \bar{\partial}_i g_i(x) = - \sum_{i=1}^n (g_i(x) - g_i(x - e_i)) \\ &= - \sum_{i=1}^n (f(x + e_i) - f(x) - (f(x) - f(x - e_i))) \\ &= 2nf(x) - \sum_{i=1}^n (f(x + e_i) + f(x - e_i)). \end{aligned}$$

Thus, for 0-forms, the Hodge Laplacian is the usual discrete Laplace operator. The following lemma shows that this extends to k -forms in a simple way.

Lemma C.8.7. *Let R be a ring and*

$$f = \sum_{1 \leq i_1 < \dots < i_k \leq n} f_{i_1 \dots i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}$$

be an R -valued k -form on $\mathbb{Z}^n$. Then

$$\Delta f = \sum_{1 \leq i_1 < \dots < i_k \leq n} \Delta f_{i_1 \dots i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

where the Δ on the right stands for the usual discrete Laplacian.

Proof. Since we have already proved the claim for $k = 0$ above, let us take $1 \leq k \leq n$ and a k -form f . Let $g := df$ and $h := \delta g = \delta df$. If $k = n$, then $df = 0$ and therefore $h = 0$.

Suppose that $k < n$. Then by the formula (C.8.2),

$$g_{i_1 \dots i_{k+1}}(x) = \sum_{j=1}^{k+1} (-1)^{j-1} \partial_{i_j} f_{i_1 \dots \widehat{i_j} \dots i_{k+1}}(x).$$

Thus, by the formula (C.8.9),

$$\begin{aligned} h_{i_1 \dots i_k}(x) &= - \sum_{i=1}^n \bar{\partial}_i g_{ii_1 \dots i_k}(x) \\ &= - \sum_{i=1}^n \bar{\partial}_i \left(\partial_i f_{i_1 \dots i_k}(x) + \sum_{j=1}^k (-1)^j \partial_{i_j} f_{ii_1 \dots \widehat{i_j} \dots i_k}(x) \right) \\ &= \Delta f_{i_1 \dots i_k}(x) - \sum_{i=1}^n \sum_{j=1}^k (-1)^j \bar{\partial}_i \partial_{i_j} f_{ii_1 \dots \widehat{i_j} \dots i_k}(x). \end{aligned}$$

Next, let $u := \delta f$ and $v := du = d\delta f$. Then

$$u_{i_1 \dots i_{k-1}}(x) = - \sum_{i=1}^n \bar{\partial}_i f_{ii_1 \dots i_{k-1}}(x),$$

and therefore,

$$\begin{aligned} v_{i_1 \dots i_k}(x) &= \sum_{j=1}^k (-1)^{j-1} \partial_{i_j} u_{i_1 \dots \widehat{i_j} \dots i_k}(x) \\ &= - \sum_{j=1}^k (-1)^{j-1} \partial_{i_j} \left(\sum_{i=1}^n \bar{\partial}_i f_{ii_1 \dots \widehat{i_j} \dots i_k}(x) \right) \\ &= \sum_{j=1}^k \sum_{i=1}^n (-1)^j \partial_{i_j} \bar{\partial}_i f_{ii_1 \dots \widehat{i_j} \dots i_k}(x). \end{aligned}$$

Now, an easy verification shows that the operators ∂_i and $\bar{\partial}_j$ commute for any i, j . Thus, the above calculations show that $\Delta f = h + v$ has the required form. Lastly, if $k = n$, then

$$v_{1 \dots n}(x) = \sum_{i,j=1}^n (-1)^j \partial_j \bar{\partial}_i f_{i_1 \dots \widehat{i_j} \dots n}(x).$$

But

$$f_{i_1 \dots \widehat{i_j} \dots n} = \begin{cases} 0 & \text{if } i \neq j, \\ (-1)^{i-1} f_{1 \dots n} & \text{if } i = j. \end{cases}$$

Thus,

$$v_{1 \dots n}(x) = - \sum_{i=1}^n \partial_i \bar{\partial}_i f_{1 \dots n}(x) = \Delta f_{1 \dots n}(x).$$

This completes the proof. $\square$

Let Λ be a hypercube in $\mathbb{Z}^n$. Let $\partial\Lambda$ be the boundary of Λ , meaning the set of vertices in Λ that have at least one neighbor outside Λ . We will say that a k -cell c is in Λ if all of its vertices are in Λ . A k -cell of Λ will be called a ‘boundary k -cell’ if all of its vertices are in $\partial\Lambda$. The remaining k -cells of Λ will be called ‘interior k -cells’. Let Π_Λ denote the following operator on R -valued differential forms. For any k -form α , $\Pi_\Lambda\alpha$ is the k -form

$$\Pi_\Lambda\alpha(c) := \begin{cases} \alpha(c) & \text{if } c \text{ is an interior } k\text{-cell of } \Lambda, \\ 0 & \text{otherwise.} \end{cases}$$

It is not hard to see that for a given k , Π_Λ is the orthogonal projection (with respect to the inner product defined in Subsection C.8.8) on to the subspace of k -forms which vanish on all k -cells that are not interior k -cells of Λ . Define the restriction of Δ to Λ under zero boundary condition as

$$\Delta_\Lambda := \Pi_\Lambda\Delta.$$

It turns out that the Δ_Λ^{-1} is well-defined on the range of Π_Λ , as shown by the following lemma.

Lemma C.8.8. *For any $0 \leq k \leq n$, the operator Δ_Λ is a linear bijection on the space of all R -valued k -forms that are in the range of Π_Λ .*

Proof. Clearly, Δ_Λ maps into the range of Π_Λ . Since $\dim(\text{range } \Pi_\Lambda)$ is finite, it suffices to show that Δ_Λ is injective. Suppose that $\Delta_\Lambda\alpha = 0$ for some $\alpha \in \text{range } \Pi_\Lambda$. Then $(\alpha, \Delta_\Lambda\alpha) = 0$. But

$$\begin{aligned} (\alpha, \Delta_\Lambda\alpha) &= (\alpha, \Pi_\Lambda\Delta\alpha) \\ &= (\alpha, \Delta\alpha) = (\alpha, (d\delta + \delta d)\alpha) \\ &= \|\delta\alpha\|^2 + \|d\alpha\|^2, \end{aligned}$$

where the identity holds even if $k = 0$ or $k = n$. Thus, $\delta\alpha = 0$ and $d\alpha = 0$. Suppose that $1 \leq k \leq n - 1$. By the Poincaré lemmas for d and δ , this means that there is a $(k + 1)$ -form τ and a $(k - 1)$ -form ξ such that $\alpha = \delta\tau$ and $\alpha = d\xi$. Thus,

$$\|\alpha\|^2 = (\delta\tau, d\xi) = (\tau, dd\xi) = 0,$$

which completes the proof. If $k = 0$, then the fact that $d\alpha = 0$ implies that $\alpha = 0$ (by the Poincaré lemma). Similarly, if $k = n$, then the fact that $\delta\alpha = 0$ implies that $\alpha = 0$. $\square$

Appendix D

Functional Analysis

D.1 Some basic results

D.1.1 The Baire category theorem

Throughout this chapter, all Hilbert spaces and Banach spaces are over the field of complex numbers. We will need the following basic fact about Banach spaces.

Theorem D.1.1 (Baire category theorem for Banach spaces). *Let $\mathcal{B}$ be a Banach space. Suppose that $\mathcal{B}$ is the union of a countable sequence of closed sets $\{F_n\}_{n \geq 1}$. Then at least one F_n has nonempty interior.*

Proof. Let $U_n := \mathcal{B} \setminus F_n$. Then each U_n is open. Suppose that each F_n has empty interior. Then each U_n is dense. Take any $x_1 \in U_1$ and a radius $r_1 \in (0, 1)$ so small that $B(x_1, r_1) \subseteq U_1$ (where $B(x, r)$ denotes the open ball of radius r and center x). Since U_2 is dense, there exists a point $x_2 \in U_2 \cap B(x_1, r_1/2)$. Find $r_2 \in (0, r_1/2)$ so small that $B(x_2, r_2) \subseteq U_2$. Continuing this way, we find sequences $\{x_n\}_{n \geq 1}$ and $\{r_n\}_{n \geq 1}$ such that for each n , $x_{n+1} \in U_{n+1} \cap B(x_n, r_n/2)$, $r_{n+1} \in (0, r_n/2)$, and $B(x_n, r_n) \subseteq U_n$. It is easy to see that $\{x_n\}_{n \geq 1}$ is a Cauchy sequence, and therefore, has a limit $x \in \mathcal{B}$. Now, for each n ,

$$\begin{aligned} \|x_n - x\| &\leq \sum_{k=n}^{\infty} \|x_k - x_{k+1}\| \\ &\leq \frac{1}{2} \sum_{k=n}^{\infty} r_k \leq \frac{r_n}{2} \sum_{j=0}^{\infty} 2^{-j} = \frac{r_n}{2}. \end{aligned}$$

Consequently, $x \in B(x_n, r_n) \subseteq U_n$ for each n . This contradicts the fact that the intersection of the U_n 's is empty. $\square$

D.1.2 The open mapping theorem

The Baire category theorem implies the following result.

Theorem D.1.2 (Open mapping theorem). *Let $\mathcal{B}_1, \mathcal{B}_2$ be two Banach spaces and $A : \mathcal{B}_1 \rightarrow \mathcal{B}_2$ be a continuous and surjective linear map. Then A maps open sets to open sets.*

Proof. Let $F_n := \overline{A(B(0, n))}$ be the closure of the image of the open ball $B(0, n) \subseteq \mathcal{B}_1$ under A . Since A is surjective, the union of the F_n 's equals $\mathcal{B}_2$. Thus, by the Baire category theorem, there is some n , some $y \in \mathcal{B}_2$, and some $\delta > 0$ such that $B(y, \delta) \subseteq F_n$. This implies that $B(0, \delta) \subseteq F_n - y$. Now, if $x, x' \in F_n$, then there are sequences $\{z_k\}_{k \geq 1}$ and $\{z'_k\}_{k \geq 1}$ in $\mathcal{B}_1$ such that $z_k, z'_k \in B(0, n)$ for every n , and $Az_k \rightarrow x$, $Az'_k \rightarrow x'$. But $z_k - z'_k \in B(0, 2n)$ for every k , and $A(z_k - z'_k) \rightarrow x - x'$. Thus, $x - x' \in F_{2n}$. In particular, $B(0, \delta) \subseteq F_n - y \subseteq F_{2n}$.

We claim that $B(0, \delta) \subseteq A(B(0, 5n))$. To see this, take any $x \in B(0, \delta)$. Since $B(0, \delta) \subseteq F_{2n}$, there is some $y_1 \in B(0, 2n)$ such that

$$\|x - Ay_1\| < \frac{\delta}{2}.$$

Now, the above display shows that $2(x - Ay_1) \in B(0, \delta)$. Thus, there is some $y_2 \in B(0, 2n)$ such that

$$\|2(x - Ay_1) - Ay_2\| < \frac{\delta}{2}.$$

Continuing like this, we find a sequence of points $\{y_k\}_{k \geq 1}$ in $B(0, 2n)$ such that for each k ,

$$\|2^k x - 2^k Ay_1 - 2^{k-1} Ay_2 - \cdots - Ay_{k+1}\| < \frac{\delta}{2}.$$

Dividing both sides by 2^k and sending $k \rightarrow \infty$, we see that

$$x = \lim_{k \rightarrow \infty} \sum_{j=1}^k 2^{-j+1} Ay_j. \quad (\text{D.1.1})$$

But note that if we define

$$z_k := \sum_{j=1}^k 2^{-j+1} y_j,$$

then $\{z_k\}_{k \geq 1}$ is a Cauchy sequence, since the y_j 's are all in $B(0, 2n)$. Thus, $z_k \rightarrow z$ for some z . Moreover,

$$\|z\| \leq \sum_{j=1}^{\infty} 2^{-j+1} \|y_j\| \leq 4n.$$

Consequently, $z \in B(0, 5n)$. But since A is a continuous map, equation (D.1.1) shows that

$$x = \lim_{k \rightarrow \infty} Az_k = Az.$$

Thus, $x \in A(B(0, 5n))$. This proves the claim that $B(0, \delta) \subseteq A(B(0, 5n))$.

Now take any open set $U \subseteq \mathcal{B}_1$. Take any $x \in U$ and a radius r such that $B(x, r) \subseteq U$. We claim that $B(Ax, \delta r/5n) \subseteq A(B(x, r)) \subseteq A(U)$. To see this, take any $y \in B(Ax, \delta r/5n)$. Then $(5n/r)(y - Ax) \in B(0, \delta)$. Thus, there is some $z \in B(0, 5n) \subseteq \mathcal{B}_1$ such that

$Az = (5n/r)(y - Ax)$. But then,

$$y = Ax + \frac{r}{5n}Az = A\left(x + \frac{r}{5n}z\right) \in A(B(x, r)),$$

since $z \in B(0, 5n)$. This proves that $A(U)$ is open. $\square$

D.1.3 The closed graph theorem

An important consequence of the open mapping theorem is the following. Recall that for a Banach space $\mathcal{B}$, the *graph* of a linear map $A : \mathcal{B} \rightarrow \mathcal{B}$ is the set $\{(x, Ax) : x \in \mathcal{B}\} \subseteq \mathcal{B} \times \mathcal{B}$.

Theorem D.1.3 (Closed graph theorem for Banach spaces). *Let $\mathcal{B}$ be a Banach space and $A : \mathcal{B} \rightarrow \mathcal{B}$ be a linear map whose graph is a closed set in the product topology of $\mathcal{B} \times \mathcal{B}$. Then A is continuous.*

Proof. Let $\|\cdot\|$ denote the norm on $\mathcal{B}$. Define a new norm

$$\|x\|' := \|x\| + \|Ax\|.$$

It is easy to see that this is a norm. We claim that this norm is complete. To see this, take a sequence $\{x_n\}_{n \geq 1}$ that is Cauchy in the norm $\|\cdot\|'$. Then $\{x_n\}_{n \geq 1}$ and $\{Ax_n\}_{n \geq 1}$ are both Cauchy in the norm $\|\cdot\|$. Since the graph of A is closed, it follows that there is some $x \in \mathcal{B}$ such that $x_n \rightarrow x$ and $Ax_n \rightarrow Ax$. Thus, $\|x_n - x\|' \rightarrow 0$.

Now, it is clear that the identity map $I : (\mathcal{B}, \|\cdot\|') \rightarrow (\mathcal{B}, \|\cdot\|)$ is continuous and surjective. Thus, by the open mapping theorem, its inverse $I^{-1} : (\mathcal{B}, \|\cdot\|) \rightarrow (\mathcal{B}, \|\cdot\|')$ is continuous. Since I^{-1} is a linear map, continuity implies that it is bounded. Thus, there is some $C \geq 0$ such that $\|I^{-1}x\|' \leq C\|x\|$ for all $x \in \mathcal{B}$. But

$$\|I^{-1}x\|' = \|x\|' = \|x\| + \|Ax\|.$$

This proves that A is bounded, and hence, continuous. $\square$

D.1.4 The uniform boundedness principle

Another basic tool that we will need is the following. The very short proof given here is due to Alan Sokal.

Theorem D.1.4 (Uniform boundedness principle). *Let $\{T_n\}_{n \geq 1}$ be a sequence of bounded operators from a Banach space $\mathcal{B}$ into itself. Suppose that for each $x \in \mathcal{B}$, $\sup_{n \geq 1} \|T_n x\|$ is finite. Then $\sup_{n \geq 1} \|T_n\|$ is finite.*

Proof. Suppose that the hypothesis holds but $\sup_{n \geq 1} \|T_n\| = \infty$. Then passing to a subsequence if necessary, let us assume that $\|T_n\| \geq 4^n$ for each n . Let $x_0 = 0$, and inductively define $x_1, x_2, \dots$ as follows. Suppose we have chosen up to x_{n-1} . Note that for

any y ,

$$\begin{aligned}\|T_n y\| &= \frac{1}{2} \|T_n(x_{n-1} + y) - T_n(x_{n-1} - y)\| \\ &\leq \frac{1}{2} (\|T_n(x_{n-1} + y)\| + \|T_n(x_{n-1} - y)\|) \\ &\leq \max\{\|T_n(x_{n-1} + y)\|, \|T_n(x_{n-1} - y)\|\}.\end{aligned}$$

But the supremum of the left side over all y with $\|y\| \leq 3^{-n}$ is equal to $\|T_n\|3^{-n}$. This shows that there exists y with $\|y\| \leq 3^{-n}$ and $\|T_n(x_{n-1} + y)\| \geq \frac{2}{3}\|T_n\|3^{-n}$. Let $x_n := x_{n-1} + y$. Thus, we get a sequence $\{x_n\}_{n \geq 1}$ such that $\|x_n - x_{n-1}\| \leq 3^{-n}$ and $\|T_n x_n\| \geq \frac{2}{3}\|T_n\|3^{-n}$ for each n . Clearly, this is a Cauchy sequence. Let x denote its limit. Then for each n ,

$$\|x - x_n\| \leq \sum_{k=n}^{\infty} \|x_{k+1} - x_k\| \leq \sum_{k=n}^{\infty} 3^{-k} = \frac{1}{2}3^{-n}.$$

Thus, we get

$$\|T_n x\| \geq \|T_n x_n\| - \|T_n(x - x_n)\| \geq \frac{2}{3}\|T_n\|3^{-n} - \frac{1}{2}\|T_n\|3^{-n} = \frac{1}{6}\|T_n\|3^{-n}.$$

Since $\|T_n\| \geq 4^n$ for each n , this shows that $\sup_{n \geq 1} \|T_n x\| = \infty$, contradicting the hypothesis. $\square$

D.1.5 Riesz representation theorem for Hilbert spaces

The following result is known as the Riesz representation theorem for Hilbert spaces.

Theorem D.1.5 (Riesz representation theorem). *Let $\mathcal{H}$ be a Hilbert space and $\lambda : \mathcal{H} \rightarrow \mathbb{C}$ be a continuous linear map. Then there exists unique $x \in \mathcal{H}$ such that $\lambda(y) = \langle y, x \rangle$ for all $y \in \mathcal{H}$.*

Proof. Let $K := \ker \lambda$. Since λ is continuous, it is easy to see that K is a closed subspace of $\mathcal{H}$. If $K = \mathcal{H}$, there is nothing more to prove, since we can simply take $x = 0$. Suppose that $K \neq \mathcal{H}$, and take any $z \in \mathcal{H} \setminus K$. Let $d := \inf_{x \in K} \|z - x\|$. Since K is closed and $z \notin K$, d must be nonzero. Let $\{w_n\}_{n \geq 1}$ be a sequence of points in K such that $\|z - w_n\| \rightarrow d$. By the parallelogram identity,

$$\|w_m - w_n\|^2 = 2\|z - w_m\|^2 + 2\|z - w_n\|^2 - \|2z - (w_m + w_n)\|^2$$

for each m, n . Since $\frac{1}{2}(w_m + w_n) \in K$, this implies that

$$\|w_m - w_n\|^2 \leq 2\|z - w_m\|^2 + 2\|z - w_n\|^2 - 4d^2.$$

Taking $\limsup$ on both sides as $m, n \rightarrow \infty$, we get

$$\limsup_{m, n \rightarrow \infty} \|w_m - w_n\|^2 \leq 0.$$

Thus, $\{w_n\}_{n \geq 1}$ is a Cauchy sequence, and therefore, has a limit w . Since K is closed, $w \in K$. Let $u := z - w$. We claim that $\langle u, x \rangle = 0$ for all $x \in K$. To see this, take any $x \in K$ and note that for any $\alpha \in \mathbb{R}$, $w + \alpha x \in K$. Thus,

$$\|z - (w + \alpha x)\| \geq d = \|z - w\|.$$

But

$$\|z - (w + \alpha x)\|^2 = \|z - w\|^2 - 2\alpha \langle z - w, x \rangle + \alpha^2 \|x\|^2.$$

Therefore, the above quadratic polynomial in α is minimized at $\alpha = 0$. This implies that its derivative at zero must vanish. This is equivalent to the claim that $\langle u, x \rangle = 0$.

Since $z \neq w$, we conclude that u is nonzero. Note that for any $y \in \mathcal{H}$, the linearity of λ implies that

$$\lambda(\lambda(u)y - \lambda(y)u) = 0.$$

Thus, $\lambda(u)y - \lambda(y)u \in K$. Consequently,

$$0 = \langle \lambda(u)y - \lambda(y)u, u \rangle = \lambda(u) \langle y, u \rangle - \lambda(y) \|u\|^2.$$

Thus, if we take

$$x := \frac{\overline{\lambda(u)}}{\|u\|^2} u,$$

then for any $y \in \mathcal{H}$,

$$\lambda(y) = \langle y, x \rangle.$$

This proves the existence of x . To prove uniqueness, simply observe that if x' is another candidate, then the identities

$$\langle x - x', x \rangle = \lambda(x - x') = \langle x - x', x' \rangle$$

imply that $x = x'$. □

D.1.6 The Hellinger–Toeplitz theorem

We will also need the following less well-known result.

Theorem D.1.6 (Hellinger–Toeplitz theorem). *Let $\mathcal{H}$ be a Hilbert space and $A : \mathcal{H} \rightarrow \mathcal{H}$ be a symmetric linear operator. Then A is continuous.*

Proof. Since A is defined everywhere on $\mathcal{H}$ and is symmetric, we see that A is self-adjoint with domain $\mathcal{H}$. But by Lemma D.2.2, A^* is a closed operator. Thus, A is a closed operator. By the closed graph theorem, this implies that A is a bounded operator. □

D.2 Self-adjoint operators

D.2.1 Self-adjoint operators on a Hilbert space

Let $\mathcal{H}$ be a complex Hilbert space with inner product $\langle \cdot, \cdot \rangle$ that is linear in the first argument and conjugate linear in the second argument. (This needs to be spelled out, since physics textbooks sometimes take the opposite convention.) A *densely defined* linear operator on $\mathcal{H}$ is a linear map $A : \mathcal{D}(A) \rightarrow \mathcal{H}$, where $\mathcal{D}(A)$ is a dense subspace of $\mathcal{H}$, called the ‘domain of A ’. Given a densely defined operator A , the *adjoint* A^* of A is defined as follows. First, we define the domain of A^* . Let $\mathcal{D}(A^*)$ be the set of all $x \in \mathcal{H}$ such that there is some $z \in \mathcal{H}$ satisfying the condition that for all $y \in \mathcal{D}(A)$,

$$\langle Ay, x \rangle = \langle y, z \rangle.$$

In case such a z exists (and hence, $x \in \mathcal{D}(A^*)$), we define $A^*x := z$, so that we have

$$\langle Ay, x \rangle = \langle y, A^*x \rangle.$$

Note that A^*x is well-defined, since if there exist z and z' satisfying the above criterion, then $\langle y, z - z' \rangle = 0$ for all $y \in \mathcal{D}(A)$. But $\mathcal{D}(A)$ is dense, and therefore $z = z'$. Similarly, it is easy to verify that A^* is linear. Lastly, it is easy to see that $\mathcal{D}(A^*)$ is a subspace of $\mathcal{H}$. Note that if $\mathcal{D}(A) = \mathcal{H}$ and A is continuous, then it is a simple consequence of the Riesz representation theorem for Hilbert spaces that $\mathcal{D}(A^*) = \mathcal{H}$.

A densely defined operator A is called *symmetric* (or Hermitian) if for all $x, y \in \mathcal{D}(A)$, we have

$$\langle Ax, y \rangle = \langle x, Ay \rangle.$$

Note that if A is symmetric, then $\mathcal{D}(A) \subseteq \mathcal{D}(A^*)$. This is because the above equation shows that for any $y \in \mathcal{D}(A)$, the vector Ay can serve as A^*y . In particular, $Ay = A^*y$ for all $y \in \mathcal{D}(A)$ if A is symmetric.

Let A be a symmetric linear operator on a Hilbert space $\mathcal{H}$, defined on a domain $\mathcal{D}(A)$ that is a dense subspace of $\mathcal{H}$. We say that A is self-adjoint if $\mathcal{D}(A^*) = \mathcal{D}(A)$. A trivial but important observation is that if the domain of a symmetric operator A is the whole of $\mathcal{H}$, then A is self-adjoint.

D.2.2 Criteria for an operator to be self-adjoint

For a densely defined operator A , let $\ker A$ denote the kernel of A and $\text{range } A$ denote its range. For any subset $S \subseteq \mathcal{H}$, let

$$S^\perp := \{x \in \mathcal{H} : \langle x, y \rangle = 0 \text{ for all } y \in S\}$$

denote its orthogonal complement. It is easy to see that for any subset S , the set $S^\perp$ is closed subspace of $\mathcal{H}$. The following lemma is basic.

Lemma D.2.1. *Let A be a densely defined linear operator on a Hilbert space $\mathcal{H}$. Then*

$$(\text{range } A)^\perp = \ker A^*.$$

Proof. Take any $x \in (\text{range } A)^\perp$. Then $\langle Ay, x \rangle = 0 = \langle y, 0 \rangle$ for all $y \in \mathcal{D}(A)$. This shows that $x \in \mathcal{D}(A^*)$ and $A^*x = 0$. Thus, $x \in \ker A^*$. Conversely, take any $x \in \ker A^*$. Then $A^*x = 0$, and hence, $\langle Ay, x \rangle = \langle Ay, 0 \rangle = 0$ for all $y \in \mathcal{D}(A)$. Thus, $x \in (\text{range } A)^\perp$. $\square$

A linear operator A defined on a subspace $\mathcal{D}(A)$ of a Hilbert space $\mathcal{H}$ (which is not necessarily dense in $\mathcal{H}$) is said to be *closed* if for any sequence $\{x_n\}_{n \geq 1}$ in $\mathcal{D}(A)$ such that $x_n \rightarrow x \in \mathcal{H}$ and $Ax_n \rightarrow y \in \mathcal{H}$, we have that $x \in \mathcal{D}(A)$ and $y = Ax$. Equivalently, A is closed if its graph $\{(x, Ax) : x \in \mathcal{D}(A)\}$ is a closed subset of $\mathcal{H} \times \mathcal{H}$ in the product topology. The following lemma shows that the adjoint of any densely defined operator is closed.

Lemma D.2.2. *For any densely defined operator A on a Hilbert space $\mathcal{H}$, its adjoint A^* is a closed operator.*

Proof. Define an inner product on $\mathcal{H} \times \mathcal{H}$ as

$$\langle (x, y), (x', y') \rangle := \langle x, x' \rangle + \langle y, y' \rangle.$$

It is easy to verify that this is indeed an inner product and that $\mathcal{H} \times \mathcal{H}$ is a Hilbert space under this inner product. Let $S \subseteq \mathcal{H} \times \mathcal{H}$ be the set

$$S := \{(-Ax, x) : x \in \mathcal{D}(A)\}.$$

We claim that the graph of A^* is the set $S^\perp$. To see this, first take any $(x, y) \in S^\perp$. Then for all $z \in \mathcal{D}(A)$, we have

$$0 = \langle (x, y), (-Az, z) \rangle = -\langle x, Az \rangle + \langle y, z \rangle.$$

Thus, for all $z \in \mathcal{D}(A)$,

$$\langle Az, x \rangle = \langle z, y \rangle.$$

Consequently, $x \in \mathcal{D}(A^*)$ and $A^*x = y$. Thus, (x, y) is in the graph of A^* . Conversely, if (x, y) is in the graph of A^* , then $x \in \mathcal{D}(A^*)$ and $y = A^*x$. Thus, $\langle Az, x \rangle = \langle z, y \rangle$ for all $z \in \mathcal{D}(A)$, and we can retrace the steps backwards to deduce that $(x, y) \in S^\perp$. Since the orthogonal complement of any set is closed, this shows that A^* is a closed operator. $\square$

The following theorem gives two criteria for checking that a densely defined symmetric operator is self-adjoint. It is useful because it is usually not easy to directly verify using only the definition that a given operator is self-adjoint. Below, I is used to denote the identity operator.

Theorem D.2.3. *Let A be a densely defined symmetric operator on a Hilbert space $\mathcal{H}$. Then the following are equivalent:*

1. A is self-adjoint.
2. A is closed, and $\ker(A^* \pm iI) = \{0\}$.
3. $\text{range}(A \pm iI) = \mathcal{H}$.

Proof. First, let us prove that 1 implies 2. Let A be self-adjoint. Since $A = A^*$, Lemma D.2.2 shows that A is closed. Next, if $x \in \mathcal{D}(A^*) = \mathcal{D}(A)$ is such that $A^*x = ix$, then

$$\begin{aligned} i\|x\|^2 &= \langle ix, x \rangle = \langle A^*x, x \rangle \\ &= \langle Ax, x \rangle = \langle x, A^*x \rangle \\ &= \langle x, ix \rangle = -i\|x\|^2. \end{aligned}$$

Thus, $x = 0$. A similar argument shows that if $A^*x = -ix$, then $x = 0$. Thus, 1 implies 2.

Next, suppose that condition 2 holds. Then by Lemma D.2.1,

$$\text{range}(A \pm iI)^\perp = \ker((A \pm iI)^*).$$

Now, it is easy to show that $(A \pm iI)^* = A^* \mp iI$. Thus, $\text{range}(A \pm iI)^\perp = \{0\}$. Therefore, to prove condition 3, we only need to show that $\text{range}(A \pm iI)$ is closed. Let us prove it for $\text{range}(A + iI)$. The proof of the other case is similar.

Let $\{x_n\}_{n \geq 1}$ be a sequence in $\text{range}(A + iI)$ converging to some $x \in \mathcal{H}$. Choose $y_n \in \mathcal{D}(A)$ such that $Ay_n + iy_n = x_n$. Now, note that for $z \in \mathcal{D}(A)$,

$$\begin{aligned} \|Az + iz\|^2 &= \langle Az + iz, Az + iz \rangle \\ &= \langle Az, Az \rangle + i\langle z, Az \rangle - i\langle Az, z \rangle - i^2\langle z, z \rangle. \end{aligned}$$

Now, since A is a symmetric operator and $z \in \mathcal{D}(A)$, $\langle Az, z \rangle = \langle z, Az \rangle$. Thus, the above expression shows that

$$\|Az + iz\|^2 = \|Az\|^2 + \|z\|^2.$$

In particular, for each m and n

$$\|x_m - x_n\|^2 = \|Ay_m - Ay_n\|^2 + \|y_m - y_n\|^2.$$

This shows that both $\{y_n\}_{n \geq 1}$ and $\{Ay_n\}_{n \geq 1}$ are Cauchy sequences, and hence, convergent. Suppose that $y_n \rightarrow y$ and $Ay_n \rightarrow z$. By condition 2, A is closed. Thus, $y \in \mathcal{D}(A)$ and $z = Ay$. Consequently,

$$x = \lim_{n \rightarrow \infty} (Ay_n + iy_n) = Ay + iy \in \text{range}(A + iI).$$

This proves that 2 implies 3. Finally, let us show that 3 implies 1. Assume that condition 3 holds. Since A is symmetric, we only need to show that $\mathcal{D}(A^*) \subseteq \mathcal{D}(A)$. Take any $x \in \mathcal{D}(A^*)$, and let $y := A^*x - ix$. By condition 3, there exists $z \in \mathcal{D}(A)$ such that $y = Az - iz$. Since A is symmetric and $z \in \mathcal{D}(A)$, we have that $Az = A^*z$, by one of our

previous observations. Combining, we get

$$A^*x - ix = A^*z - iz.$$

Thus, $x - z \in \ker(A^* - iI)$. But by Lemma D.2.1 and condition 3,

$$\ker(A^* - iI) = \ker((A + iI)^*) = \text{range}(A + iI)^\perp = \{0\}.$$

Thus, $x = z \in \mathcal{D}(A)$. This completes the proof. $\square$

D.2.3 Spectrum of a self-adjoint operator

Let A be a densely defined linear operator on a Hilbert space $\mathcal{H}$. We say that A is invertible if $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ is a bijection. The inverse of A is denoted by A^{-1} . Clearly, A^{-1} is a linear operator. A complex number α is said to be in the resolvent set $\rho(A)$ of a densely defined linear operator A if the operator $\alpha I - A : \mathcal{D}(A) \rightarrow \mathcal{H}$ has a bounded inverse. The complement of $\rho(A)$ in $\mathbb{C}$, denoted $\sigma(A)$, is called the spectrum of A . If $\alpha \in \rho(A)$ then the inverse of $\alpha I - A$, often denoted by $R(\alpha)$, is called the *resolvent* of A at α . Note that $R(\alpha)$ is defined on the whole of $\mathcal{H}$, and is bounded.

Theorem D.2.4. *Let A be a self-adjoint linear operator on a Hilbert space $\mathcal{H}$. Then $\sigma(A) \subseteq \mathbb{R}$. Moreover, for any $\alpha \in \mathbb{C} \setminus \mathbb{R}$, we have the bound*

$$\|R(\alpha)\| \leq \frac{1}{|\text{Im}(\alpha)|}.$$

Proof. Take any $\lambda \in \mathbb{R} \setminus \{0\}$. It is easy to see that $\lambda^{-1}A$ is also self-adjoint. Thus, by Theorem D.2.3, $\ker((\lambda^{-1}A)^* - iI) = \{0\}$. But $(\lambda^{-1}A)^* = \lambda^{-1}A$. Thus,

$$\ker(A + i\lambda I) = \ker(\lambda^{-1}A - iI) = \{0\}.$$

Also by Theorem D.2.3,

$$\mathcal{H} = \text{range}(\lambda^{-1}A - iI) = \text{range}(A - i\lambda I).$$

This shows that $A - i\lambda I : \mathcal{D}(A) \rightarrow \mathcal{H}$ is invertible for any $\lambda \in \mathbb{R} \setminus \{0\}$. Now, for any $\mu \in \mathbb{R}$, it is easy to see that the operator $A - \mu I$ is self-adjoint. Thus, repeating the above argument with $A - \mu I$ in place of A , we see that for any $\lambda \in \mathbb{R} \setminus \{0\}$, $A - (\mu + i\lambda)I$ is invertible. Let $\alpha := \mu + i\lambda$ and $R(\alpha) := (\alpha I - A)^{-1}$. We claim that $\|R(\alpha)\|$ is finite and satisfies the claimed upper bound. To see this, note that for any $x \in \mathcal{D}(A)$,

$$\begin{aligned} \|(\alpha I - A)x\|^2 &= \langle (\alpha I - A)x, (\alpha I - A)x \rangle \\ &= |\alpha|^2 \|x\|^2 - \bar{\alpha} \langle Ax, x \rangle - \alpha \langle x, Ax \rangle + \|Ax\|^2. \end{aligned}$$

Since A is self-adjoint $\langle Ax, x \rangle = \langle x, Ax \rangle$. Thus, we get

$$\begin{aligned} \|(\alpha I - A)x\|^2 &= (\mu^2 + \lambda^2)\|x\|^2 - 2\mu\langle Ax, x \rangle + \|Ax\|^2 \\ &\geq (\mu^2 + \lambda^2)\|x\|^2 - 2|\mu|\|x\|\|Ax\| + \|Ax\|^2 \\ &= \lambda^2\|x\|^2 + (|\mu|\|x\| - \|Ax\|)^2 \\ &\geq \lambda^2\|x\|^2. \end{aligned}$$

Thus, for any $y \in \mathcal{H}$, taking $x := R(\alpha)y$, we get

$$\|y\|^2 = \|(\alpha I - A)x\|^2 \geq \lambda^2\|x\|^2 = \lambda^2\|R(\alpha)y\|^2.$$

This proves that $\|R(\alpha)\| \leq 1/|\lambda|$. $\square$

D.3 Stone's theorem

Recall that a one parameter group of unitary operators on a Hilbert space $\mathcal{H}$ is a family $(U(t))_{t \in \mathbb{R}}$ such that $U(t)$ is unitary for each t (meaning that $U(t) : \mathcal{H} \rightarrow \mathcal{H}$ is surjective, and $\langle U(t)x, U(t)y \rangle = \langle x, y \rangle$ for all $x, y \in \mathcal{H}$) and $U(s+t) = U(s)U(t)$ for all $s, t \in \mathbb{R}$. Recall also that such a group is called strongly continuous if $t \mapsto U(t)x$ is continuous for any x . Stone's theorem is an important result that says strongly continuous one parameter groups of unitary operators are in a one-to-one correspondence with self-adjoint operators.

Theorem D.3.1 (Stone's theorem). *Let $(U(t))_{t \in \mathbb{R}}$ be a strongly continuous one parameter group of unitary operators on a Hilbert space $\mathcal{H}$. Let*

$$\mathcal{D} := \left\{ x \in \mathcal{H} : \lim_{t \rightarrow 0} \frac{U(t)x - x}{-it} \text{ exists} \right\}.$$

Then $\mathcal{D}$ is a dense subspace on $\mathcal{H}$. Furthermore, if we define an operator $H : \mathcal{D} \rightarrow \mathcal{H}$ as the limit displayed above, then H is self-adjoint. Conversely, given any densely defined self-adjoint operator H on $\mathcal{H}$, there is a unique strongly continuous one parameter group of unitary operators $(U(t))_{t \in \mathbb{R}}$ such that H and $\mathcal{D}$ arise from this group in the above manner, with $\mathcal{D} = \mathcal{D}(H)$.

The operator H , as obtained above, is called the 'infinitesimal generator' of the group $(U(t))_{t \in \mathbb{R}}$. In physics language, it is the Hamiltonian of the group. We write $U(t) = e^{-itH}$, to reflect that for any $x \in \mathcal{H}$, $U(t)x$ is in $\mathcal{D}$ and satisfies the differential equation

$$\frac{d}{dt}U(t)x = -iHU(t)x. \quad (\text{D.3.1})$$

(The assertion $U(t) = e^{-itH}$ also has an equivalent meaning via the spectral theorem, but we omit that discussion.) To see that the above equation holds, first note that if $x \in \mathcal{D}$,

then for any t ,

$$\begin{aligned} \lim_{s \rightarrow 0} \frac{U(s)U(t)x - U(t)x}{-is} &= \lim_{s \rightarrow 0} \frac{U(t)(U(s)x - x)}{-is} \\ &= U(t) \left(\lim_{s \rightarrow 0} \frac{U(s)x - x}{-is} \right) = U(t)Hx, \end{aligned}$$

where $U(t)$ could be brought outside the limit because $U(t)$ is a unitary operator (which implies that $\|U(t)x\| = \|x\|$ for all x), and $U(s)U(t) = U(t)U(s)$ because they are both equal to $U(s+t)$. This shows that $U(t)x \in \mathcal{D}$. Thus,

$$\begin{aligned} \frac{d}{dt}U(t)x &= \lim_{s \rightarrow 0} \frac{U(t+s)x - U(t)x}{s} \\ &= \lim_{s \rightarrow 0} \frac{U(s)U(t)x - U(t)x}{s} = -iHU(t)x, \end{aligned}$$

where the last identity follows from the definition of H .

Lemma D.3.2. *The set $\mathcal{D}$ is a dense subspace of $\mathcal{H}$ and H is a linear operator on $\mathcal{D}$.*

Proof. Take any $x \in \mathcal{H}$. First, we claim that $t \mapsto U(t)x$ is a uniformly continuous map on any bounded interval. To see this, let $\{t_n\}_{n \geq 1}$ and $\{s_n\}_{n \geq 1}$ be two sequences in an interval $[a, b]$ such that $t_n - s_n \rightarrow 0$. Passing to a subsequence if necessary, let us assume that both sequences converge to some $t \in [a, b]$. Then note that

$$\begin{aligned} U(t_n)x - U(s_n)x &= (U(t_n - s_n) - I)U(s_n)x \\ &= (U(t_n - s_n) - I)U(s)x + (U(t_n - s_n) - I)(U(s_n) - U(s))x \end{aligned}$$

The first term on the right tends to zero because of strong continuity. For the second term, note that

$$\|(U(t_n - s_n) - I)(U(s_n) - U(s))x\| \leq 2\|(U(s_n) - U(s))x\| \rightarrow 0,$$

again by strong continuity. Thus, $(U(t_n) - U(s_n))x \rightarrow 0$. This proves that $t \mapsto U(t)x$ is uniformly continuous on any bounded interval.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a smooth function with compact support. Let $[a, b]$ be a bounded interval containing the support of f . For a partition $\mathcal{P} = \{t_0, \dots, t_n\}$ of $[a, b]$ (where $a = t_0 < t_1 < \dots < t_n = b$), let

$$I(\mathcal{P}, f) := \frac{1}{n} \sum_{i=0}^{n-1} (t_{i+1} - t_i) f(t_i) U(t_i)x.$$

By the uniform continuity of $t \mapsto U(t)x$ on $[a, b]$, it is easy to show that $I(\mathcal{P}, f)$ has a limit as we take a sequence of partitions with $\max_{0 \leq i \leq n-1} (t_{i+1} - t_i)$ tending to zero, just as in

the construction of ordinary Riemann integrals. Let us denote this limit by

$$x_f := \int_{-\infty}^{\infty} f(t)U(t)x dt. \quad (\text{D.3.2})$$

Next, we claim that $x_f \in \mathcal{D}$ for any compactly supported smooth f . To see this, first note that by a simple argument involving the definition of the above integral, it follows that

$$\begin{aligned} U(s)x_f &= \int_{-\infty}^{\infty} f(t)U(s)U(t)x dt \\ &= \int_{-\infty}^{\infty} f(t)U(s+t)x dt = \int_{-\infty}^{\infty} f(t-s)U(t)x dt. \end{aligned} \quad (\text{D.3.3})$$

By the above identity, we get that for any $s \neq 0$,

$$\frac{U(s)x_f - x_f}{-is} = \int_{-\infty}^{\infty} \frac{f(t-s) - f(t)}{-is} U(t)x dt.$$

Again, using the definition of the integral and the above identity, it is not hard to show that

$$\lim_{s \rightarrow 0} \frac{U(s)x_f - x_f}{-is} = \int_{-\infty}^{\infty} \lim_{s \rightarrow 0} \left(\frac{f(t-s) - f(t)}{-is} \right) U(t)x dt = -i \int_{-\infty}^{\infty} f'(t)U(t)x dt. \quad (\text{D.3.4})$$

Thus, $x_f \in \mathcal{D}$ for any compactly supported smooth function f . Finally, using the uniform continuity of $t \mapsto U(t)x$ and the definition of the integral, one can show that for any compactly supported nonnegative smooth function f which integrates to 1, we have $x_{f_n} \rightarrow x$, where $f_n(t) := nf(nt)$. This shows that $\mathcal{D}$ is dense in $\mathcal{H}$. $\square$

We are now ready to prove the forward direction of Theorem D.3.1.

Proof of the forward direction of Theorem D.3.1. Let $(U(t))_{t \in \mathbb{R}}$ be a strongly continuous one parameter group of unitary operators on $\mathcal{H}$. By Lemma D.3.2, $\mathcal{D}$ is dense in $\mathcal{H}$ and H is a linear operator with domain $\mathcal{D}$. We claim that H is self-adjoint. To show this, first observe that for any $x, y \in \mathcal{D}$,

$$\langle Hx, y \rangle = \left\langle \lim_{t \rightarrow 0} \frac{U(t)x - x}{-it}, y \right\rangle = \lim_{t \rightarrow 0} \frac{1}{-it} (\langle U(t)x, y \rangle - \langle x, y \rangle),$$

where the second identity holds because the map $x \mapsto \langle x, y \rangle$ is continuous under the norm topology on $\mathcal{H}$. Since $U(t)$ is unitary, it is invertible and $U(t)^{-1} = U(t)^*$. Since $U(0) = U(0)^2$, we see that $U(0) = I$. But then $U(t)^{-1} = U(-t)$, since $U(t)U(-t) = U(0) = I$. Thus, $U(t)^* = U(-t)$. This shows that

$$\langle U(t)x, y \rangle = \langle x, U(t)^*y \rangle = \langle x, U(-t)y \rangle.$$

Plugging this into the previous display (and recalling that the inner product is conjugate

linear in the second argument), we get

$$\begin{aligned}\langle Hx, y \rangle &= \lim_{t \rightarrow 0} \frac{1}{-it} (\langle x, U(-t)y \rangle - \langle x, y \rangle) \\ &= \left\langle x, \lim_{t \rightarrow 0} \frac{U(-t)y - y}{it} \right\rangle = \langle x, Hy \rangle.\end{aligned}$$

Thus, H is a symmetric operator. We will use Theorem D.2.3 to show that H is self-adjoint. Specifically, we will use the criterion 3 from Theorem D.2.3, that is, show that $\text{range}(H \pm iI) = \mathcal{H}$.

Recall that in the proof of Lemma D.3.2, we defined integrals of the form

$$\int_{-\infty}^{\infty} f(t)U(t)dt,$$

for smooth compactly supported $f : \mathbb{R} \rightarrow \mathbb{R}$. By the same argument (and using the unitary nature of $U(t)$), it is easy to show that we can define a linear operator $R : \mathcal{H} \rightarrow \mathcal{H}$ as

$$R := \int_0^{\infty} e^{-t}U(t)dt \tag{D.3.5}$$

as a limit of Riemann sums. It is easy to justify the steps below to show that for any $x \in \mathcal{H}$ and $s \in \mathbb{R}$,

$$\begin{aligned}(U(s) - I)Rx &= \int_0^{\infty} e^{-t}U(s)U(t)xdt - \int_0^{\infty} e^{-t}U(t)xdt \\ &= \int_0^{\infty} e^{-t}U(s+t)xdt - \int_0^{\infty} e^{-t}U(t)xdt \\ &= \int_s^{\infty} e^{-(t-s)}U(t)xdt - \int_0^{\infty} e^{-t}U(t)xdt \\ &= (e^s - 1) \left(Rx - \int_0^s e^{-t}U(t)xdt \right) - \int_0^s e^{-t}U(t)xdt,\end{aligned} \tag{D.3.6}$$

where $\int_0^s$ means $-\int_s^0$ when $s < 0$. Let us now divide both sides by $-is$ and take $s \rightarrow 0$. Then note that

$$\frac{e^s - 1}{-is} \rightarrow i.$$

Since $\|U(t)x\| = \|x\|$,

$$\left\| \int_0^s e^{-t}U(t)xdt \right\| \leq \int_0^s \|e^{-t}U(t)x\|dt \leq s\|x\|.$$

Thus,

$$\lim_{s \rightarrow 0} \frac{1}{-is} (e^s - 1) \left(Rx - \int_0^s e^{-t}U(t)xdt \right) = iRx. \tag{D.3.7}$$

Next, note that

$$\frac{1}{-is} \left(\int_0^s e^{-t} U(t)x dt - sx \right) = \frac{1}{-is} \left(\int_0^s e^{-t} (U(t)x - x) dt + (1 - e^{-s} - s)x \right).$$

Now, $1 - e^{-s} - s = o(s)$ as $s \rightarrow 0$, and by strong continuity,

$$\sup_{0 \leq t \leq s} \|U(t)x - x\| = o(1)$$

as $s \rightarrow 0$. Using this in the previous display, we see that

$$\lim_{s \rightarrow 0} \frac{1}{-is} \int_0^s e^{-t} U(t)x dt = ix. \quad (\text{D.3.8})$$

Using equations (D.3.7) and (D.3.8) in equation (D.3.6), we get

$$\lim_{t \rightarrow 0} \frac{(U(s) - I)Rx}{-is} = iRx - ix.$$

Thus, $Rx \in \mathcal{D}$, and $HRx = iRx - ix$. This can be rewritten as $(H - iI)R(ix) = x$. This proves that $\text{range}(H - iI) = \mathcal{H}$. Similarly, taking

$$S := \int_0^\infty e^{-t} U(-t) dt,$$

one can show that $(H + iI)S(-ix) = x$, proving that $\text{range}(H + iI) = \mathcal{H}$. This proves that H is self-adjoint, thus completing the proof of the forward direction of the theorem. $\square$

The converse direction requires some preparation. Suppose we are given a self-adjoint operator H with domain $\mathcal{D}$. Our task is to show that there is a unique strongly continuous one parameter group of unitary operators $(U(t))_{t \in \mathbb{R}}$ such that H is the infinitesimal generator of this group. We will construct $U(t)$ through a sequence of approximations. For each $n \geq 1$, define a linear operator $T_n : \mathcal{H} \rightarrow \mathcal{H}$ as

$$T_n := (I + in^{-1}H)^{-1}. \quad (\text{D.3.9})$$

Note that this is well-defined on $\mathcal{H}$ by Theorem D.2.4, and moreover, by the same theorem,

$$\|T_n\| = n \|((-in)I + H)^{-1}\| \leq 1. \quad (\text{D.3.10})$$

For each $n \geq 1$ and $t \geq 0$, let

$$U_n(t) = e^{-nt} e^{ntT_n} := e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k T_n^k}{k!}. \quad (\text{D.3.11})$$

Note that the right side is well-defined because T_n is a bounded operator. Moreover, by

equation (D.3.10), we have

$$\begin{aligned} \|U_n(t)\| &\leq e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k}{k!} \|T_n^k\| \\ &\leq e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k}{k!} \|T_n\|^k \leq e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k}{k!} = 1. \end{aligned} \quad (\text{D.3.12})$$

The idea is very simple. Since we can expect that $T_n \approx I - in^{-1}H$ when n is large, we can also expect that

$$U_n(t) = e^{nt(T_n - I)} \approx e^{-itH}.$$

However, making this precise takes a considerable amount of work. The starting point is the following lemma.

Lemma D.3.3. *For any k, m, n and t , the operators T_k, T_m and $U_n(t)$ commute with each other. Moreover, these operators map $\mathcal{D}$ into $\mathcal{D}$, and H commutes with these operators on $\mathcal{D}$.*

Proof. Take any $x \in \mathcal{H}$ and any m, n . Let $y := T_n x$, so that $y \in \mathcal{D}$ and $(I + in^{-1}H)y = x$. Let $z := T_m y$, so that $z \in \mathcal{D}$ and $(I + im^{-1}H)z = y$. Then note that $z = T_m T_n x$, and

$$\begin{aligned} x &= (I + in^{-1}H)y \\ &= (I + in^{-1}H)(I + im^{-1}H)z \\ &= (I + im^{-1}H)(I + in^{-1}H)z, \end{aligned}$$

which shows that $z = T_n T_m x$. Thus, T_m and T_n commute for any m and n . From this and the definition of $U_n(t)$, it follows that T_k, T_m and $U_n(t)$ commute for any k, m, n, t . Lastly, take any $x \in \mathcal{D}$. Let $y := Hx$ and $z := T_n y$. Then note that $z \in \mathcal{D}$ and $y = (I - in^{-1}H)z$, so that

$$Hx = (I - in^{-1}H)z.$$

Now, let $w := T_n x$, so that $w \in \mathcal{D}$ (because T_n maps $\mathcal{H}$ into $\mathcal{D}$) and $(I + in^{-1}H)w = x$. Thus, from the above equation, we get

$$(I + in^{-1}H)z = H(I + in^{-1}H)w = (I + in^{-1}H)Hw.$$

Applying T_n on both sides, we get $z = Hw$. That is, $T_n Hx = HT_n x$. This proves that for any n , H commutes with T_n on $\mathcal{D}$. To show that $U_n(t)$ maps $\mathcal{D}$ into $\mathcal{D}$ and H commutes

with $U_n(t)$ on $\mathcal{D}$, take any $x \in \mathcal{D}$. Then note that since H commutes with T_n ,

$$\begin{aligned} U_n(t)Hx &= e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k T_n^k Hx}{k!} \\ &= e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k H T_n^k x}{k!} \\ &= \lim_{m \rightarrow \infty} H y_m, \end{aligned}$$

where

$$y_m := e^{-nt} \sum_{k=0}^m \frac{(nt)^k T_n^k x}{k!}.$$

By Theorem D.2.3, H is a closed operator. Since $y_m \rightarrow U_n(t)x$ and $H y_m \rightarrow U_n(t)Hx$ as $m \rightarrow \infty$, this shows that $U_n(t)x \in \mathcal{D}$ and $U_n(t)Hx = H U_n(t)x$. This completes the proof of the lemma. $\square$

Since $T_n(I + in^{-1}H) = I$, we have that for any $x \in \mathcal{D}$,

$$T_n(I + in^{-1}H)x = x.$$

This can be rewritten as

$$T_n Hx = in(T_n - I)x. \quad (\text{D.3.13})$$

An immediate consequence of the above identity is the following.

Lemma D.3.4. *For each $x \in \mathcal{H}$, $T_n x \rightarrow x$ as $n \rightarrow \infty$.*

Proof. From the identity (D.3.13) and the inequality (D.3.10), we have that for any $y \in \mathcal{D}$,

$$\|(T_n - I)y\| = n^{-1} \|T_n H y\| \leq n^{-1} \|H y\|,$$

and thus, $T_n y \rightarrow y$ as $n \rightarrow \infty$. Since $\mathcal{D}$ is dense and $\|T_n\| \leq 1$ for all n , this allows us to conclude that $T_n x \rightarrow x$ as $n \rightarrow \infty$ for every $x \in \mathcal{H}$, as follows. For any $x \in \mathcal{H}$ and $y \in \mathcal{D}$, and any n ,

$$\|T_n x - x\| \leq \|T_n(x - y)\| + \|T_n y - y\| + \|y - x\| \leq 2\|x - y\| + \|T_n y - y\|.$$

This implies that $\limsup_{n \rightarrow \infty} \|T_n x - x\| \leq 2\|x - y\|$. Taking $y \rightarrow x$ through $\mathcal{D}$ shows that $\lim_{n \rightarrow \infty} \|T_n x - x\| = 0$. $\square$

From the definition of $U_n(t)$, we have $U_n(t) = e^{nt(T_n - I)}$ (interpreted as a power series). A simple calculation using the power series expansion shows that for any $x \in \mathcal{H}$,

$$\frac{d}{dt} U_n(t)x = n(T_n - I)U_n(t)x. \quad (\text{D.3.14})$$

Combining (D.3.14) and (D.3.13), and recalling from Lemma D.3.3 that $U_n(t)x \in \mathcal{D}$ for any $x \in \mathcal{D}$, we get that for any $x \in \mathcal{D}$,

$$\frac{d}{dt}U_n(t)x = -iT_n H U_n(t)x. \quad (\text{D.3.15})$$

This formula for the time derivative of $U_n(t)x$ allows us to construct $U(t)$ as the limit of $U_n(t)$ via the following lemma.

Lemma D.3.5. *For each $t \geq 0$, there is a linear operator $U(t) : \mathcal{H} \rightarrow \mathcal{H}$ such that $\|U(t)\| \leq 1$, and for each $x \in \mathcal{H}$, $U_n(t)x \rightarrow U(t)x$ uniformly over t belonging to any bounded interval in $[0, \infty)$.*

Proof. Take any $x \in \mathcal{D}$. For any $0 \leq s \leq t$ and any m, n , it is easy to see from Lemma D.3.3 and the identity (D.3.15) that

$$\frac{d}{ds}U_m(t-s)U_n(s)x = iT_m H U_m(t-s)U_n(s)x - iU_m(t-s)T_n H U_n(s)x.$$

Lemma D.3.3 and the above identity imply that

$$\begin{aligned} \frac{d}{ds}U_m(t-s)U_n(s)x &= iU_m(t-s)U_n(s)T_m H x - iU_m(t-s)U_n(s)T_n H x \\ &= iU_m(t-s)U_n(s)(T_m - T_n)H x. \end{aligned} \quad (\text{D.3.16})$$

It is easy to justify rigorously (using that $U_n(0) = U_m(0) = I$) that

$$U_m(t)x - U_n(t)x = - \int_0^t \frac{d}{ds}U_m(t-s)U_n(s)x ds. \quad (\text{D.3.17})$$

Again, using this, it is easy to justify that

$$\|U_m(t)x - U_n(t)x\| \leq \int_0^t \left\| \frac{d}{ds}U_m(t-s)U_n(s)x \right\| ds. \quad (\text{D.3.18})$$

But by (D.3.12) and (D.3.16),

$$\left\| \frac{d}{ds}U_m(t-s)U_n(s)x \right\| \leq \|(T_m - T_n)H x\|.$$

Thus, we get

$$\|U_m(t)x - U_n(t)x\| \leq t\|(T_m - T_n)H x\|. \quad (\text{D.3.19})$$

But by Lemma D.3.4, we have that for any $x \in \mathcal{D}$, $\|(T_n - T_m)H x\| \rightarrow 0$ as $m, n \rightarrow \infty$. Together with equation (D.3.19), this shows that for any $s \geq 0$ and $x \in \mathcal{D}$, there exists a point $U(s)x \in \mathcal{H}$ such that

$$\lim_{n \rightarrow \infty} \sup_{0 \leq s \leq t} \|U_n(s)x - U(s)x\| = 0. \quad (\text{D.3.20})$$

It is easy to see that the map $x \mapsto U(t)x$ is linear for any t , since each $U_n(t)$ is linear.

Now, since $\|U_n(t)\| \leq 1$ for each n (by the inequality (D.3.12)), we have $\|U(t)x\| \leq \|x\|$ for each $x \in \mathcal{D}$. Since $\mathcal{D}$ is dense in $\mathcal{H}$, it is now easy to argue that $U(t)$ admits an extension to $\mathcal{H}$ that satisfies $\|U(t)\| \leq 1$, and equation (D.3.20) holds for all $x \in \mathcal{H}$. $\square$

Having constructed the family of operators $(U(t))_{t \geq 0}$, we now want to show that it has the required properties. The following lemma establishes some of the properties.

Lemma D.3.6. *The family $(U(t))_{t \geq 0}$ satisfies the following properties:*

1. $U(0) = I$.
2. For each $s, t \geq 0$, $U(s+t) = U(s)U(t)$.
3. For each $x \in \mathcal{H}$, the map $t \mapsto U(t)x$ is continuous.

Proof. Since $U_n(0) = I$ for each n , we have $U(0) = I$. It is easy to see from the definition of $U_n(t)$ that $U_n(s+t) = U_n(s)U_n(t)$ for all $s, t \geq 0$. From this and Lemma D.3.5, it follows that $U(s+t) = U(s)U(t)$. Also, it is easy to verify that $t \mapsto U_n(t)x$ is a continuous map for any x and n . For any $s, s' \in [0, t]$, $n \geq 1$ and $x \in \mathcal{H}$,

$$\begin{aligned} \|U(s)x - U(s')x\| &\leq \|U(s)x - U_n(s)x\| + \|U_n(s)x - U_n(s')x\| + \|U_n(s')x - U(s')x\| \\ &\leq \|U_n(s)x - U_n(s')x\| + 2 \sup_{0 \leq r \leq t} \|U_n(r)x - U(r)x\|. \end{aligned}$$

Thus,

$$\limsup_{s' \rightarrow s} \|U(s)x - U(s')x\| \leq 2 \sup_{0 \leq r \leq t} \|U_n(r)x - U(r)x\|.$$

Taking $n \rightarrow \infty$ on the right and using Lemma D.3.5, we see that $U(s')x \rightarrow U(s)x$ as $s' \rightarrow s$. $\square$

The next lemma shows that H is the infinitesimal generator of the family $(U(t))_{t \geq 0}$.

Lemma D.3.7. *The operator H is the infinitesimal generator of $(U(t))_{t \geq 0}$, in the sense that the set*

$$\mathcal{D}' := \left\{ x \in \mathcal{H} : \lim_{t \downarrow 0} \frac{U(t)x - x}{-it} \text{ exists} \right\} \quad (\text{D.3.21})$$

is equal to the set $\mathcal{D}$, and the limit displayed above is equal to Hx .

Proof. For $x \in \mathcal{D}'$, let $H'x$ denote the limit displayed in equation (D.3.21). We have to show that $\mathcal{D} = \mathcal{D}'$ and $H = H'$. Take any $x \in \mathcal{D}$. The identity (D.3.15) and Lemma D.3.3

imply that

$$\begin{aligned}
 U(t)x - x &= \lim_{n \rightarrow \infty} (U_n(t)x - x) \\
 &= \lim_{n \rightarrow \infty} \int_0^t \frac{d}{ds} (U_n(s)x) ds \\
 &= -i \lim_{n \rightarrow \infty} \int_0^t T_n H U_n(s)x ds \\
 &= -i \lim_{n \rightarrow \infty} \int_0^t U_n(s) T_n H x ds.
 \end{aligned}$$

Next, note that for any $s \in [0, t]$,

$$\begin{aligned}
 \|U_n(s)T_n H x - U(s)H x\| &\leq \|U_n(s)(T_n - I)H x\| + \|(U_n(s) - U(s))H x\| \\
 &\leq \|(T_n - I)H x\| + \sup_{0 \leq r \leq t} \|(U_n(s) - U(s))H x\|.
 \end{aligned}$$

Since $T_n y \rightarrow y$ for all $y \in \mathcal{H}$ (by Lemma D.3.4), and the second term on the right tends to zero as $n \rightarrow \infty$ (by Lemma (D.3.5)), this shows that

$$U(t)x - x = -i \int_0^t U(s)H x ds.$$

Consequently,

$$\|U(t)x - x - (-it)H x\| \leq \int_0^t \|(U(s) - I)H x\| ds.$$

Since $U(s)H x \rightarrow H x$ as $s \rightarrow 0$, this proves that the point x is in $\mathcal{D}'$, and $H'x = Hx$. To summarize, we have shown that $\mathcal{D} \subseteq \mathcal{D}'$, and $H = H'$ on $\mathcal{D}$. So it only remains to show that $\mathcal{D}' \subseteq \mathcal{D}$. To show this, take any $y \in \mathcal{D}'$. By Theorem D.2.3, $I + iH$ is an invertible map. In particular, there exists $x \in \mathcal{D}$ such that $(I + iH')y = (I + iH)x$, which can be rewritten as

$$x = (I + iH)^{-1}(I + iH')y = T_1(I + iH')y. \quad (\text{D.3.22})$$

Now, since T_1 is bounded, we have

$$\begin{aligned}
 T_1 H' y &= T_1 \left(\lim_{t \downarrow 0} \frac{U(t)y - y}{-it} \right) \\
 &= \lim_{t \downarrow 0} \frac{T_1 U(t)y - T_1 y}{-it}.
 \end{aligned}$$

But by Lemma D.3.3, T_1 commutes with each $U_n(t)$. Since T_1 is continuous, this shows that T_1 commutes with $U(t)$. Consequently, the above expression shows that

$$T_1 H' y = \lim_{t \downarrow 0} \frac{U(t)T_1 y - T_1 y}{-it}.$$

In particular, $T_1 y \in \mathcal{D}'$ and $T_1 H' y = H' T_1 y$. But we also know that T_1 maps into $\mathcal{D}$ (by Lemma D.3.3), and we have proved that $H = H'$ on $\mathcal{D}$. Thus, $T_1 H' y = H T_1 y$. Using this in equation (D.3.22), we get that

$$x = (I + iH)T_1 y.$$

But $T_1 = (I + iH)^{-1}$. Thus, $x = y$. This shows that $y \in \mathcal{D}$, and completes the proof. $\square$

The next lemma shows that $U(t)$ preserves inner products.

Lemma D.3.8. *For each t , the operator $U(t)$ preserves inner products, and hence, $U(t)^* U(t) = I$.*

Proof. It is easy to check that the properties of $(U(t))_{t \geq 0}$ established so far are sufficient to prove, using the same argument as the one used to prove equation (D.3.1), that equation (D.3.1) holds (provided that we interpret the derivative at 0 to be the right derivative). The only ingredient we do not have is that $U(t)$ is unitary, but we know from Lemma D.3.6 that $\|U(t)\| \leq 1$ for each t , and this suffices for the proof. Using equation (D.3.1), we get that for any $x, y \in \mathcal{H}$,

$$\begin{aligned} \frac{d}{dt} \langle U(t)x, U(t)y \rangle &= \left\langle \frac{d}{dt} U(t)x, U(t)y \right\rangle + \left\langle U(t)x, \frac{d}{dt} U(t)y \right\rangle \\ &= \langle -iH U(t)x, U(t)y \rangle + \langle U(t)x, -iH U(t)y \rangle \\ &= -i \langle H U(t)x, U(t)y \rangle + i \langle U(t)x, H U(t)y \rangle. \end{aligned}$$

Since H is a self-adjoint operator and $U(t)x, U(t)y \in \mathcal{D}$, the last expression is zero. Since $U(0) = I$, this shows that

$$\langle U(t)x, U(t)y \rangle = \langle U(0)x, U(0)y \rangle = \langle x, y \rangle.$$

Thus, $U(t)$ preserves inner products. This can be alternately written as

$$\langle x, U(t)^* U(t)y \rangle = \langle x, y \rangle$$

for all x, y . From this, we conclude that $U(t)^* U(t) = I$. $\square$

So, we have established all required properties except that $U(t)U(t)^* = I$ (which would complete the proof that $U(t)$ is unitary), and that $U(t)$ is yet undefined for $t < 0$. Towards achieving both of these goals, we define $(U(-t))_{t \geq 0}$ using $-H$ exactly in the same way that we defined $(U(t))_{t \geq 0}$ using H . It is then easy to see that the family of bounded operators $(U(t))_{t \in \mathbb{R}}$ is strongly continuous, satisfies $U(0) = I$ and $U(t)^* U(t) = I$ for each t , and the limit

$$\lim_{t \rightarrow 0} \frac{U(t)x - x}{-it}$$

exists if and only if $x \in \mathcal{D}$, and in that case, the limit is equal to Hx . Now, we need to verify that $U(s+t) = U(s)U(t)$ for all $s, t \in \mathbb{R}$. We already know this when s and t have

the same sign. For proving that this holds for all s, t , a moment's thought shows that it suffices to prove that $U(t)$ is invertible and $U(-t) = U(t)^{-1}$. To prove this, take any $t \geq 0$, and consider the sequences $\{U_n(t)\}_{n \geq 1}$ and $\{U_n(-t)\}_{n \geq 1}$ approximating $U(t)$ and $U(-t)$. Recall that for $t \geq 0$,

$$U_n(\pm t) = e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k (I \pm in^{-1}H)^{-k}}{k!}. \quad (\text{D.3.23})$$

Since H is self-adjoint, it is easy to see that the domain of the adjoint of $I - in^{-1}H$ is equal to $\mathcal{D}$, and on this domain, we have

$$(I - in^{-1}H)^* = I + in^{-1}H.$$

This implies that for any $x, y \in \mathcal{H}$,

$$\begin{aligned} \langle (I + in^{-1}H)^{-1}x, y \rangle &= \langle (I + in^{-1}H)^{-1}x, (I - in^{-1}H)(I - in^{-1}H)^{-1}y \rangle \\ &= \langle (I + in^{-1}H)(I + in^{-1}H)^{-1}x, (I - in^{-1}H)^{-1}y \rangle \\ &= \langle x, (I - in^{-1}H)^{-1}y \rangle. \end{aligned}$$

Thus, the adjoint of $(I + in^{-1}H)^{-1}$ is $(I - in^{-1}H)^{-1}$, with domain $\mathcal{H}$. By equation (D.3.23), this shows that for any $x \in \mathcal{H}$,

$$\begin{aligned} U_n(t)^*x &= e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k ((I + in^{-1}H)^{-k})^*}{k!} x \\ &= e^{-nt} \sum_{k=0}^{\infty} \frac{(nt)^k (I - in^{-1}H)^{-k}}{k!} x = U_n(-t)x. \end{aligned}$$

Taking $n \rightarrow \infty$ on both sides proves that $U(t)^*x = U(-t)x$. Thus, $U(t)^* = U(-t)$ for any $t \geq 0$. By a similar argument, $U(-t)^* = U(t)$ for all $t \geq 0$. Thus, by Lemma D.3.8 applied to $U(-t)$, we get

$$U(t)U(t)^* = U(-t)^*U(-t) = I.$$

This shows that $U(t)$ is invertible and $U(t)^{-1} = U(t)^* = U(-t)$.

This completes the construction of a strongly continuous one parameter group of unitary operators that has H as its infinitesimal generator. The only thing remaining to prove is that this group is unique. To show this, suppose that $(V(t))_{t \in \mathbb{R}}$ is another such group. Take any $x \in \mathcal{H}$. Let $w(t) := U(t)x - V(t)x$. Then by equation (D.3.1),

$$\frac{dw}{dt} = -iHw(t).$$

Thus, we get

$$\begin{aligned} \frac{d}{dt} \|w(t)\|^2 &= \left\langle \frac{dw}{dt}, w(t) \right\rangle + \left\langle w(t), \frac{dw}{dt} \right\rangle \\ &= \langle -iHw(t), w(t) \rangle + \langle w(t), -iHw(t) \rangle \\ &= -i\langle Hw(t), w(t) \rangle + i\langle w(t), Hw(t) \rangle. \end{aligned}$$

Since H is self-adjoint, the last expression is zero. But $w(0) = 0$, since $U(0) = V(0) = I$. Thus, $w(t) = 0$ for each t . This completes the proof of the converse direction of Theorem D.3.1.

D.4 Semigroups of symmetric operators

D.4.1 The Hille–Yosida theorem

Let $\mathcal{H}$ be a Hilbert space and $(P(t))_{t \geq 0}$ be a family of linear maps from $\mathcal{H}$ into $\mathcal{H}$. The family is called a *semigroup* if $P(0) = I$ and $P(s+t) = P(s)P(t)$ for all $s, t \geq 0$. The semigroup is called strongly continuous if for each $x \in \mathcal{H}$, the map $t \mapsto P(t)x$ is continuous. Our goal in this section is to prove the following theorem.

Theorem D.4.1 (One direction of the Hille–Yosida theorem). *Let $(P(t))_{t \geq 0}$ be a strongly continuous semigroup of symmetric operators defined everywhere on a Hilbert space $\mathcal{H}$. Then the set*

$$\mathcal{D} := \left\{ x \in \mathcal{H} : \lim_{t \downarrow 0} \frac{P(t)x - x}{-t} \text{ exists} \right\}$$

is a dense subspace of $\mathcal{H}$. Moreover, if we denote the above limit by Hx for $x \in \mathcal{D}$, then H is a self-adjoint linear operator with domain $\mathcal{D}$.

The operator H defined in the theorem is called the infinitesimal generator of the semigroup. Notice that the main difference with unitary groups of operators is that we keep $-t$ instead of $-it$ in the denominator when taking the limit. Accordingly, we write $P(t) = e^{-tH}$ instead of e^{-itH} .

The first step in the proof of Theorem D.4.1 is to establish the following bound on the growth rate of $\|P(t)\|$ using the uniform boundedness principle.

Lemma D.4.2. *Let $(P(t))_{t \geq 0}$ be as in Theorem D.4.1. Then there are finite positive constants C_1 and C_2 such that $\|P(t)\| \leq C_1 e^{C_2 t}$ for all t .*

Proof. By the Hellinger–Toeplitz theorem, each $P(t)$ is a bounded operator. First, we claim that there is some $t_0 > 0$ such that $\sup_{0 \leq t \leq t_0} \|P(t)\| < \infty$. Suppose that this is not true. Then we can find a sequence $t_n \rightarrow 0$ such that $\|P(t_n)\| \rightarrow \infty$. But for any $x \in \mathcal{H}$, strong continuity and the fact that $P(0) = I$ imply that $P(t_n)x \rightarrow x$. In particular, $\sup_{n \geq 1} \|P(t_n)x\| < \infty$. But then, by the uniform boundedness principle, $\sup_{n \geq 1} \|P(t_n)\|$ must be finite, giving us a contradiction. This proves the claim.

Let t_0 be as above. Take any $t > 0$. Then $t = nt_0 + s$, where $n := \lfloor t/t_0 \rfloor$ and $s := t - nt_0 \in [0, t_0)$. Thus, by the semigroup property,

$$\|P(t)\| \leq \|P(t_0)^n P(s)\| \leq \|P(t_0)\|^n \|P(s)\| \leq K^{n+1},$$

where $K := \sup_{0 \leq s \leq t_0} \|P(s)\|$. Note that $K \geq 1$, since $P(0) = I$. Since $n \leq t/t_0$, this proves the lemma. $\square$

Lemma D.4.2 allows us to prove the following analogue of Lemma D.3.2.

Lemma D.4.3. *The domain $\mathcal{D}$ defined in Theorem D.4.1 is dense in $\mathcal{H}$.*

Proof. The bound given by Lemma D.4.2, together with the strong continuity of the semigroup, show that for any $x \in \mathcal{H}$, the map $t \mapsto P(t)x$ is uniformly continuous on any bounded time interval. The proof is exactly the same as the proof of the analogous claim in the proof of Lemma D.3.2. The rest of the proof also goes through as before. $\square$

Clearly, H is a linear operator on the domain $\mathcal{D}$. Moreover, for any $x, y \in \mathcal{D}$, the fact that $P(t)$ is symmetric for all t implies that

$$\begin{aligned} \langle Hx, y \rangle &= \left\langle \lim_{t \rightarrow 0} \frac{P(t)x - x}{-t}, y \right\rangle = \lim_{t \rightarrow 0} \frac{1}{-t} (\langle P(t)x, y \rangle - \langle x, y \rangle) \\ &= \lim_{t \rightarrow 0} \frac{1}{-t} (\langle x, P(t)y \rangle - \langle x, y \rangle) \\ &= \left\langle x, \lim_{t \rightarrow 0} \frac{P(t)y - y}{-t} \right\rangle = \langle x, Hy \rangle. \end{aligned}$$

Thus, H is a symmetric operator on the domain $\mathcal{D}$.

Now, let C_1 and C_2 be as in Lemma D.4.2. Let $\alpha := C_2 + 1$. By Riemann sum approximation as in the proof of Lemma D.3.2, we can define the operator

$$\int_0^t e^{is-\alpha s} P(s) ds$$

for any given $t \geq 0$. Since

$$\|e^{is-\alpha s} P(s)\| = e^{-\alpha s} \|P(s)\| \leq C_1 e^{-s},$$

we can now take $t \rightarrow \infty$ and show that the operator

$$R := \int_0^\infty e^{is-\alpha s} P(s) ds$$

is well-defined bounded linear operator on $\mathcal{H}$.

Lemma D.4.4. *For any $x \in \mathcal{H}$, we have $Rx \in \mathcal{D}$ and $HRx = (i - \alpha)Rx + x$.*

Proof. Let $\beta := i - \alpha$. Then note that for any $x \in \mathcal{H}$ and $s \geq 0$,

$$\begin{aligned} (P(s) - I)Rx &= \int_0^\infty e^{\beta t} P(s)P(t)x dt - \int_0^\infty e^{\beta t} P(t)x dt \\ &= \int_0^\infty e^{\beta t} P(s+t)x dt - \int_0^\infty e^{\beta t} P(t)x dt \\ &= \int_s^\infty e^{\beta(t-s)} P(t)x dt - \int_0^\infty e^{\beta t} P(t)x dt \\ &= (e^{-\beta s} - 1) \left(Rx - \int_0^s e^{\beta t} P(t)x dt \right) - \int_0^s e^{\beta t} P(t)x dt. \end{aligned}$$

Dividing both sides by $-s$, taking $s \downarrow 0$, and using the strong continuity of the semigroup and the bound given by Lemma D.4.2, we get the desired result. $\square$

We are now ready to complete the proof of Theorem D.4.1.

Proof of Theorem D.4.1. Take any $x \in \mathcal{H}$. Then by Lemma D.4.4,

$$x = (H + \alpha I - iI)Rx.$$

Thus, $\text{range}(H + \alpha I - iI) = \mathcal{H}$. By a similar argument, $\text{range}(H + \alpha I + iI) = \mathcal{H}$. But $H + \alpha I$ is a symmetric linear map defined on the domain $\mathcal{D}$, which is dense by Lemma D.4.3. By Theorem D.2.3, this shows that $H + \alpha I$ is self-adjoint. From this, it is easy to deduce that H is self-adjoint. $\square$

We end this section with the following lemma, which is often useful for proving strong continuity of operator semigroups.

Lemma D.4.5. *Let $(P(t))_{t \geq 0}$ be any collection of linear operators from a Hilbert space $\mathcal{H}$ into itself. Let S be a subset of $\mathcal{H}$ whose span is dense in $\mathcal{H}$. Suppose that (a) for any $T \geq 0$, $\sup_{0 \leq t \leq T} \|P(t)\| < \infty$, and (b) for any $x \in S$, the map $t \mapsto P(t)x$ is continuous. Then $t \mapsto P(t)x$ is continuous for any $x \in \mathcal{H}$.*

Proof. Take any $x \in \mathcal{H}$, $t \geq 0$, and a sequence of nonnegative real numbers $\{t_n\}_{n \geq 1}$ converging to t . Let T be so large that $t_n \in [0, T]$ for all n , and let $C := \sup_{0 \leq s \leq T} \|P(s)\|$. Take any $\epsilon > 0$, and any y in the span of S such that $\|x - y\| \leq \epsilon$. Then for any n ,

$$\begin{aligned} \|P(t_n)x - P(t)x\| &\leq \|P(t_n)x - P(t_n)y\| + \|P(t_n)y - P(t)y\| + \|P(t)y - P(t)x\| \\ &\leq C\|x - y\| + \|P(t_n)y - P(t)y\| + C\|y - x\| \\ &\leq 2C\epsilon + \|P(t_n)y - P(t)y\|. \end{aligned}$$

Since y is a finite linear combination of the elements of S , it follows from the hypothesis that $P(t_n)y \rightarrow P(t)y$ as $n \rightarrow \infty$. Thus

$$\limsup_{n \rightarrow \infty} \|P(t_n)x - P(t)x\| \leq 2C\epsilon.$$

Since ϵ is arbitrary, this completes the proof. $\square$

D.4.2 Analytic continuation of semigroups

Let $(P(t))_{t \geq 0}$ be a strongly continuous semigroup of symmetric operators on a Hilbert space $\mathcal{H}$. By Theorem D.4.1, we know that the semigroup has a generator H , which is a densely defined self-adjoint operator on $\mathcal{H}$. By Stone's theorem, this implies that one can construct a strongly continuous one-parameter group of unitary operators $(U(t))_{t \in \mathbb{R}}$ on $\mathcal{H}$ which also has H as its generator. Is there a simple way to calculate the action of $U(t)$ on a vector x if we know how to calculate the action of $P(t)$ on x ? The formal representations $P(t) = e^{-tH}$ and $U(t) = e^{-itH}$ suggest that one should somehow try to 'analytically continue' $P(t)x$, as a function of t , to the imaginary axis to obtain $U(t)x$. The following theorem makes this precise.

Theorem D.4.6. *Let $(P(t))_{t \geq 0}$ be a strongly continuous semigroup of symmetric operators on a Hilbert space $\mathcal{H}$. Let H be its generator, and let $U(t) := e^{-itH}$, as obtained from Stone's theorem. Then for any $x, y \in \mathcal{H}$, the map $t \mapsto \langle P(t)x, y \rangle$ has an analytic continuation to the open right half-plane $\Omega := \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$. This analytic continuation extends continuously to the boundary of Ω — that is, the imaginary axis — as the map $it \mapsto \langle U(t)x, y \rangle$. For any given $t \in \mathbb{R}$ and $x \in \mathcal{H}$, the values of $\langle U(t)x, y \rangle$ obtained by this procedure, taken over all $y \in \mathcal{H}$, jointly determine $U(t)x$ by the Riesz representation theorem for Hilbert spaces.*

We need several lemmas. Throughout, $\mathcal{D}$ will denote the domain of H .

Lemma D.4.7. *For any $t \geq 0$, the operator $P(t)$ maps $\mathcal{D}$ into $\mathcal{D}$. Similarly, for any $t \in \mathbb{R}$, $U(t)$ maps $\mathcal{D}$ into $\mathcal{D}$.*

Proof. Take any $x \in \mathcal{D}$ and $t \geq 0$. Let $y := P(t)x$. Then for any $s > 0$,

$$\frac{P(s)y - y}{-s} = \frac{P(s)P(t)x - P(t)x}{-s} = \frac{P(t)(P(s)x - x)}{-s}.$$

By the Hellinger–Toeplitz theorem, $P(t)$ is a continuous map. Thus, the above identity implies that

$$\lim_{s \downarrow 0} \frac{P(s)y - y}{-s} = P(t) \left(\lim_{s \downarrow 0} \frac{P(s)x - x}{-s} \right) = P(t)Hx.$$

Thus, $y \in \mathcal{D}$. A similar argument shows that $U(t)x \in \mathcal{D}$. □

The above lemma implies that the following statement makes sense.

Lemma D.4.8. *For any $s \geq 0$ and $t \in \mathbb{R}$, the operators $P(s)$, $U(t)$ and H commute with each other on $\mathcal{D}$.*

Proof. From the proof of Lemma D.4.7, we already have that H commutes with $P(s)$ and $U(t)$ on $\mathcal{D}$. Next, recall the operators T_n and $U_n(t)$ defined in the proof of Stone's theorem (for $t \geq 0$), in equations (D.3.9) and (D.3.11), respectively. We first claim that

$P(s)$ commutes with T_n on $\mathcal{D}$. To see this, take any $x \in \mathcal{D}$. Let $y := T_n x = (I + in^{-1}H)^{-1}x$, so that $y \in \mathcal{D}$ and $x = (I + in^{-1}H)y$. Since $P(s)$ commutes with H on $\mathcal{D}$, we have

$$P(s)x = P(s)(I + in^{-1}H)y = (I + in^{-1}H)P(s)y = (I + in^{-1}H)P(s)T_n x.$$

Applying T_n on both sides, we get $T_n P(s)x = P(s)T_n x$. Thus, $P(s)$ commutes with T_n . Since $U_n(t)$ is the limit of linear combinations of powers of T_n , this shows that $P(s)$ commutes with $U_n(t)$. But by Lemma D.3.5, $U_n(t)x \rightarrow U(t)x$ as $n \rightarrow \infty$, for any $x \in \mathcal{H}$. Therefore, $U_n(t)P(s)x \rightarrow U(t)P(s)x$, and by the continuity of $P(s)$, $P(s)U_n(t)x \rightarrow P(s)U(t)x$. This proves that $P(s)$ commutes with $U(t)$ on $\mathcal{D}$ for $t \geq 0$. A similar argument works for $t \leq 0$. $\square$

We are now ready to prove Theorem D.4.6.

Proof of Theorem D.4.6. Fix some $x, y \in \mathcal{H}$. For any point $u = s + it \in \overline{\Omega}$, define the operator

$$P(u) := P(s)U(t).$$

Let $f(u) := \langle P(u)x, y \rangle$. We claim that f is analytic on Ω . To prove this, take any $u = s + it \in \Omega$. First, let us prove the claim under the assumption that $x \in \mathcal{D}$. For $\epsilon > 0$, define

$$\begin{aligned} R(\epsilon) &:= \|P(\epsilon)x - x + \epsilon Hx\|, \quad S(\epsilon) := \|U(\epsilon)x - x + i\epsilon Hx\|, \\ A(\epsilon) &:= \|P(\epsilon)Hx - Hx\|, \quad B(\epsilon) := \|U(\epsilon)Hx - Hx\|. \end{aligned}$$

By Lemma D.4.2, there is a finite constant C such that $\|P(s')\| \leq C$ for all $s' \in [0, s+1]$. Thus, for any $0 \leq a \leq b \leq s+1$,

$$\begin{aligned} \|P(b)x - P(a)x + (b-a)HP(a)x\| &= \|P(a)(P(b-a)x - x + (b-a)Hx)\| \\ &\leq CR(b-a). \end{aligned}$$

On the other hand, if $0 \leq b \leq a \leq s+1$, then by the above inequality, we get

$$\begin{aligned} &\|P(b)x - P(a)x + (b-a)HP(a)x\| \\ &\leq \|P(a)x - P(b)x + (a-b)HP(b)x\| + |a-b|\|(P(a) - P(b))Hx\| \\ &\leq CR(a-b) + |a-b|\|P(b)(P(a-b) - I)Hx\| \\ &\leq CR(a-b) + C|a-b|A(|a-b|). \end{aligned}$$

Combining the above inequalities, we see that for any $a, b \in [0, s+1]$,

$$\|P(b)x - P(a)x + (b-a)HP(a)x\| \leq CR(|a-b|) + C|a-b|A(|a-b|). \quad (\text{D.4.1})$$

Next, since $U(a)$ is a unitary operator for each a , we have that for all $-\infty < a \leq b < \infty$,

$$\|U(b)x - U(a)x + i(b-a)HU(a)x\| = \|U(a)(U(b-a)x - x + i(b-a)Hx)\| \leq S(b-a).$$

On the other hand, if $-\infty < b \leq a < \infty$, then the above inequality gives

$$\begin{aligned} & \|U(b)x - U(a)x + i(b-a)HU(a)x\| \\ & \leq \|U(a)x - U(b)x + i(a-b)HU(b)x\| + |a-b|\|(U(a) - U(b))Hx\| \\ & \leq S(a-b) + |a-b|\|U(b)(U(a-b) - I)Hx\| \\ & \leq S(a-b) + |a-b|\|(U(a-b) - I)Hx\| \leq S(a-b) + |a-b|B(a-b). \end{aligned}$$

Combining, we get that for any $a, b \in \mathbb{R}$,

$$\|U(b)x - U(a)x + i(b-a)HU(a)x\| \leq S(|a-b|) + |a-b|B(|a-b|). \quad (\text{D.4.2})$$

Then for any $u' = s' + it' \in \Omega$, where $s' \in (0, s+1]$, we have by Lemma D.4.8 that

$$\begin{aligned} P(u')x - P(u)x &= P(s')U(t')x - P(s)U(t)x \\ &= P(s')(U(t')x - U(t)x) + U(t)(P(s')x - P(s)x). \end{aligned}$$

By Lemma D.4.8) and the inequalities (D.4.1) and (D.4.2), this implies that

$$\begin{aligned} & \|P(u')x - P(u)x + (u' - u)HP(u)x\| \\ &= \|P(s')(U(t')x - U(t)x) + U(t)(P(s')x - P(s)x) \\ & \quad + (s' - s)HU(t)P(s)x + i(t' - t)HP(s)U(t)x\| \\ &\leq \|P(s')(U(t')x - U(t)x + i(t' - t)HU(t)x)\| + \|i(t' - t)U(t)(P(s') - P(s))Hx\| \\ & \quad + \|U(t)(P(s')x - P(s)x + (s' - s)HP(s)x)\| \\ &\leq CS(|t' - t|) + C|t' - t|B(|t' - t|) + |t' - t|A(|s' - s|) + CR(|s' - s|) + C|s' - s|A(|s' - s|). \end{aligned}$$

Since $R(\epsilon)$ and $S(\epsilon)$ are $o(\epsilon)$ as $\epsilon \downarrow 0$, and $A(\epsilon)$ and $B(\epsilon)$ are $o(1)$ as $\epsilon \rightarrow 0$, the above bound is $o(|u' - u|)$ as $u' \rightarrow u$. This shows that f is differentiable at u when $x \in \mathcal{D}$, with

$$f'(u) = -\langle HP(u)x, y \rangle.$$

Next, take any $x \in \mathcal{H}$. Since $\mathcal{D}$ is dense in $\mathcal{H}$, we can find a sequence $\{x_n\}_{n \geq 1}$ converging to x . Let $f_n(u) := \langle P(u)x_n, y \rangle$, so that each f_n is analytic on Ω . But for any compact set $K \subseteq \Omega$,

$$\sup_{u \in K} |f_n(u) - f(u)| \leq \|P(u)(x_n - x)\| \|y\| \leq C \|x_n - x\| \|y\|,$$

where $C := \sup_{u \in K} \|P(u)\|$. Since C is finite, this shows that $f_n \rightarrow f$ uniformly on K . Since this is true for any compact set $K \subseteq \Omega$, we conclude that f is analytic on Ω .

Lastly, let us show that f is continuous at the boundary of Ω . Take any $t \in \mathbb{R}$ and

$u' = s' + it' \in \Omega$ such that $s' \leq 1$. Then

$$\begin{aligned} |f(it) - f(u')| &\leq \|U(t)x - U(t')P(s')x\| \|y\| \\ &\leq \|U(t)x - U(t')x\| \|y\| + \|U(t')(P(s')x - x)\| \|y\| \\ &= \|U(t)x - U(t')x\| \|y\| + \|P(s')x - x\| \|y\|. \end{aligned}$$

By the strong continuity of U , $U(t')x \rightarrow U(t)x$ as $t' \rightarrow t$, and by the strong continuity of P , $P(s')x \rightarrow x$ as $s' \rightarrow 0$. Thus, $f(u') \rightarrow f(it)$ as $u' \rightarrow it$.

Lastly, to apply the Riesz representation theorem, simply recall that for any given x , the map $y \mapsto \langle U(t)x, y \rangle$ is linear and continuous. The above procedure produces this continuous linear map, which determines the value of $U(t)x$. $\square$

D.4.3 Contracting semigroups

Let $\mathcal{H}$ be a Hilbert space and let $(P(t))_{t \geq 0}$ be a strongly continuous semigroup of symmetric operators on $\mathcal{H}$. By Theorem D.4.1, such a semigroup has a self-adjoint generator H . The semigroup is called *contracting* if $\|P(t)\| \leq 1$ for each $t \geq 0$. Contracting semigroups arise from Markov processes (as shown by Theorem B.5.5) and are important in rigorous constructions of quantum field theories. The following result shows that generators of contracting semigroups are positive operators.

Theorem D.4.9. *Let $(P(t))_{t \geq 0}$ be a strongly continuous semigroup of symmetric, contracting operators on $\mathcal{H}$, and let H be its generator. Then $\langle Hx, x \rangle \geq 0$ for all x in the domain of H , and the spectrum of H is contained in $[0, \infty)$.*

Proof. Let $\mathcal{D}$ denote the domain of H . Take any $\alpha < 0$. We have to show that $H - \alpha I : \mathcal{D} \rightarrow \mathcal{H}$ is an invertible map with a bounded inverse. Since $\|P(t)\| \leq 1$ for each $t \geq 0$, it is easy to show that the following operator is well-defined:

$$R := \int_0^\infty e^{\alpha t} P(t) dt.$$

Then, as in the proof of Lemma D.4.4, we have that for any $x \in \mathcal{H}$,

$$(P(s) - I)Rx = (e^{-\alpha s} - 1) \left(Rx - \int_0^s e^{\alpha t} P(t)x dt \right) - \int_0^s e^{\alpha t} P(t)x dt.$$

Dividing both sides by $-s$, taking $s \downarrow 0$, and using the strong continuity of the semigroup and the contraction property of the semigroup, we get that $Rx \in \mathcal{D}$ and $HRx = \alpha Rx + x$. Thus,

$$(H - \alpha I)Rx = x, \tag{D.4.3}$$

which shows that $H - \alpha I$ is a surjective map from $\mathcal{D}$ onto $\mathcal{H}$. Next, let us show that $H - \alpha I$

is injective. Note that for any $t > 0$ and any $x \in \mathcal{H}$,

$$\langle P(t)x, x \rangle \leq \|P(t)x\| \|x\| \leq \|x\|^2.$$

Thus, we have

$$\left\langle \frac{P(t)x - x}{-t}, x \right\rangle = \frac{\langle P(t)x, x \rangle - \|x\|^2}{-t} \geq 0.$$

Taking $t \rightarrow 0$ and applying the continuity of the inner product in the first argument, we get that $\langle Hx, x \rangle \geq 0$ for any $x \in \mathcal{D}$. In particular, if $x \in \ker(H - \alpha I)$, then this shows that

$$\alpha \|x\|^2 = \langle Hx, x \rangle \geq 0.$$

But $\alpha < 0$. Thus, x must be zero. This shows that $H - \alpha I$ is injective. Finally, note that by equation (D.4.3), $R : \mathcal{H} \rightarrow \mathcal{D}$ is the inverse of $H - \alpha I$, and by the contracting property of $P(t)$,

$$\|R\| \leq \int_0^\infty e^{\alpha t} dt < \infty.$$

This completes the proof. □

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